1

INTRODUCTION TO INTEGRATED SCIENCE

Specific Objectives

After completing this chapter, you will be able to:

- Explain the various interrelated branches of science and their importance to life.
- Apply the scientific method to arrive at scientific solutions to everyday problems.
- Follow basic safety precautions in the laboratory.

INTRODUCTION

Science refers to a system of acquiring knowledge through observation and experimentation to describe and explain natural activities and phenomena. The term science also refers to the organized body of knowledge people have gained using that system. Science often describes any systematic field of study or the knowledge gained from it.

There are two major fields of science:

- Pure or basic science, involves scientists working primarily at academic institutions to pursue research simply to satisfy the thirst for knowledge.

- Applied science, involves scientists working on practical issues based on researches conducted.

Our discussions will focus mostly on the pure science, which includes:

Physics: Deals with the fundamental constituents of the universe, the forces they exert on one another, and the results produced by these forces. Branches under Physics include Mechanics, Electricity, Quantum Mechanics, Nuclear Physics, High-Energy Physics, etc.

Chemistry: The study of the composition, structure, properties, and interactions of matter.

Some branches under Chemistry are Biochemistry, Organic Chemistry, inorganic Chemistry, Analytical Chemistry, Physical Chemistry etc.

Biology: This is referred to as the science of life. It is the study of living organisms (i.e. how they live and interact with each other). In studying living organisms (which are made up of plants and animals) biologists put them into groups known as classification. Branches under Biology are Genetics, Ecology, Zoology, Taxonomy, Entomology etc.

Earth Science: The field of study concerned with the planet Earth or one or more of its parts. It includes the study of the atmosphere, biosphere (i.e. the zone at or near Earth's surface that supports life), hydrosphere (i.e. the ice, water, and water vapour at or near Earth's surface), lithosphere (i.e. the solid portion of Earth) and space beyond the atmosphere. Branches under Earth Science include Environmental Science, Meteorology, Geography, Hydrology, Oceanography, etc.

THE CONCEPT OF INTEGRATED

As the name implies, Integrated Science as a subjects taught at the senior high school level combines the major areas of science, which includes biology, chemistry, physics, mathematics and Agricultural Science. Gaining basic knowledge in the above areas helps one to solve some basic problems.

For example, in controlling a malaria epidemic, the knowledge in Biology will help to know the cause of the disease. The knowledge in Chemistry will be used in creating a vaccine. In making the vaccine presentable, the knowledge in Physics is used, and then finally, coming out with a dosage involves the calculative knowledge in Mathematics.

In Integrated Science, these areas collectively have been further grouped into five sections, namely:

1. Diversity of Matter
2. Cycles
3. Systems
4. Energy
5. Interactions in Nature

GROUP ACTIVITY:

Students to work in groups to design and draw diagrams to show the inter-relationship between the various branches of Science and Technology.

Groups to display their work in class for discussion.

CAREERS IN SCIENCE AND TECHNOLOGY

Arguably, there are more career paths and specialties under science and technology than there are in any other discipline. Some of the careers open to science and technology are as follows:

Engineering

Aerospace Engineer, Electrical and Electronic Engineer, Civil Engineer, Environmental Engineer, Industrial Engineer, Marine Architect Engineer, Material Scientist and Engineer, Nuclear Engineer, Mechanical Engineer, Petroleum Engineer, etc

Fig 1.0: An engineer in a laboratory

Mathematics and Computer Science

Computer Hardware Engineer, Mathematician, Multimedia Artist, Network Systems and Data Communications Analyst, Software Quality Assurance Engineer, Statistician, etc

Life Science

Animal Breeder, Medical Doctor, Anthropologist, Biochemist, Biologist; Dietitian or Nutritionist, Natural Sciences Epidemiologist, Health Educator, Physician, Microbiologist, Pharmacist, Physical Therapist, Nuclear Medicine Technologist, Veterinarian, Zoologist, Wildlife Biologist, Plant Scientist, etc.

Physical Science

Astronomer; Audio and Video Equipment; Technician; Aviation Inspector; Chemist; Chemical Technician; Electrician; Film and Video Editor; Food Science Technician; Pilot; Food Scientist or Technologist; Physicist; Forensic Science Technician; Nuclear Power Reactor Operator; Occupational Health and Safety Specialists; Power Distributors and Dispatcher; Power Plant Operator; Ship and Boat Captain; Sound Engineering Technician

Earth & Environmental Science

Aquacultural Manager; Diver; Geographer Emergency Management Specialist; Hydrologist; Environmental Compliance Inspector; Geoscientist; Industrial Health & Safety Engineer; Meteorologist; Park Range; Soil Scientist Soil and Water Conservationist; Surveyor; etc.

All these and many more are the career opportunities available in science.

SOME PROMINENT SCIENTISTS

In Ghana:

- Professor Nii Narku Quaynor: A scientist and engineer who has played an important role in the introduction and development of the Internet throughout Africa.

- Professor Francis Kofi Ampenyin Allotey: Physicist, Mathematician. Pioneered the introduction of computer science in Ghana. Worked as the chairman of Ghana Atomic Energy Commission and at the International Atomic Energy Agency (IAEA). Also chaired the Council of Scientific and Industrial Research.

- Dr. Isaiah Blankson: Physicist; engineers of jet and rocket powered engines. Works at the National Aeronautic and Space Administration (NASA).

- Prof. Kwesi Andam: Civil and Structural Engineer. Former Vice Chancellor of KNUST.

- Prof Ivan Addae-Mensah: former Vice Chancellor of the University of Ghana. He chaired the National Petroleum Authority.

Prominent International Scientists

- Sir Isaac Newton: Discovered gravity and the law of gravitation. He also propounded the three famous laws of motion and the laws of friction.

- Albert Einstein: He was responsible for the special and general theory of relativity, and his work on the photoelectric effect. He also invented a few devices like Einstein calculator.

- Charles Babbage: He was an English mathematician, philosopher, inventor and mechanical engineer who came up with the concept of a programmable computer and is said to have invented the first mechanical computer.

- Thomas Edison: An American inventor who developed several devices like the electric light bulb and the phonograph.

THE SCIENTIFIC METHOD

The scientific method is a step by step experimentation process that is used to explore observations and answer questions. Scientists use the scientific method to search for cause and effect relationships in nature. In other words, they design an experiment so that changes to one item cause something else to vary in a predictable way.

The steps to the scientific method are:

i. **State the problem:** You cannot solve a problem until you know exactly what it is. For example My Problem is – my back-pack is too heavy.

ii. **Research the problem:** What will it take to solve my problem? What do I know, and need to know, about my problem?
To solve my problem, "I have to examine the content of the bag".

iii. **Form a hypothesis:** A possible solution to my problem. The simplest solution is often the best solution!
"the cause of the overweight is the extra books I bought today".

iv. **Test the hypothesis:** Perform an experiment to see if your hypothesis works.
"remove the extra books from the bag".

v. **Draw conclusions from the data:** Data are the results of an experiment. In its simplest form, there are only two possibilities:

If your hypothesis was correct, you will now have a lighter bag. PROBLEM SOLVED!
On the other hand, if your hypothesis was incorrect, the experiment failed.
DON'T GIVE UP! DO MORE RESEARCH!

What was wrong with your original hypothesis?
Did you make a poor selection? Was your experiment flawed? Form another hypothesis based on additional research.
Test your new hypothesis.
Continue this process until the problem is solved.

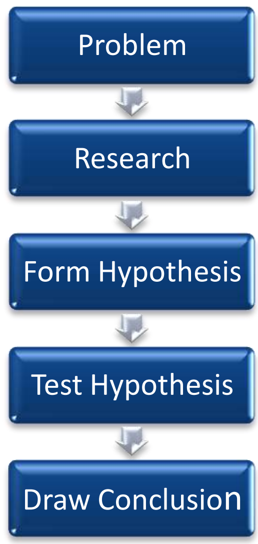


Fig. 1.1: The scientific method

Application of scientific method

- ❖ ***State the problem*:** People get sick from drinking from the well.
- ❖ ***Research the problem***: Is it really the water from the well that is making them sick? Test the water from the well.
- ❖ ***Form a hypothesis***: The cause of the disease is the water from the well.
- ❖ ***Test the hypothesis*:** Get a sample of the water from the well, treat it and distribute it.

- ***Draw conclusion***: The disease stopped after the people had drunk the treated water. Therefore, the hypothesis is correct – the water from the well is not well treated to be drunk.

If after treating the water sample and given it out to the people the disease still persists, another hypothesis has to be drawn.

The scientific method has been used over the years by scientists in coming out with great inventions and discoveries.

The discovery of penicillin

The research of Alexander Fleming in 1928 led to the discovery of penicillin, an important antibiotic derived from the mould *Penicillum notatum.*

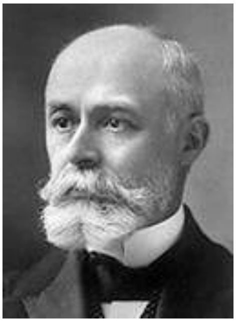
Alexander Fleming used the scientific method in discovering penicillin. His observation that the mould contaminating one of his culture plates had destroyed the bacteria laid the basis for the development of penicillin therapy.

Sir Alexander Fleming

Penicillin is effective against a wide range of disease-causing bacteria. Penicillin acts by killing bacteria directly or by inhibiting their growth.

The discovery of radioactivity

The discovery of radioactivity was another breakthrough in science. The phenomenon was discovered in 1896 by the French physicist Antoine Henri Becquerel when he observed that the element uranium can blacken a photographic plate, although separated from it by glass or black paper. He also observed that the rays that produce the darkening are capable of discharging an electroscope, indicating that the rays possess an electric charge.



Antoine Henri Becquerel

In 1898 the French chemists Marie Curie and Pierre Curie deduced that radioactivity is a phenomenon associated with atoms, independent of their physical or chemical state. They also deduced that because the uranium-containing ore pitchblende is more intensely radioactive than the uranium salts that were used by Becquerel, other radioactive elements must be in the ore. They carried through a series of chemical treatments of the pitchblende that resulted in the discovery of two new radioactive elements, polonium and radium. Marie Curie also discovered that the element thorium is

Marie Curie

radioactive, and in 1899 the radioactive element actinium was discovered by the French chemist André Louis Debierne

In that same year the discovery of the radioactive gas radon was made by the British physicists Ernest Rutherford and Frederick Soddy, who observed it in association with thorium, actinium, and radium.

The discovery of flotation

Archimedes, a Greek mathematician and inventor, live around 287-212 B.C. The era when science was not very popular. Archimedes is credited with important contributions to the development of physics. He is known for applying science to everyday life, developing practical inventions such as the lever and the screw. These simple machines have found uses as diverse as warfare and irrigation. Archimedes is said to have discovered the principle of water displacement while taking a bath, shouting "Eureka!" when he realized why his body caused the level of the water to rise.

In all the discoveries discussed above, the scientific method was invariably employed. It goes to show that (real) scientists cannot do away with the scientific method.

PROJECT WORK
Students to form small groups to investigate a common problem in the community e.g. frequent flooding, and apply scientific method to arrive at a solution to the problem. Students can also select problems of interest to them and investigate.

SAFETY PRECAUTIONS IN THE LABORATORY

For an experiment to be successful, one needs a well equipped laboratory. Even though they cannot be done without; nevertheless, equipments, structures and chemicals in the laboratory may be hazardous to humans. Therefore care should be taken at the laboratory.

The following guidelines give some precautions to be taken at the laboratory:

1) Wear protective clothing: laboratory coat, shoes and goggles are to be worn to protect the skin, feet and eyes respectively.

2) Do not eat or drink in the laboratory: food samples in the laboratory may be treated with poisonous substances which when eaten may harm the body.

3) Do not smell an unknown gas: some gases are very poisonous or choking and if smelled may cause some discomfort to the body.

4) Point the mouth and opening of test tubes and bottles away from yourself or other people: this is to prevent spilling chemicals and other substances on people.

5) Do not touch chemicals with bare hands: some chemicals are corrosive and if they come in contact with the body may burn the skin.

6) Do not touch hot substances with bare hands: a test tube holder, tong or cloth is the best apparatus for holding hot thing at the laboratory. This will prevent burns.

7) Wipe off any spilled substance: if a substance spills on the floor, make sure to wipe it dry in order to prevent slippage and falling.

8) Rinse off any chemical which comes in contact with your skin: If a chemical is accidentally spilled on you, or you accidentally touch a chemical, make sure that it is rinsed off with plenty of water.

9) Study and read all safety signs, warning and precautions at the laboratory before using it.

10) Follow every instruction to the latter: always listen to the teacher or instructor before doing or touching anything.

PROJECT WORK
Carry out demonstrations on safety precautions in the use of the Laboratory; laboratory equipment, chemicals etc.

TEST QUESTIONS

1. State five precautions to be taken at the laboratory.

2. (a) What is the scientific method?
   (b) Mention the steps in the scientific method.
   (c) Discuss how you will use the scientific method in controlling congestion in our streets.

3. Discuss the concept of integrated science.

4. Give one reason why each of the following precautions should be taken in the laboratory:
   (a) Do not eat or drink in the laboratory.
   (b) Do not smell an unknown gas.
   (c) Do not touch chemicals with bare hands.
   (d) Wipe off any spilled substance.

(e) Point the mouth and opening of test tubes and bottles away from yourself or other people.

(f) Rinse off any chemical which comes in contact with your skin

2

MEASUREMENT

Specific Objectives

After completing this chapter, you will be able to:

- Use the SI unit in all measurements.
- Use scientific measuring instruments accurately.
- Measure density and relative density.

INTRODUCTION

If all the countries in the world are ever going to speak one language then that will be promoted by measurement.

Measurement is the process of calculating the size, rate, effect etc. of something by comparing their quantity to a fixed quantity.

Quantities which can be measured include length, mass, temperature, time, etc. These are known as ***physical quantities***.
Knowing the correct measurements of some quantities was a global problem until the introduction and adoption of a new system of units measurement known in French as the **Le *Systēme International d'Unitēs*** **(International System of Units)**, abbreviated to **SI.**

SI units are described as ***standard units***. A standard unit is a unit that is understood and accepted globally.

BASIC AND DERIVED UNITS

In the S.I system, there are two main units of measurements – the basic and the derived units.

Basic units of the SI system

Basic units are the units from which other units can be obtained.

They are also known as the ***fundamental units***. In the SI system, there are seven basic units.
Each of the basic unit has a symbol and an SI unit in which it is measured.

Table 1.0: Base quantities, their corresponding SI units, and their symbols

Base quantity	SI unit	SI unit symbol
Length	metre	m
mass	kilogram	kg
time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Amount of substance	mole	mol

Derived units

Apart from the base units, there are other units which are derived from the base units. These are called the derived units. ***Derived units are units which are obtained from the base units.***

It is a combination of two or more base units or a base unit and a derived unit.

Table 1.1: Derived quantities and their SI units

Quantity	SI unit	Symbol	Expressed in SI unit
Force	Newton	N	$N = kg.m.s^{-2}$
Work and energy	joule	J	J= N.m
Power	watt	W	$W = J.s^{-1}$
Quantity of electricity	coulomb	C	C = A.s
Electric potential	volt	V	$V = W.A^{-1}$
Electric resistance	ohm	Ω	$W = V. A^{-1}$
Volume	cubic metre	m^3	V = m.m.m

USING MEASURING INSTRUMENTS

Different quantities have different instruments with which they are measured.

The instruments used to measure length are the metre rule, the surveyor's tape, measuring tape, ruler etc.

The vernier calliper measure short distances, internal and external diameters of hollow objects. It measures to the precision of 0.1 mm.

The micrometer screw gauge on the other hand is used to measure extremely short distances to the precision of 0.01 mm. It can therefore be used to measure the thickness of paper, cloth, etc. The vernier calliper and the micrometer screw gauge are known as the precision instruments.

Thermometers are used to know the temperature of an object.
Hydrometers measure the densities of liquids.

Balances are used to measure the mass of an object. Some examples of balances are beam balance, top pan balance, electronic balance.
The volumes of liquids are measured with measuring cylinder, pipette, burette, volumetric flask, beaker etc.

Table 1.2: Some quantities and the instruments used to measure them

Quantity	Measuring instruments	Unit	Sub-unit
Length	Metre rule surveyor's tape Vernier calliper micrometer screw gauge	metre (m)	centimetre (cm) millimetre (mm)
Mass	Beam balance Electronic balance Lever balance	kilogram (kg)	gram (g)
Time	Stop clocks and watches Wrist watch Wall clock	second (s)	millisecond (ms)
Volume	Pipette burette Measuring cylinder	cubic centimetre (cm^3) or milliliter (ml)	cubic millimetre (mm^3)
Temperature	Absolute thermometer Celsius thermometer Clinical thermometer	Kelvin (K) degrees celsius (C)	
Atmospheric pressure	Mercury barometer Aneroid barometer	pascal (Pa)	
Electric potential	voltmeter	volt (V)	millivolt (mV)
Electric current	ammeter	ampere (A)	milliampere (mA)
Luminous intensity	photometer	Candela (cd)	
Amount of substance		mole (mol)	

Concept of replication of results

It is very important for experiments to be repeated by various scientists. This is because a mistake or miscalculation by one scientist will hardly be repeated by others. Therefore replication of result is necessary to:

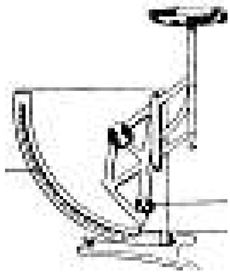
- Make better predictions
- Add to scientific knowledge
- Ensure accurate and consistent result
- Arrive at hypothesis

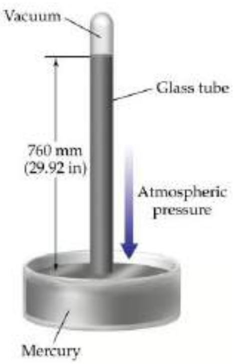
Human values that are of importance to science

1. Honesty
2. Integrity
3. Truthfulness
4. Resourcefulness
5. Patience
6. Amicability


beam balance


electronic balance


lever balance

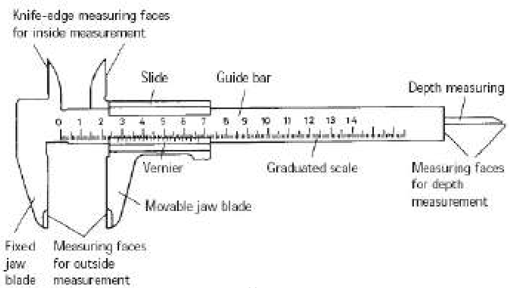


mercury barometer


stop watch


ammeter

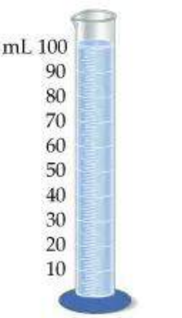
voltmeter



vernier calliper

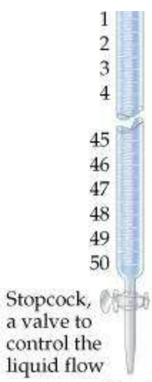

micrometer screw gauge


surveyor's tape



Graduated cylinder

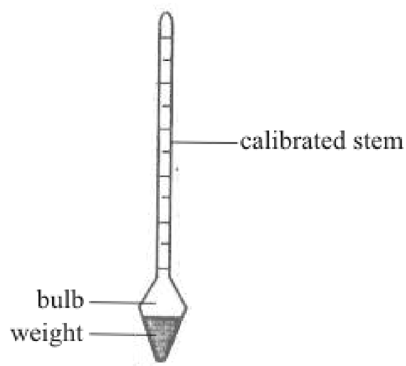

Syringe



Buret


Pipet


Volumetric flask



hydrometer

Fig. 1.3: Basic measuring instruments

How to obtain the unit of some quantities

Volume of a:

Rectangular block = $l \times b \times h = m \times m \times m = m^3$

Cylinder = $\pi r^2 h = m \times m \times m = m^3$

Sphere = $\frac{4}{3}\pi r^3 h = m^3$

Cone = $\frac{1}{3}\pi r^2 h = m^3$

Work (J) = *force* x *distance* = *Nm*

Potential difference (V) = $\frac{work\ done}{charge} = \frac{J}{C}$

Quantity of electricity (C) = *Current* x *Time*= *As*

Electrical resistance (Ω) = $\frac{potential\ difference}{current}$

$= \frac{V}{A}$

Power (W) = $\frac{workdone}{time} = \frac{J}{s}$

Force (N)= *mass* x *acceleration* = *kg* x ms^{-2}

Area = *length* x *breadth* = $l \times b = m^2$

Velocity = $\frac{change\ of\ displacement}{time} = ms^{-1}$

PROJECT WORK

Use a ruler, balances, stop watches, thermometer, ammeter, measuring cylinder, calipers, pipettes, burette, hydrometer etc. to measure quantities in various units.

DENSITY

The density of a substance is it mass per unit volume.

Density is expressed mathematically as:

$$Density = \frac{mass}{volume} \quad \text{or} \quad \rho = \frac{M}{V}$$

Where, ρ = density, M = mass, and V = Volume.

Density is measured in kilogram per cubic metre (kg/m^3 or kgm^{-3}).

Remember that the SI unit of mass is kg, and that of volume is m^3. Therefore mass per volume (mass/volume) is kg/m^3.

Measuring the Density of Liquid

- Measure the mass of a clean measuring cylinder as M_1
- Pour a known volume of liquid into the measuring cylinder.
- Record that as V.
- Measure the mass of both beaker and liquid. Record it as M_2

Hence:

Mass of liquid (M) = Mass of beaker and liquid (M_2) – Mass of beaker (M_1)

i.e. $M = M_2 - M_1$

Volume of liquid = V

Therefore, density of liquid (ρ) = $\frac{Mass}{volume}$

$$\rho = \frac{(M2 - M1)}{V}$$

Density of regular objects

A regular object is an object which has uniform dimensions that can be determined with a ruler.

To find the density of a regular object, example a book,

- Find its volume by multiplying its length, breadth and height.
- With the aid of a balance, measure the mass of the book.

Hence, $density = \dfrac{mass\ \ of\ \ the\ book}{volume\ \ of\ \ the\ book}$

Density of Irregular Objects

Irregular objects are objects which do not have definite dimensions. Examples are stone, fragment of glass or metal, a piece of coal, etc.

Procedure:

- Measure and record the mass of the object as *M*
- Pour water into a measuring cylinder and record the volume as V_1
- Tie a string around the irregular object and lower it gently into the measuring cylinder.
- Record the level of water as V_2.

Calculation

Mass of the object = *M*

Volume of the object = $V_2 - V_1$

Density (ρ) of the irregular object =

Mass (*M*) of the object/ Volume ($V_2 - V_1$)

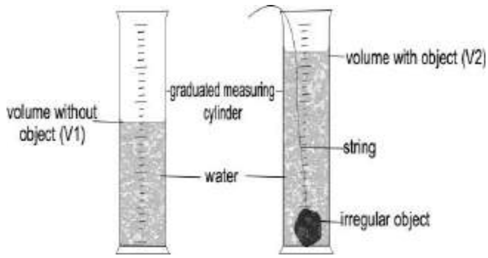


Fig. 1.4*:* Determining density of irregular objects

Alternative Method

- Measure and record the mass of the irregular object as *M*.
- Fill a Eureka can with water till it overflows.
- Place an empty measuring cylinder below the sprout of the Eureka can.
- Tie a string around the irregular object and lower it gently into the Eureka can.
- Record the volume of water that overflows into the measuring cylinder as *V*.

Calculation

Density (ρ) of the irregular object

$= \dfrac{mass\ \ (M)}{\text{Volume}\ \ (\text{V})}$

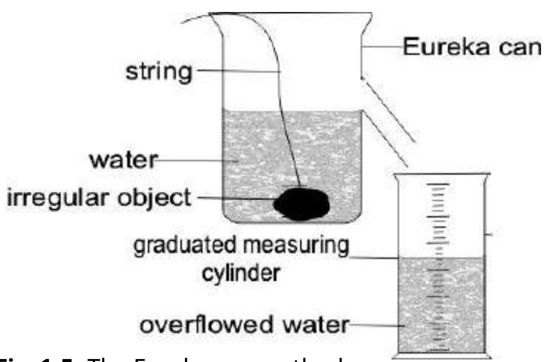


Fig. 1.5*:* The Eureka can method

Density of a Soil Sample

- Measure and record the mass of an empty rectangular box as M_1.
- Fill up the rectangular box with the soil sample.
- Measure and record the mass of the box containing the soil sample as M_2.
- Hence, the mass (M) of the soil sample $= M_2 - M_1$

The volume is determined by measuring the length, breath and height of the inside of the rectangular box. This is done because the box is filled up with the soil sample. Hence, the volume (V) of the soil sample = Lx B x H of the inside of the rectangular box.

Density of soils sample = $\frac{Mass}{\text{Volume}}$

$$\rho = \frac{(M_2 - M_1)}{V}$$

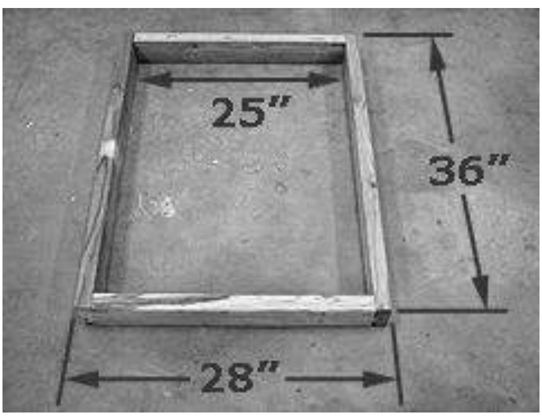


Fig. 1.6: Determining density of soil sample

Alternative method

- Measure and record the mass of the soil sample as M.
- Pour water into a measuring cylinder.
- Record the volume of water as V_1.
- Pour the soil sample into the water and record the new volume as V_2.
- The volume, V, of the soil sample $= V_2 - V_1$

Density of the soil sample = $\frac{Mass}{\text{Volume}}$

$$\rho = \frac{M}{V_2 - V_1}$$

NB: To find the mass of a substance, make mass the subject of the relation $\rho = M/V$ by multiplying through by V. i.e. $M = \rho V$

To make volume the subject, multiply through the relation by V and divide through by ρ; i.e. $V = M / \rho$

Solved questions

1. Find the density of a piece of rock which measures 70 g, and when put into a measuring cylinder containing 32 cm^3 of water, the water level rose to 67 cm^3.

Solution:

Mass of the rock = 70 g

Volume of water without rock = 32 cm^3 (V_1)

Volume of water with rock = 67 cm^3 (V_2)

Volume of stone = $V_2 - V_1$

= 67 - 32 = 35 cm^3

Hence, the density of the rock = $\frac{mass}{volume}$

Substituting:

Density (ρ) = $\frac{70}{35}$ = 2 gcm^{-3}

2. The mass of a ball bearing is 2.0 kg, its volume is recorded as 1.6 m^3. Calculate its density.

Solution:

Mass of the ball bearing = 2.0 kg

Volume of ball bearing = 1.6 m^3

Therefore, density of the ball berry

$= \frac{Mass\ of\ ball\ bearing}{\text{volume of ball bearing}}$ or ρ = $\frac{M}{V}$

$\rho = \frac{20}{1.6}$ = 1.25 kgm^{-3}

3. A sphere of diameter 0.8 m, has a mass of 16 kg. Determine its density. (Volume of sphere = $\frac{4}{3\pi r^3}$;' $\pi = \frac{22}{7}$)

Solution:

Half of a diameter is a radius. Therefore, to find the radius of the sphere, divide its diameter by 2.

Thus, $\frac{0.8}{2}$ = 0.4 m

Hence, volume of the sphere

$= \frac{4}{3} \text{ x } \frac{22}{7} \text{ x } 0.4^3 = 0.268 \text{ m}^3$

Density = $\frac{mass}{volume}$

$= \frac{16}{0.268}$

= 59.70 kg/m^3

4. The bulb of a simple pendulum has a volume and a mass of 38 cm^3 and 19 g. Find its density in gcm^{-3} and kgm^{-3}.

Solution:

Density = $\frac{mass}{volume} = \frac{19}{38}$ = 0.5 gcm^{-3}

To convert gcm-^{3} to kgm-^{3}, multiply by 1000.

Thus 0.5 gcm-^{3} to kgm^{-3} will be

0.5 x 1000 = 500

Therefore density in kgm^{-3} = 500 kgm^{-3}

Alternative method

Density in kgm^{-3}.

Mass = 19 g = $\frac{19}{1000} = 19 \times 10^{-3}$

Volume = 38 cm^3 = $\frac{30}{1000000} = 30 \times 10^{-6}$

$\therefore$ Density = $\frac{19 \times 10^{-3}}{38 \times 10^{-6}}$ = 500 kgm^{-3}

5. The density of a bag of cement which has a mass of 50 kg is 3.2 kg/m^3. Find the volume of the bag of cement.

Solution:

Mass = 50 kg

Volume= ?

Density = 3.2 kg/m^3

$$\text{Density}(\rho) = \frac{mass}{volume}$$

We have to find the volume so we make *V* the subject of the relation.

$$\therefore \text{Volume } (V) = \frac{mass}{density}$$

$$V = \frac{50}{3.2} \quad = 15.625 \text{ m}^3$$

RELATIVE DENSITY

Relative density is the ratio of the density of a substance to the density of equal volume of water.

Relative density of a substance is the comparison of the density of the substance to the density of water. Mathematically,

$$\textbf{Relative density} = \frac{\textbf{density of substance}}{\textbf{density of water}}$$

Given the volume of a substance and water only, relative density can be expressed as:

$$\textbf{Relative density} = \frac{\textbf{volume of substance}}{\textbf{volume of water}}$$

Since volume substance is the same as volume of water, it can be expressed as:

$$\textbf{Relative density} = \frac{\textbf{mass of substance}}{\textbf{mass of water}}$$

Hence,

Relative density can be defined as ratio of mass of a substance to the mass of equal volume of water.

Mass of a body is proportional to its weight, therefore, with that it can be said that:

Relative density is the ratio of the weight of a substance to the weight of equal volume of water.

$$\textbf{Relative density} = \frac{\textbf{weight of substance}}{\textbf{weight of water}}$$

Relative density has no units.

Measuring relative density of an irregular object

Procedure:

- Measure and record the mass of the irregular object as M_1.
- Measure and record the mass of an empty beaker as M_2.
- Fill a Eureka can to overflow.
- Place the empty beaker below the spout of a Eureka can.
- Tie the irregular object with a thread and lower it gently into the Eureka can.
- Measure the mass of the beaker and the overflowed water as M_3.

Calculation:

$$\text{Relative density} = \frac{mass\ of\ object}{mass\ of\ water}$$

$$\text{RD} = \frac{M1}{(M3 - M2)}$$

Some substances, such as liquids and powdered substances, do not have definite shapes, therefore, the relative density bottle is used to measure their relative densities.

- ❖ The relative density bottle has a stopper with a fine hole through it which enables excess liquids to run through it when the stopper is inserted.
- ❖ If the density bottle is used with the same liquid level at the top of the hole, the volume will be the same no matter the liquid that is put in, provided there is a constant temperature.

Fig. 1.7: RD bottle

NB: To ensure accurate measurements, the relative density bottle must always be wiped clean of any liquid, and must not be handled with warm hands to avoid the lost of liquid through expansion. Get rid of air bubbles by shaking, tapping or rotating the bottle gently.

Measuring Relative Density of a Liquid

- ❖ Weigh and record the mass of an empty relative density bottle as M_1.
- ❖ Fill the bottle with the liquid, put the stopper in place and wipe the bottle dry. Measure and record the mass of both liquid and bottle as M_2.
- ❖ Pour out the liquid from the bottle, rinse it with water, fill it with water, put the stopper on and wipe it dry.
- ❖ Measure and record the mass of the bottle with water as M_3.

Calculation:

- ❖ Relative density of liquid = mass of liquid / mass of water
- ❖ But, mass of empty bottle = M_1
- ❖ mass of bottle and liquid = M_2
- ❖ mass of bottle and water = M_3

Therefore, mass of liquid = $M_2 - M_1$

Mass of water = $M_3 - M_1$

Hence, Relative density = $\frac{(M2 - M1)}{(M3 - M1)}$

Measuring the Relative Density of Powdered Solid or Granule

- ❖ Measure and record the mass of a empty relative density bottle as M_1.
- ❖ Fill the bottle about 1/3 full with the powdered solid, measure and record the mass as M_2.
- ❖ Add water to the content of the bottle to the full; measure the bottle with its contents and record the mass as M_3.
- ❖ Empty the bottle of its contents, rinse with water, fill it with water, measure and record the mass as M_4.

Calculation:

- ❖ Relative density = $\frac{mass\ of\ solid}{mass\ of\ water}$

- Mass of powdered solid = $M_2 - M_1$
- Mass of water = $M_4 - M_1$
- Mass of water = $M_3 - M_2$
- Mass of equal volume of water = $(M_4 - M_1) - (M_3 - M_2)$

Relative density of powdered solid = $\frac{(M2 - M1)}{(M4 - M1) - (M3 - M2)}$

NB: Density of water is usually expressed as 1 gcm^{-3}.
1 gcm^{-3} = 1000 kgm^{-3} = 1.0x 10^3 kgm^{-3}

Solved questions

1. A relative density bottle which usually measures 2000 g was found to measure 5000 g when filled with alcohol. It was later found to measure 7000 g when filled with water. Find the density of alcohol in kilograms?
   (Density of water = 1000 kgm^{-3})

Solution

Relative density of alcohol

$= \frac{mass\ \ of\ \ alcohol}{Mass\ \ of\ \ equal\ \ volume\ \ of\ \ water}$

Mass of alcohol = 7000 – 2000 = 5000g
Mass of water = 5000 – 2000 = 3000g

$$RD\ of\ alcohol = \frac{5000}{3000} = 1.67$$

Since relative density = $\frac{density\ \ of\ \ substance}{density\ \ of\ \ water}$

Density of alcohol =
relative density x *density of water*
= 1.67 x 1000
= 1670 gcm^{-3}

Converting 1670 g into kilograms,

$$\frac{1670}{1000} = 1.67\ kg$$

Therefore the density of alcohol
= 1.67 kgm^{-3}

2. The weight of an empty relative density bottle is 15 N. if the weight increases to 32 N when filled with pure water, and 37 N when filled with kerosene. Determine the:
   i. Volume of the bottle filled with water
   ii. Density of the kerosene.

(Density of water = 1000 kgm^{-3}; g = $10m^{-2}$)

Solution:

Weight of pure water = 32 – 15 = 17 N
Weight of kerosene = 37 – 15 = 22 N

i. $$Density = \frac{mass\ \ of\ \ substance}{volume\ \ of\ \ substance}$$

$$Volume = \frac{mass}{density}$$

Since we do not know the mass of the liquid we will use the following formula to find it

Weight = mass x gravity

Weight of liquid = 17 N;

gravity = 10 ms^{-1}

$$\text{Mass} = \frac{weight}{gravity} = \frac{17}{10}$$

M = 1.7 kg

Now back to the formula:

$$Volume = \frac{Mass}{Density}$$

$$V = \frac{1.7}{1000}$$

V = 0.0017 m^3 or 1.7 x 10^{-3}

ii. Density of kerosene

Relative density =

$$\frac{density\ \ of\ substance}{density\ \ of\ water}$$

However, since we have weight instead:

$$Relative\ density = \frac{Weight\ of\ substance}{weight\ of\ water}$$

$RD = \frac{17}{22}$; $RD = 0.772$

Back to the formula:

Relative density =

$$\frac{density\ \ of\ substance\ \ (DS)}{density\ \ of\ water\ \ (DW)}$$

Density of substance (DS) = RD x DW

DS = 0.772 x 1000

Density of the kerosene = 772. 72 kg/m^3

Table 1.3: Differences between density and relative density

Density	**Relative density**
Has a unit (kgm^{-3})	Has no unit
A measured quantity	Comparison between two measured quantities
Mass per unit volume of a substance	Mass of substance compared to that of equal volume of water
Involves only one substance	Involves two substances
Calculations may contain errors	Calculations are often accurate

TEST QUESTIONS

1. (a) Distinguish between basic and derived units.
   (b) State four basic and four derived units with their corresponding SI units.

2. Mention the uses of the measuring instruments in the table below

Measuring Instrument	*Uses*
Micrometer screw gauge	
Stop clock	
Metre rule	
Thermometer	
Barometer	
Hydrometer	
Beam balance	
Ammeter	
Newtonmetre	

Voltmeter	
Photometer	
Measuring cylinder	
Vernier caliper	

3. (a) Mention two differences between density and relative density.
   (b) Describe briefly how you will determine the density of a stone.
4. A piece of rock with mass 15 g was placed in a measuring cylinder containing 62 cm^3 of water. Supposing the water level increased to 85cm^3, what is the density of the stone:
   (a) In gram
   (b) In kilogram

5. A clean and empty relative density bottle of mass 26.0 g weighs 42 g when filled with and unknown liquid of density 6.3 g/cm^3.
   (a) Calculate the volume of the bottle.
   (b) If another liquid of density 3.6 g/cm^3 is used to fill the bottle, find the mass of the liquid that will fill the bottle.
6. State the S.I unit and an instrument you will use to measure the following quantities:
   (a) length
   (b) volume
   (c) mass
   (d) mass
   (e) temperature
   (f) electric current
   (g) luminous intensity
   (h) voltage
   (i) electric resistance

7. Describe how you will:
   (a) Determine the density of an irregular object using a eureka bottle;
   (b) Calculate the relative density of an unknown liquid.

3

DIVERSITY OF LIVING AND NON-LIVING THINGS

Specific Objectives

After completing this chapter, you will be able to:

- Use the SI unit in all measurements.
- Use scientific measuring instruments accurately.
- Measure density and relative density.

INTRODUCTION

Everything in the world falls under either of the two categories – living and non-living things, based on their characteristics. All objects which fall into the category of living things must have life in them. In other words, they must be able to go through basic life processes like respiration, nutrition, excretion, etc. Examples of living thing are: man, lion, chicken, cat, goat, dog, etc

On the other hand, all objects which fall under the non-living things category are referred to as inanimate (not living). Example of non-living things are: rock, table, car, shoe, etc.

LIFE PROCESSES

Living things go through seven basic life processes or activities. They are:

Respiration: The process by which living organisms break down food substances in the body to release energy with or without oxygen.

Nutrition: The process by which an organism gets and uses food.

Excretion: The process by which an organism removes metabolic waste products from the body.

Movement: The ability of an organism to leave one place to another or to change position of its body or parts of the body.

Irritability/ Sensitivity: The ability of living organisms to react to physical changes around them.

Growth: The irreversible increase in size and weight of living organisms.

Reproduction: The ability of organisms to give rise to young ones of the same kind.

Table 1.4: Differences between living and non-living things

Living things	Non-Living things
Respires to release energy	Do not respire
Feed	Do not feed
Move on their own	Cannot move on their own
Get rid of metabolic waste products	Do not excrete
Respond to external stimuli.	Do not respond to stimuli
Grow	Cannot grow
Give rise to young ones of the same kind.	Do not reproduce

Differences between plants and animals

When we talk of living organisms, we are talking mainly of plants and animals. Despite their similarities, plants and animals have differences.

Table 1.5: Differences between plants and animals

Plants	Animals
Take in carbon dioxide during the day and give out oxygen	Take in oxygen, and give out carbon dioxide at all time
Prepare their own food through photosynthesis	Get their food from plants and other animals
Only some parts can move	Can move freely from place to place
Do not have special excretory organs	Have special excretory organs for excretion
React slowly to stimuli	React quickly to stimuli
No uniform growth	All the parts grow

CLASSIFICATION OF ORGANISMS

In general, the term classification is the process of sorting out things and putting them into groups based on their common characteristics.

Biological classification, on the other hand, is the process of sorting out living organisms into groups based on their common characteristics.

Importance of classification

1. Allows things to be described using a few words.
2. Helps to easily identify and study organisms.
3. Brings order in naming and identifying organisms.

4. Helps to differentiate one organism from the other.
5. Allows new organisms to be identified.
6. Helps to know the relationship between organisms both within the same or different groups.
7. Allows easy communication among biologists.
8. Brings out potential uses of living and non-living things.

CLASSIFICATION IN BIOLOGY

There are different methods of classifying biological information. Modern systems of biological classification descend from the thought presented by the Greek philosopher ***Aristotle***, who was the first scientist to put organisms into groups.

Aristotle classified organisms based on their physical characteristics such as shape, size, colour etc. He also grouped animals according to their modes of movement. That is, he placed all land animals (such as man, dog, rabbit, lion etc.) in one group and flying animals (such birds, bats, insetcs etc into another.

The most commonly used system of classification today is the Linnaeus' classification system called ***the natural system of classification***, which was developed by *Carolus Linnaeus*, a Swedish biologist.

His system is based on natural relationship and brought together organisms according to the things they have in common. Organisms are grouped on the basis of their body structure.

According to Linnaeus' system, every organism has got its own two-part Latin name called the ***Binomial System*** of nomenclature (naming). The first part of the name is the *genus* to which the organism belongs; and it always begins with a capital letter. The second part of the name is the *species* to which the organism belongs. It begins with a small letter. Table 1.6 gives examples of the two-part Latin name.

Table 1.6: Examples of the binomial system

Common Name	Genus	Species
Man	Homo	sapiens
Domestic dog	Carnies	familiaris
House fowl	Gallus	domesticus
Lion	Panthera	loo
Mango	Magnifera	indica
Maize	Zea	mays
Tiger	Panthera	tigris
African elephant	Loxodota	cyclotis / africana
Chimpanzee	Pan	troglodytes/ paniscus
Hippopotamus	Hippopotamus	amphibius

Under the natural system of classification, all organisms belong to seven ranks. The ranks are arranged in increasing order of similarities among organisms as it descends. The organisms at the top level (kingdom) have low resemblance whiles the organisms at the bottom level (species) have high resemblance.

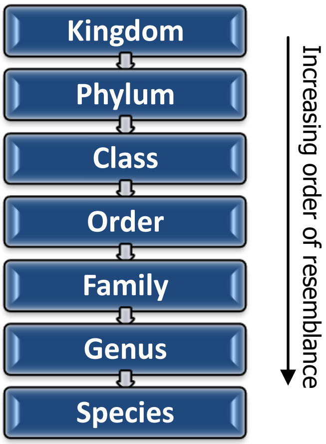


Fig. 1.8: Order of classification

General characteristics in biological classification

Another system of classification puts organisms into five groups:

1. Kingdom Prokaryotae
2. Kingdom Protoctista
3. Kingdom Fungi
4. Kingdom Plantae
5. Kingdom Animalia

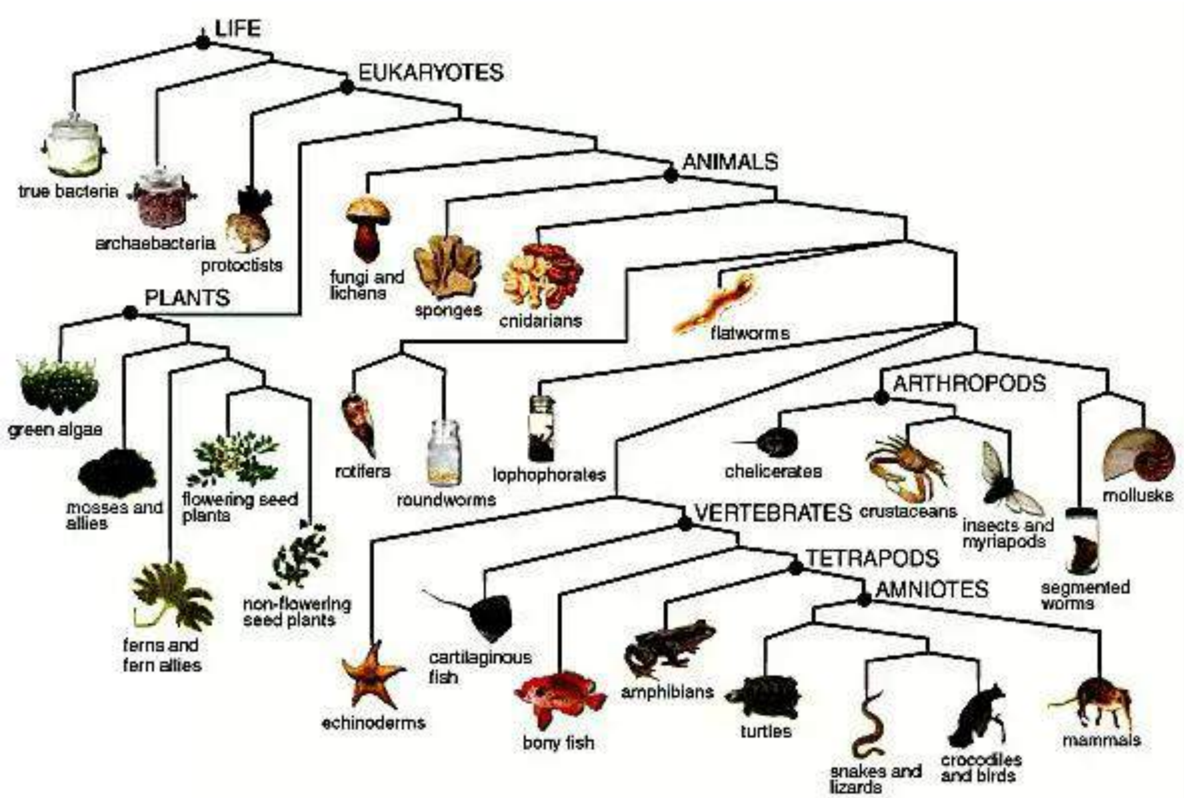


Fig. 1.9: Classification of organisms

Kingdom Prokaryotae (Monera)

Examples are bacterial, blue-green algae.

Characteristics of organisms in Kingdom Prokaryotae

1. They are unicellular (single celled)
2. They are microscopic (Very tiny)
3. Have no definite nucleus
4. Cytoplasm is surrounded by membrane
5. Have no membrane around the nucleus

Kingdom Protoctista (Protista)

Examples are Amoeba, Paramecium, plasmodium, Spirogyra, Euglena, green algae, slime mould, Trypanosoma.

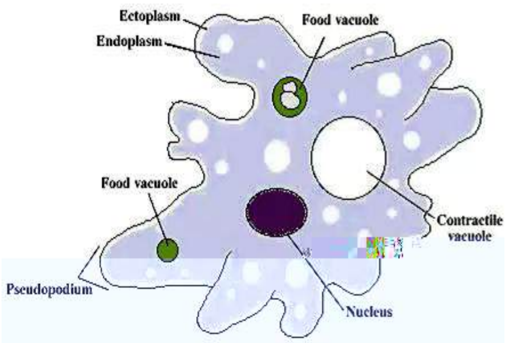


Fig. 2.0: Amoeba

Characteristics of organisms in Kingdom Protoctista

1. Mostly unicellular (few are multicellular)
2. Have membrane around nucleus (Eukaryotic)
3. Have normal cells, but no tissue or organs.

Kingdom Fungi

Examples are fungus, moulds, mushrooms, rhizopus, penicillium, mucor, yeast.

Characteristics of organisms in Kingdom Fungi

1. They are multicellular
2. They have no roots, stem or leaves
3. Have no chlorophyll
4. Are not photosynthetic (cannot produce their own food).
5. Reproduce by mean of spore formation
6. Vegetative part, called mycelium consist of hyphae.

Fig. 2.1: Mushroom

Kingdom Plantae

Examples of organisms in Kingdom Plantae are: flowering plants, ferns, shrubs, mosses etc.

Characteristics of organisms in Kingdom Plantae

1. They are multicellular (have many cells)
2. Have chlorophyll

3. Have definite nucleus
4. Cell wall made of cellulose
5. They are photosynthetic (prepare their own food through photosynthesis)

Kingdom Animalia

Examples of organisms in Kingdom Animalia are: man, birds, fish, snakes

Characteristics of organisms in Kingdom Animalia

1. They are multicellular
2. Are not photosynthetic
3. Have no cell wall
4. Have nucleus bound by a membrane
5. Do not have chlorophyll

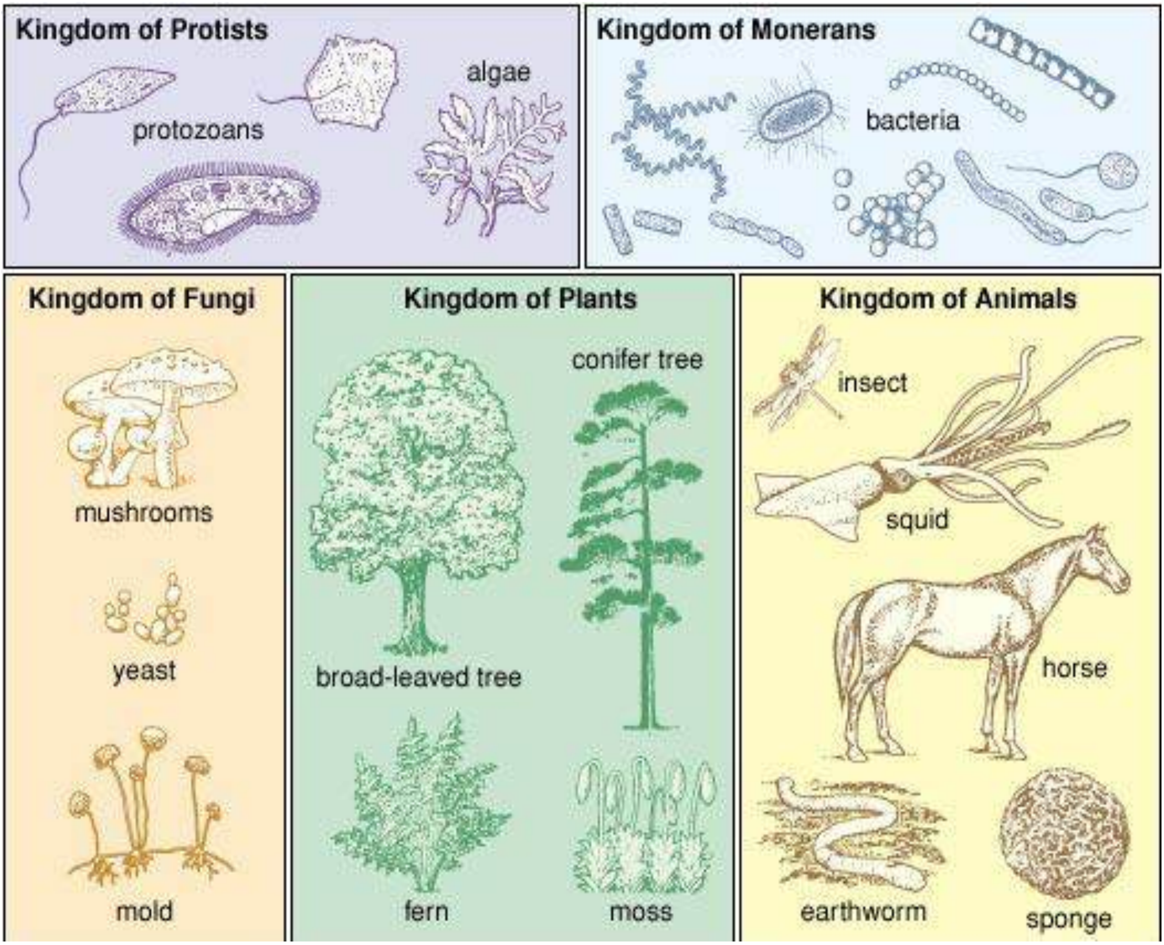


Fig. 1.9: Classification of organisms

VIRUSES

Viruses are tiny organisms that exhibit the characteristics of both living and non-living things. They are so small that only microscopes can be used to see them.

Viruses do not fall into any of the kingdoms above, because they are not entirely living organisms.

A virus is made up of a strand of deoxyribonucleic acid (DNA) or ribonucleic acid (RNA), contained in a sheath of protein.

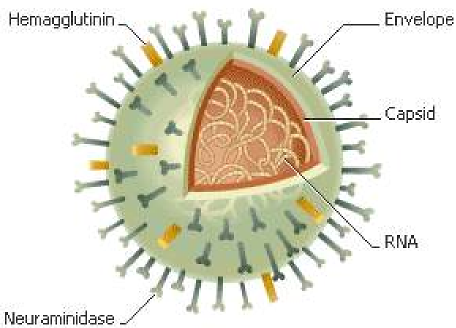


Fig. 2.0: Structure of a virus

Characteristics of viruses

1. They reproduce when they are within other living cells.
2. They are made up of protein and either DNA or RNA.
3. They cannot grow or reproduce on their own.
4. They crystallize when outside a living host or cell.
5. They lack definite nucleus.

All known viruses cause diseases in both plants and animals; examples include poliomyelitis, yellow fever, ebola, foot and mouth disease in cattle, mosaic disease in cassava and tobacco.

Classification in chemistry

In chemistry the system of classification of elements is called the ***periodic system***. The elements are arranged in a table called the ***periodic table***. The elements are arranged in order of increasing atomic number. Elements with similar properties are place in vertical columns (called groups) and horizontal rows (called periods). Elements have been classified as metal, non-metals and semi-metals.

Classification of social groups

A group of humans may also be classified into social groups. Human societies can be grouped as family, clan, ethnic group, class, etc.

Family: A social group which consist of a husband a wife and child or children.

Clan: A number of families who believe to have come from a common ancestor.

Ethnic group/Tribe: A collection of clans. Member in an ethnic group may live in the same area, speak the same language, have the same culture, etc.

Class: A collection of people who have a common characteristics like education, wealth, job, power, etc.

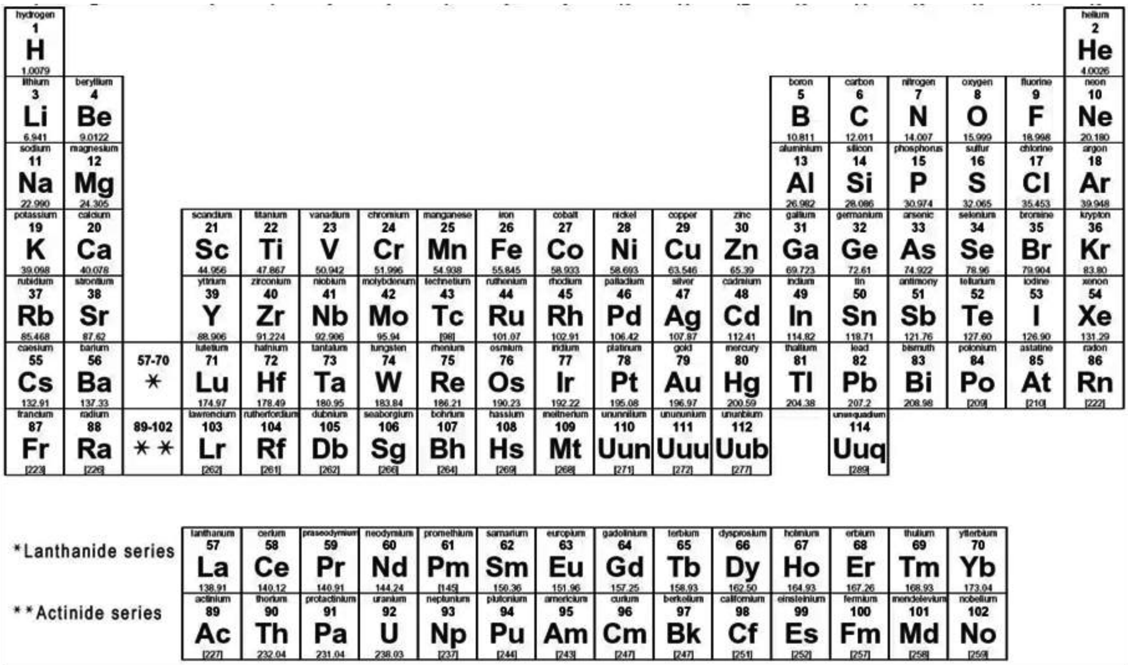


Fig. 2.1: Periodic table of elements

The importance of other organisms to humans

Humans rely mostly on other organisms to satisfy their living in the following areas:

1. Food
2. Shelter
3. Fuel
4. Drugs
5. Clothing
6. Industrial raw materials
7. Jewellery
8. Furniture

TEST QUESTIONS

1. (a) Define biological classification.

   (b) State four importance of classification.

1. (a) Mention four similarities between plants and animals.

   (b) State seven differences between living and non-living things.

2. Itemize the five kingdoms in the natural system of classification and give one example of organisms found in each kingdom.

3. Mention two differences between Kingdom Animalia and Kingdom Plantae.

4. State and explain the seven ranks of living organisms.

5. (a) What are viruses?
   (b) Mention three characteristics of viruses.

6. Itemize the following under living and non-living things:
   man, rock, mosquito, tomato, neem tree, sock, clock, eel, fan, computer

7. Classify the following organisms under the appropriate kingdoms:
   maize, octopus, mould, euglena, centipede, sparrow, bacteria,

4

MATTER

Specific Objectives

After completing this chapter, you will be able to:

- Describe the different building blocks of matter.
- Differentiate between elements, compounds and mixtures.
- Describe the formation of covalent and ionic compound.
- Relate atomic numbers, mass numbers, isotopes and relative atomic mass among each other.
- Perform calculations using the mole concept.
- Prepare solutions of given concentrations.

INTRODUCTION

Everything around us that has the attribute of gravity and inertia (weight) and the ability to occupy space (volume) is said to be matter. Therefore by way of definition, ***matter is anything that has weight and volume.***

Some matter are visible while others such as air can only be felt but not seen.

STATES / TYPES OF MATTER

There are three states of matter. They are:

- Solid
- Liquid
- Gas

Characteristics of Solids

1. Solids have fixed shapes.
2. They have fixed volumes.
3. They are very difficult to compress.
4. They have very low rate of diffusion.
5. The molecules that make up solids have high attraction among them.
6. The molecules are very closely packed
7. The molecules spin and vibrate

Examples of solids are stones, wood, books, man etc.

Characteristics of Liquids

1. Liquids do not have fixed shapes but take the shape of the containing vessel
2. They have fixed volumes (this is because they have enough molecular attraction to resist force tending to change their volumes.)

3. They have fast rate of diffusion
4. They are fairly easy to compress.
5. Molecules are less closely packed together.
6. They flow.
7. Molecules move fast.
8. Molecules move at random.

Examples of liquids are water, fruit juice, urine, petrol, kerosene

Characteristics of Gases

1. Gases have no fixed shape.
2. They have no fixed volume.
3. They have very high rate of diffusion.
4. They are very easy to compress.
5. Molecules have no attraction between them.
6. Molecules move in all directions.
7. Molecules move very fast.

Examples of gases are oxygen, water vapour, hydrogen, smoke.

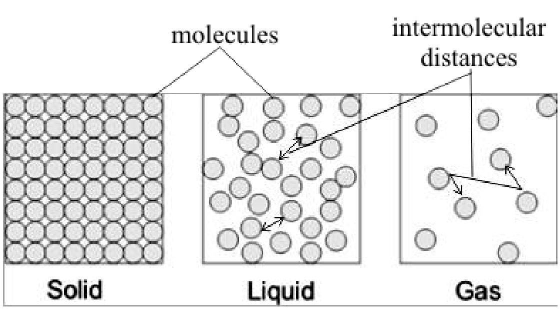


Fig. 2.5*:* Intermolecular distances between the states of matter

THE BUILDING BLOCKS OF MATTER

All matter, weather living things or non-living things are made up of three particles called the building blocks or the basic units of matter.

The particles are:

- Atom
- molecule
- Ion

ATOM

An atom is the smallest particle of an element that shows all the properties of the element and can take part in a chemical reaction.

An atom is the tiniest and the most basic building block of all matter.

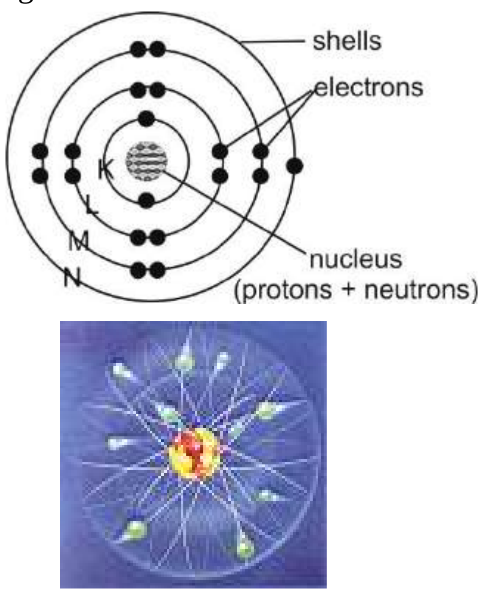


Fig. 2.6*:* Structure of an atom

An atom is made up of smaller particles, namely, electrons, protons, and neutrons, called the subatomic particles. An atom consists of a cloud of electrons that surrounding a small dense nucleus of protons and neutrons. The electrons and protons have a property called electric charge which affects the way they interact with each other electrically charged particles.

Electrons and shells

Electrons are tiny negatively charged particle that form a cloud around the nucleus of an atom. Properties of electrons include:

- They have negative charges (-1)
- They are found around the nucleus of an atom.
- They travel on an orbit or shell or energy level.
- They have mass of 1/1840

Electrons travel on shells or orbit, and each electron has its own shell; therefore, they do not crash as they move around the nucleus. An atom can have one or more shells depending on the number of electrons present. However, there is maximum number of electrons that each shell carries. For the sake of clarity, the shells have been labelled (K, L, M, N).

The shell closest to the nucleus is the K shell; and it carries not more than 2 electrons. It is filled first. The next shell is the L shell which carries a maximum of 8 electrons. The third shell, the M shell carries a maximum of 18 electrons. The N shell carries a maximum of 32 electrons. The number of electrons per shell increases as the number of shells increases.

Protons

Protons are found in the nucleus of an atom, and carry positive charges. Properties of protons are:

- They have positive charges (+1)
- They are located in the nucleus
- They have a mass of 1.

Table 1.7: Differences between protons and electrons

Protons	Electrons
Located in the nucleus	Located around the nucleus
Have positive charge	Have negative charge
Have a mass of 1	Have a mass of 1/1840
Form mass numbers with neutrons	Do not form mass numbers

Similarities between protons and electrons

i. They are both charged.
ii. The number of protons in a neutral atom is equal to the number of electrons. However, they have different charges.

Neutrons

Neutrons are also found in the nucleus of an atom. Unlike protons and electrons, neutrons have no charge, therefore they are said to be neutral. Properties of neutrons are:

- They are found in the nucleus of an atom
- They have no charge (0)
- They have a mass of 1

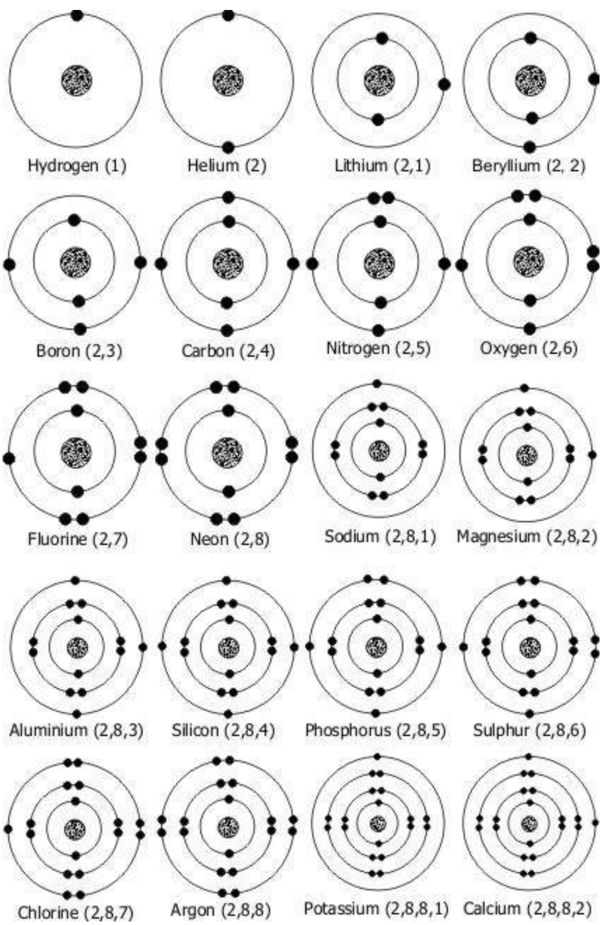


Fig. 2.7: Electronic structure of the first twenty elements

Characteristics / Properties of Atom

Atoms have special characteristic which help distinguish one atom from another and determine how atoms change under certain conditions the characteristics are atomic number, mass number, isotope, relative atomic mass, radioactivity.

Atomic number (proton number)

It is the number of protons in the nucleus of an atom. The number of protons is the same as the number of electrons. The symbol for atomic number is ***Z***. each element has a unique atomic number.

Mass number

Mass number is the number of protons and neutrons in an atom. The symbol for mass number is ***A***. The protons and neutrons are called the nucleons. The number of neutrons in an atom is the neutron number.

For an element 'W' there is ${}^{A}_{Z}\mathbf{W}$ where A is the mass number and Z, the atomic number. The mass number is written at the top left while the atomic number is written at the bottom left of the elements as shown above.

For example, Sodium which has anatomic number of 11 and a mass number of 23 is represented as:

Mass No.(A) = 23
Atomic No. (Z) = 11 Na

This means that Sodium has 11 protons and since atomic number (or proton number) is the same as the number electrons, it has 11 electrons.

Because the mass number is protons plus neutron, to find the number of neutrons, subtract the number of protons (or atomic number) from the mass number. Therefore we have:

23 – 11= 12. Hence the number of neutrons is 12.

Example 2

Potassium atom has atomic number of 19 and mass number of 39. To represent that, we have:

$${}^{A\,=\,39}_{Z\,=\,19}K$$

From the above, it can be deduced that Potassium has 19 protons, 19 electrons and 20 neutrons (A – Z = 20).

Question:

Find the number of (a) protons, (b) electrons and (c) neutrons of an element *Si* which has an atomic number of 16 and a mass number of 32.

Solution:

Mass number (A) = 32;

Atomic number (Z) = 16

a) The number of protons is 16 (number of protons is the same as atomic number).
b) The number of electrons is 16 (number of electrons = number of protons)
c) The number of neutrons is mass number (A) – atomic number (Z).
   Substituting, we have 32 – 16 = 16. Therefore, the number of neutrons is 16.

Isotopes

Isotopes are atoms of the same atomic number but different mass numbers. The difference in the mass numbers is due to the difference in the number of neutrons. Many elements have two or more isotopes. Examples of isotopes are:

Carbon ${}^{12}_{6}C$ ${}^{13}_{6}C$ ${}^{14}_{6}C$

Oxygen ${}^{16}_{8}O$ ${}^{17}_{8}O$ ${}^{18}_{8}O$

Hydrogen ${}^{1}_{1}H$ ${}^{2}_{1}H$ ${}^{3}_{1}H$

Trial Question:

The element oxygen has 3 isotopes with mass numbers 16, 17 and 18, with an atomic number of 8. Find the number of neutrons in each mass number.

Solution:

To find the number of neutrons in each mass number, subtract the atomic number from the mass numbers. Therefore, it will be 8, 9 and 10 respectively.

Relative atomic mass (A_r)

Atoms, though very tiny and their masses cannot be measured directly, however their masses can be compared. Atoms of different elements have different masses. One atom is selected and atoms of different elements are compared to it. Therefore, the masses of the other atoms are in relation to the mass of the selected atom, hence the term relative atomic mass.

The carbon-12 isotope is currently used to determine the masses of other atoms and the scale of atomic masses obtained is called carbon-12 scale.

The relative atomic mass of an element is the average mass of one atom of an element compared to 1/12 of the mass of one atom of carbon-12.

Mathematically,

$$A_r = \frac{\frac{1}{12} mass\ of\ one\ atom\ of\ carbon-12}{average\ mass\ of\ one\ atom\ of\ element}$$

Relative atomic mass has no unit

To illustrate, let's take an atom which has two isotopes X and Y with mass numbers A and B, with relative abundance of Z and 100-Z respectively, then their relative atomic mass is expressed as:

$$A_r = \frac{A\ X\ Z}{Z} + \frac{B\ X\ (100-Z)}{X\ x\ Z}\ A \text{ x } X + \frac{B \times (100-Z)}{Z}$$

Isotope X and Y can, therefore be as expressed as $^{A}_{C}$X and $^{B}_{C}$Y where ***C*** is the atomic number.

Sample questions with solution

Carbon has two main isotopes, Carbon-13, $^{A}_{C}13$ and carbon-12, $^{B}_{C}12$. The relative abundance of Carbon-13 is 1.11% and that of Carbon-12 is 98.89%. Find the relative atomic mass of carbon.

Solution

Relative atomic mass = mass due to carbon-13 + mass due to carbon-12.

$$= \frac{13 \times 1.11}{100} + \frac{12 \times 98.89}{100} \quad = 12.011$$

1. Copper (Cu) which has two isotopes ^{63}**Cu** and ^{65}**Cu** with relative abundance of 62.9 and 64.9 respectively. Calculate the percentage abundance of each isotope if the relative atomic mass of the naturally occurring copper is 63.55.

Solution:

Cu = 63.55; ^{63}***Cu*** = 62.9; ^{65}***Cu*** = 64.9

If the fractional abundance of ^{63}***Cu*** = p, then that of ^{65}***Cu*** will be (1- p) because there are only two isotopes available.

Hence, Cu = (64.9 x p) + 62.9 (1-p)

63.55 = 64.9p + 62.9 (1-p)

0.65 = 2p

p = 0.325

The percentage abundance of ^{63}***Cu*** = 32.5 therefore, the percentage abundance of

^{65}***Cu*** = 100-32.5 = 67.5%

2. Chlorine has two isotopes ^{35}**Cl** and ^{37}**Cl**. The relative proportions of ^{35}**Cl** to ^{37}**Cl** is 3 : 1. Calculate the relative atomic mass of chlorine.

Solution

The relative proportion is 3:1

Therefore, the relative atomic mass of chlorine:

$$= \frac{35 \times 3}{4} + \frac{37 \times 1}{4} \quad = \frac{142}{4} = 35.5$$

In examinations, the relative atomic masses are provided. For example, (H = 1; O = 16)

Table 1.9: Some elements with their relative atomic masses

Element	Symbol	Relative atomic mass
hydrogen	H	1.0
helium	He	4.0
lithium	Li	6.9
beryllium	Be	9.0
boron	B	10.8
carbon	C	12.0
nitrogen	N	14.0
oxygen	O	16.0
fluorine	F	19.0
neon	Ne	20.2
sodium	Na	23.0
magnesium	Mg	24.3
aluminum	Al	27.0
silicon	Si	28.1
phosphorus	P	31.0
sulphur	S	32.1
chlorine	Cl	35.5
argon	Ar	39.9
potassium	K	39.1
calcium	Ca	40.1
manganese	Mn	54.9
iron	Fe	55.9
cobalt	Co	58.9
nickel	Ni	58.7
copper	Cu	63.5
zinc	Zn	65.4

Molecule

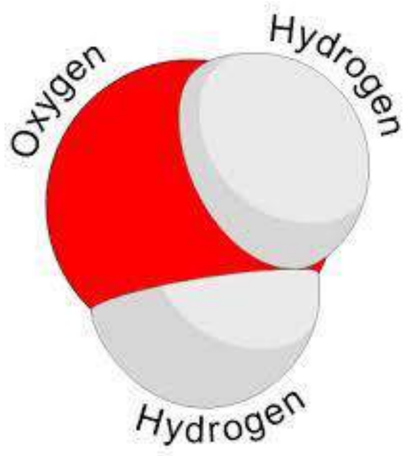
A molecule is formed when two or more atoms are chemically combined.

OR

A molecule is the smallest possible amount of a particular substance that has all the characteristics of that substance.

Molecules are electrically neutral group of two or more atoms held together by chemical bonds. A molecule may be ***homo-nuclear***, that is, it consists of atoms of a single chemical element, as with oxygen (O_2); or it may be a chemical compound composed of more than one element, as with water (H_2O). Other examples of molecules include NaCl, KOH, $CaCO_3$, CO_2, etc.

The smallest molecule is the diatomic hydrogen (H_2), with a bond length of 0.74Å whiles the largest is *Mesoporous silica* have been produced with a diameter of 1000 Å (100 nm).



Water molecule

Characteristics of molecules

1. Molecules are composed of two or more atoms
2. The atoms are chemically combined.
3. They are electrically neutral.
4. Diatomic molecules have the characteristics of the constituent atoms.
5. The constituent atoms of a molecule can only be separated by chemical means.

Ion

An ion is an atom which is either positively or negatively charged.

In other words the total number of electrons is not equal to the total number of protons, giving the atom a net positive or negative electrical charge. Ions can be created by both chemical and physical means. An ion consisting of a single atom is an atomic or monatomic ion; if it consists of two or more atoms, it is a molecular ion.

Types of ion

There are two types of ions. They are cations and anions.

Cation

A cation is created if a neutral atom loses one or more electrons. This gives it more protons than electrons, hence the atom is positively charged. Examples of cations are calcium (Ca), potassium (K), sodium (Na), magnesium (mg) etc.

Anion

An anion is created if an atom gains electrons. Such an atom has more negatively charged electrons than protons, therefore making the atom negatively charged. Examples of anions are chlorine (Cl), carbon (C), sulphur (S), fluorine (F), etc.

THE MOLE

All substances are made up of tiny particles known as atoms, ions and molecules. These particles are so small that they cannot be seen with the naked eye nor counted. However, a collection of millions of them can be seen and counted.

As we have a collection of objects like, *dozen* (collection of 12), *cent* (collection of 100), *gross* (collection of 144) etc. the collective name for millions of elementary

particles (atoms, molecules and ions) is the unit called the Avogadro's constant which is made up of 602 000 000 000 000 000 000 000 or 6.02×10^{23} of particles.
The unit of measurement for amount of substance (n) is the *mole.*

A mole of a substance is made up of 6.02×10^{23} particles of that substance. The SI unit of mole is *mol*.
NB: One mole of each of the elementary particles is 6.02×10^{23} particles.
Now, if $L = 6.02 \times 10^{23}$, then
$AL = 6.02 \times 10^{23} A$

$$\frac{1}{2} L = \frac{6.02 \times 10^{23}}{2} = 3.01 \times 10^{23}$$
$$2L = 2 \times 6.02 \times 10^{23} = 12.04 \times 10^{23}$$

Examples

1. Given that Avogadro's constant is 6.02×10^{23}, calculate the number of atoms in 0.01 mole of carbon.

Solution

1 mole of carbon contains 6.02×10^{23} atoms
Therefore, 0.01 mole of carbon will contain $0.01 \times 6.02 \times 10^{23}$
$= 6.02 \times 10^{21}$
Hence, 0.01 mole of carbon contains 6.02×10^{23} atoms

2. Find the number of atoms in 6 moles of oxygen. ($n = 6.02 \times 10^{23}$)

Solution:

1 mole of oxygen contains 6.02×10^{23} atoms.
∴ 6 moles of oxygen will have 6 $6.02 \times 10^{23} = 36.02 \times 10^{23}$ atoms

MOLAR MASS (M)

Molar mass is the mass of 1 mole (or 6.02×10^{23} particles) of a substance. The SI unit of, molar mass is gram per mol (g/mol). Numerically, the molar mass is equal to the relative atomic mass of an atom. For example, the relative atomic mass (Ar) of Nitrogen is 14, therefore its molar mass is 14 g/mol.
The relative mass of (A_r) Helium (He) is 4;
∴ its molar mass = 4 g/mol.

In the same way if you want to find the molar mass of a compound, you add all the relative atomic masses of the constituent atoms together. For example, to calculate the molar mass of H_2O, (relative atomic masses are H = 1, O = 16)
We will have:
(1 x 2) + 16 = 18 g/mol.

Questions

Calculate the molar mass of the following:
1. NaCl, **2**. H_2SO_4, **3**. HCl, **4**. $C_{12}H_{22}O_{11}$
(Relative atomic masses are Na = 23, O = 16, C = 12, H = 1, Cl =35.5, S = 32)

Solution

1. Molar mass of NaCl = 23 + 35.5 = 58.5 g/mol

2. Molar mass of H_2SO_4 = (1 x 2) + 32 + (16 x 4)
= 98 g/mol

3. Molar mass of HCl = 1 + 35.5 = 36.5 g/mol

4. Molar mass of $C_{12}H_{22}O11$
= (12 × 12) + (1 × 12) + (16 × 11)
= 342 g/mol

AMOUNT OF SUBSTANCE IN MASS OF SUBSTANCES

The following formula is used to find the number of moles in mass of substances:

Amount of substance (n) =

$$\frac{Mass\ of\ Substance\ \ (m)}{Molar\ \ mass\ \ of\ \ substance\ \ \ (M)}$$

The SI unit of amount of substance is *mol*.

Thus, $n = \frac{m}{M}$ is the mol

Example

1. Calculate the amount of substance in 9 g of aluminium.
2. Find the number of moles of carbon dioxide molecules in 14 g of carbon dioxide.
3. What is the mass of 0.5 mol of H_2O?
(Molar mass of Al = 27; CO_2 = 44; H_2O = 18 g/mol)

Solution

1. Amount of substance in Al =

$$\frac{Mass\ \ of\ \ Substance\ \ \ (m)}{Molar\ \ mass\ \ of\ \ substance\ \ \ (M)}$$

$$n = \frac{9}{27} = \ \ 0.33 \text{ mol}$$

2. Number of moles (n) of CO_2=

$$\frac{Mass\ of\ Substance\ (m)}{Molar\ mass\ of\ substance\ (M)}$$

$$n = \frac{14}{44} = \ \ 0.318 \text{ mol}$$

3. Amount of substance (n) =

$$\frac{Mass\ of\ Substance\ (m)}{Molar\ mass\ of\ substance\ (M)}$$

Since we are supposed to find the mass, we will make mass of substance (m) the subject of the relation.

Hence, $m = n \times M$

$m = 9$ g

RELATIVE MOLECULAR MASS / FORMULA MASS

The relative molecular mass is the sum of the relative atomic masses of all the atoms in a molecule. For example, the relative molecular mass of potassium hydroxide, KOH is 56. This is calculated as follows:

K = 39, O = 16, H = 1

So by adding them together, we have

39 +16 + 1 = 56

In the same way to calculate the relative molecular mass of sodium chloride, NaCl, we will have:

Na + Cl = 23 + 35 = 58

To find the relative molecular mass of $CaCO_3$, add all the constituent atoms as follows:

A_r of calcium = 40

A_r of carbon = 12

A_r of oxygen = 16 (for the three oxygen Ar = 16 x 3 = 48)

$\therefore$ the relative molecular mass of $CaCO_3$ = 40 + 12 + 48 = 100

NB: Relative molecular mass has no unit.

It can be deduced from above that relative molecular mass or formula mass is numerically equal to molar mass. Thus:

Molar Mass = Relative Molecular Mass = Formula Mass

ELEMENTS, COMPOUNDS AND MIXTURES

Elements

An element is smallest form of a substance that cannot be broken down into smaller particle.

There are many known and unknown elements arranged on the periodic table according to their physical and chemical properties such as metals and non-metals. For easy identification, elements have unique symbols. For example oxygen (O), gold (Au), argon (Ar), lead (Pb) etc.

Compounds

A compound is a substance which contains two or more elements chemically combined together.

Individual elements which make up a compound are in fixed amount. E.g. carbon dioxide (CO_2) is a compound which contains one element of carbon and two elements of oxygen.

The newly formed substance has completely different properties from the constituent elements. E.g. hydrogen and oxygen are both gases, yet, when they come together they form water (H_2O) which is a liquid.

There are two types of compounds – ionic and covalent compounds

Table 2.0: Differences between Elements and Compounds

Element	Compound
Cannot be broken down	Can be broken down
Made up of atoms of the same kind	Made up of atoms of different kind
Has the same properties as the atom	The constituent atoms have different properties from the compound

Mixtures

***A mixture is made up of two or more elements or compounds that can be separated by physical means*.**

Mixtures can be separated by physical means because the constituents are not chemically combined.

Table 2.1: Differences between Compound and Mixture

Compounds	Mixtures
Constituents cannot be separated by a physical means	Constituents can be separated by a physical means
New substances are formed	No new substances are formed
Constituents have fixed amounts	Constituents do not have fixed amounts
Have different properties from the constituent elements	Have the same properties as the constituent elements
Constituents become a single substance	Constituents do not become a single substance
Energy is usually involved	No energy involved

Table 2.2: Differences between Elements and Mixture

Elements	Mixtures
Made up of the same kind of atom	Made up of different kinds of elements
Have fixed physical properties	Do not have fixed physical properties
Have definite composition	Do not have definite composition

Types of Mixtures

1. ***Solid – Solid mixtures:*** These mixtures are made up of two or more physically combined solid substances. Examples are: Sand and iron filing; Sugar and rice; Brass (copper and zinc); Bronze (copper and tin)

2. ***Solid - Liquid Mixtures:*** Made up of solid and liquid in which the solid is soluble (i.e. can dissolve). Example are: Sugar and water; Salt and water; Blood; Chalk and water

3. ***Liquid – Liquid Mixtures***: Made up of two or more liquids which are miscible. (Miscibility is a measure of how easily different liquids will dissolve when mixed together.). Examples are: Ethanol and water; Petrol and kerosene

4. ***Liquid – Gas Mixture:*** Made up of liquid and gas . examples are: Fog; foam

5. ***Gas – Gas mixtures:*** Made up of two or more gases. An example is the atmospheric air

6. ***Solid – Gas Mixtures:*** Made up of a solid and a gas. An example is harmattan.

IONIC AND COVALENT COMPOUNDS

Ionic or electrovalent compounds are formed by the attraction of positive and negative ions.

The ions are held by a strong electrostatic bond known as ***ionic*** or ***electrovalent*** bond. Ionic compound have high melting and boiling points. This is as a result of the strong force of attraction between the positive and the negative ions; therefore, large amount of energy is required to break this strong electrostatic force between the ions. Examples of ionic compounds are, NaCl; MgCl; NaOH, etc

Covalent compounds are form when two or more atoms of non-metals which are unable to form stable ions share electrons in order to be stable.

Covalent compounds have weak force of attraction between the molecules. Example of covalent compounds are O_2; H_2; $Cl_{2;}$; HCl; NH_3; H_2O.

Chemical bonds

A chemical bond is a force that holds atoms, ions or molecules together.

One major importance of chemical bonds is that it allows atoms to acquire stable electronic configuration similar to that of a noble gas.

A noble gas has an outer shell filled with either eight or two electrons. Examples are, helium (2), neon (2,8) and argon (2,8,8). An atom is only stable when it has two or eight electrons in the outer shell.

How atoms achieve stable electronic configuration

i. Metal atoms with one, two or three electrons in the outermost shell lose the electrons to form positively charged ions called ***cations***.

ii. Non-metal atoms with five, six or seven electrons in the outer shell gain three, two or one electrons respectively to form negative charged ions called ***anions.***

iii. Two non-metallic elements with four to seven outer electrons may gain electrons by sharing them with each other.

Types of Chemical Bond

There are two main types of chemical bonds – ionic bond and covalent bond.

Ionic / electrovalent bond

This is the type of bond that forms an electrostatic force of attraction between negative and positive ions and holds them together.

Ionic bond results in the formation of ionic compounds

In the formation of ionic compounds, metal atoms lose their outermost electron(s), thereby forming cations (positive ions). Non-metals, on the other hand, gain electrons to fill their outermost electron shell, thereby becoming anions (negative ions).

For example, in the formation of sodium chloride (NaCl), sodium which is a metal atom with an electronic configuration of 2,8,1 loses it last electron so that it will become stable. Chlorine, a non-metal atom with electronic configuration of 2,8,7, accepts that electron to its outer shell in order to be stable. Hence, sodium will become Na^+ and chlorine will become Cl^-.

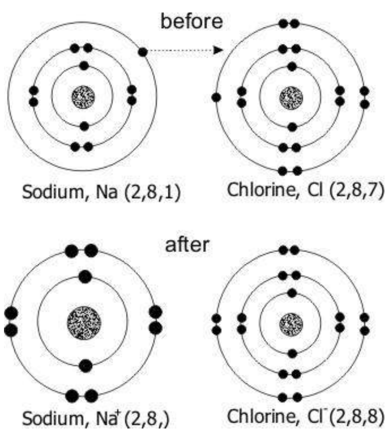


Fig. 2.8*:* Formation of sodium chloride, NaCl

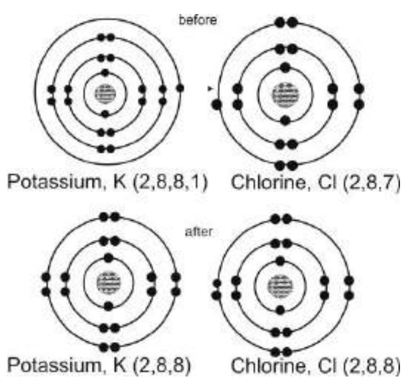


Fig. 2.9*:* Formation of potassium chloride (KCl*)*

Covalent Bond

***Covalent bond is formed when two or more atoms share a pair of electrons in order to be stable*.**

In simple covalent bonding, each of the two combining atoms contributes one electron to the bond. Thus, the bond consists of two electrons shared between the two atoms. The attraction between the nuclei of the combining atoms and the shared pair(s) of electrons provide the binding force that holds the atoms together. In a single covalent bond, the bonds are shown as a straight line between the symbols of the atoms involved. For example, H-H, Cl-Cl. When the atoms involved give two more electrons each, double or triple covalent bonds are formed. For example, O_2 could be shown as O=O; and N_2 shown as N=N.

The following illustrates the formation of covalent bonds.

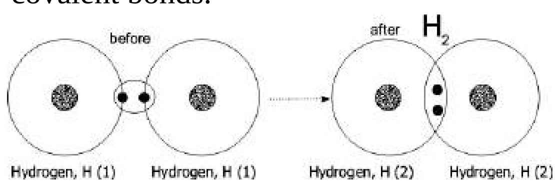
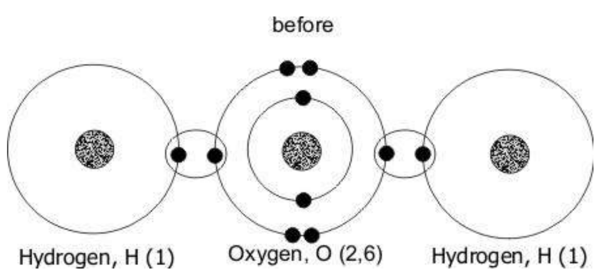


Fig. 3.0: Single covalent bond formed by two hydrogen atoms



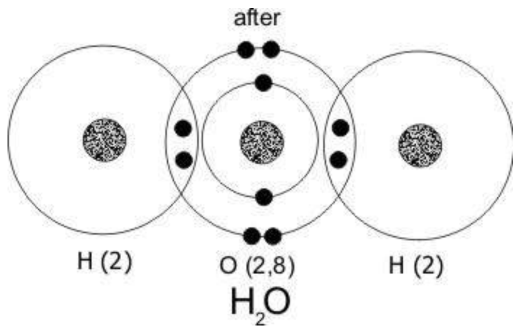


Fig. 3.1: Formation of water

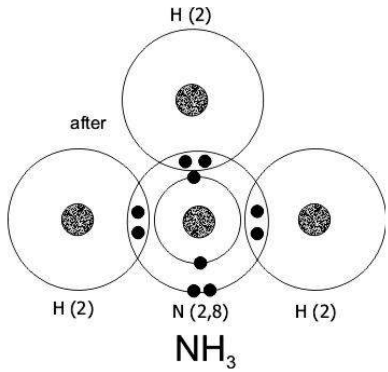
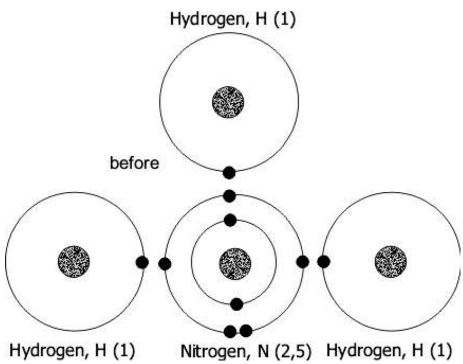


Fig. 3.2*: A covalent bond formed by three hydrogen atoms and one nitrogen atom (ammonia)*

Characteristics or properties of ionic compounds

1. They have high melting point.
2. They have high boiling point.
3. They react quickly with other substances or elements.
4. They are formed by a strong electrostatic force.
5. They are mostly soluble in water.
6. When molten, they conduct electricity.
7. They conduct electricity in aqueous solution (solution with water).

Examples of ionic compounds are sodium chloride (NaCl), calcium carbonate ($CaCO_3$), magnesium oxide (MgO), potassium chloride (KCl) etc.



Characteristics of covalent compounds

1. They have low melting point.
2. They have low boiling point
3. They react slowly with other substances or elements.
4. They are formed by a weak force of attraction.
5. They are mostly insoluble in water.
6. They do not conduct electricity.
7. They are usually liquid or gas at room temperature.

Example of covalent compounds are carbon dioxide (CO_2), water (H_2O), ammonia (NH_3), hydrogen chloride (HCl), etc.

NAMING OF COMPOUNDS

The system used in naming compounds is called IUPAC nomenclature. This is a system adopted by the International Union of Pure and Applied Chemistry (IUPAC).

The oxidation number of an element determines its naming.

The oxidation number of an element is the electrical charge it carries in its pure state or in its compound.

Rules for Determining the Oxidation Number of Substances

a. The oxidation number of an element in its atomic or molecular state (i.e. not combined with any other element) is zero (0). For example, the oxidation numbers of oxygen, potassium, sodium etc are zero.
The oxidation number of an ion of a single atom is equal to the charge on the atom. For example, the oxidation numbers of H^+ is +1, Cl^- is -1, O^{2-} is -2, Al^{3+} is +3, etc.

b. The oxidation number of an oxygen is -2 e.g. CO_2, H_2O, OH, except in peroxides e.g. H_2O_2 and K_2O_2 where it is -1, and superoxides e.g. KO_2 where it is $-^1/_2$.

c. The oxidation number of hydrogen is +1 e.g. H_20, HCl, OH, except in metal hydrides where it is -1, e.g. NaH, KH, CaH, etc.

d. The total of the oxidation numbers of atoms in a neutral compound is zero. For example, the sum of the oxidation numbers of the elements in the compound NaCl is zero. This also means that the charge a neutral compound is zero.

e. The total sum of the oxidation numbers of all atoms of a radical (i.e. atoms that stay together as a charged unit), is equal to the charge of the radical, e.g. the oxidation numbers of NO^-_3, CO_2, SO^{-2}_4 are -1, 2 and -2, respectively.

f. Some elements such as Fe, Cu, Pb ,Zn, etc have variable (more than one) oxidation numbers

g. All metals show a positive oxidation state, e.g. Mg^2+, Mn^7+, etc.

h. The charge of a substance is written as a right superscript followed by its positive or negative sign, e.g. Cl^-, Al^{3+}, O^{2-}, etc.

i. In substances with oxidation number of +1 or -1, only the sign is written, e.g. H+, F-, etc

How to Determine Oxidation Numbers

What are the oxidation numbers of the following ions?

1.C 2. Li 3. Be 4. H

Solution:

1.0 2. 0 3. 0 4. 0

This is according to the first rule ‘the oxidation number of an element in its atomic state is zero’.

What are the oxidation numbers of the following ions?

1. Al^{3} 2. O^{2-} 3. F 4. Mn^{7+}

Solution:

1. +3 2. -2 3. -1 4. +7

This is according to rule two, oxidation number of an atom = charge.

Find the oxidation numbers of the underlined elements in the following:

1. BrO^{-}_{3} 2. SO^{2}_{-4} 3. OH^{-} 4. NO^{-}_{3}

Solution:

Lets represent the oxidation numbers of the underlined elements by ***x***

1. Oxidation number of BrO_3^-

   $x + (3\text{ x}-2) = -1 \Rightarrow x = -1 + 6$

   $x = 5$

2. Oxidation number of SO_4^{2-}

   $x + (4\text{ x }-2) = -2 \Rightarrow x = -2 + 8$

   $x = 6$

3. Oxidation number of NO_3^-

   $x + (3\text{ x }-2) = -1 \Rightarrow x = -1 + 6$

   $x = 5$

   (This is according to rule ***f)***

4. Oxidation number of OH^- is

   $x + 1 = -1 \quad x = -2$

(a) Binary compounds

Binary compounds are compounds which consist of only two elements. One of the elements is more electronegative (attract more electrons to itself) than the other. The more electronegative element has a negative oxidation number. Examples of binary compounds are, NaCl, CO_2, CaCl.

Rules for Naming Binary Compounds

a. The suffix '-ide' is replaces the last two or three letters In the name of the more electronegative element, e.g. oxygen – oxide, chlorine – chloride, hydrogen – hydride, etc.
b. The modified name of the more electronegative element is written second, e.g. sodium hydride, calcium chloride, etc.
c. The name of the less electronegative element is not modified and is written first. If it oxidation number is more than one, it is shown in capital roman numerals inside a bracket after the name of the elements, e.g. copper (II) oxide, phosphorus (III) chloride.

Example:

Let's consider CO.

The first element is carbon which has variable oxidation number, therefore, let's represent it by x.

$X + (1\text{ x }-2) = 0 \Rightarrow x = 0 + 2 \Rightarrow x = 2$

Hence CO is named carbon (II) oxide (according to rules a and c)

Table 2.3: Binary compounds, their common and IUPAC names

Compound	Common name	IUPAC name
CO_2	carbon dioxide	carbon (ii) oxide
SO_3	sulphur trioxide	sulphur (vi) oxide
HCl	hydrogen chloride	hydrochloric acid
PbO	lead monoxide	lead (ii) oxide
PCl_3	phosphorus trioxide	phosphorus (iii)oxide
K_2O	potassium oxide	potassium oxide
H_2O_2	hydrogen peroxide	potassium peroxide
KO_2	potassium superoxide	potassium superoxide
PCl_5	phosphorus pentachloride	phosphorus (v)
NH_3	ammonia	ammonia

(b) Ions

The following rules are used to name simple ions

i. For cations of elements with only one oxidation number, the name of element is written with the word ion without the oxidation number.
For example, hydrogen ion (H^+)

For cations of elements with viable oxidation numbers, the oxidation numbers are written in Roman capital numerals, in brackets after the name of the element, e.g. iron (II) ion (Fe^{2+}), lead (II) ion (Pb^{2+})

ii. Anions are suffixed "-ide" plus the word ion, e.g. sulphide ion.

(c) Oxoanions

Oxoanions are group of ions that contain oxygen atoms. Oxoanions are negatively charged. E.g. CO_3^{2-}, ClO^-, etc.

Rules for Naming Oxoanions

a. The suffix '-ate' replaces the last two or three letters in the name of the first or the middle element , e.g. carbon becomes carbonate.
b. The number of oxygen atoms is placed before the name of the middle atom as*, dioxo, tetraoxo, heptaoxo*, etc. for two, four,or seven oxygen atoms respectively. If the oxygen atom is one, it is not prefixed (mono).
c. The oxidation number of the middle atom is written in Roman capital numerals in brackets, after its name.
d. The word ion is added to the name.

Examples:

1. Let's consider CO_3^{2-}

It contains three oxygen atoms, so according to rule ***b,*** the name starts with 'trioxo'. Followed by carbon with the suffix '–ate'. (rule *a*)

Now the oxidation number of carbon is calculated as follows:

$x + (3x-2) = -2$; $x = -2 + 6$; $x = +4$

$\therefore CO_3^{2-}$ is named as trioxocarbonate (IV) ion

2. Let's consider SO_4^{2-}

   It contains four oxygen atoms, therefore its name begins with tetra. Followed by sulphate, and then the oxidation number which is calculated as follow:

   $x + (4 \text{ x } -2) = -2 \quad x = -2 + 8 \quad x = +6$

$\therefore SO_4^{2-}$ is name as tetraoxosulphate (VI) ion.

Table 2.4: Oxoanions, their common and IUPAC names

Oxoanions	Common name	IUPAC name
ClO^-	hypochlorite ion	oxochlorate (i) ion
NO_3^-	nitrate ion	trioxonitrate (v) ion
MnO_4^-	permanganate ion	tetraoxomanganate (vii) ion
NO_2^-	nitrite ion	dioxonitrate (iii) ion
HCO_3^-	hydrogen carbonate ion	Hydrogntrioxo-carbonate(iv) ion

(d) Oxoacids

Oxoacids are compounds that contain hydrogen atoms. They are named in the same way as oxoanions but the word *acid* replaces the word *ion* in the name of oxoanions. Examples are tetraoxosulphate (VI) acid (H_2SO_4), trioxocarbonate (V) acid (H_2CO_3), etc.

Table 2.5: Examples of oxoacids

Oxoacids	Common name	IUPAC name
HCl	hydrochloric acid	hydrochloric acid
HNO_2	nitrous acid	dioxonitrate(iii) acid
H_2SO_3	sulphuric acid	trioxosulphate(iv) acid

(e) Bases

There are two types of salts- oxides and hydroxide. Oxides follow the naming sequence of common compounds. For example, carbon (II) oxide CO^2, potassium oxide (K_2O), etc.

Hydroxide, on the other hand, is made up of a cation and an hydrogen ion, (OH^-). They also follow the same naming sequence of binary compounds. E.g. calcium hydroxide ($Ca(OH)_2$, sodium hydroxide (NaOH), etc.

Table 2.6: Examples of bases

Base	Common name	IUPAC name
$Fe(OH)_2$	ferrous hydroxide	iron (ii) hydroxide
KOH	potassium hydroxide	potassium hydroxide
$Zn(OH)_2$	zinc hydroxide	zinc(ii) oxide

(f) Salts

Salt is formed when the hydrogen(s) of an acid is replaced by ammonium or metal cation.

There are two types of salts - binary salts and oxoacid salts.
Binary compounds are named by combining the names of the cations and the anions.

For example, iron (II) sulphide (FeS), ammonium chloride (NH_4Cl), sodium chloride (NaCl).

Oxoacid salts are named by writing the cation, followed by the name of the oxoanion and then finally its oxidation number.

For example, $MgSO_4$ is named by fist writing the cation, ***magnesium,*** followed by ***tetraoxosulphat****e*, and then its oxidation number if its variable, which is calculated as follows:

If oxidation number is x, then,
$x + (4x - 2) = 0 \qquad x = 0 + 6 \quad x = +6$
Hence, $MgSO_4$ is named as magnesium tetraoxosulphate(VI).

Table 2.7: Examples of salts

Salt	Common name	IUPAC name
Na_2CO_3	sodium carbonate	sodium trioxocarbonate (v)
$CaCO_3$	calcium carbonate	calcium trioxocarbonate (iv)
$CuSO_4$	cupric sulphate	copper (ii)tetraoxosulphte (vi)

SOLUTION

A solution is a uniform mixture of solute and solvent, where the solute dissolves in the solvent.

For a solution to be uniform, there has to be more solvent than solute.

Thus, **solution = solute + solvent**

A solute is a (solid) substance that dissolves in a given solvent.

A solvent is a (liquid) substance that dissolves solutes.

Water is often considered as a ***universal solvent*** because it is capable of dissolving many solutes.
The solution in which water is the solvent is known as ***aqueous solution***.
For example, sugar which is a solute when added to water which is a solvent dissolves, forming an aqueous sugar solution.

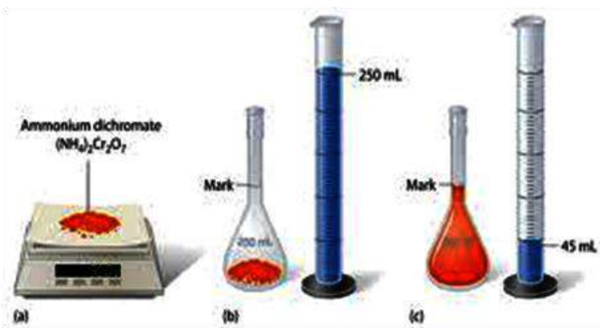


Fig. 3.3: Preparing a solution with ammonium dichromate

Concentration of solutions

***Concentration is a measure of how much solute there is in a solution*.**

A solution which has a very high amount of solute is called a ***concentrated solution***; compared to a ***dilute solution*** which contains less solute than solvent.

Calculating Concentration of Solutions

As we found out,

Amount of substance (n) =
$$\frac{Mass\ of\ Substance\ (m)}{Molar\ mass\ of\ substance\ (M)}$$

The quantities of substances in a solution can either be expressed in *mol* or *gram*. When expressed in *mol per cubic decimetre (dm^3)* it is referred to as concentration (C), and is given by the formula:

Concentration (c) = $\frac{\boldsymbol{Amount\ of\ substance\ (n)}}{\boldsymbol{Volume\ of\ substance\ (v)}}$

Example:

What s the concentration of 0.60 mol of sodium chloride in 0.700 dm^3 of water and sodium chloride solution.

Solution:

Concentration (c) = $\frac{Amount\ of\ substance\ (n)}{Volume\ of\ substance\ (v)}$

$$c = \frac{0.60}{0.700} = 0.857\ \text{mol/dm}^3$$

If the mass is given, the amount of substance is first calculated in mole before solving for concentration. If the volume is in cm^3, convert it to dm^3 (1 dm^3 = 1000 cm^3)

When 1 mol of solute is dissolved in a solvent to make 1 dm^3 of solution, the concentration is 1 mol/dm^3 (or 1 M, a shortened for of mol/dm^3). A concentration of 1 M of any substance is called a ***molar solution***.

When concentration is expressed in grams per cubic decimetre, g/dm^3, the following formula is used:

Concentration (c) = $\frac{mass\ of\ substance\ (m)}{Volume\ of\ solution\ (v)}$

Example:

What is the concentration of a solution which contains 5 g of sugar in a 20 dm^3 of solution?

Solution:

Concentration (c) = $\frac{mass\ of\ substance\ (m)}{Volume\ of\ solution\ (v)}$

$$c = \frac{5}{20} = 0.25\ \text{g/dm}^3$$

Sometimes the mass of the solution would be given instead of the volume. In this case the following formula will be used:

Concentration (c) =
$$\frac{mass\ of\ substance\ (m)}{mass\ of\ solution\ (m)} \times 100$$

Example:

20 g of salt was dissolved in a solution of 300 g. Calculate the concentration of the salt solution.

Solution:

Concentration (c) =

$$\frac{mass\ of\ substance\ (m)}{Volume\ of\ solution\ (v)} \text{ x } 100$$

$$c = \frac{20}{300} \text{ x } 100 = 6.67\ \%$$

PREPARATION OF STANDARD SOLUTIONS

A standard solution is a solution whose concentration is known.

The following are the apparatus needed to prepare a standard solution:

- ❖ Beaker
- ❖ Funnel
- ❖ Spatula
- ❖ Stirring rod
- ❖ Washing bottle
- ❖ Balance (electronic, beam, etc.)

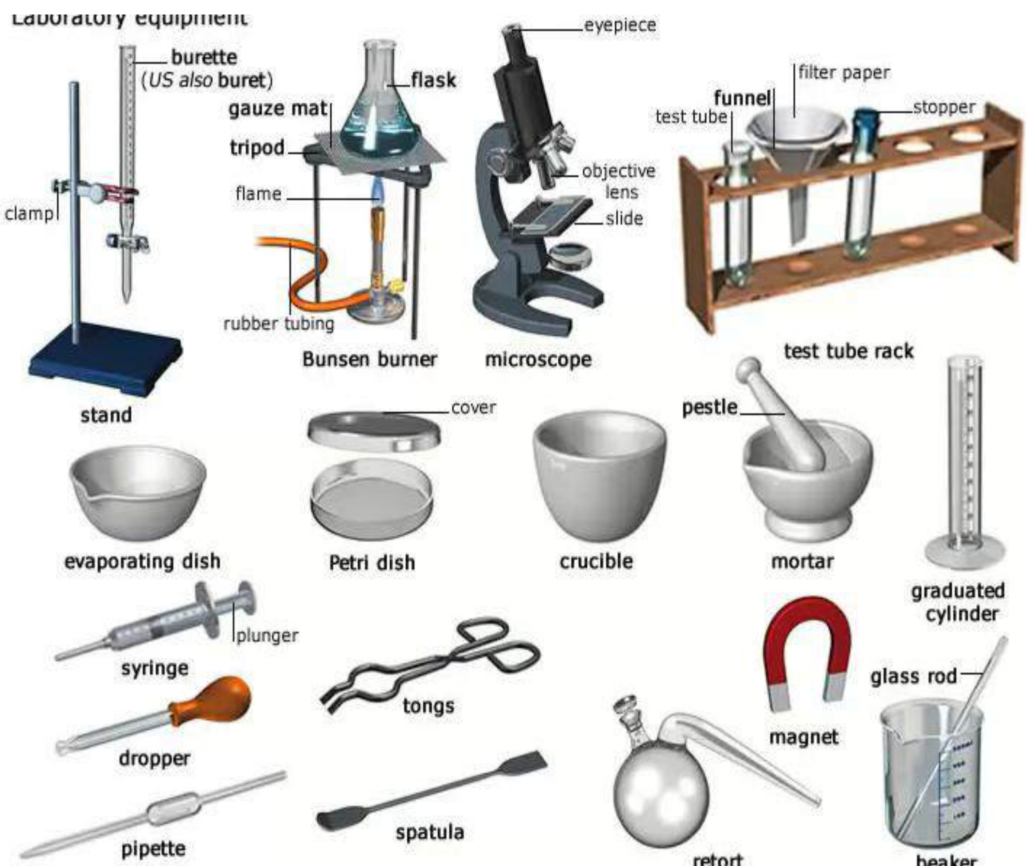


Fig. 3.4: Laboratory apparatus

To prepare a standard solution:

- ❖ Calculate the mass and the molar mass of the substance you need to prepare the solution with.
- ❖ Measure out the known mass of the substance into a beaker containing distilled water.
- ❖ Transfer the solution into the required volumetric flask.
- ❖ Add more water until it reaches the graduation mark.
- ❖ Put cork stopper on flask and shake it well to get a uniform solution.

How to prepare 1 dm^3 1m of an aqueous sugar solution

Apparatus to use include a beaker, standard volumetric flask (250 cm^3), a funnel, a stirring rod and a balance.

Procedure

- ❖ Calculate the molar mass of the sugar. The formula for sugar is $C_{12}H_{22}O_{11}$.
  The relative atomic mass of the constituent elements are C = 12; H = 1; O = 16. Therefore the molar mass of $C_{12}H_{22}O_{11}$ is 342 g/mol.
  Hence 1 mole of $C_{12}H_{22}O_{11}$ = 342 g which means 342 g of sugar will be used to prepare 1 dm^3 of 1 M solution.

- ❖ Measure 342 g of sugar into a beaker containing distilled water; and stir to dissolve the sugar.
- ❖ With the aid of a funnel, transfer the solution from the beaker into a clean 1000 cm^3 volumetric flask.

- ❖ Rinse the inside of the beaker with distilled water and add to the solution in the volumetric flask.

- ❖ Add more water to the solution until it reaches the 1000 cm^3 graduation mark. Put a cork stopper on the volumetric flask and shake it to get a uniform solution.

Example

Prepare 500 cm^3 of 1M solution of NaCl.

Solution

Relative atomic mass of Na = 23; Cl = 35.5

Molar mass of NaCl = 58.5 g/mol

Hence 58.5 g of NaCl will be used to prepare 500 cm^3 of 1 M solution

1 dm^3 = 1000 cm^3, therefore 500 cm^3 of 1M solution = $\frac{50}{1000}$ = 0.5 mol

1 mol of NaCl = 58.5g

$\therefore$ 0.5 mol NaCl = 0.5 x 58.5 = 29.25 g.

Hence 29.25 g of NaCl will be used to prepare 500 cm^3 1 M solution.

Measure 29.25 g into a beaker containing distilled water.

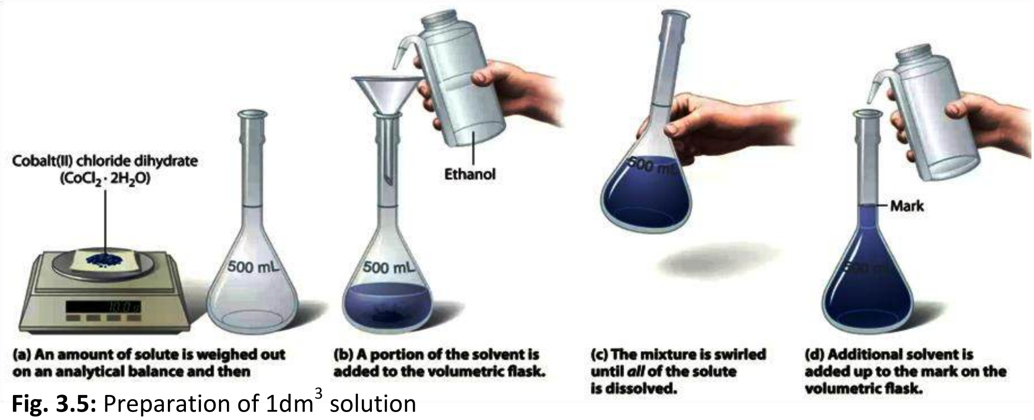


Fig. 3.5: Preparation of 1dm^3 solution

Transfer the solution into a 1000 cm^3 volumetric flask

Top up the solution to 1000 cm^3 graduation mark

Put on a cork stopper and shake to mix well.

DILUTION OF SOLUTIONS

Dilution of a solution occurs when more solvent is added to the solution.

When a solution is diluted only the volume of the solvent increases but the amount of substance in the solution remains the same. Thus:

$n_{\text{concentrated solution}} = n_{\text{dilute solution}}$

$n = c \times V$

where

n = amount of substance;

c = concentration;

V =volume of solution

$c_{conc} \times V_{conc} = c_{dil} \times (V_{conc} + V_{water})$

This relationship can be used to determine the volume of water to be added to a more concentrated solution to produce solution of lower concentration.

Uses of Dilution

Dilution is very useful in homes, industries, hospitals, schools etc. Some of the uses of dilution include:

- Preparation of food
- Preparation of drugs
- Production of paint
- Production of cosmetics

TEST QUESTIONS

1. Write a short note on the following terms:
   a. *Atom*
   b. *Molecule*
   c. *Ion*
   d. *Atomic number*
   e. *Mass number*
   f. *Isotope*

2. Mention three differences between compounds and mixtures.

3. The nucleon number and the proton number of a neutral atom are 53 and 25 respectively. Calculate the:
   a. Number of neutrons
   b. Number of electrons
   c. Atomic number
   d. Draw the electronic configuration of the atom.

4. An atom has three isotopes with mass number 14, 15 and 16. If the number of protons in the atom is 7, find the number of neutrons and electrons present in each isotope.

5. What is meant by the term mole?

6. Calculate the amount of substance in 42.0 g of NaCl.
   (A_r of Na = 23.0, Cl = 35.5)

7. Prepare 250 cm^3 of 1M solution of NaOH. (Ar Na =23.0; O=16.0; H= 1)

8. (a)Explain the term *solution.*
   (b) An aqueous solution of volume 5.0 dm^3 contains 36.4 g of sodium chloride. Calculate the concentration of the solution in mol dm^{-3}.
   [NaCl = 58.5]

9. (i) Explain the term *oxidation number* of an atom.
   (ii) Determine the oxidation number of manganese (*Mn*) in potassium permanganate ($KMnO_4$).

10. Magnesium ribbon of mass 4.0 g is placed in dilute hydrochloric acid contained in a beaker. Calculate the number of moles of hydrochloric acid that would be required to react completely with the ribbon.
    [Mg = 24, H = 1, Cl = 35.5]

11. (a) Study the table below and answer the questions that follow.

Atom/ Element	*Atomic number*	*Mass number*
V	10	18
W	18	39
X	15	33
Y	10	20
Z	11	23

(j) How many neutrons are there in atom **X**?
(ii) Which of the atoms will readily form an ion?
(iii) Which of the atoms will form an ion with a positive charge of +3?
(iv) (α) Indicate the atoms that are isotopes.
(β) Explain your answer in (a) (iv) above.

12. A stock solution of 1M HCl was prepared by dissolving 7.30 g of HCl in 200 cm^3 of distilled water. 5 cm^3 of stock solution was put into each of the four test tubes labelled **C**, **D**, **E** and **F** and diluted with distilled water to 8 cm^3, 10 cm^3 and 15 cm^3 respectively.

Equal amounts (*W g*) of zinc (Zn) granules were gently put added to each of the test tubes as illustrated in figures 3.

Study the figure carefully and answer the questions that follow.

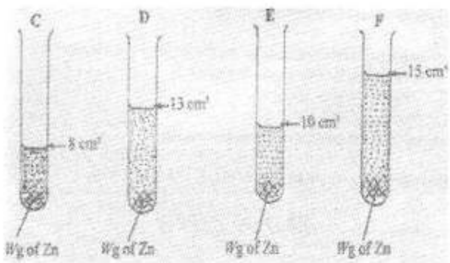


Fig. 3

(a) State what would be observed in this experiment.

(b) (i) Arrange the test tubes **C**, **D**, **E** and **F** in order of increasing reactivity.

(ii) Give an explanation for your answer in (b((i).

(iii) State three other factors that could affect the rate of reaction in the test tubes illustrated above.

(c) Calculate the concentration of the solution in test tube E in g/dm^3.

(d)(i) What is a standard solution?

(ii) A solution is made by dissolving 47.5 g of potassium iodide (KI) in 300 cm^3 of water. Calculate the concentration of KI in mole/litre. [K = 39, I = 27]

13. From the following compounds indicate which are covalent or ionic
    i) ethanol
    ii) potassium oxide
    iii) hydrogen chloride
    iv) copper (II) trioxonitrate (V)
    v) sulphide

14. **Fig. 4** is an illustration of a set of glassware used in preparation of exactly 100 cm^3 of 1.0 molar solution of sodium chloride in the laboratory.

Study the figure and the information below carefully and answer the questions that follow.

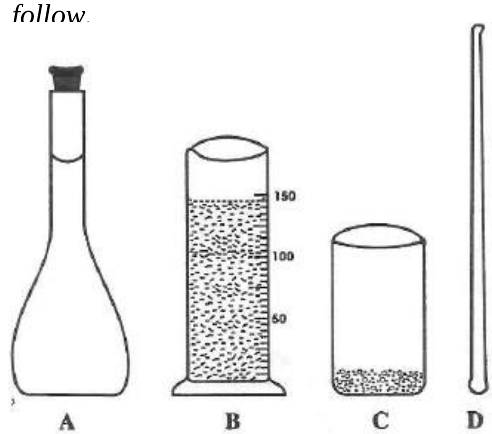


Fig 4

The capacity of **A** is 100 cm^3.
B contains distilled water.
C contains weighed amount of sodium chloride (NaCl).
The molar mass of NaCl is 58.5.

a) Identify each of the glassware A, B, C and D

b) Read and record the volume of distilled water in B.

c) Determine the mass of sodium chloride in C for the preparation of the 1.0 M NaCl solution.

d) Describe how 100 cm^3 of the 1.0 M solution of the sodium chloride is prepared using the materials provided.

e) Determine the appropriate volume of

distilled water that will remain in B in C for the preparation of the after the preparation.

15. **Fig. 2** below illustrates four steps taken to perform a simple experiment to determine the density of an irregular solid, **A.**

Study the figure carefully and answer the questions that follow.

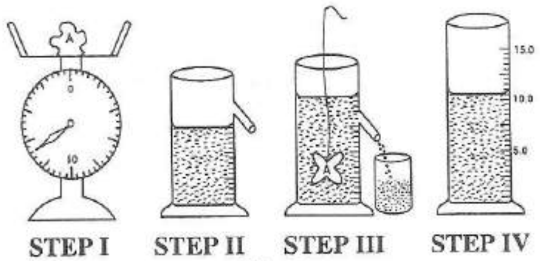


Fig 2

b) Describe briefly each of the four steps.

c) i) Record the readings in step I and step IV.

ii) Hence determine the density of the solid A.

d) State two precautions to be taken when carrying out the experiment.

5

CELLS AND CELL DIVISIONS

Specific Objectives

After completing this chapter, you will be able to:

- Describe the structure and function of plant and animal cells.
- Explain the process of cell division.

INTRODUCTION

Every living organism is made up of units of life called cells.

A cell can therefore be defined as the basic unit of life.

OR

A cell is the structural or functional unit of a living organism.

Each cell is independent of the other cells in the organism, and can undergo all the life activities such as reproduction, excretion, respiration, growth, etc.

Some living things are made up of one cell and are called ***unicellular organisms***. e.g. bacterial, amoeba,. Trypanosome, etc. Other organisms on the other hand are made up of many cells, and they are known as ***multi-cellular organisms***. e.g. plants and animals.

STRUCTURE OF THE PLANT AND ANIMAL CELLS

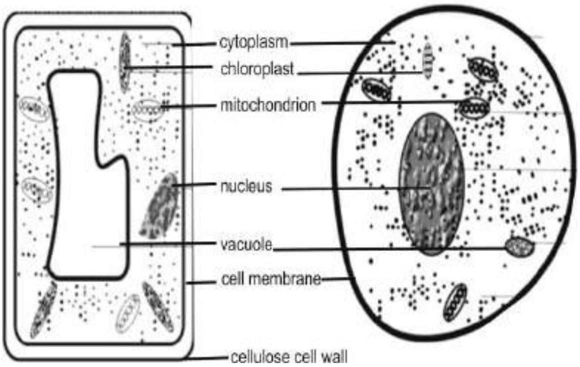


Fig. 3.6: Structure of plant and animal cells

Cell membrane: Also called plasma membrane, is a thin flexible semi-permeable membrane that encloses the contents of the cell.

Functions

- Keeps the contents of the cell in place
- Controls the movements of materials in and out of the cell.

Cellulose cell wall: It is found only in plant cells, and is made up of cellulose, which is a non-living material.

Functions

- ❖ Supports and protects the cell
- ❖ Gives shape to the cell
- ❖ It is permeable, and allows materials to pass in and out of the cell.

Centrioles: They are found only in plant cells. They are a pair of small rod-like structures in the cytoplasm near the nucleus. Centrioles can only be seen under high-power microscopes.

Functions

- ❖ Responsible for the formation of spindle in cell division.
- ❖ Aids in the formation of cilia and flagella.

Chloroplast: They are found only in plant cells. They are large organelles bounded by a double membrane and contain a green pigment called chloroplast.

Function

- ❖ Chlorophyll absorbs sunlight for photosynthesis

Chromosomes: They are hereditary materials in the form of DNA. They can be seen only under high-power microscopes.

Functions

- ❖ Determines the characteristics of the organism.
- ❖ Controls inheritance

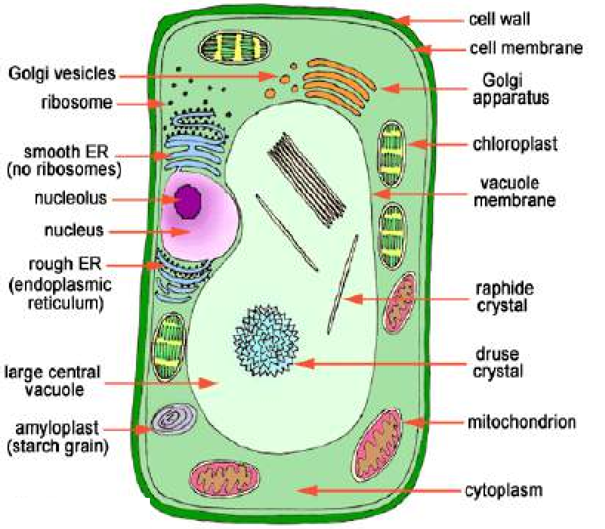


Fig. 3.7: A typical plant cell

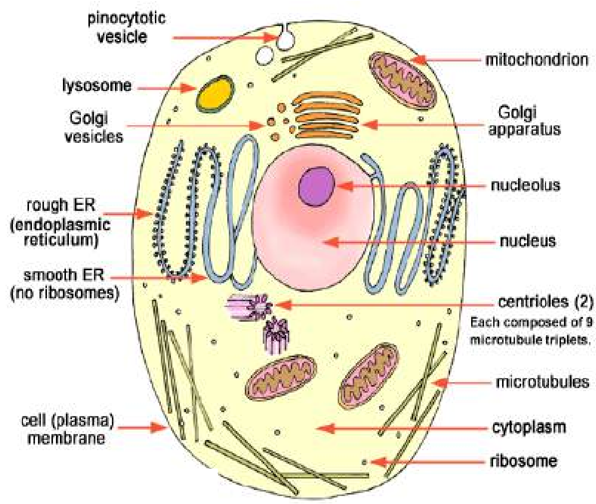


Fig. 3.8: A typical animal cell

Cytoplasm: Transparent watery fluid which contains various organelles and chemical substances.

Functions

- ❖ Site for most chemical reactions
- ❖ Carries the organelles in the cell

Endoplasmic reticulum: They are small interconnected channels surrounded by a membrane. There are two types – smooth endospermic reticulum and rough endospermic reticulum. They are only visible under high-power microscopes.

Functions

- ❖ Connects the plasma membrane and the nuclear membrane
- ❖ Gives mechanical support to the cytoplasm
- ❖ Serves as a pathway for the transport of materials in and out of the cell.
- ❖ Rough endospermic reticulum supports the ribosomes and transports proteins produced by the ribosomes.
- ❖ Provide rooms for chemical reaction in the cell.

Golgi body: They are strands of small flattened sacs surrounded by a membrane. They have a cluster of small bodies containing secretory material called vesicle.

Functions

- ❖ Produce and repair cell membranes
- ❖ Involved in the packaging and secretion of proteins and complex carbohydrates.
- ❖ Synthesize complex carbohydrates

Lysosome: They are small, dark, spherical structures filled with fluid and surrounded by a single membrane. They are more abundant in animal cells than plant cells. They are visible under high-power microscopes.

Functions

- ❖ Contain enzymes involved in digestion of materials in the cell
- ❖ Produce enzymes which destroys old and worn-out cells as part of the cell replacement, growth and repair of tissues
- ❖ Produce enzymes which defends the cell against virus, bacterial and other poisonous substances.

Mitochondrion: They are small spherical structures surrounded by two membranes. The number of mitochondria in a cell depends on the cell's energy requirement; this is because mitochondria are the site for the release of energy from respiration.

Functions

- ❖ Contains enzymes for respiration
- ❖ Energy released is used by the cell for life activities. For this reason, mitochondria are referred to as the powerhouse of the cell.

Organelles: Small membrane bound structures in the cytoplasm. Each organelle performs a specific function.

Ribosomes: They are small organelles found in large numbers in cells. They are made up of RNA (ribonucleic acid).

Function

- ❖ Site for protein synthesis in the cell.

Vacuole: They are sacs filled with fluid and surrounded by a single membrane called ***tonoplast***. Vacuoles are found mainly in plant cells where they occupy about 80% of the cell volume.

Functions

- Responsible for the control of waste content of the cell.
- provide turgidity in plant cells
- Temporary storage for food substances.

Table 2.8: Differences between plant and animal cells

Plant Cell	Animal Cell
Cell wall present	No cell wall
Contains chloroplast	No chloroplast
Large and permanent vacuole present	Small and temporary vacuole
Protoplasm is less dense	Dense protoplasm
Stores starch as carbohydrate	Stores glycogen as carbohydrate
Has a fixed shape	Has no fixed shape

Similarities between plant and animal cells

1. Nucleus is present in both
2. Both have cell membrane
3. Both have cytoplasm
4. They both possess mitochondrion
5. Vacuole is present in both
6. Both have endospermic reticulum
7. Both possess ribosome

SPECIALIZED CELLS

Different cells, both in plants and animals, have different structures and perform different functions. Each cell is specialized to perform a specific task**.**

A specialized cell can be defined as a cell which is adapted to perform a particular function.

Examples of specialized cells are those listed below under types of plant and animal cell.

Types of plant cell

Leaf epidermal cells: Transport and allow light to enter the leaf to the photosynthetic tissue below them.

Root tip cells: Have the ability to divide and give rise to different tissues.

Palisade cells: Have chloroplast with chlorophyll to absorb sunlight for photosynthesis.

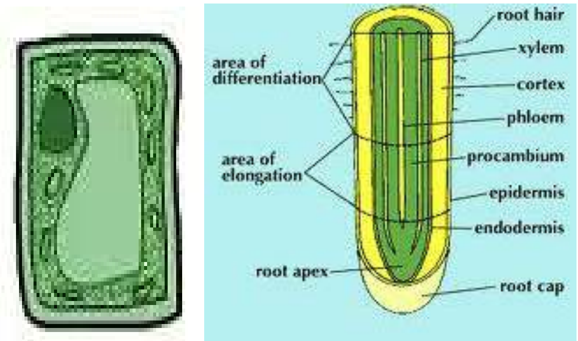


Fig. 3.9: Palisade (Left) cell and root cell (Right)

Types of animal cells

Red blood cells: Carry oxygen to other blood cells through the blood stream.

Muscle cells: Have the ability to contract because of the thin cylindrical or spindle shapes.

Sperm cells: Possess long tail to help them swim to the female egg to fertilize it.

Nerve cells: Have long and thin to help them transmit nerve impulses from one part of the body another.

Phagocyte: Kind of white blood cell, or leukocyte that destroys foreign substances in the body, such as bacteria.

Lymphocyte: A group of white blood cells which protect the body from diseases and infections

CELL DIVISION

Cell division is the process whereby cells in living organisms break up to form new cells.

In cell division, the DNA of the parent cell is carried on to the newly created cells depending on the type.

There are two types of cell division – meiosis and mitosis.

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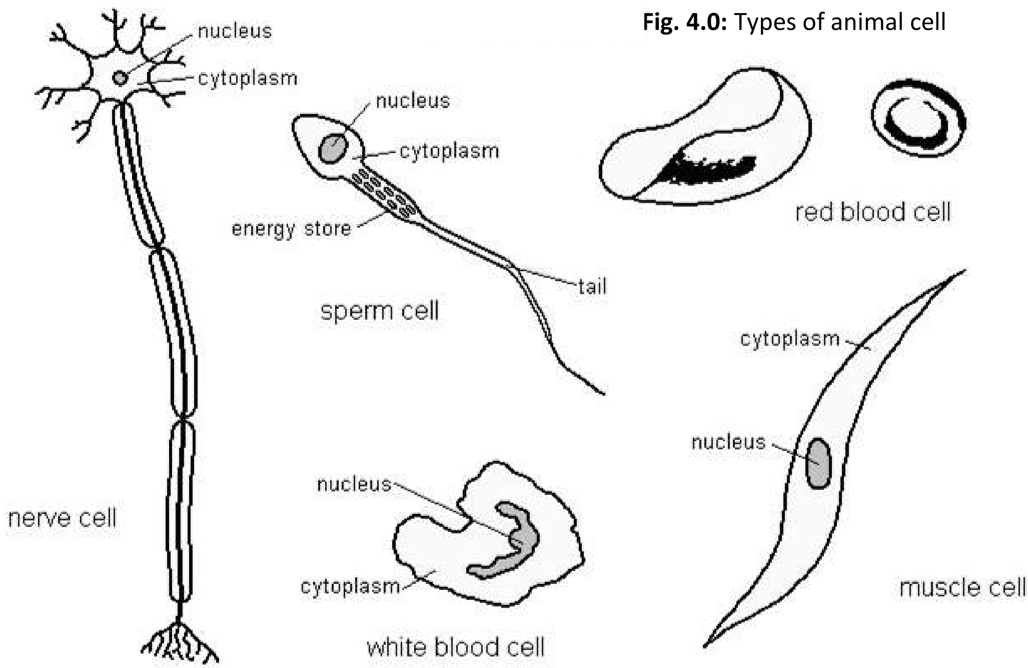

Fig. 4.0: Types of animal cell

Meiosis

***Meiosis is the process of cell division in organisms that reproduce sexually*.**

In meiosis the nucleus divides into four nuclei, each of which contains half the usual number of chromosomes.

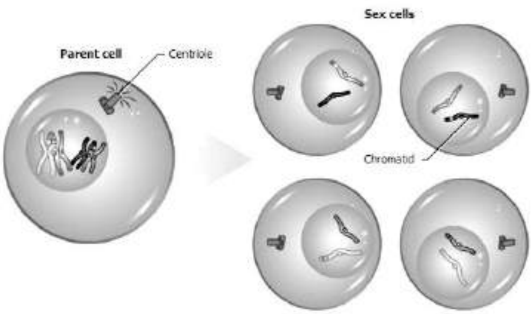

Fig. 4.1: Meiosis

How meiosis occurs

- The cell now has a chromosome and a centriole at each end.
- The cell membrane pinches inwards and divides into two.
- Membrane forms around the two newly created nuclei which have half the number of chromosomes as the parent cell.
- The centrioles copy themselves again and move apart to the opposite ends of the cells
- The cell membranes pinch inwards and break up.
- Meiosis ends with the creation of two more cells, making four, each with half the number of parent chromosomes and each with a different combination of genetic information.

Mitosis

***Mitosis is the type of cell division in which a cell divides into two daughter cells each of which has the same number of chromosomes as the original cell*.**

Mitosis results in two cells that are genetically similar. Mitosis is vital for growth; for repair and replacement of damaged or worn out cells; and for asexual reproduction, or reproduction without eggs and sperm.

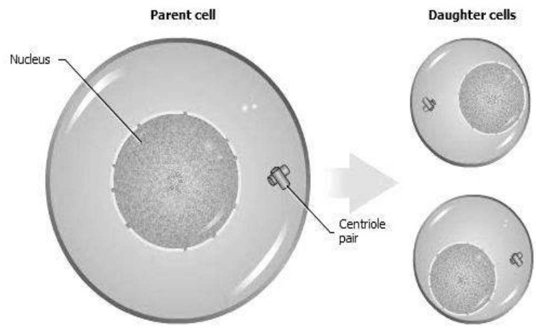

Fig. 4.2: Mitosis

How mitosis occurs

- The membrane surrounding the nucleus become less distinct and the chromosomes become more distinct
- Centrioles position themselves at the opposite ends of the cell
- Chromatids separate from chromosomes and move to either of the centrioles

- ❖ The cell now has a chromosome and a centriole at each end.
- ❖ The cell membrane divides into two.
- ❖ New nuclei are formed in each division.
- ❖ That completes the formation of two new cells similar to the original cell.

Table 2.8: Differences between meiosis and mitosis

Meiosis	Mitosis
Each cell has half the number of chromosomes as the parent cell	Each cell has the same number of chromosomes as the parent cell
Four dissimilar cells are created at the end	Ends with he creation of two similar cells
Occurs in organisms that reproduce asexually	Occurs in organisms that reproduce sexually
Nucleus divides into four	Nucleus divides into two
No crossing over	Crossing over occurs
No exchange or genes	Genes are exchanged

Level of organization in living organisms

All multicellular organisms have five levels of organization. They are:

Cell *is the basic unit of life.*

There are two principal types of cell – animal cell and plant cell.

Tissue is a group of similar cells which perform the same function.

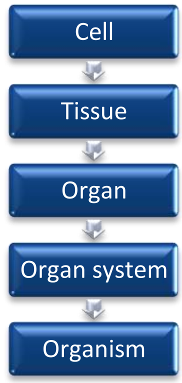

Fig. 4.2: Level of organization

Table 2.9: Examples of plant and animal tissues

Name of tissue	Cells forming them	Main function
Animal tissues		
Nerve tissue	Nerve cells (neurons)	Co-ordinate and conduct nerve impulses
Blood tissue	Red blood cells	Carries oxygen around the body.
Skeletal tissue	Ostecytes	Provide support
Muscle tissue	Muscle cells	Contract to bring about movement
Plant tissues		
Epidermal tissue	Epidermal cells	Covers the surface of plants
Photosyntheti c tissue	Palisade mesophyll cell	Produce food
Vascular tissue	Xylem vessels and phloem sieve tube elements	Transports water and organic food
Strengthenin g tissue	Sclerenchyma cells of the pericycle	Provides mechanical support

Some organs perform one function whiles others perform two or more functions.

Organ: A group of different tissues that perform the same functions.

Table 3.0: Examples of organs in animals and plants

Name of organ	Name of tissue forming organ	Main function
Animal organs		
Heart	Papillary muscles, cardiac muscle	Pumps food around the body
Kidneys	Cortex, medulla	Excretes metabolic waste substances
Plant organs		
Leaf	Epidermis, palisade mesophyll, spongy mesophyll, vascular bundles	prepares food, exchange gases, transpires
Root	Root hair, epidermis, cortex, xylem, phloem, pericycle, piliferous layer	Absorbs water and mineral salt from the soil and supports the plant
Stem	Epidermis, vascular bundles, parenchyma, collenchymas	Transports water and mineral, supports he flowers, fruits and leaves

Organ system: It is a collection of two or more organs which work together.

Organs systems are structured to work in a co-ordinate manner that an organism can undergo all the life activities and live independent of others.

Table 3.1: Examples of organ systems

Name of system	Main organ	Main functions
In animals		
Sensory system	Eyes, skin, ear, nose, tongue	Detects stimuli
Digestive system	Stomach, liver, intestines, pancreas	Digests and absorbs food
Circulatory system	Heart, arteries, veins, capillaries	Carries food and oxygen around the body
Reproductive system	Testis, uterus, ovaries	Produces offspring
In plants		
Shoot system	Stems, laves, buds, flowers, fruits	Makes and transports food, water, and minerals
Root system	Roots and their branches	Absorbs water and mineral, supports the plant

***Organism: Two or more organ systems come together to form an organism*.**

For example, the various systems such as the digestive system, respiratory system, reproductive system, nervous system, excretory system etc. work together to form one organism, man.
In plants, the shoot and the root systems come together to form a particular plant.

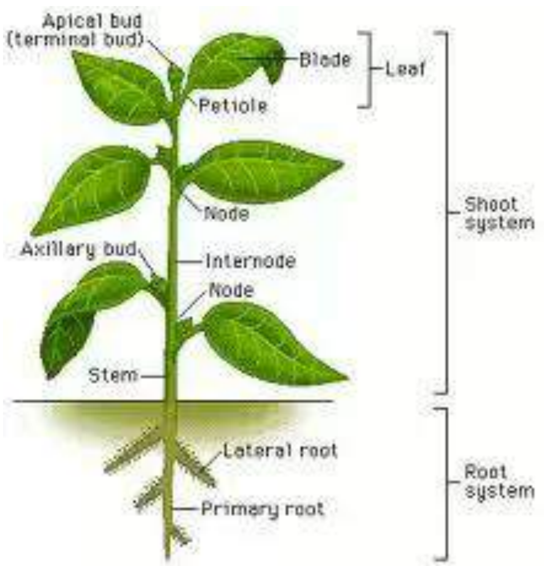

Fig. 4.4: *Shoot and root systems of a plant*

TEST QUESTIONS

1. (a) (i) What is a tissue?
   (ii) List three types of tissues in plant and animals
   (b) State one function of each of the tissues mentioned in (a) (ii) above.

2. (a) (i) Define the term cell.
   (ii) draw and label an animal cell.
   (iii) state three differences and two similarities between plant and animal cells.
   (b) (i) What are specialized cells?
   (ii) Mention three plants and animal cells.

3. (a) Define the following and explain how they occur:
   (i) meiosis,
   (ii) mitosis.
   (b) State three differences between meiosis and mitosis.

4. (a) State and explain the level organization in living organisms.
   (b) Mention two organ systems each in plants and animals and state their functions.

5. (a) Explain the term cell division.
   (b) What is the importance of cell division
   (c) Describe the types of cell division.

6. (a) Sate one function of each of the following organelles:
   (i) nucleus
   (ii) mitochondrion
   (iii) vacuole
   (iv) chloroplast
   (v) cell wall
   (vi) endoplasmic reticulum
   (vii) cytoplasm
   (viii) Golgi body
   (ix) cell membrane
   (x) chromosome
   (b) Which of the organelles in 6 (a) above are only found in plant cells?

6
ROCKS

Specific Objectives

After completing this chapter, you will be able to:

- Describe the major types of rocks, their formation and characteristics.
- Explain the process of weathering of rocks.

INTRODUCTION

***Rocks are naturally occurring solid material consisting of one or more minerals*.**

Minerals are solid chemical elements or compounds that are homogenous (have a definite chemical composition) and a very regular arrangement of atoms.

Rocks are everywhere, in the ground, forming mountains, and at the bottom of oceans. Earth's outer layer, or crust, is made mostly of rock. Some common rocks include granite and basalt.

TYPES OF ROCKS

Rocks are classified into three main types – igneous, sedimentary and metamorphic rocks, based mainly on their mode of formation.

Igneous rocks

Igneous rocks are hard, large, crystallize or glass-like rocks formed from a molten or partly molten material called magma.

Formation of Igneous Rocks

Magma forms deep underground when rock that was once solid melts. Overlying rock presses down on the magma, and the less dense magma rises through cracks in the rock. As magma moves upward, it cools and solidifies. Magma that solidifies underground usually cools slowly, allowing large crystals to form. Magma that reaches Earth's surface is called lava. Lava loses heat to the atmosphere or ocean very quickly and therefore solidifies very rapidly, forming very small crystals or glass. When lava erupts at the surface again and again, it can form mountains called volcanoes.

Fig. 4.5: Molten magma

Characteristics of Igneous Rocks

1. They are hard, large and heavy
2. They contain many crystals
3. Mostly resistant to weathering
4. They do not form layers
5. They do not contain fossils
6. Igneous rocks commonly contain the minerals feldspar, quartz, mica, pyroxene, amphibole, and olivine.

   Examples of igneous rocks are granite, pegmatite, rhyolite, gabbro, and basalt.

Fig. 4.6: Pegmatite

Sedimentary rocks

Sedimentary rocks are form when loose sediments or fragments of other rocks harden in layers.

Sedimentary rocks are also referred to as ***stratified rocks***.

Formation of Sedimentary Rocks

Sedimentary rock forms when sediments of other weathered rocks pile up into layers. As the sediments pile up, the weight of the layers of sediment presses down and compacts the layers underneath. The sediments become cemented together into a hard rock when minerals (most commonly quartz or calcite) precipitate, or harden, from water in the spaces between grains of sediment, binding the grains together.

Characteristics of Sedimentary Rocks

1. Sedimentary rocks weather easily.
2. They form in layers.
3. They are relatively soft.
4. They have lines of weakness between strata (layers).
5. They usually contain hardened remains of prehistoric animals called fossils.

Types of Sedimentary Rocks

Geologists place sedimentary rocks into three broad categories:

a. **Clastic rocks**, which form from ***clasts***, or broken fragments, of pre-existing rocks and minerals. Examples are: sandstone and shale

b. **Chemical rocks**, which form when minerals precipitate, or solidify, from a solution, usually seawater or lake water. The most common types of chemical rocks are called ***evaporates***, because they form by evaporation of seawater or lake water. Examples are: gypsum and halite

c. **Organic rocks**, which form from accumulations of animal and plant remains. Examples are limestone and coal.

Fig. 4.7*:* Sandstone

Metamorphic rocks

***Metamorphic rocks are formed when either igneous or sedimentary rocks change**, **(in a process called rock metamorphosis**)*.

Heat and pressure are the main causes.

Formation of Metamorphic Rocks

When igneous rocks or sedimentary rocks are exposed to great heat and pressure, they change their forms; and metamorphic rock is the result. If a mineral is heated or compressed beyond its stability range, it breaks down and forms another mineral.

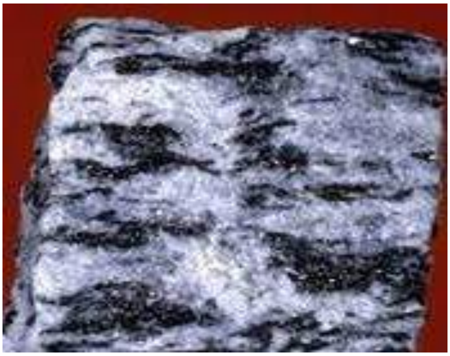

Fig. 4.8*:* Serpentine

Characteristics of Metamorphic Rocks

1. They are hard.
2. They have rough appearance and texture.
3. They are heavy.
4. Mineral distribution tends to be uneven sometimes in a process known as ***foliation*** (example gneiss).

Examples of metamorphic rocks formed from igneous rocks are gneiss, and serpentine (from granite and gabbro respectively).

Examples of metamorphic rocks formed from sedimentary rocks are marble, slate, and quartzite (from limestone, shale and sandstone respectively).

WEATHERING OF ROCKS

Weathering of rocks is the gradual breakdown of rocks into smaller particles over a period of time.

Types of Weathering

There are three types of weathering of rock, namely, Physical weathering Chemical weathering and Biological weathering.

Physical (mechanical) weathering

This is the type of weathering which results in the breakdown of rocks and the minerals they contain but does not change the chemical composition.

The mains causes of physical weathering are temperature, water and wind.

Temperature: Rocks expand at high temperature and contract at low temperature. The continuous expansion and contraction cause rocks to breakdown.

Water: Fast moving water exerts high pressure on rocks and causes them to breakdown. In some instances, the moving water may carry some rock particles which strike against others rocks to breakdown particle. These particles strike the surfaces of exposed rocks, causing small bits to chip off.

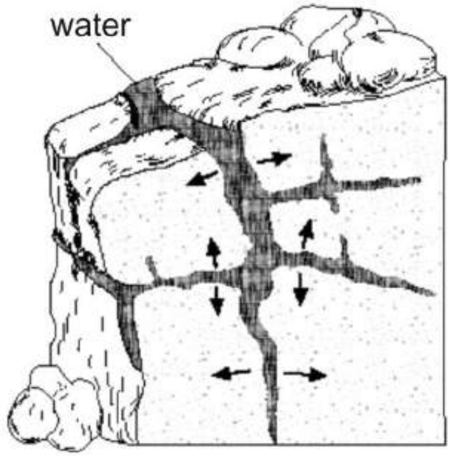

Fig. 4.9: weathering by water

Wind: Wind often carry many particles. Since wind usually travel at a high velocity, this causes the particles it carries to break off parts of larger rocks as they moves on.

Chemical weathering

Chemical weathering is the process in which existing minerals are broken down into new mineral components.

Chemical weathering is fastest in hot, moist climates and slowest in cold, dry climates. Causes of chemical weathering are oxidation, reduction, carbonation, hydration, hydrolysis and solution.

Oxidation: This is the reaction between oxygen in the atmosphere or oxygen dissolved in rain water with some rock minerals. For example the oxidation of rocks carrying iron from iron (II) oxide to iron (III) oxide. ($Fe_2O + O \rightarrow Fe_2O_3$).

This results in the breakdown of the rock along the surface where the oxidation takes place.

Reduction: This is the removal of oxygen from some rock minerals because of oxygen shortage; the rock breakdowns as a result of that.

Carbonation: This is the reaction between inorganic carbonic acid in soil water and some rock minerals. This reaction also causes the breakdown of the rock involved.

Hydration: This is the process whereby water molecules attach themselves to rock minerals, causing them to change into new minerals containing water of crystallization. The rock expands, softens, becomes porous, and as a result breaks down.

Hydrolysis: This is the breakdown of rock due to the reaction of water and rock minerals.

Solution: Some rock minerals are soluble in water. Therefore in contact with water they dissolve. This consequently weakens and breaks down the rock

Biological weathering

***This is the breakdown of rock into smaller particle due to the actions and activities of living organisms*.**

It involves the cracking and breaking down of rocks under pressure, including stress from the growing roots of plants and the action of humans in activities such as farming, road construction and building. Plants, fungi, algae and mosses cause biological weathering by growing on the surface into the rock crevices, creating considerable pressure. They also produce organic acids which dissolves rock minerals and assists in their disintegration.

Fig. 4.9: Surface of a weathering rock

TEST QUESTIONS

1. (a) State three characteristics each of igneous, sedimentary and metamorphic rocks.
   (b) Describe the formation of metamorphic rock.

2. (a) What is weathering of rocks?
   (b) Describe the following causes of chemical weathering:
   (i) hydration
   (ii) carbonation
   (iii) oxidation

3. State and explain the three types of weathering.

4. Explain how the following cause weathering of rocks
   (a) Water;
   (b) Temperature;
   (c) Wind.

7

AIR MOVEMENT

Specific Objectives

After completing this chapter, you will be able to:

- Explain the formation of land and sea breeze.
- Identify the various types of air masses and describe their pattern of movement.
- Describe the effect of moving air masses.

INTRODUCTION

Have you ever wondered why during the day the temperature on the land is hotter than that of the ocean; and why at night the temperature on the sea become hotter than that of the land? This happens as a result of land and sea breeze.

LAND AND SEA BREEZE

The differences in temperature between the land and the sea causes land and the sea breeze. As a result of the temperature differences, convection currents (air) move from the sea to the land or vice versa.

Formation of land breeze

Land breeze occurs at night, when the land cools down much quicker than the sea. Therefore, the temperature of the air over the land is lower than that of the air over the sea. Cooler air (convection currents) from the land blows towards the sea, often bringing with it clouds which usually produce rain along the coast.

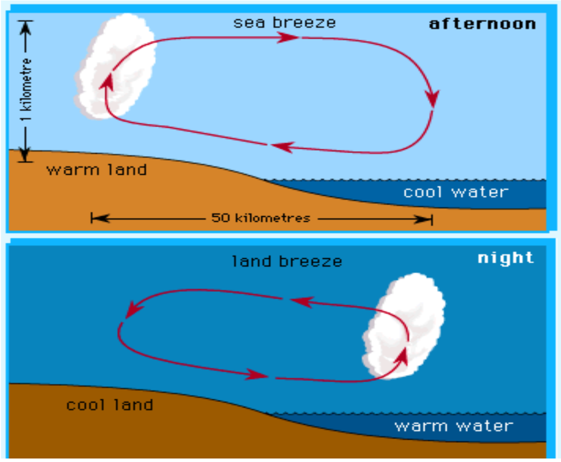

Fig. 5.0: Land and sea breeze

Formation of sea breeze

During the day, sea breeze forms when the air over the land is warmed up by the sun causing it to become less dense. This warmer, less dense air rises, and is replaced by cooler air above the sea.

AIR MASSES

An air mass is a body of air that extends over a large area and has nearly uniform temperature and humidity in any horizontal direction.

Air masses cover many hundreds or thousands of square miles, and adopt the characteristics of the surface below them. Places where air masses form are called *source regions*.

Classification and notation of air masses

Air mass classification involves three set of letters. The first set describes its moisture properties, with ***c*** used for continental air masses (dry) and ***m*** for maritime air masses (moist). The second set describes the thermal characteristic of its source region: ***T*** for Tropical, ***P*** for Polar, ***A*** for Arctic or Antarctic, ***M*** for monsoon, ***E*** for Equatorial, and ***S*** for superior air (dry air formed by significant downward motion in the atmosphere).

Formation of air masses

The uneven heating of the earth causes air masses to circulate. Air masses at the equator are heated and made lighter. That causes them to move towards the poles. A low pressure is created around the equatorial region. The air that sinks at latitude 30 degrees north and south causes a zone of high-pressure call ***horse latitudes***. As the air moves further away from the equator, it cools and sinks, creating high pressure at the poles. These cool air mass at the poles flow back towards the pressure belt at the equator.

Air currents flow outwards across the surface of the earth from the horse latitudes. The winds that blow toward the equator are called the ***trade winds***, and those that blow towards the poles are called the ***westerly winds*** or ***westerlies***. The westerlies converge with cold air currents called ***easterlies***. The easterlies are very cold and dense air moving from the poles to the equator.

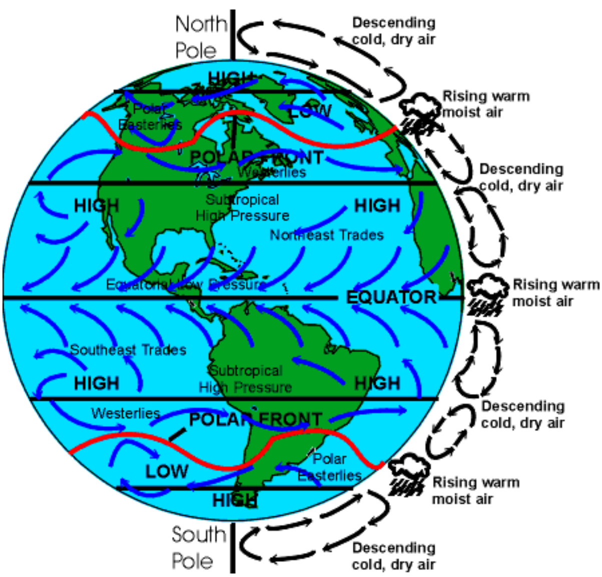

Fig. 5.1: Wind movement on the Earth

Types of air masses

Storm

A storm is an atmospheric disturbance in the form of strong winds usually accompanied by rain, snow, or other precipitation and sometimes accompanied by lightning and thunder.

Cyclone

Cyclone is an unstable weather conditions created at the transitional zones of the equator. Cyclones are strong storms that whirl while moving. This is due to the intermingling of polar and subtropical air. The rotation of the earth deflects air to the right of the natural direction in the northern hemisphere and to the south to the southern hemisphere.

Fig. 5.2: A shot of a hurricane taken by a satellite

Tornado

A tornado is a violently rotating column of air that is in contact with both the surface of the earth and a cumulonimbus cloud or, in rare cases, the base of a cumulus cloud. Tornadoes come in many shapes and sizes, but they are typically in the form of a visible condensation funnel, whose narrow end touches the earth and is often encircled by a cloud of debris and dust. Most tornadoes have wind speeds less than 177 km/h, 76 m across, and travel several kilometres before dissipating. The most extreme tornadoes can attain wind speeds of more than 483 km/h, stretch more than 3.2 km across, and stay on the ground for more than 100 km.

Fig. 5.3: A tornado

EFFECTS OF MOVING AIR MASSES

1. **Destruction of landscape**: Storms are accompanied by great winds and rains which erode lands especially in the coastal regions where storms are always stronger.

2. **Flood:** Torrential rains (heavy rains) triggered by storms cause areas, including non costal area to flood, which is always fatal and devastating.

3. **Pollution:** Moving air masses sweep out everything in their path including pollutants. They transport pollutants from areas where they are less harmful to places where they are more prone to cause harm to ecosystems. Industrial waste can easily be transported to residential areas.

4. **Temperature regulation:** One good effect of moving air masses is they regulate temperatures by transporting heat and energy from the tropics to the temperate latitudes. This helps maintain balance in the earth's troposphere, and helps in seasonal changes.

5. **Supply of water**: Another good effect of moving air masses is they flood areas devastated by drought by transporting water there.

Fig. 5.4: Devastating effects of tropical storms

PRECAUTIONS AGAINST THE EFFECTS OF STORMS

1. Always listen out to weather forecasts and reports on radio or television.
2. Move to a safer areas if a storm is forecast to strike in your area
3. Since storms could cause closure of stores and food joints, store enough food with you.
4. Keep emergency phone numbers handy.
5. Stay away from the sea and coastal area.
6. After the storm, makes sure everything is in order and safe. Check gas leakages and other appliances especially electronic appliances before using them; make sure they are well dry.

TEST QUESTIONS

1. Describe the formation of land and sea breezes.

2. State four precautions that should be taken against storms.

3. (a) What is an air mass?
   (b) Describe the formation of the following air masses:
      (i) tornado
      (ii) cyclone

4. (a) Explain the term trade winds.
   (b) Describe the formation of trade winds

8

NITROGEN CYCLE

> **Specific Objectives**
> After completing this chapter, you will be able to:
> - Describe the nitrogen cycle.
> - Explain the importance of nitrogen cycle to plants and animals.

INTRODUCTION

***Nitrogen Cycle is a natural cyclic process by which atmospheric nitrogen enters the soil and becomes useful to living organisms, before returning to the atmosphere*.**

Nitrogen, an essential part of the amino acids, is a basic element of life. It also makes up 78 percent of the Earth's atmosphere; however gaseous nitrogen must be converted to a chemically usable form before it can be used by living organisms. This is accomplished through the nitrogen cycle, in which gaseous nitrogen is converted to ammonia or nitrates.

The high energies provided by lightning and cosmic radiation serve to combine atmospheric nitrogen and oxygen into nitrates, which are carried to the Earth's surface in precipitation (e.g. rain).

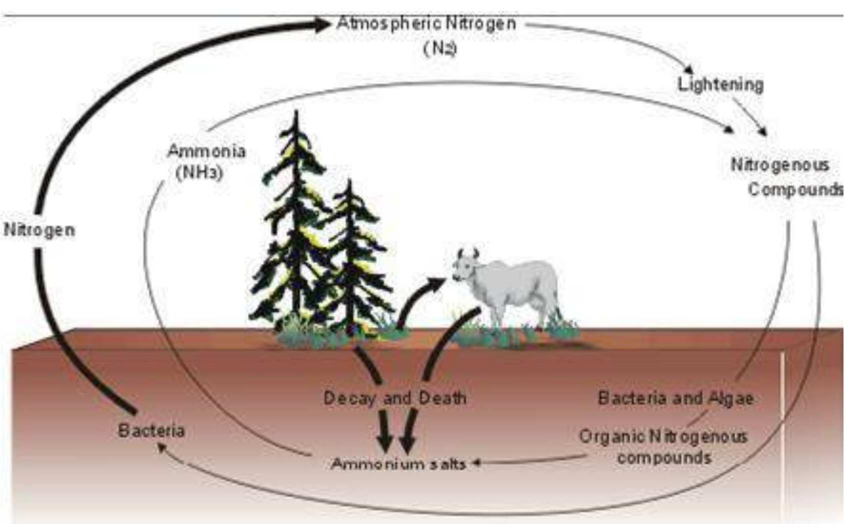

Fig. 5.5: *Nitrogen cycle*

NITROGEN FIXATION

Nitrogen Fixation is a process by which molecular atmospheric nitrogen is converted into a chemical compound that is essential for plant growth.

There are two major ways of nitrogen fixation – biological nitrogen fixation and industrial nitrogen fixation.

Biological nitrogen fixation

This is the most widely and useful form of nitrogen fixation. In this method, microbes are the agents for fixing nitrogen in the soil. Microorganisms capable of nitrogen fixation are symbiotic bacteria of the genus *Rhizobium*, which colonize and form nodules on the roots of leguminous plants such as beans, groundnut and cowpeas. These bacteria obtain food from the legume, which in turn is supplied with abundant nitrogen compounds.

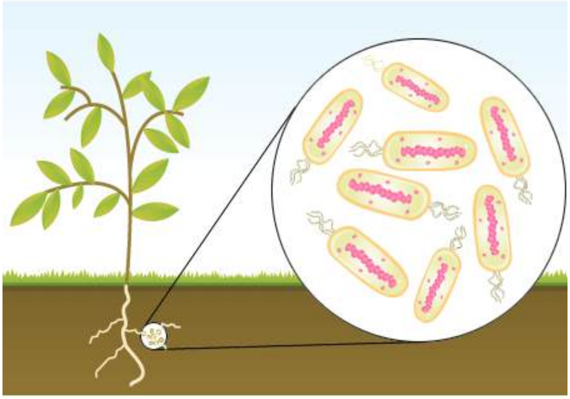

Fig. 5.6: Bacterial nitrogen fixation

Industrial nitrogen fixation

The principal industrial nitrogen-fixation process today is the production of ammonia by combining atmospheric nitrogen and hydrogen. Ammonia is then oxidized to form nitric acid, which is in turn combined with ammonia to yield ammonium nitrate, used primarily in explosives and fertilizers. In another method, cyanamide, which is used as a fertilizer or in the production of cyanides, is produced by passing atmospheric nitrogen over heated calcium carbide in.

Some processes involved in nitrogen fixation

Nitrification

The conversion of ammonia into nitrate is performed mainly by soil living bacteria. Nitrification involves two main stage:

i. conversion of ammonium ion NH_4 to nitrite ions NO_2

ii. oxidation of the nitrite ions, NO_2 to nitrate ions, NO_3

Denitrification

This is the reduction of nitrates back into the largely non-reactive nitrogen gas (N_2), which completes the nitrogen cycle process. Again, bacteria are responsible for this process.

Importance of nitrogen cycle to plants and animals

Lost nitrogen in the soil is replaced through the nitrogen cycle process. This improves the fertility of the soil. The availability of nitrogen in the soil means plants can easily get access to it for proper growth. Nitrogen, which is an important source of nutrient for animals as well, is passed on to animals from plants. This occurs when animals eat the plants that have absorbed nitrogen from the soil through their roots.

In a nutshell, nitrogen cycle is very important because:

1. It supplies plants with needed nitrogen (nitrate ions).
2. It improves the fertility of the soil.
3. It improves the productivity and structure of the soil and sustains the activities and growth of microbes.
4. It prevents the accumulation of worn-out plant and animals tissues in the soil.
5. It releases nitrogen which is stack-up in organisms.
6. Nitrogen is an important nutrient which helps plants to manufacture their food for the nourishment of other organisms.

TEST QUESTIONS

1. What is nitrogen cycle?

2. State and explain the two major ways of nitrogen fixation.

3. Describe the importance of nitrogen cycle to plants and animals.

4. With the aid of a suitable diagram, describe how the nitrogen cycle occurs.

9

THE SKELETAL SYSTEM

Specific Objectives

After completing this chapter, you will be able to:

- Describe the structure and functions of the mammalian skeleton.

INTRODUCTION - THE MAMMALIAN SKELETON

A skeleton is a large interconnected bones that supports and protects the body of living organisms.

The skeleton supports the soft tissues and provides leverage for movement. An adult man has about 206 bones in the body as opposed to 275 in a child.

Types of skeletons

There are two types of skeletons, namely, endoskeleton and exoskeleton.

Endoskeleton: This is the type of skeleton found within the bodies of vertebrate (animals with backbones). Examples of animals with endoskeleton are humans, birds, fish, snakes, etc.

Exoskeleton: This type of skeleton is found outside the bodies of invertebrates (animals without backbones). Examples of animals with exoskeleton are shrimp, insects, prawn, crab, etc.

STRUCTURE AND FUNCTIONS OF THE MAMMALIAN SKELETON

The human skeleton is divided into two distinct parts – the axial skeleton and the appendicular skeleton.

The axial skeleton

This consists of bones that form the axis of the body and support and protect the organs of the head, neck, and trunk. Examples are:

The Skull: The skull is the skeleton which is on top of the vertebral column in all vertebrates. The skull encases and protects the brain and provides attachment for the muscles of the face and mouth.

The Sternum: Also known as the breastbone, the sternum is a rigid structure which is found at the front of the chest to form the thoracic cage with the ribs

The Ribs: the ribs are long, slender bones attached to the backbone that curve around the chest cavity, or thorax. Ribs occur in pairs and are found in almost all vertebrates, or animals with backbones.

The Vertebral Column: Also called spinal column or backbone, the vertebral column is the structure of bone or cartilage surrounding and protecting the spinal cord in vertebrate

The spinal column forms the major part of the skeleton. To it are attached the skull, shoulder bones, ribs, and pelvis. The vertebral column consists approximately of thirty-three bones called *vertebrae.* Between each pair of vertebrae is a disk-shaped pad of fibrous cartilage with a jelly-like core, which is called the *intervertebral disk.*

Each pair of vertebrae is connected by a joint which stabilizes the vertebral column and allows it to move. These disks cushion the vertebrae during movement. The entire spine encloses and protects the spinal cord, which is a column of nerve tracts running from every area of the body to the brain.

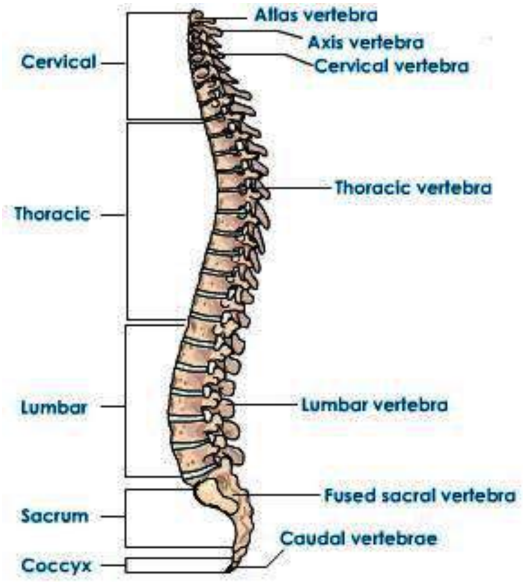

Fig. 5.7*:* The vertebral column of man

Types of vertebrae

There are five major vertebrae in mammals:

- ❖ **Cervical vertebrae**: found around the neck area
- ❖ **Thoracic vertebrae**: found around the chest area.
- ❖ **Sacral vertebrae**: found around the lower abdomen
- ❖ **Caudal vertebrae**: found around the tail area.
- ❖ **Lumber vertebrae:** found around the thoracic cavity.

The appendicular skeleton

This is composed of bones that anchor the appendages to the axial skeleton. Examples are: the shoulder girdle and the pelvic girdle. (The sacrum and coccyx are considered part of the vertebral column)

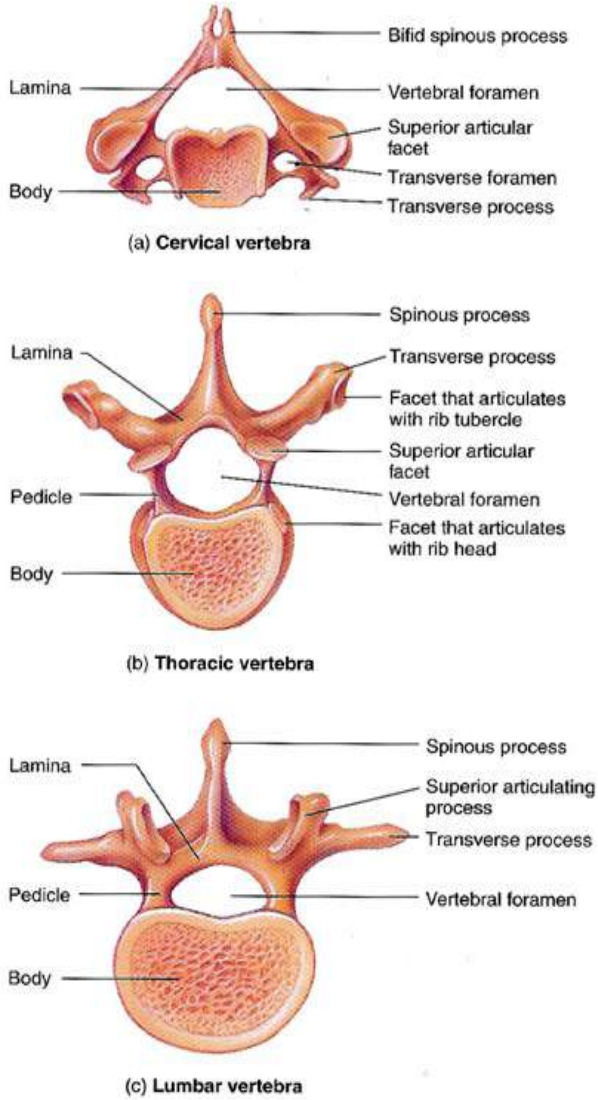

Fig. 5.8 :Superior view of the vertebrae

TYPES OF BONES

The bones of the body fall into four general categories: long bones, short bones, flat bones, and irregular bones.

Long bones: these bones are longer than they are wide and work as levers. The bones of the upper and lower extremities (examples humerus, tibia, femur, ulna, metacarpals, etc.) are of this type.

Short bones: These bones are short, cube-shaped, and found in the wrists and ankles.

Flat bones: These bones have broad surfaces for protection of organs and attachment of muscles (examples are ribs, cranial bones, bones of shoulder girdle).

Irregular bones: These are all other bones that do not fall into the above categories. They have varied shapes, sizes and surfaces features and include the bones of the vertebrae and a few bones in the skull.

Bone composition

Bones are composed of tissue that may take one or two forms. Compact or dense bone, and spongy or cancellous bone. Most bones contain both types.

Compact bone is dense, hard, and forms the protective exterior portion of all bones.

Spongy bone is inside the compact bone and is very porous (full of tiny holes). Spongy bone occurs in most bones. The bone tissue is composed of several types of bone cells embedded in a web of inorganic salts (mostly calcium and phosphorus) to give the bone strength, and collagenous fibres and ground substance to give the bone flexibility

Structure of human skeleton

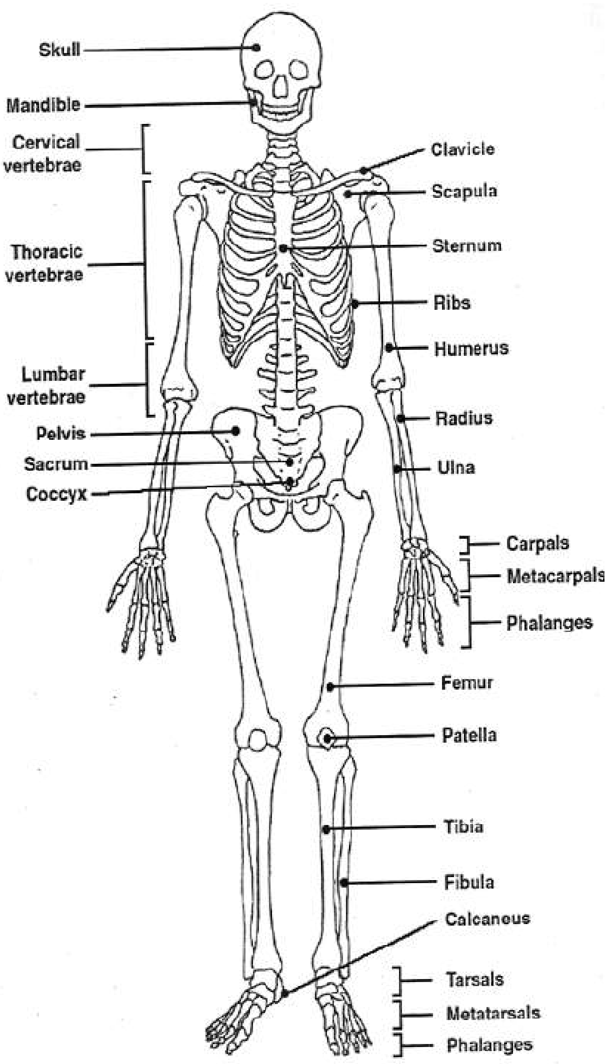

Fig. 5.9: Structure of the human skeleton

Skull

The skull is the bony framework of the head. It is comprised of the eight cranial and fourteen facial bones.

Functions:

- ❖ It encases the brain
- ❖ It protects the brain
- ❖ It provides attachment for the muscles of the face and mouth.

Scapula (the shoulder blade)

A flat, triangular bone that lies over the back of the upper ribs.

Functions:

- ❖ It serves as an attachment for some of the muscles and tendons of the arm, neck, chest and back and
- ❖ It aids in the movements of the arm and shoulder.

Spine, vertebra and disk

The spine is a column of bone and cartilage that extends from the base of the skull to the pelvis.

Functions:

- ❖ It encloses and protects the spinal cord.
- ❖ It supports the trunk of the body and the head.

The sacrum

The sacrum, at the base of the vertebral column, is wedged between the coaxial bones of the pelvis and is attached to them by fibro-cartilage at the sacroiliac joints.

Function:

- ❖ The weight of the body is transmitted to the legs through the pelvic girdle at the sacrum.

Sternum (breastbone)

A long, narrow, flat plate that forms the centre of the front of the chest.

Tarsal bone

The foot consists of an ankle, an instep, and five toes. The ankle is composed of seven tarsal bones, forming a group called the *tarsus*.

Function:

- ❖ It helps to support the weight of the body and provides an attachment for muscles that move the foot.

Carpal bones

The skeleton of the wrist consists of eight small carpal bones that are firmly bound in two rows of four bones each. .

Clavicle

The clavicle is the collarbone. There are two of these bones, each curved a little like an "f," that joins the top of the breastbone (sternum) to the shoulder blade (scapula).

The clavicles support the arms and transmit force from the arms into the central skeleton.

The coccyx

The coccyx (or tail) is the lowest part of the vertebral column and is attached by ligaments to the margins of the sacral hiatus.

When a person is sitting, pressure is exerted on the coccyx, and it moves forward, acting like a shock absorber. (Sitting down with too great a force may cause the coccyx to be fractured or dislocated).

The ribs

Ribs are flat, curved bones that form the framework of the chest and make up a cage to protect the heart, lungs and other upper organs. There are twelve pairs of ribs, each joined at the back of the cage to a vertebra in the spine. There are seven ***true ribs*** attached to the sternum (breastbone) directly by their costal cartilages. The remaining five pairs are called ***false ribs***, because their cartilages do not reach the sternum directly. Instead, the cartilages of the upper three false ribs join the cartilages attached to the ribs above, while the last rib pairs have no cartilaginous attachments to the sternum at all. These last two pairs are sometimes called ***floating ribs***.

Functions:

- ❖ Protect some vital organs such as the heart and lungs
- ❖ Aids in breathing

Phalanges

The phalanges are the small bones that make up the skeleton of the fingers, thumb and toes. Each finger and smaller toe has three phalanges; the thumb and big toe each have two. The phalange nearest the body of the hand or foot is call the ***proximal phalange***; the one at the end of each digit is the ***distal phalange***; and when there are three, the middle one is called the ***middle phalange***.

Femur

The femur is the thigh bone, the longest bone in the body. It lowers into a ball (or head of the femur) that fits into a socket in the pelvis to form the hip joint.

Fibula

The fibula is the outer and thinner of the two long bones of the lower leg. It is much narrower than the other bone (the shin), to which it runs parallel and to which it is attached at both ends by ligaments. The upper end of the fibula does not reach the knee, but the lower end descends below the shin and forms part of the ankle. Its main function is to provide attachment for muscles. It doesn't give much support or strength to the leg, which explains why the bone can safely be used for grafting onto other bones in the body.

Humerus

The humerus is the bone of the upper arm. The smooth, dome-shaped head of the bone lies at an angle to the shaft and fits into a shallow socket of the scapula (shoulder blade) to form the shoulder joint. Below the head, the bone narrows to form a cylindrical shaft. It flattens and widens at the lower end and, at its base, it joins with the bones of the lower arm (the ulna and radius) to make up the elbow.

Pelvis (or Os Coxa)

The pelvis is a ring of bones in the lower trunk of the body, which is bounded by the coccyx (tail bone) and the hip bones.
The pelvis protects abdominal organs such as the bladder, rectum and, in women, the uterus.

Tibia

The tibia is the inner and thicker of the two long bones in the lower leg. It is also called the ***shin bone***.
The tibia is the supporting bone of the lower leg and runs parallel to the other, smaller bone (the fibula) to which it is attached by ligaments.

Functions of the skeleton

1. It gives shape to the body
2. It protects the internal organs of the body
3. It aids in movement.
4. The ribs aid in breathing.
5. The skeleton provides points of attachment for the muscles.
6. Red blood cells are produced in the bone marrow.
7. The skeleton gives support to the body.

Ligaments

A ligament is a band of tissues that connect bones or cartilages.
Ligaments are tough, fibrous, slightly elastic tissue and white which in colour,.

Ligaments, especially those in the ankle joint and knee, are sometimes damaged by injury. A torn ligament usually results from twisting stress when the knee is turned while weight is on that particular leg. Minor sprains are treated with ice, bandages and sometimes physical therapy, but if the ligament is torn, the joint may be placed in a plaster cast to allow time to heal or it may require surgical repairs. If a ligament is made up of several thick bands of fibrous branches, it is called a ***collateral ligament***.

Examples of ligaments in a body are:

- Intertransverse Ligaments,
- Interclavicular Ligaments,
- Fibular Collateral Ligament,
- Ligaments of The Foot, etc.

Functions of the ligament

1. It binds the bone ends together to prevent dislocation and excessive movement that might cause breakage.
2. Ligaments also support many internal organs; including the uterus, the bladder, the liver, and the diaphragm
3. It helps in shaping and supporting the breasts.

JOINTS

Joints are areas where bones or cartilages in the skeleton meet.

There are two main types of joints in the skeleton:

i. movable joints and
ii. immovable joints.

Movable joints

Movable joints allow movement of parts of the body. They consist of an external layer of fibrous cartilage giving rise to strong ligaments that support the separate bones. The bones of movable joints are covered with smooth cartilage and are lubricated by a thick fluid, called ***synovial fluid***, produced between the bones in membranous sacs, known as ***bursae***. Bursitis, or inflammation of the bursae, is a common painful condition of movable joints.

Types of movable joints

Ball and socket joint

This is made up of a bone with a round end which fits into another with a hollow end. This joint allows movement in more than one area. Examples are the shoulder joint and the hip joint.

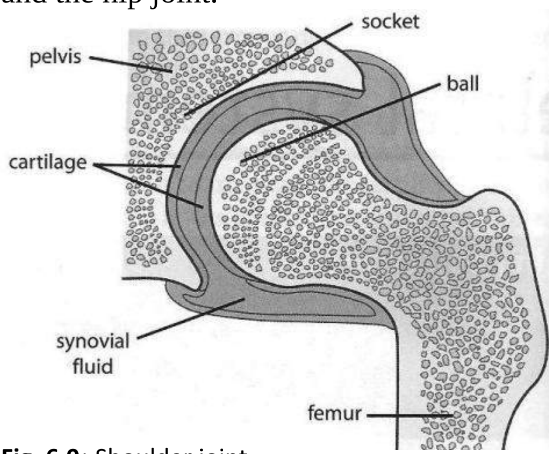

Fig. 6.0: Shoulder joint

Hinge joint

This joint allows movement in only one area. Examples are the elbow joint and the knee joint.

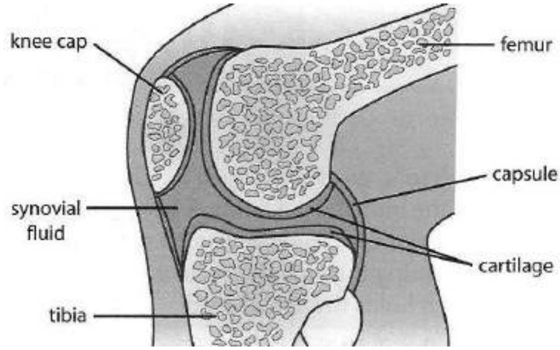

Fig. 6.1: *Knee joint*

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Gliding joint

This joint allows slide movement of bones over each other

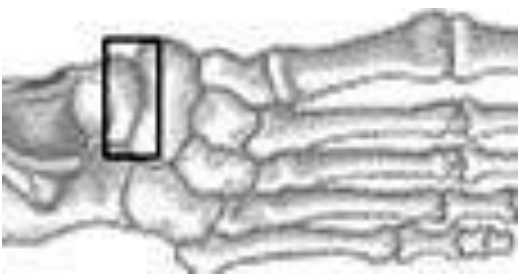


Fig. 6.2: Gliding joint

Immovable/ fixed joints

Immovable joints are held together by actual intergrowth of bone or by strong fibrous cartilage; this prevent movement at all areas. Examples are the pelvic girdle and the *suture* (joint in the skull).

TEST QUESTIONS

1. What is a skeleton?

2. Enumerate five functions of the mammalian skeleton.

3. What is a joint?

4. Mention three types of movable joints and give one example each.

5. The following are some parts of the mammalian skeleton. Place them in the table below as axial or appendicular skeleton.
   - Skull
   - Ribs
   - Pelvic girdle
   - Sternum
   - Arm
   - Leg
   - Shoulder girdle

Axial skeleton	Appendicular skeleton

10

REPRODUCTION AND GROWTH IN PLANTS

Specific Objectives

After completing this chapter, you will be able to:

- Identify parts of a flower and variations in flower structure.
- Describe the process of pollination and fertilization.
- Describe different type of fruits.
- Describe the structure of seeds and state the functions of their parts.
- Describe the mechanisms of seed and fruit dispersal.
- Describe the process and conditions necessary for germination.
- Describe vegetative (asexual) reproduction in plants.

INTRODUCTION

Reproduction is the process whereby all living organisms produce new individuals of their own kind.

Reproduction is one of the essential functions of plants, animals, and single celled organisms as far as the preservation of the species is concerned.

Types of reproduction

There are two types of reproduction – sexual and asexual reproduction.

Sexual reproduction

Sexual reproduction is the type of reproduction where the male and the female egg cells (gametes) fuse together to give rise to a new individual.

Examples of organisms which reproduce sexually are: humans, cats, birds, reptile, plants, etc.

Asexual reproduction

This type of reproduction involves the use of a part or whole of an organism to reproduce a new one. Asexual reproduction is very prominent in plants.

There are two types of asexual reproduction in plants – artificial reproduction and vegetative reproduction or vegetative propagation.

FLOWER

A flower is the sexual reproductive part of a plant which may produce fruits and seeds.

Flowers bear the sex organs that produce male and female gametes, and therefore carry out the multiple roles of sexual reproduction, seed development, and fruit production.

STRUCTURE OF A FLOWER

Flowers typically are composed of four parts, or whorls, arranged in concentric rings attached to the tip of the stem. From innermost to outermost, these whorls are the:

- gynaecium (found at the centre of the flower)
- androecium (surrounded by the corolla)
- corolla (collection of petals)
- calyx (collection of sepals)

Gynaecium

This is the innermost whore. It is made up one or more ***carpels*** or ***pistil***s. The carpel is the female organ of the flower and produces the female gametes. Each carpel has three parts:

- Ovary: contains ovules
- Style: the stalk that connects the stigma to the ovary
- Stigma: receives pollen grains during pollination.

Androecium

The androecium consists of the male sex organs called ***stamens***. Each stamen is made up of an anther and a filament. The anther produces pollen grains. The filament is a stalk that supports the anther.

Corolla

The corolla consists of petals. Petals are large, brightly coloured and usually scented to attract animal pollinators. Petals may also possess ***nectaries*** which produce nectar, a sugary liquid.

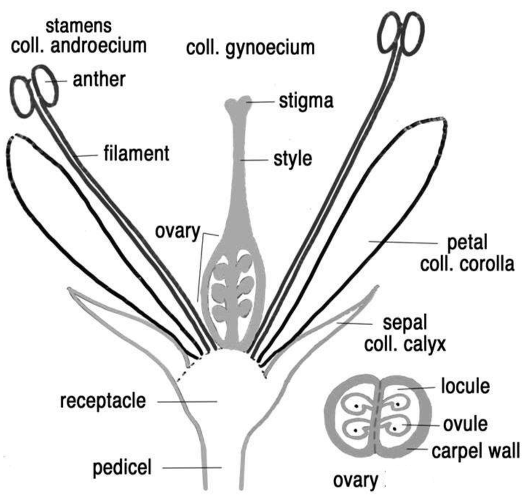


Fig. 6.2: *Structure of a flower*

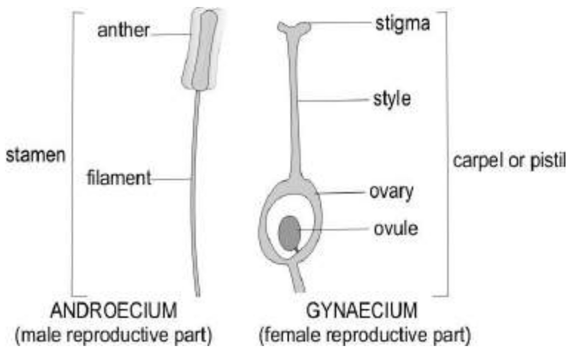


Fig. 6.3*:* Male and female reproductive parts

Calyx

The calyx is the outermost whorl and is made up of sepals. Sepals are mostly green in colour. The calyx supports the inner whorls.

Types of flowers

The variations in flowers gives rise to two main types of flowers – bisexual and unisexual flowers.

Bisexual /hermaphroditic flowers: These are the type of flower which has both male and female gamete –androecium (stamens) and gynaecium (carpels). In other words, they are complete flowers. Examples of bisexual flowers are hibiscus, flamboyant, pride of Barbados, crotalaria.

Unisexual flowers: These flowers have only one of the reproductive parts. - they have either androecium or gynaecium. They are therefore termed as incomplete. Examples of unisexual flowers are pawpaw, water melon, etc.

PROJECT WORK:

- Make a collection of about ten different flower specimens.
- Open fully to locate the main parts.
- Mount each flower on a separate sheet of paper and the parts separated out and labeled.
- Enclose each flower with a plastic sheet and transparent tape.
- Display work by groups on a bulletin board or science table for award of marks.

POLLINATION

Pollination is the transfer of pollen grains from the anther of a flower to the stigma of a flower.

The stigma could be on the same flower or another flower on the same plant (*autogamy*) or another plant of the same species (*geitogamy*).

Types of pollination

Pollination comes in two types –self-pollination and cross pollination

Self-pollination: This is the transfer of pollen grains form the anther of a flower to the stigma of the same flower or another flower on the same plant.

Cross-pollination: It is the transfer of pollen grains from the anther of a flower to the stigma of another flower or another plant of the same species.

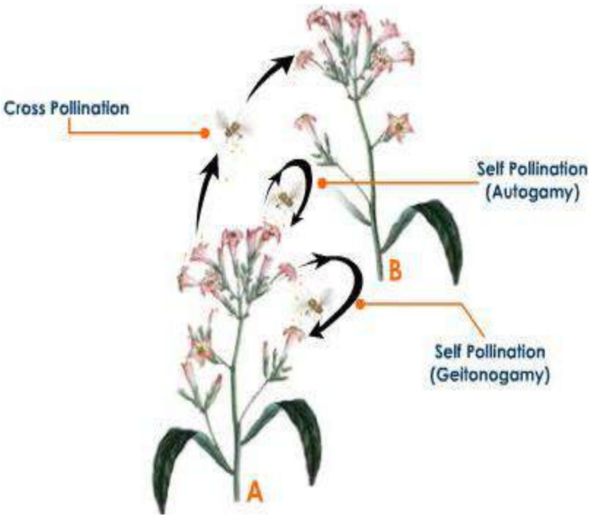


Fig. 6.4: The pollination process

Adaptations of Plants for Self Pollination

1. Each flower is bisexual (has both male and female gametes)
2. Flowers have the anther and stigma maturing at the same time (***homogamy***)
3. Flowers remain closed until self-pollination has occurred. (***cleistogamy***)

Adaptations of Plants for Cross Pollination

1. Flowers are unisexual.
2. Male and female flowers grow on different plants
3. Anther and stigma mature at different times
4. The style of the gynaecium and the filaments of androecium differ in length.
5. Pollen grains of the flower cannot fertilize their own flower

Advantages of Self Pollination

1. Does not need any agent of pollination.
2. Pollen grains are transported over a short distance.
3. Less pollen grains are wasted or fall off.

Advantages of Cross Pollination

1. The pollen grains are brought from different flowers, which ensures variety. Varied seeds are often resistant to diseases.
2. Seeds produced after cross pollination are healthier than those from one parent (self pollination).
3. Pollination is still possible even when stamen and anthers mature at different times.

Agents of Pollination

Agents of pollination are the factors which are responsible for the transfer of pollen grains from the anther to the stigma of flowers.

The agents of pollination include insects, wind, man, water, birds, and bats. Insects and wind are the prominent agents of pollination.

Fig. 6.5: Birds and flying insects as agents of pollination

Table 3.3: Characteristics of insect and wind pollinated flowers

Insect pollinated flowers	Wind pollinated flowers
Large and conspicuous	Small and inconspicuous
Brightly coloured petals	Petals are dull in colour
Petals produce nectar	Petals do not produce nectar
Small and compact stigma found inside the flower	Large , feathery stigma hanging outside the flower
Produce small quantities of pollen grains	Produce large quantities of pollen grains
Produce large, rough, heavy pollen grains	Produce small, smooth, light pollen grains
Stamens have short, thick filaments	Stamens have long, thin filaments
Usually scented	Not scented

The characteristics above are also the comparisons between insect and wind pollinated flowers.

Variations in flower structure

Flowers display many variations in their structure. Most flowers have all four whorls—gynaecium, androecium, corolla, and calyx. These are called ***complete flowers***. However, some flowers are incomplete, meaning they lack one or more whorls. ***Incomplete flowers*** are most common in plants whose pollen is dispersed by the wind or water. Since these flowers do not need to attract pollinators, most have no petals, and some even lack sepals.

Flowers that lack either stamens or a pistil are said to be ***imperfect***. Imperfect flowers can still function in sexual reproduction. Flowers that have only stamens are termed ***staminate***, and flowers that have only a pistil are called ***pistillate***.

Although a single flower can be either staminate or pistillate, a plant species must have both to reproduce sexually. In some species with imperfect flowers, the staminate and pistillate flowers occur on the same plant. Such plants, known as ***monoecious species***, include maize.

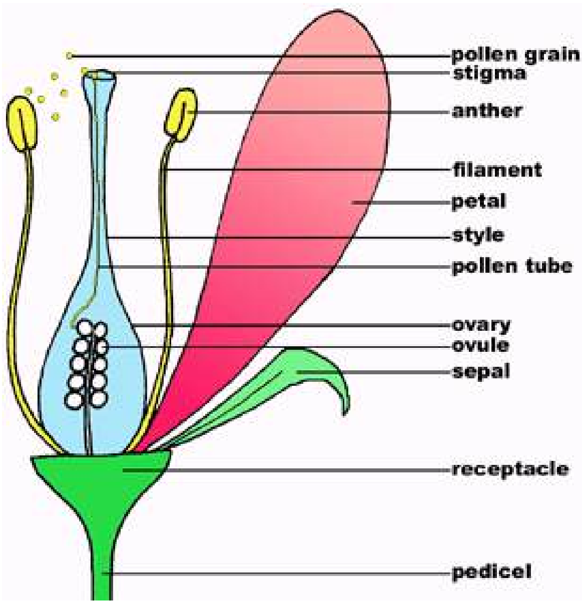


Fig. 6.6: A monoecious flower

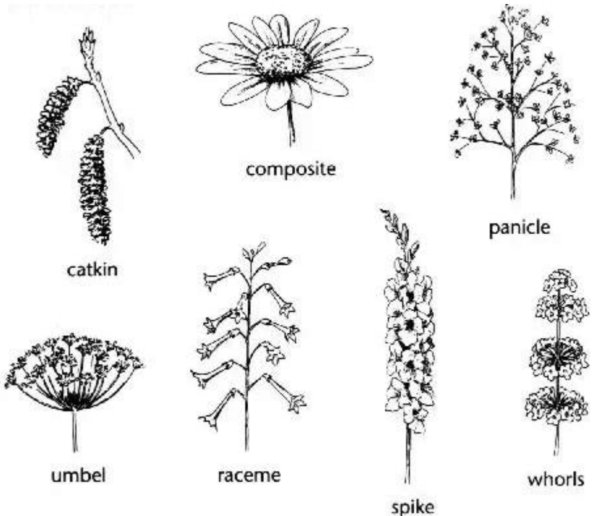


Fig. 6.7: Some types of inflorescence

In ***dioecious species*** such as date, willow, and hemp, staminate and pistillate flowers are found on different plants. A date tree, for example, will develop male or female flowers but not both. In dioecious species, at least two plants, one bearing staminate flowers and one bearing pistillate flowers, are needed for pollination and fertilization.

A group of flowers with a common stalk is called an ***inflorescence***. The stalk of an inflorescence flower is called the ***peduncle.*** Each flower has its own stalk called ***pedicel.*** Some flowers lack a stalk and are describe ***as sessile flowers***.

FERTILIZATION

Fertilization is the process whereby the male and the female gametes fuse together to form a zygote.

Fertilization occurs in the ovule, which contains the female gametes known as ***ovum***.

Mechanism of Fertilization in Plants

- ❖ When a mature pollen grain is deposited on a mature stigma, it absorbs water and nutrients from the stigma and swells.
- ❖ The nucleus of the pollen grain divides into two unequal sizes. The larger one is called the generative nucleus and the smaller one the vegetative nucleus or pollen tube.

- The walls of the pollen grain slits and a pollen tube protrudes from it. The pollen tube then penetrates the stigma and grows through its tissue into the style.
- The nucleus of the pollen tube moves to the tip of the pollen tube, followed by the generative nucleus.
- The pollen tube grows through the style toward the ovary. It gets nutrients from the style. As the pollen tube gets closer to the ovary, the generative nucleus divides into two male nuclei.
- The ovary contains one or more ovules each consist of an embryo sac and two protective coats called inner and outer integuments.
- A small pore called ***micropyle*** is found at one end of the ovule. The pollen tube enters the ovary grows to an ovule. It then grows through the micropyle, and enters the embryo sac. The pollen tube nucleus disintegrates and disappears.
- The two male nuclei are released into the embryo sac. One male nucleus fuses with the egg nucleus to form zygote.
- The other male nucleus moves to the centre of the embryo sac and fuses with the polar nuclei to form the primary endosperm nucleus. This is known as double penetration, and occurs only in flowering plants.

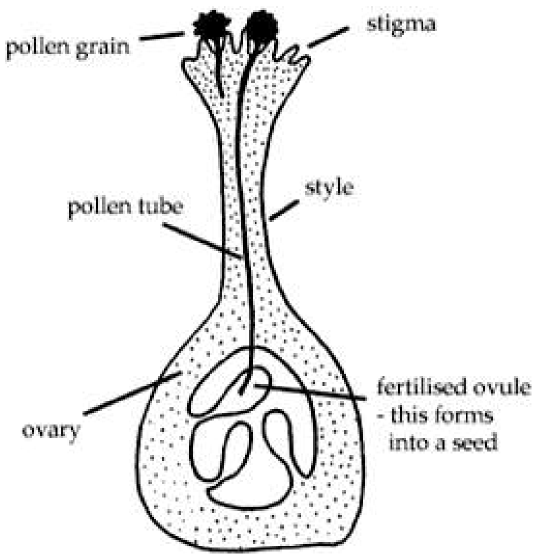


Fig. 6.8: The fertilization process

FRUITS

A fruit is a fertilized ovary which protects and disperses seeds

Type of Fruits

There are two types of fruits: true fruits and false fruits

True fruits: These are the types of fruits which form only from the ovary of the flower. Most fruits fall into the category of true fruits. Examples are orange, tomato, cowpea, pawpaw, sunflower, etc.

False fruits: These are those which form from the ovary and some other part of the flower. Examples are cowpea, pineapple, apple.

True fruits are further considered as simple, aggregate, and multiple fruits.

Simple fruits are formed from a flower with either a single carpel or many carpels fused together. Examples are orange, pawpaw, tomato, cowpea.

Aggregate fruits are formed from one flower which has many free carpels. Each carpel forms a simple fruit called a ***fruitlet***. Examples are raspberry and strawberry.

Multiple/compound fruits are formed from many flowers, or a whole inflorescence. An example is pineapple.

Further classification places simple fruits into fleshy fruits and dry fruits.

Fleshy/succulent fruits are fruits with soft, juicy pericaps. Examples are tomato, apple.

Dry fruits are fruits with dry, hard and woody pericaps. Examples are beans, flamboyant, pride of Barbados, cotton, okro (okra).

Succulent/fleshy fruits are classed as berry or drupe. The pericap of succulent fruits consists of three layers: an outer ***epicarp***, a middle ***mesocarp*** and an inner ***endocarp***. Part or all of the pericarp is fleshy and can be eaten.

Berry

A berry has a thin epicarp, fleshy mesocarp and a thin fleshy endocarp. It also has many carpels and contains many seeds attached to the placenta at the centre of the fruit. Example are guava, tomato, pepper.

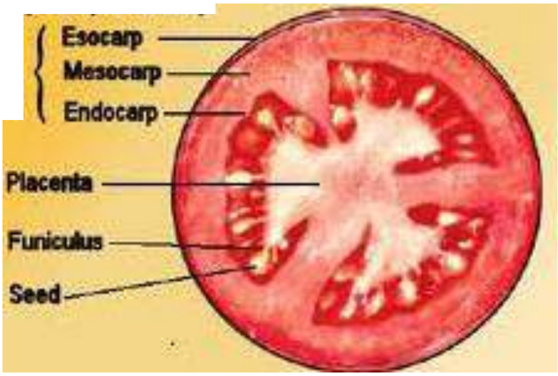


Fig. 6.9: *Berry of tomato*

Drupe

A drupe has a thin epicarp, a thick, fleshy or fibrous mesocarp, and a hard endocarp. Examples are mango, coconut.

Dry fruits are described as dehiscent and indehiscent fruits.

Dry dehiscent fruits are fruits whose pericaps split to release seeds. They are classified as:

- ***Legume (pod***), examples, cowpea, flamboyant, pride of Barbados.
- ***Capsule***, examples, okro, cotton, poppy.
- ***Follicle***, example, cola, cnestis.
- ***Schizocarp***, example, Cassia, Desmodium.

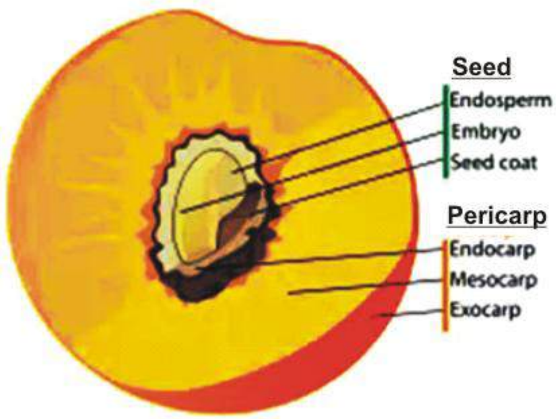


Fig. 7.0: *Longitudinal section of a drupe*

Dry indehiscent fruits are classified as:

- ❖ ***Achene***, an example is sunflower.
- ❖ ***Cypsela*** an example is tridax
- ❖ ***Caryopsis*** an example is maize grain.
- ❖ ***Samara*** examples are combretum, pteocarpus
- ❖ ***Nut*** an example is cashew fruit.

SEED

A seed is a fertilized ovary found in a fruit.

The structure of a seed

Seed is made up of a seed coat, an embryo, and an endosperm.

Seed coat

Two layers make up the seed coat – testa and tegmen. The testa is formed from the outer integument, while the tegmen is formed from the inner integument of the ovule.

Functions:

- ❖ Protects the inner part of the seed
- ❖ Allows water and oxygen to enter the seed

Embryo

The embryo consists of three parts – the plumule (embryonic shoot), radicle (embryonic root) and cotyledons (seed leave). A stalk holds each cotyledon to the embryo.

Functions:

- ❖ The plumule develops into the shoot system.
- ❖ The radicle develops into the shoot system of the plant.
- ❖ The cotyledon stores food for the use and growth of the embryo. It also encloses and protects other parts of the embryo.

Endosperm

The endosperm is a large tissue found only in some seeds.

Function:

- ❖ Stores food. (In most seeds the cotyledon takes the place and function of the endosperm).

Cotyledon

Also called seed leaf, is the part of the seed which stores food for growth of the embryo. A seed with one cotyledon, such

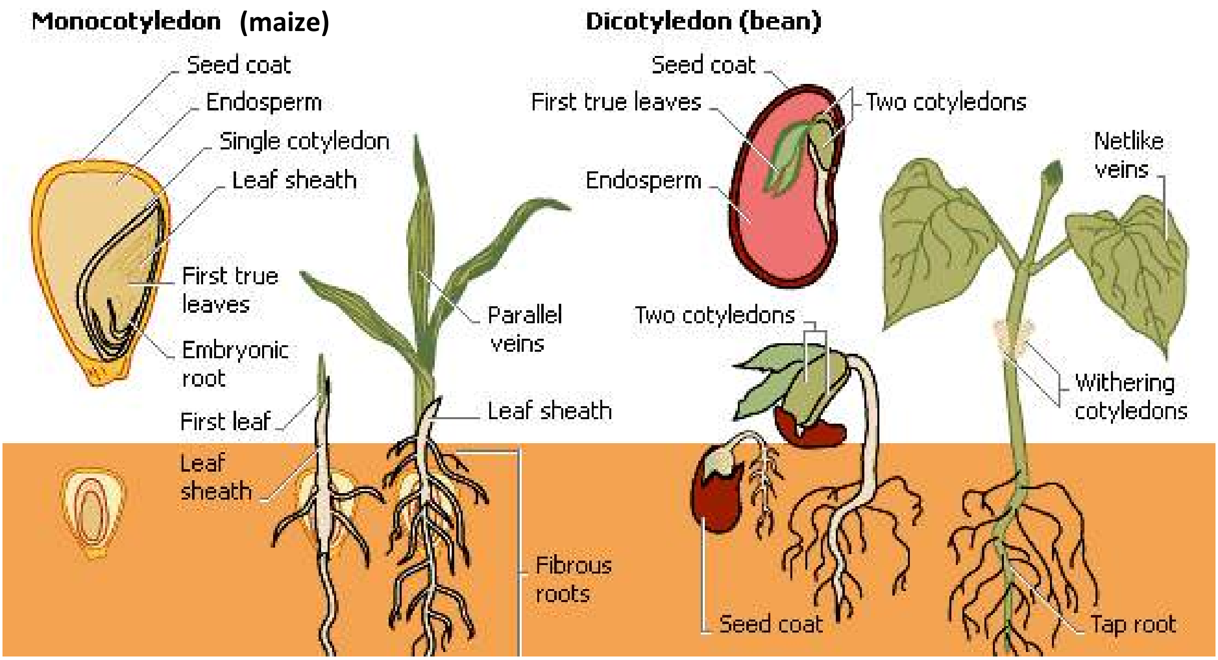


Fig. 7.1: Structure of seeds

as maize, is known as ***monocotyledonous***, while a seed with two cotyledons, such as beans, is known as ***dicotyledonous***.

Differences between Fruits and Seeds

Fruits	Seeds
They are fertilised ovary	They are fertilized ovule
They contain seeds	They contained in fruits
They have two scars	They have one scar

DISPERSAL OF FRUITS AND SEEDS

Dispersal is the scattering of fruits and seeds from the parent plants to new places by agents such as wind, water and animals.

Agents of Dispersal

Agents of dispersal are the factors which bring about dispersal of fruits and seeds. They are wind, water, and animals. Some fruits disperse the seeds themselves by a method known as *explosive mechanism* or *self-dispersal*.

Wind Dispersal

For a fruit or seed to be dispersed by wind, it must have the following features:

1. Dry and light in weight
2. Small in size e.g. orchids
3. Have floss (a mass of silk thread). e.g. cotton
4. Have wings (flattened and extended pericarp). e.g. tecoma, dutchman's pipe, combretum
5. Have pappus (parachute-like hair).e.g. tridax

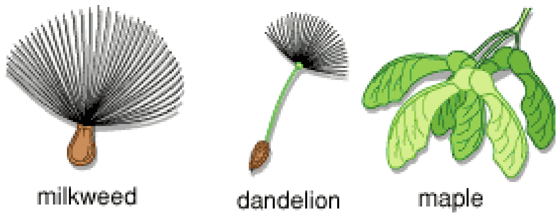


Fig. 7.2: Examples of wind dispersed fruits

Water Dispersal

Features of fruits and seeds dispersed by water are:

1. Thick and fibrous mesocarp with air spaces between the fibres. e.g. coconut.
2. Spongy seed coat containing several air spaces. e.g. whit mangrove.

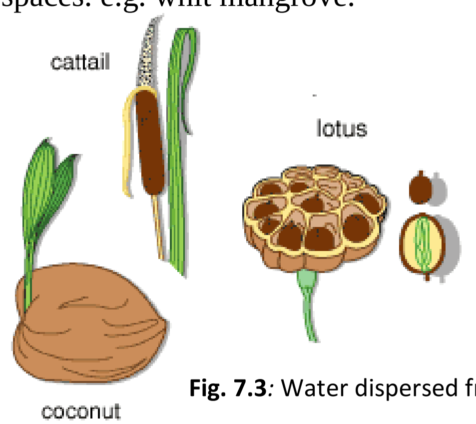


Fig. 7.3: Water dispersed fruit

Animal Dispersal

Features of fruits dispersed by animal are:

1. Edible (eatable).
2. Brightly coloured to attract animals. e.g. mango, guava, pawpaw and tomato.
3. Fruits and seed with hooks to attach themselves to the fur of mammals and clothes of humans. e.g. Desmodium, Pupalia seed and bidens.
4. Fruits with sticky hairs to attach themselves to animals and people. e.g. boerhaavia and plumbago.

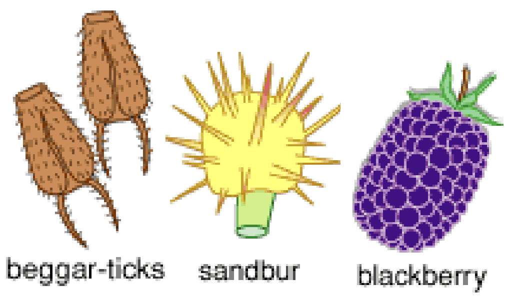


Fig. 7.4: Example of animal dispersed fruit

Self-Dispersal (Explosive Mechanism)

Feature of fruits and seeds which undergo self-dispersal are:

1. Uneven drying of the seed coat. Some parts of the fruits dry faster than other parts creating tension which causes the fruits to split open suddenly with an explosion. e.g. cowpea, flamboyant, pride of Barbados.
2. Turgidity of the seed coat. e.g. Balsam plants.

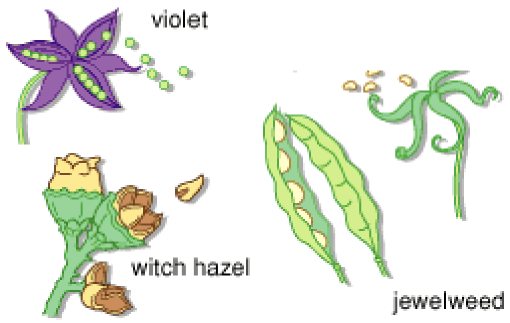


Fig. 7.5: *Examples of self-dispersed fruits*

Advantages of Fruits and Seeds Dispersal

1. Reduces overcrowding of plants.
2. Reduces competition for light.
3. Reduces competition for nutrients.
4. Helps plants to move to new locations.
5. Reduces epidemic diseases among plants.
6. The rate of destruction of a species by fire, flood, etc. is reduced.

Disadvantages of Fruits and Seeds Dispersal

1. Some of the fruits or seeds may land on an arid land (unproductive land) and die.
2. Some of them may be in a colony of herbivores and therefore be eaten.
3. Some could land in an overcrowded area and die from nutrients and sunlight starvation.

GROUP ACTIVITY:

- Make a collection of different seeds from the community.
- Study the features, draw and label the longitudinal section of two of the seeds e.g. maize and *Jatropha sp.*
- List the functions of the parts.
- Keep information on each seed including name of collector, date of collection, differences in size, shape, colour and uses.

GERMINATION OF SEEDS

Germination is the process whereby the embryo emerges from the seed coat as a result of growth of the seed into a seedling.

Seeds may germinate immediately or undergo a period of ***dormancy***, after their dispersal.

Dormancy is the stage during which growing stops temporarily and metabolism is reduced to the lowest rate.
Dormant seeds may survive adverse conditions like drought, flood, and irregular temperatures.

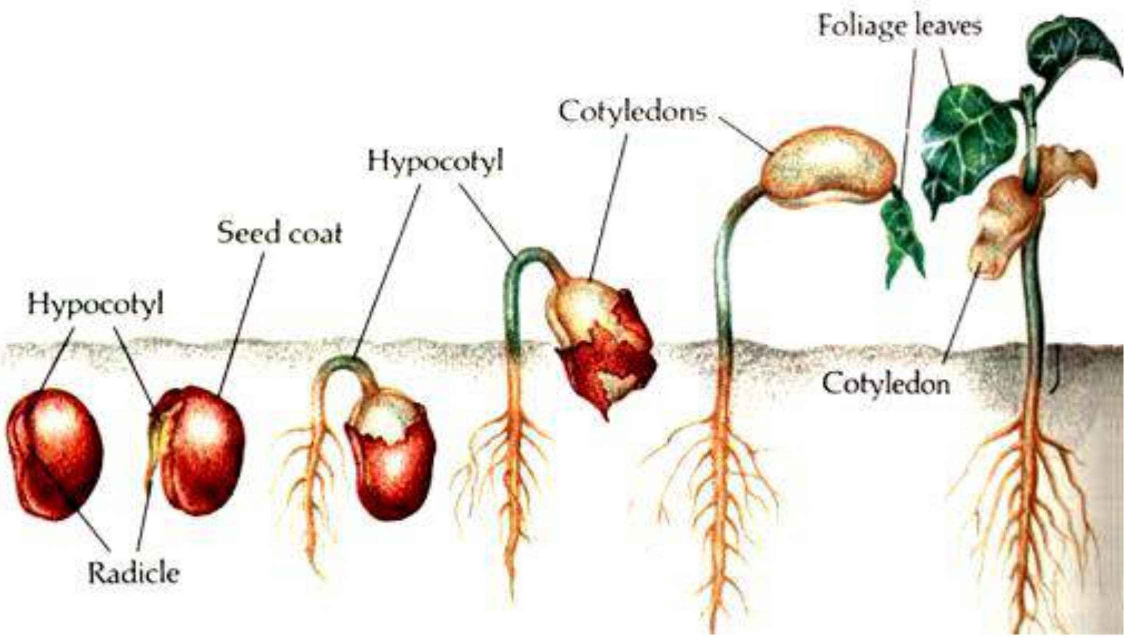


Fig. 7.6: The germination process

The Germination Process

- The testa which is permeable to water and air absorbs water.
- Digestive enzymes in the seed become active and break down the food substances in the seed to soluble forms.
- The soluble food substances are transported to the plumle and radicle.
- The radicle grows downwards into the soil to form the root system.
- The plumle grows upwards through the soil to become the shoot system.
- The young plant is called a seedling.

Conditions Necessary for Seed Germination

For a seed to germinate it must first be viable; that is it should be ideal and well suited for germination. The environment were the seed is plays an important role in its germination. Before a seed germinates, it must be exposed to some environmental conditions such as:

1. Moisture (water)
2. Oxygen (air)
3. Warmth (suitable temperature)

To sum it up, the condition necessary for seed germination include moisture, oxygen, warmth and the viability of the seed. Though the environmental conditions for seed germination are moisture, oxygen, and warmth, some seeds, such as tobacco seeds require exposure to light before they germinate.

Demonstrating the conditions necessary for germination

- Label four test tubes **A**, **B**, **C** and **D**.
- Put a piece of cotton wool and four viable bean seeds into each test tube.
- Pour a few drops of water into test tubes **A** and **D**.
- Leave test tube **A** at room temperature and place test tube **D** in ice cubes or refrigerator to get rid of warmth.
- Leave the contents of test tube **B** as it is making sure that they are dry.
- Heat water to eliminate oxygen, allow the water to cool and pour a few drops into test tube **C**. Pour oil onto the water to form a layer on the surface to prevent oxygen from entering.
- Leave the test tubes for seven days and observe what happens.

Observation

It would be observed that only the seeds in test tube A germinate. The others will not germinate.

Conclusion

The seeds in test tube A, which have all the conditions for germination, germinate while the other seeds which lack one condition appease do not germinate.

This shows that a viable seed needs all the conditions – water, air and warmth – in order to germinate.

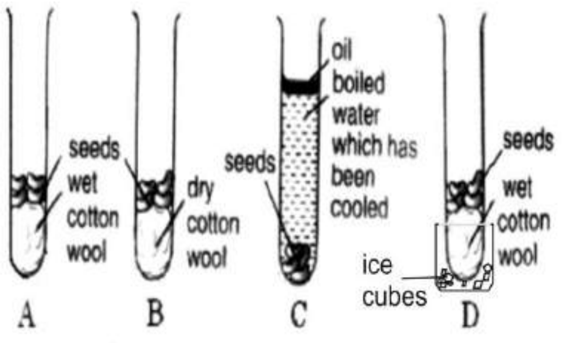


Fig. 7.7

Types of germination

There are two types of germination: epigeal and hypogeal germination.

Epigeal germination

This is the type of germination which occurs when the cotyledon appears above the ground.

This is caused by the elongation of the radicle. Examples of plants which undergo epigeal germination are mango, groundnut, cowpea, red beans, etc.

Hypogeal germination

This is the type of germination which occurs when the cotyledon remains below the ground.

This is caused by the elongation of the plumule.

Table 3.4: Differences between epigeal and hypogeal germination

Epigeal germination	Hypogeal germination
Cotyledons rise from the ground	Cotyledons remain in the soil
Normally has small cotyledons	Normally has larger cotyledons
Cotyledons store less food	Cotyledons store more food

VEGETATIVE (ASEXUAL) REPRODUCTION IN PLANTS

Vegetative propagation or reproduction is the use of parts or the whole of an organism to give rise to new organisms.

Vegetative reproduction occurs only in plants. Most of the plants which reproduce asexually do not have flowers which contain the sexual parts (androecium and gynaecium) of a plant. Therefore, they reproduce from vegetative parts such as leaves, stems or roots.

Methods of vegetative propagation

Plants which undergo vegetative propagation have modified stems underground, such as bulbs, rhizomes, suckers corms, runners and stem tubers.

Propagation by bulb

A bulb has a short vertical stem. It has a terminal bud, which develops into an aerial shoot. It also has axilary buds, which develop into daughter bulbs. The leaves are arranged in a concentric around the short stem. The fleshy inner scale leaves store food and they are covered and protected by the dry, brown outer leaves.

It has a lot of fibrous roots which rise from the base of the short stem.

Procedure

Propagation of bulbs is done by planting the entire bulb in the soil. The axilary buds develop into daughter buds underground and later sprouts into young bulbs. Examples of bulbs are onions, garlic, spider lily, etc.

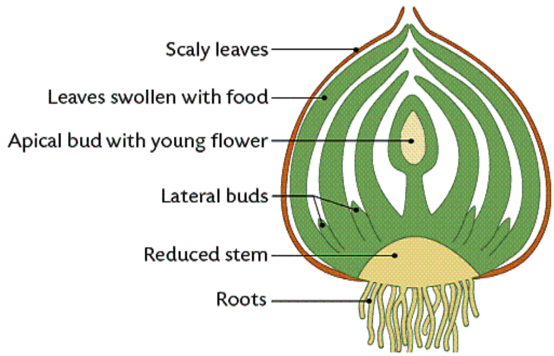


Fig. 7.8: An onion bulb

Propagation by rhizome

A rhizome is a thick stem which grows horizontally in the soil.

It has a terminal and a lateral (axillary) bud that develops into aerial shoot and stem branch respectively. Food is stored in the stem. Rhizomes have adventitious roots which develop from the nodes.

Procedure

Propagation of rhizome is done by cutting the stem into pieces, with each piece having two to three buds and planting them in the soil. The buds sprout and grow into an aerial shoot.

Examples of rhizomes are ginger, canna lily.

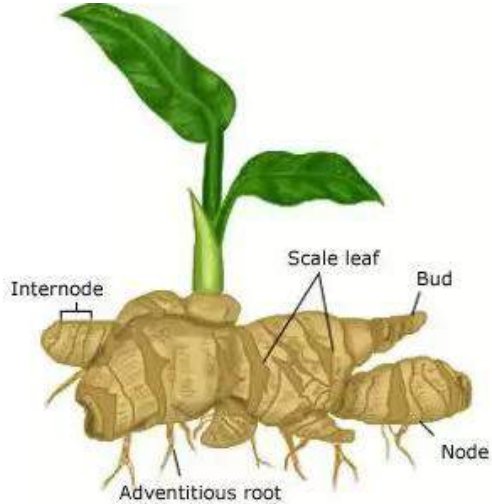


Fig. 7.9: *Propagation by rhizome*

Propagation by suckers

Suckers are lateral underground stems which bear a terminal bud, scale leaves and adventitious leaves. The tips of the stems which give rise to the aerial shoots emerge above the soil level. A sucker depends on the parent plant for nourishment, since it cannot store food.

Procedure

A sucker is propagated by cutting the young sucker close to the parent plant and replanting it in the soil.

Examples of suckers are plantain and banana.

Fig. 8.0: *Propagation by sucker*

Propagation by corms

A corm is a short bulging stem, which grows vertically underground. It has a terminal bud which gives rise to an aerial shoot and an axillary bud which develops into the axils of the scale leaves. Internodes with adventitious roots rising from the nodes are also found on the stem. The stem stores food.

Procedure

Propagation of corms is done by cutting the whole corm into pieces such that each piece has a number of buds. Each piece is planted in the soil. The axillary buds develop into daughter corms.

Examples of corm are cocoyam, Caladium sp.

Fig. 8.1: Propagation by corm

Runners

A runner is a thin creeping stem which grows horizontally along the surface of the ground. The stem has long internodes and produces adventitious roots at its nodes. These roots go into the soil and fix the weak stem firmly to the surface of the soil.

The nodes opposite the adventitious roots develop into aerial shoots.

Examples of runners are sweet potato, strawberry etc.

Stem tuber: A stem tuber is a bulging underground stem which has adventitious buds and stores food. Stem tubers do not have adventitious roots.

Stem tubers can be propagated by cutting the stem into pieces, with each stem having a bud which will germinate to the shoot system. Examples of stem tubers are yam, cassava, potato, etc.

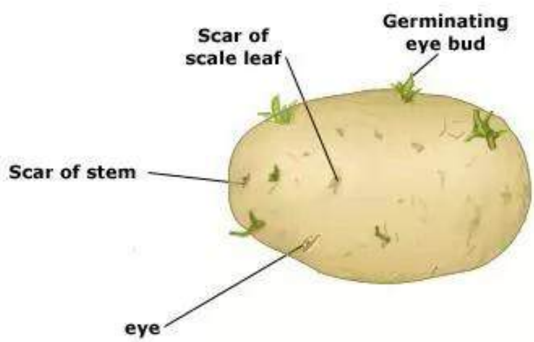


Fig. 8.3: Propagation by corm

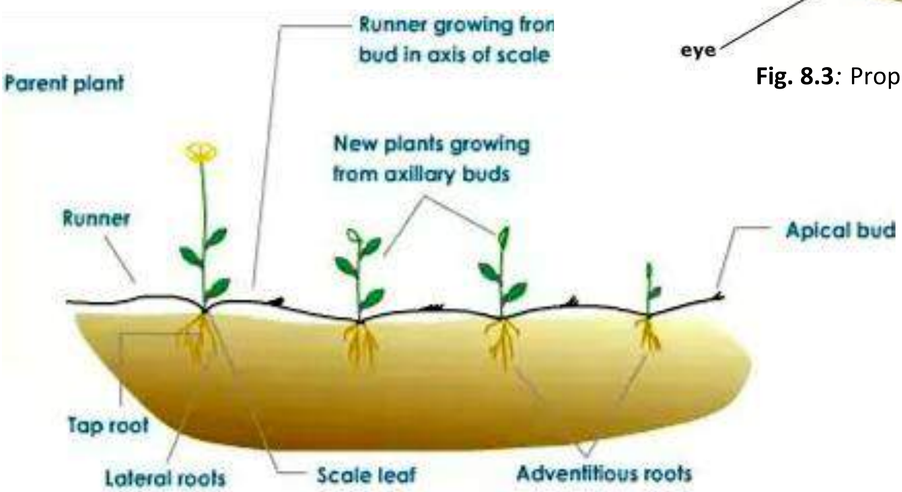


Fig. 8.4: Propagation by runners

ARTIFICIAL PROPAGATION

The methods mentioned above are also known as the ***natural methods of propagation***. Apart from these methods of propagation, there are other methods which are used to propagate plants. These are stem cutting, budding, grafting, and layering.

Stem cutting

This can be used for plants that produce flowers but do not form seeds. The stems are cut from an angle just below a node. Each cutting must have three to five nodes. The lower end of the stem is planted in the soil with at least one node below the surface of the soil. Adventitious roots develop from the node in the soil. Examples of plants propagated by stem cutting are sugar cane, cassava, hibiscus and runners (e.g. sweet potato).

Budding

In budding, a dormant bud on a sliver of stem called the ***scion*** is carefully removed from one plant using a sharp pen knife. It is inserted into a T-shaped cut in the back of another plant called the ***stock***. The stock must have a well established rooting system. The cambium of the scion must be exposed by carefully scraping off the woody part of the stem attached to it. The scion is carefully placed in the T-shaped cut of the stock so that the cambia of the stock and scion make contact. The scion and stock are firmly bound together, and the area is covered with a waterproof material such as polythene sheets. The bud on the scion develops and bears the fruits of the plant from which it is removed. Examples of plants which can be propagated by budding are citrus trees (e.g. orange) and roses.

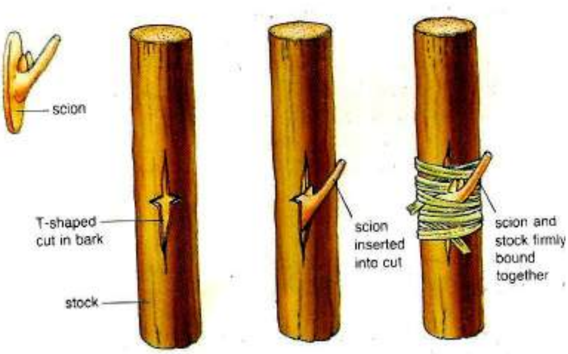


Fig. 8.5: Propagation by budding

Grafting

Vegetative propagation by grafting involves joining parts on one plant into another plant of different variety of the same species. The scion is a short length of stem with at least one bud. It is joined to the cut-end of the stem of another plant, the stock. The cambia of the stock and scion must make contact. To achieve this, the scion may be wedge-shaped to fit into a V-shaped stock, or slanted to fit a slanted stock. The scion and stock are then bound together. The cut heals soon and the stock and scion continue to grow as one plant and bear the fruit of the plant from which the scion was removed.

Examples of plants propagated by grafting are citrus (e.g. orange, grape fruit, tangerine), mango, cocoa.

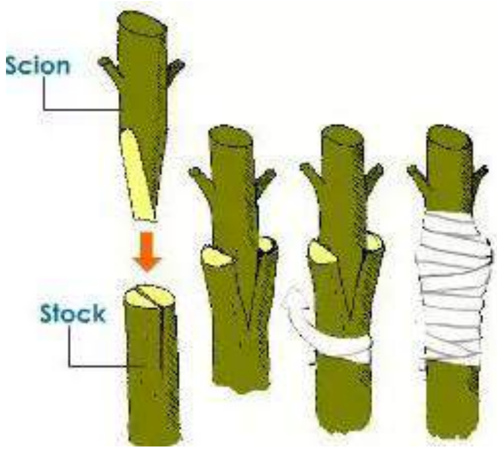


Fig. 8.6: *Propagation by grafting*

Layering

This method of propagation is used to propagate plants that do not produce seeds and will not always grow from stem cutting either. A branch of the stem close to the ground is carefully bent over so that a part of it having a number of nodes touches the ground. The end of the branch is tied to a firm support. A slit is made with a penknife at a node that touched the ground. This part of the branch is pushed into the soil and held in place with two pegs. It is covered with a heap of fertile soil and a heavy object is placed on top.

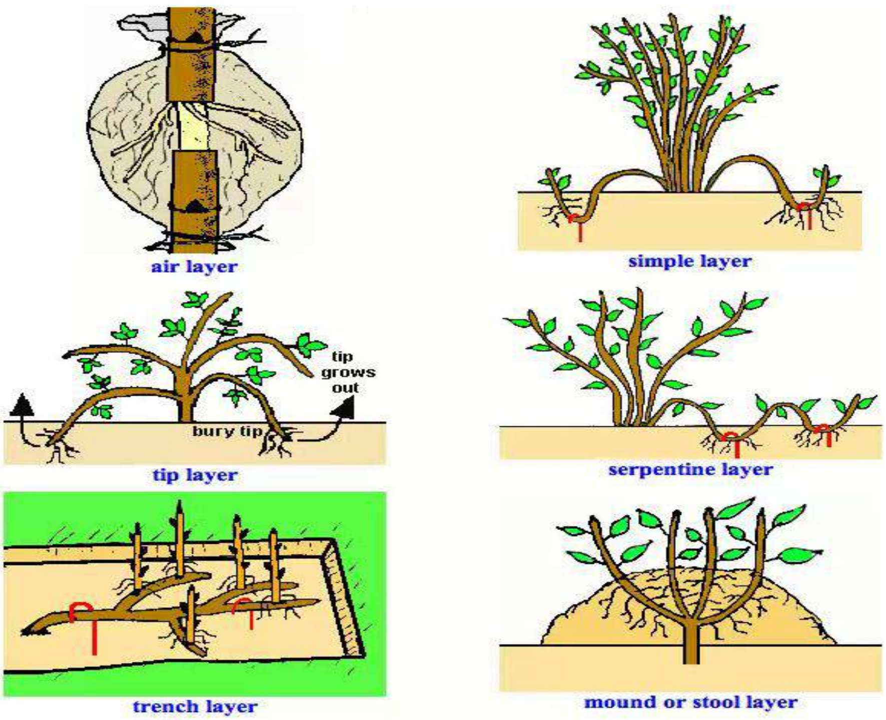


Fig. 8.7: types of layering

The soil is irrigated regularly and dug out when adventitious roots develop. The rooted twig is cut off and planted in a fertile soil.

Other methods of layering include:

- ❖ Air layer
- ❖ Tip layer
- ❖ Serpentine layer
- ❖ Trench layer
- ❖ Mound layer

Advantages of Asexual Propagation

1. Only one parent is needed to produce offspring.
2. Faster growth of plants.
3. Good qualities of parents are maintained by offspring.
4. Used to propagate plants which cannot produce seeds.
5. Parent material provides nutrients for the young plants.
6. The young plants can withstand adverse conditions.
7. Early maturity of plant is attained.

Disadvantages of Vegetative Propagation

1. Offspring and parent plant may compete for nutrients and sunlight due to overcrowding
2. There is lack of variety
3. Many plants may be destroyed by disaster in an area.
4. Offspring are produced close to parent plant, therefore colonization of new localities is unlikely
5. Any disease of parent plant is passed on to the offspring.

Table 3.5: Differences between Sexual and Asexual propagation

Sexual propagation	Asexual propagation
Requires two parents	Requires only one parent
New varieties are produce	No new variety produced
Involves the fusion of male and female gametes to form a zygote	Does not involve fusion of gametes
Growth is slow	Growth is fast
Parent and offspring may not compete for nutrients	Parent and offspring always compete for nutrients

TEST QUESTIONS

1. (a) Draw and label a fully developed flower.
   (b) State the functions of the parts labelled

2. (a) What is pollination?
   (b) State four differences between insect pollinated flower and wind pollinated flower.
3. (a) Define reproduction.

(b) Distinguish between sexual reproduction and asexual reproduction

4. (a) Explain the term germination of plants.
   (b) Differentiate between epigeal and hypogeal germination.

5. Describe three methods each of asexual and vegetative propagations.

6. (a) What are fruit and seed dispersal?
   (b) Explain three methods of fruit and seed dispersal and give two examples of the agents involved.

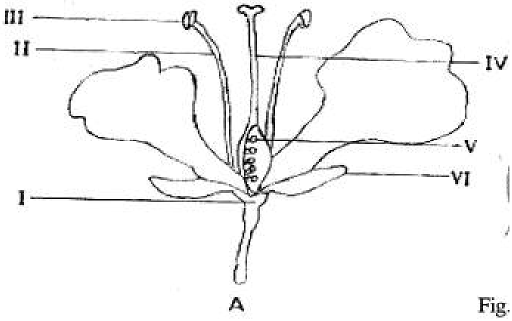
7. (a) Describe the germination process.
   (b) State the conditions necessary for germination.

8. Write a short note on the following methods of vegetative propagation:
   (a) budding
   (b) layering
   (c) grafting

9. (a) Explain the term vegetative propagation.
   (b) Distinguish between sexual reproduction and asexual reproduction.

10. Fig. **3** is an illustration of structures associated with plants.

Study the figure carefully and answer the questions that follow.



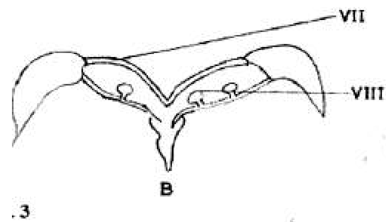


Fig. 3

(i) Identify the structures **A** and **B**.
(ii) (α) Name each of the parts labelled I, II, III, IV, V, VI, VII and VIII.
   (β) State one function each of the parts labelled II and IV.
(iii) State the relationship between structure **A** and structure **B**.
(iv) State the mode of dispersal of the part labelled VIII.
(v) Name one plant which produces structures similar to **B**.

11. (a) Mention the method of propagation for the following plants.
   (i) ginger
   (ii) garlic

(iii) sweet potato
(iv) potato
(v) cassava
(vi) plantain
(vii) orange
(vii) hibiscus flower
(ix) carrot
(x) taro

(b) Describe two natural methods of vegetative propagation.

12. (a) What is a fruit?
(b) State two differences between fruits and seeds.
(c)` Describe the following types of fruits:
(i) true fruits
(ii) false fruits

13. Figure three illustrates three different methods of crop propagation.
Study the figure carefully and answer the questions that follow.

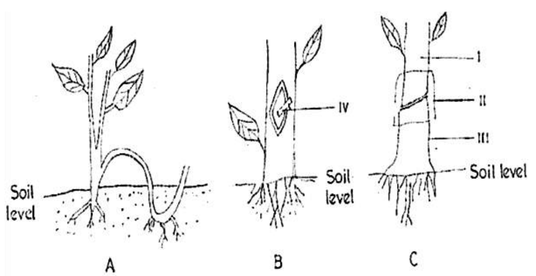


Fig. 3

(a) Name the parts labelled I, II, III and IV.
(b) Identify each of the methods of propagation in **A**, **B** and **C**.
(c) Given a citrus seedling, a mature citrus plant, a knife and a wrapping tape, describe how the propagation method would be performed as illustrated in set-up **B** above.
(d) State three factors that influence the success of the method of propagation illustrated in **C** above.
(e) Name one ornamental plant propagated by the method illustrated in A.
(f) State four advantages of the method of propagation illustrated above.

14. Describe the following methods of fruit and seed dispersal and give two examples each:
(a) explosive mechanism
(b) Water dispersal
(c) Animals dispersal
(d) Wind dispersal

11

FOOD AND NUTRITION

Specific Objectives

After completing this chapter, you will be able to:

- Outline the different classes of food and describe a balanced diet.
- State the effects of malnutrition.
- Explain the need to fortify and enrich food.
- Outline the health benefits of water.

INTRODUCTION

All living organisms need energy to carry out basic life processes. The number one source of energy is food. Some living organisms (e.g. green plants) can prepare their own food with sunlight in a process known as ***photosynthesis***. These organisms are described as ***autotrophic*** (self feeders) because they make their own food. Some other living organisms do not produce their own food. They are described as ***heterotrophic*** (other feeders). Therefore, they feed on the food prepared by green plants for energy.

Food is any substance that living organisms consume to gain energy.

Nutrition is the process whereby living organisms get food.

CLASSES OF FOOD

The different classes of food are:

- Carbohydrates
- Proteins
- Fats and oils (lipid)
- Minerals
- Vitamins

Carbohydrates

Carbohydrates are compound of hydrogen, carbon, and oxygen.

The ratio of hydrogen to oxygen is 2:1. Carbohydrates are the major source of energy for living organisms.

There are three types of carbohydrates: monosaccharides, disaccharides and polysaccharide.

Monosaccharides

Monosaccharides are known as simple sugars. This is because they have a reduction action on Benedict's and Fehling's solution. They are therefore called reducing sugars, and their general formula is $C_6H_{12}O_6$. Monsaccharides are sweet and soluble in water.

Examples of monosaccharides are glucose in fruits, galactose in milk, fructose, flower nectar, fruits and honey.

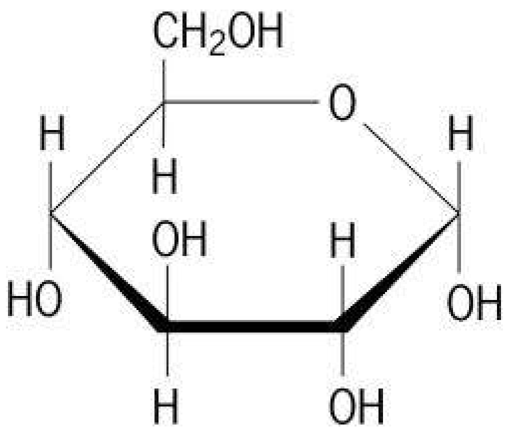


Fig. 8.8*:* Molecular structure of monosaccharide

Disaccharides

Disaccharides are referred to as complex sugar. A disaccharide is made up of two monosaccharides joined together in a chemical process known as ***condensation***. The general formula for disaccharides is $C_{12}H_{22}O_{11}$. Disaccharides can be broken down into monosaccharides in a chemical process known as ***hydrolysis.*** Disaccharides are sweet and soluble in water.

Examples of disaccharides are sucrose in sugar cane, lactose (milk sugar), maltose from germinating grains (e.g. maize). All disaccharides except sucrose (non-reducing sugar) are reducing sugar.

Polysaccharides

Polysaccharides are made up of several monosaccharides chemically combined. The general formula for polysaccharides is $(C_6H_{10}O_5)n$ where ***n*** is a large number, about 300 for starch and 300 to 3000 for cellulose. Polysaccharides are not sweet and usually insoluble in water. Examples are starch, cellulose, glycogen, and chitin.

Test for starch

- Put a sample of the food into a test tube.
- Add some drops of iodine solution.

It would be observed that the colour of the food sample changes to blue-black, showing the presence of starch in the food.

Test for reducing sugar

- Put a food sample, e.g. glucose into a test tube.
- Add a few drops of Fehling's solution (A and B mixed) or Benedict's solution
- Heat the test tube gently for a few minutes.

It would be observed that the colour of the food sample changes to orange or brick-red indicating the presence of reducing sugar.

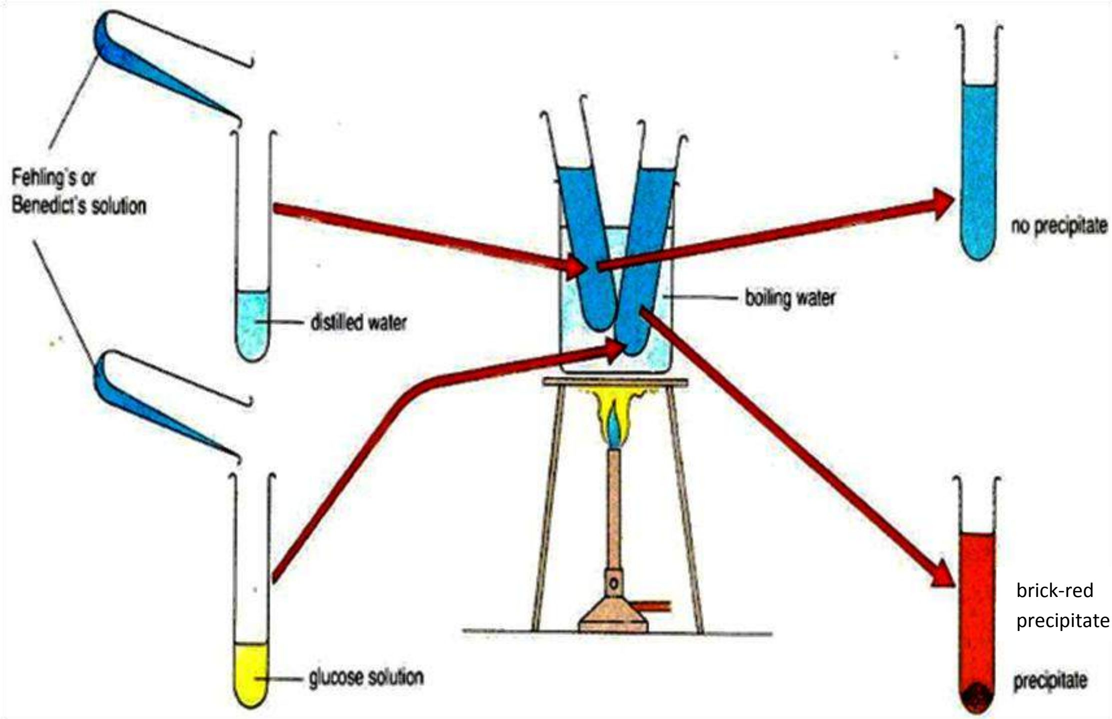


Fig. 8.9: Test for reducing sugar

Test for non-reducing sugar

- Put a piece of the food sample into a test tube
- Add a few drops of dilute hydrochloric acid (HCl)
- Heat gently for a few minutes in a water bath.
- Cool the mixture and solid sodium hydrogen carbonate ($NaHCO_3$) slowly until the fizzing stops.
- Add a few drops of Benedict's solution and heat for a five minutes

It would be observed that the colour of the food sample changes to orange or brick-red showing the presence of non-reducing sugar.

Importance of Carbohydrates

1. It serves as a major source of energy.
2. It serves as the source for other organic molecules such as amino acids.
3. It maintains the structure of the cell wall.
4. It serves as food storage compound.
5. Cellulose is important for the cell walls.

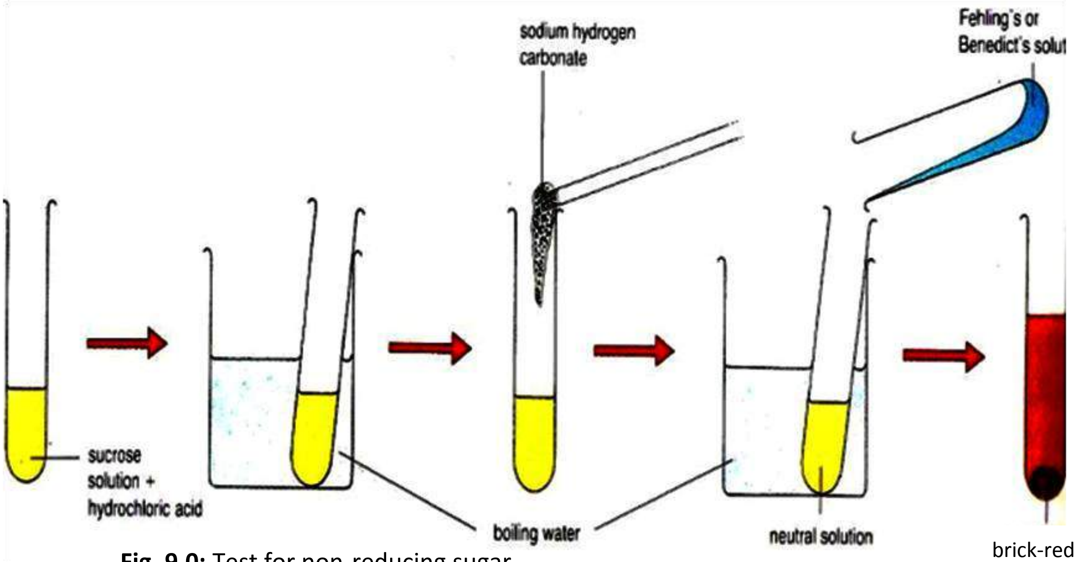


Fig. 9.0: Test for non-reducing sugar

Proteins

***Proteins are compounds which contain the elements carbon, hydrogen, oxygen and nitrogen*.**

Some proteins contain sulphur and phosphorus as well. The smallest unit of protein is called an ***amino acid***. One protein molecule contains thousands of amino acids, chemically combined together by peptide bond and peptide linkage. Proteins are destroyed by temperatures higher than 60^{o}C. A destroyed protein loses it chemical structure and physical texture, e.g. boiled egg.

Classes of protein

Proteins are classed into two – first class proteins and second class proteins.

First class proteins: These type of proteins contain all the essential amino acids needed by the body.

Second class proteins: These are plant proteins and lack some of the amino acids needed by the body.

Sources of proteins are fish, meat, egg, beans etc.

Test for proteins

- Put a piece of the food e.g. mashed meat, into a test tube
- Add a few drops of dilute sodium hydroxide (NaOH) solution
- Add a few drops of copper sulphate ($CuSO_4$) drop by drop; shake the test tube after each drop.

It would be observed that the colour of the food sample changes to violet indication the presence of proteins.

Alternative method

- ❖ Add a few drops of Millon's reagent to the food sample in a test tube
- ❖ Heat for one minute
- ❖ The colour changes to violet indicating the presence of proteins.

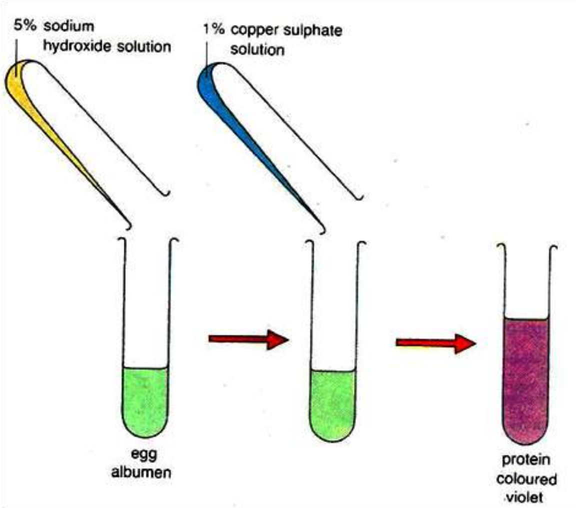


Fig. 9.1: Test for protein

Importance of Proteins

1. Repair damaged tissues in the body.
2. Help animals to grow.
3. Provide energy to the body.
4. Control the rate of metabolism.
5. Responsible for the formation of hormones and enzymes.

Lipids (fats and oils)

Lipids are compounds containing carbon, hydrogen and oxygen.

The amount of oxygen atom in the structure of lipids is far lower than in carbohydrates. Lipids, which are fat and oil, are formed from the chemical combination of glycerol and fatty acids. Fat is stored by animals while oil is stored by plants.

Fats are solid at room temperature, while oils are liquid. Fats and oils are soluble in organic solvents, e.g. ethanol, but insoluble in inorganic solvents e.g. water.

Test for lipids

a) Grease spot test

- ❖ Dip a finger in oil and place it on a sheet of paper.
- ❖ Dip another finger in water and place it on the same paper.
- ❖ Leave the paper for 10 minutes. Afterward, hold it up against light.
- ❖ It would be observed that the water has dried out whiles the oil stain remains visible on the paper.

a) Emulsion test

- ❖ Put 2 cm^3 of ethanol in a test tube
- ❖ Add 2 cm^3 of oil and shake vigorously
- ❖ Pour the mixture into another test tube containing 2 cm^3 of water.

❖ A cloudy, white emulsion develops showing the presence of lipids.

Importance of Lipids

1. Speed nerve transmission
2. Take part in the formation of membranes.
3. Some hormones are produced from lipids
4. Dissolve fat-soluble vitamins.
5. Provide energy to the body in the absence of carbohydrates.

Vitamins

Vitamins are organic food substances needed by the human body in very small quantities for growth and development.

Vitamins keep the body healthy and prevent some deficiency diseases. Examples of vitamins are vitamins A, B, C, D, E and K. These names are assigned to them in order of their discovery. Some vitamins have submultiples because of their chemical structures, e.g. vitamin B_1, B_2 and B_{12}.

Table 3.6: Vitamins, their source, functions and symptoms of deficiency

Vitamin	Source from food	Functions	Symptom of deficiency
A (retinol)	Egg yolk, liver green vegetable, carrot, cocoa, red palm oil	• Builds good eye sight • Protects the surface of the eye • Maintains membrane • Necessary for cell respiration	• Night blindness • Unhealthy skin • Reduced night vision
B_1 (thiamine)	Beans, egg yolk, cereal, bread, lean meat, unpolished cereal, palm wine	• Necessary for cell respiration	• Retarded growth • Beriberi • Nerve inflammation • Weak paralysis
B_2 (riboflavin)	Yeast, fish, green vegetables, meat cereals,	• Produces energy from food • Healthy skin	• Skin disorders, dermatitis disease • Sores on mouth and eye
B_{12} (cynobalamin)	liver	• Helps in the formation of red blood cells	• Anaemia
C (ascorbic acid)	Raw vegetable, fresh fruits, citrus fruits	• Forms strong skins • Helps wounds to heal • Good for some body tissues	• Gum bleeding, • scurvy disease, • immune disorders
D (calciferol)	Fat, fish, liver, egg yolk, sunlight	• Forms strong bones and teeth	• Rickets (deformed bones)

Table 3.6: Vitamins, their source, functions and symptoms of deficiency (Continued)

Vitamin	Source from food	Functions	Symptom of deficiency
E (tocopherol)	Green vegetables, liver	• Prevents haemolysis of red blood cells in rats	• Anaemia
K (phylloquinone)	Green vegetables, liver, egg yolk, unpolished cereal	• Synthesizes proteins which help the blood to clot	• Prolonged bleeding

Minerals

Minerals are minute amounts of metallic elements that are vital for the healthy growth of teeth and bones.

They also help in such cellular activity as enzyme action, muscle contraction, nerve reaction, and blood clotting. Minerals are classified as major elements (calcium, chlorine, magnesium, phosphorus, potassium, sodium, and sulphur) and minor or trace elements (chromium, copper, fluoride, iodine, iron, selenium, and zinc).

Table 3.7: Minerals, their sources, functions and symptoms of deficiency

Mineral	Source in food	Function	Symptoms of deficiency
Calcium (Ca)	Milk, cheese, green vegetable	• Formation of bones and teeth Enhances blood clotting • Needed for muscle contraction	Rickets (poor bones and teeth formation)
Chlorine (Cl) **Sodium (Na)**	Table salt	• Maintenance of tissue fluid, blood and lymph	Muscle cramps
Iodine (I)	Iodised table salt, sea food, cheese	• Formation of thyroxin • Prevents goitre	Goitre Reduced body growth
Iron (Fe)	Green vegetables, liver, yeast, eggs, kidney	• Forms haemoglobin in red blood cells	Anaemia
Copper (Cu)	Fruits, liver, nuts green vegetables, grains	• Essential for the utilization of irons in the body	Anaemia
Phosphorus (P) **Nitrogen (N)** **Sulphur (S)**	Fish, egg, lean meat, milk	• For protein synthesis • Formation of ATP	Rickets
Fluorine (F)	Water, toothpastes	• Promotes healthy teeth	Tooth decay, toothache

BALANCED DIET

Diet is the variety of food and drink which a person takes into the body.

A balanced diet is the diet which contains all the food nutrients needed by the body in their right proportions or amount.

The nutrients are carbohydrates, proteins, lipids, vitamins, minerals, roughage and water. A diet that has too little or too much of one or more food substance is described as an ***unbalanced diet***.

Percentage of Foods in a Balanced Diet

Carbohydrates	55%
Proteins	20%
Fats/oils	15%
Water, vitamins and Minerals	10%

Importance of Balanced Diet

1. A balanced diet helps in maintaining proper growth of the body.
2. It makes the body healthy.
3. It prevents deficiency diseases.

MALNUTRITION

Malnutrition occurs when an organism has inadequate amounts of certain nutrients required by the body.

Inadequate amounts of essential nutrients leads to diseases known as ***deficiency diseases***.

Fig: 9.2: Kids suffering from kwashiorkor

Effects of Malnutrition

1. It results in slow growth.
2. Calcium deficiency causes rickets.
3. Protein deficiency causes kwashiorkor.
4. The deficiency of vitamin A causes night blindness.
5. Malnutrition may eventual result in death.

Fig: 9.3: X ray film showing legs with rickets

Over-Nutrition

While having too little of essential nutrients is bad, having too much of some nutrients is equally bad. It results in a condition known as ***over-nutrition***. Over-nutrition may also lead to some serious diseases such as:

1. High blood pressure (hypertension), which is caused by excess fat in the body.
2. Heart diseases, which normally results from hypertension
3. Stroke, caused by the bursting of arteries as a result of heart diseases.
4. Diabetes, caused by eating too much refined sugar
5. Obesity (overweight), caused by overeating.

Apart from eating a well balanced diet, regular exercise will also keep the body fit and free from diseases.

Fig: 9.4: An obese woman

FOOD FORTIFICATION AND ENRICHMENT

Food fortification or enrichment is the process of adding micronutrients (essential trace elements and vitamins) to food.

It can be purely a commercial choice to provide extra nutrients in a food, or sometimes it is a public health policy which aims to reduce numbers of people with dietary deficiencies in a population.

Diets that lack variety can be deficient in certain nutrients. Sometimes the staple foods of a region can lack particular nutrients, due to the soil of a region, or because of the inherent inadequacy of the normal diet. Addition of micronutrients to staples and condiments (seasonings) can prevent large-scale deficiency diseases in these cases.

While it is true that both fortification and enrichment refer to the addition of nutrients to food, the true definitions do slightly vary.

Difference between food fortification and food enhancement

Food fortification refers to the practice of deliberately increasing the content of an essential micronutrient, i.e. vitamins and minerals in a food irrespective of whether the nutrients were originally in the food before processing or not, so as to improve the nutritional quality of the food supply and to provide a public health benefit with minimal risk to health.

Food enrichment refers to the addition of micronutrients to a food which are lost during processing.

Examples of fortified food

The most common fortified foods include:

1. Cereals and cereal based products
2. Milk and milk products
3. Fats and oils
4. Accessory food items
5. Tea and other beverages
6. Infant formulas

Body mass index

The body mass index (BMI) is a measure of body fat based on height and weight.

BMI does not actually measure the percentage of body fat.

Body mass index is expressed as the individual's body mass divided by the square of his or her height. The formula universally used in medicine produces a unit of measure of kg/m^2.

$$BMI = \frac{Body\ weight\ (kg)}{Height^2\ (m^2)}$$

BMI can also be determined using a BMI chart, which displays BMI as a function of weight (horizontal axis) and height (vertical axis) using contour lines for different values of BMI or colours for different BMI categories.

- BMI of less than 18.5 = underweight
- BMI of 18.5 – 24.9 = normal weight
- BMI of 25 – 29.9 = overweight
- BMI of more than 30 = obesity.

Roughage (dietary fibre)

Roughage is fibrous material that is indigestible and consists mostly of cellulose.

It is actually not a food substance, nonetheless, very essential in diets. Roughage can be obtained from solid fibre parts of fruits such as oranges, pineapple, apples, vegetables and cereals.

Importance of Roughage

1. Facilitates free bowel movement and prevents constipation, thereby reducing the risk of bowel cancer.
2. Helps in easy movement of food through the gut.
3. The cellulose in roughage is digested by some gut bacteria in humans to release important vitamins.

Health benefits of water

Water plays a major role in all organisms. Some of the roles water plays in humans are:

1. Regulates the temperature of the body
2. Dissolves soluble substances in the body

3. Transports various substances in the body
4. Major component of digestive juice
5. Medium for the elimination of some excretory products from the body

TEST QUESTIONS

1. Write a short note on any three classes of food.

2. Describe how you will test for the following food classes:
   a. Starch
   b. Protein
   c. Lipids

3. (a) What are vitamins?
   (b) Mention three known vitamins and state the symptoms of the deficiency.

4. The following table shows some minerals. State their functions and the effects of their deficiencies.

Mineral	*functions*	*Effect of deficiency*
Calcium		
Fluorine		
Iodine		
Iron		

5. (a) What is a balanced diet?
   (b) What are the importance of a balanced diet?
   (c) State three effects each of malnutrition and over-nutrition.

6. Food substances labelled J, K and L, were tested with various chemicals school laboratory. The results obtained are tabulated below.

Food substance	***Test***	***Observation/ Result***
J	2 cm 3 each of fehling's solution A and B added to J. Mixture is boiled for 2 mnutes in water bath.	Brick-red precipitate formed
K	Iodine solution added drop by drop to K	Blue-black colur obtained
L	a) two drops of dilute hydrochloric acid is added to L. Mixture is boiled for 2 minutes.	Yellow precipitate obtained
	b) 2 cm3 of Benedicts solution added to mixture in (a) above. Mixure is boiled for 2 minutes.	No change in colour

a) What inference would you make about the food substances from the results above?

b) i) What changes would take place

when the food substance K is chewed for about 3 minutes?

ii) What observation would be made when Benedict's solution is added to the chewed substances K in a test tube.

iii) What conclusion can be drawn from the nature of substance Benedict's solution reacted with?

7. (a) What are *vitamins*?

(b) Name one source of each of the following vitamins:

(α) vitamin A;

(β) vitamin B;

(γ) vitamin K.

12

DENTITION, FEEDING AND DIGESTION IN MAMMALS

Specific Objectives

After completing this chapter, you will be able to:

- Identify the different types of teeth in mammals and relate them to their function.
- Enumerate various ways of preventing dental problems.
- Draw and label the digestive system of mammals.
- Describe the process of digestion, absorption and assimilation in mammals.
- Mention some diseases and disorders associated with the digestive system of humans.

INTRODUCTION - DENTITION IN MAMMALS

Dentition is the number, type and arrangement of teeth in an organism.

The diet of an animal, generally determines its dentition. For example, the dentition of a carnivore is different from that of a herbivore or omnivore.

The teeth play a major role in the digestion of food. They help to divide food into smaller pieces.

TYPES OF TEETH AND THEIR RELATION TO FOOD

The characteristics of teeth make up dentition. In mammals, for example, the teeth vary in size, shape and function – this is known as ***heterodont dentition***. Vertebrates such as fish, amphibians and reptiles have teeth of the same size, shape and function – this is described as ***homodont dentition.***

Dentition of carnivores

Carnivores are fresh eating animals, e.g. lion, dog, cat. They have highly specialized teeth.

The characteristics of dentition in carnivore are:

1. Small, chisel-shaped incisors for seizing prey.

2. Long, conical, pointed and slightly inward-curved canines, used to hold and kill the prey and tear the fresh.
3. Small premolars and molars, except for the first lower molar and third upper molar which are very large. They are called ***carnassial teeth.***
4. Smooth sides and sharp cups and edges that run along the jaw line, used for scraping fresh off bones.
5. Molars have blunt cusps for crushing bones.

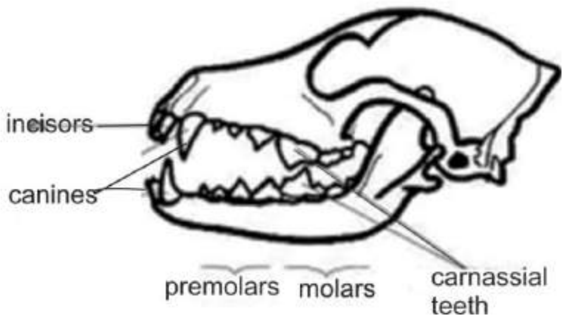


Fig. 9.5*:* Skull of a carnivore

Dentition of herbivore

Herbivores are animals that eat only plants, e.g. goat, cattle, rabbit.

Characteristics of carnivore dentition

1. Long, chisel shaped incisors for cutting and gnawing vegetation.
2. File-like premolars and molars with flat-ridged surfaces for grinding plant materials.
3. Absence of canine but instead, a large gap called a ***diastema*** between the incisors and premolars.
4. No canines in the upper jaw with poorly developed canines in the lower jaw.

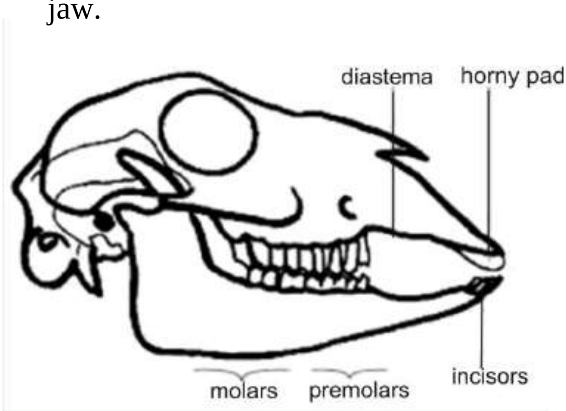


Fig. 9.6: Skull of a herbivore

Dentition of omnivores

Omnivores are animals that feed on both animals and plants. Their teeth are not highly specialised.

THE DENTAL FORMULA

Dental formula is the summary of the number, type and arrangement of teeth in the half of the lower and upper jaw of an animal.

The 32 teeth in an adult humans, for example, can be summarized as ***i*** 2/2, ***c*** 1/1, ***pm*** 2/2, ***m*** 3/3, where the initials are incisors (i), canines (c), premolars (pm), and molars (m) respectively. The figures on the lines represent the teeth in the upper jaw and those below the line are the teeth in the lower jaw. The total number of teeth in an adult human can therefore be obtained by multiplying the sum of the figures indicated in the dental formula by 2, i.e. 2 (4+2+4+6) = 32.

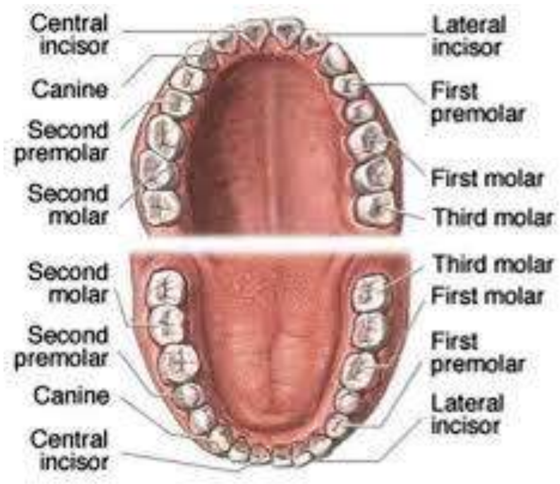


Fig. 9.7: Dental formula of man

STRUCTURE OF THE TOOTH

Teeth develop from a group of cells in the skin overlying the jawbones. Humans and other animals have different kinds of teeth. Although teeth differ is size and shape and perform different function, their internal structure is similar.

Parts of the tooth

Crown: This is the part that projects above the gum

Root: The part which is buried in a socket in the jawbone. Some teeth have a single root while others have two or three roots.

Enamel:, This is a layer of thin, hard, shiny material composed mainly of calcium phosphate. It covers the crown.

Dentine: The dentine is located beneath the enamel and extended into the root. It makes up the greatest bulk of the tooth. Dentine is a very hard bone, but is not as hard as the enamel. Fine channels, containing living materials, pass through the dentine. The dentine encloses a central ***pulp cavity.***

The pulp cavity: The pulp is made up of blood vessels, tooth-forming cells and nerve fibres with sensory nerve endings, which penetrates the dentine, making it sensitive to pain and temperature changes.

Cement: Teeth are fixed in the lower bone by the cement, which is a hard bony substance that covers the outside of the root.

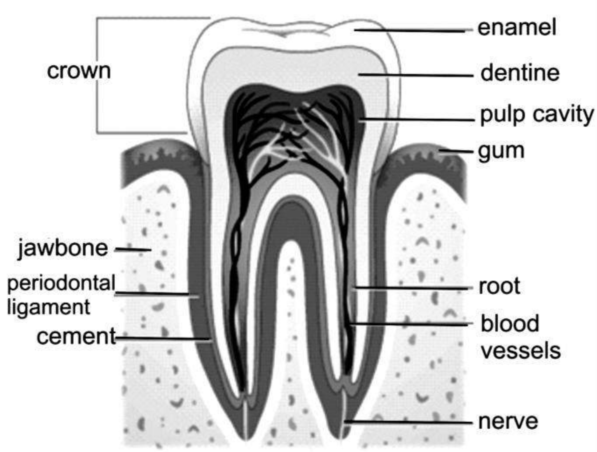


Fig. 9.8: Structure of a tooth (molar)

Periodontal fibres: Attached to the cement are tough periodontal fibres which run into the jaw bone. These fibres hold the tooth in its socket and permit slight tooth movements that cushion the tooth from excessive jarring when it hits a hard object.

Types of teeth and their functions in mammals

Teeth in mammals come in four different types – incisors, canines, premolars and molars.

Incisors

These are the two pairs of teeth found in the front part of each jaw.

Features of incisors

1. They are chisel shaped.
2. They have broad, flat cutting edge for biting and cutting food.
3. They have a single root.

Canines

There are two canines next to the incisors.

Features of canines

1. They are bluntly pointed.
2. They are used for cutting and biting solid foods.

Premolars

These are found in either half of the jaws.

Features of premolars

1. They have flat tops.
2. They have cusps (pointed ridges).
3. They have one or two roots.

Molars

They make the three final teeth on each half of each jaw.

Features of molars

1. They have flat tops.
2. They have four or five cusps.
3. They have three roots (upper molars) and two roots (lower molars).

The premolars and molars are collectively known as ***cheek teeth.***

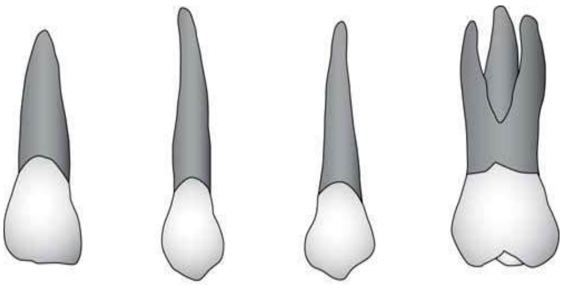


Fig. 9.9: Types of teeth in humans

Set of teeth in humans

Mammals have two sets of teeth in their lifetime. The first set of 20 teeth is called ***milk teeth*** and is found in young humans. They are made up of incisors, canines and premolars. These teeth fall out one by one in early life and are replaced by the ***permanent teeth***. The last molar teeth in each jaw is called ***wisdom teeth*** because they are the last to appear when the person is between 17 and 25 years old.

Care of the teeth in humans

In order to prevent and control dental problems, it is important for us to take good care of the teeth. Lack of proper care of the teeth results in dental caries (tooth decay) and periodontal disease.

Dental caries (tooth decay)

This is a dental condition caused when bacteria feed on sugar and starch in the mouth. Their enzymes react with the sugar and starch and produce acid.

The acid reacts with calcium in the enamel and dentine and breaks down the enamel and dentine creating a hole in the tooth. Bacteria move on to feed on the pulp and the sensitive nerve ending which causes toothache. At the early stages, the tooth can be repaired.

Causes of dental caries

1. Lack of hard food
2. Lack of vitamin D
3. Eating too much sugar
4. Lack of calcium
5. Improper cleaning of teeth

Periodontal disease:

This is caused by the accumulation of food debris, saliva and bacteria to form a hard thin coating on the surface of the teeth called ***plaque***. The plaque spreads down to the gum causing the gum to swell, breed, recede and give off bad breath. The recession of the gum causes the teeth to fall out.

Causes of periodontal disease

1. Lack of vitamin A and C
2. Improper cleaning of teeth
3. Lack of massage of the gum

Ways of preventing dental problems

The following are necessary measures to prevent the above diseases:

1. Cleaning the teeth regularly
2. Regular visit to the dentist
3. Eating food rich in minerals and vitamins
4. Using the teeth properly; for example, not using the teeth to open bottles or screws.

DIGESTIVE SYSTEM

Digestive system is a network of organs and enzymes which converts the complex food (i.e. carbohydrates, proteins, lipids etc) eaten into small and absorbable forms for the use of the body.

Or

Digestion is the breakdown of complex food substances into simple, soluble and absorbable forms.

Structure and functions of the digestive system of mammals

The digestive system of mammals is made up of the gut or alimentary canal and other organs and glands. The alimentary canal is a muscular tube made up of the following parts:

Mouth

Digestion starts from the mouth, where food is chewed with the teeth. Chewing (or mastication) breaks down solid food into smaller particles. Salivary glands produce saliva which:

- ❖ contains the an enzyme called ***saliva amylase*** that converts starch into ***maltose***.
- ❖ is slightly alkaline and keeps the pH of the mouth approximately neutral which help in the action of saliva amylase.
- ❖ mixes with the food during chewing, making it soft and easier to swallow.

Oesophagus

A muscular tube that connects the mouth to the stomach. Food passes down the oesophagus through contraction and relaxation of the circular muscles of the gut. This action is called ***peristalsis.***

The stomach

The stomach is a sac-like organ that produces ***gastric juice*** which contains:

- ❖ ***hydrochloric*** acid (sterilizes food, ends action of saliva amylase, provide acidic pH for action of the enzyme
- ❖ ***pepsin*** which converts proteins to ***polypeptides***,
- ❖ ***mucus*** which forms a protective inner lining to the animal wall and
- ❖ ***rennin*** which solidifies liquid proteins so that they will remain in the stomach for pepsin to work on.

The contractions of the stomach wall muscles churn the food into ***chyme*** (a creamy-white mixture of food substance). At this stage glucose and alcohol are absorbed from the food into the blood stream.

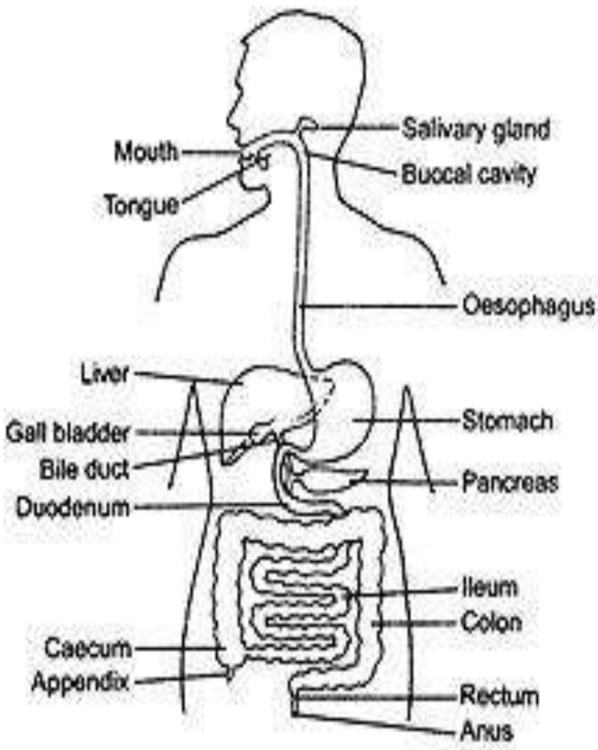


Fig. 10.0: Digestive system of man

Duodenum

The chyme from the stomach passes the pyrolic sphincter into the duodenum where the hormone ***secretin*** is released into the bloodstream. Secretin then stimulates the gall bladder to release bile and the pancreas to release ***pancreatic juice*** into the duodenum.

Liver

The liver is one important organ which is responsible for most chemical activities in the body.

Functions of the lever

1. Production of bile which emulsifies or breaks down fats, converts glucose to glycogen, produces urea (the main substance of urine), makes certain amino acids (the building blocks of proteins).
2. Filtration of harmful substances from the blood (such as alcohol).
3. Storage of vitamins and minerals (vitamins A, D, K and B_{12})
4. Maintaining the proper level or glucose in the blood.
5. Production of cholesterol; it produces about 80% of the cholesterol in the body
6. Production of heat to the body through chemical activities.
7. Production of some proteins such as fibrinogen in the blood plasma.
8. Destruction and stopping of hormones from unlimited action.
9. Conversion of excess fats into glycerol and fatty acids.
10. Excretion of excess cholesterol and old red blood cells from the body.

Bile

Bile is alkaline and raises the pH of the chyme. It emulsifies fat by changing large fat globules into small fat droplets. ***Emulsification*** increases the surface area of the fat making the digestion of fat by enzymes more efficient.

Pancreas

The pancreas secrets ***pancreatic juice*** which is alkaline and therefore raises the pH of the chyme. The pancreatic juice contains

- ***amylase*** which converts starch to maltose,
- ***trypsin*** which converts protein to peptides and polypeptides and
- ***lipase*** which converts fat to fatty acids and glycerol.

All these enzymes digest food as is passes though the duodenum into the ileum.

Ileum (small intestines)

The digestion process ends in the ileum. The walls of ileum secrets:

- ***lipase*** which converts all remaining fats into fatty acids and glycerol;
- ***maltase*** which converts maltose to glucose;

- ***lactase*** which converts lactose to glucose and galactose,
- ***sucrase*** which converts sucrose to glucose and fructose,
- ***erepsin*** which converts peptides to amino acids and
- ***enterokinase*** which converts trypsinogen to trypsin.

The small molecules of glucose, amino acids, fatty acids, and glycerol are the end products of digestion in mammals and are absorbed into the bloodstream by a process known as ***diffusion***.

The ileum contains thousands of microscopic projections called ***villi*** (singular, villus), which increases the surface area of the ileum and facilitates the absorption process. Undigested food passes into the colon.

Colon (large intestines)

The colon is responsible for the absorption of water from the undigested food. It contains some bacteria which digest the cellulose and produce vitamin B and K which are absorbed into the body. The remains of the undigested food pass on to become faeces.

Table 3.8: Organs and enzymes of digestion

Organ/ gland	Secreted enzyme	Enzyme action
Mouth (salivary glands)	• Salivary amylase	• Starch to maltose
Oesophagus	• Passage of food to stomach	
Stomach	• Pepsin • Rennin • Hydrochloric acid	• Polypeptides • Solidifies protein • sterilizes food; provide acidic pH for action of the enzyme
Duodenum	• Secretin	• Stimulates gall bladder to release bile and pancreas to release pancreatic juice
Liver	• Bile	• Emulsifies starch
Pancreas (pancreatic juice)	• Amylase • Trypsin • Lipase	• Starch to maltose • Protein to peptides and polypeptides • Fatty acids to glycerol
Ileum	• Lipase • Maltase • Lactase • Sucrose • Erepsin	• Fats into fatty acids and glycerol • Maltose to glucose • Lactose to glucose • Sucrose to glucose • Peptides to amino acids

Table 3.8: Organs and enzymes of digestion (continued)

	• Enterokinase	• Trypsinogen to trypsin
Colon		Absorption of water from undigested food
Rectum		Temporary storage for faeces

Rectum

The faeces move into the rectum, where they are stored and released periodically through the anus. The removal of food from the rectum is known as ***egestion***. Not to be confused with ***ingestion***, which is the act of putting food into the mouth?

DIGESTION IN MAMMALS

The term digestion refers to the breakdown of complex food into smaller form which can be absorbed by the body. Digestion occurs as a result of chemical or mechanical processes in the body. Digestion is very important because it allows the body to gain energy through ingested food.

Different food substances have different modes of digestion.

Digestion of carbohydrates

- ❖ Carbohydrate digestion starts in the mouth.
- ❖ Saliva contains ptyalin (salivary amylase) which converts starch to maltose.
- ❖ In the stomach, HCl in gastric juice sterilizes the food.
- ❖ The food is churned into liquid paste called chyme.
- ❖ In the duodenum, the bile secreted by the liver neutralizes the chime.
- ❖ Pancreatic amylase secreted by the pancreas converts starch into maltose sugar.
- ❖ In the ileum, the enzyme maltase converts maltose into glucose, which is the end-product of starch digestion.

Digestion of proteins

- ❖ Protein digestion starts in the stomach.
- ❖ Pepsin converts proteins to peptones.
- ❖ HCl sterilizes it; it is churned to chime and sent to duodenum where the bile neutralizes it.
- ❖ In the duodenum, the typsin which is secreted by the pancreas converts peptones into amino acids.

Digestion of fats

- ❖ The digestion of fat starts in the duodenum.
- ❖ Gall bladder releases bile into the duodenum.
- ❖ Bile neutralizes and emulsifies the fat.
- ❖ The enzyme lipase secreted by the pancreas and the wall of the small intestines converts emulsified fat into fatty acids and glycerol, which is the end-product of fat digestion.

Table 3.9: End-products of digestion

Food	End-product
Carbohydrates	Glucose
Proteins	Amino acids
Fats	Fatty acids and glycerol

Uses of Digested Food

The end-product of digestion is ***assimilated*** (taken up) by the cells in various parts of the body. Glucose and others move into the blood by ***diffusion*** and enter the liver through the hepatic portal vein. In the liver excess glucose is converted to ***glycogen*** and stored. Some of the monosaccharides enter body cells where they are respired.

Amino acids are also assimilated into the liver though the hepatic portal vein. Some enter the body cells where they are used to form protein for growth, repair of damaged and worn-out tissues and formation of enzymes and hormones. Excess amino acids are excreted out of the body as ***urea***.

Fatty acids and glycerol come together again to form fats, which enters the lymph system before entering the bloodstream. They are stored as fat or respired to release energy.

Assimilation is the process whereby digested food substances are absorbed into the bloodstream.

DISEASES AND DISORDERS ASSOCIATED WITH THE DIGESTIVE SYSTEM OF HUMANS

Constipation

This is caused by not going to the toilet regularly. Faeces overstay in the rectum and more water is absorbed from them. This hardens and dries it making it difficult to pass out. Constipation is also caused by adequate chewing of food.

Prevention of constipation

Constipation can be prevented by including roughage in the diet, chewing food properly and visiting the toilet regularly.

Indigestion

Indigestion is caused by eating too quickly. This causes the gastric juice to produce excess acid. The acid climbs up the oesophagus, when a person belches. This gives a burning sensation in the heart (heartburns).

Prevention of indigestion

Slowing down the rate and speed of eating can prevent indigestion.

Diarrhoea

This occurs when faeces move through the colon too quickly before water can be absorbed from them. This causes the faeces to come out in a more liquid form. It is caused by bacteria in the gut.

Prevention of diarrhoea

Staying away from food for a short period will cause the bacteria to be driven away with the faeces. A lot of water should be drunk to prevent dehydration.

Stomach ulcer

This is caused by an attack on the stomach wall by excess hydrochloric acid in the stomach; this makes is sore.

Prevention of stomach ulcer

Eating at normal intervals can prevent stomach ulcer.

Jaundice

Jaundice, also known as ***icterus***, is a yellowish discoloration of the skin, the conjunctival membranes over the sclerae (whites of the eyes), and other mucous membranes caused by increased levels of bilirubin in the blood.

Treatment

Infected persons must seek medical attention immediately.

Cirrhosis

Cirrhosis is a condition in which the liver slowly deteriorates and malfunctions due to chronic injury. Scar tissue replaces healthy liver tissue, partially blocking the flow of blood through the liver. Scarring also impairs the liver's ability to:

- control infections
- remove bacteria and toxins from the blood
- process nutrients, hormones, and drugs
- make proteins that regulate blood clotting
- produce bile to help absorb fats—including cholesterol—and fat-soluble vitamins

A healthy liver is able to regenerate most of its own cells when they become damaged. With end-stage cirrhosis, the liver can no longer effectively replace damaged cells. A healthy liver is necessary for survival.

Cirrhosis is caused by heavy alcohol consumption, obesity, hepatitis B, C and D, drug and toxins

Many people with cirrhosis have no symptoms at the early stages of the disease. However, as the disease progresses, a person may experience the following symptoms:

1. weakness
2. fatigue
3. loss of appetite
4. nausea
5. vomiting
6. weight loss
7. abdominal pain and bloating when fluid accumulates in the abdomen
8. itching
9. spiderlike blood vessels on the skin

Treatment and prevention

Medical attention is necessary if infected with cirrhosis. It is also advisable not to drink or engage in drugs. Nutritious food must be eaten always.

Hepatitis

Hepatitis (plural hepatitides) implies inflammation of the liver characterized by the presence of inflammatory cells in the tissue of the organ. A group of viruses known as the *hepatitis viruses* cause most cases of hepatitis worldwide, but it can also be due to toxins (notably alcohol, certain medications and plants), other infections and autoimmune diseases.

Treatment and prevention

Reducing the intake of alcohol, eating nutritious food and not doing drug may help prevent hepatitis. If infected by hepatitis, it is advisable to seek medical attention.

DIGESTIVE SYSTEM OF RUMINANTS

Ruminants are mammals which have a stomach system consisting of four chambers.

The stomach system are rumen, reticulum, omasum and abomasums.

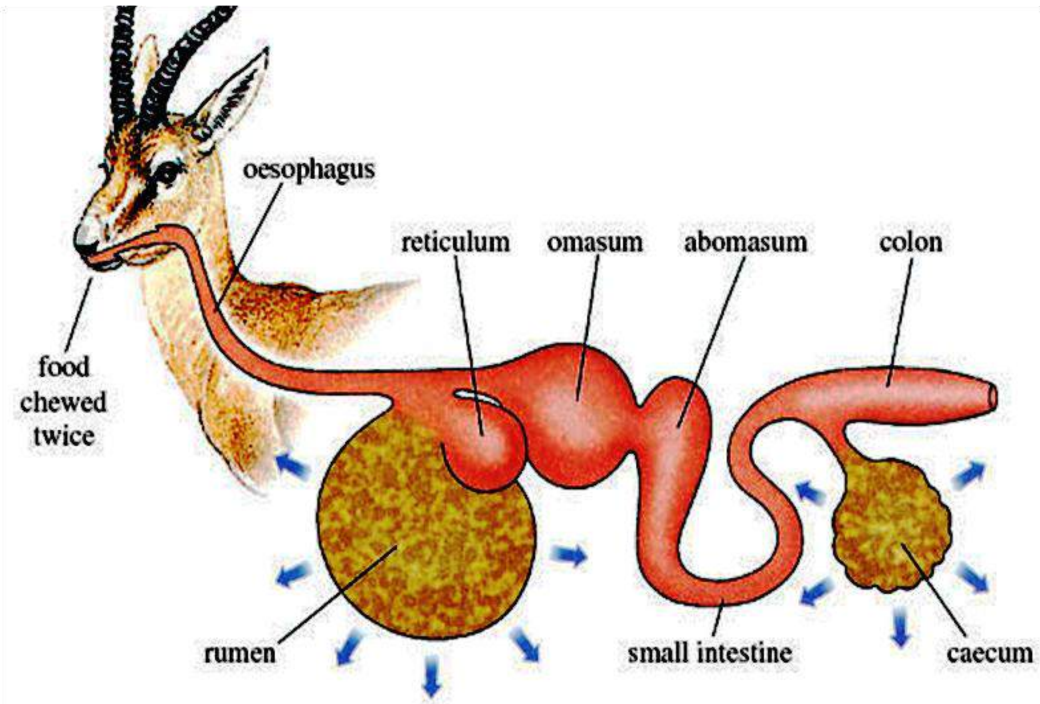


Fig. 10.1: Digestive system of a ruminant

Rumen

This is the first and largest chamber. Swallowed food enters this chamber where it is fermented by anaerobic bacteria which breakdown the cellulose.

Reticulum

When the food reaches the reticulum, it is regurgitated (brought back to the mouth) and chewed again. Food in the reticulum is known as ***cud***.

Omasum

Much of the water in the cud is reabsorbed in the omasum.

Abomasum

This is also known as the true stomach of a ruminant. Gastric secretion in the abomasum digests the proteins in the food. Chyme is formed which moves on to the duodenum

Digestion in birds

The stomach of birds is divided into two – ***crop*** and ***gizzard***. Food is stored in the crop and ground up in the gizzard with the aid of small stones, before it is passed into the intestines.

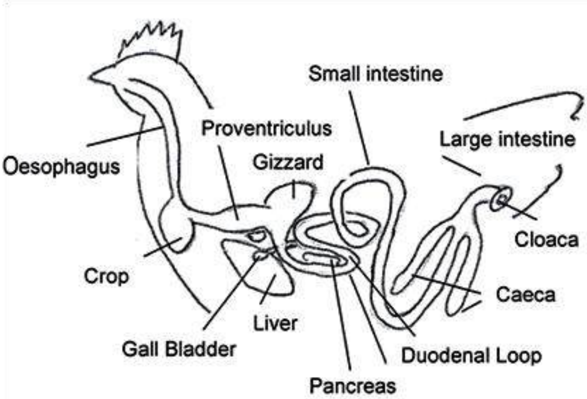
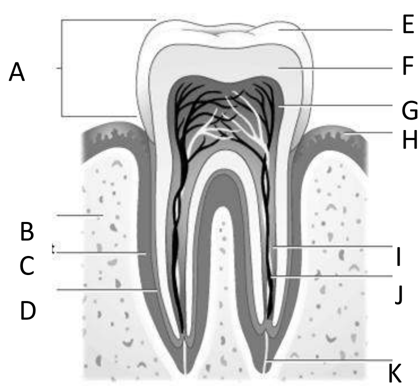


Fig. 10.2: Digestive system of a bird

TEST QUESTIONS

1. (a) What is dentition?
   (b) List three characteristics of the following types of dentition:
      i. dentition of a carnivore.
      ii. dentition of a herbivore.
      iii. dentition of an omnivore.
2. State three causes of dental problems and three ways of preventing them.
3. Describe the digestive system of a mammal.
4. The diagram below is a longitudinal section of through a molar tooth.



i. Name the parts labeled A-K.
ii What is the function the tooth?
iii. What is the function of the parts labeled E, G and H.

5. Explain three diseases associated with the digestive system of humans.

6. Fig. **5** is an illustration of the digestive system of a farm animal.

Study the figure carefully and answer the questions that follow.

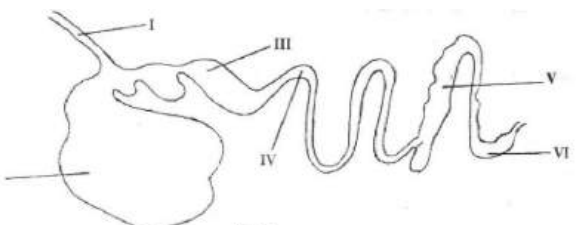


Fig 5

(a) (i) Name the parts labelled **I**, **II**, **III**, **IV**, **V** and **VI**
(ii) State one function each of the parts labelled **I**, **V** and **VI**.
(b) (i) Name two farm animals that possess the digestive system illustrated .
(ii) What is the general name given to the farm animal in (i)?
(c) (i) Name one parasite that affect the part labelled **IV**.
(ii) State two ways of controlling the parasite named in (i).

7. (a) State
(i) two causes of tooth decay.
(ii) two ways of preventin tooth decay.

8. **Fig. 3** below is an illustration of the human body.
Study the figure carefully and answer the questions that follow.

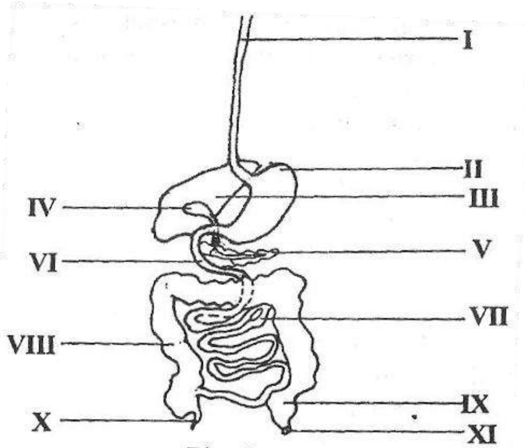
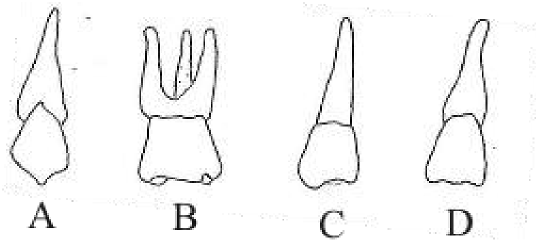


Fig. 3

a) Identify the part of the human body illustrated.
b) Name the parts labelled I, II, III, IV, V, VI, VII, VIII, IX and X.
c) Describe the digestive process that occurs in VI.
d) i) List two enzymes secreted by the part labelled II.
ii) State one function of each of the two enzymes you have listed in (d)(i) above.

9. Figure 3 is an illustration of different types of teeth.
Study the figure carefully and answer the questions that follow.



a) Identify each of the types of teeth illustrated.
b) Describe the structure of tooth C.
c) Give three structural differences between tooth A and tooth B.
d) State one function of each type of teeth.
e) i) Name the group of animals which possess all the types of teeth illustrated in figure 3.
   ii) Support your answer with reasons.

10. a) State two functions of each of the following organs associated with digestion:
    i) liver;
    ii) pancreas.
    (b) Describe the process of fat digestion in humans.

11. **Fig. 2** below is an illustration of two types of upper jaws, A and B of two group of mammals.
    Study the figure carefully and answer the questions that follow.

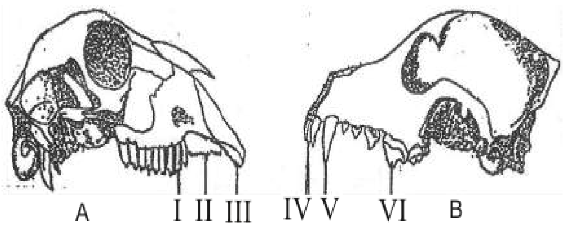


Fig. 2

a) Name the group of mammals which possess upper jaw type **A** and upper jaw type **B** respectively.
b) Give two examples each of mammals which possess upper jaw type **A** and upper jaw type **B**.
c) Name the parts labelled I, II, III, IV, V and VI.
d) State one function of each of the parts labelled II, IV and VI.
e) Tabulate four differences between upper jaw A and upper jaw **B**.

13

RESPIRATORY SYSTEM

Specific Objectives

After completing this chapter, you will be able to:

- Define respiration and explain its importance.
- Distinguish between aerobic and anaerobic respiration.
- Identify the respiratory organs of the respiratory system of a mammal and describe their functions..
- Explain the mechanisms of inhalation and exhalation in humans
- Enumerate some problems and disorders associated with the respiratory system of humans.
- Explain how respiratory gases are taken in and out of plants.

INTRODUCTION

Respiration is the breakdown of organic food substances in the living cells to release energy.

The breakdown of organic food substances is a complex process involving many enzymes acting as catalysts. The energy released is in a form of ATP (adenosine triphosphate).

Respiration is also known as tissue, cell or internal respiration.

NOTE: Tissue or internal respiration is different from breathing (external respiration), which is physical and does not occur in the cells.

IMPORTANCE OF RESPIRATION

The chemical energy released is used in:

1. cell division
2. transmission of nerve impulses.
3. transportation of substances in and out of the cells.
4. muscular contraction which aids in movement.
5. production of heat.
6. production of chemical substances in the cells. e.g. protein synthesis.
7. breakdown of chemical substances in the cells (biodegradation). e.g. glucose.
8. production of energy for life processes.
9. generation of energy for day to day activities.

TISSUE OR CELL RESPIRATION

Tissue respiration is the breakdown of food substances inside the cells to release energy.

There are two types of tissue respiration, namely aerobic respiration and anaerobic respiration.

Aerobic respiration

Aerobic respiration is the type of respiration which takes place in the presence of oxygen.

A lot of energy is produced in this type of respiration. Carbon dioxide and water are produced as by-products.
Aerobic respiration can be expressed as:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy
glucose + oxygen = carbon dioxide + water + energy

NOTE: The importance of aerobic respiration is the same as the importance of respiration above.

Anaerobic respiration

This is the type of respiration which occurs in the absence of oxygen.

Small amount of energy is produced. It occurs in yeast, with alcohol and carbon dioxide produced as by-products.
Anaerobic respiration is also known as ***alcohol fermentation***.

It can be expressed as:

$C_6H_{12}O_6 \rightarrow C_2H_5OH + 2CO_2$ + energy
glucose = alcohol + carbon dioxide + small energy

Anaerobic respiration also takes place in some bacteria and skeletal muscles of mammals. Lactate (lactic acid) is produced as a by-product. This is known as ***lactate fermentation***. It can be expressed as:

$C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$ + energy
glucose = alcohol + small energy

Importance of Anaerobic Respiration

1. Used in bread preparation.
2. Used in brewing of alcohol.
3. Used in the production of antibodies.
4. Used in food preparation.

Table 4.0: Differences between aerobic and anaerobic respiration

Aerobic respiration	Anaerobic respiration
Takes place in the presence of oxygen	Does not take place in the presence of oxygen
A lot of energy is released	Less amount of energy is produced
Carbon dioxide and water are produced as by-products	Lactate, alcohol and carbon dioxide are produced as by-products
Takes place in mitochondrion	Takes place in cytoplasm

Similarities between Aerobic and Anaerobic Respiration

1. Energy is produced in both.
2. Carbon dioxide is produced as by product in both.

Experiment to demonstrate the release of energy during respiration or germination of seeds

- Wash some germinating seeds in disinfectants to prevent growth of microbes.
- Put some of the seeds in a vacuum flask and label that **A**.
- Boil the rest of the seeds to kill them and stop germination, wash them in formalin to prevent decay and put them in vacuum flask labelled **B**.
- Plug both flasks with cotton wool and insert a thermometer through the cotton wool.
- Check and record the initial reading of both thermometers.
- Recheck the reading periodically for 12 hours.

Observation

It would be observed that the temperature in flask **A** increases gradually, while the temperature in flask **B** either remains the same or drops.

Conclusion

This shows that germinating seeds release heat energy.

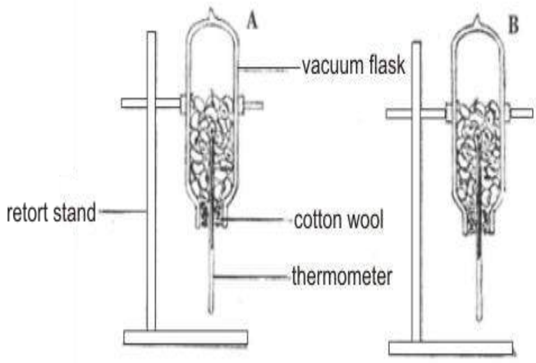


Fig. 10.3

***NOTE*:** It is important to sterilize the seeds in flask **A** and wash the seeds in flask **B** in formalin. This is because decay causing microorganisms produce heat which will interfere with the reading on the thermometers.

The germinating seeds can be replaced with insects when demonstrating the release of heat by small animals. The insects should be released within an hour to prevent suffocation.

THE STRUCTURE AND FUNCTIONS OF THE RESPIRATORY SYSTEM OF MAMMAL

The respiratory system consists of the ***lungs***, (which is a pair of elastic organs housed in the chest cavity), and the air passages leading to them.

Air enters the respiratory system through the ***nose*** or ***mouth***. It then travels through the ***larynx*** (voice box) and into the ***trachea*** (windpipe). At about the middle of the chest, the trachea divides into two tubes, the right and left ***bronchi***. The right ***bronchus*** carries air to the three lobes of the right lung. The left bronchus supplies air to the two lobes of the left lung.

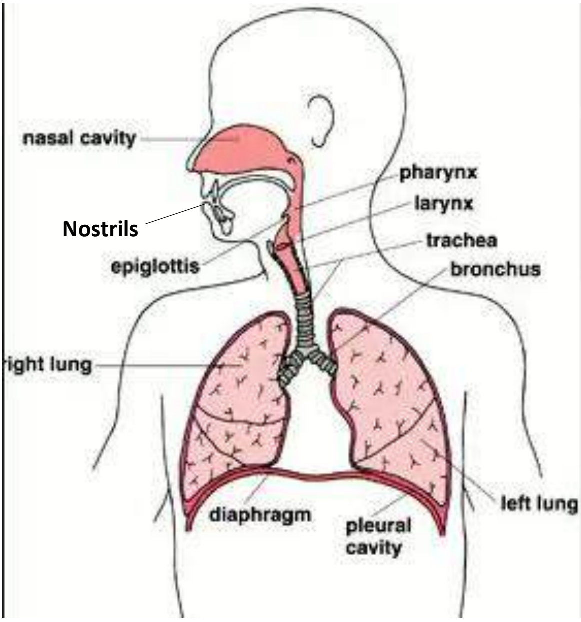


Fig. 10.4: Respiratory system of man

Inside the lungs the bronchi divide into smaller branches called ***bronchioles***, which eventually empty into thousands of minute sacs called ***alveoli.*** The alveoli are surrounded by thin-walled capillaries. The air in the alveoli passes through to the blood cells within the capillaries. At the same time, carbon dioxide from these blood cells passes into the alveoli. From the alveoli, air containing carbon dioxide travels out of the lungs and is exhaled through the nose or mouth.

Parts of the respiratory system

Nostrils

The passage for outside air into the body. The nostrils contain hair which traps dust particles from inhaled air. The nostrils are lined with mucus which traps dust particles in inhaled air.

Nasal cavity

The nasal cavity warms inhaled air. It contains hair and mucus which traps dust in inhaled air. It is also responsible for smelling.

Larynx (voice box)

The larynx is a hollow chamber in which the voice is produced.

The air inhaled into the lungs provides oxygen to cells throughout the body. Air forced out of the lungs removes carbon dioxide from the body.

Trachea

The trachea, commonly called windpipe, is made up of numerous cartilaginous half-rings. Passage for inhaled and exhaled air in and out of the lungs

It is lined internally with a ciliated mucous membrane.

Diaphragm

The diaphragm is a wide muscular partition separating the thoracic, or chest cavity, from the abdominal cavity. It contracts and relaxes to aid breathing.

Ribs
The ribs form a protective cage for the lungs. They aid in gaseous exchange.

Lungs
This is the main organ for gaseous exchange.

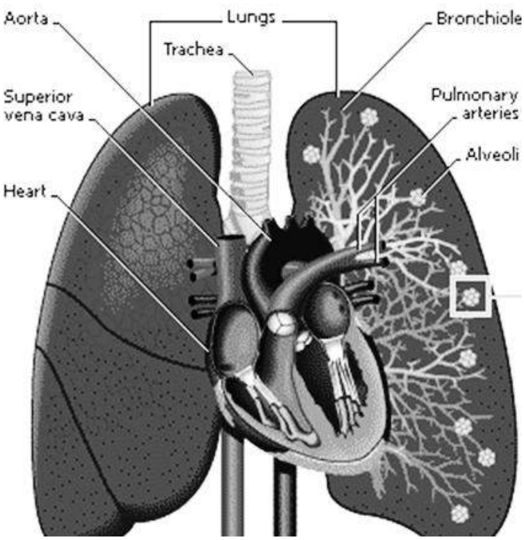


Fig. 10.5: Structure of the lungs

BREATHING MECHANISMS

Breathing is the act of taking in oxygen and releasing carbon dioxide.

It is also referred to as gaseous exchange or external respiration (to distinguish it from internal or cellular respiration). The structure and mechanisms of breathing varies from one organism to another. For example, mammals, reptiles, amphibians and bird have lungs for breathing, whiles fish have gills.

Breathing involves inhalation (breathing in) and exhalation (breathing out).

Mechanism of Inhalation (Breathing In)
Inhalation occurs when the external intercostal muscles contract and the internal intercostal muscles relax. These actions cause the ribs and sternum to move upwards and outwards. At the same time, the diaphragm muscles contract which causes the diaphragm (which usually has a dome shape) to flatten. The movements increase the volume of the thoracic cavity. The pressure of the thoracic cavity falls below that of external atmosphere. Air therefore moves under pressure from the external atmosphere into the lungs through the respiratory tubes.

Mechanism of Exhalation (Breathing Out)
The internal intercostal muscles contract whiles the external intercostal muscles relax. The ribs and the sternum then move downwards and inwards to the original positions. At the same time the muscles of the diaphragm relax and the diaphragm returns to its original dome shape. The volume of the thoracic cavity decreases. The air pressure in the thoracic rises above that of the external atmosphere.

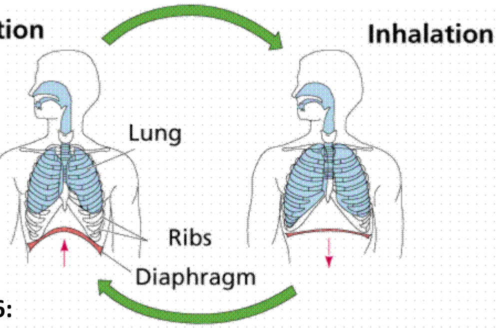


Fig. 10.6:

Table 4.1: Differences between inhalation and exhalation

Inhalation	Exhalation
External intercostal muscles contract	External intercostal muscles relax
Internal intercostal muscles relax	Internal intercostal muscles contract
Diaphragm contracts	Diaphragm relaxes
Diaphragm becomes flat	Diaphragm returns to its usual dome shape
Ribs and sternum move upwards and outwards	Ribs and sternum move downwards and inwards
Thoracic cavity's volume increases	Thoracic cavity's volume decreases
Air moves into the lungs	Air moves out of the lungs

Gas exchange in the lungs

The alveoli of the lungs are the spot where the exchange of gases takes place. Oxygen diffuses into the blood through the blood capillaries, surrounding the alveoli. Carbon dioxide diffuses out of the blood capillaries into the alveoli. Water also evaporates from the moist alveoli so that the alveolar air becomes saturated with vapour. The air breathed out from the lungs is therefore always moist.

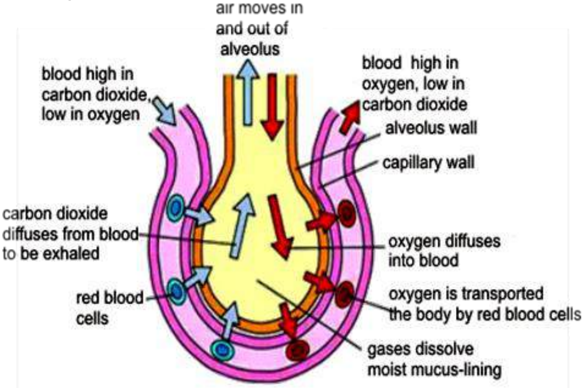


Fig. 10.7: Gas exchange in the alveoli

Table 4.2: Comparison of inhaled and exhaled air

Component	Inhaled air	Exhaled air
Carbon dioxide (CO_2)	0.03 %	3.5 %
Oxygen (O_2)	21 %	17 %
Water vapour	variable	Saturated
Temperature	Atmospheric temperature (25^oC)	Body temperature (37^oC)

Comparing the amount of carbon dioxide in inhaled and exhaled air

- Arrange two test tubes with delivery tubes as below.
- Put your mouth over the mouthpiece
- Breathe in and out gently
- Note which test tube the inhaled and exhaled air bubbles through.
- Repeat this for a number of times and record your observation.

Observation

It would be observed that the limewater in test tube **B**, which bubbles when exhaled into the mouthpiece, turns milky, while the limewater in the test tube **A**, which bubbles during inhalation remains clear.

Conclusion

Exhaled air contains more carbon dioxide than inhaled air.

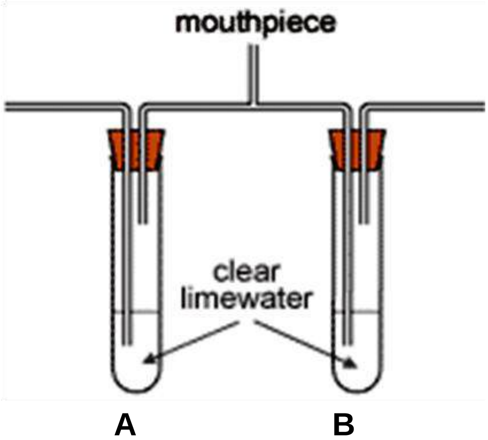


Fig. 10.8

Table 4.3: Differences between respiration and breathing

Respiration	Breathing
Energy is released	No energy is released
Takes place inside the cells	Takes place outside the cells
Involves oxidation of glucose into carbon dioxide, water and energy	Involves exchange of oxygen and carbon dioxide
A complex biochemical process	Mechanical process

Problems and disorders associated with the respiratory system

The human respiratory system may be exposed to certain disorders due to germs or improper functioning of certain parts of the respiratory system.

The problems and disorders of the respiratory system include:

1. Asthma
2. Tuberculosis
3. Lung cancer
4. Pneumonia
5. Bronchitis
6. Whooping cough (pertusis)

EXCHANGE OF RESPIRATORY GASES IN PLANTS

Gas exchange takes place on the surface of the spongy mesophyll cells in the leaves of plants. The gases enter and leave the leaf through tiny pores called ***stomata*** in the lower surface of the leaf.

Unlike animals, plants take in carbon dioxide during the day and oxygen at nights (in the absence of light).

During the day, carbon dioxide from the atmosphere passes through the stomata into the leaf, and diffuses into the mesophyll cells where it is used for photosynthesis.

At night, oxygen from the atmosphere passes through the stomata into the mesophyll cells of the leaf where it is used in aerobic respiration.

At the same time, carbon dioxide produced in the mesophyll cells as a by-product of respiration diffuses out of the cells and passes through the stomata into the atmosphere. Other areas where gas exchange takes place are the epidermal cells of the leaf and stem, pores on the stems of woody plants called ***laticel***, as well as roots of plants.

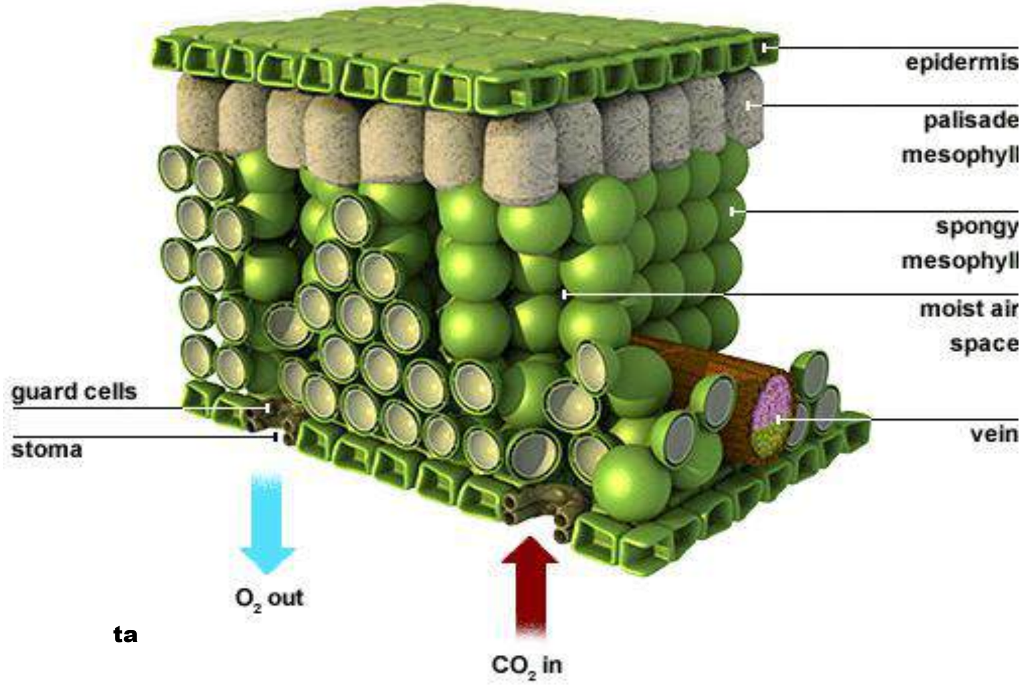


Fig. 10.9: Exchange of gases in plant leaf

Table 4.4: Differences between respiration in mammals and that of plants (photosynthesis)

Respiration in animals	Photosynthesis
Occurs at all time	Occurs only during the day in the presence of sunlight
Energy is released	Energy is used up
Oxygen is needed as raw material	Carbon dioxide is the raw material
Carbon dioxide and water are the by-products	Oxygen is the by-product
Takes place in all living cells	Takes place in plants only

Experiment to demonstrate the release of carbon dioxide during respiration by small animals and germination of seeds

- Set up four glass jars labelled **A, B, C** and **D** connected together by a delivery tube.
- **A** contains caustic soda;
- **B** contains limewater;
- **C**, a bell jar containing a live animal; the animal can be replaced with germinating seeds.
- **D** contains limewater.
- Caustic soda solution in **A** absorbs carbon dioxide.

- When carbon dioxide gets to jar **B**, the limewater turns milky. All the carbon dioxide is absorbed by the limewater.
- When the animal respires, carbon dioxide passes through the delivery tube and turns the limewater in jar **D** milky.
- This shows that animals release carbon dioxide during respiration.

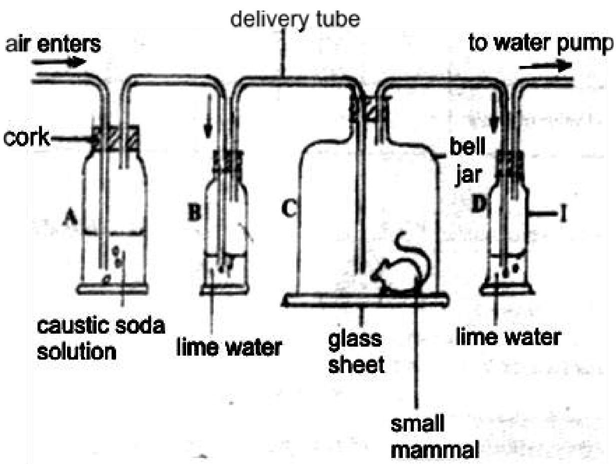


Fig. 11.0

TEST QUESTIONS

1. (a) (i) Explain the term respiration.
   (ii) State four importance of respiration to an organism.
   (b) Differentiate between aerobic and anaerobic respiration and write an appropriate equation for each.

2. (a) Describe the mechanisms of inhalation an exhalation in humans.
   (b) Mention three differences between inhaled air and exhaled air.

3. (a) Describe how respiratory gases are exchanged in plants.
   (b) State four disorders associated with the respiratory system of humans.

4. (a) Draw and label the reparatory system of man.
   (b) State the function of four of the parts of the reparatory system of man.

5. (a) Describe an experiment to show that animals release carbon dioxide during exhalation.
   (b) State four differences between respiration and photosynthesis.

6. Fig. 4 is an experimental set-up that illustrates respiration of a small mammal.
   Study the set-up carefully and answer the questions that follow.

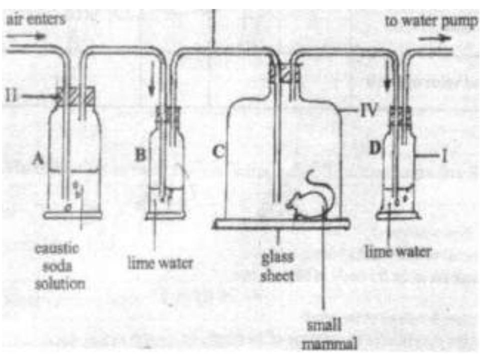


Fig. 4

(a) Name the parts labelled I, II, III and IV.

(b) Explain the role played by the solution in A.

(i) Which of the set-up serves as the control.

(ii) Give one reason for your answer in (c)(i).

(c) (i) What happens to the lime water in D at the end of the experiment?

(ii) Give one reason for your answer in (d)(i).

(d) State one precaution to be taken during the experiment.

(e) Suggest an aim for the experiment.

7. Fig. **4** is an illustration of an experimental set-up to show that heat is given out during germination of bean seeds.

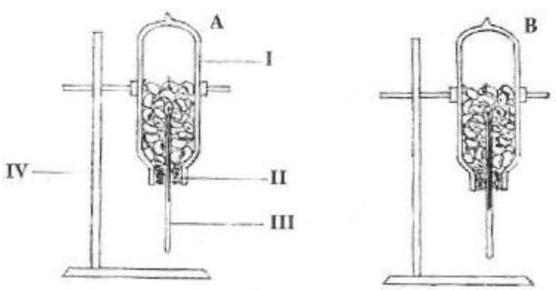


Fig. 4

Equal quantities of seeds were put in **A** and **B**. The seeds in **A** were boiled for few minutes and seeds in **B** were soaked in distilled water to for few hours. Both seeds in **A** and **B** were sprinkled with formalin and were left for 24 hours.

(a) Name the parts labelled I, II, III and V.

(b) what would be observed in each of set-ups **A** and **B** after 24 hours?

(c) Give reasons for the observation in (b).

(d) State two precautions that would be taken during the experiment.

14

TRANSPORT – DIFFUSION AND OSMOSIS

Specific Objectives

After completing this chapter, you will be able to:

- Explain the process of diffusion, osmosis and plasmolysis.
- Sate the importance of the xylem and phloem tissues in plan

INTRODUCTION

Multi-cellular organisms have a very great network of cells, tissues, organs and organ systems. These structures communicate with each other by exchanging materials. The network of structures responsible for that is the ***transport system.***

In humans there are two types of transport systems – the blood circulatory system (made up of the heart, blood vessels and blood.) and the lymphatic system (carries materials around the body).

Plants also have their own transport system – xylem and phloem tissues.

The xylem tissue transports water and dissolved minerals.

The phloem tissue transports food substances through the plant.

Importance of Transport System

1. It connects different parts of the body with one another.
2. It transports food materials to all cells, tissues and organs.
3. It distributes water to various parts of the body.
4. It transports waste products from different parts of the body to the excretory organs.
5. It transports raw materials such as oxygen to the cells, tissues and organs that need them.

MOVEMENT OF SUBSTANCES IN AND OUT OF CELLS

Multi-cellular organisms have cells which allow substances in solution to pass into and out of them. This happens in two major ways – ***diffusion*** and ***osmosis***.

DIFFUSION

Diffusion is the movement of molecules from a region of high molecular concentration to the region of low molecular concentration until the molecules are evenly distributed.

The difference in concentration is called ***concentration gradient***.
The higher the gradient the faster the rate of diffusion. In other words diffusion stops when the concentration gradient of the two regions is the same.

Heating and stirring increase the movement of molecules, and therefore, can speed up the rate of diffusion.

Diffusion is very prominent in gases because the molecules in gases have wide spaces between them. Diffusion is slower in liquids because the molecules are closely packed together than they are in gases. The molecules in solids are tightly packed and therefore the molecules do not have enough rooms to move around, hence diffusion is absent in solids.

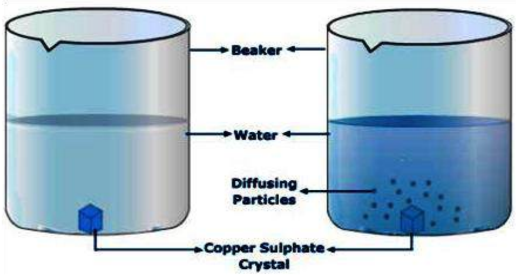


Fig. 11.1: Diffusion in a liquid

Examples of diffusion in living organisms

1. Diffusion of oxygen into the blood and carbon dioxide out of the blood in the lung of mammals.
2. Diffusion of hormones out of the endocrine glands.
3. Absorption of end products of digestion, e.g. glucose, amino acids in the ileum (small intestines) of mammals.
4. Diffusion of carbon dioxide into the blood and oxygen out of the blood in the tissue of vertebrates.
5. Diffusion of oxygen into the cells and carbon dioxide out of the cells of unicellular organisms such as amoeba.
6. Diffusion of carbon dioxide into the leaves and oxygen out of the leaf through the stomata.
7. Diffusion of mineral salt from the soil water into root hair cells.
8. Diffusion of oxygen into the leaves and carbon dioxide out of the leaves of plants at night.

OSMOSIS

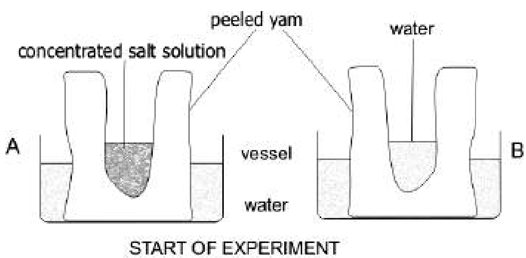
Osmosis is the movement of water molecules from a dilute solution to a more concentrated solution across a semi-permeable membrane.

The higher the difference in concentration between the two solutions, the faster the rate of osmosis. A rise in temperature also speeds up the rate of osmosis.

A ***semi-permeable*** membrane is a membrane that allows certain substances to pass through and blocks the passage of some other substance depending on the nature and size of the substance.

Demonstrating Osmosis in a Semi-Permeable Membrane

- Peel off the skin of a yam tuber
- Cut it into two
- Make a cavity at the centre of each half and flatten the bottom
- Label one **A** and fill it with known volume of a concentrated salt solution
- Label the other one **B** and fill it with water
- Fill the two vessel with water to about 1/3 full and record the volumes
- Sit the yam cups into each of the vessels
- Leave the whole experimental setup for 24 hours.



Observation

After 24 hours, it would be observed that the level of concentrated salt solution in yam cup **A** has increased in volume whiles the water in the vessel has decreased. Compared to **B**, the volume of water in both yam cup and the vessel will remain the same.

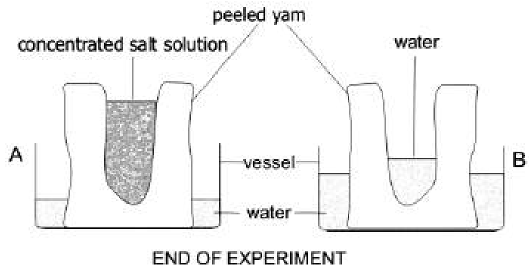


Fig. 11.2: Osmosis in semi-permeable membrane

Conclusion

Water molecules in the vessel of setup **A** have moved from a low concentration to a high concentration (salt solution) through the yam which acts as a semi-permeable membrane.

In setup **B**, water molecules in both the vessel and the yam cavity have the same (low) concentration. Therefore, there was no movement of molecules from one region to the other.

Demonstrating Osmosis in Non-Living Tissue

- Cover the ends of two thistle funnels, labelled **A** and **B**, with cellophane paper.
- Fill funnel **A** with sugar solution and **B** with water.
- Clamp them both upside down into a beaker of water.
- Leave the setup for 24 hours.

Observation

After 24 hours, it would be observed that the level of liquid in test tube **A** would rise but that of test tube **B** would remain the same.

Conclusion

Water molecules have moved from a lower concentration to a higher concentration through the cellophane paper which acts as a semi-permeable membrane.

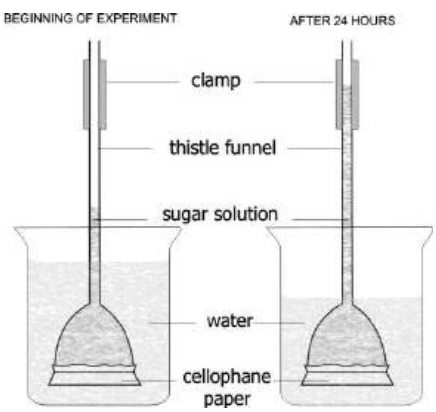


Fig. 11.3: Osmosis in non-living tissue

Examples of Osmosis

1. Absorption of water from the soil by roots of plant.
2. Movement of water across the cell of plants.
3. Movement of water from the root hair into the cortex.
4. Absorption of water in the colon (large intestines) of mammals.
5. Absorption of water in the nephron of the kidney of mammals.
6. Entering of water into the cells of amoeba.

Table 4.5: Differences between diffusion and osmosis

Diffusion	Osmosis
Occurs in both gases and liquids	Occurs in liquids only
Does not require partial membrane	Requires a partial membrane
Involves movement of any substance (both solvent and solute)	Involves movement of only solvent particles (e.g. water molecules)

PROJECT:
Group of student to apply the principles of osmosis in the following activities:

i. Salting of fish for preservation e.g. *Tilapia* ("koobi")

ii. Preservation of liquid food products e.g. fruit juice concentrate

SOME TERMS TO NOTE

Hypertonic solution

This is a solution which has higher concentration than the one it is being compared to.

If a plant or animal cell is introduced to hypotonic solution, the solution absorbs the liquid content of the cell, thereby causing it to become flaccid (shrink and wrinkle). If this continues for long, the cell will become ***plasmolysed***.

Hypotonic solution

This is a solution which has lower concentration than the one it is being compared to.

If a cell is introduced to a hypotonic solution, it absorbs the solution. This causes it to become turgid (swell up) and eventually burst.

Isotonic solution

This is a solution which has the same concentration as the one it is being compared to.

If a cell is introduced to an isotonic solution, the level, size, and structure of both cell and solution will remain the same since they have the same concentration.

Plasmolysis

Plasmolysis is the process whereby a cell looses water by osmosis and shrinks because of the cell being surrounded by a hypertonic solution (i.e. a solution which has higher concentration than the cell sap).

Plant cells in hypertonic, isotonic and hypotonic solutions

- If a plant cell is put in a hypertonic solution (e.g. water), it swells up and becomes turgid.
- If the plant cell is put in a hypertonic solution (e.g. sugar solution), it loses water and shrinks with the cell content moving away from the cell wall.
- If a plant cell in placed in an isotonic solution, it remains the same.

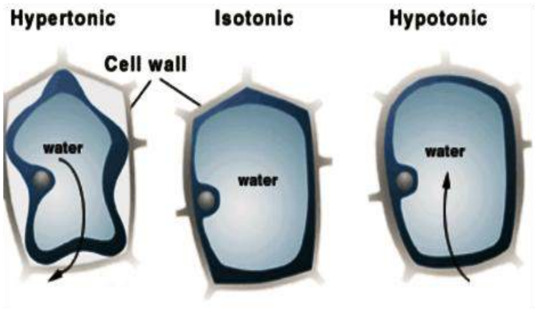


Fig. 11.4: Osmosis in plant cells

Animal cells in hypertonic, isotonic and hypotonic solutions

- If an animal cell is placed in a hypotonic solution, it swells and bursts.
- If it is placed in a hypertonic solution, it loses water, shrinks and wrinkles.

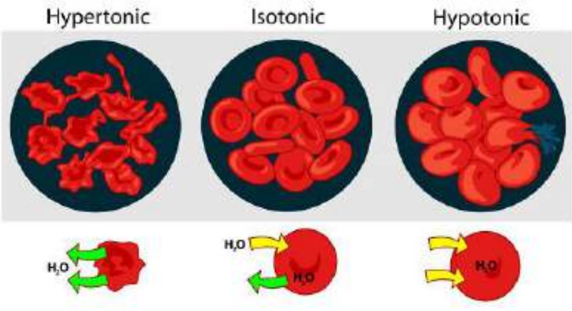


Fig. 11.5

Active transport

Active transport is the movement of substances from a low concentration region to a high concentration region across a living cell membrane using energy from the cell.

The molecules move in an opposite direction to the direction they would go during diffusion. Thus, the molecules move against the concentration gradient.

Examples of Active Transport

1. Absorption of digested food into the blood in the intestines of mammals.
2. Movement of glucose into the phloem tissues of plants.
3. Re-absorption of glucose, amino acids and ions in the kidney of mammals.
4. Absorption of mineral salt from the soil by plant roots.

Table 4.6: Differences between diffusion and active transport

Diffusion	Active transport
Does not require energy	Requires energy
Involves movement of all substances	Involves movement of liquids only
Molecular movement is against concentration gradient	Molecular movement is along with concentration gradient

TRANSPORT IN PLANTS

In plants, the transport of various substances such as water and mineral salt from the soil through the roots and stems to the leaves and the movement of food produced through photosynthesis from the leaves to other parts such as the storage organs and growing part of the plants keep the plants alive.

Absorption and transport of water and mineral salt in plants

1. Water enters the root hair and then the cells of plants by osmosis. This is because vacuoles in the root hair contain higher concentration of solutes that the surrounding soil solution.
2. Water then moves from the root cortex from one cell to another by three different pathways – mostly through the cellulose cell wall, some through the cytoplasm and the rest pass through one vacuole to another.
3. Water finally enters the xylem vessels.

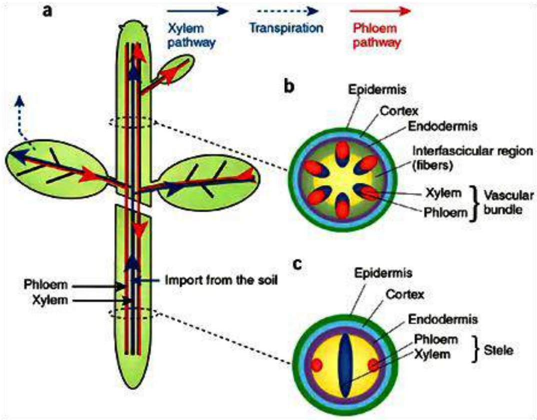


Fig. 11.6: Transverse sections of a stem

The absorption of mineral salt occurs in two major ways – diffusion and active transport

Diffusion

Diffusion occurs only when the concentration of a particular mineral element is higher in the soil solution than in the root hair. Many mineral elements are found in higher concentration in cell saps than in the soil solution. These must be taken up actively by the plants

Active transport

This requires the use of energy produced during respiration, which is in the form of ATP, to absorb mineral ions into the root hair cells against their concentration gradient. The minerals ions then diffuse through the cortex cells and enter the xylem with the water.

Transport of manufactured food (Translocation)

Translocation is the movement of dissolved products of photosynthesis.

Translocation occurs mostly in phloem tissue within the sieve tubes.

Translocation is necessary because the leaves which synthesize organic materials are far away from other parts of the plant which need a constant supply of the manufactured food.

Sugar and amino acids normally move from their manufacture sites in the leaves to growing regions such as buds, or to storage organs such as fruits, seeds or tubers. They may also move from storage organs to growing regions.

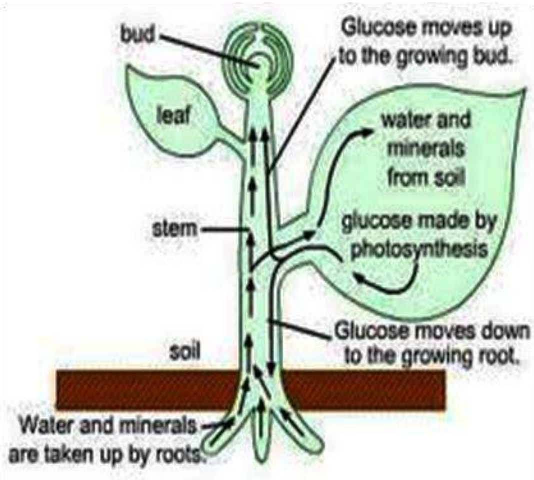


Fig. 11.7: Translocation

Experiment to show that the phloem tissue is responsible for translocation (the ringing experiment)

- Remove a ring from the bark of a young plant.
- Leave it for some 24 hours.

Observation

It would be observed that the area just above the ring starts to bulge, because food substances, which have to move down to the roots, accumulate there. The whole plant dies of starvation after a few weeks.

Conclusion:

This shows that the phloem tissues are responsible for the transport of food substances in plants.

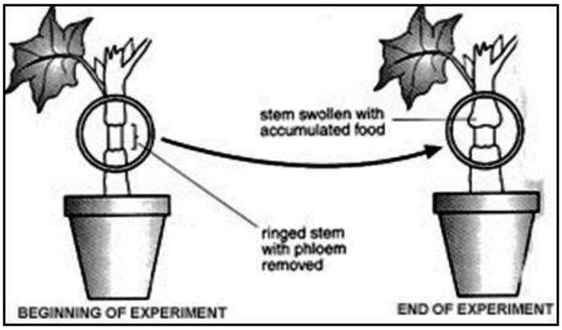


Fig. 11.8: The ring method

TEST QUESTIONS

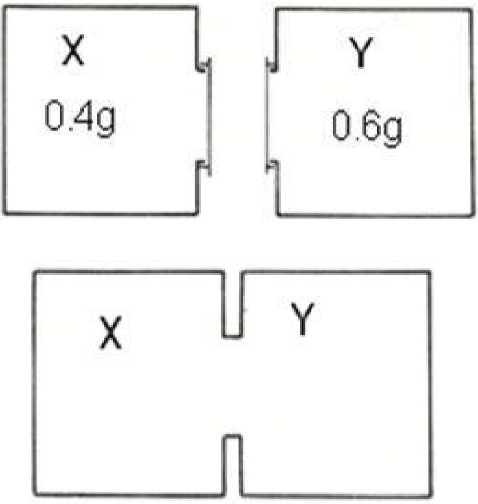
1. (a) What do you understand by the following terms:
   i) Diffusion;
   ii) Osmosis;
   iii) Plasmolysis.

   (b) Mention three differences between osmosis and diffusion.

2. (a) Define the following terms:
   (i) hypertonic solution;
   (ii) hypotonic solution
   (iii) isotonic solution

   (b) Describe what will happen to an animal cell when placed in each of the solutions in (a).

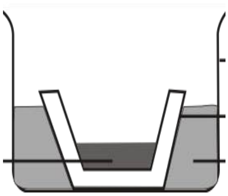
3. Two containers X and Y each hold one litre of air. X also contains 0.4g of a gas and Y contains 0.6 g of the same gas. The two containers are connected together as shown in the diagram.



(a) Which way will the gas diffuse?
(b) After a long period of time, what will be the concentration of the gas (in grams per litre) in each container?

4. Describe an experiment to demonstrate that the phloem tissue is responsible for translocation in plants.

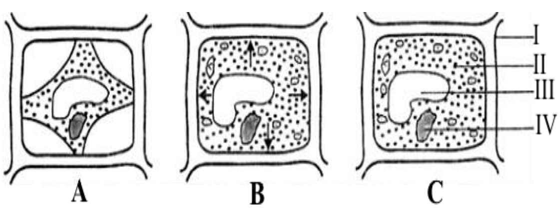
5. The figure below represent the beginning of an experiment to demonstrate osmosis in living cell using yam tissue.



a) Draw and label a diagram to illustrate what would be observed if the set up is allowed to stand for 24 hours.
b) What does the yam cup represent?
c) Explain the principle involved in the experiment.
d) How would you set up control of the experiment above?
e) Give one example of the osmotic process in each of the following living organisms.
   i) Flowering plant;
   ii) Humans.

6. (a) Explain the importance of translocation.
(b) (i) State three examples of active transport.
(ii) Mention three differences between active transport and diffusion.

7. In an experiment to investigate a physiological phenomenon, a student peeled off stripe of the epidermis of the leaf of Rheo discolours. The strips were then placed in three solutions of different concentration for 30 minutes after which the strips were removed and examined under a light microscope. The figure below shows representative cells **A**, **B** and **C** of strips from the three solutions. Study the cells carefully.



a) Name the parts labelled I, II, III and
IV.
b) What physiological phenomenon did the student demonstrate?
c) Name four materials the student needed for the demonstration.
d) (i) State the three types of solutions required for the demonstration.

ii) Match the different states of the cells **A**, **B** and **C** against the solutions you have stated.

(e) What do the arrows in B imply?

(f) Explain the results that would have been observed if red blood cells had been used for the experiment in place of the epidermal cells of Rheo discolour.

15

FORMS OF ENERGY AND ENERGY TRANSFORMATION

Specific Objectives

After completing this chapter, you will be able to:

- Distinguish between various forms of energy and modes of transformation.
- Discuss the conservation of energy and efficiency of energy conversion.

INTRODUCTION

Energy is defined as the ability to do work.

In other words, for a person to do a simple task such as raising the hand, nodding the head and flipping the pages of a book, they must possess energy.

Different activities require different forms and amounts of energy. For example, the energy required to lift a load of 50 kg would definitely be greater than the energy that would be needed to raise a 5 kg load. Animals, including humans, obtain energy from food. When we eat food, energy is stored in our bodies for immediate and future usage. Energy is necessary for some basic life activities such as keeping the body warm, heartbeat, making decisions, etc.

When work is done, energy is never used up, but is converted from one form to another.

The SI unit of energy is ***joules (J).***

FORMS / TYPES OF ENERGY

Energy exists in many different forms. Some forms of energy are:

Light energy

This is the type of energy that makes vision possible. E.g. the Sun, fluorescent tube, etc.

Chemical energy

This is the type of energy gained when atoms or molecules of a substance rearrange to form a new substance. e.g. battery, food, fossil fuel, etc.

Electrical energy
This type of energy is acquired when electrical charges move through a conductor. Electrical energy can be obtained from solar panel, dry cells, generators etc.

Solar energy
This is the type of energy obtained from the sun. The sun is the major source of energy needed for most life activities.

Heat energy
This is the energy obtained as a result of the rise in temperature. E.g. sun, fire, electric iron etc.

Nuclear energy
This is the type of energy released during the splitting or fusing of atomic nuclei. E.g. Hydrogen, Uranium, radium, etc.

Sound energy
This is the energy produced when objects vibrate. E.g. bell, drums, etc.

Mechanical energy
This is the energy possessed by a body due to its position, state or motion. There are two types of mechanical energy:

- ❖ Potential energy
- ❖ Kinetic energy

Potential energy (P.E.): This is the energy a body has as a result of its position or state. Examples of potential energy include:

i. A cat lying on a mat
ii. A piece of cloth hanging on line
iii. A girl sitting on a chair

The formula for potential energy is given as

P.E. = mgh

where m = mass, g = acceleration due to gravity and h = height of the body

Example

A mango fruit of mass 0.06 kg hangs 4.0 m above the ground. Calculate its potential energy.
[g = 10 ms^{-2}]

Solution

P.E. = mgh
m = 0.06 kg, h = 4 m, g = 10 ms^{-2}
P.E. = 0.06 x 10 x 4
P.E. = 2.4 J

Kinetic energy (K.E.)

This is the energy possessed by a body due to its motion.
Examples of K.E. include:

i. A boy riding a bicycle
ii. A fan whirling around
iii. A car speeding

Kinetic energy is expressed as:

$$K.E. = \frac{1}{2}mv^2$$

where m = mass and v = velocity

Example

A particle of mass 20 g moves with a velocity of 3 ms^{-1}. What is its kinetic energy?

Solution

$K.E. = \frac{1}{2}mv^2$

$m = 20\ g = \frac{20}{1000} = 0.02\ kg,\ v = 3\ ms^{-1}$

$K.E. = \frac{1}{2} \times 0.02 \times 3^2$

$K.E. = 0.09\ J$

ENERGY TRANSFORMATION

Energy transformation is the process of changing energy from one form to another. This process is happening all the time, both in the world and within people. When people consume food, the body utilizes the chemical energy in the bonds of the food and transforms it into mechanical energy, a new form of chemical energy, or thermal energy. The following show various energy transformations.

1. Photosynthesis

SOLAR ENERGY
↓
CHEMICAL ENERGY

2. Voltaic cells

CHEMICAL ENERGY
↓
ELECTRICAL ENERGY
↓
MECHANICAL ENERGY
↓
HEAT ENERGY

3. Solar cells

SOLAR ENERGY
↓
ELECTRICAL ENERGY
↓
MECHANICAL ENERGY

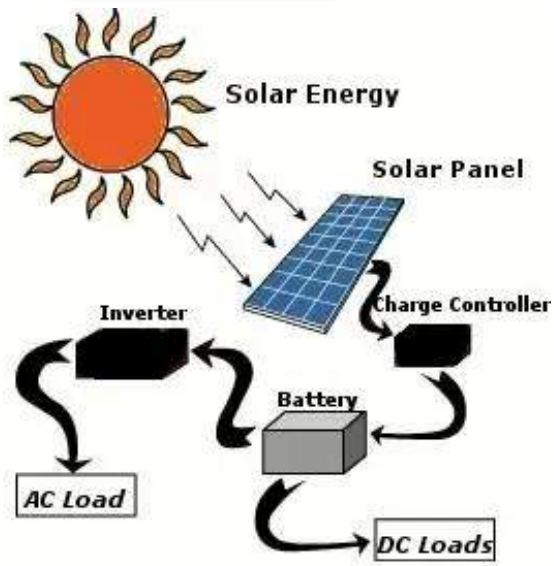


Fig. 11.9: Converting solar energy into other forms of energy

4. Electric bulb

CHEMICAL ENERGY

↓

ELECTRICAL ENERGY

↓

HEAT ENERGY

5. Fossil fuel

CHEMICAL ENERGY

↓

THERMAL ENERGY

↓

ELECTRICAL ENERGY

6. Falling object

CHEMICAL ENERGY

↓

ELECTRICAL ENERGY

↓

POTENCIAL ENERGY

7. Cellular respiration

CHEMICAL ENERGY

↓

ELECTRICAL ENERGY

↓

POTENCILA ENERGY

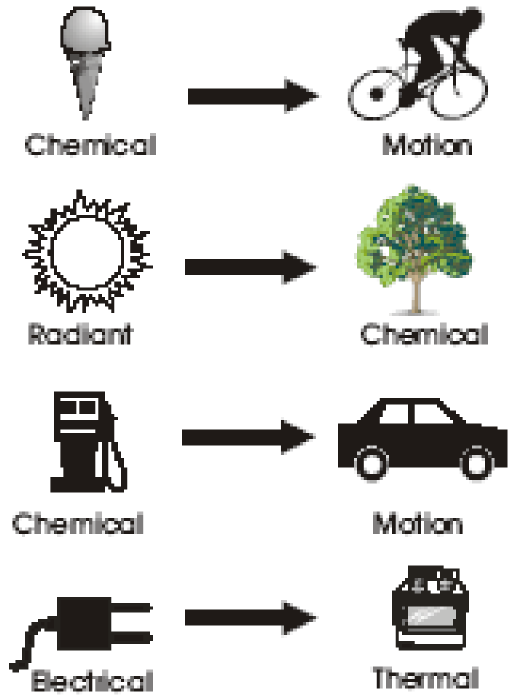


Fig. 12.0: Examples of energy transformation

CONSERVATION OF ENERGY

The law of conservation of energy states that energy cannot be made or lost, but can be changed from one form to another.

How to conserve energy

1. Switch off all electrical appliances that are not in use.
2. Do not leave doors of fridges and freezers open.
3. Do not put hot food in fridges and freezers.
4. Turn off and remove mobile phone chargers from the power outlets after charging mobile phones

5. Iron clothes in bulk.
6. Close all doors and windows when using air-conditioners.
7. Use energy saving lamps.
8. Open doors and windows to allow in fresh air instead of electric fans and air-conditioners.
9. Turn off light during the (bright and sunny) day.
10. Have your electric metre earthed.

EFFICIENCY OF ENERGY

Efficiency of energy is the ratio of energy output to the energy input.

Efficiency is expressed in percentage.
In other words, it is the ratio of energy put into a work to the energy obtained from it.

Mathematically,

$$\text{Efficiency} = \frac{energy\ \ output}{\text{energy input}} \times 100\%$$

Efficiency of energy is never 100%. This is because parts of the energy put into work are used to overcome friction, inertia and gravity. This makes the energy output always less than the energy input.

Sample question:
A machine was supplied with 90 J of energy. It could do 75 J of work. Calculate its efficiency.

Solution:
Energy output = 75 J, energy input = 90 J

$$\text{Efficiency} = \frac{\text{energy output}}{\text{energy input}} \times 100\%$$

$$= \frac{75}{90} \times 100 \quad = 83\%$$

TEST QUESTIONS

1. (a) State and explain any three forms of energy.
   (b) Explain the concept of energy transformation.

2. State the energy transformation that takes place in the following:
   a. Photosynthesis
   b. When an orange drops from a tree.
   c. Cellular respiration.

3. (a) (i) State the law of conservation of energy.
   (ii) state five methods of conserving energy in the house.
   (b) (i) explain the term efficiency of energy.
   (ii) Why is the efficiency of energy always less than 100% in machines.

4. (a) Mention five ways of conserving energy.
   (b) With a aid of a flow chart,

demonstrate how chemical energy in voltaic cell is converted to heat energy.

5. (a) Explain why efficiency is never 100%.

   (b) Find the efficiency of a simple machine that was supplied with 1800 J of energy and could do 1200 J of work.

16

SOLAR ENERGY

Specific Objectives

After completing this chapter, you will be able to:

- ❖ Identify the uses of solar energy.
- ❖ Document the uses of solar energy in daily activities.

INTRODUCTION

Solar energy is the radiation produced (by nuclear fusion reactions) deep in the sun's core.

The sun provides almost all the heat and light the Earth receives and therefore sustains every living being.

USES OF SOLAR ENERGY

The possible uses of solar energy fall into three categories: photochemical processes, photoelectric processes and thermal processes.

Photochemical processes

Photochemical processes are those in which light energy causes a chemical process.

Photoelectric processes

Photoelectric processes involve a direct conversion of radiation to electrical energy.

Thermal processes

In thermal processes, the radiant energy is absorbed as heat by a receiver or receiving substance, which then undergoes an increase in temperature, vaporization, or other heat-absorbing process.

The most commonly considered uses of solar energy are those which are classed as thermal processes. They include

1. house-heating,
2. distillation of seawater to produce potable water,
3. refrigeration and air conditioning, power production by solar-generated steam,
4. cooking,
5. water heating,

6. the use of solar furnaces to produce high temperatures for experimental studies.

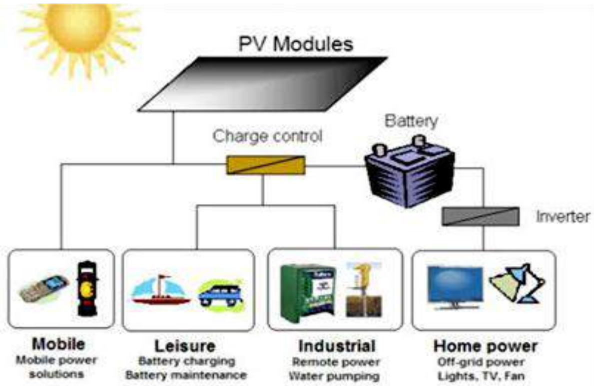


Fig. 12.1: Some everyday uses of solar energy

Power/ Electricity Generation

The production of mechanical or electrical energy from solar energy has been the object of numerous studies. Photoelectric cells are used in calculator, flashlight, radios and other devices to convert solar energy into electricity.

Photovoltaic cells are a means of directly converting radiation from the sun into electricity. These cells, made from thin slices of semiconductor materials, appear to produce significant amounts of energy at little cost.

Fig. 12.2: Photovoltaic cells

House Heating

Solar house heating has received attention, especially in our modern era. In it application, heat must be collected during sunny periods and stored for use in non-sunny periods. Solar house heating is very common in the template zone, and is very effective during the winter season. The following diagram illustrates how solar house heating is done.

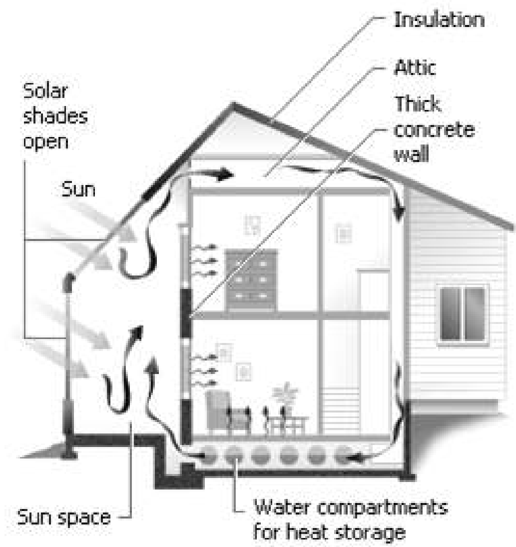


Fig. 12.3: Solar-house heating

Distillation of Salt Water

Solar distillation of sea or brackish waters to produce potable water has been the object of much research. The most common type of experimental solar still utilizes a shallow pan of water covered with a glass roof in the form of an inverted V. Solar energy heats the water, causing it to evaporate; the vapour condenses on the underside of the inclined glass covers and run down into collecting troughs at their lower edges.

Solar Furnaces

These devices are accurate parabolic reflectors which are mounted so that they precisely track the sun and focus solar radiation on small areas. Thus, they concentrate the usually low-energy flux as much as 50,000 times. With them, targets of appropriate design can be heated to high temperatures, up to approximately 4426°C. Several furnaces have been used for laboratory studies of the properties of metals and ceramics and of various chemical reactions.

The largest of these is a paraboloid which is 34 feet in diameter and utilizing a movable flat mirror to reflect radiation into the paraboloid. It has been called a semi-industrial tool by its designer and builder, F. Trombe of the Mont-Louis Laboratory in France.

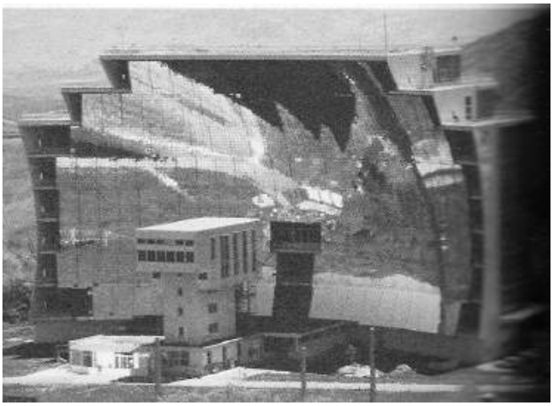


Fig. 12.4: Solar furnace

Solar Cookers

One of the fascinating uses of solar energy is solar cookers. With solar cooker, you can cook food without lighting fire.

You can construct your own solar cooker.

- Get an aluminium sheet, and shape it into a parabola (a dome-like shape).
- This causes the sun's rays to be concentrated on the saucepan.
- The concentration of the sun's rays on the saucepan causes it to heat-up, thereby heating it contents.

Water Heater

Flat plate collectors utilize the sun's energy to warm a carrier fluid, which in turn provides usable heat to a household. The carrier fluid, which in this case is water, flows through copper tubing in the solar collector, and in the process absorbs some of the sun's energy.

Next, the carrier fluid moves to the heat exchange, where the carrier fluid warms water that is used by the household. Finally, a pump moves the carrier fluid back to the solar collector to repeat the cycle.

Applications of solar energy in practical daily activities

1. Drying clothes
2. Cooking food using solar cookers
3. Heating water for bathing
4. Boiling egg
5. Drying crops (for preservation)

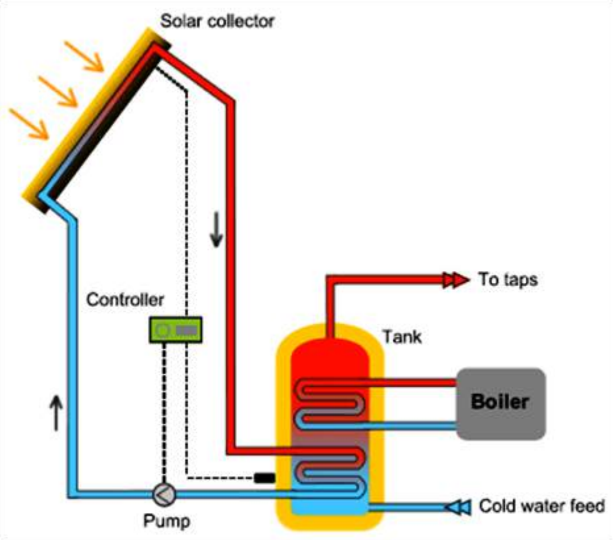


Fig. 12.5: *Solar-water heating system*

Importance of solar energy

1. Solar energy is renewable; does not get finished like fossil fuels.
2. It emits no or relatively small pollutants.
3. There is absence of incessant power outages and low shedding.
4. It saves money. After the initial cost of the installation, there is no or little cost for maintenance.
5. No additional fuel is needed to produce the energy.
6. Usage of the solar energy contributes in reducing the threats to the health costs.
7. This source is very environmentally friendly and does not aggravate the global warming as compared to fossil fuels.
8. Solar panels operate silently and do not produce noise pollution and unbearable smells.
9. Solar energy can be produced on or off the grid. (i.e. either controlled by a power company or not).
10. Because solar does not rely on constantly mining raw materials, it does not result in the destruction of forests and eco-systems that occurs with most fossil fuel operations.

Fig. 12.6: Percentage use of solar energy by household appliances

Disadvantages of solar energy

1. Initial costs for installing a home solar system or building a solar farm are high.
2. Solar panels are bulky and may be difficult to transport.

3. Requires constant sunlight and the ability to store energy for the night.
4. Areas which remains mostly cloudy and foggy will produce electricity but at a reduced rate and may require more panels to generate enough electricity for your home.
5. Houses which are covered by trees, landscapes or other buildings may not be suitable enough to produce solar power.

TEST QUESTIONS

1. (a) Define the term solar energy.
   (b) List four daily applications of solar energy.

2. Describe four uses of solar energy.

3. Explain how solar energy may be used for the following purposes:
   (a) Solar furnaces;
   (b) Electricity production;
   (c) Water heating system.

4. Describe the processes into which solar energy is categorized and harnessed.

5. (a) State five advantages of solar energy over other forms of energy.
   (b) List four short comings of solar energy.

6. Discuss how solar energy can be made accessible to all Ghanaians.

7. Discuss the environmental impact of solar energy.

17

PHOTOSYNTHESIS

Specific Objectives

After completing this chapter, you will be able to:

- Describe the process of photosynthesis.
- Explain the process of transformation of energy that occurs during photosynthesis

INTRODUCTION – THE PROCESS OF PHOTOSYNTHESIS

Photosynthesis is the process whereby green plants prepare their own food with carbon dioxide and water, in the presence of sunlight, which is absorbed by chlorophyll.

Photosynthesis provides the basic energy source for almost all organisms. An extremely important by-product of photosynthesis is oxygen, on which most living organisms depend.

The process of photosynthesis can be expressed as:

Carbon dioxide + water = glucose + oxygen

OR

$$6CO_2 + 6H_20 \rightarrow C_6H_{12}O_6 + 6O_2$$

Water is obtained from the soil, carbon dioxide from the atmosphere and the sunlight is attracted by chlorophyll which is produced by the chloroplast in the leaves.

CONDITIONS FOR PHOTOSYNTHESIS

The conditions necessary for photosynthesis are:

Carbon dioxide

Plants make use of the carbon dioxide in the atmosphere. The atmosphere contains only 0.04% carbon dioxide.

Water

The root hairs of plants absorb water, which is then transported to the stems and leaves through the xylem vessels.

Chlorophyll

This is a green pigment that gives leaves their green colour. Chlorophyll is produce by chloroplast and is responsible for attracting sunlight onto the leaves. Chloroplast contains other enzymes that help produce sugar which is converted into starch.

Light

Sunlight is a very essential source of energy. Artificial light is also effective in photosynthesis, but it is not as productive as sunlight. Here light or solar energy is converted to chemical energy.

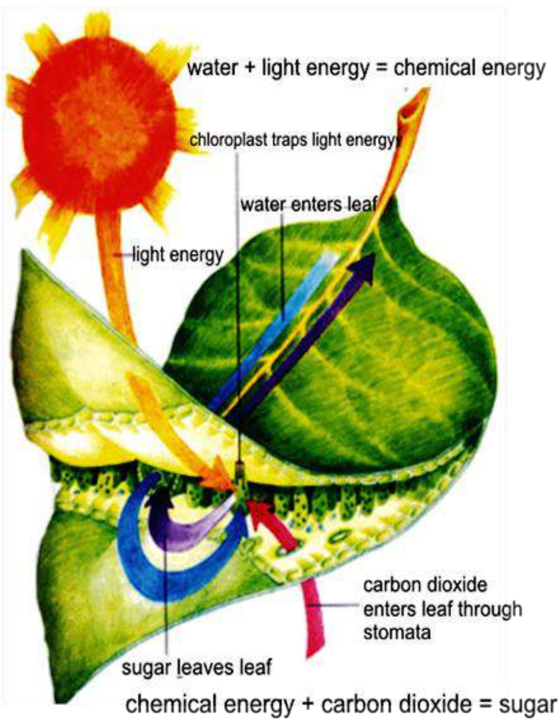


Fig. 12.7

Energy Transformation in Photosynthesis

In photosynthesis, light energy (either from the sun or artificial) is converted to chemical energy in food.

Photosynthesis

SOLAR ENERGY

↓

CHEMICAL ENERGY

Testing for starch in a green leaf

- Label two potted plants, **A** and **B** and put them in the dark for a few days. This will cause them to use all the starch they have stored in their leaves.
- Put plant **A** in the light and leave plant **B** in the dark for a few days.
- Take a leaf from each plant and run a test for starch on them.
- Dip each leaf in boiling water for one minute to kill and soften it.
- Dip each leaf in hot ethanol for one minutes to remove the green pigment
- Wash the leaves in hot water
- Add dilute iodine solution to each leaf.

Observation

It would be observed that leaf **A** would turn dark blue, which shows the presence of starch, while leaf **B** would turn brown, which shows the absence of starch because it was kept in the dark.

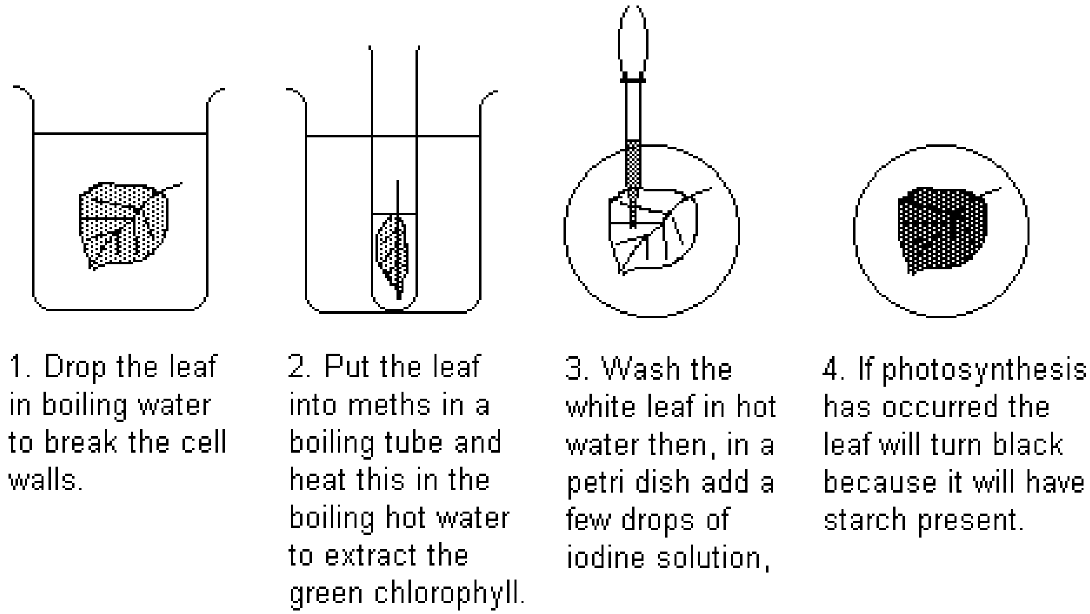


Fig. 12.8: Test for starch in leaf

An experiment to show the necessity of light for photosynthesis

- Cut a shape out from a piece of aluminium foil, making a stencil.
- Attach it to a de-starched leaf.
- After 4 to 6 hours of daylight, detach the leaf and test for starch.

Observation

Only areas which received sunlight turn blue with iodine.

Conclusion

Since starch has not accumulated in the areas without light, it is evident that light plays an important role in photosynthesis.

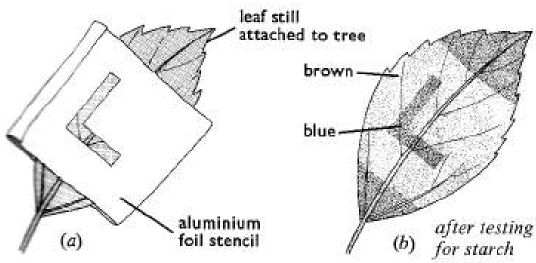


Fig. 12.9: The need for light in photosynthesis

Experiment to show that carbon dioxide is essential for photosynthesis

- Water two de-starched potted plants.
- Enclose the shoots in polythene bags, one of which contains soda-lime to absorb carbon dioxide from the air and the other sodium hydrogen-carbonate

(sodium trioxocarbonate (IV)) solution to produce extra oxygen.

- Place both plants in sunlight or fluorescent light for several hours.
- Detach a leaf from each plant and test for starch.

Observation

The leaf deprived of carbon dioxide will not turn blue while that enriched with carbon dioxide will turn blue.

Conclusion

Since no starch is made in the leaf deprived of carbon dioxide suggests that carbon dioxide is necessary for photosynthesis.

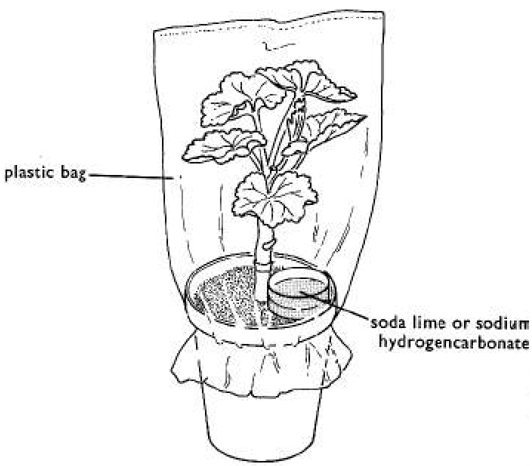


Fig. 13.0: Experiment showing that CO_2 is necessary for photosynthesis

Experiment to show that oxygen is produced during photosynthesis

- Place a short-stemmed funnel over some plant shoots (e.g. *ceratophyllum*) in a beaker of water.
- Invert a test tube filled with water over the funnel stem.
- Raise the funnel above the bottom of the beaker to allow for free circulation of water.
- Place the set-up in sunlight. Bubbles of gas appear from the cut stems and is collected in the test-tube.
- Remove the test-tube and introduce a glowing splint.

Observation

The glowing splint bursts into flames.

Conclusion

It could be deduced that the plant has given off a gas which is considerably richer in oxygen than in atmospheric air.

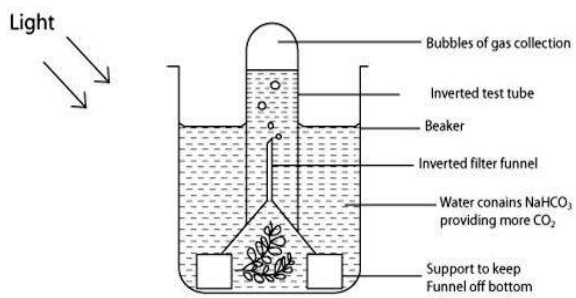


Fig. 13.1: Oxygen is produced during photosynthesis

A controlled experiment should be setup in a similar manner but the plant should be placed in a dark cupboard.

It would be observed that little or no gas should collect.

Experiment to show that chlorophyll is necessary for photosynthesis

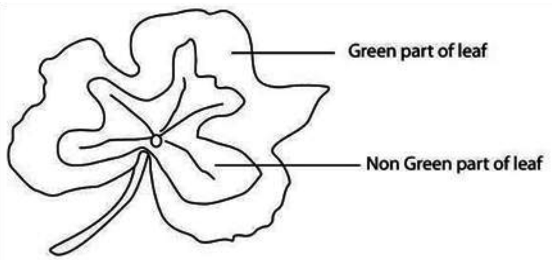
- ❖ Leave a plant with variegated leaves (leaves with diverse colours) in light for 2-6 hours.
- ❖ Remove one leaf and make a labelled diagram to show the green and non-green parts.
- ❖ Test the leaf for starch.

Observation

When the variegated leaf is tested for starch, the green parts (which contain chlorophyll) turn blue-black while the non-green parts (which lack chlorophyll) show no colour change.

Conclusion

Starch is made only in areas of the leaf with chlorophyll, showing that chlorophyll is necessary for photosynthesis.



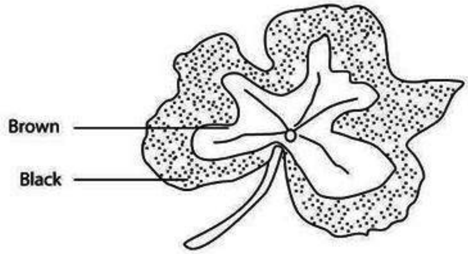


Fig. 13.2: Presence of chlorophyll in green leaf

Adaptation of Leaves for Photosynthesis

1. Leaves have broad and flat shapes which provide them with large surface area for absorption of carbon dioxide.
2. The thin nature of leaves enables carbon dioxide to diffuse easily into the mesophyll cells.
3. The presence of stomata helps in gas exchange.
4. Chloroplast secrets chlorophyll which attracts light into the leaves.
5. Phloem and xylem tissues transport manufactured food from the leaves to other parts of the plants and minerals salt from the roots to the leaves.
6. Leaves are arranged in a regular pattern along the stem which prevents overshadowing and allows each leaf to receive light.

Factors which affect the rate of photosynthesis

When plenty of carbon dioxide, sunlight and water are provided to a plant, photosynthesis will be at its maximum rate.

Sometimes the rate of photosynthesis is not as high as expected due to inadequacy of factors that include light intensity, water, temperature and carbon dioxide concentration.

Providing plenty of each of these factors to a plant increases the rate of photosynthesis. However, if one of these factors is not adequate, the rate of photosynthesis may become low. A factor that is inadequate is called a ***limiting factor***.

Factors which affect the rate of photosynthesis include:

1. **Light intensity**: In case of dim light, the rate of photosynthesis is low. As light intensity increases, the rate of photosynthesis increases.

2. **Carbon dioxide concentration**: The more carbon dioxide a plant is given, the faster is the rate of photosynthesis, until a maximum is reached.

3. **Temperature**: Photosynthesis is an enzyme-controlled reaction. Increase in temperature increases the rate of photosynthesis while a decrease lowers it.

4. **Stomatal opening** and **closing:** This **r**egulates the amount of carbon dioxide entering a plant which it uses for photosynthesis. If the stomata are closed then photosynthesis cannot take place and when they are open, carbon dioxide enters and the rate of photosynthesis increases.

5. **Mineral salt:** Several elements are known for normal plant growth and development. Some of these elements are Carbon, Hydrogen and Oxygen.

6. Other elements are Nitrogen, Sulphur, Phosphorus, Potassium, Calcium and Magnesium. There are other elements called ***trace elements*** which are needed in small quantities. Plants obtain mineral salts from the soil. The effects of these chemical elements can be discovered by growing plants in water solutions containing balanced amounts of salts necessary for healthy plant growth. This solution is called a ***culture solution*.**

Importance of Photosynthesis

1. It provides food for some living organisms such as herbivores and omnivores (carnivores do not eat plants).
2. Oxygen given off by green plants during photosynthesis is used by animals for respiration.
3. Green plants take in carbon dioxide during photosynthesis, thereby reducing the amount of carbon dioxide in the atmosphere.

4. Photosynthesis produces materials for protein and starch formation.
5. It helps to maintain the oxygen, carbon and water cycle in the atmosphere.

TEST QUESTIONS

1. (a) (i) What is photosynthesis?
   (ii) describe the conditions necessary for photosynthesis.
   (b) (i) state four adaptations of leaves for photosynthesis.
   (ii) mention four importance of photosynthesis.

2. (a) Describe how you will test for starch in a leaf.
   (b) Explain four factors which affect the rate of photosynthesis in a leaf.

3. Describe an experiment to demonstrate the need for the following conditions of photosynthesis:
   (a) Sunlight
   (b) Carbon dioxide
   (c) Chlorophyll

4. (a) (i) Define the term photosynthesis
   (ii) Give three reasons why photosynthesis is important in all living organisms.
   (b) List four structural adaptation of leaf for photosynthesis and state how each of them facilitate the process.

5. (a) (i) Write the chemical equation for photosynthesis.
   (ii) What are the conditions necessary for photosynthesis?
   (iii) List four importance of photosynthesis.

18

ELECTRONICS I

Specific Objectives

After completing this chapter, you will be able to:

- Observe the behaviour of discrete electronic components in a d.c. and a.c. electronic circuit.

INTRODUCTION

Electronics is the field of engineering and applied physics dealing with the design and application of devices, usually electronic circuits, the operation of which depends on the flow of electrons for the generation, transmission, reception, and storage of information.

The information can consist of voice or music (audio signals) in a radio receiver, a picture on a television screen, or numbers and other data in a computer.

BEHAVIOR OF DISCRETE ELECTRONIC COMPONENTS

Electronic components are specially designed electronic elements which have leads with which they are connected to the circuit board.

Electronic circuits consist of inter-connections of electronic components. Components are classified into two categories—active or passive.

Passive components are components that never supply more energy than they absorb. Examples are resistors, capacitors and inductors.

Active components are components that can supply more energy than they absorb. Examples are batteries, generators, vacuum tubes, and transistors.

Resistors

Resistors are made from carbon mixtures, metal films, or resistance wire and have two connecting wires attached. They are used for current control in electronic circuit.

Variable resistors, with an adjustable sliding contact arm, are often used to control volume on radios and television sets.

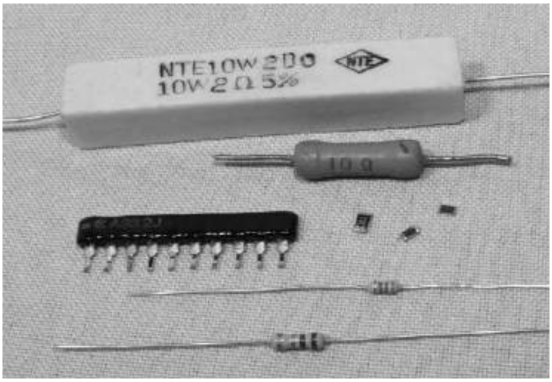


Fig. 13.3: Some types of resistors

Capacitors

Capacitors consist of two metal plates that are separated by an insulating material. If a battery is connected to both plates, an electric charge will flow for a short time and accumulate on each plate. If the battery is disconnected, the capacitor retains the charge and the voltage associated with it. Rapidly changing voltages, such as caused by an audio or radio signal, produce larger current flows to and from the plates; the capacitor then functions as a conductor for the changing current.

This effect can be used, for example, to separate an audio or radio signal from a direct current in order to connect the output of one amplifier stage to the input of the next amplifier stage.

Fig. 13.4: Some types of capacitors

Inductors

Inductors consist of a conducting wire wound into the form of a coil. When a current passes through the coil, a magnetic field is set up around it that tends to oppose rapid changes in current intensity.

As a capacitor, an inductor can be used to distinguish between rapidly and slowly changing signals. When an inductor is used in conjunction with a capacitor, the voltage in the inductor reaches a maximal value for a specific frequency.

This principle is used in a radio receiver, where a specific frequency is selected by a variable capacitor.

Fig. 13.5: Inductors

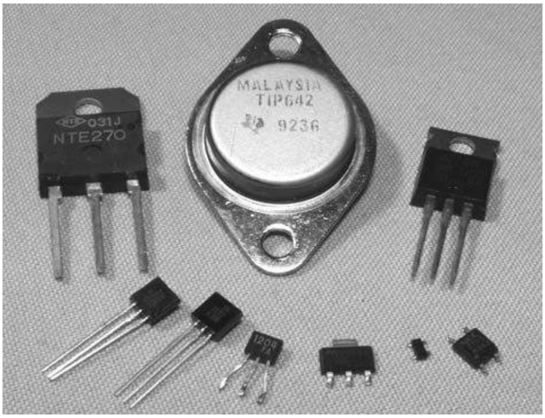
Transistors

Transistors are made from semiconductors. These are materials, such as silicon or germanium that are *doped* (have minute amounts of foreign elements added) so that either an abundance or a lack of free electrons exists. In the former case, the semiconductor is called *n-type*, and in the latter case, *p-type.*

By combining n-type and p-type materials, a diode can be produced. When this diode is connected to a battery so that the p-type material is positive and the n-type negative, electrons are repelled from the negative battery terminal and pass unimpeded to the p-region, which lacks electrons.

With the battery reversed, the electrons arriving in the p-material can pass only with difficulty to the n-material, which is already filled with free electrons, and the current is almost zero.

Fig. 13.7: Some discrete electronics components in a circuit



Transistors

Light Emitting Diode (LED)

Light emitting diodes (LEDs) are discrete electronic components which convert electrical signals into light.

They usually have low resistances and easily damaged if connected to a high voltage. A voltage of 1.5 can light an LED. LEDs come in diverse colours and are normally used as indicators.

An LED has two terminals – positive and negative terminals. In a circuit, the positive terminal connects the positive hand of the cell, while the negative terminal takes the negative hand of the cell. If the terminals are reversed there will be no light.

ACTIVITY:

- Build a circuit that emits a light through LED
- Build a circuit that flashes using just LED and RC components
- Build a circuit that produces a sound through a buzzer (door bell)
- Build a liitle device that will make a sound with variable frequency

Behaviour of Light Emitting Diode (LED) in circuits

Needed items

- 1 LED (any colour)
- 1 capacitor 1uF
- 1 PNP resistor
- 1 NPN resistor
- 5 resistors (1 M OTHER VALUES)
- LED circuits

To observe light emitting diode in action we shall build a very basic electric circuit that will show the use of LEDs. First build a circuit that is shown below:

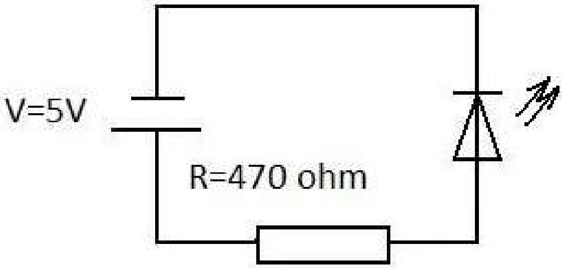


Fig. 13.8

Once connected, the LED emits photons that can be detected by the eye.

Let's build another circuit to explore the properties of LED.

To observe the properties of a diode let us connect it to 3 V battery or DC signal generator. You will notice that only one of the LEDs emits light

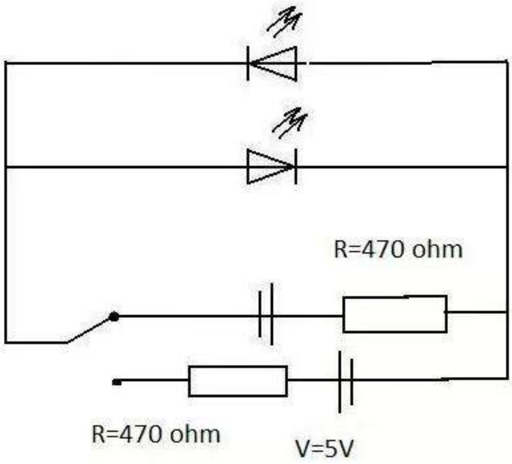


Fig. 13.9

Now let us reverse the polarity of the battery or signal generator. You probably have noticed that now the LED that was emitting light before is now switched off, while the other one is on.

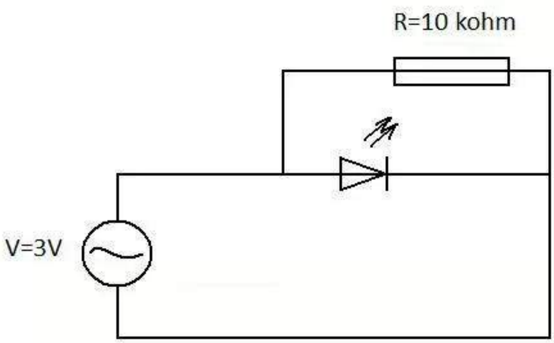


Fig. 14.0

The circuit above is connected to a low frequency AC signal generator.

(NOTE: the battery is not suitable as it produces DC voltage). If the frequency is low enough LED should be flashing. If the frequency is increased the flash rate will increase too. Now try to generate a signal with a frequency of FREQUENCY Hz. It appears that LEDs is lit up constantly, but that is not exactly correct. LED is still flashing, just the frequency is too high for humans eye to detect this flash.

Capacitor in LED circuit

Now we will introduce a capacitor to show the effect of delayed flash. It is known that capacitor can store a charge which can be released through resistor.

Build the following circuit to observe its behaviour. (Tip: switch circuit on and off and observe the behaviour of LED)

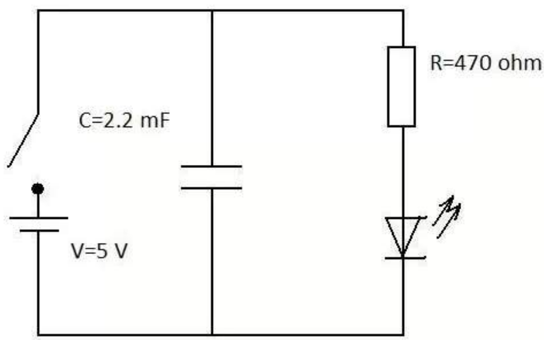


Fig. 14.1

From the circuit above it can be seen that when the switch is on it charges the capacitor as well as emits light through the LED. However, when the switch is off, the LED does not die straight away. It still keeps emitting light after some time after the circuit was unplugged. Now let us change the resistor with the one that has R= OHMS Ohms. Switch circuit on for a few seconds and then switch it off.

Notice the longer time it takes the LED to die as compared to using resistor with smaller resistance. The water analogy would be straight forward. Imagine having a tank full of water with a plug that would have some sort of mechanical resistance. It is quite obvious that the bigger resistance the more time required for the tank to empty.

Transistors in LED circuit

For the next circuit you are require to familiarise yourself with the basic principles of transistors. The transistors we will be using are bipolar PNP and NPN and the main purpose for it just to act as an electronic switch.

An example with flashing LED required AC voltage with variable frequency in order to observe these flashes. Using a combination of transistor and capacitor we can build a circuit that will make LED flash even using a direct current (DC). Build the circuit that is shown below.
Once this circuit is connected to a battery the LED flashes. The rate can be defined by changing resistances as well as capacitance of capacitor.

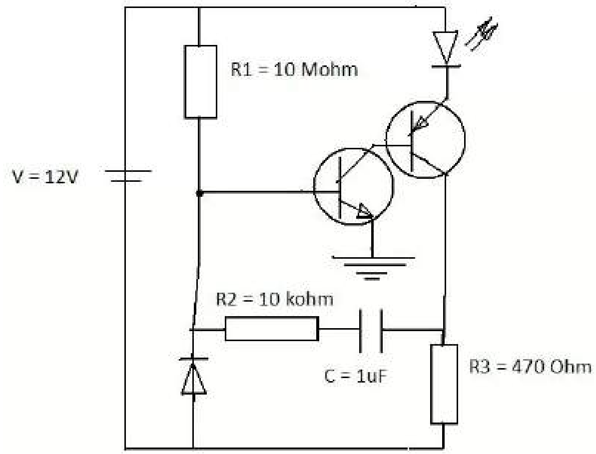


Fig. 14.2

Building a door bell

Now let's play with the sound. You will require a buzzer and a push switch. Try to connect the circuit below.

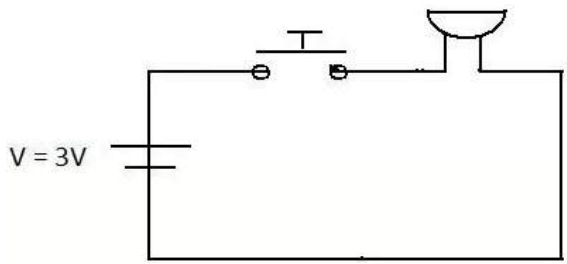


Fig. 14.3

Combining LED and buzzer in one circuit can be very fun. Try to build the following circuit and explore its behaviour.

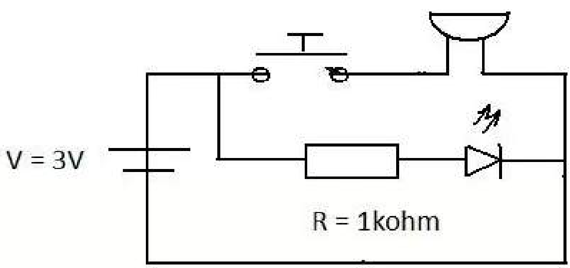


Fig. 14.4

TEST QUESTIONS

1. (a) Distinguish between active and passive components.
   (b) Describe the behaviour of an LED in circuit.

2. Write a short note on the following discrete components:
   (a) Resistor;
   (b) Capacitor;
   (c) Transistor
   (d) Inductor

3. (a) Describe two uses of light emitting diodes.
   (b) Compare the behaviour of the following pairs of components in a circuit.
   (i) capacitor and inductor
   (ii) transistor and resistor

4. Classify the following components as either active or passive in the table below and state one use each:
   accumulator; capacitor; resistor; dry cell; vacuum tube; inductor; diode; transistor; LED.

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ECOSYSTEM

Specific Objectives

After completing this chapter, you will be able to:

- Define basic ecological terms.
- Classify ecosystems and identify their components.
- Explain food chain and food web and identify the components.

INTRODUCTION

An ecosystem is made up of living and non-living organisms in a particular habitat and their interaction with the physical environment. The habitat could be a forest, coral reef, river, etc.

BASIC ECOLOGICAL TERMS

Ecology

Ecology is the study of the relationship among living things and to their surroundings.

Species

Species is a distinct kind of organism, with a characteristic shape, size, behaviour, and habitat that remains constant from year to year. Organisms of one species mate and produce offspring with one another, but do not breed with other populations.

Population

This term refers to all organisms of the same species living in a particular habitat. E.g., a tree could house a population of birds.

Community

A community consists of all the population of all the different species in a habitat. Thus, a community is made up of different species of organisms living together. For example, the ocean could be a community for different species of fish, crabs, coral, etc.

Ecosphere

Ecosphere is the whole of the earth's atmosphere, rivers, lakes and sea that is inhabited by living organism. The word ecosphere is interchangeable with biosphere. ***Biosphere*** can be divided into ***biomes***.

Biomes
This is a large natural area which has a particular climate. Examples are tropical rain forest, grassland, desert, swamp etc.

Habitat
A habitat is the natural environment of an animal or plant. Different organisms have different habitats in which they adapt. An organism has to have some basic features in order to fit into a particular habitat. Habitats are categorized as terrestrial (land and forest) and aquatic (freshwater and marine)

Niche
A niche is an area within an ecosystem in which a particular group of organisms live in. A niche could provide a habitat for organisms of the same or different species. For example, a pond can be a home for different species of fish.

Parasitism
Parasitism is the relationship between two organism in which one (called ***parasite***) lives on and causes harm to the other (called ***host***). For example a flee (parasite) on a dog (host)

Communalism
This is a situation where one organism lives on another but does not harm or benefit from it.

TYPES OF ECOSYSTEM

Ecosystem can be classified into two main types – terrestrial and aquatic.

Terrestrial (land) ecosystem
There are different types of terrestrial ecosystem or habitat, according to the vegetation present. For example, in forests there are large trees whiles deserts contain little vegetation. The type of vegetation is also dependent on physical factors such as temperature, the amount of rainfall and the type of soil. Examples of terrestrial habitat are rain forest, desert and savannah.

Sampling methods in terrestrial ecosystem
Small aquatic animals, such as insects, may be sampled for research. There are a number of ways of sampling animals, including the use of the following:

Pooter

A pooter is made up of a corked-tube or cylinder. The cork has two holes on top for two flexible plastic tubes. One tube leads to the mouth where suction is applied, while the other sucks the animals into the cylinder.

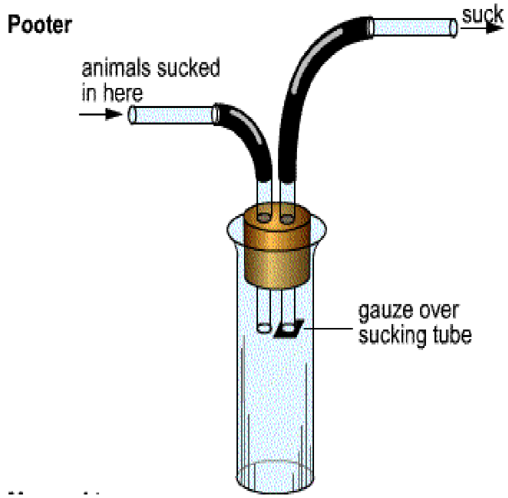


Fig. 14.5: A pooter

Pitfall trap

This is simply a jar or vase dug into the ground and partially covered with a wooden or stone slab or slate so that small animals can fall into the jar. A straight-sided jar is used to prevent the animals from clawing back out.

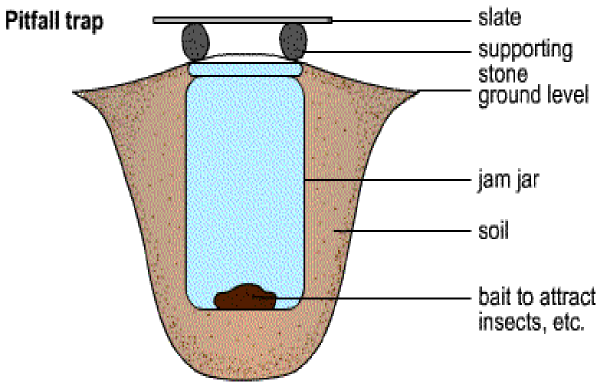


Fig. 14.6: A pitfall trap

Sweep net and butterfly net

These nets generally used to catch flying or winged insects. They are also used to catch or trap clawing insect. They may also be used to sample aquatic organisms.

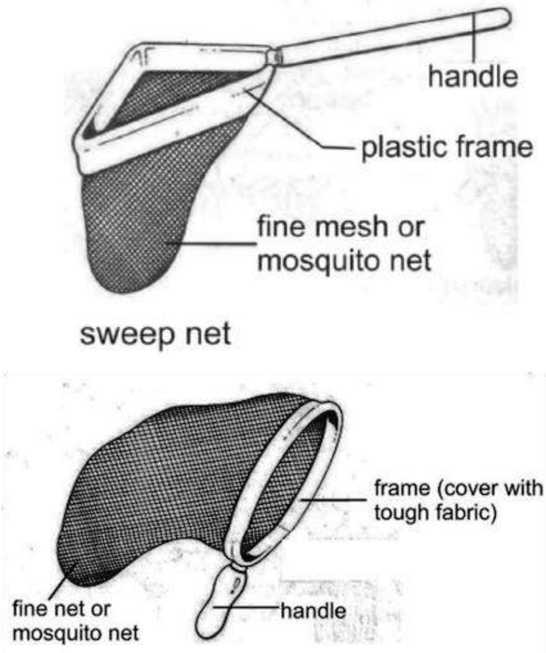


Fig. 14.9: A sweep net and butterfly net

Mammal trap

A mammal trap is used to trap small mammals such as rodents. It usually has a springy door or shut, which shuts with little agitation. A bate is put in the trap to attract animals.

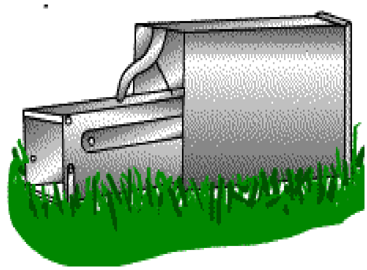


Fig. 14.8: A mammal trap

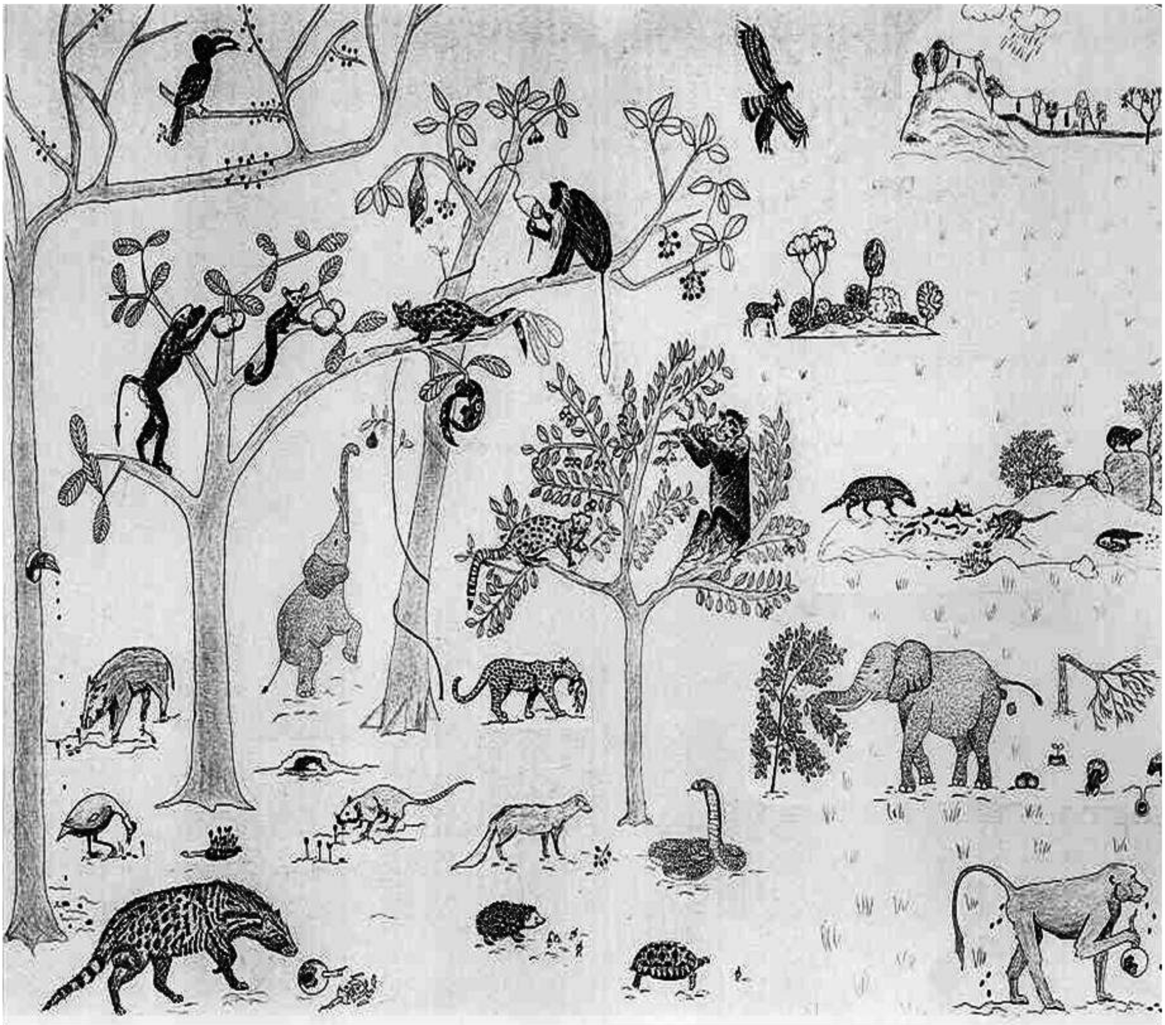


Fig. 15.1: A forest ecosystem

Quadrant

A quadrant is a square or rectangular wooden frame divided into sections with thin wires. It is used to sample vegetation. This is done by randomly and carefully throwing the quadrant around in a habitat. The plants which fall into the quadrant are carefully studied after which the next throw is done.

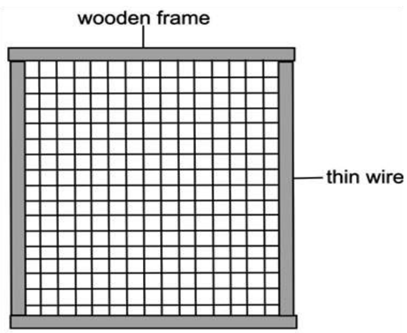


Fig. 15.0: A quadrant

Aquatic (water) ecosystem

There are two types of aquatic ecosystem – fresh water and marine (salt water).

Fresh water ecosystem: This is a watery environment where there is no salt (salt free). Examples are lakes, rivers, ponds, etc.

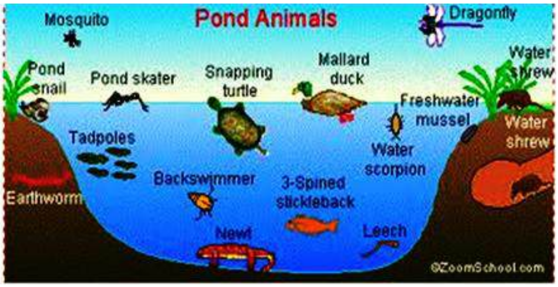


Fig. 15.2: An aquatic habitat

Marine ecosystem: This ecosystem consist of the sea, lagoons, oceans, and estuaries. These habitats contain salty water.

Sampling methods in aquatic ecosystem

***Wicker work trap*:** This is a cylindrical tube made from hard wire or palm branches wove together. It has an enclosure which carries the bait and a one-way opening where the aquatic animals enter. The wicker work trap is normally used to trap shrimps, crabs, lobsters and other small fish. This is done by dropping and leaving the trap in the water for a period of time.

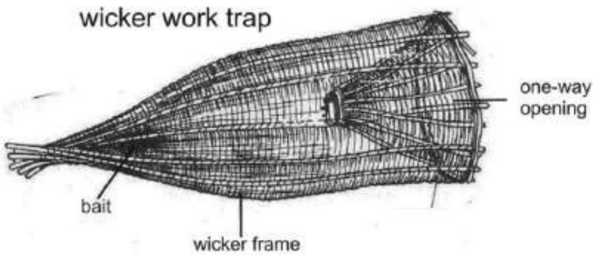


Fig. 15.3:

Plankton net: The plankton net is a tube made from a fine mesh with an enclosed end to capture plankton and other smaller aquatic organisms.

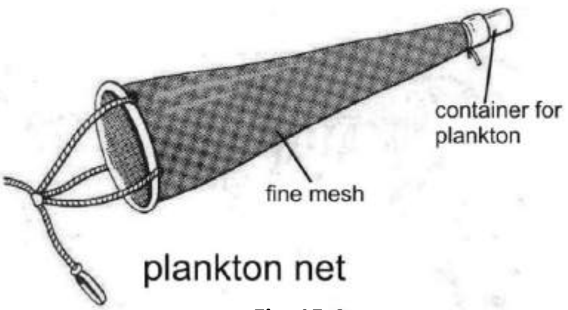


Fig. 15.4

***Long handled net*:** As its name implies, the long handled net is made up of a tough mesh with a long handle which makes it easier to be swept through a pool, lake, etc.

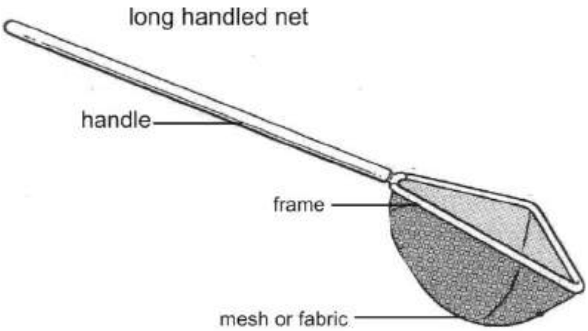


Fig. 15.5

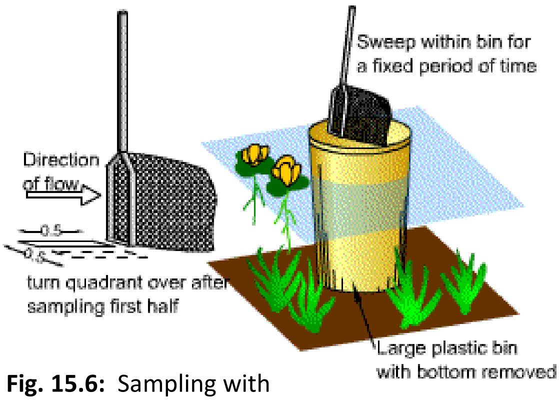


Fig. 15.6: Sampling with Long handled and a sweep net

Grapnel: Submerged plants can be sampled using a grapnel. This is thrown into the water and pulled in again. Population densities of plants cannot be obtained but this method is an indicator of abundance. Line transects can be used for plants on the edge of the water.

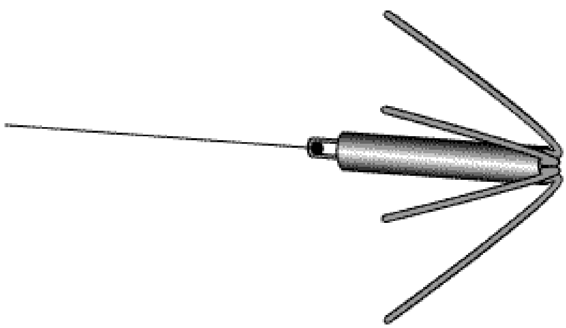


Fig. 15.7: A grapnel

Classifications of ecosystems

Ecosystems can further be classified as natural or artificial.

Natural ecosystem is made up of fresh water, marine, estuaries, lake, rain forest, savannah and desert.

Artificial ecosystem is ecosystem which has been influenced by human activities. Examples of artificial ecosystem include farmland, man-made lakes, roads etc.

Components of ecosystem

Components of ecosystem include both living and non-living things. Hence, we have biotic (living) components which consist of all living organisms, e.g. animals and plants and abiotic (non-living) components which include soil, air and water.

ECOLOGICAL FACTORS

Ecological factors are factors which directly or indirectly affect organisms within an environment. There are two types of ecological factor – biotic and abiotic factors.

Biotic factors

Biotic factors are factors that affect the way living organisms interact in an environment. Biotic factors are also referred to as living factors.

Examples of biotic factors

1. **Predator:** A carnivorous animal that hunts, kills and eats other animals in order to survive, or any other organism that behaves in a similar manner. This process is known as ***predation***.

2. **Symbiosis**: A situation in which two organisms live with and are dependent on each other, to the advantage of both.

3. **Competition:** When a shared resource is in short supply, organisms compete, and those that are more successful survive.

4. **Epiphytism:** This occurs when epiphytes (plants that do not have roots in the soil) attach themselves to other plants for supports. Epiphytes do not take any food from their hosts, and do not harm them.

Examples of useful biotic factors

1. Bacterial fixing nitrogen in the soil.
2. Birds acting as agents of pollination.
3. Fungi and bacteria decomposing organic matter.
4. Animals acting as agents of fruits and seeds dispersal.
5. Macro-organisms aerating the soil.

Harmful biotic factors

1. Herbivores such as goats, sheep and cattle feeding on plants
2. Carnivores (e.g. lions, tigers etc) feeding on other animals
3. Parasites living on their hosts.

Abiotic factors

These are physical factors that affect non-living organisms in an environment. They are called no-living factors.

Examples of abiotic factors

❖ Climate (rainfall, temperature, humidity, etc)
❖ Salinity (measurement of the mass of dissolved solids present in a given amount of water)
❖ Altitude (the height of something above sea level) slope of land etc.

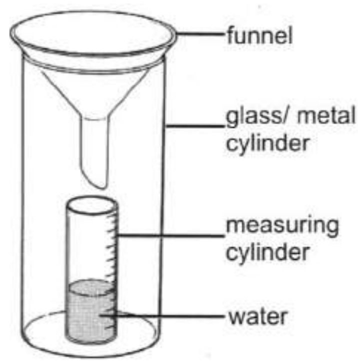


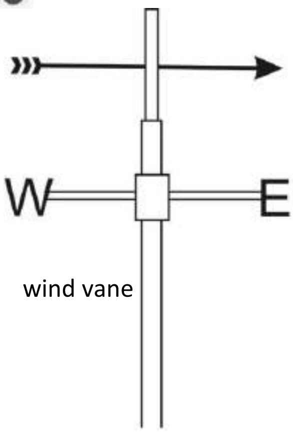
Fig. 15.8
A rain gauge

Table 4.7: Some abiotic factors and the instruments used to measure them

Abiotic factor	Instruments use to measure them
Amount of rain fall	Rain gauge
Temperature	Thermometer
Wind speed	Anemometer
Wind direction	Wind vane
Atmospheric pressure	Barometer
Relative humidity	Hydrometer
Light intensity on land	Photometer
Acidity or alkalinity of a substance	pH indicator

thermometer

aneroid barometer



wind vane

Fig. 15.9: Instruments used to measure abiotic factors

FOOD CHAIN

One feature of organisms within a given habitat is their inter-dependence in their feeding habit. These inter-dependence or interaction is known as ***food chain***.

A food chain is a string of organisms within an ecosystem which are connected through their mode of feeding.

In a food chain, one organism feeds on the one before it. Energy is transferred from one organism to the one which consumes or eats it.

For example plants are eaten by butterflies who are eaten by frogs who are in turn eaten by snakes and then finally, snakes are eaten by eagles. (see fig 16.0 below).

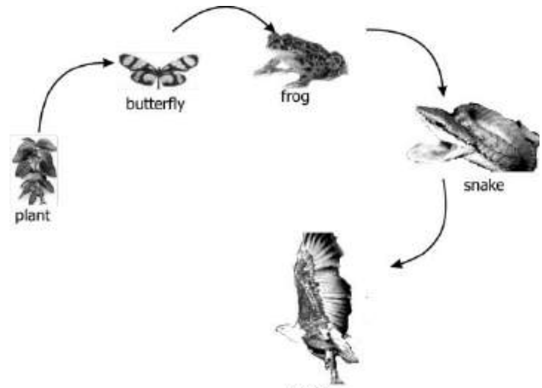


Fig. 16.0: A Food chain (plant, butterfly, frog, snake and eagle)

Components of Food Chain

Plants use light, the energy from the sun to prepare their own food in a process known as ***photosynthesis***. In a food chain, plants are called ***producers.***

The energy they get from the sun is passed on to the organism which feeds on the plants. These organisms are known as ***primary consumers***, they in turn pass it on to the ***secondary consumer***s, which pass it on to ***the tertiary consumers***.

The energy transfer ends with ***decomposers*** which feed on dead plants and animals.

In each stage of the transfer, most of the energy is lost. This is because only some parts of various organisms are eaten. For example, primary consumers eat only the leaves and sometimes soft stems, leaving the roots and hard stems. Therefore the energy transfer between producers and primary consumers is about 10% efficient.

Depending on the number of organisms present, a food chain can contain a few or more organisms, with plants always being the producers.

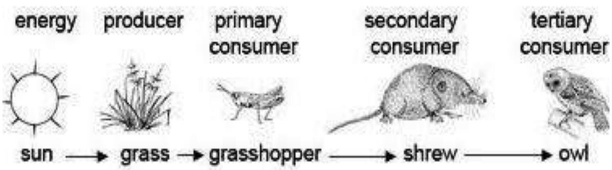


Fig. 16.1: A food chain

Terms to note

Producers

These are organisms which produce their own food through photosynthesis. Only green plants do that. Examples are tomato, yam, maize, mango, etc.

Primary consumers

These are organisms which feed on the producers. They are mostly herbivores. Examples include butterfly, grasshoppers, goat, deer etc.

Secondary consumer:

These organisms feed on and obtain energy from primary consumers. They are mostly carnivores. Examples include snake, cat, shrew, etc

Tertiary consumers: These consumers feed on secondary consumers, and hence, obtain energy from them. They are either carnivores or omnivores. Examples include man, owl, lion, etc.

FOOD WEB

A food web is a number of food chains linked together.

In a food web, an organism could feed or is fed on by more than one organism.

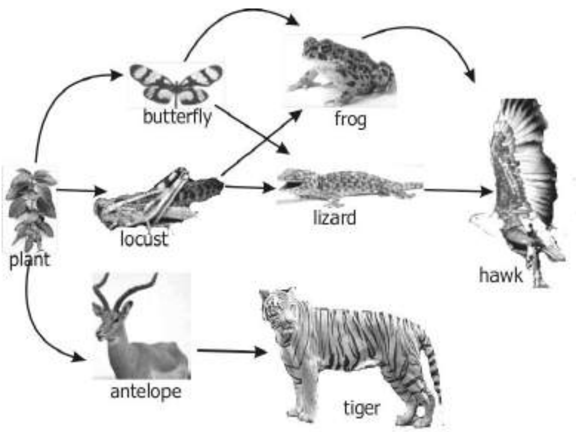


Fig. 16.1: A Food web

A food web can even grow to be more complex with so many organisms. In fact, food chains are less common in a given habitat since there is always a competition for nutrients and energy among organisms, and most organism have similar feeding habit.

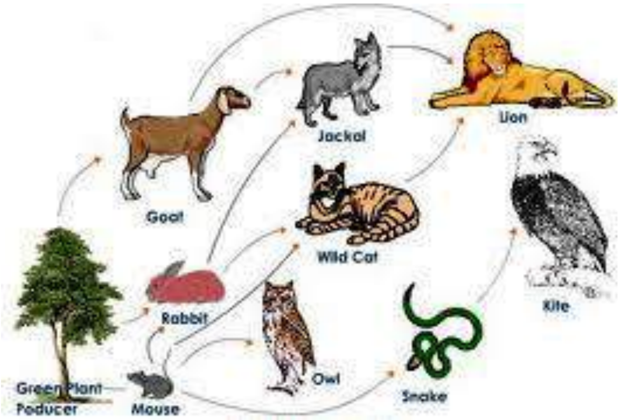


Table 4.8: Difference between food chain and food web

Food chain	Food web
Contains fewer organism	Contains many organisms
Linear feeding relationship	Complex feeding relationship
Low rate of energy transfer	High rate of energy transfer
One chain or string	Many chains combined
Found in smaller communities	Found in larger communities

TEST QUESTIONS

1. Write a short note on the following terms:
   (a) Ecosystem
   (b) Habitat
   (c) Ecosphere
   (d) Biosphere
   (e) Community
   (f) Population
   (g) Epiphitism
   (h) Species
   (i) Commensalism
   (j) Symbiosis

2. (a) Explain the following types of ecosystem and give one example each:
   (i) Terrestrial ecosystem
   (ii) Aquatic ecosystem
   (b) describe two sampling methods used in both terrestrial and aquatic habitats

3. Differentiate between the following ecological terms:
   (a) ecosystem and ecology
   (b) ecosystem and ecosphere
   (c) species and population
   (d) population and community

4. (a) Construct a food chain with microscopic plants as shown below.
   I $\rightarrow$ II $\rightarrow$ III $\rightarrow$ IV $\rightarrow$ V
   I. microscopic plants.
   II. ....
   III. ....
   IV. ....
   V. ....
   (b) Give three major advantages of an aquatic habitat.

5. (a) Distinguish between biotic factors and abiotic factors and state three examples each.
   (b) Mention three useful and three harmful biotic factors.

6. **Fig. 3** below is an illustration of three instruments used in ecological studies. *Study the figure carefully and answer the questions that follow*.

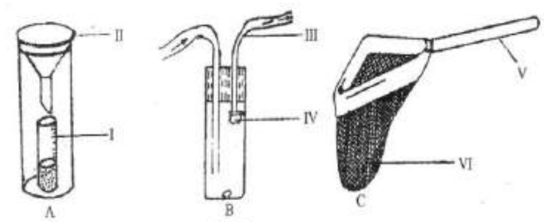


Fig 3

(a) Identify the instruments **A**, **B** and **C**.
(b) Name the parts labelled I, II, III, IV, V and VI.
(c) State one use each of **A**, **B** and **C**.
(d) State one function each of the pats labelled
 (i) II;
 (ii) III;
 (iii)VI.
(e) Describe how **A** could be used for measuring an abiotic factor in a field trip.

7. Describe how to use the following sampling methods:
 (a) Wicker work trap
 (b) Grapnel
 (c) Sweep net
 (d) Rain gauge
 (e) Pitfall trap

8. (a) Explain the following terms:
 (i) food chain
 (ii) food web
 (b) State four differences between food chain and food web.

20

ATMOSPHERE AND CLIMATE CHANGE

Specific Objectives

After completing this chapter, you will be able to:

- Describe the various regions of the atmosphere.
- Outline the effects of human activities on the atmosphere.
- Describe the major pollutants of the atmosphere and their effects.
- Explain greenhouse effect on the climate change.
- Explain the causes and effects of the depletion of the ozone layer.
- Explain the causes and effects of acid rain.

INTRODUCTION

The atmosphere of the Earth is composed of a mixture of gases. These gases consist primarily of the elements nitrogen (78 %), oxygen (21 %), argon, and smaller amounts of hydrogen, carbon dioxide, water vapour, helium, neon, krypton, xenon, and others.

The atmosphere is kept in place by a strong gravitational force that keeps the gases from escaping.

The most important attribute of the atmosphere is its life-sustaining property. Human and animal life would not be possible without oxygen in the atmosphere. In addition to providing life-sustaining properties, the various atmospheric gases can be isolated from air and used in industrial and scientific applications, ranging from steelmaking to the manufacture of semiconductors.

REGIONS OF THE ATMOSPHERE

The atmosphere is made up of four main layers. They are:

- Troposphere
- Stratosphere
- Mesosphere
- Thermosphere

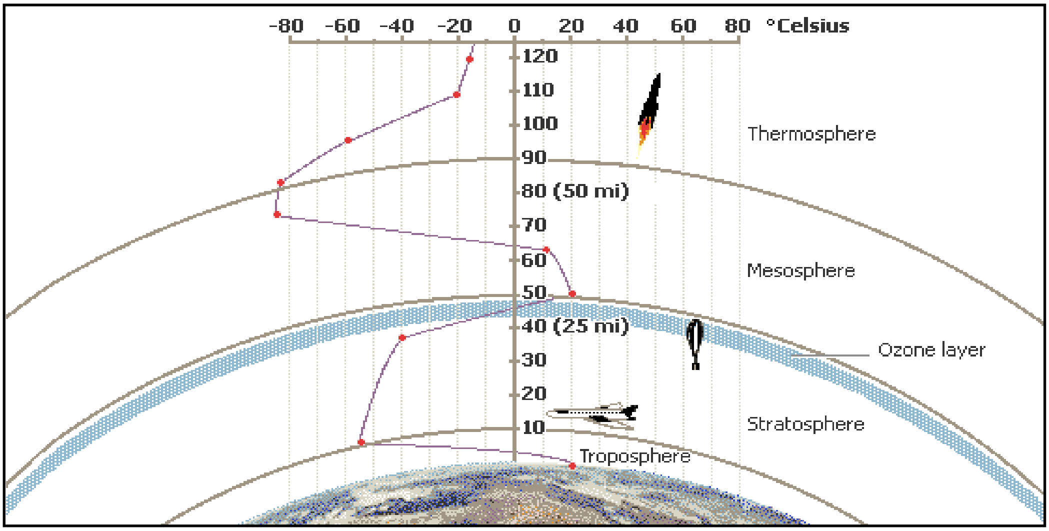


Fig. 16.2: Regions of the atmosphere

Troposphere

The Troposphere is the lowest layer of the earth's atmosphere and site of all weather changes on the earth. The troposphere is bounded on the top by a layer of air called the ***tropopause***, which separates the troposphere from the stratosphere and on the bottom by the surface of the earth. The troposphere is wider at the equator (16 km) than at the poles (8 km). The temperature of the troposphere is warmest at the tropical climatic zone and coldest at the polar climatic zone.

Stratosphere

Stratosphere is the upper layer of the atmosphere with a height of 8 to 16 km and extending upward to about 50 km. In the lower portion of the stratosphere, the temperature remains nearly constant with height, but in the upper portion the temperature increases rapidly with height because of absorption of sunlight by ozone. The stratosphere is almost completely free of clouds or other forms of weather.

Mesosphere

The Mesosphere is found above the ozone-rich stratosphere, where air and temperature, again, decrease with height. The mesosphere is the coldest layer of the atmosphere and extends from an altitude of about 50 km to about 85 km. The mesosphere has no weather.

Thermosphere

Above the mesosphere lies the hot thermosphere, where air temperatures can exceed 1000° C, primarily due to oxygen absorbing the sun’s energetic rays. Within the thermosphere is a region called

ionosphere, which is made up of ionized air. The region beyond the thermosphere is called the ***exosphere***, which extends to about 9,600 km the outer limit of the atmosphere.

HUMAN ACTIVITIES AND THEIR EFFECTS ON THE ATMOSPHERE

The quest for growth and development has also resulted in the increased in the concentration of greenhouse gases such as carbon dioxide, water vapour, methane, chlorofluorocarbons (CFCs), nitrous oxide and ozone in the atmosphere.

Greenhouse gases are those gases that contribute to the *greenhouse effect*. Human activities such as electricity generation, transportation, agriculture and industrial processes contribute greatly to the release of greenhouse gases which results in global warming.

Electricity generation

Humans depend on electricity for many things. Most of the electricity we get in our homes, schools, work places, etc is generated from plants which require coal, oil or other fossil fuels to run. These fuels when burnt release carbon dioxide, which is a greenhouse into the atmosphere.

Transportation

Another sector which contributes to the release of greenhouse gases into the atmosphere is transportation. Planes, trains, buses and cars run on petrol, diesel, kerosene and natural gas which produce greenhouse gases when they are used up.

Agriculture

The use of agricultural chemicals such as fertilizers, pesticides and weed-killers could destroy vegetative cover and pollute water bodies. These gases also release nitric oxide into the atmosphere. Farm animals also contribute to this ordeal by releasing methane into the atmosphere.

Industrial processes

Some of the industrial activities such as the use and the manufacture of chemicals only result in further polluting our environment. The machines and plants which help in production mostly run on fossil fuels, and the result of that is not avoidable.

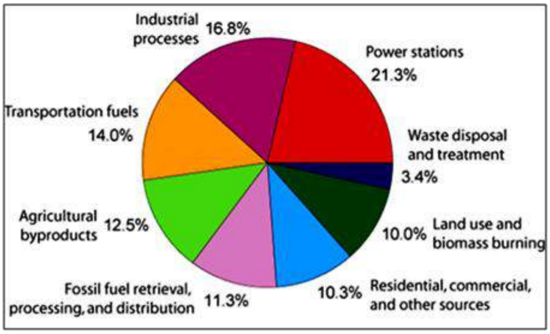


Fig. 16.3: Pollution by various sources

Other human activities such as defence, the manufacture and use of heavy weapons, land use and deforestation heighten the release of greenhouse gases.

MAJOR POLLUTANTS OF THE ATMOSPHERE

An atmospheric pollutant is a substance in the atmosphere that can cause harm to humans and the environment.

Since the atmosphere is made up of a mixture of gases, most of its pollutants are also gases.

The table below gives some major pollutants of the atmosphere, the sources and effects.

Table 4.9: *Pollutants, their sources and effects*

Pollutant	Source	Effect
Carbon monoxide (CO)	• Motor-vehicle exhaust; • Some industrial processes	• Headaches, dizziness in humans. • Long exposure may lead to death
Sulphur dioxide (SO_2)	• Heat and power generation facilities that use oil or coal containing sulphur; • sulphuric acid plants	• Makes breathing more difficult by causing the finer air tubes of the lung to constrict. • Causes acid rain.
Lead (Pb)	• Motor-vehicle exhaust; • lead smelters; • battery plants	• Damages nerve cells in humans, • Causes learning disabilities in children
Nitrogen oxides (NO_2, NO, N_2O)	• Motor-vehicle exhaust; • heat and power generation; • nitric acid; • explosives; • fertilizers	• Causes acid rain, discomfort to human eye, headaches destroys plants, Has a role in reducing stratospheric ozone.
Ozone (O_3)	• Motor vehicle exhaust and industrial emissions, • hydrocarbons, and sunlight	• Destroys the ozone layer causing global warming.
Chlorofluorocarbons (CFCs)	• CFC-containing aerosols, refrigeration equipment and some foam.	• Damage the Earth's ozone layer, which protects the earth's surface from harmful ultra-violet radiation. • CFCs are also powerful greenhouse gases which contribute to global warming.
Carbon dioxide	• Combustion of fossil fuels such as coal, oil and gas in power plants, automobiles, industrial facilities	• Contributes to global warming
Halons (Bromofluorocarbons	• Fire extinguishers, refrigerants, propellants in spray, pesticides	• Causes Throat, eye, and nasal irritation, • voice change, • cough • light-headedness

GREENHOUSE EFFECT AND CLIMATE CHANGE

Life on earth depends on energy from the sun. About 30 percent of the sunlight that beams toward Earth is deflected by the outer atmosphere and scattered back into space. The rest reaches the planet's surface and is reflected upward again as a type of slow-moving energy called ***infrared radiation***. The heat caused by infrared radiation is absorbed by greenhouse gases such as water vapour, carbon dioxide, ozone and methane, which slows its escape from the atmosphere.

Although greenhouse gases make up only about one percent of the Earth's atmosphere, they regulate our climate by trapping heat and holding it in a kind of warm-air blanket that surrounds the planet. This phenomenon is known as ***greenhouse effect***. Without it, scientists estimate that the average temperature on Earth would be colder by approximately 30 degrees Celsius, far too cold to sustain our current ecosystem.

Greenhouse effect is a process by which radiated energy leaving the atmosphere is absorbed back into the atmosphere by some gases, called greenhouse gases.

Examples of greenhouse gases by their contribution to the greenhouse effect are:
Water vapour, 36 –70%
Carbon dioxide, 9 –26%
Methane, 4 –9%
Ozone, 3 –7%
Greenhouse effect results in global warming.

Global warming is the increase in the average temperature of the Earth.

Effects of global warming

1. Desertification
2. Drought
3. Melting of polar ice caps
4. Rise in sea level, which will intensify the strength and increase the number of hurricanes and other environmental disasters.
5. Extinction of plants and animal species.
6. Destruction of fertile land resulting in low crop yield leading to famine.
7. Flood because of irregular atmospheric balance.

Remedies for global warming

1. Avoid cutting trees which absorb excess carbon dioxide in the atmosphere
2. Use energy efficient electronic appliances to reduce the amount of electricity used.
3. Develop the habit of walking, riding bicycles, skating, etc over short

distances instead of doing so by cars or buses.
4. Use renewable source of energy such as the solar energy (solar cells), wind mills etc.
5. Avoid the use of inorganic fertilizers, pesticides and weed killers.
6. Proper disposal of refuse to reduce the emission of methane into the atmosphere.

CAUSES AND EFFECTS OF THE DEPLETION OF THE OZONE

The ozone layer is a region in the stratosphere (i.e. the second layer of the atmosphere) that absorbs harmful ultra-violet radiation from the sun, thereby protecting life on earth.

The ozone forms there by the action of sunlight on oxygen. This action has been taking place for many millions of years, but increase in the concentration of some greenhouse gases has resulted in the gradual depletion or destruction of the ozone layer.

Ozone layer depletion is the process whereby greenhouse gases mostly chlorofluorocarbons (CFCs) and halons break up the chemical composition of the ozone layer causing holes to develop in it.

The depletion of the ozone layer would lead to increase in ultra-violet radiation which may have serious negative effects on plants and animals,

Effects of the depletion of the ozone layer
1. Skin problems such as skin cancer, wrinkled skin.
2. Depression of the immune system.
3. Corneal cataracts (an eye disease that often leads to blindness).
4. Massive die-off of phytoplankton and which leads to increased global warming.
5. Destruction of chlorophyll in green leaves resulting in low no or low crop yields.

ACID RAIN AND ITS EFFECTS

Acid rain is a form of air pollution in which air-borne acids produced by electric utility plants and other sources fall to Earth in distant regions.

The corrosive nature of acid rain causes widespread damage to the environment. The problem begins with the production of sulphur dioxide and nitrogen oxides from the burning of fossil fuels, such as coal, natural gas, and oil, and from certain kinds of manufacturing. Sulphur dioxide and nitrogen oxides react with water and other chemicals in the air to form sulphuric acid, nitric acid, and other pollutants. These acid pollutants reach high into the atmosphere, travel with the wind for hundreds of kilometres, and eventually return to the ground by way of rain, snow, or fog, and as invisible "dry" forms.

Effects of acid rain

1. **In soil**
   Acid rain dissolves and washes away nutrients needed by plants. It can also dissolve toxic substances, such as aluminium and mercury, which are naturally present in some soils, freeing these toxins to pollute water or to poison plants that absorb them.

2. **On plants and animals**
   The effects of acid rain on wildlife can be far-reaching. If a population of one plant or animal is adversely affected by acid rain, animals that feed on that organism may also suffer. Ultimately, an entire ecosystem may become endangered. Some species that live in water are very sensitive to acidity, some less so. Freshwater clams and mayfly young, for instance, begin dying when the water pH reaches 6.0. Frogs can generally survive more acidic water, but if their supply of mayflies is destroyed by acid rain, frog populations may also decline. Fish eggs of most species stop hatching at a pH of 5.0. Below a pH of 4.5, water is nearly sterile, unable to support any wildlife.

Fig. 16.4: Forest destroyed by acid rain

3. **On buildings**
   Acid rain and the dry deposition of acidic particles damage buildings, statues, and other structures made of stone. Acid rain has been the cause of the destruction of some famous historic sites such as pyramids of Giza in Egypt.

Fig.16.4 (a)

Fig. 16.4(b): Effect of acid rain on buildings and stone structures

4. **On metal**

Acid rain is very corrosive and easily destroys metallic objects and structures.

TEST QUESTIONS

1. Explain the following regions of the atmosphere:
   a. Troposphere
   b. Stratosphere
   c. Mesosphere
   d. Thermosphere

2. (a) Explain three human activities that contribute to global warming.
   (b) Mention four effect and four remedies for global warming.

3. (a) What is acid rain?
   (b) Discuss the effects of acid rain on the following:
   i. Buildings
   ii. Soil
   iii. Plant and animals
   iv. metal

4. Discuss the sources and the effects of the following pollutants:
   (a) Oxides of lead
   (b) nitrogen and sulphur
   (c) ozone
   (d) halons/ freons

5. Differentiate between the troposphere and stratosphere in terms of the following:
   (a) temperature,
   (b) air composition
   (c) pressure

6. Briefly describe how the following human activities affect the atmosphere:
   (a) Transport
   (b) Industrial processes;
   (c) Agriculture
   (d) Electricity production.

7. List four pollutants of the atmosphere, their sources and effects.

8. (a) What is the ozone layer?
   (b) Mention three effects of the depletion of the ozone layer

9. (a) What is global warming?
   (b) State four effects of global warming.
   (c) State four remedies of global warming.

21

TECTONIC MOVEMENT

Specific Objectives

After completing this chapter, you will be able to:

- ❖ Identify the layers of the earth.
- ❖ Describe the significance of plate tectonics.
- ❖ Explain how plate tectonics cause earth tremors, earthquakes, volcanoes and tsunamis.
- ❖ Give examples of recent land formations and submerging lands.

INTRODUCTION

The individual continents and other landmasses are said to have shifted a couple of times from their original positions. How, when and why did that happen? Can it happen again? In this chapter we shall consider the movement of the Earth's surface and its effects.

LAYERS OF THE EARTH

The surface of the earth is made up of 70 percent water and 30 percent land. However, within the earth, there are four different layers. They are:

- ❖ Crust
- ❖ Mantle
- ❖ Outer core
- ❖ Inner core

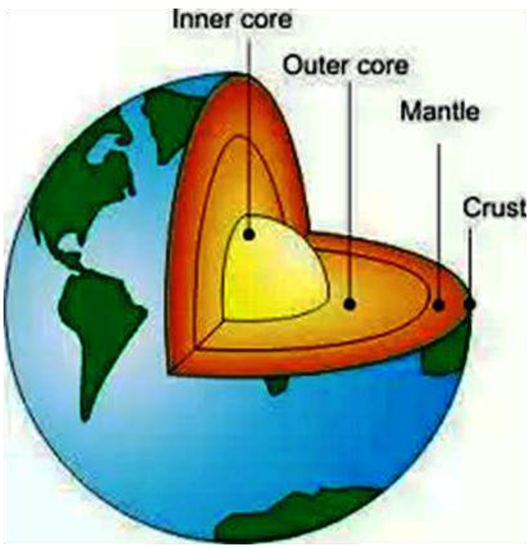


Fig. 16.5: Structure of the Earth

Crust

The crust is the outer layer of the earth. It is on the crust that we live. Compared to the other layers, the crust is much thinner.

The crust is made up of a solid material but this material is not the same everywhere. There is an Oceanic crust and a Continental crust. The first one is about 6-11 km thick and mainly consists of heavy rocks, like basalt. The Continental crust is thicker than the Oceanic crust, about 30 km thick. It is mainly made up of light material like granite.

Mantle

The mantle begins about 40 km below the crust. It is about 2,900 km thick and makes up nearly 80 percent of the Earth's total volume. Science deals with the structure of the mantle in two different ways. One way is based on its chemical construction (the material), the other on the way layers stream or move. The mantle consists of two main parts – the ***inner mantle*** and the ***outer mantle***. The inner mantle (also known as asthenosphere) can be found between 300 km and 2,890 km below the earth's surface. The average temperature is 3000ºC, nevertheless the rock is solid because of the high pressures. The outer mantle (also called lithosphere) is a lot thinner than the inner mantle. It can be found between 10 km and 300 km below the surface of the earth.

Core

The inner part of the earth is the core. This part of the earth is about 2,900 km below the earth's surface. The core is a dense ball of the elements iron and nickel. It is divided into two layers, the ***inner core*** and the ***outer core***. The inner core is the centre of earth. It is solid and about 1,250 km thick. The outer core is so hot that the metal is always molten, but the inner core pressures are so great that it cannot melt, even though temperatures there reach 3700ºC. The outer core is about 2,200 km thick. Because the earth rotates, the outer core spins around the inner core and that causes the earth's magnetism.

PLATE TECTONIC

The earth's crust consists of a number of moving pieces or plates that are always colliding or pulling apart. The Lithosphere consists of nine large plates and twelve smaller ones. The continents are imbedded in continental plates; the oceanic plates make up much of the sea floor. The study of tectonic plates is called ***Plate Tectonics***. It helps to explain continental drift, the spreading of the sea floor, volcanic eruptions and how mountains are formed. The force that causes the movement of the tectonic plates may be the slow churning of the mantle beneath them. Mantle rock is constantly moved upwards to the surface by the high temperatures below and then sinks by cooling. This cycle takes millions of years.

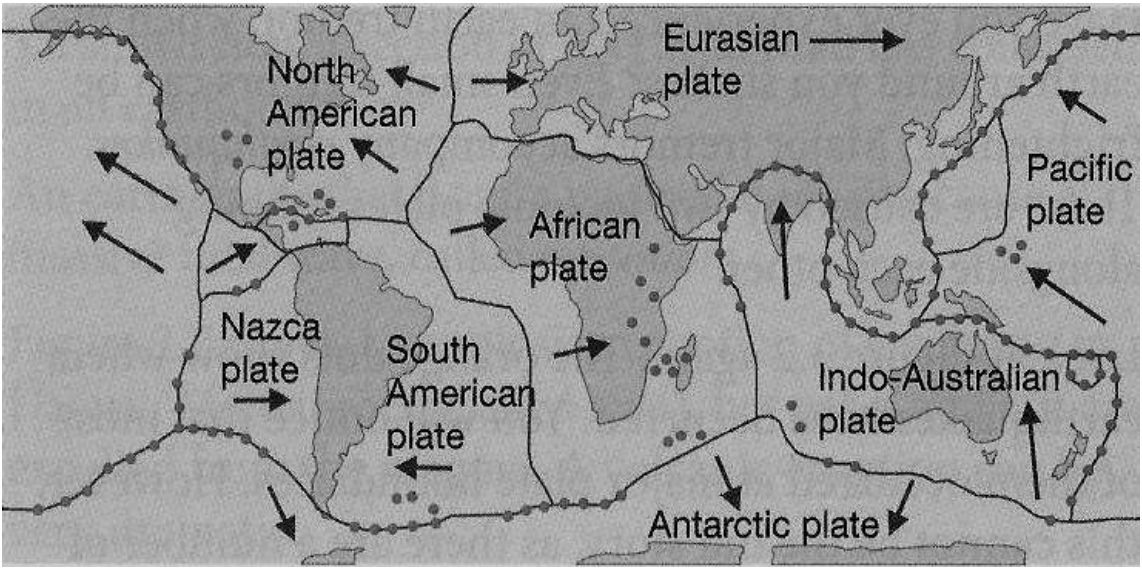


Fig. 16.6: The tectonic plates

Plate tectonics arose from an earlier theory proposed by German scientist Alfred Wegener in 1912. Looking at the shapes of the continents, Wegener found that they fit together like a jigsaw puzzle. Using this observation, along with geological evidence he found on different continents, he developed the theory of ***continental drift***, which states that today's continents were once joined together into one large landmass.

Effects of Plate Tectonics

Earth quakes

An Earthquake is the shaking of the ground caused by sudden movements in the earth's crust. The biggest earthquakes are set off by the movement of tectonic plates. Some plates slide past each other gently, but others can cause a heavy pressure on the rocks, so they finally crack and slide past each other. By this, vibrations or shock waves are caused, which go through the ground. It is these vibrations or seismic waves which cause an earthquake. The closer to the source of the earthquake (the focus or hypocenter), the more damage occurs. Earthquakes are classified according to the depth of the focus.

Effects of earthquakes

Loss of lives*:* One of the evils of earthquakes is the large number of people and animals which die or get injured as a result. Most of the earthquakes experienced have resulted in massive casualties.

Destruction of structures: The shaking of the earth causes buildings, bridges, masts, pylons etc to collapse. Roads develop large, gaping cracks, thereby making them difficult to use.

Fig. 16.7: Effect of Earthquake

***Destruction of towns and cities*:** Beautiful towns and cities could turn to a dump of collapsed buildings, dead people and animals making them unsightly.

Fire outbreak: During earthquakes gas and petrol tanks are crushed, electric poles are broken down, etc. All these can easily cause fire which can easily spread.

Tsunamis

A tsunami is a very large sea wave that is generated by a disturbance along the ocean floor. This disturbance can be an earthquake, a landslide, or a volcanic eruption. A tsunami is undetectable far out in the ocean, but once it reaches shallow water, this fast-travelling wave grows very large.

Effects of Tsunamis

Loss of lives: Tsunamis are known by the huge number of deaths they cause. The worst tsunami in history which occurred in 24th December, 2004 in the Indian Ocean claimed over 250,000 lives.

***Destruction of properties*:** The waves of tsunamis wash away everything in their path. They break down building, wash away roads, destroy towns etc.

Fig. 16.8: Tsunami waves

Volcanoes

Volcanoes are mountains or hills formed when molten rocks called ***magma*** flows from the semi solid magma onto the surface of the land. A volcano consists of a magma chamber, pipes and vents. The magma chamber is where magma from deep within the planet pools, while pipes are channels that lead to surface vents, openings in the volcano's surface through which lava is ejected during an eruption.

Effects of volcanoes

1. Volcanoes release heavy amount of greenhouse gases into the atmosphere.
2. The flow of lava destroys everything in its path.
3. Volcanic activities are often accompanied by earthquakes, landslides, hot springs, etc; all of which pose hazard to humans.

Fig. 16.9: An erupting volcano

NEW LAND FORMATIONS AND SUBMERGING LANDS

Another effect of tectonic movement is the formation of new land forms. Mountains, deserts, islands etc are still created as a result of the frequent movements of the tectonic plates. Magma from volcanoes also hardens to form solid rocks. The accumulation of these can form mountains, hills and islands.

TEST QUESTIONS

1. Write a short note on the following layers of the earth:
   a. Crust
   b. Mantle
   c. Outer core
   d. Inner core

2. (a) Explain plate tectonic.
   (b) Discuss three effects of plate tectonic.

3. (a) Draw and lable the structure of the Earth.
   (b) Describe the parts labelled

4. Decsribe the formation and effects of the following:
   (a) earthquake;
   (b) volcano

22

INFECTIONS AND DISEASES

Specific Objectives

After completing this chapter, you will be able to:

- ❖ Describe the causes of diseases.
- ❖ Describe the modes of transmission, symptoms and modes of control of common diseases

INTRODUCTION

Disease is any harmful change that interferes with the normal appearance, structure, or function of the body or any of its parts.

Since time immemorial, disease has played a role in the history of societies. It has affected—and been affected by economic conditions, wars, and natural disasters. Indeed, the impact of disease can be far greater than better-known calamities. For example, the epidemic of influenza that swept the globe in 1918 killed from 20 million to 50 million people.

CAUSES OF DISEASES

Diseases affect man, animals and plants. The microorganisms that cause diseases are known as ***pathogens***. There are different types of pathogenic diseases. They are:

- ❖ bacterial diseases,
- ❖ viral diseases,
- ❖ fungal diseases,
- ❖ rickettsial diseases
- ❖ protozoan diseases etc.

Bacterial diseases

Bacteria are one-celled organisms visible only through a microscope. Bacteria live all around us and within us. The air is filled with bacteria, and they have even entered outer space in spacecraft. Bacteria live in the deepest parts of the ocean and deep within Earth. They are in the soil, in our food, and on plants and animals. Even our bodies are home to many different kinds of bacteria. Our lives are closely intertwined with theirs, and the health of our planet depends very much on their activities. Though some bacteria live in our bodies and prevent some diseases, most bacteria are the carriers of the diseases.

Some of the diseases caused by bacteria are cholera, tuberculosis, whooping cough, typhoid fever, diphtheria, tetanus, and cerebrospinal meningitis.

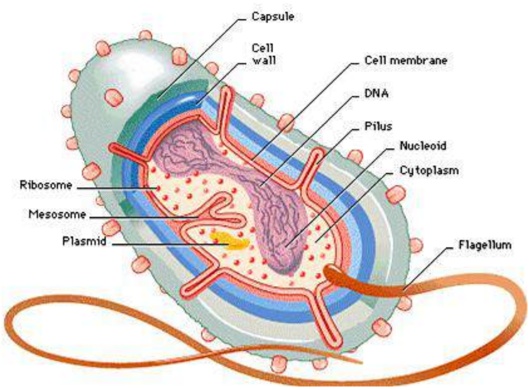


Fig. 17.0: A bacterial cell

Viral diseases

A virus is an infectious agent found in almost all life forms, including humans, animals, plants, fungi, and bacteria. Viruses consist of genetic material—either deoxyribonucleic acid (DNA) or ribonucleic acid (RNA)—surrounded by a protective coating of protein, called a *capsid.*

Viruses are not considered free-living, since they cannot reproduce outside of a living cell. Viruses often damage or kill the cells that they infect, causing diseases in infected organisms. Viral diseases are difficult to control or cure.

Diseases caused by viruses include measles, rabies, and poliomyelitis.

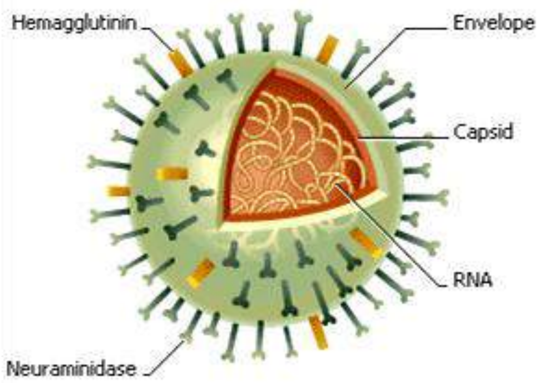


Fig. 17.1: Structure of a virus

Fungal diseases

Fungus is any member of a diverse group of organisms that, unlike plants and animals, obtain food by absorbing nutrients from an external source. Fungal diseases are diseases caused by the growth of fungi in or on the body. In most healthy people fungal infections are mild, involving only the skin, hair, nails, or other superficial sites, and they clear up spontaneously. They include ringworm and athlete's foot. In someone with an impaired immune system, however, such infections, called *dermatophytoses*, can persist for long periods.

Fig. 17.2: Ring worm caused by fungus

Rickettial diseases

Rickettsia are micro-organisms which are intermediate in size between viruses and most bacteria. Like viruses, they can survive and multiply only inside cells. They are found in a great variety of mammals and cause several diseases in humans. They are generally carried by arthropods, such as fleas, lice, mites, and ticks. One exception infects domestic animals such as cattle and sheep, and can be transmitted to humans through ingestion of infected raw milk, or through inhalation of airborne particles of the milk, urine, faeces, or tissues of infected animals. Numerous insect-infesting species of rickettsia cause no mammalian animal disease.

Symptoms of rickettsial diseases

The rickettsial diseases are commonly characterized by sudden onset and cause lethargy, high fever, headache, muscle aches, skin rashes in most cases, and damage to the lining of the blood vessels; damage to tissues of the central nervous system often follows. Rickettsial diseases include epidemic typhus; endemic typhus; Rocky Mountain spotted fever; tsutsugamushi fever, or scrub typhus; and trench fever. Q fever and rickettsial pox are also rickettsial diseases.

Diagnosis, treatment and prevention of rickettsial diseases

Diagnosis of most of the rickettsial diseases facilitated by the development in the blood of infected patients of specific antibodies that can be detected by serologic tests.

Prevention of rickettsial diseases involves eradication of the arthropod carriers, injection with antibiotics, such as the tetracycline drugs and chloramphenicol.

Protozoan diseases

Protozoa is a collective name for animal-like, single-celled organisms, some of which may form colonies. Most species of protozoa are found in aquatic habitats such as oceans, lakes, rivers, and ponds. Some of the diseases caused by protozoa are malaria, typanosomiasis and giardiasis.

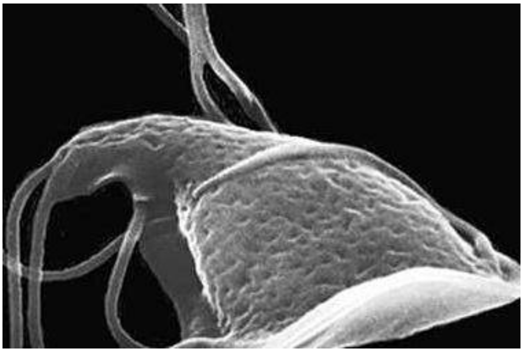


Fig. 17.3: Giardia lamblia (protozoa)

Table 5.0: Some pathogenic diseases, their mode of transmission, symptoms and methods of prevention and control

Disease	Mode of transmission	Symptoms	Method of control
Bacterium diseases			
Tuberculosis *(Mycobacterium tuberculosis)*	Droplet infection (airborne), contact over long periods	Loss of weight, continuous coughing resulting in chest pains, breathlessness, fever.	Isolation of patient, antibiotics, vaccination with BCG vaccine
Tetanus *(Clostridium tetani)*	Causative bacteria in the blood stream	Muscular spasms, nervous disorders, lockjaw	Keep wounds covered with clean materials and dressed continuously, vaccination with toxoid, antibodies
Cholera *(Vibrio cholera)*	Contaminated food and water	Diarrhoea, dehydration, severe vomiting and abdominal pains	Personal hygiene, good sanitation, antibiotics, vaccination
Typhoid *(Salmonella typhi)*	Contaminated food and water	High fever, diarrhoea, vomiting, abdominal pains	Personal hygiene, clean drinking water, antibiotics, good sanitation and immunization
Cerebrospinal Meningitis (CSM)	Airborne	Headache, severe fever, convulsion	Vaccination, good ventilation in a room
Dysentery *(Shigella)*	Contaminated food and water	diarrhoea	Clean drinking water, general hygiene, antibiotics
Leprosy *(Mycobacterium leprae)*	Droplet infection	Ulcer, deformities, raised blotches on skin	Antibiotics, vaccination
Viral diseases			
Poliomyelitis	Contaminated food and water, inhalation	Fever, destruction of the nervous system, paralysis	Vaccination, proper sanitation, personal hygiene
Measles	Airborne and droplet infection	Rash, fever, sore throat	Vaccination, isolation of patients.
Rabies	Bite from infected animal, especially dogs or cats	Fever, headache, body pains paralysis vomiting	Vaccination (immediate), immunization of dogs and cats. Elimination of infected animals

Influenza	Droplet infection, airborne	Fever, headache, body pains, respiratory disorder	Vaccination, antibodies
Fungal diseases			
Ringworm *(Tibea capitis)*	Contact with contaminated objects	Ring-shaped sport which increases in size with time	Application of fungicidal ointment at the infected area. Avoid sharing some basic things such as comb or brush with an infected person
Athlete' s foot (*Tibea pedis)*	Contact with infected person or contaminated objects such as flip-flops, sandals, etc.	Development of blisters, cracking and peeling of skin between the toes	Disinfection of contaminated objects, avoid sharing shoes with infected person. Fungicidal ointment
Protozoan diseases			
Malaria *plasmodium*	Bite from infected female anopheles mosquito	Heavy fever	Using anti-malarial drugs
Trypanosomiasis (*trypanosoma)*	Bite from infected tsetse fly	Muscular spasm, nervous and lymphatic system disorder	Antibodies, vaccination
Dysentery (*Entamoebia histolitica)*	Contaminated food and water	Diarrhoea, profuse bleeding	Antibodies, good sanitation, personal hygiene
Giardiasis (*Giardia lamblia)*	Contact with infected person, improper personal hygiene, contaminated food and water	Diarrhoea, nausea, poor appetite, fatigue, abdominal cramps	Careful personal hygiene, proper sanitation, proper filtration of water

NON-PATHOGENIC DISEASES

Apart from pathogenic diseases, there are other diseases which are caused by some other factors such as poor nutrition, genetic, stress conditions, poor sanitation etc. These diseases are collectively called non-pathogenic diseases or non-infectious diseases.

Table 5.1: Some non-pathogenic diseases, their causes and possible controls

Disease	Causes/type	Prevention/ control
Diabetes	Hormonal: inability of the pancreas to sufficient amount of insulin which helps body cells to absorb glucose so it can be used as source of energy	Diet control (eating food which is low in sugar), exercise, weight reduction
Kwashiorkor	Malnutrition: low-protein and high-starchy food (especially in children)	Eating more protein and less starch
Rickets	Malnutrition: lack of vitamin D	Eating food rich in calcium and phosphorus, having enough sunlight
Anaemia	Malnutrition: Lack of iron or vitamin B_2	Eating food rich in iron and vitamin B_2
Night blindness	Malnutrition: Vitamin A deficiency	More vitamin A in diet
Lung cancer	Environmental: tobacco smoke	Avoid smoking
Liver and brain damage	Environmental: excessive drinking	Avoid drinking too much alcohol
Repetitive strain injury (RSI)	Environmental/stress: working continuously	Take a break from time to time from work
Sickle-cell anaemia	Genetic/hereditary: mutated genes from parent(s)	
Birth defect	Genetic: inherited from parents or develop when the baby is in its mother's womb or during delivery	
Allergy	Immunological: malfunctioning of the immune system towards some food	Avoid eating food which you are allergic to

MODES OF TRANSMISSION OF DISEASES

When a person is infected with a particular disease, it means the pathogens of that disease has entered his or her system interfering with the normal functioning of\the cells and other organs. Different diseases have different modes through which they are transmitted.

Some of the modes of disease transmission include:

1. Air borne
2. Water related
3. Insect borne (vectors)
4. Food contaminated (food poisoning)
5. Nutrition related
6. Sexually transmitted
7. Communicable
8. Zoonotic diseases

Air borne diseases

These are diseases that are transmitted through the air. The pathogens are in tiny droplets that get mixed up in the surrounding air and are easily carried by moving air. Air born diseases easily engulf the air and their rates of transmission are high in over-crowded, low ventilated and highly humid areas.

Examples of airborne diseases include tuberculosis (TB), whooping cough, measles, influenza, chicken pox, pneumonia, etc.

Water related diseases

The causative organisms of water related diseases are transmitted through water. Drinking, swimming or walking in contaminated water normally results in the contraction of the diseases.

Examples of water borne diseases include bilharzias (schistosomiasis), cholera, Guinea worm, poliomyelitis etc.

Insect born or vector diseases

Most insects are carriers or vectors of disease causing pathogens. These diseases are transmitted to people or farm animals when they are bitten by those insects. Examples of insect borne diseases include malaria (transmitted by mosquitoes), trypanosomiasis or sleeping sickness (transmitted by tsetse fly), river blindness or onchocerchiasis (transmitted by black fly), etc.

Food contamination or poisoning

Food contamination normally results from food spoilage. Spoiled foods become a habitat for disease causing organisms. Another cause of food contamination is eating with contaminated hand, spoon or fork.

The diseases are transmitted to humans and animals when they eat or feed on the contaminated food.

Prevention and control of food contamination or poisoning

1. Washing hands with soap before eating or handling food.
2. Cooking food properly before eating
3. Washing dishes properly before using them to cook or eat.
4. Keeping food in hygienic conditions.
5. Properly covering all food to prevent contact with flies.
6. Disposing off expired or spoiled foods.
7. Keeping left-over food in fridge and freezers.
8. Properly heating left-over food before eating.

Nutrition related

Humans and animals need some essential nutrients for their growth and survival; the lack of these nutrients in a person or animal results in a deficiency disease. Examples of deficiency diseases include night blindness, rickets, kwashiorkor, etc

Communicable diseases

These are diseases which are easily transmitted from one person to another. Communicable diseases can be contracted through direct contact or through other modes such as air, water, vectors etc.

Zoonotic diseases

Zoonotic diseases are diseases in animals that can be transmitted to humans. They can be transmitted through animal bites or other means such as air or vectors. Examples include rabies, avian flu (or bird flu), anthrax, salmonellosis, etc.

Sexually transmitted diseases (STIs)

STIs are diseases which are transmitted through sexual intercourse in human and mating in animals. The disease is passed on from an infected person to a healthy person mostly as a result of the exchange of body fluid. Examples of STIs include AIDS, gonorrhoea, syphilis etc.

LIFE-STYLES THAT MAKE PEOPLE PRONE TO HIV/AIDS INFECTION

Having unprotected sex with an infected person

HIV transmission occurs most commonly during intimate sexual contact with an infected person, including genital, anal, and oral sex. The virus is present in the infected person’s semen or vaginal fluids. During sexual intercourse, the virus gains access to the bloodstream of the uninfected person by passing through openings in the mucous membrane, the protective tissue layer that lines the mouth, vagina, and rectum, and through breaks in the skin of the penis. The best policy to avoid HVI is to follow the ABC plan

- **A**bstinence;
- **B**e faithful;
- **C**ondom use

Contact with infected blood. Direct contact with HIV-infected blood occurs when people who use heroin or other injected drugs share hypodermic needles or syringes contaminated with infected blood.

HIV infection also results when health professionals accidentally stick themselves with needles containing HIV-infected blood or expose an open cut to contaminated blood. Transfusion of infected blood to a patient is another disturbing case.

Mother to baby

HIV can be transmitted from an infected mother to her baby while the baby is still in the woman's uterus or, more commonly, during childbirth. The virus can also be transmitted through the mother's breast milk during breast-feeding. Mother-to-child transmission accounts for 90 percent of all cases of AIDS in children. Mother-to-child transmission is particularly prevalent in Africa.

TEST QUESTIONS

1. Describe the following pathogenic diseases:
   (a) bacterial diseases
   (b) viral diseases
   (c) fungal diseases
   (d) protozoan diseases
   (e) rickettial diseases

2. (a) What are non-pathogenic diseases?
   (b) Mention four non-pathogenic diseases, their symptoms and methods of control.

3. Describe five modes of disease transmission.

4. Copy and complete the table on diseases below:

Diseases	Causative organism	One symptom	Control method
Malaria			
Cholera			
Measles			
Tuberculosis			
Poliomyelitis			
Ring worm			
Rabies			
Chicken pox			
Avian flue			
Cholera			

5. (a) What are sexually transmitted diseases?

(b) State and explain three life styles that make people prone to HIV/ AIDS.

6. Write a short note on the following modes of disease transmission:
   (a) airborne;
   (b) water related;
   (c) food poisoning;
   (d) sexually transmitted

(a)

23

ACIDS, BASES AND SALTS

Specific Objectives

After completing this chapter, you will be able to:

- ❖ Define acids, bases and salts and state their properties.
- ❖ Identify common chemical substances as acids, bases or salts and classify them according to the sources.
- ❖ Prepare salts
- ❖ Describe the effect of acid – base indicators.
- ❖ Use universal indicators and the pH-scale to determine the pH of given solutions.

INTRODUCTION

Acids and bases are two classes of chemical compounds that display generally opposite characteristics. They are so opposite that when they react with each other in a process known as ***neutralization reaction,*** a completely new product with entirely different characteristics is formed. This reaction is characteristically very rapid and generally produces salt and water.

ACIDS

Acids in food

Acids can be found in many substances, including food such as fruits. Citrus fruits like orange, grape fruit and lemon contain an acid called ***citric acid***. Acid contained in fruits is called ***ascorbic acid***. Sour milk, palm oil and bees contain lactic acid, palmolic acid and methanoic (or formic) acid respectively.

Acids in animals and plants

Acids can also be found in animals. Mammals, for example, have different acids in different parts of the body. The acid found in the stomach is known as ***hydrochloric acid***, which helps in the digestion of food. Acids found in plants and animals are known as ***organic acids***. These acids like many others are everyday acids.

Types of acids

Most organic acids are also referred to as ***weak acids*** because they partly dissolve or dissociate in water. They have low hydrogen concentration.

Weak acids *are acids which dissolve or dissociate in water.*

Apart from these there are also laboratory acids, which include hydrochloric acid, sulphuric acid and nitric acid. These acids are called ***inorganic acids*** and are normally obtained from mineral sources. They are called laboratory acids because they are commonly produced or used in laboratories.

Most inorganic acids are also referred to as ***strong acids*** because they dissolve or ionize completely in water. They have high hydrogen concentration.

Strong acids *are acids which do not dissolve or ionize completely in water.*

Definition of acids

By definition, acids are chemical substances that turn blue litmus paper red. Acids can also be defined as substances that react with some metals to produce hydrogen gas.

Brønsted-Lowry theory about acids

A more satisfactory theory was proposed in 1923 by the Danish chemist Johannes Brønsted and independently by Thomas Lowry, a British chemist. This theory is usually known as Brønsted-Lowry concept. With the Brønsted-Lowry concept, we usually refer to a hydrogen ion as a proton. That is because a proton is all that is left when a hydrogen atom loses an electron to become an ion.

Brønsted-Lowry definition

Brønsted-Lowry concept states that an acid is an acid because it provides or donates a proton to something else.

An acid is a substance that donates or transfers a proton.

When an acid reacts, the proton is transferred from one chemical to another. As will be noted later, the chemical which accepts the proton is a base.
When an acid dissolves or dissociates in water it gives a proton to the water.

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

The Brønsted-Lowry view is that the acid (HCl) gives a proton to water to make two ions, one of which is H_3O^+. H_3O^+ is called ***hydronium ion***.

H_2SO_4 can also give a proton to water.

$$H_2SO_4 + H_2O \rightarrow H_3O^+ + HSO_4^-$$

In this case, the product HSO_4^- still has a proton that can be donated to another water molecule.

$HSO_4^- + H_2O \rightarrow H_3O^+ + SO_4^{2}$

HCl can also give a proton to a hydroxide ion (OH^-) rather than water.

$HCl + OH^- \rightarrow H_2O + Cl^-$

The first chemical in each of these equations is an acid because they are each giving a proton to something else.

Table 5.2: Some acids and their sources

Acid	Formula	Source
Acetic acid	$HC_2H_3O_2$	Vinegar
Citric acid	$H_3C_6H_5O_7$	Lemon juice, citrus fruits
Ascorbic acid	$H_2C_6H_6O_6$	Vitamin C
Hydrochloric acid	HCl	Car battery Gastric juices
Sulfuric acid	H_2SO_4	Batteries
Acetylsalicylic acid	$HC_9H_7O_4$	Aspirin

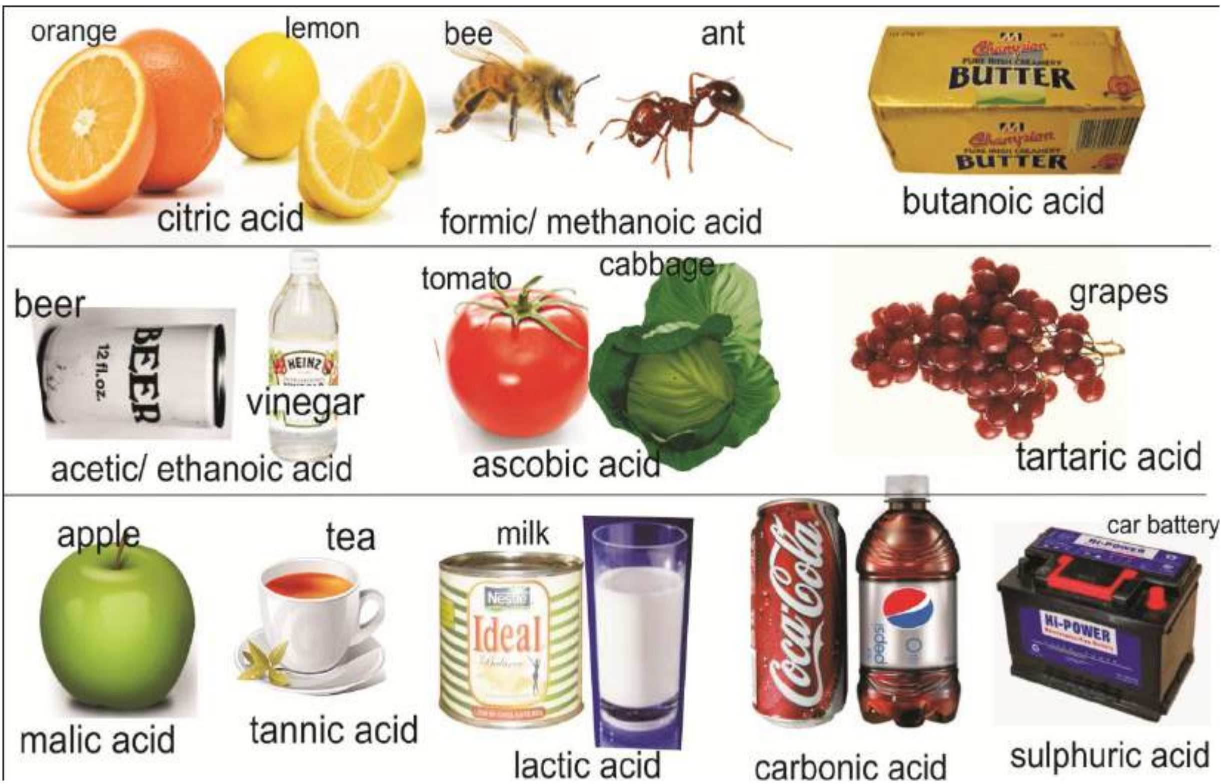


Fig. 17.4: Sources of acids

Properties of Acids

Acids have a number of properties. These properties could be physical or chemical and make acidic substances easy to distinguish.

Physical properties of acids

1. Acids are corrosive.
2. They have a sour taste.
3. Acids have a pH less than 7.
4. Acids change litmus (a dye extracted from lichens) red.
5. Acids can conduct electricity.
6. They are soluble in water.

Chemical properties of acids

1. Acids react with bases to form salt and water. This process is called ***neutralisation***.

   For example, Hydrochloric acid reacts with sodium hydroxide to produce sodium chloride (salt) and water.

   $$HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$
   Acid + Base Salt + Water

   Copper oxide reacts with tetraoxosulphate (VI) acid to produce copper (ii) tetraoxosulphate (VI) (salt) and water.

   $$CuO_{(s)} + H_2SO_{4(aq)} \rightarrow CuSO_{4(aq)} + H_2O_{(l)}$$
   Base + Acid = Salt + Water

2. Acids react with metals (the more reactive metals such as magnesium, calcium and iron) to form hydrogen gas and a salt.

   For example, hydrochloric acid reacts with magnesium to form hydrogen gas and magnesium chloride salt.

   $$2HCl_{(aq)} + Mg_{(s)} \rightarrow H_{2(g)} + MgCl_{2(aq)}$$
   Acid + metal hydrogen + salt

   Tetraoxosulphate (VI) acid reacts with iron to produce hydrogen gas and iron (II) tetraoxosulphate

   $$H_2SO_{4(aq)} + Fe_{(S)} \rightarrow H_{2(g)} + FeSO_{4(aq)}$$
   Acid + metal = hydrogen + salt

   The magnesium and iron metals replace the hydrogen atom of the hydrochloric and tetraoxosulphate acids thereby producing magnesium chloride and iron (II) tetraoxosulphate salts.

3. Acids react with carbonates to form carbon dioxide gas, water and a salt.

   For example, hydrochloric acid reacts with calcium trioxocarbonate to produce carbon (II) oxide, water and calcium chloride salt.

$$2HCl_{(aq)} + CaCO_{3(s)} \rightarrow CO_{2(g)} + H_2O_{(l)} + CaCl_{2(aq)}$$
Acid + calcium trioxocarbonate = carbon (II) oxide + water + salt

   Trioxonitrate(V) acid reacts with sodium hydrogentrioxocarbonate (IV) to produce carbon(II) oxide, sodium trioxonitrate(V) salt and water.

$2HNO_3 + NaHCO_3 \rightarrow CO_3 + Na(NO_3)_2 + 2H_2O$

Acid +sodium trioxocarbonate = sodium (II) oxide + water + salt

Uses of acids

1. Acids are normally used in industries, household cleaning product, daily products, and soft drinks.
2. We can get acidic or sour taste from most food items such as lemons, vinegar, cream, yogurt, cottage cheese, etc.
3. A wasp sting, which is alkali, can be neutralized with a weak acid such as vinegar and lemon juice.
4. Sulphuric acid is widely used in car batteries as wet cell battery electrolyte.
5. Phosphoric acid is normally used in cola soft drinks.
6. Dilute hydrochloric acid in the stomach helps in digestion.
7. Industrial hydrochloric acid is used to make glue and for cleaning steel.
8. Sulphuric acid is used to manufacture fertilizers, paints and dyes, detergents, plastics, paper and explosives.
9. Sulphuric acid is also used in refining petroleum.
10. Nitric acid is used in cleaning and purifying gold and silver.
11. It is also used in manufacturing fertilizers, drugs, paints and dyes.

BASES

Bases are the opposite of acids. Bases are basic because they take or accept protons. Hydroxide ion, for example can accept a proton to form water.

$$OH^- + H^+ \rightarrow H_2O$$

Brønsted-Lowry theory about bases

Brønsted and Lowry realized that not all bases had to have a hydroxide ion. As long as something can accept a proton it is a base.

So anything, hydroxide or not, that can accept a proton is a base under the Brønsted-Lowry definition. The water molecules that accept protons when HCl dissolves in water are acting as bases.

Ammonia can accept or react with hydrogen ion to give ammonium ion NH_4^+.

$$NH_3 + H^+ \rightarrow NH_4^+$$

Carbonate ion can accept a hydrogen ion, or accept a proton, to become bicarbonate ion.

$$CO_3^{2-} + H^+ \rightarrow HCO_3^-$$

Also, water molecules, as mentioned before, can act as a base by accepting protons.

$$H_2O + H^+ \rightarrow H_3O^+$$

Hydroxide, ammonia, carbonate and water are all Brønsted-Lowry bases.

Be sure to note the distinction between ***ammonia*** and ***ammonium***. NH_3 is ammonia and NH_4^+ is ammonium. They sound very much the same and their formulas are very similar, but their chemical properties are far different. They are different because one has one more proton than the other. Ammonia is a base and ammonium is an acid.

Examples of bases

Some bases are often found in household cleaners, they help clean grease from windows and floors and are found in the soap we use everyday.

- Some other examples of basic substances are toothpaste, egg whites, dishwashing liquids and household ammonia.
- Bases like potassium hydroxide are normally obtained when wood, cocoa pod, plants etc are burned.
- Other bases, such sodium hydroxide are produced industrially from salt.

Alkalis

Bases which are soluble in water are called ***alkalis***. Examples of alkalis are ammonium hydroxide, sodium hydroxide, potassium hydroxide and calcium hydroxide.

A base is any substance that accepts protons.

Types of bases

Bases can be organic or inorganic.

Organic bases: These are bases obtained from natural source, such as ashes of plant, cocoa pods etc. Most organic bases are weak bases.

Inorganic bases: These are bases obtained from mineral sources; examples are calcium hydroxide ($Ca(OH)_2$), aluminium oxide (Al_2O_3), and milk of magnesia ($Mg(OH)_2$). Inorganic bases such as $NaOH$, $Ca(OH)_2$ etc. are strong bases.

Properties of bases

Like acids, bases also have properties which are either physical or chemical.

Physical properties of bases

1. Bases taste bitter.
2. Bases feel soapy and slippery.
3. Bases have a pH over 7.
4. They change litmus paper to blue.
5. They are very corrosive in concentrated or solid form.

Chemical properties of bases

1. Bases react with acids to form a salt and water. This process is called ***neutralization reaction***.

For example, calcium hydroxide reacts with hydrochloric acid to produce calcium chloride and water,

$$CaOH_{(aq)} + HCl_{(aq)} \rightarrow CaCl_{(aq)} + H_2O_{(l)}$$

Base + acid = salt + water

2. Bases react with ammonium salt to release ammonia gas, salt and water. For example, ammonium chloride reacts with calcium hydroxide to produce ammonia gas and salt.

$$2NH_4Cl_{(s)} + Ca(OH)_{2(s)} \rightarrow 2NH_{3(g)} + CaCl_{2(s)} + 2H_2O_{(l)}$$

Ammonium + base = ammonia gas + salt + water

Uses of bases

3. Ammonium hydroxide, frequently called ammonia, is used in the preparation of important related compounds such as nitric acid and ammonium chloride.
4. Ammonia is also used as a cleaning agent, refrigerant and for softening hard water.
5. Sodium hydroxide is used in the manufacture of soap, rayon, and paper. Strong solutions of this base are very caustic; that is, they are extremely harmful to the skin.
6. Calcium hydroxide, commonly known as slaked lime, is used in the preparation of plaster and mortar.
7. It is also used to neutralize acidic soil.
8. Water solutions of calcium hydroxide, called limewater, can be used in the lab as a test for the presence of carbon dioxide.

Fig. 17.5: Common bases used at home

SALTS

Many chemical compounds may be classified as salts. The salt most familiar to most people is table salt - sodium chloride (NaCl). Baking soda is the salt sodium bicarbonate. Magnesium sulphate, also called *Epsom salts*, is often found in the home.

Salts are ionic compounds that are composed of metallic ions and non-metallic ions.

For example, sodium chloride (NaCl) is composed of metallic sodium ions and non-metallic chloride ions. Some salts are composed of metallic polyatomic ions and non-metallic polyatomic ions (ammonium nitrate is composed of ammonium ions and nitrate ions).

Properties of Salts

Physical properties

1. Salts are soluble in water.
2. Salts are usually odourless.
3. Molten salts conduct electricity.

4. Normal salt (NaCl) is colourless. However, different salts exist in different colours. (For example, sodium chromate is yellow, potassium dichromate is orange, nickel chloride hexahydrate is green, etc.).

5. *Salts are crystalline in nature.*
   That is they are like shards of broken glasses. This is because the ions in a solid salt are usually arranged in a definite crystalline structure; each positive ion is associated with a fixed number of negative ions and vice versa.

Chemical properties

1. *Salts are ionic compounds.*
   That is they are made up of ions rather than molecules. In NaCl, for example, a fixed proportions of sodium ion (Na^+) and Chlorine ion (Cl^-) come together to form the salt.

2. *They are composed of related number of cations (positively charged ions) and anions (negatively charged ions).*

3. *Salt are electrically neutral.* That is they have no charge.

4. *Salts dissolve or dissociate in water.* Salts consist of tightly bonded ions. In water, these bonds are weakened and the ions become mobile. In water, for example, sodium chloride ionizes, or dissociates like this:
   $NaCl_{(s)} \rightarrow Na^{1+}_{(aq)} + Cl^{1-}_{(aq)}$

5. *Salts react with water to form acids and bases in a process called* ***hydrolysis reaction***.
   For example, when sodium carbonate (Na_2CO_3) is dissolved in water, carbonic acid (H_2CO_3) and sodium hydroxide (2NaOH) are formed.

$$Na_2CO_{3\,(aq)} + 2\,H_2O_{(l)} \rightarrow H_2CO_{3\,(aq)} + 2NaOH_{(aq)}$$

NB: Since acids and bases react to form water and salt (neutralization reactions) hydrolysis reactions are the reverse of neutralization reactions.

A salt that includes water in it solid crystalline form is called a ***hydrate***.

Salt in the sea

The salty taste of sea water is due to the presence of salts such as sodium chloride and magnesium bromide. There are many different salts present in sea water:

Table 5.3: Percentage of salts found in seawater

Salt in seawater	Formula	Percentage in seawater
sodium chloride	NaCl	2.72 %
magnesium chloride	$MgCl_2$	0.38%
calcium sulphate	$CaSO_4$	0.13 %
magnesium sulphate	$MgSO_4$	0.17 %
calcium carbonate	$CaCO_3$	0.01 %
potassium chloride	KCl	0.09 %
magnesium bromide	$MgBr_2$	0.01 %

Types of salts

Based on the above properties of salts, we can deduce four main types of salts: acidic salts, basic salts, normal (neutral) salts and double salts.

Acidic salts

This is the type of salt in which not all the protons or hydrogen ions are replaced or donated. Examples are:

1. Potassium hydrogen tetraoxosulphate (VI), $KHSO_4$
2. Sodium hydroxide trioxocarbonate (IV), $NaHCO_3$
3. Sodium dihydrogen tetraoxosulphate (IV), NaH_2PO_4
4. Sodium hydrogen chloride, NaHCl

Basic salts

Basic salts are form when there is a neutralization reaction between a strong base and a weak acid. Examples are:

1. Sodium hydroxide carbonate, $Na(OH)CO_3$
2. Magnesium hydroxide chloride, Mg(OH)Cl
3. Calcium hydroxide carbonate, $Ca(OH)CO_3$
4. Lead hydroxide carbonate, $Pb(OH)NO_3$

Normal salts

Normal salts are formed when there is a perfect neutralization. That is when the strength of both the acids and bases involved are the same and all the protons are donated. Examples are:

1. Sodium chloride, NaCl
2. Magnesium Chloride, MgCl
3. Potassium carbonate, K_2CO_3
4. Potassium nitrate, KNO_3
5. Ammonium chloride, NH_4Cl
6. Sodium sulphate, Na_2SO_4

Double salts

These types of salts are formed when two salts are mixed together. They are usually referred to as alums and generally assume the properties of the constituent salts. Examples are:

1. Calcium magnesium carbonate $CaMg(CO_3)_2$

2. Potassium sodium tartrate and bromide
3. Calcium potassium alum, $Ca(SO_4)K_2SO_424H_2O$
4. Aluminium potassium alum, $AlSO_4K_2SO_424H_2O$

Complex salts

This type of salt contains two different types of metal ions one of which does not dissociate or dissolve in water. Examples include:

1. Potassium ferricyanide, $K_3Fe(CN)_6$
2. Potassium mercury iodide,K_2HgI_4
3. Chromium ammonium chloride, $Cr(NH3)_6Cl_3$

Table 5.4: Uses of Salts

Name of salt	Formula	Uses
ammonium chloride	NH_4Cl	• in soldering; • as electrolyte in dry cells
sodium bicarbonate	$NaHCO_3$	• in baking powder; • in the manufacture of glass
sodium chloride	NaCl	• for seasoning and preserving food; • essential in life processes
calcium chloride	$CaCl_2$	• as a drying agent to absorb moisture; • in freezing mixtures
silver bromide	AgBr	• in making photographic film
potassium nitrate	KNO_3	• in the manufacture of explosives; • fertilizer production
sodium nitrate	$NaNO_3$	• fertilizer production; • source of nitric acid

Preparation of Salts

As you might have observed, all the chemical properties of acid discussed above give rise to the formation of salt. You may have to take another look at those properties. This shows that there are at least three methods of preparing salt:

Neutralization of acid and base

When an acid and base react, they counteract each other; that is, they neutralize each other. Such a reaction known as a ***neutralization reaction*** results in the formation of water and a salt.

For example, when sodium hydroxide and hydrochloric acid react, water and the salt sodium chloride are formed.

$$NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$$

Base + Acid = Salt + Water

This occurs because the hydrochloric acid and the sodium hydroxide first ionize, and then react. The compounds ionize releasing hydrogen, chloride, sodium, and hydroxide ions.

Since these are mobile in solution, hydrogen ions meet hydroxide ions and unite to form water. At the same time, sodium ions and chloride ions remain as aqueous salt. This method of preparing salt is also called the ***titration method***.

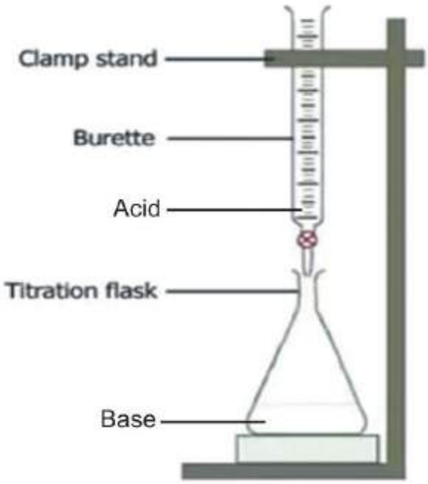


Fig. 17.6: Titration method of salt preparation

Acid and metal

Acids react with metals (the more reactive metals such as magnesium, calcium and iron) to form hydrogen gas and a salt.

$2HCl_{(aq)} + Mg_{(s)} \rightarrow H_{2(g)} + MgCl_{2(aq)}$
Acid + metal = hydrogen + salt

Metal oxide and acid

When a metal oxide reacts with an acid, a salt is formed. For example, when calcium oxide reacts with nitric acid, the salt calcium nitrate is formed.

$CaO + 2HNO_3 \rightarrow Ca(NO_2)_3 + H_2O$
Metal oxide + acid = salt + water

Direct combination

When a metal reacts with a non-metal, a salt is generally formed. For example, when the metal magnesium is burned in chlorine gas, the salt magnesium chloride is formed.

$Mg + Cl_2 \rightarrow MgCl_2$

Salts produced by the above methods are known as ***soluble salts.***

***Precipitation* method**

This method is used to produce salt which is insoluble in water. A solutions of two soluble salts which contain the ions of insoluble salts, (example lead), are mixed together and the precipitate formed is washed with distilled water and dried. For example lead nitrate ($PbNO_3$) and sodium carbonate ($NaCO_3$), both of which are soluble salts, give rise to the insoluble salt lead carbonate.

$PbNO_{3(aq)} + NaCO_{3(aq)} \rightarrow PbCO_{3(s)} + NaCO_{3(aq)}$

Other examples of insoluble salts are calcium tetraoxosulphate (VI), $CaSO_4$; lead (II) chloride, $PbCl_2$; silver chloride, AgCl; and lead (II) tetraoxsulphate (VI), $PbSO_4$. The precipitation method is usually referred to as the ***double decomposition***. That is:

$AB + YZ \rightarrow AZ + BY$

ACID-BASE INDICATORS

The concentration of hydrogen ion is a measure of the acidity and the basicity or alkalinity of a solution. The concentration of hydrogen ion may be expressed in terms of the molarity, M of the acid or base solution; however, it is frequently more convenient to express the concentration as a function of the hydrogen ion concentration, pH (hydrogen potenz or hydrogen power).

pH is a measure of the acidity or alkalinity (basicity) of a solution.

The pH of a solution may be defined as the exponent of the hydrogen ion concentration.

This definition may be stated mathematically as:

$pH = -\log [H^+]$

where $[H^+]$ is the molar hydrogen ion concentration.

pH scale

pH scale is a range or scale used to determine the acidity or alkalinity of a solution.

The pH scale ranges from 1 to 14 only. The pH range from 1 to 6 is acidic. The lower the value, the more acidic it is. That is, the pH of 1 is more acidic than the pH of 6. pH of 7 is neutral. The pH range from 8 to 14 is basic or alkaline; the higher the value, the stronger the alkalinity.

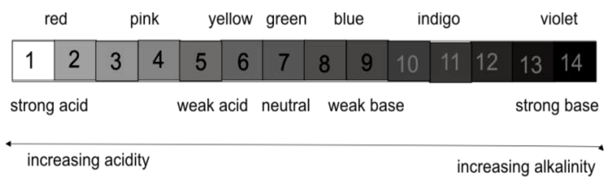


Fig. 17.7: The pH scale

Table 5.5: Range of pH

pH range	Strength
1 - 2	Strong acid
3 – 4	moderate acid
5 - 6	Weak acid
7	Neutral
8 - 9	Weak base/ alkali
10 – 11	Moderate base/ alkali
12 - 14	Strong base/ alkali

Universal indicators

Indicators have been developed in order to assist in the determination of the pH of a solution. These indicators are weak organic acids or bases which have the property of changing colour in solution when the hydrogen ion concentration reaches a definite value. An acid indicator may be represented by the equation:

$$\mathbf{HIn} = \mathbf{H^+} + \mathbf{In^-}$$

The anion, In^-, represents a complex organic group which has changed its structure due to the loss of a hydrogen ion. The loss of the hydrogen ion is accompanied by a change in colour. Since an indicator reaction is an equilibrium reaction, the addition of hydrogen ions would force the above reaction to the left and a colour indicating an acid solution would result. The addition of hydroxide ions would cause the reaction to go to the right and a colour associated to a basic solution would result.

The pH ranges of some indicators are given in table 5.6 below. With this table, you can estimate the pH of a solution. Suppose phenolphthalein is introduced into a solution and the colour of the solution becomes red; this red colour indicates that the pH of the solution is 10.0 or higher. If indigo carmine is added to a new sample of the same solution and a blue colour results, the pH will be narrowed to a range of 10.0 to 11.4, since the lower limit of colour change for indigo carmine is blue. By using an additional indicator or indicators and a new sample of the solution, the pH of the solution can be narrowed to a small range.

Fig. 17.8: acidity or alkalinity of food samples

Table 5.6: Universal indicators and their ranges

Indicator	pH range	Colour range
Methyl violet	0.0 – 1.6	Yellow to blue
Thymol blue	1.2 – 2.8	Red to yellow
Methyl orange	3.2 – 4.4	Red to yellow
Congo red	3.0 – 5.0	Blue to red
Methyl red	4.8 – 6.0	Red to yellow
Phenol red	6.6 – 8.0	Red to blue
Litmus	4.7 – 8.2	Red to blue
Cresol red	7.4 – 8.6	Yellow to red
Phenolphthalein	8.2–10.0	Colourless to red
Thymolphthalein	9.4– 10.6	Colourless to blue
Alizarin yellow R	10.1 – 12.0	Yellow to red
Indigo carmine	11.4 – 14.0	Blue to yellow

Importance of pH

1. The pH of a solution can be used to describe the acidic or basic qualities of food and beverages.
2. It is used to check the pH of soil.
3. It is used to determine the pH of drugs in the pharmaceutical industry.
4. It is used to ensure that soaps are neither acidic nor basic, hence neutral.
5. It enables cosmetic manufacturers to check the pH of their products.

TEST QUESTIONS

1. a) Define the following salts and state one example of each and the reaction that takes place in their formation.
   i) normal salt
   ii) acidic salt
   iii) basic salt
   iv) double salt

   b) Write a short note on acids and bases, based on the Bronsted-Lowry concept.

   c) State three physical and two chemical properties each of acids and bases.

4. (a) Write an equation for the reaction that takes place and the products formed when:
   i) acid reacts with base
   ii) acid reacts with metal
   iii) acid reacts with carbonate
   iv) base reacts with ammonium salt.

   (b) Write a short note on double decomposition.

5. (a) Explain how neutralization reaction occurs.

   (b) Write three equations to justify your answer in (a) above.

   (c) Tetraoxosulphate(VI) reacts with calcium hydroxide to form calcium tetraoxosulphate and water.
   i) Write an equation for this reaction.
   ii) Explain this reaction in terms of proton donors and proton acceptors.

7. a) Name two examples of each of the following:
   i) Strong acid
   ii) Weak acid
   iii) Strong base

iv) Weak base

(b) Name two common items at home which can be made from

i) Sodium chloride

ii) Sodium hydroxide

iii) Sulphuric acid

8. a) What is pH scale?

b) Give the range of a pH scale and explain its significance

c) Compare the H+ ion concentration of 0.1MCHOOH with 0.1M HCl.

8. Study the reaction of liquid A with the named substance in the table and complete the inference column.

Experiment	***Observation***	***Inference***
i) liquid A + red litmus liquid A + blue litmus	Litmus remains red Litmus turns red	Liquid is............
ii) liquid A + sodium trioxocarbo-nate (IV)	Effervescence of colourless gas. Gas turns limewater milky	The gas is............
iii) liquid A + zinc	Bubbles of colourless gas. The gas burns with 'pop' sound.	The gas is ....

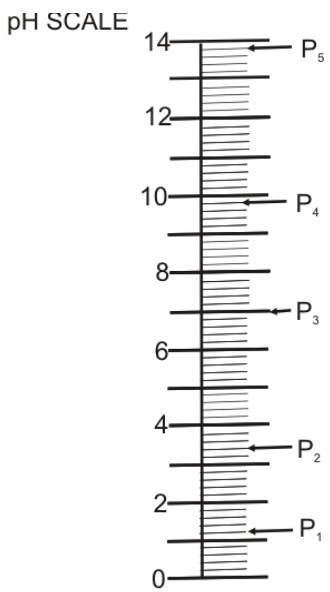
b) Name a compound which reacts in a similar way as liquid A.

c) State the colour of phenolphthalein in

i) calcium hydroxide solution;

ii) lemon juice.

9. Using the universal indicator, a student measured the pH of water, ethanoic acid, dilute sodium hydroxide solution, dilute hydrochloric acid and sodium bicarbonate solution and recorded the value on a vertical scale. The pH value P = P_1, P_2, P_3, P_4, P_5 which were obtained are shown on the vertical scale below.



(a) Read and record each of the pH values, P = P_1, P_2, P_3, P_4 and P_5.

(b) Match each of the pH values P = P_1, P_2, P_3, P_4 and P_5 against the appropriate liquid. Present your answer in a tabular form.

(c) Give a brief reason in each case for assigning each liquid a particular pH value.

24

SOIL CONSERVATION

Specific Objectives

After completing this chapter, you will be able to:

- Explain the principles of soil and water conservation.
- Distinguish between macro and micro nutrients.
- State the functions of some soil nutrients and their deficiencies symptoms.
- Describe methods of maintaining soil fertility.
- Classify fertilizers into organic and inorganic fertilizers.
- Outline factors which lead to the depletion of soil nutrients.

INTRODUCTION

Soil is the loose material that covers the land surfaces of the Earth and supports the growth of plants.

In general, soil is an unconsolidated or loose combination of *inorganic* and *organic* materials.

The inorganic components of soil are principally the products of rocks and minerals that have been gradually broken down by weather, chemical action, and other natural processes. The organic materials are composed of debris from plants and from the decomposition of the many tiny life forms that inhabit the soil.

PRINCIPLES OF SOIL AND WATER CONSERVATION

Soil actually constitutes a living system, combining with air, water, and sunlight to sustain plant life.

The essential process of photosynthesis, in which plants convert sunlight into energy, depends on exchanges that take place within the soil. Plants, in turn, serve as a vital part of the food chain for living things, including humans.

Without soil there would be no vegetation: no crops for food, no forests, flowers, or grasslands. To a great extent, life on Earth depends on soil. In view of this, it is very essential for the soil to be conserved to sustain life.

To conserve the soil is to prevent the soil from being destroyed.

Soil conservation is the act of protecting the soil and preventing it from destruction.

The elements of destruction could be natural or man-made. There is a variety of ways to conserve soil water and maintain soil fertility.

CONSERVING SOIL MOISTURE

To conserve the moisture or water content in the soil thereby ensuring that plant get ample supply for photosynthesis and other activities, the following practices must be undertaken.

Afforestation

Planting trees, shrubs etc. on a dry land are a sure way to improve the moisture content of the soil. Plants prevent direct rays from the sun on the soil, which minimizes the rate by which the soil loses water. Soil that is under a vegetation cover has hardly any chance of getting eroded as the vegetation cover acts as a wind barrier as well.

Terraces

Terracing is one of the very good methods of soil conservation. A terrace is a levelled section of a hilly cultivated area. Owing to its unique structure, it prevents the rapid surface runoff of water. Terracing gives the landmass a stepped appearance, thus slowing the easy washing down of the soil.

No-till farming

When soil is prepared for farming by ploughing it, the process is known as ***tilling***. No-till farming is a way of growing crops without disturbing it through tillage. The process of tilling is beneficial in mixing fertilizers in the soil, shaping it into rows and preparing a surface for sowing. However, the tilling activity can lead to compaction of soil, loss of organic matter in soil and the death of the organisms in soil. No-till farming is a way to prevent the soil from being affected by these adversities.

Contour ploughing

The practice of farming across the slopes takes into account the slope gradient and the elevation of soil across the slope. It is the method of ploughing across the contour lines of a slope. This method helps in slowing the water runoff and prevents the soil from being washed away along the slope. Contour ploughing also helps in the percolation of water into the soil.

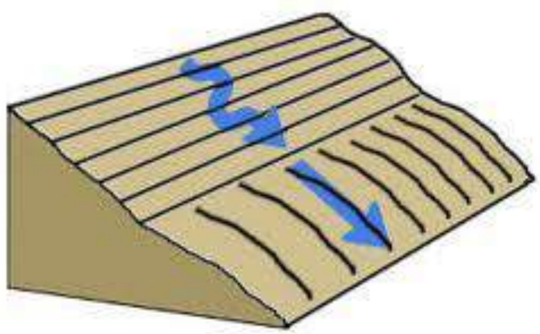


Fig. 17.9: Contour ploughing

Crop rotation

Some pathogens tend to build up in soil if the same crops are cultivated consecutively. Continuous cultivation of the same crop also leads to an imbalance in the fertility demands of the soil. To prevent these adverse effects from taking place, crop rotation is practiced. It is a method of growing a series of dissimilar crops in an area sequentially. Crop rotation also helps in the improvement of soil structure and fertility.

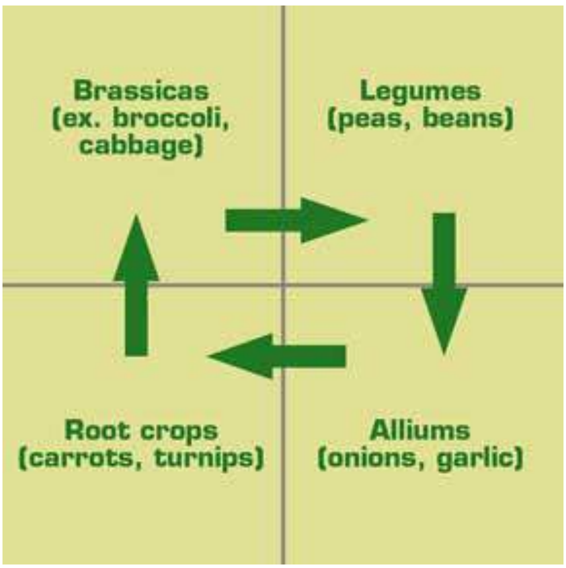


Fig. 18.0: Example of crop rotation

Soil pH

The contamination of soil by addition of acidic or basic pollutants and acid rains has an adverse effect on the pH of the soil. Soil pH is one of the determinants of the availability of nutrients in soil. The uptake of nutrients in plants is also governed to a certain extent, by the soil pH. The maintenance of the most suitable value of pH is thus essential for the conservation of the soil.

Irrigation

Irrigation is an artificial application of water to the soil. Irrigation is most convenient during the dry season, when the soil loses most of it moisture to heat from the sun. Watering the soil along with the plants is a way to prevent soil erosion caused by wind.

Soil organisms

Organisms like earthworms and others benefiting the soil should be promoted. Earthworms, through aeration of soil, enhance the availability of macronutrients in the soil. They also enhance the porosity of the soil. The helpful organisms of the soil promote its fertility and form an element in the conservation of soil.

Indigenous Crops

Planting of native crops is known to be beneficial for soil conservation. If non-native plants are grown, the fields should be bordered by indigenous crops to prevent soil erosion and achieve soil conservation.

Mulching

This is the act of covering the surface of the land with some protective materials to retain moisture, reduce erosion, provide nutrients, suppress weed growth and enhance seed germination.

Mulch may be manure, sawdust, leaves and grass, cereal chaff, peat moss, straw, or even stones. Organic materials used for mulching, in addition to protecting the plants, decay in time and enrich the soil. Natural mulch is formed by fallen leaves and by decaying non-woody plant parts.

Addition of organic matter

Organic matter in the soil sticks the loose particles of the soil together, thereby helping the soil maintain or improve its water holding ability.

Cover cropping

Cover crops are special crops which are planted primarily to maintain the moisture level of the soil, and by so doing making the soil more fertile.

NB: *The above soil and water conservation methods also help in maintaining the fertility of the soil.*

SOIL NUTRIENTS

The crop products we are consuming today meet our nutritional needs as a result of nutrients applied to soils as fertilizer, livestock manure, or crop residues.

Soil nutrients are substances that are added to the soil to improve its fertility.

Classification of soil nutrients

The essential nutrients needed by the soil have been grouped into two – ***macro-nutrients*** and ***micro-nutrients***.

Macro/ major nutrients

Macro-nutrients, also known as major nutrients are the nutrients needed by plants in large quantities. Examples of macro nutrients are nitrogen (N), phosphorous (P), potassium (K). Calcium (Ca), magnesium (Mg), and sulphur (S). Complete fertilizers contain at least nitrogen (N), phosphorus (P) and potassium (K). These are the most essential macro nutrients.

Bags of complete fertilizers contain three numbers, such as 5-3-3, for example. Each number represents a percentage of N-P-K in that bag, as measured by weight. In this case, a bag of 5-3-3 fertilizer contains 5% nitrogen, 3% phosphorous and 3% potassium.

Each of these three nutrients plays a critical role in plant growth and development.

Table 5.7: Soil nutrients, their functions and deficiency symptoms

Macro-nutrient	Function	Deficiency symptoms
nitrogen (N)	• Important component of amino acids and proteins	• Old leaves turn yellow • Plant growth retarded; small leaves. • Too much nitrogen leads to overgrown plants which are highly susceptible to diseases.
potassium (K)	• Important in maintaining cell turgor, phloem transport, cell growth and cell wall development	• K deficiency leads to susceptibility to pests because cell walls are weakened. • Older leaves show first chlorotic, later necrotic borders. • Younger leaves remain small
phosphorus (P)	• Provides energy (ATP) • Helps intransport of assimilates during photosynthesis • Important functions in fruit ripening	• Small plants with erect growth habit; thin stems, slow growth. • Leaves appear dirty grey-green, sometimes red.
calcium (Ca)	• Stabilizes cell membranes and cell walls • Interacts with plant hormones • Ca is extremely immobile and can only be taken up through young, unlignified roots.	• Deficiency is often only visible in retarded growth.
magnesium (Mg)	• Component of chlorophyll– photosynthesis is hindered when deficient. • Binds ATP to enzymes. • Important for protein synthesis.	• Old leaves become chlorotic from middle or between veins, rarely necrotic. Leaves turn orange-yellow • Leaves drop prematurely.
sulphur (S)	• Component of etheric oils, vitamin B, vitamin H, amino acids • has important functions in protein synthesis	• Similar to N-deficiency but symptoms show first on young leaves.
Micro-nutrients	**Function**	**• Deficiency symptoms**
iron (Fe)	• Component of chloroplasts • Part of the redox system in the electron transport during assimilation • important for RNA synthesis.	• Young leaves turn yellow and then, white.
manganese (Mn)	• Important for enzyme activation, *photolysis*.	• When deficient, protein synthesis and carbohydrate formation are hindered. • Youngest leaves show chlorotic spots, later they grow into • Necrotic areas parallel to the veins.
zinc (Zn)	• Has enzyme activating function,	• Small leaves and short internodes;

	e.g. starch synthetase • Found in chloroplasts.	thin shoots
copper (Cu)	• Found in chloroplasts. Important for carbohydrate synthesis and protein synthesis	• Youngest leaves are chlorotic or necrotic, fruit set is insufficient.
Molybdenum (Mo)	• Important component of enzymes, specifically nitrate reductase and nitrogenase. • Essential element for all nitrogen-fixing plants.	• Old leaves develop necrotic borders, often the symptoms are caused by secondary nitrogen deficiency.
chlorine (Cl)	• Important in maintaining cell tugor, increases sugar content in fruits.	• Deficiency symptoms occur only • In halophytes (salt-loving plants), mainly as loss in turgor
boron (B)	• Found in cell walls • Important for transport of assimilates and cell growth • If deficient, shoot tip dries.	• Youngest leaves are deformed, thick, dark green to grayish • Root system development is hindered.

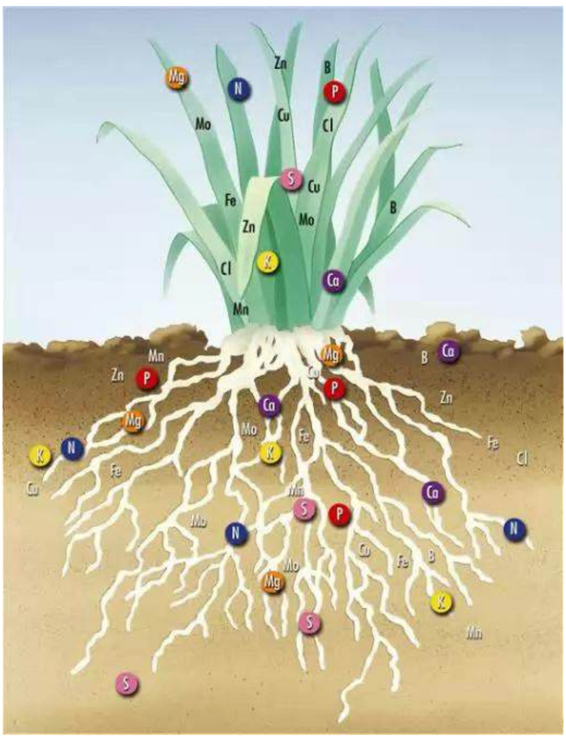


Fig. 18.1: Movement of nutrients in a plant

SOIL FERTILITY

Soil fertility refers to the amount of nutrients in the soil which is sufficient to support plant life.

A fertile soil produces high plant yield.

Characteristics of a fertile soil

To be fertile soil needs:

1. macro-nutrients and micro-nutrients
2. organic matter
3. suitable ph value
4. good moisture
5. good aeration
6. micro and macro-organisms
7. to be well drained

Methods of maintaining soil fertility

The fertility of the soil can be maintained or improved. The following are some of the measures to take in order to maintain the fertility of the soil.

Application of organic and inorganic manure/ fertilizer

Soil used for growing crops must have plant nutrients and organic matter added in order to maintain the fertility and quality of the soil. Soil which is well cared for will continually produce good yields. If plant nutrients and organic matter are not added the soil will become exhausted after a few years and crop yields will drop.

Crop rotation

This is the act of growing different types of crops on the same piece of land in succession. Crop rotation is important because some plants add nutrients to the soil; for example, leguminous plants add nitrogen to the soil. Other plants take more nutrients from the soil. Different plants have different regions in the soil from which they acquire nutrients. Therefore, employing crop rotation ensures that a particular group of plant does not suck all the nutrients from the region where it gets its nutrients.

Cover cropping

There are some periods in a year, (especially during the dry season), when most plants do not grow well. This is the time to grow cover crops. Cover cropping involves growing special crops to protect and maintain the moisture of the land. If a leguminous crop is used, it fixes nitrogen in the soil. Cover crops also serve as a source of green manure.

Green Manuring

Some crops, also known as green-manure crops, are grown solely to be ploughed into the soil to increase the organic-matter content of the soil. This is termed as green manuring. Although no yield is expected of a green-manure crop, it is supposed to increase the yield of subsequent crops planted on the same fields.

Liming

Most plants are acidic, and as they grow in the soil, they add more acids to the soil. Acid rain also increases the acidity of the soil. Since a soil needs a suitable pH in order to be fertile, it is advisable to reduce the acidic content of the soil. Calcium hydroxide (CaOH), also known as lime, is normally added or ploughed into the soil to neutralise the excess acid. The practice of adding Calcium hydroxide or lime to the soil is termed as ***liming***.

Other methods of maintaining the fertility of the soil, such as mulching, irrigation, afforestation, terracing, etc. have been explained above under ways of conserving soil.

FERTILIZERS

Fertilizers are natural or artificial substances or mixtures used to enrich soil so as to promote plant growth.

Plants require different chemical elements and these elements must be present in such forms as to allow an adequate availability for plant use.

Classification of fertilizers

Fertilizers are generally classified as ***organic*** or ***inorganic.***

Organic fertilizers

Organic fertilizers are fertilizer compounds that contain one or more kinds of organic matter. The ingredients may be animal or vegetable matter or a combination of both. It is possible to purchase commercial brands of organic rich fertilizer as well as prepare organic fertilizer at home by building a compost heap. That is, the fertilizer is composed of elements that are produced in a completely natural manner, without the aid of any artificially manufactured components or additives. Organic fertilizers are also known as ***organic manure***.

Organically prepared fertilizing agents can be used to grow vegetables, raise flowers, produce a green lawn, etc.

Application of organic fertilizers

As with any type of fertilizing product, the organic fertilizer is spread evenly across the expanse of the yard. Often, the fertilizer application process takes place before the seeds are sown. Once the soil is properly tilled and mixed with the organic fertilizer, the seeds are distributed and the area is watered. The presence of the natural materials helps the seeds to sprout and take root.

Examples of organic fertilizers

1. Green manure
2. Farmyard manure
3. Compost
4. Bone meal
5. Fish meal
6. Guano

Advantages of organic fertilizers

1. Organic fertilizers are not easily leached out by rain or irrigation.
2. They are relatively cheaper.
3. Nutrients stay in the soil for long.
4. They help the soil retain water.
5. They aerate compact soils and bind loose soils.
6. They serve as a source of food for micro-organisms.
7. Their application does not require an expert to do.
8. Over use of organic fertilizers does not harm the soil.

Disadvantages of organic fertilizers

1. Organic fertilizers are bulky and are difficult to transport.
2. Nutrients take a long time to reach the crop.
3. They normally contain weeds, disease pathogens or unwanted seeds.
4. The exact amount of nutrients they contain cannot be determined.
5. Large quantities are needed at a time.

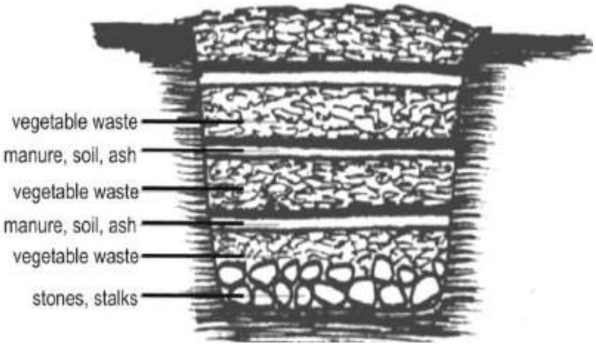


Fig. 18.2: Compost heap

Compost

Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment. Compost is a key ingredient in organic farming. At the simplest level, the process of composting simply requires making a heap of wetted organic matter (leaves, "green" food waste) and waiting for the materials to break down into humus after a period of weeks or months.

The decomposition process is aided by shredding the plant matter, adding water and ensuring proper aeration by regularly turning the mixture. Worms and fungi further break up the material. Aerobic bacteria manage the chemical process by converting the inputs into heat, carbon dioxide and ammonium. The ammonium is further converted by bacteria into plant-nourishing nitrites and nitrates through the process of ***nitrification***.

Importance of compost

1. It is used in gardens, landscaping, horticulture, and agriculture.
2. It is beneficial to the land in many ways, including as a soil conditioner, a fertilizer and addition of vital humus or humic acids.
3. In ecosystems, compost is useful for erosion control, land and stream reclamation, wetland construction and as landfill cover.
4. Organic ingredients intended for composting can alternatively be used to generate biogas through anaerobic digestion.

Inorganic fertilizers

Inorganic fertilizers**,** also known as chemical fertilizers, are made with different chemical compounds to suit specified uses.

Chemical fertilizer usually comes in either granular or powder form in bags and boxes, or in liquid formulations in bottles.

The different types of chemical fertilizers are usually classified according to the three principal elements, namely *Nitrogen* (N), *Phosphorous* (P) and *Potassium* (K), and may, therefore, be included in more than one group.

Inorganic fertilizers come in two main types:

- single fertilizers
- compound fertilizers

Single or straight fertilizers

They are also referred to as simple or straight fertilizers and contain only one macro nutrient, such as nitrogen, N; phosphorus, P; or potassium, K.

Examples of straight fertilizer

1. Urea, ammonium nitrate, sodium nitrate (that contain nitrogen)
2. Single, double or triple superphosphate, rock phosphate (contain phosphorus)
3. Potassium sulphate, murate of potash (contain potassium)

Compound fertilizers

They are also called mixed or complex fertilizers. While single fertilizers contain only one macro-nutrient, compound fertilizers have the advantage of containing either all or at least two of the macro-nutrients in specific amounts.

Examples of compound fertilizer

1. NPK 15-15-15 (contain 15% nitrogen N, 15% phosphorus P and 15% potassium K)
2. NPK 10-26-26 (26% N, 26% P, 26% K)
3. NPK 15-15-0 (15% N, 15% P, 0% P)
4. Mono potassium phosphate
5. Urea ammonium phosphate
6. Nitrophosphate
7. Ammonium phosphate sulphate nitrate

Advantages of inorganic fertilizers

1. Inorganic fertilizers are less bulky and are easier to transport.
2. They are available to the plant relatively quickly.
3. They normally contain no weeds, no disease pathogens or unwanted seeds.
4. The exact amount of nutrients needed by the plant is calculated and given out.
5. Only a small quantity is needed at a time

Disadvantages of inorganic fertilizers

1. Inorganic fertilizer, particularly nitrogen, is easily leached out by rain or irrigation.
2. They are relatively expensive.
3. Nutrients do not stay in the soil for long.
4. Improper application and measurement can destroy the crop and pollute the land.

5. They can pollute surrounding water bodies and alter the soil pH.

Fig. 18.3: A bag of compound fertilizer

Fertilizer application

Fertilizer application is a necessary part of successfully growing many plants, especially when the yield of the crop is important for food production or flower growth. Fertilizers add nutrients to the soil, which are then absorbed by plant.

Factors to consider when selecting fertilizer application method

i. Rooting characteristic of the crop to be planted.
ii. Crop demand for various nutrients at different stages of growth.
iii. Physical and chemical characteristics of the soil.
iv. Physical and chemical characteristics of the fertilizer material to be applied.
v. Availability of moisture.
vi. Type of irrigation system used if irrigation is the only or major source of water.
vii. Frequency and rate of irrigation water to be applied.

Methods of fertilizer application

Broadcasting

Broadcasting is among the simplest ways to apply fertilizer to your plants in a medium to large area. This process involves distributing the fertilizer over the top of the soil. This method can be used with liquid or solid fertilizers and is typically done using a spreading machine or by hand. Broadcasting is normally done before or immediately after the crops start germinating.

Injection/ band placement

In the band placement method, the fertilizer is put directly into the soil in order to make immediate contact with the root system.

Side dressing

This involves injecting the fertilizer into the soil on one or both sides of the plants which are growing in rows. This method minimizes nutrient loss by leaching.

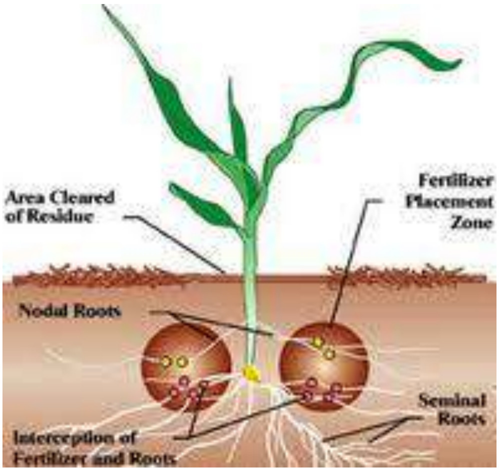


Fig. 18.4: Side dressing fertilizer application

Foliar application

Also known as top dressing, entails applying the fertilizer on the foliage (leaves) of the crops. Foliar application is more effective with liquid fertilizers, which can easily be sprayed or sprinkled onto the growing crops.

Ring method

The fertilizer is placed in a small circular furrow made in the soil around each plant. The furrow is then covered with soil.

Fig. 18.5: A farmer using the ring method

FACTORS WHICH LEAD TO THE DEPLETION OF SOIL

1. Erosion
2. Overgrazing
3. Poor farming and tillage methods
4. Dumping of non- biodegradable waste on land
5. Improper irrigation and drainage practices
6. Surface mining and quarrying.
7. Deforestation.
8. Excessive use of fertilizer

Soil erosion

Soil erosion is the washing away of the nutrient-rich top soil from one place to another.

Though soil erosion is a natural process caused by agents such as wind, running water, rain, ice or even gravity, man's activities such as deforestation, construction, etc. have increased it impacts on the environment.

Types of soil erosion

There are two main types of soil erosion, named after the two principal causes of soil erosion - wind and water. Thus, wind erosion and water erosion.

Wind erosion

In arid or dry climates, the main cause of erosion is wind. There are two main effects of wind erosion.

First, wind causes small particles to be lifted and therefore moved to another region. This is called ***deflation*.**

Second, these suspended particles may impact on solid objects causing erosion by ***abrasion*.**

Wind erosion generally occurs in areas with little or no vegetation, often in areas where there is insufficient rainfall to support vegetation.

Water erosion

The effects of water erosion are seen all around us. It is caused by fast running water overland or rain splashing upon the earth.

Based on the mode of erosion, water erosion has been split into four main groups. They are splash erosion, sheet erosion, rill erosion and gulley erosion.

Types of water erosion

Splash erosion

This is caused by the impact of raindrops on soil. When rain drops splash on the soil, soil particles are detached and easily carried away from their original place.

Sheet erosion

This normally occurs down slope and usually follows splash erosion. When splash erosion occurs and the soil particles are loose, fast running water from rain or river spanning a wider area wash the loose soil particles away. Sheet erosion generally travel shorts distances and last only for a short time.

Rill erosion

Rill erosion refers to the development of small, gutter-like flow paths, caused as a

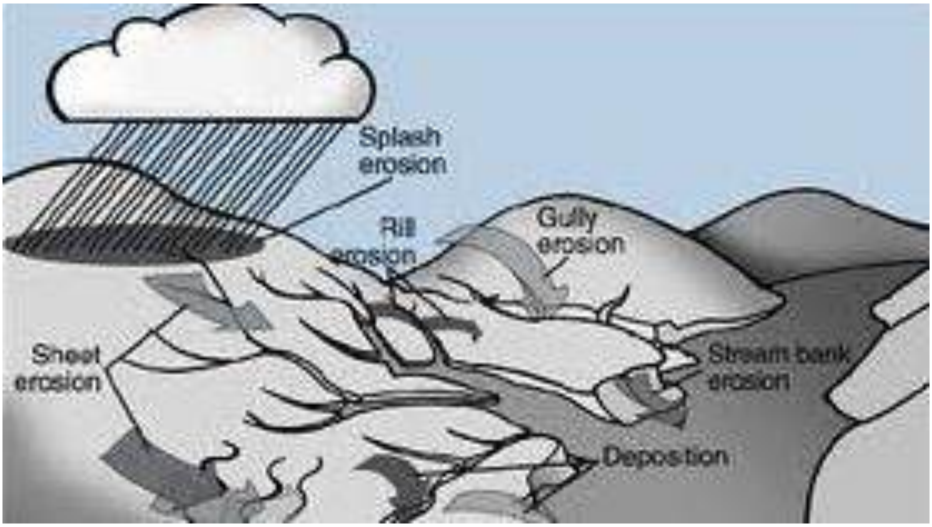


Fig. 18.6: Types of water erosion

result of streams of running water moving in different ways mostly parallel to each other. Rill eroded land can be worsened by gully erosion

Gully erosion

This occurs when water flows in narrow channels caused by rill erosion during or immediately after heavy rains or melting snow. This is particularly noticeable in the formation of hollow ways or large gutters. Gully erosion normally distorts the topography of the land, making the land undulated.

Fig. 18.7: Gully erosion

Apart from the two main types of erosion, there are other types of soil erosion such as

- ❖ Ice erosion
- ❖ Shoreline erosion
- ❖ Gravity erosion
- ❖ Water bank erosion

Causes of soil erosion

Natural causes of soil erosion include:

1. Water (especially rain)
2. Wind
3. Ice
4. Gravity
5. Nature of the land (soil type, slope)

Human activities that cause soil erosion

1. Deforestation
2. Overgrazing by farm animals
3. Sand winning
4. Construction activities
5. Bush burning
6. Monoculture or over-cropping
7. Over-population

Effects of soil erosion

1. Loss of soil fertility
2. Deformation of land's topography
3. Eroded soil can pollute water bodies
4. Destruction of buildings, roads and other structures
5. Roots of plant may be exposed causing the plant to fall down or die

Ways to control soil erosion

1. Terracing
2. Mulching
3. Crop rotation
4. Addition of organic fertilizers to the soil
5. Contour ploughing
6. Avoidance of bush fires

7. Afforestation
8. Proper drainage system

TEST QUESTIONS

1. a) Define soil.
   b) Explain the concept of soil conservation.
   c) Outline five activities that farmers can undertake to enhance soil productivity.

2 a) Write a short note on four methods by which soil can be conserved.
   b) Outline four ways by which soil moisture can be conserved.
   c) Outline four methods of maintaining soil productivity.

3 Copy and complete the table below

Nutrient	*Function*	*Deficiency symptom*
Nitrogen		
Potassium		
Phosphorus		
Iron		
Manganese		
Boron		

4 a) What are soil nutrients?
   b) Distinguish between macro and micro nutrients.
   c) List four macro and four micro nutrients, their functions and symptoms of their deficiency.

5 a) What are fertilizers?
   b) Write a short note on the two classes of fertilizers.
   c) State three advantages and three disadvantages each of organic and inorganic fertilizers,

6 a) Write on three methods of applying fertilizers.
   b) Mention four factors to consider when selecting a fertilizer application method.
   c) Enumerate five factors that lead to the depletion of soil nutrients.

7. a) What is soil erosion?
   b) State six causes of soil erosion.
   c) Explain the four types of water erosion.

8 a) Mention four effects of soil erosion.
   b) Itemize five ways of controlling soil erosion.

8. a) What is soil fertility?
   b) Mention five characteristics of a fertile soil.
   c) State four ways by which soil loses its fertility.
   d) Enumerate four ways by which the fertility of the soil can be maintained.

9. a) What are macronutrients?
   b) A farmer bought a bag of compound fertilizer labelled NPK 20–0-20. Describe the nature of the fertilizer in the bag.

10. Describe four methods that a crop farmer can adopt to improve the fertility of his land.

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WATER

Specific Objectives

After completing this chapter, you will be able to:

- ❖ Describe the physical and chemical properties of water.
- ❖ Distinguish between hard and soft water.
- ❖ Demonstrate how to soften hard water.
- ❖ Describe the steps involved in the treatment of water for public consumption.

INTRODUCTION

Water is the commonest compounds everyone might have come across, covering about 70% of the earth. If you have seen, tasted, felt or smelled water before, you may easily describe water as being colourless, tasteless, or odourless. Technically, water is a compound of two atoms of hydrogen and an atom of oxygen; expressed as H_2O.

Sources of water

Water can be obtained from:

1. The sea
2. Lakes
3. Rivers
4. Lagoons
5. Rain
6. Wells
7. Pipe-borne/ taps

PROPERTIES OF WATER

Physical properties

1. Water is colourless, tasteless, odourless.
2. It boils at 100^oC and freezes at 0^oC.
3. It is neutral, hence has a pH of 7.
4. It has a maximum density of $1gcm^{-1}$ at 4^oC.
5. It expands when heated from -4^oC and 0^oC and contracts when melted from 0^oC to 4^oC. This is the reason why ice floats on water; ice has lower relative density.
6. Water has a high *specific heat.*

7. It conducts heat more easily than any liquid except mercury. This fact causes large bodies of water, like lakes and oceans, to have essentially a uniform vertical temperature profile.

8. Its molecules exist in liquid form over an important range of temperature from 0^oC - 100°C. This range allows water molecules to exist as a liquid in most places on our planet.

9. It is a universal solvent. It is able to dissolve a large number of different chemical compounds. This feature also enables water to carry solvent nutrients in runoff, infiltration, groundwater flow and living organisms.

10. Water has a high surface tension. In other words, water is adhesive and elastic, and tends to aggregate in drops rather than spread out over a surface as a thin film. This phenomenon also causes water to stick to the sides of vertical structures despite gravity's downward pull. Water's high surface tension allows for the formation of water droplets and waves, allows plants to move water (and dissolved nutrients) from their roots to their leaves, and the movement of blood through tiny vessels in the bodies of some animals.

Determining the boiling point of water

- ❖ Pour water into a clamped boiling tube.
- ❖ Put a thermometer in the water.
- ❖ Record the reading on the thermometer.
- ❖ Heat the setup for two minutes and note how the reading on the thermometer rises as the water heats up.
- ❖ Note the temperature reading when the water begins to boil.

Observation

It would be observed that when the water starts to boil the temperature reading will be 100^oC and remain so no matter how long the water boils. This shows that the boiling point of water of water is 100^oC.

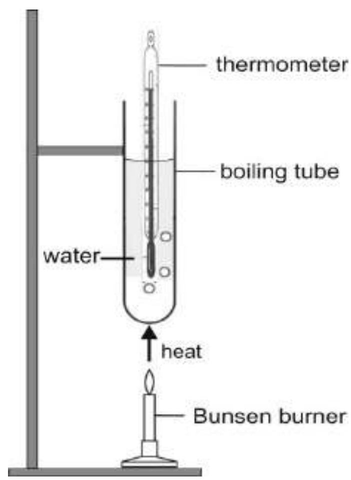

Fig. 18.8

Chemical properties of water

1. Water reacts with metallic oxides to form alkaline solutions and hydrogen gas. For example:
   $CaO + 2H_2O \rightarrow 2CaOH + H_2$

2. Water reacts with non-metallic oxides to form acidic solutions. For example:
   $H_2O + SO_2 \rightarrow H_2SO_3$

3. Water reacts with metals to form metal oxides and give off hydrogen gas. For example:
   $2H_2O + 2K \rightarrow 2KHO + H_2$

4. Anhydrous white copper (II) tetraoxsulphate (VI) reacts with water to form blue pentahydrate salt.

   $CuSO_4 + 5H_2O \rightarrow CuSO_4.5H_2O$

5. Dry blue cobalt (II) chloride reacts with water to form red or pink hexahydrate salt.
   $CoCl_2 + 6H_2O \rightarrow CoCl_2.6H_2O$

Test for water

Not all liquids are or contain water; and a solid substance may contain water which is not observable with the naked eye. A Test for water must be used to identify water.

Testing whether a liquid is water

- Add a drop of the liquid to white anhydrous copper sulphate.
- The spot where the water droplet fell turns blue immediately.
- This shows that the liquid is or contain water.

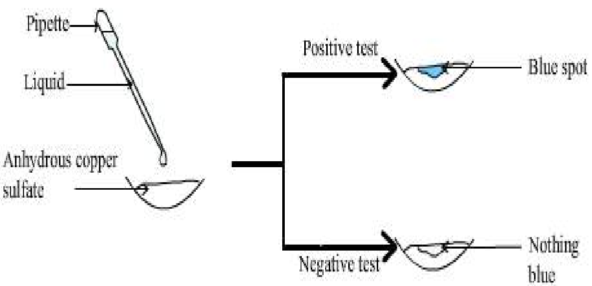

Fig. 18.9

Testing for water in a solid

- Put a sample of anhydrous copper sulphate on the solid.
- The area turns blue.
- This shows that the solid contains water.

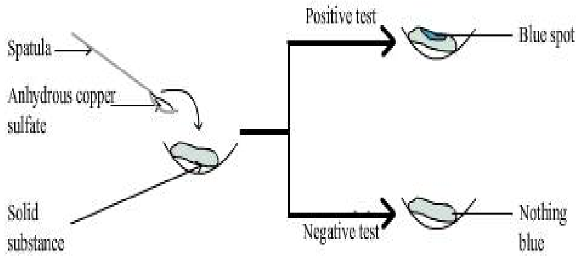

Fig. 19.0

HARDNESS AND SOFTNESS OF WATER

How does water become hard or soft? You might ask. Well, since water is such a good solvent, during its journey from its source it dissolves and picks up some compound such as calcium, magnesium or iron compounds and other impurities. These compound and impurities affect the properties of water and may make it be hard.

Hard water

Water is said to be hard if it does not lather easily with soap.

Such water usually contains magnesium, Mg^{+}, calcium, Ca^{+} or iron (III), Fe^{+} ions. Hard water does not lather because the ions react with the soap to form an insoluble scum.

Soft water

Soft water is the water which lathers readily with soap.

Since soft water does not contain dissolved ions which interfere with the action of soap, it lathers easily with soap.

Types of hard water

There are two types of hard water, namely:

- Temporal hard water
- Permanent hard water

Temporal hard water

This is the type of hardness caused by the presence of dissolved calcium hydrogen carbonate, ($Ca(HCO_3)_2$ in the water.

This occurs when rainwater, which contains dissolved atmospheric carbon (II) oxide, comes in contact with limestone, calcium carbonate, $CaCO_3$, as it runs through rocks or soil which contains the compound ($CaCO_3$). This reaction can be expressed as:

$$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$$

Removal of temporary hardness

By boiling or heating

When temporary hard water is heated, the reaction that causes it is reversed. That is calcium hydrogen carbonate, $Ca(HCO_3)$, reverses to form a white solid insoluble calcium trioxocarbonate (IV), $CaCO_3$, carbon dioxide, CO_2 and water, H_2O. The reaction is given as:

$$Ca(HCO_3) \rightarrow CaCO_3 + CO_2 + H_2O$$

By adding calcium hydroxide

A calculated amount of lime, calcium hydroxide, can be added to temporary hard water to cause calcium carbonate to float on the water, which can be filtered off easily. The equation is given as;

$$Ca(OH)_2 + Ca(HCO_3)_2 \rightarrow 2CaCO_3 + 2H_2O$$

Permanent hard water

Permanent hardness is caused by dissolved calcium, Ca^{+}, magnesium mg^{+} or iron Fe^{+} ions. The above ions are contained in salts such as $CaSO_4$, $MgSO_4$ or $FeSO_4$, which are contained in rocks. When runoff water comes into contact with these salts, the ions dissolve and flow into water sources.

Permanent hardness cannot be removed by boiling, but by chemical means.

Fig. 19.1: Scale formed by hard water on heating element of kettle

Removing or softening permanent hardness in water

1. Distillation

When water is boiled, it vaporizes. The vapour can then be condensed or cooled back into pure water in another container, leaving the ions that cause hardness in the original container.

2. Addition of washing soda, Na_2CO_3

Sodium carbonate removes soluble calcium and magnesium ions as insoluble calcium or magnesium carbonate ($CaCO_3$ or $MgCO_3$) that float on the surface of the water and can easily be filtered off. The reaction is given as:

$$Na_2CO_3 + Ca^{2+} \rightarrow CaCO_3 + 2Na^{+}$$

3. Addition of caustic soda, NaOH

NaOH removes soluble calcium and magnesium ions as insoluble hydroxides. The equation is given as:

$$2NaOH + Ca^{2+} \rightarrow Ca(OH)2 + 2Na^{+}$$

4. De-ionization or ion exchange

This is the use of a special device known as ***ion exchanger*** to remove the ions, e.g. Ca^{2+}, Mg^{+}, responsible for hardness by exchanging them for other ions that do not cause hardness such as sodium, Na^{+} ion.

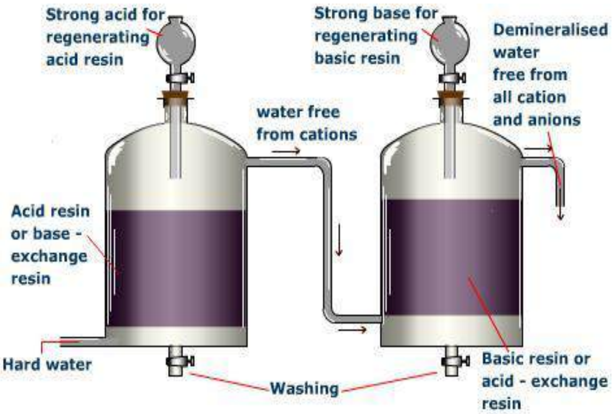

Fig. 19.2: Ion exchange process

Testing for hardness in water

- Fill a beaker with sample of water from a river.
- Add some drops of liquid soap to the water and stir or shake it.
- Check whether the water lathers.
- Boil the solution and add and a few drops of liquid soap to it.
- Compare the lathering ability of the water before and after boiling.
- Check the beaker for scum (white powdery substance).

Observation

If the water lathers more after heating then it means its hardness is temporary. If the water forms scum in the beaker and does not lather, then its hardness is permanent.

A control experiment can be setup with a sample of distilled water.

ADVANTAGES AND DISADVANTAGES OF HARD AND SOFT WATER

Table 5.9: Advantages of soft water over hard water

Soft water advantages	Hard water disadvantages
It lathers easily with soap (saves soap)	Does not lather with soap (wastes soap)
Does not form scale in taps, kettles, etc.	Forms scale in taps, kettles, boilers etc.
Suitable for dyeing and tanning	Not suitable for dyeing and tanning (prevents chemicals from reacting efficiently

Table 6.0: Advantages of hard water over soft water

Hard water advantages	Soft water disadvantages
It has a pleasant taste	Does not taste so good
Dissolved calcium ion helps build strong bones and teeth	Does not build strong bones and teeth
Does not dissolve lead in pipes to cause lead poisoning	Dissolves lead in pipes and can cause lead poisoning
Calcium ion helps in blood clotting	Does not help in blood clotting
Reduces risk of heart diseases	Increases risk f heart diseases
Calcium and magnesium ions are essential plant nutrients	Does not contain essential plant nutrients

TREATMENT OF WATER FOR PUBLIC CONSUMPTION

Pure water is not found in nature. Chemicals, bacteria, and suspended sediment particles enter the water through exposure to air and runoff. Before arriving at your tap, raw water is treated to eliminate the presence of harmful bacteria and unpleasant coloration, taste, and odour.

The Stages in large-scale water treatment

1. Preliminary Treatment

Preliminary treatment or pre-treatment is any physical, chemical or mechanical process used on water before it undergoes the main treatment process. During preliminary treatment:

- screens may be used to remove rocks, sticks, leaves and other debris;
- chemicals may be added to control the growth of algae;
- pre-sedimentation stage can settle out sand, grit and gravel from raw water.

2. Coagulation

After preliminary treatment, the next step is coagulation. Coagulation removes small particles that are made up of microbes, silt and other suspended materials in the water. Treatment chemicals such as alum are added to the water and mixed rapidly in a large basin. The chemicals cause small particles to clump together (coagulate). Gentle mixing brings smaller clumps of particles together to form larger groups called ***floc***. Some of the floc begins to settle during this stage.

3. Flocculation

During the flocculation stage, the heavy, dense floc settles to the bottom of the water in large tanks. As you can imagine, this can be a slow process. Once the floc settles, the water is ready for the next stage of treatment.

4. Clarification/ sedimentation

Clarification occurs in a large basin where water is again allowed to flow very slowly. Sludge, a residue of solids and water, accumulates at the basin's bottom and is pumped or scraped out for eventual disposal. Clarification is sometimes called sedimentation.

5. Softening and Stabilization

When water is too hard (i.e. contains too much calcium, magnesium or other minerals), it forms scale and causes a variety of problems in pipes. Hard water can also result in laundering and washing problems, because it reduces the effectiveness of soaps and detergents.

Conversely, when too many of these minerals are removed, water can become too soft. Soft water can cause corrosion in pipes.

Drinking water plants attempt to maintain a desirable balance between hardness and softness. This is accomplished by adding minerals to soft water and removing them from hard water.

6. Filtration

Turbidity is a physical characteristic that makes water appear cloudy when suspended matter is present. The filtration process removes suspended matter, which can consist of floc, microorganisms (including protozoan cysts such as *Giardia* and *Cyrptosporidium*), algae, silt, iron, and manganese precipitates from ground-water sources, as well as precipitants which remain after the softening process.

These suspended materials are filtered out when water passes through beds of granular material, usually composed of layers of sand, gravel, coal, garnet, or related substances.

7. Chlorination & Disinfection

Chlorine is fed into the water system as either a dry powder or in solution. During disinfection, disease-causing organisms are destroyed or disabled. Chlorine is normally used because it is economical and rapid. It is important to add the right amount of chlorine at the water treatment plant to make sure disinfection continues while the water is flowing through the distribution system.

8. Storage

Finished water (the term water treatment professionals use) is stored in holding tanks. The tanks provide a water reserve to

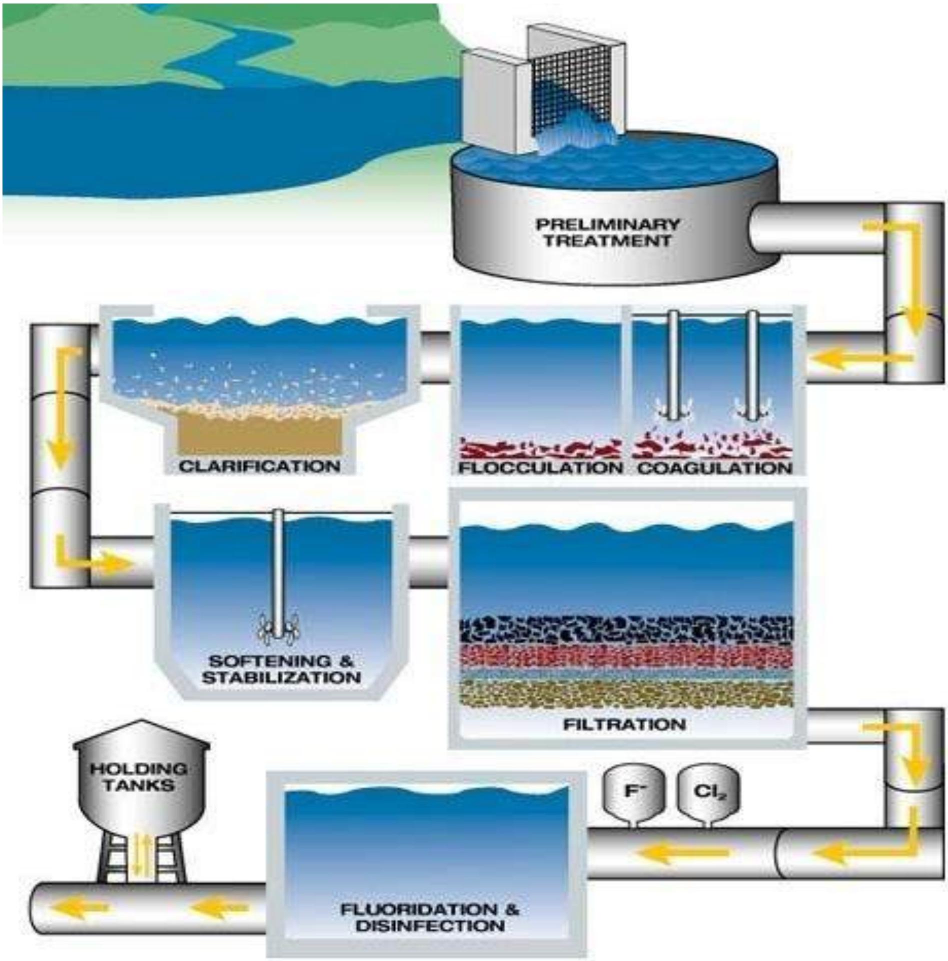

Fig. 19.3: Treatment of water for public consumption

meet the changing water demands of the communities they serve.

Summary large scale water treatment

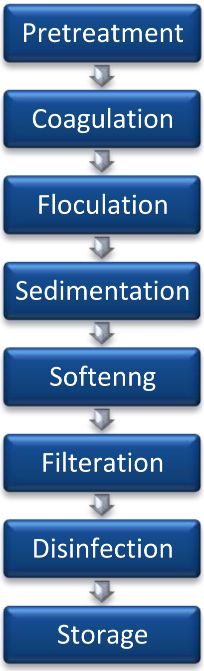

Fig. 19.4: Summary of public water treatment procedure

Small-scale water treatment

In our homes, school etc, we can still treat impure water for our own consumption. The following are some methods of purifying water on a small scale:

1. Boiling

Raw water can be boiled to kill bacteria and other microbe in it. Boiling is a very effective water treatment method, but can only be done on a small scale because of the cost involved if it is done on a large scale to serve a town or community.

2. Filtration

Suspended particles on the surface of the water can be filtered off with a water filter or fine mesh. Filtration can be done before or after boiling, and can be done a couple of time to ensure that the water is free of all suspended particles.

3. Sedimentation

Water which contains sediments or suspending particles can also be allowed to stand for some time in order to allow all the particles to settle at the bottom.

4. Distillation

Another efficient way of purifying water is distillation. Here, water is boiled, and the vapour is condensed into another container leaving the impurities in the original container.

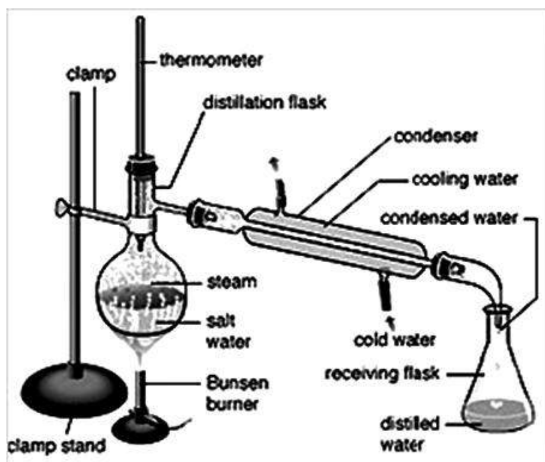

Fig. 19.5: Distillation process

Importance or uses of water

Domestic use – At home, school, etc. water is used for drinking, washing, bathing, flushing toilet etc.

Energy production –- Hydroelectric plants capture the kinetic energy of falling water to make electricity. This is done with a dam. The dam forces the water level to go up so that the water will have more power when falling. The force of the falling water pressing against turbines' blades causes them to spin. The spinning turbines transmit the kinetic energy of the falling water to generators. The generators spin when the turbines spin generating electricity that will be transmitted on the power lines to homes, schools and businesses.

Fig. 19.6: A hydroelectric power dam

Plants and animals – Plants and animals cannot exist without water. Plants need water to prepare their food which animals depend on.

Agricultural use -- Water is used for irrigation, spreading fertilizers, herbicides, and pesticides, etc.

Industrial use – Water is used for cooling heavy duty machines and washing equipments. It is an important element in many products like chemicals, drugs, lotions, shampoos, cosmetics, cleaners and beverages. Water is used in processing food and in innumerable factories and industrial processes including the manufacturing of paper.

TEST QUESTIONS

2. a) What is the chemical formula for water?
b) Mention five sources of water.
c) List three physical and three chemical properties of water.

3. a) Write a short note on the following:
i) hard water
ii) soft water
b) Explain three methods for softening permanent hard water.
c) How can boiling remove temporary hardness in water?

4. a) Give three advantages and three disadvantages each of hard and soft water.
b) What are the causes of permanent

and temporary hardness in water.

c) Differentiate between hard water and soft water.

5. a) Mention four pollutants of water.

b) Outline the steps involved in the treatment of water for public consumption.

c) explain four ways in which water is useful.

5. a) Perennial water shortage and frequent disruptions in water supply in most communities in Ghana can be traced to erosion of human values. Discuss.

b) Outline the role of Ghana Water Company in the control of water supplies in cities, towns and villages.

6. Explain how the following methods help in water treatment:

(a) coagulation

(b) flocculation

(c) sedimentation

(d) filtration

(e) disinfection

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WATER MOVEMENT

Specific Objectives

After completing this chapter, you will be able to:

- Explain the causes of high and low tide.
- Outline the effects of tidal waves.

INTRODUCTION - TIDES

Tides are the rise and fall of sea levels caused by the combined effects of the gravitational forces exerted by the Moon and the Sun and the rotation of the Earth.

Most places in the ocean usually experience two high tides and two low tides each day (***semidiurnal tide***), but some locations experience only one high and one low tide each day (***diurnal tide)***.

The times and amplitude of the tides at the coast are influenced by the alignment of the Sun and Moon, by the pattern of tides in the deep ocean and by the shape of the coastline and near-shore bathymetry.

Tides vary on timescales ranging from hours to years due to numerous influences.

Tidal phenomena are not limited to the oceans, but can occur in other systems, such as lakes whenever a gravitational field (that varies in time and space) is present.

Fig. 19.7: High and low tides

CAUSES OF TIDES

As the moon revolves around the earth, it exerts a gravitational pull on the earth. The ***moon's gravitational force*** pulls on water in the oceans so that there are ***bulges*** in the ocean on both sides of the planet. The moon pulls water toward it, and this causes the bulge toward the moon. The bulge on the side of the Earth opposite the moon is caused by the moon ***pulling the Earth away*** from the water on that side.

High and low tides

High tide occurs when the level of the sea rises to the maximum.

If you are on the coast and the moon is directly overhead, you should experience a high tide.

Low tide

Low tide occurs when the sea drops to the lowest level.

If the moon is directly overhead on the opposite side of the planet, you should experience a low tide. Because tidal currents cease it is also called ***slack tide***.

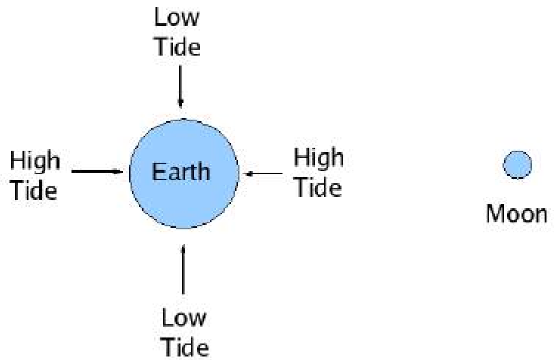

Fig. 19.8: Low and high tides

During the day, the Earth rotates 180 degrees in 12 hours. The moon, meanwhile, rotates 6 degrees around the earth in 12 hours. The twin bulges and the moon's rotation mean that any given coastal city experiences a high tide every 12 hours and 25 minutes. The sea level rises and falls over several times.

Flood tide or ***incoming tide*** is formed when the water rises up the beach. When the level falls, ***ebb tide*** is formed.

Spring tides

Spring tides are especially strong tides. They occur when the Earth, the Sun, and the Moon are in line. The gravitational forces of the Moon and the Sun both contribute to the tides. Spring tides occur during the full moon and the new moon.

Neap tides

Neap tides are especially weak tides. They occur when the gravitational forces of the Moon and the Sun are perpendicular to one another (with respect to the Earth). Neap tides occur during quarter moons.

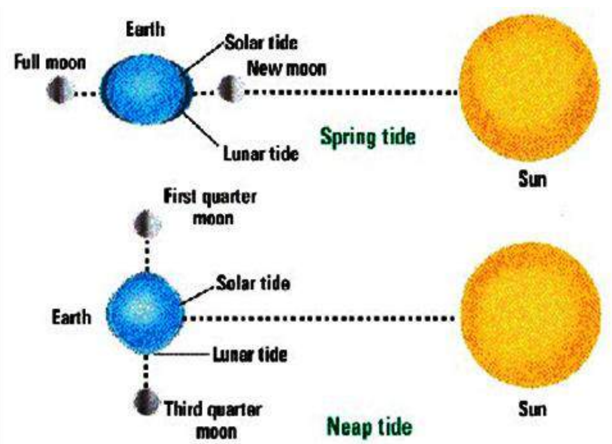

Fig. 19.9: Spring and neep tides

EFFECTS OF TIDAL WAVES

Tidal waves, also known as ***tsunami***, can be generated by an undersea earthquake, an undersea landslide, the eruption of an undersea volcano, or by the force of an asteroid crashing into the ocean.

Tidal waves are known for the damage they cause when they hit. Some effects of tidal waves are:

i) Loss of lives
ii) Damage to coastline
iii) Destruction of properties
iv) Pollution of water and land

In Ghana, the effects of tidal waves are immense on coastal area such as Keta, Ada, Elmina, etc.

Effects of tidal waves in Ghana

1. Tidal waves have eroded the coastline and damaged most buildings and other structures.
2. They have made the soil infertile by depositing infertile soil and chemicals on fertile land.
3. They have washed away most road networks closed to the sea.
4. They have made coastal areas inhabitable.
5. Tidal waves have destroyed the livelihood of people, especially fishermen who rely on the sea for their source of food and income.

TEST QUESTIONS

1. a) What are tides?
   b) Explain how tides occur.
   c) With the aid of a suitable diagram, explain the differences between spring and neap tides.

2. a) What are the causes of tidal waves?
   b) State three effects of tidal waves
   c) Mention three effects of tidal waves on Elmina.

3. a) Differentiate between high tide and flood tide.
   b) What is the difference between low tide and neap tide?
   c) Describe how the moon and the sun bring about low and high tides.

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HYDROLOGICAL CYCLE

Specific Objectives

After completing this chapter, you will be able to:

- ❖ Explain the distribution of the earth's water.
- ❖ Explain the relevance of the hydrological cycle to plants and animals.
- ❖ Outline the main causes of water contamination and the effects on humans.
- ❖ Describe household water conservation methods.

INTRODUCTION

Hydrologic Cycle (or water cycle), is a series of movements of water above, on, and below the surface of the earth.

The water cycle consists of four distinct stages:

- ❖ storage
- ❖ evaporation
- ❖ precipitation
- ❖ runoff

Water may be stored temporarily in the ground; in oceans, lakes, and rivers; and in ice caps and glaciers. It evaporates from the earth's surface, condenses in clouds, falls back to the earth as precipitation (rain, sleet or snow), and eventually either runs into the seas or re-evaporates into the atmosphere.

Almost all the water on the earth has passed through the water cycle countless times. Very little water has been created or lost over the past millions of years.

DISTRIBUTION OF THE EARTH'S WATER

Water is the most widespread substance to be found in the natural environment and it is the source of all life on earth; it covers 70% of the earth's surface.

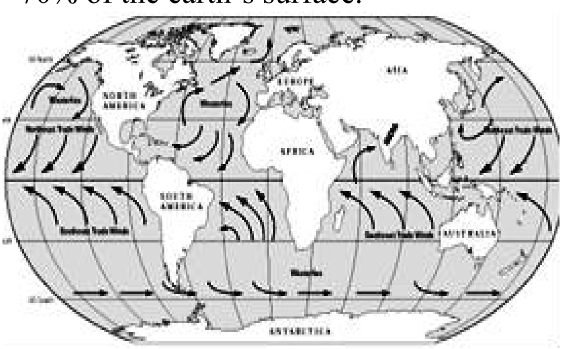


Fig. 20.0: Water covers 70% of Earth's surface

Water exists in three states: *liquid, solid* and *gas* (*vapour)*. It forms the oceans, seas, lakes, rivers and the underground waters found in the top layers of the earth's crust and soil cover. In a solid state, it exists as ice and snow cover in polar and alpine regions. A certain amount of water is contained in the air as water vapour, water droplets and ice crystals, as well as in the biosphere. Huge amounts of water are bound up in the composition of the different minerals of the earth's crust and core.

Current estimates are that the earth's *hydrosphere* contains a huge amount of water - about 1386 million cubic kilometres. However, 97.5% of this amount exists as *saline* (salty) waters and only 2.5% as fresh water.

The greatest portion of the fresh water (68.7%) is in the form of ice and permanent snow cover in the Antarctic, the Arctic and in the mountainous regions. 29.9% exists as fresh *ground waters*. Only 0.26% of the total amount of fresh water on the earth is concentrated in lakes, reservoirs and river system, where it is most easily accessible for our economic needs and vital for water ecosystems.

RELEVANCE OF HYDROLOGICAL CYCLE

i. Hydrological cycle helps regulate the temperatures of animals through perspiration and evaporation.
ii. Through transpiration, plants lose excess water to the atmosphere thereby regulating their temperature.

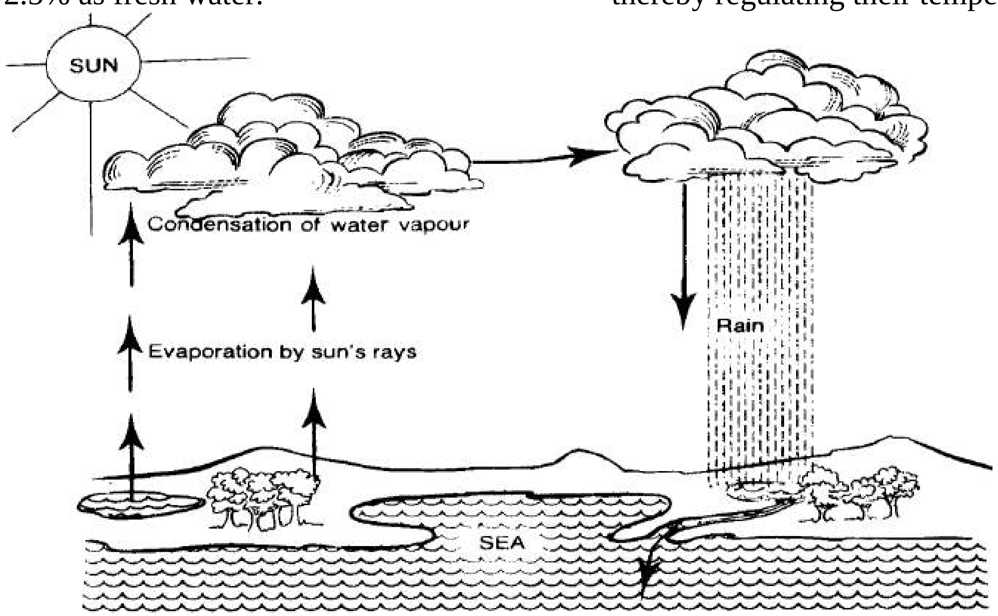


Fig. 20.1: Hydrological cycle

iii. Without hydrological cycle, the total amount of water on the earth might be used up. This means animals and plants that depend on water would also be no more, and life on earth would cease.

iv. Evaporation and precipitation cool the earth's atmosphere preventing it from overheating.

SOURCES OF WATER CONTAMINATION

Water contamination occurs when harmful substances are discharged directly or indirectly into water bodies without adequate treatment to remove harmful compounds.

Water pollution affects plants and organisms living in these bodies of water and in almost all cases the effect is damaging not only to individual species and populations but also to the natural biological communities.

Some sources of water contamination

❖ **Industrial waste**

Dumping of industrial wastes containing heavy metals, harmful chemicals, by-products, organic toxins and oils, into nearby source of water is one of the visible causes of water pollution. Effluents from factories, refineries, injection wells and sewage treatment plants are dumped into urban water supplies, leading to water pollution.

❖ **Domestic waste**

Another cause of contamination of water is the improper disposal of human and animal wastes as well as other pollutants from household wastes.

❖ **Agricultural waste**

The residue of agricultural practices, including fertilizers and pesticides, are some of the major sources of water pollution.

❖ **Mining waste**

Chemicals used in the mining industries are allowed to drain to nearby water bodies polluting it. Some small scale miners even wash their minerals in rivers and streams with chemicals.

❖ **Radioactive waste**

Radioactive wastes are often discharged into the ocean and other water body. These wastes have serious effects on plants and animals if not properly disposed of.

❖ **Special waste**

Some other wastes such as wastes from hospitals and clinics are some of the major contaminants found in water bodies. Proper disposal of these wastes would save people from going to hospitals and clinics in the first place.

HARMFUL EFFECTS OF WATER POLLUTION

1. A number of waterborne diseases are produced by the pathogens present in polluted water, inffecting humans and animals alike.

2. Pollution affects the chemistry of water. The pollutants, including toxic chemicals, can alter the acidity, conductivity and temperature of water.

3. Polluted municipal water supplies are found to pose a threat to the health of people using them.

4. The concentration of bacteria and viruses in polluted water causes increase in solids suspended in the water body which in turn leads to health problems.

5. Marine life becomes deteriorated due to water pollution. Killing of fish and aquatic plants in rivers, oceans and seas is an after effect of water contamination.

6. Diseases affecting the heart, poor circulation of blood and the nervous system and ailments like skin lesion, cholera and diarrhoea are often linked to the harmful effects of water pollution.

7. Carcinogenic pollutants found in polluted water might cause cancer.

8. Alteration in the chromosomal makeup of the future generation is foreseen as a result of water pollution.

9. Discharges from power stations reduce the availability of oxygen in the water body in which they are dumped.

10. The flora and fauna (plants and animals) of rivers, sea and oceans are adversely affected by water pollution.

WATER CONSERVATION METHODS

It is very vital to save or conserve water. Some water conservation methods include:

1. Wash fruits and vegetables in a pan of water instead of running water from the tap.
2. Use a broom or brush instead of a hose to clean driveway and sidewalk and save water every time.
3. When washing dishes by hand, do not let the water run while rinsing. Fill one sink with wash water and the other with rinse water.

4. Shorten your shower by a minute or two and you'll save up to 150 gallons per month.
5. Try keeping a pitcher or bottle of water in the refrigerator for cool drinking water. Running tap water to cool it off for drinking water is wasteful.
6. If you do not drink all of the water in your glass rather than dumping the rest down the drain use it to water a plant or something useful.
7. Repair leaks in pipes. Often taps and pipes can have leaks and they are not even noticed. Even the smallest leak can waste up to 20 gallons of water a day. Leaks wastewater 24 hours a day, seven days a week, and can often be repaired with only an inexpensive washer.
8. Put a little food colouring into your toilet tank. If the colour begins to appear in the bowl, without flushing, you have a leak that should be repaired as soon as possible.
9. Every time you flush a facial tissue or other small bit of trash, you waste five to seven gallons of water. Therefore put waste in bins instead of toilet.
10. Turn off the water while brushing your teeth. There is no need to keep water pouring down the drain. Just wet your brush and turn off the tap. You can turn the water back on when you are ready to rinse.

Rainwater harvesting

Rainwater harvesting is the accumulation and storing, of rainwater.

It has been used to provide drinking water, water for livestock, water for irrigation or to refill aquifers in a process called ***groundwater recharge***. Rainwater collected from the roofs of houses, tents and local institutions can make an important contribution to the availability of drinking water.

Water collected from the ground, sometimes from areas which are especially prepared for this purpose, is called ***Storm water harvesting***. In some cases rainwater may be the only available or economical, water source.

Rainwater harvesting systems can be simple to construct from inexpensive local materials, and are potentially successful in most habitable locations. Roof rainwater can be of good quality and may not require treatment before consumption. Although some rooftop materials may produce rainwater that is harmful to human health, it can be useful in flushing toilets, washing clothes, watering the garden and washing cars. These uses alone reduce the amount of treated water used by a typical home.

Fig. 20.2: Rainwater harvesting

TEST QUESTIONS

1. a) Explain the term hydrological cycle.
   b) How is the Earth's water distributed?
   c) State four importance of hydrological cycle to living things.

2. a) Explain four sources of water contamination.
   b) State five harmful effects of water contamination.
   c) Mention five water conservation methods.
3. a) Write a short note on rainwater harvesting.
   b) Discuss the importance of rainwater harvesting.
   c) Describe how you will construct a simple rainwater harvesting system.

28

GENERAL PRINCIPLES OF FARM ANIMAL PRODUCTION

Specific Objectives

After completing this chapter, you will be able to:

- Outline the main activities involved in the production cycle of farm animals.
- Describe ruminant production.
- Describe non-ruminant production.

INTRODUCTION

Farm animals are domestic animals raised on the farm mostly for economic purposes.

Farm animal production involves the breeding, feeding, and management of animals, or livestock, for the production of food, fibre, work, and pleasure. Farm animals raised on the farm include ruminants (e.g. cattle, goat, sheep, etc.) and non- ruminants (e.g. chicken, ducks, guinea, rabbit, pig etc.). Animals which may also be raised on the farm are fish, snails, grass cutters etc.

Importance or benefits of animal husbandry

1. Food (e.g. meat, milk, egg)
2. Income for farmers
3. Employment
4. Clothing from skin, wool and hide
5. Transportation (e.g. horse and donkey)
6. Labour (e.g. cattle for ploughing)
7. Recreation or entertainment (e.g. horse race)

MAIN ACTIVITIES INVOLVED IN FARM ANIMAL PRODUCTION

- Selection of suitable breeds
- Choice of management system
- Breeding systems and care of the young
- Feeding
- Finishing, processing and marketing produce.

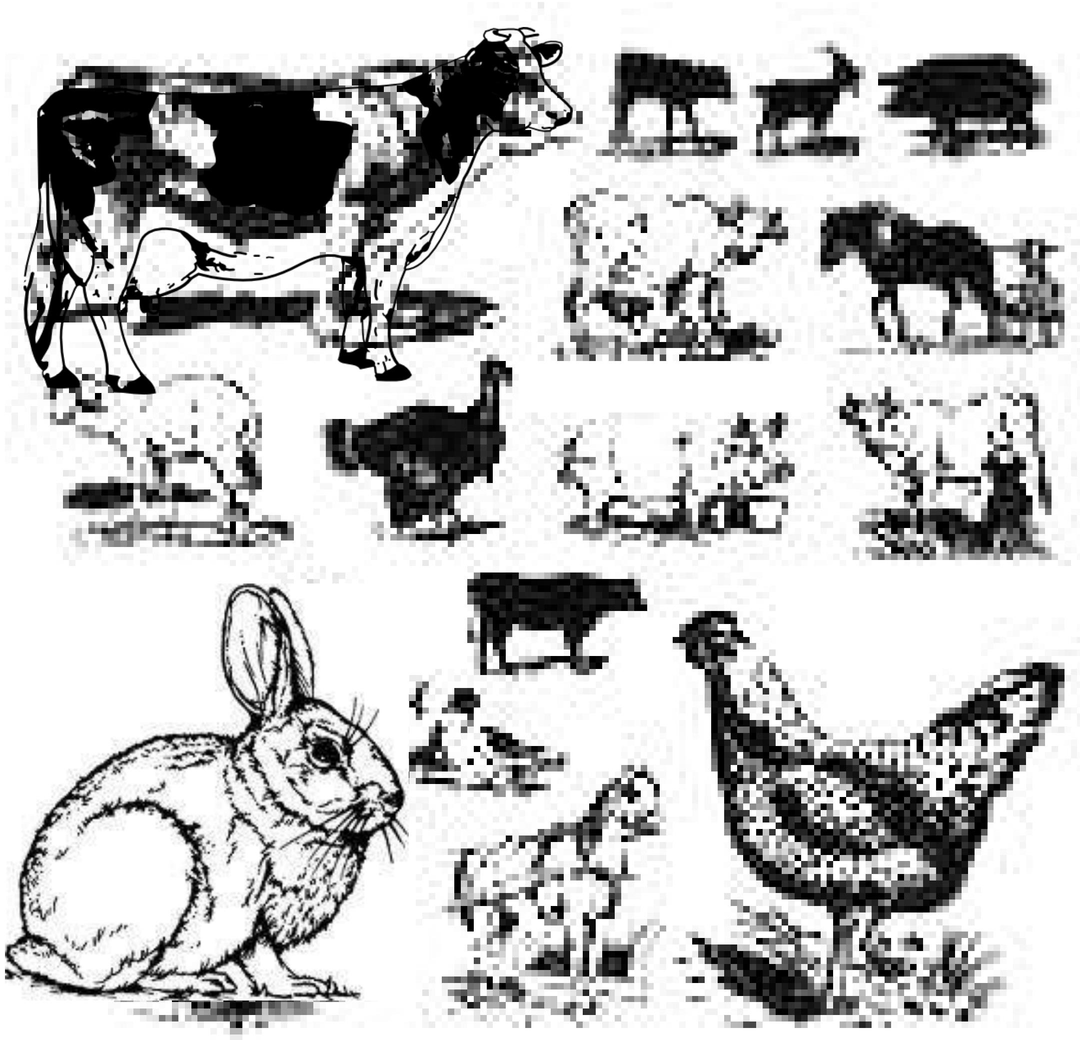


Fig. 20.3: Farm animals

Selection of suitable breeds

The first thing the farmer takes into consideration is the type and breed of animal he wants to raise. If for example, the farmer wants to raise cattle, he should consider the pedigree record (i.e. the performance of that breed of cattle in the past), the individual performance of the selected animals or cattle, their appearance (i.e. body conformation, absence of defects etc.) and performance of siblings and offspring.

Choice of management system

How are the animals going to be housed? Will the farmer use the expensive intensive system which requires expensive housing and feeding routine; the semi-intensive system which require some simple housing or the extensive system which requires little or no housing and feeding at all, and hence inexpensive?

The farmer's choice of the above-mentioned management systems is influenced by factors such as cost of housing and feeding, type of scale of enterprise (large or small-scale), the skills of the farmer and the location of the farm.

Conditions for good animal housing

1. Protection from pests, parasites and predators
2. Clean and dry floor or litter
3. Good ventilation
4. Good drainage system
5. Availability of feeding and water trough
6. Good environmental sanitation
7. Adequate floor space to control overcrowding
8. Availability of footpath at the entrance of the house
9. Availability of essential equipment to meet the production needs of the animals, such as wallows, egg laying nests, etc.
10. Absence of sharp and pointed objects

Breeding systems and care of the young ones

In animal production,

Breeding is the mating or pairing of male and female animals of a selected breeding stock to produce young ones with desirable qualities.

Importance of animal breeding

1. It increases the genetic potential of the animals to produce more.
2. It improves the resistance to pests and diseases.
3. It increases the quality of products.
4. It improves the fertility of the animals.
5. It increases growth rate.
6. It improves tolerance to adverse environmental conditions.
7. It improves fecundity (i.e. the ability of a pregnant animal to successfully carry and develop an embryo to foetus).

Some important breeding methods include:

i. introduction or upgrading
ii. cross-breeding
iii. mating

Introduction or upgrading

This involves the transportation of animals with desirable traits from one geographical area to another where they did not exist previously. The animals introduced to this new area are not supposed to replace the animals there but to upgrade their performance.

The animals for transportation are first quarantine to determine their health status before introducing them to their new location. It is important to ensure that the conditions in the new environment are similar to those of the old one.

Cross-breeding

Cross-breeding, also known as ***hybrid*** is the mating of two animals of different breeds to produce offspring with special qualities. It is also known as ***heterosis***. Animals being mated should be pure breeds or inbred lines in order to obtain better results.

Mating

There are three main methods of mating farm animals. They are pasture mating, hand mating and artificial insemination.

Pasture mating

This is the natural mating of male and female animals while grazing on pasture. Pasture mating is common among livestock kept under the extensive and semi-intensive systems and normally produces undesirable results.

Hand mating

This involves the use of a proven male animal to mate female animal on heat, one after the other. Hand mating is practiced mainly under the intensive system. Mated female animals are described as having been serviced by the males.

Importance of hand mating

- Hand mating allows for the implementation of desired breeding programmes.
- It also enables mating to be planned carefully so that the pregnant animal gives birth at a suitable time.

Artificial insemination

This is the use of a syringe to carefully collect and deposit chemically treated sperms of proven male into the womb or vagina of a female animal.

This method requires expert skills and high degree of organisation. It is relatively costly as it involves other sophisticated equipment.

Advantages of artificial insemination

1. The semen containing the sperm of proven male can easily be transported from one area to another.
2. Semen can be stored by freezing; hence even if the animal gets sick or dies its semen would still be useful.
3. The dilute semen can be used to service very large number of female animals.
4. It controls the spread of diseases as the semen is chemically treated.

5. Mating is mostly successful when the female animal is on heat or on her oestrus cycle.

Signs of animals on heat

1. They become restless.
2. The vulva swells and become reddish in colour.
3. They lose appetite
4. They make grunting noise.
5. They sniff at other animals and attempt to mount them.
6. They discharge a slimy substance from the vulva.

Care of the young

It is very essential to care for the young animals as they are more vulnerable or exposed to disease and pest attacks. They also have small and weak digestive system which are sensitive to food, even though they require the best of nutrition for growth and development and to resist disease and pest attacks. They should also be shielded from adverse environmental conditions.

Management practices

Management practices are the use of production resources, techniques and methods intelligently aimed at increasing productivity and profit.

They may be carried out periodically or routinely. Management practices include:

1. selecting good breeding stock and breeding programmes;
2. providing good housing;
3. controlling pests and diseases;
4. ensuring good environmental sanitation;
5. providing adequate quality and appropriate nutrition to animals;
6. using appropriate management systems.

Feeding

Like humans, animals need the right quantities of food, rich in nutrient for their growth and development. Animals must be fed a well-balanced diet regularly and in a sanitary environment.

Importance of good feeding of animals

1. It enables them to grow faster and stronger.
2. It helps them to produce more quality products.
3. It helps them to resist diseases.
4. It helps them to work better.
5. It helps them to reproduce better.

Since young animals are easily prone to diseases and also have sensitive digestive system, they should be carefully fed.

Conditions for feeding young animals

1. They should have access to colostrums (the first milk after birth which is rich in vitamins, minerals and antibodies).
2. They should be provided with adequate clean and fresh water.
3. They should be fed with balanced diets.
4. They should be fed in a sanitary environment.

Finishing, processing and marketing produce

Farm animals are reared because of their produce. It is very necessary to care for them, particularly regarding nutrition, to improve their product and carcass qualities for higher market value.
The animal produce and sometimes the live animals are sold in the farm or market.

Pre-slaughter care

Before an animal is slaughtered they should be starved and rested for about 24 hours. *This pre-slaughter care* empties the gut of faeces and undigested food which may contaminate the meat. It also relaxes them and reduces nervous excitement. Another pre-slaughter care is veterinary inspection to avoid the slaughter of sick animals.

Slaughtering or slaughter care

Animals can be safely slaughtered by first stunning them with electric shot or a captive shock in the head to reduce struggle, before cutting their throat with sharp knife.

Post-slaughter treatment

Slaughtered animals should be bled thoroughly by hanging them upside down. This avoids the retention of blood in the meat which could render it less tasty or unattractive or even serve as a medium for bacterial growth resulting in rotting of meat.

The carcass is then flayed (removal of skin and hide) and eviscerated (removal of intestines).
It could then be cooked and eaten or preserved by freezing, smoking, drying, salting or canning for it to remain wholesome over a period of time.

RUMINANT PRODUCTION

Ruminants are a class of farm animals which chew cud.

Chewing of the cud, also known as ***cudding***, is a more thorough chewing of partially chewed, swallowed and regurgitated herbage in a relative state of comfort and relaxation. Examples of ruminants are cattle, sheep and goats. These animals have cloven (divided) hooves.

Cattle

There are two principal species of cattle reared globally. They are *Bos indicus* of Asia and *Bos taurus* of Europe. In West Africa, the commonest species reared is the *Bos indicus* also known as the ***Zebu cattle***. This is because Zebu cattle can tolerate warm and dry climate

Characteristics of Zebu cattle

1. They have a hump (enlarged muscle) on their shoulders.
2. They have a long face.
3. They have short, upright horns.
4. They have drooping ears.
5. They also have thin legs.
6. They have dewlap (large fold of loose skin) hanging from the throat.

Examples of Zebu cattle include White Fulani (Adamawa), Red Fulani, Red Bororo and Sokoto Gudali.

Fig. 20.4: Zebu cattle

Breeds and types of cattle

Cattle are classified into four types according to their body configuration and uses. They are:

- beef cattle
- dairy cattle
- dual-purpose cattle
- work or drought cattle

Beef cattle

Beef cattle are mainly kept for meat production. They generally mature earlier and produce less milk. Meat from mature cattle is called ***beef*** and meet from a calf less than three months old is called ***veal***.

Examples of local meet cattle are N'dama and Muturu, known collectively as ***West African Shorthorns***.

Examples of exotic beef cattle are Aberdeen Angus, Charolais, Beef Friesian, Brahma and Red Devon.

Characteristics of Beef Cattle

1. They have higher growth rate
2. They produce less milk
3. They have square-shaped body conformation.
4. They have short, strong and straight legs.
5. They are stocky and well filled with fresh.

Fig. 20.5: N'dama cattle

Fig. 20.6: Muturu cattle

Dairy Cattle

The primary purpose of keeping dairy cattle is milk production. They have the ability to convert food into milk rather than fat or flesh. Milk is synthesised and stored in the udder. With the exception of Zebu cattle, most dairy cattle in West Africa are exotic. Examples include Jersey, Holstein-Friesian, Guernsey. Ayrshire and Brown Swiss cattle.

Characteristics of dairy cattle

1. They have slender bodies with prominent bones.
2. They have wedge-shaped bodies.
3. They have large and well-developed udders.
4. They have longer legs
5. They have wide space between hind legs.
6. Their veins are well-developed.

Dual-purpose cattle

This breed of cattle produces both milk and meet. However, they do not produce large quantities of milk as dairy cattle neither do they produce good meet as meet cattle. Examples include Red Poll and dairy Shorthorn.

Work or drought cattle

These cattle are raise to help with menial jobs on the farm such as ploughing, and carting of produce and equipment. They are mostly strong, muscular and well-built, are usually quiet and have strong legs

Work cattle ploughing a paddy field

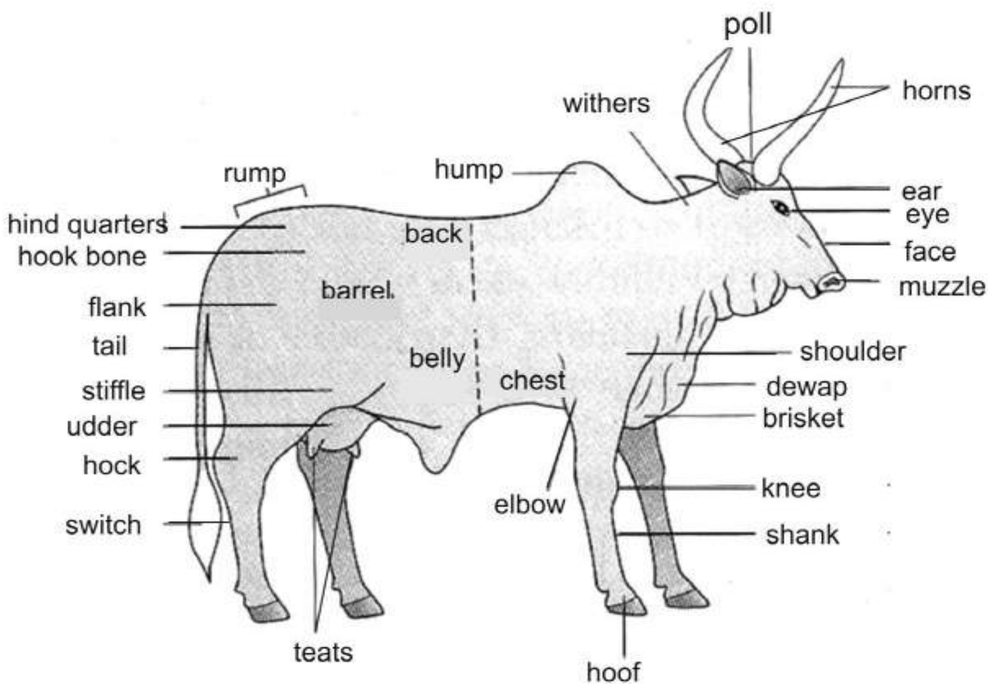


Fig. 20.7: External features of cattle

Management practices

The proper management practices of cattle rearing involve effective housing, good healthcare, proper feeding and accurate record-keeping.

Housing

Cattle are kept under any of the three management practices –

- ❖ intensive system
- ❖ semi-intensive system
- ❖ extensive system

Intensive system

The intensive system of managing ruminants is characterized by total confinement of the livestock in housing units known as ***sheds*** or ***pens***. These housing units have concrete floors and walls with vents near the roof.

The intensive system is the best system for the production of high quality livestock because of the following advantages:

Advantages of intensive system

1. The animals are well fed.
2. They are protected from pests, diseases, theft and predators.

3. They are also protected from rain, strong wind and heat.
4. Management practices such as vaccination, mating, de-worming etc. can be easily done.
5. Records are properly kept.
6. It ensures the production of high quality livestock.
7. Sick animals are easily detected and culled or treated, reducing the spread of diseases.
8. Animal confinement prevents them from going astray and causing mayhem or destroying people's properties.
9. Farmyard manure is easily obtained for farm crop.

Disadvantages of intensive system

1. Intensive system is expensive to run.
2. It is labour intensive.
3. Rate of spread of diseases among animals is quick if not detected quickly.
4. Animals have little room for movement, thus are less active which exposes them to ill-health.

Semi-intensive system

This system requires partial confinement of animals especially at nights and during harsh weather conditions. The animals are allowed to graze on improved pastures.

Advantages of semi-intensive system

1. It is relatively less expensive as compared to the intensive system.
2. It requires less labour.
3. Farmyard manure can be obtained.
4. Animals are active as they have room for movement.
5. Some husbandry and management practices such as dehorning, castration, identification etc. can be carried out on livestock.
6. Animals are protected against harsh weather conditions.

Disadvantages of semi-intensive system

1. Since animals are kept in pens only a part of the day there is little time to carry out all husbandry services.
2. Grazing on pastures can lead to overgrazing if not controlled.

Extensive system

The extensive system is the commonest system in Ghana. In this system the cattle are housed in an open-fenced space, called ***kraal*** and are exposed to all sorts of whether conditions. There are no or few facilities for drinking, feeding and other management services. They are normally driven to the field to graze under the care of herdsmen.

Advantages of extensive system

1. It requires very little money to run.
2. It is not labour intensive.
3. It requires little managerial practices.
4. Little time and attention is needed.

5. Animal are more active as they move about freely.
6. The spread of diseases among livestock is minimal.
7. Animals have access to herbage and hence can meet their nutritional requirement.

Disadvantages of extensive system

1. Animals are exposed to all sorts of danger.
2. They are easily attacked by pests and diseases.
3. Livestock usually go astray and cause destruction on other people's properties.
4. They are exposed to extreme weather conditions.
5. It is hard to keep record of animals.
6. Animals obtain food by feeding on pastures, which may lead to overgrazing.

Fig. 20.8: Cattle in a kraal

Table 6.1: Differences between intensive and extensive systems

Intensive system	Extensive system
Animals are confined	Animals are not confined
They are well fed	They search for their own food
They are sheltered from harsh weather conditions	They are exposed to harsh weather conditions
Expensive to run	Cheap to run
Requires more attention	Requires little attention
Records are properly kept	Records are not properly kept
Labour intensive	Not labour intensive
Management practices are properly observed	Management practices are not properly observed

Feeding

Cattle are herbivores (plant-eating only) and normally graze on pastures.

Pasture is a piece of grassland or a mixture of grass-legume which has high nutritional values.

Feeding methods for cattle

❖ Under the intensive system and semi-intensive systems, the cattle are sometimes ***zero-grazed*** on forage or on hay and silage, especially when there are little or no green pastures available.

❖ Like all mammals, cattle need water and should have ample supply of clean water, especially during the dry season when the pastures are often dry.

- They should also be provided with salt to provide the necessary minerals they require.
- The cattle should be well fed on an adequate ration of a balanced diet. Pregnant cows should be given steamed-up ration. This is rich in proteins and vitamins and helps the cow and its foetus.
- Flushing, where cows are given extra rich food before mating to increase their pregnancy rate, should be observed.

Fig. 20.9: Cattle grazing

Hay

Hay is dry forage which is conserved for use by livestock in the future. It is usually a mixture of grass and legumes and may contain cereals such as maize or sorghum. It is best prepared during the dry season when there is sufficient sunlight to cure the collection of grass and legume (called ***swath***).

Preparation of hay

- First harvest the hay crop at the early flowering stage when the crop is leafy, more nutritious and has less water content.
- Remove foreign materials and weeds from the swath.
- Leave it in the sun to dry the moisture content.
- Gather and store the well-cured hay on stacks in a well ventilated room or shed.

Fig. 30.0: A hay bail

Qualities of a well prepared hay

1. It has slightly green colour.
2. It has low water content.
3. It is less fibrous and easily digestible.
4. It contains no dust or mould.
5. It is free from weed and other foreign materials such as stones.
6. It can be stored for a very long time without deteriorating.

Silage

Silage is fresh green forage conserved by partial fermentation. Preparation of silage is also referred to as ***ensiling***. Ensiling involves cutting, chopping and compacting the silage crop for about two to three days.

Preparation of silage

- Harvest the silage crop at the early flowering stage.
- Chop the swath into pieces and put it into a pit or silo.
- Compact the material to removed air.
- Cover the pit or silo with tarpaulin to protect it from rain and sun.

Qualities of a well prepared silage

1. It should be green in colour.
2. It should have a moisture content of about 70-80%
3. It should be leafy.
4. It should be free all foreign materials.
5. It should be free of mould.

Reasons why silage is preferred to hay

1. Ensiling is cheap, less laborious and quicker to prepare.
2. It has a high proportion of water, leaf and nutrient.
3. Silage has a sweet aroma.
4. It helps prevent constipation.

Table 6.2: Differences between hay and silage

Hay	Silage
Low moisture content (about 20%)	High moisture content (about 70%)
Dry, dusty and less green	Wet and green
Does not prevent constipation	Prevents constipation
Low acid content	High acid content
Strong aroma	Weak aroma
Less palatable	Highly palatable
High vitamin D content	Low vitamin D content

Husbandry practices

These are practices routinely performed to improve the quality, productivity and good health of farm animals.

The most important husbandry practices include identification, dehorning, de-worming, record keeping, castration and dipping.

Identification

Identification is the act of marking livestock in order to ensure recognition and prevent loss of animals, as well as proper management of breeding programmes.
Identification methods include branding, tattooing, ear notching etc.

Branding **-** This requires the use of a hot iron with a desired mark to superficially burn off the skin of the animal leaving the mark permanently after healing. ***Cold-branding,*** in which liquid nitrogen is used in the branding iron to kill hair follicles on animals, leaves hairless insignia.

Tattooing - In tattooing, a special instrument with chosen letter, number or symbol designed from metal pins is used to prick and dye the letter or number permanently on the skin.

Ear notching – In this method of identification, the ears of the animals are notches with coded numbers.

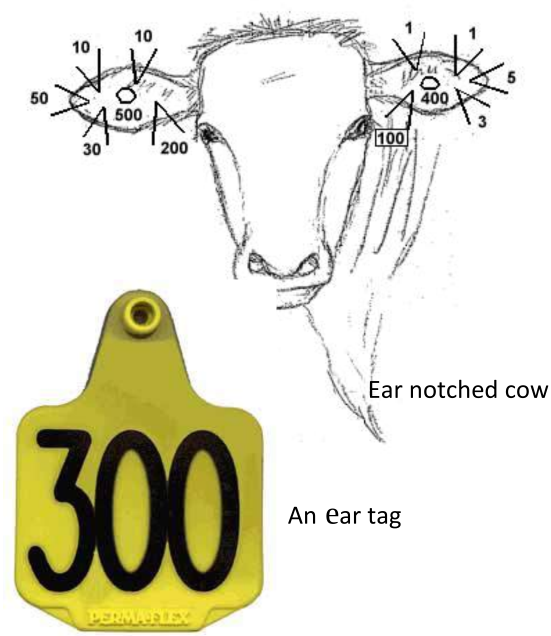


Fig. 30.1

Ear tagging – This involves the use of plastic or metallic tag with identification symbols and clipping them into the ears of the animals.

Dehorning

This is the removal of horns from livestock. It may be done by sawing off the horns or preventing the horns from growing in young animals by burning off the horn in the process called ***disbudding***. Disbudding is done with hot iron dehorners or sodium stickers.

Importance of dehorning

1. It prevents livestock from hurting each other.
2. It reduces the space required to house the animals.
3. It makes animals easier to handle.
4. It reduces damages done to equipment on the farm.

Castration

Castration refers to the removal of the testes, the spermatic cord or any method which may render male animals incapable of producing sperm for reproduction. Female animals may also have their ovaries removed surgically to prevent egg production in a practice known as ***ovariectomy.***

Methods used in castration include:

Ring method, where a strong circular rubber band called *elastrator*, is mounted on the scrotum above the testicles;

Injection method, where chemicals are injected into the animals to stop sperm production;

Burdizzo method, where a Burdizzo clamp (a plier-like instrument) is used to crush the spermatic cord of the animal.

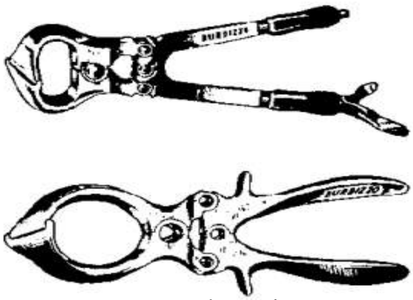


Fig. 30.2: Burdizzo clamps

Importance of castration

1. It prevents indiscriminate mating among livestock.
2. Castrated animals become gentle, docile and easier to manage.
3. It prevents the spread of venerable diseases.
4. Castrated animals normally grow faster.
5. The meat of castrated animals is tender.
6. Castration removes bad smell from animals.

Deworming

This is the use of suitable drugs called ***dewormer*** to remove or kill worms in farm animals. Deworming ensures that animals have good appetite and are not prone to diseases posed by the endoparasitic worms.

Dipping

This involves the provision of footbath with disinfectant at the entrance of the pen so that animals, as well as visitors, may dip their feet before entering. Dipping controls ectoparasites.

Common pests, parasites and diseases of ruminants

The commonest pests of livestock include tsetse fly and mosquitoes. A tsetse fly is a flying insect which carries the parasites trypanosome, the transmitter of trypanosomiasis, otherwise known as *nagana* in cattle and sleeping sickness in humans.

Mosquitoes on the other hand are the carriers of plasmodium which causes malaria. Malaria is more prominent in humans, nevertheless, may render livestock less active or less productive.

Control of tsetse fly and mosquitoes

1. Remove bushes and undergrowth along streams to reduce their population.
2. Spray infested areas with recommended insecticides.
3. Spray livestock with insect repellents.
4. Drench animals with recommended drugs such as prophylactics to kill parasites in them.
5. Use gamma rays to make male tsetse fly sterile in order to prevent them from reproducing.

6. Introduce fish in lakes to feed on tsetse fly and mosquito lavae.

Other ectoparasites (i.e. parasites which live outside the body) of livestock include flea, tick and lice may be controlled

Table 6.3: Common diseases of cattle

Disease	Transmission	Symptoms	Prevention
Foot and mouth disease (*viral disease*)	• Through infected feed, water and milk	• High fever • Blisters in and around the mouth and hoofs	• Quarantine and vaccination • Killing infected animals and burning their bodies
Anthrax (*bacterium disease*)	• Contaminated feed and water. • Infected equipment	• High fever, • swelling in the neck, genitals and lower abdomen, • discharge of blood from dead animals	• Vaccination and culling of infected animals, • Burning the bodies of infected animals, • Disinfecting farm equipment
Rinderpest (*viral disease*)	• Airborne • contact with infected animal	• Ulcer in the mouth and tongue of animal, • fever, • running eyes and nostrils	• Quarantine and immunization of animal • Isolation and vaccination of infected animals
Bloat	• Accumulation of gases produced from the fermentation of starchy forage in animal	• Laboured breathing, • profuse salivation, • staggering	• Grazing on fresh pasture for short periods, • providing roughage such as hay, zero grazing
Mastitis (*bacterium disease*)	• Wound on the udder, • exposure of udder to wet surfaces and cold	• Inflamed udder which produces yellowish milk, • Sores on udder	• Disinfection of equipment, • use of antibodies such as tetracycline and streptomycin
Trypanosomiasis (protozoan disease)	• Bites from tsetse fly	• Sleepiness • dullness, • high body temperature	• Spraying infected areas with insecticides
Brucellosis (bacterium disease)	• Contaminated feed and pasture	• Inflammation of testes and epididymis causing sterility in male animals, • Aborting in female animals	• Periodic vaccination of animals • Killing and burying infected animals.

dipping, hand-picking, rotational grazing and general cleaning.

Endoparasites (i.e. parasites which live inside the body) of livestock include worms. Worms are best controlled by deworming.

Sheep

Sheep are ruminants normally kept for their meet and sometimes for their milk, skin and wool. In Ghana, sheep are mostly kept in rural areas and in subsistence. Because of financial constraints in those areas, the management system used is commonly the extensive system.

Breeds of sheep

Common breeds of sheep include Nungua blackhead, Yankasa, West African dwarf and Ouda.

These breeds are o classified based on the following qualities:

- ❖ size and shape of body
- ❖ length of leg (long or short)
- ❖ tail size and coat
- ❖ function or use (meet, milk or wool)

Characteristics of some common breeds of sheep

West African dwarf

1. They are short and small.
2. They have small, short and erect ears.
3. The ewes are polled (without horn).
4. The rams have curved horns.
5. They are normally resistant to trypanosomiasis.

Fig. 30.3: West African dwarf sheep

Yankasa

1. They have short neck.
2. Their hair is short.
3. They are normally white in colour with black patches on the face.
4. They are generally resistant to trypanosomiasis.
5. They have non-pendulous ears.

Ouda

1. They are generally tall
2. They possess short hair.
3. Their face is convex.
4. They have long and pendulous ears.
5. They have dewlaps (folds of loose skin hanging from the throat)

Nungua blackhead

1. They are hybrids between the West African dwarf and the black head Persian breed from South Africa.

2. Their body colour is white with black head.

Goats

Goats are some of the commonest livestock found anywhere in the world. In Ghana, the popularity of goats is attributed to the fact that they are easy to rear as they require little attention and capital and very little management practices. They can withstand pests, parasites and diseases.
Goats are mainly kept because of their provision of meat, milk, wool and skin.

Breeds of goat

Common breeds of goat found in Ghana include the Sokoto Red, Bornu Red, Kano Brown, African Dwarf breeds and Bauchi.

Characteristics of Sokoto Red

1. They are generally small in size.
2. They have red glossy coats.
3. They have light, thin and supple skins.

Fig. 30.4: A Sokoto red goat

Characteristics of Bornu Red

1. They are large in size.
2. They have large pendulous ears.
3. They have long thin legs.
4. Their horns are thin and pointed.
5. They have coarse skins

Characteristics of Kano Brown

1. Their coats are red and glossy.
2. They have straight and horizontal ears.
3. They possess coarse skins.

Characteristics of West African Dwarf breeds

1. They are small in size.
2. They can withstand intense heat.
3. They are resistant to trypanosomiasis.

Characteristics of Bauchi

1. The coat colour is generally red and white.
2. They have short, thick-set frames.
3. They have coarse bones

Management of goats

Goats normally prefer the outdoors and are generally kept in the extensive and the semi-intensive systems. They are good browsers and can survive harsh conditions with little feed. However, when left on pasture for long they often overgraze to the point of exposing the soil to erosion.

In rural areas, unsheltered and unattended goats are often found causing disturbance on other peoples' properties. Large farms may employ a herdsman or two to guide the goats to pasture.

Billy (male) goats are generally ready for mating after a year or two. One billy goat can serve about 50 nanny (female) goats. With this other billy goats of unfavourable traits can be castrated.

NON-RUMINANT PRODUCTION

Non-ruminants are farm animals which do not chew the cud or do not possess four-chamber stomach. Examples of non-ruminants are poultry, pigs and rabbits.

Production of poultry

Poultry is a collective term for all domestic birds kept mainly for their egg and meat.

Types of poultry include chicken (fowl), duck, turkey, pigeon, guinea fowl and ostrich. Our discussion will be centred on chicken since it is the commonest and most profitable poultry reared.

Reasons for keeping poultry

1. For egg and meat production.
2. Income for farmers
3. Employment for people
4. Feathers for decoration, feeds for farm animals, filling pillows and processing into fertilizers
5. Manure for vegetable garden and fish pond

Chicken (fowl)

Chickens belong to the family *Phasianidae* and are classified as *Gallus gallus.* They originated from Southeast Asia and are considered descendants of a single wild species, the red jungle fowl, which is found in the wild state of India in Southeast Asia. Genetic analyses have shown that every breed of domestic chicken can be traced to the red jungle fowl.

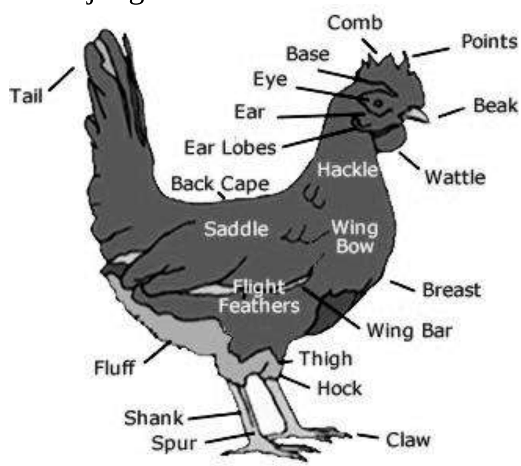


Fig. 30.5: External features of a chicken

Breeds of chicken

In Ghana, both local and exotic breeds are kept. Local breeds are mostly kept for subsistence while exotic breeds are found in large poultry farms.

Local or indigenous breeds

These breeds of chicken originated in Africa and have undergone little changes over time.

Characteristics of local breeds

1. They are usually small in size.
2. They have varying colours – brown, black, white, red, multi-coloured.
3. Their meat is tough.
4. They have a slow growing rate.
5. They have low egg laying capacity (20 to 30 eggs per year).
6. They are resistant to diseases.
7. They tend to be broody (sit on their eggs).
8. They can withstand adverse weather conditions.

Exotic breeds

These are breeds of chicken that have been introduced into Africa from other countries. Exotic breeds are usually kept in large quantities for commercial purposes. They are often the preferred choice because of the following characteristics they possess:

Characteristics of exotic breeds

1. They have large body sizes
2. They have faster growth rate.
3. They have good egg laying capacity (200 to 270 per year).
4. Their meat is tender.
5. They have distinct colour depending on the breed.

Types of chicken

Chickens, especially the exotic breeds are classified as layers, broilers, dual purpose and breeders. This classification is based on the main purpose for which they are kept.

Layers

Layers are female chickens kept for their egg production. They are not often broody, are normally small in size and difficult to confine.

Examples of layers are White and Brown Leghorns.

Characteristics of good layers

1. Each should be able to lay more than 15 eggs per month.
2. Their vents are enlarged, oval and moist.
3. They have bright eyes.
4. They have large, glossy, bright and warm combs.

Broilers

Broilers are chickens raised mainly for their meat production. Broilers could be male or females.

Examples of broilers are Rhode Island Red, New Hampshire, Plymouth Rock and Dark Cornish.

Characteristics of broilers

1. They have tender meat.

2. They have flexible breast-bone cartilage.
3. They have dull eyes

Dual purpose

These breed of fowl are kept for both meat and egg production.

Examples of dual purpose fowls are Light Sussex, Rhode Island Red and Black Austrolop.

Breeders

Breeders are fowls (male and female) of desired qualities raised to produce good eggs. These eggs are normally hatched into day-old chicks through artificial incubation.

Plymouth Rock

Rhode Island Red

White Leghorn

Light Sussex

Fig. 30.6: Breeds of fowls

Management of chickens

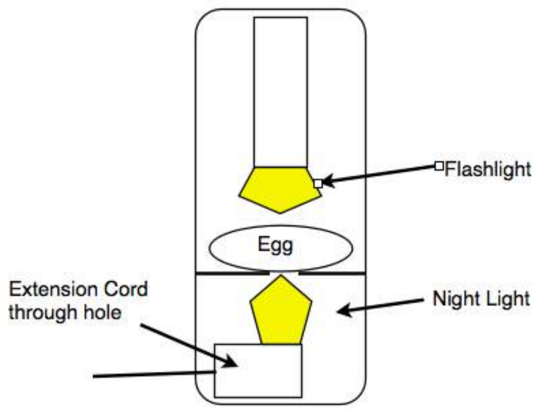
Only fertile eggs may be hatched into chicks. When these are required, the breeders can be kept in a deep-litter house together at a ratio of about 15 females to 1 male.

Testing for egg fertility

The fertility of the eggs may be tested between the ninth and fifteenth day. This can be done by holding a candle behind the egg in a dark room to illuminate the contents. This process is known as ***candling***. A special egg candler, which consists of a box with a strong electric light shown through a small opening over which the egg may be rotated to view the content, may be used. Fertile eggs are called ***clears.***

Structure of the egg

The shell of a good egg is strong and porous which allows the developing chick to carry on respiration. The shell is lined



Egg Candler

with two membranes which separate at the end and leaves an air-space. The egg white (albumen) lies inside the membranes and make up about 60 % of the egg's weight. The yolk, which carries the nucleus or germ spot of the ovum (egg cell), is attached to each end of the egg by a twisted rope of thick white material called *chalaza*.

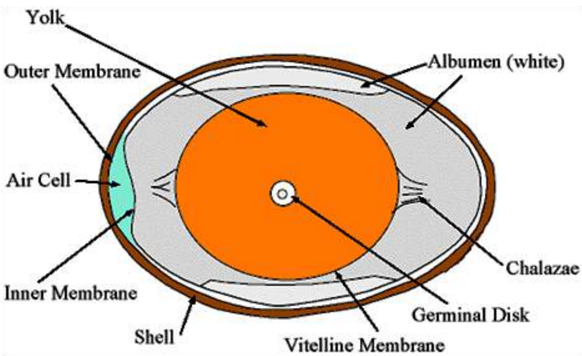


Fig. 30.7: Internal structure of an egg

Incubation

Incubation is the process of keeping fertile eggs under favourable conditions for the embryo to develop into chicks within a period of time (incubation period).

Eggs for breeding are to be kept at a warm temperature of about 38°C to 39°C for about three weeks for them to develop into chicks. This process may be done naturally or artificially. In ***natural incubation*** the hen sits on her eggs to provide the needed conditions. In ***artificial incubation***, which is a preferred choice in brooder houses, incubators are used to hatch large number of eggs at a time, depending on the size of the incubator.

Fig. 30.8: A poultry egg incubator

Conditions necessary for incubation

1. Warmth of about 38°C to 39 °C
2. Humidity of about 60% and 90% for the first two weeks and the last week respectively
3. Adequate ventilation
4. Continuous turning of eggs

Brooding

Day-old chicks are usually left in the incubator without feeding for a day after hatching. They are then moved to a brooder also called a brooder house.

***A brooder is a specially made structure equipped with feeding and drinking facilities as well as warmth for the chicks*.**

It could range from a complex structure housing a thousand chicks to a simple home-made structure containing a few chicks.

The chicks require much attention in the first few weeks and therefore, brooder house management practices, routinely taken to ensure their health, comfort and growth, must be observed. The temperature of the brooder must be critically taken into consideration. The temperature in the house should be about 35^{o}C in the first week and continuously lowered until the sixth week when the chick would have developed enough feathers. The temperature is regulated with electric light with a hover to concentrate the heat.

Fig. 30.9: A simple brooder

Actions of chicks in a brooder

The farmer should look out for the following actions of the chicks to determine whether they are comfortable with the temperature in the brooder house:

- ❖ If the chicks cluster together under the light (hover), it means the temperature is too cold.
- ❖ If they spread out with their beaks open, it means the temperature is too hot.
- ❖ If they spread themselves uniformly under the hover, it means they are comfortable with the temperature.

Qualities of a good brooder house

1. It should have a suitable temperature.
2. There must have adequate ventilation.
3. It should be built with thick insulating walls.
4. It should contain feeding and water containers.
5. It should be provided with footbath.
6. The floor should be cemented and littered with suitable materials such as wood shavings or straw.

NB: Fine sawdust should not be used for littering as it can block the nostrils of the chicks.

Feeding

Young chicks need protein, vitamins and minerals. They should therefore be fed well with *starter mash*, which contain these nutrients. When they are between 12 and 24 weeks old, the starter mash can be replaced with grower's mash and then later layer's or breeder's mash. They should be supplied with fresh water all the time.

Vaccination

Chicks have weak immune system and could be affected by many infections. Vaccinating them against those infections is a sure way to get them going. The vaccines can be administered orally through water or through injection.

Debeaking

Debeaking involves cutting off about half of the upper mandible and the tip of the lower mandible of the beak of a poultry bird.

Debeaking is done in order to control cannibalism in the form of vent pecking and egg eating. It should be done by an expert, because if the lower mandible is severely cut, the bird may not be able to eat well. It is done with a hot knife or an electric debeaker when the birds are about a week or two old.

Fig. 40.0: A debeaked fowl

Culling

Culling is the removal of sick, unproductive and poor performing birds from the flock. Poor performing or unproductive bird can be sold off or slaughtered for food, while sick birds may be isolated and treated or in worst cases disposed of. Culling of birds should normally be done at night when the birds are quiet and less frightened.

Importance of culling

1. It reduces the spread of disease and parasites.
2. It creates more space for the rest of the flock.
3. It saves the cost of feeding unproductive birds.

Poultry management systems

Poultry is kept under three management systems:

- ❖ Intensive system
- ❖ Semi-intensive system
- ❖ Extensive system

Intensive system

Under the intensive system, the birds are kept permanently in a poultry house. There are two types of intensive system of poultry keeping – the deep-litter system and the battery-cage system.

The deep-litter system

In this system birds are confined in a well-ventilated house with a litter of wood shavings, cut straw or sawdust. Birds are supplied with water and feeding troughs as well as egg-laying nests if layers are kept.

Fig. 40.1: Deep-litter system

Advantages of deep-litter system

1. There is increase in production because of more efficient management.
2. It requires less labour
3. Birds are protected from predators and theft.
4. Birds are protected from harsh weather conditions.
5. The eggs are kept intact.
6. Birds are comfortably fed.
7. Disease outbreak is minimized.

Disadvantages of deep-litter system

1. It requires high capital for housing and poultry construction.
2. Birds have no access to natural vegetation, insect or natural light.
3. Overstocking or overcrowding may lead to cannibalism.
4. Dirty litter may soil the eggs.
5. It is hard to keep record on individual birds.
6. The birds have little exercise as they do not move about much.
7. Sick and unproductive birds are hard to detect.

Battery-cage system

The battery-cage system is the the most preferred intensive system. Here, birds are kept in wire-mesh, galvanised or wooden cages for the whole of their productive life. The cages are large enough to allow the birds to stand, sit or move a little. The cages may be arranged in one, two or three tiers. Each cage can house one or two bird with water and food troughs. Extended racks at the front and back of the cages collect the eggs and droppings respectfully.

Fig. 40.2: Battery-cage system

Advantages of battery-cage system

1. It requires less labour.
2. It does not take up much space.
3. There is low incidence of disease and parasites.
4. Birds have easy access to feed and water.
5. Cannibalism is controlled.
6. Eggs are clean and free from brakes.
7. Records on each bird are easily kept.
8. Sick and unproductive birds are easily detected.
9. Less noise is produced because of little interaction between the birds.

Disadvantages of the battery-cage system

1. The cages are expensive to build and maintain.
2. Battery-cages cannot be used for brooding chicks.
3. The battery-cages easily experience technical fault and need constant repairs.
4. Birds have little access to natural food.
5. Birds suffer from cage fatigue due to little exercise.
6. Birds normally experience weak legs and wounds as a result of long stay in the cage.
7. Eggs with thin shells may easily break when dropped on the metal racks.

Semi-intensive system

The semi-intensive system is an intermediary between the intensive and the extensive systems, in that, birds are provided with some form of housing as well as freedom to roam in a limited (normally fenced) area during the day. There are two types of semi-intensive system – the run unit system and the fold unit system.

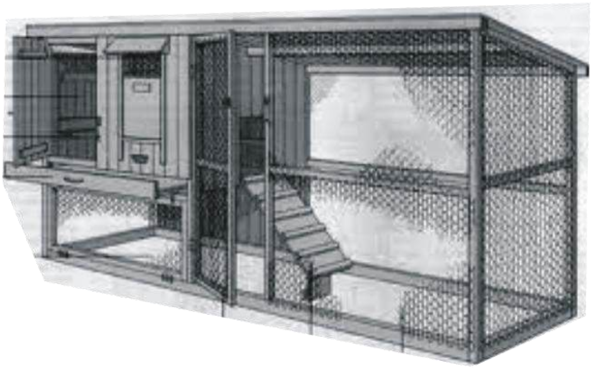
The run unit system

This system has fixed housing surrounded on either side by fenced grasslands known as ***runs***. The birds are allowed on one run at a time, rotated in six to twelve month cycle, to feed on the vegetation, worms and grit during the day. They are allowed into the house at nights and during extreme weather.

Run unit system

The fold unit system

Under this system, the birds are housed in a movable house with enclosed runs attached to it. The runs are covered from below with wire net to bar off predators. The unit is moved to a new and cleaner place of grassland each day.



A fold unit

Advantages of the semi-intensive system

1. It requires relatively low capital.
2. The birds are protected from predators and harsh weather.
3. It is good for egg production.
4. Birds have access to natural vegetation.
5. It is ideal for backyard poultry keeping.
6. It is suitable for birds of all ages

Disadvantages of semi-intensive system

1. It requires constant labour in moving the fold unit.
2. It is only suitable for flat land.
3. Cannibalism and feather picking may occur.

Extensive system

This system of poultry keeping is mostly practiced on subsistence and is mostly used with local breeds of chicken. It normally has a low stocking rate with a maximum of about 750 to 1000 fowls per acre. Birds are left to move about all the time in search of their own food. In some cases, there may be a little housing facility to shelter the birds at night or in extreme weather. There are two types of the extensive system – the free running system and the free range system.

The free running system

In this system, birds are often found roosting on trees and walls at night.
Occasionally, a simple coop made from cheap materials such as wood or bamboo, with some supplementary food may be provided by the owner. The coop may be in two or three tiers and serves as the brooder.

Free running system

The free range system

This extensive system is characterized by the fowls allowed to move about on a large, fenced backyard. They are normally provided with a simple housing to shelter them at nights. Supplementary ration may be left around the compound. It is better than the free running system in that the fencing prevents the birds from going astray or being stolen.

Free range system

Parasites

Endoparasites such as tapeworms and round worms can be controlled with recommended dewormers. Ectoparasites such as tick, lice, fleas, mites and flies can be controlled with insecticides. Sodium fluoride is very effective against tick and lice when dusted on birds.

Processing, storage and marketing of poultry products

Pre-slaughter activities

Only healthy fowls should be slaughtered for food. Sick birds culled and killed must be disposed of. Before healthy fowls are slaughtered, they should not be fed for 24 hours. This is to ensure that the digestive tracts of the birds are clean of faeces, which may render the meat unwholesome.

Post-slaughter activities

Slaughtered fowls should be allowed to bleed fully. This improves the quality of the meat and also reduces microbial activities on the meat. The carcass is dressed by:

Scalding – dipping the carcass in water at about 65^{o}C for three to five minutes; this soaks the base of the feathers and makes them easier to remove.

Plucking – removal of feathers either by hand or with a feather plucker.

Singeing – exposing the carcass to flame to burn off the hair-like filoplume feathers. The carcass is then washed clean.

Evisceration - cutting open and removing the internal organs. Some organs such as the liver, the heart and gizzard are packed together as giblets and sold or inserted in the carcass, while the other organs such as the pancreas, the gall-bladder and the intestines are thrown away. The legs are sometimes kept while the head is thrown away. It is then cooked and eaten or sealed in a plastic bag and refrigerated or sold.

Table 6.4: Diseases and parasites of poultry

Disease	Symptom	Prevention and control
Newcastle disease (*viral disease*)	• Infected Birds become blind. • They become paralysed. • Their necks are often twisted. • Discharge from the nose	• Proper sanitation and scheduled vaccination • Infected birds should be culled and burnt or buried
Coccidiosis (*protozoan disease*)	• Loss of appetite • Blood-stained dropping • Low egg production rate • Eyes continuously closed	• Proper sanitation • Burning old litters • Disposing of infected birds
Gumboro disease (viral disease)	• Birds are often nervous. • They become sleepy always. • They become dehydrated.	• Good sanitation • Vaccination • Use of antibiotics in water • Culling infected birds
Avian flue (viral disease)	• Nervous breakdown • Difficulty breathing • Sudden death of bird	• Scheduled vaccination • Culling and burning of infected bird
Pullorum disease (*bacterial disease*)	• Ruffled feathers • Laboured breathing • Droppings stick together • Chicks huddle together	• Good sanitation • Fumigation of incubators • Infected birds should be treated with antibiotics

Storage of eggs

Egg spoilage occurs mostly because of:

❖ Microbial infections; example,
❖ penicillin, salmonella, pseudomonas etc
❖ Loss of moisture
❖ Loss of carbon dioxide

Methods of preserving eggs

1. ***Refrigeration (cold storage)*** – keeping eggs at a temperature of -2^{o}C to 0^{o}C in a refrigerator
2. ***Egg power*** – drying eggs and crushing into powder and then packing into air-tight containers
3. ***Freezing liquid eggs***- breaking and removing yolk from the albumen and keeping them separately in a freezer
4. ***Sodium silicate solution*** – putting eggs without cracks in 10% sodium silicate solution.

Pig

Pigs are animals of the swine family, extensively raised in almost every part of the world as a food animal. The adult pig has a heavy, rounded body; a comparatively long, flexible snout; short legs with cloven hooves; and a short tail. The thick but sensitive skin is partly covered with coarse bristles and exhibits a wide range of colour patterns.

Pigs are well adapted for the production of meat because they grow and mature rapidly. They have a short gestation period of about 114 days, and they produce large numbers of young each time they give birth.

They are omnivorous and can scavenge a wide range of foods. As food sources, they convert cereal grains and legumes such as soybeans into meat.

Reasons for keeping pigs

1. Meat production
2. Leather from skin
3. Hair as bristles for brushes
4. They are also primary source of edible fat

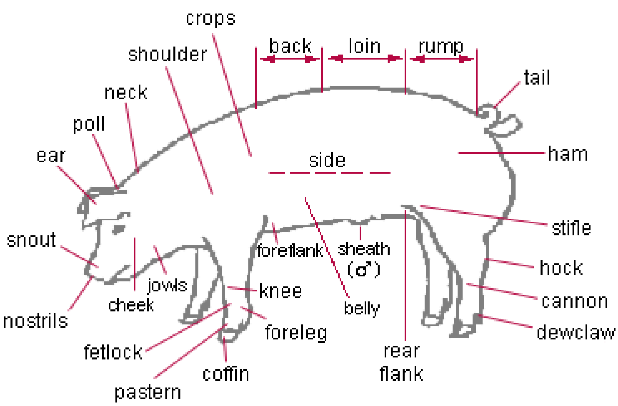


Fig. 40.3: External features of a pig

Breeds of pigs

Breeds of pigs reared in Ghana include both exotic and local breeds.

Local breed

The main local breed reared is the Ashanti Dwarf pig; which is small, black, has great resistance to pest, diseases and harsh environmental conditions. They are normally kept for subsistence as they are poor performers in reproduction.

Fig. 40.4: Ashanti dwarf

Exotic breed

Exotic breeds are preferred for commercial purposes as they usually perform better. Members include:

Yorkshire (Large White) – big stomach, wide chest, muscular body, strong legs, large and erect ears and white coat colour.

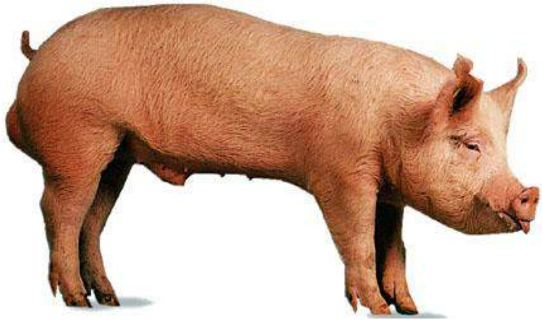
Hampshire – short and erect ears, black coat colour with white belt around the legs and shoulders.

Landrace – long and slender body, slightly concave back, drooping ears and white coat colour.

Poland China – drooping ears, black coat colour, sometimes with white markings. Good for the production of meat and lard.

Duroc – medium size with almost horizontal back, long snouts and ears, dark red or red-brown coat.

Tamworth – straight head and long neck, long body, snout and legs, erect ears, dark red and golden red coat colour.



Yorkshire

Poland China

Hampshire

Landrace

Fig. 40.5: Breeds of pig

Management systems

Pigs are housed in a unit called ***sty***. A sty should have a hard concrete, slightly slanting floor to facilitate easy drainage. It should be well ventilated and spacious to avoid overcrowding. There should be a separate feeding area with feeding and water troughs. The sty should also be provided with wallows (mud pool) as wells as farrowing pen or rail to keep the piglets in. A number of sties in an enclosed farm is called a ***piggery***.

Table 6.5: Diseases and pathogens of pigs

Disease	Symptom	Prevention/ control
Rabies (*viral*)	• Paralysis, • dumbness, • profuse salivation, • dislike for water	• Vaccination, • quarantine of infected pigs • in extreme cases, killing the animal
African swine fever (*viral*)	• High fever, • diarrhoea, • vomiting, • bad-smelling discharge from nose and eyes	• Good sanitation, • culling of infected animals,
Foot rot (*bacterial*)	• Swollen foot, • sore hooves, • high body temperature, • lameness	• Good hygiene, • use of antibiotics, • quarantine of infected animal
Tetanus (*bacterial*)	• Stiff limbs and jaws, • profuse sweating, • insensitive skin	• Immunization, • prevent wound infection, • use tranquilizers to relax muscles
Redwater (*protozoan*)	• Red urine, • loss of appetite, • abortion in pregnant sows, • salivation	• Good sanitation, • use antibiotics, • vaccination
Heartwater (*rickettsial*)	• Paralysis, walking in circles, protruding tongue,	• Insecticide against tick, vaccination, • use of antibiotics
Brooder pneumonia (*fungal*)	• Difficulty in breathing, • dull skin	• Proper sanitation, • use of antibiotics, • culling of infected animals

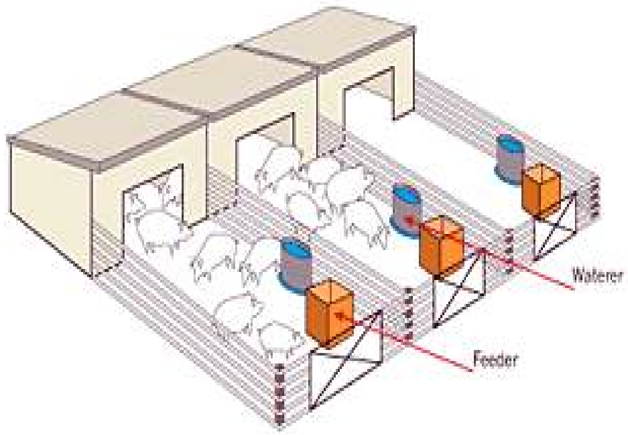


Fig. 40.6: A piggery

Like all farm animals, pigs are kept under three management systems - the intensive system, the semi-intensive system and the extensive system.

The intensive system

The intensive system is characterized by total confinement of the pigs in a sty or piggery, with feed and water provided.

The semi-intensive system

The semi-intensive system is characterized by the pigs allowed to move around during the day and confined at night.

The extensive system

The extensive system is normally practiced for non-commercial purposes and involves the animals being allowed to move about to search for the own food and wallows.

Finishing, processing and marketing

After six month, pigs become mature enough for slaughter. Only healthy pigs should be slaughtered. The slaughtered pig should be allowed to bleed well, after which some post-slaughter activities such as scalding, singeing and evisceration are carried out.

The carcass is then washed and cut into pieces – the head, shoulder, ham, loin and bacon. The fresh of a pig is called *pork*.
The cut pork may be frozen, salted, dried, canned, cooked or sold fresh.

Rabbit

Rabbits are small furry herbivorous and non-ruminant mammals with long ears and short tails. In the wild, rabbits live in colonies in underground burrows. Domestically, rabbits are kept for their meet, fur, skin (pelt) and sometimes recreation and pleasure.

Breeds of rabbits

Most of the breeds of rabbits reared in Ghana are exotic breeds. Common exotic breeds kept by rabbit farmers include California White, Chinchilla, Flemish Giant and New Zealand White.

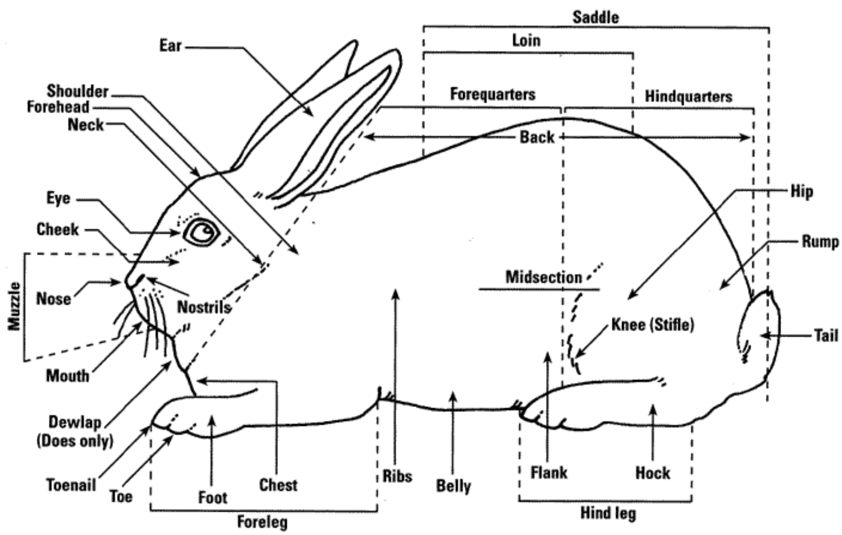


Fig. 40.7: External features of a rabbit

California White – white in colour, medium body size, straight back, erect and backward slanting ears.

Chinchilla – grey or greyish-blue in colour, long body, erect and backward slanting ears.

Flemish Giant – grey or greyish-brown in colour, large body, raised back

New Zealand White – white in colour, broad shoulders and back, pinkish eyes, large backward slanting ears

California White

Chinchilla

Flemish giant

Fig. 40.8: Breeds of rabbits

Breeding

Rabbits mature sexually after six to seven months. A doe that shows signs of heat is taken to a buck (male rabbit) for mating. It is always advisable to take a doe to a buck and not the other way round as that normally causes fighting among the does. A buck can service about ten does.

A buck should not be allowed to mate for more than four times in a week. This is to conserve its vitality.

A doe normally becomes pregnant two weeks after a successful mating.

Signs shown by a pregnant doe

1. refusal to mate;
2. increased body weight;
3. frequent sleeping;
4. loss of appetite.

Caring for the young rabbits

After a gestation period of 30 days, a pregnant doe gives birth to her bunnies in a process known as ***kindling*** or ***littering***. Bunnies are born with no fur on and their eyes closed. The mother rabbit (dam) pulls furs from her belly and uses it to cover the bunnies until they develop their own fur after, 20 days after kindling. The young rabbits suck the dam's breast for the first month until they are introduced to green feed. They are normally weaned after two to months.

Housing for rabbits

Rabbits are housed in a structure known as a ***hutch***. A rabbit farm or rabbitry is made up of a number of hutches. A hutch is simply constructed from a wood or wire gauze with a backward slanting roof. The floor which is made of wire mesh is covered with straw to provide warmth for the rabbits. Water and feeding troughs, as well as nestle boxes are provided in the hutches. Rabbits are normally fed on fresh forage such as rabbit weed, guinea glass and waterleaf. Cereals, legumes, lettuce and cabbage also serve as important sources of food for rabbits.

Pests and parasites of rabbits

Common ectoparasites of rabbits include lice, mites and ticks; with tapeworm and roundworm being the major endoparasites.

Fig. 40.9: Farmers tending their rabbits

The ectoparasites can be controlled by practices such as good sanitation and hygiene as well as using recommended insecticides. The endoparasites are controlled by keeping carriers of worm such as cats, snails, etc. away from the hutch.

Finishing, processing and marketing

Before their average maturity period of five to six months, the rabbits should be well fed to fatten them, thereby given them a higher market value. Those due for slaughter should be starved for a day in order to get rid of faeces in their guts. When slaughtered, the carcass should be allowed to bleed well to improve the taste of the meat and also reduce the activities of microorganisms on it. It should further be washed and hanged for about an hour to drain it before it is stored, cured or packaged for sale.

Table 6.6: Diseases of rabbits

Disease	Symptom	Prevention and control
Ear mange/ canker	• Brown or yellow scaly crust in the ears. • Scratching or shaking or head frequently	• Proper sanitation, • applying a mixture of iodine and palm oil on infected ears
Coccidiosis	• Diarrhoea, • animals become emaciated and develops big belly	• Use of antibiotics, • Culling of infected rabbits
Myxomatosis	• Inflammation and swelling of the eyes, ears, nose and genital organs of infected animals, • loss of appetite, high fever	• Good hygiene and sanitation • Destruction of infected animals
Rabbit fever/ tularemia	• Fever, chills, • enlarged gland at the elbow	• Isolation of infected animals; • Use of antibiotics

Importance of keeping rabbits

1. Requires little capital to start
2. Requires little housing space
3. They are prolific animals – each doe (adult female rabbit) produces from 30 to 40 bunnies (young rabbits) yearly
4. They are highly resistant to many diseases.
5. Their droppings are good source of organic manure.
6. Their feed is easy to acquire.
7. They are easy to handle.
8. They can be transported easily.

TEST QUESTIONS

1. (a) What are farm animals?
   (b) State five importance of keeping farm animals.
   (c) Describe four activities involved in farm animal production.

2. (a) Define the term breeding.
   (b) State five importance of breeding in animal husbandry.
   (c) Explain the following methods of breeding in animal husbandry:
   (i) mating;
   (ii) introduction;
   (iii) cross-breeding.

3. (a) What is artificial insemination?
   (b) Mention three advantages of artificial insemination.
   (c) State four signs shown by animals on heat.

4. Write a short note on the following types of cattle:
   (a) beef cattle;
   (b) diary cattle;
   (c) dual purpose cattle;
   (d) work cattle.

5. Describe the following management practices in animal husbandry:
   (a) intensive system;
   (b) semi-intensive system;
   (c) extensive system.
6. State four advantages and four disadvantages of the following management systems:
   (a) Intensive system;
   (b) Extensive system.
   (c) Mention four differences between intensive and extensive systems of animal keeping.

7. (a) Define hay and silage.
   (b) State four differences between hay and silage.
   (c) State three qualities each of a well prepared hay and silage.
   (d) Mention three reasons why silage is preferred to hay.

8. (a) What is identification in animal husbandry.
   (b) Describe the following methods of animal husbandry:
   (i) branding;
   (ii) tattooing;
   (iii) ear notching.

9. (a) What is castration?
   (c) Explain two methods of castrating a farm animal.

10. Describe the following husbandry practices:
    (a) dehorning;
    (b) deworming;
    (c) dipping.

11. (a) Define the term poultry.
    (b) State four reasons for keeping poultry.
    (c) State three differences between indigenous and exotic breeds of chicken.

12. (a) Describe the following types of chicken:
    (i) layers;
    (ii) broilers;
    (iii) breeders.
    (b) Mention two characteristic each of broilers and layers.

13. (a) Define the term incubation in poultry production.
    (b) state the conditions necessary for incubation.
    (c) List four methods of preserving eggs.
14. (a) Define the term brooding in poultry.
    (b) mention four qualities of a god

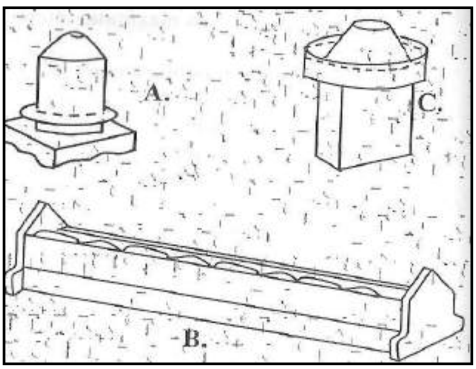
brooder house.

15. (a) What is culling?
 (b) State three importance of culling.

16. Mention the causative organism, symptom and methods of control of the following diseases in poultry:
 (a) Newcastle disease;
 (b) Coccidiosis disease;
 (c) Avian flu;
 (d) Gumboro disease

17. Explain three management practices that are used to control and prevent diseases and pests in farm animals.

18. **Fig. 3** below is an illustration of two devices **A** and **B**, used in the pen for fowls. Examine the diagram carefully and answer the questions that follow.



a) Identify each of the devices **A** and **B**.
b) State the function of each of the devices.
c) Explain why device A is placed on a platform.
d) Describe how each of the devices A and B is maintained.
e) Explain a possible problem that a poultry farmer may encounter due to the design of device B.

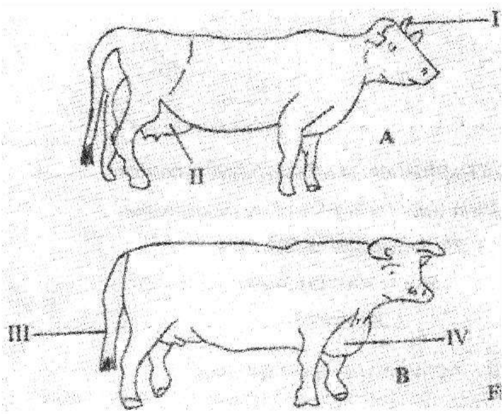
19. The diagram below is an illustration of the head of a fowl on which a management practice has been carried out.
 Study the diagram carefully and answer the questions that follow.

a) State the management practice that has been carried out on the fowl.
b) Name the parts labelled I, II, III and IV.
c) i) Give three reasons for carrying out the management practice.
 ii) Name three tools used for carrying out the management practice.
d) State three precautions to be taken when carrying out the management practice.
e) i) Name one disease that affects the

part labelled III.

ii) State one method of preventing the disease you have named in(e) (i).

20. **Fig. 1** is an illustration of to breeds of cattle **A** and **B**.
Study the diagram carefully and answer the questions that follow.



(a) Name the parts labelled I, II, III and IV.
(b) State three uses of the part labelled I to the animal.
(c) State three differences in the body conformation between cattle breeds A and B.
(d) Name the major product for which each breed is reared.
(e) (i) What is the most appropriate system of keeping cattle breed A?
(ii) State three advantages of the system named in (e)(i) above.
(f) (i) Name one ectoparasite of cattle breeds **A** and **B**.
(ii) State two ways of controlling the ectoparasite you have named in (f) (i).

21. (a) mention four breeds of pigs and two characteristics each.
(b) describe the management systems of pigs.

22. The following are diseases that affect pigs.
Copy and complete the table.

Disease	*Symptom*	*Control*
Rabies		
African swine fever		
Brooder pneumonia		
Red water disease		

23. (a) State five advantages of keeping rabbits.
(b) List three breeds of rabbits.
(c) Describe how you will prepare a rabbit for the market.

24. A farmer brought n 500 day-old chicks to be reared on his farm. During brooding, the variation of mortality of the chicks with temperature was determined over a five-week period.

The farm record available over the period is shown in the table below.
Study the table carefully and answer the questions that follow.

Week	*Temperature*	*Number of chicks that died (Mortality)*
2	34.0	28
2	31.2	10
3	28.4	6
4	25.6	2
5	22.8	2

(a) (i) How many chickens died during the five-week period?
(ii) Calculate the percentage mortality at the end of the fifth week.
(b) Explain why the temperature in the brooder house was reduced in each successive week.
(c) (i) Suggest three factors responsible for the high chicken mortality during the first week.
(ii) State two ways of reducing mortality among the chicks during the first week.
(d) (i) Name three sources of heat for a brooder house.
(ii) State two behaviours of chicks in a brooder house when the temperature is too high.

25. The table below illustrates a farm record kept by a backyard farmer rearing local sheep.

Study the table carefully and answer the questions that follow.

Pen name Name of sheep (ewe) Source: Year of Birth		4 Nungua 4 Nungua Farms, Accra July, 2000		
Date of service	Date of birth	Number of lams per litter	Number of lambs per born dead	Number of lambs per born dead before weaning
5/9/2001 20/8/2002	5/9/2001 20/1.2003	3 3	2 3	1 0

a) Name the type of record displayed in the table above.
b) State five ways in which having such a record is beneficial to the farmer.
c) From the table, determine the duration of pregnancy of the ewe.
d) i) State four possible reasons for the result obtained in the table above.
ii) What conclusion can you draw from the result?

29

EXCRETORY SYSTEM

Specific Objectives

After completing this chapter, you will be able to:

- Explain the meaning of excretion
- Identify organs of the excretory system.
- Mention some disorders of the urinary system in humans and their remedies.
- State excretory products of flowering plants and explain how they are excreted.

INTRODUCTION

All living things are made up of cells. Within each cell are two basic chemical processes – ***respiration***, where food is broken down to releases energy. The other chemical process is known as ***metabolism***, where substances are built up (*anabolism*) and broken down (*catabolism*).

During metabolism, both useful and harmful products are produced. The useful products are used in the body while the harmful ones are removed from the cells.

Some of the harmful products are poisonous, and their presence in the cells could cause some disturbance. It is therefore very important that they are removed. The harmful products are referred to as ***metabolic waste*** products and the process by which they are removed or eliminated from the body is called ***excretion***.

EXCRETION IN MAMMALS

Excretion is the removal of metabolic waste products from the body or cells of living organisms.

Excretion is normally confused with *egestion*.

Egestion is the removal of waste products of digestion as faeces.

Table 6.7: Differences between excretion and egestion

Excretion	Egestion
Removal of metabolic waste from the body	Removal of waste products of digestion
Waste products are removed from the cells	Undigested food substances are removed from the alimentary canal.
Excretory products are removed through special excretory organs eg. lungs, skin, kidney and liver.	Products or egestion are removed through the anus.
Excretory products include carbon dioxide, urea, bile pigment, water etc.	Egested products are faeces and water.

ORGANS OF EXCRETORY SYSTEM AND PRODUCTS

In mammals, metabolic waste products are produced in many forms. This requires different organs to keep the process going. The organs with their associated structures form *the excretory system*. There are four main excretory organs in mammals. They are the lungs, liver, skin and kidneys.

THE LUNGS

The lungs are responsible for the excretion of carbon dioxide (which is a waste product of cellular respiration) from the body. Carbon dioxide travels form the cells to the lungs through the blood stream, diffuses into the alveolar and is finally removed from the body when the mammal exhales.

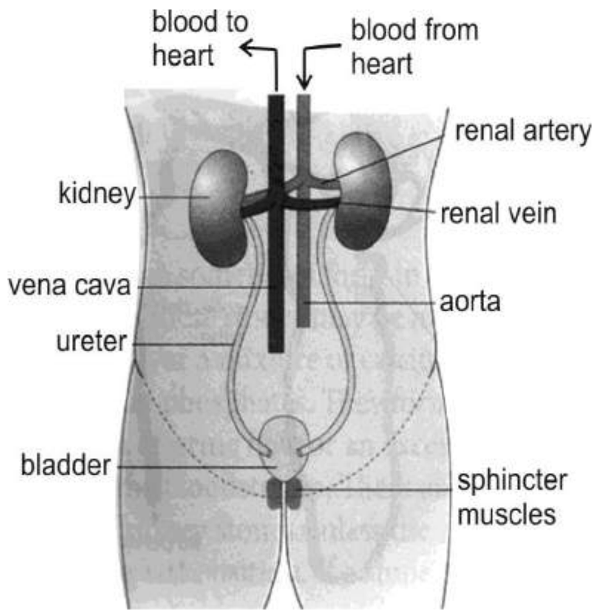


Fig. 50.0: Excretory System

THE SKIN

Excess water, mineral salt, urea and carbon dioxide are excreted from the body by the skin, mostly in a form of sweat.

Structure of the skin

The skin is made up of two layers, the *epidermis* and the *dermis.*

Epidermis

The epidermis is the upper or outer layer of the skin. It is a tough, waterproof, protective layer. It is made up of three distinct layers or regions:

Cornified or horny layer, which is the thin upper layer, made up of flat, dead cell which flake off continuously.

Granular layer, which lies below the cornified layer and consists of cells produced by Malpighian layer below. The cells are slowly pushed outwards as new cells accumulate underneath them. They replace the cornified layer as they become flat when they lose their nuclei and die.

Malpighian layer, which is a one or two-cell thick continuous layer. These cells produce new epidermis as they divide repeatedly. The Malpighian layer contains the pigment melanin which determines the skin colour.

Dermis

The dermis, or inner layer, is thicker than the epidermis and gives the skin its strength and elasticity. The dermis contains nerve endings, blood capillaries and

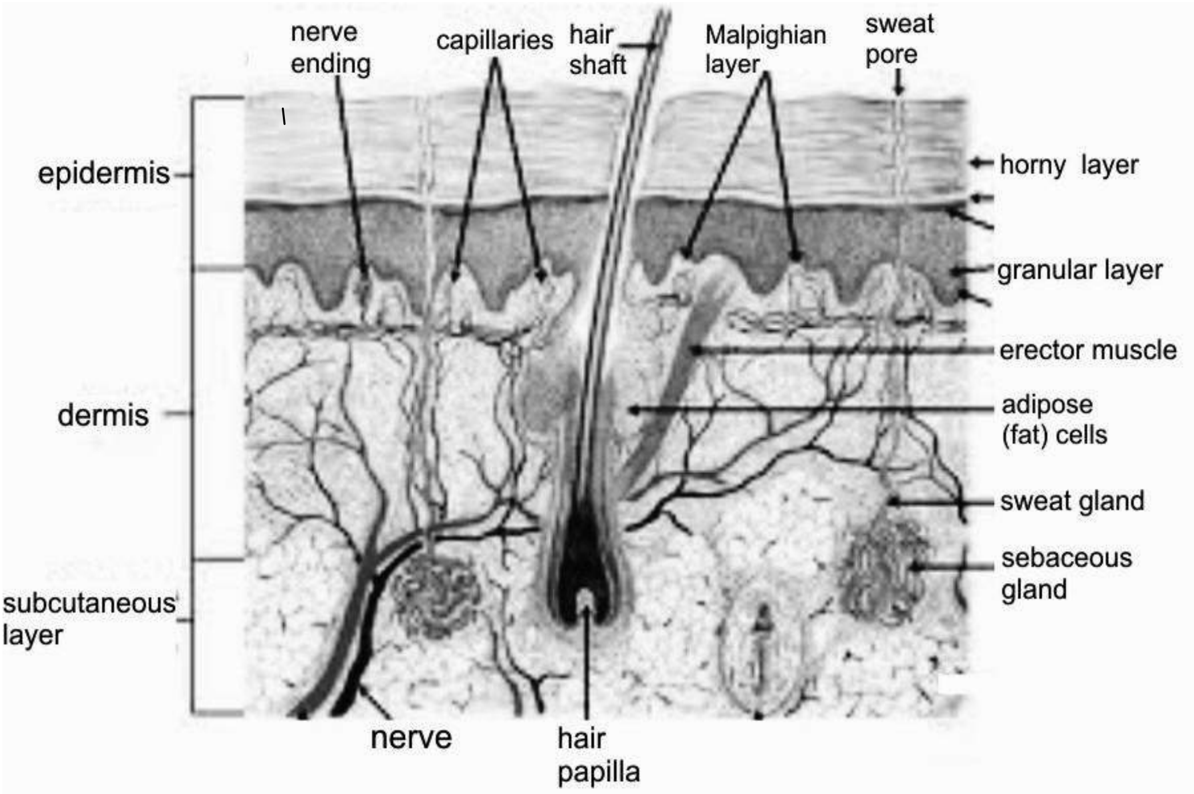


Fig. 50.1: Structure of skin

lymphatic vessels. It also houses hair follicles and sweat glands produced by the epidermis.

Subcutaneous layer

Below the dermis is the subcutaneous layer, a layer of tissue composed of protein fibres and adipose tissue (fat). Although not part of the skin itself, the subcutaneous layer contains glands and other skin structures, as well as sensory receptors involved in the sense of touch.

The two layers of the skin (i.e. epidermis and dermis) are anchored to one another by a thin but complex layer of tissue, known as the ***basement membrane***. This tissue is composed of a series of elaborately interconnecting molecules that act as ropes and grappling hooks to hold the skin together.

Functions of the skin

The primary functions of the skin are:

a) **Protection**
The skin is an organ of double-layered tissue stretched over the surface of the body and protecting it from drying or losing fluid, from harmful external substances, and from extremes of temperature. Fat cells in the dermis insulate the body, and oil glands lubricate the epidermis.

b) **Thermoregulatory**
The blood vessels of the dermis supplement temperature regulation by contracting to preserve body heat and expanding to dissipate it. The epidermis also contains pores that allow sweat to be evaporated from the body

c) **Sensory organ**
The skin has sensory nerve endings and separate kinds of receptors which convey pressure, temperature, and pain. This enables a person to detect and free from danger when subjected to it.

d) **Excretion**
Urea, mineral salt and excess water are removed from the body by the sweat gland in the form of sweat. Carbon dioxide is also excreted from the body through the skin.

THE KIDNEYS

The kidneys excrete water and urea. Urea is produced mostly in the liver as the end-product of protein metabolism. The nitrogen in urea, which constitutes most of the nitrogen in the urine, is produced mainly from food protein, but parts come from the breakdown of body cells.

The structure of the kidney

The outermost layer of the kidney is called the ***cortex***. Beneath the cortex lies the

medulla, an area that contains between 8 and 18 cone-shaped sections known as ***pyramids,*** which are formed almost entirely of bundles of microscopic tubules. The tips of these pyramids point toward the centre of the kidney.

The cortex extends into the spaces between the pyramids, forming structures called ***renal columns***. At the centre of the kidney is a cavity called the ***renal pelvis*.**

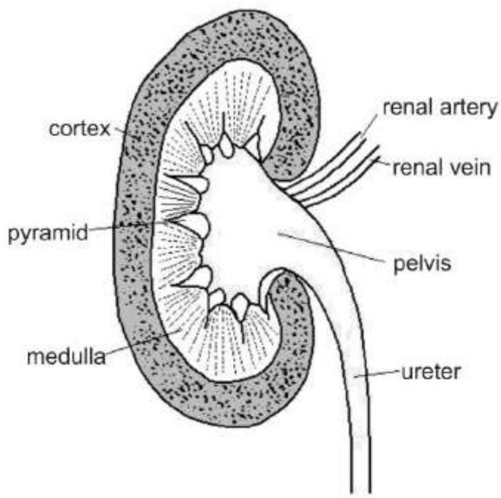


Fig. 50.2: Internal s*tructure of the kidney*

Nephron

The task of cleaning or filtering, the blood is performed by millions of ***nephrons.*** The nephrons are remarkable structures that extend between the cortex and the medulla. Under magnification, nephrons look like tangles of tiny vessels or tubules, but each nephron actually has an orderly arrangement that makes filtration of wastes from the blood possible.

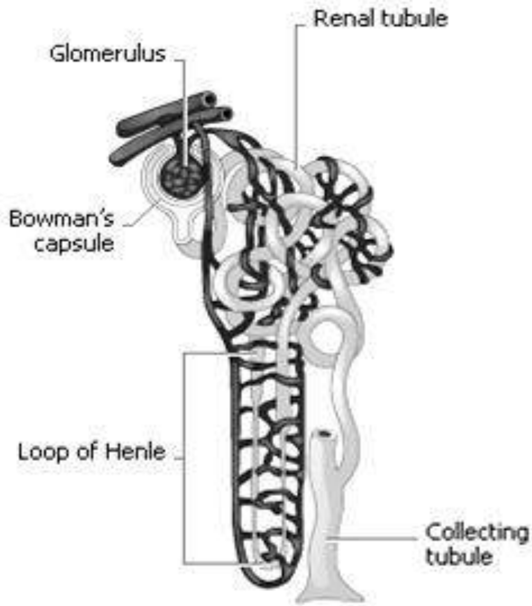


Fig. 50.3: Structure of nephron

Structure of the nephron

The primary structure in this filtering system is the ***glomerulus,*** a network of extremely thin blood vessels called ***capillaries***.

The glomerulus is contained in a cuplike structure called ***Bowman's capsule***, from which extends a narrow vessel, called the ***renal tubule***. This tube twists and turns until it drains into a collecting tubule that carries urine toward the renal pelvis.

Part of the renal tubule, called the ***loop of Henle***, becomes extremely narrow, extending down away from Bowman's capsule and then back up again in a **U** shape.

Surrounding the loop of Henle and the other parts of the renal tubule is a network of capillaries which are formed from a small blood vessel that branches out from the glomerulus.

Functions of the kidneys

1. The kidneys regulate the amount of water contained in the blood. This process is influenced by ***antidiuretic hormone*** (ADH).

2. The kidneys interact with the hormone *aldosterone* to regulate the blood's sodium and potassium content.

3. The kidneys adjust the body's acid-base balance to prevent blood disorders such as acidosis and alkalosis, both of which impair the functioning of the central nervous system. If the blood is too acidic, meaning that there is an excess of hydrogen ions, the kidney moves these ions to the urine through the process of tubular secretion.

4. The kidneys process vitamin D and convert it to an active form that stimulates bone development.

5. The kidneys produce several hormones such as *erythropoietin*. Erythropoietin influences the production of red blood cells in the bone marrow. When the kidney detects that the number of red blood cells in the body is declining, it secretes erythropoietin. This hormone travels in the bloodstream to the bone marrow, stimulating the production and release of more red cells.

Urine production

Urine production occurs as a result of three precise processes in the kidneys involving millions of nephrons. The processes are:

- filtration
- reabsorption
- secretion

Filtration

Urine formation begins with the process of filtration, which goes on continually in the renal corpuscles.

As blood passes through the glomeruli, much of its fluid, containing both useful chemicals and dissolved waste materials, soaks out of the blood through the membranes (by osmosis and diffusion) where it is filtered and then flows into the Bowman's capsule. This process is called ***glomerular filtration***.

The water, waste products, salt, glucose, and other chemicals that have been filtered out of the blood are known collectively as ***glomerular filtrate***.

The glomerular filtrate consists primarily of water, excess salts, glucose, and a waste product of the body called ***urea***.

Urea is formed in the body to eliminate the very toxic ammonia products that are formed in the liver from amino acids. Since humans cannot excrete ammonia, it is converted to the less dangerous urea and then filtered out of the blood.

Reabsorption

Reabsorption is the movement of substances out of the renal tubules back into the blood capillaries located around the tubules.

Substances reabsorbed are water, glucose and other nutrients, sodium and other ions.

Reabsorption begins in the proximal convoluted tubules and continues in the loop of Henle, distal convoluted tubules, and collecting tubules.

Large amounts of substances are reabsorbed back into the bloodstream from the proximal tubules. Glucose is entirely reabsorbed back into the blood from the proximal tubules. Sodium ions (Na^+) and other ions are only partially reabsorbed from the renal tubules back into the blood. For the most part, however, sodium ions are actively transported back into the blood from the tubular fluid.

Secretion

Secretion is the process by which substances move into the distal and collecting tubules from blood in the capillaries around these tubules.

In this respect, secretion is reabsorption in reverse. Whereas reabsorption moves substances out of the tubules and into the blood, secretion moves substances out of the blood and into the tubules where they mix with water and other wastes and are converted into urine.

These substances are secreted through either an active transport mechanism or as a result of diffusion across the membrane. Substances secreted are hydrogen ions (H^+), potassium ions (K^+), ammonia (NH^3), and certain drugs.

Kidney tubule secretion plays a crucial role in maintaining the body's acid-base balance.

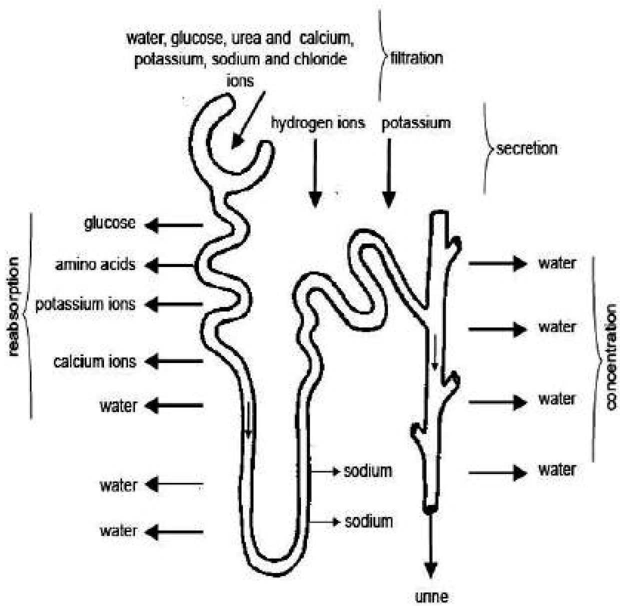


Fig. 50.4: Urine production in the nephron

Summary of urine production

- ❖ Filtration of water and dissolved substances out of the blood in the glomeruli and into Bowman's capsule;

- Reabsorption of water and dissolved substances out of the kidney tubules back into the blood (note that this process prevents substances needed by the body from being lost in the urine).

- Secretion of hydrogen ions (H^+), potassium ions (K^+), ammonia (NH^3), and certain drugs out of the blood and into the kidney tubules, where they are eventually eliminated in the urine.

LIVER

The liver is a large, multipurpose gland in mammals. It plays a vital role in excretion of many substances.
The liver changes the decomposed haemoglobin of the worn out red blood corpuscles into bile pigments (bilirubin and biliverdin). These pigments are passed into the alimentary canal with the bile for elimination in the faeces.

The liver also excretes cholesterol, steroid hormones, certain vitamins and drugs through the bile.

Urea is formed in the liver by a cyclic process called as ***urea cycle***. The amino acids that are not needed in the body are ***deaminated*** by an enzyme oxidase, producing ammonia NH^3. Ammonia, hung toxic, is quickly changed to urea.

DISORDERS OF THE URINAL SYSTEM IN HUMANS

Bed wetting

Bed wetting is common in children, and mostly attributed to over eating at night and laziness to get out of bet to urinate. In some cases, bed wetting can be traced to some medical factors. These factors can be treated surgically or with antibiotics. In some situations, muscular exercises are required to relax sphincter muscles surrounding the entrance of the bladder.

Urine retention

The main causes of urine retention are enlarged prostate gland in males, kidney stones and sexually transmitted diseases (STIs) such as gonorrhea. These causes may be treated either by drugs or surgical operation.

Kidney stones

A kidney stone is a solid mass made up of tiny crystals of protein. One or more stones can be in the kidney or ureter at the same time and obstruct the flow of urine. The stones cause spasms of pain which are normally felt from the lower back of the groin. Smaller stones may pass down the ureter and out through the bladder. Larger stones may either require a surgical operation or be shattered in fragment by laser treatments which are then passed out through the urine.

Cystitis

Cystitis is an inflammation of the urinary bladder caused by bacteria infection of the bladder. Cystitis is accompanied by frequent and painful urination and cloudy urine. Later symptoms include bloody urine, fever and backache. Cystitis can be treated with antibiotics. Drinking plenty of fluid speeds recovery and relieves symptoms. It also prevents people with bladder infection from contracting cystitis.

Pyelonephritis

Pyelonephritis is an inflammation caused by a bacterial infection that starts in the bladder and spreads to the kidney. Sometimes an obstruction that interferes with the flow of urine in the urinary tract can cause the disease. Symptoms of pyelonephritis include fever, chills, and back pain. Antibiotic drugs are usually given to fight the infection, which can scar the kidneys and impair their function if left untreated.

Glomerulonephritis

Glomerulonephritis is another common kidney disease characterized by inflammation of some of the kidney's glomeruli. This condition may occur when the body’s immune system is impaired. Antibodies and other substances form large particles in the bloodstream that become trapped in the glomeruli. This causes inflammation and prevents the glomeruli from working properly.

Symptoms may include blood in the urine, swelling of body tissues, and the presence of protein in the urine, as determined by laboratory tests. Glomerulonephritis often clears up without treatment.

When treatment is necessary, it may include a special diet, immunosuppressant drugs, or plasmapheresis, a procedure that removes the portion of the blood that contains antibodies.

Table 6.8: Summary of excretory organs in mammals with corresponding products

Organ	Excretory product
Lungs	Carbon dioxide and water
Skin	Water, mineral salt, carbon dioxide and urea
Kidney	Urea, mineral salt, water
Liver	Urea , cholesterol and bile pigment

EXCRETION IN FLOWERING PLANTS

In plants, breakdown of substances is much slower than in animals. Hence, accumulation of waste is much slower. Plants do not possess any specialized excretory organs for the removal of metabolic waste products. Nonetheless, plants excrete some waste products such as oxygen, carbon dioxide, water vapour, etc.

Oxygen

In the presence of sunlight, plants use carbon dioxide to prepare their food. At this time, oxygen is given out as a waste product. The oxygen diffuses out through the leaf cells to the leaf air spaces and then out to the atmosphere through the stomata.

Carbon dioxide

Carbon dioxide, a by-product of respiration, is an excretory product when liberated at night. It is removed from the tissue cells by diffusion through the system of intercellular spaces in the leaves, stems and roots. In leaves and young stems the carbon dioxide escapes through the stomata. In young roots carbon dioxide diffuses through the general root surface, especially the root hairs.

Water

Plants can get rid of excess water by ***transpiration.*** Excess water in the cells of the leaf diffuses into the air spaces in the leaf. The water molecules then diffuse through the stomata into the atmosphere as water vapour. Water is also excreted out of the plant through the lenticels of stems as water vapour.

Other waste products

Waste products may be stored in leaves that fall off. Other waste materials that are exuded or produced by some plants such as ***resins, saps, latexes***, ***gums*** etc. are forced from the interior of the plant by hydrostatic pressures inside the plant and by absorptive forces of plant cells. Plants also excrete some waste substances into the soil around them.

Other metabolic waste products may be deposited as harmless insoluble crystals in the plant body. For example, calcium oxalate is deposited in the form of crystals in the leaves and stems of many plants. In trees which shed their leaves yearly the excretory products are removed from the trees during leaf fall.

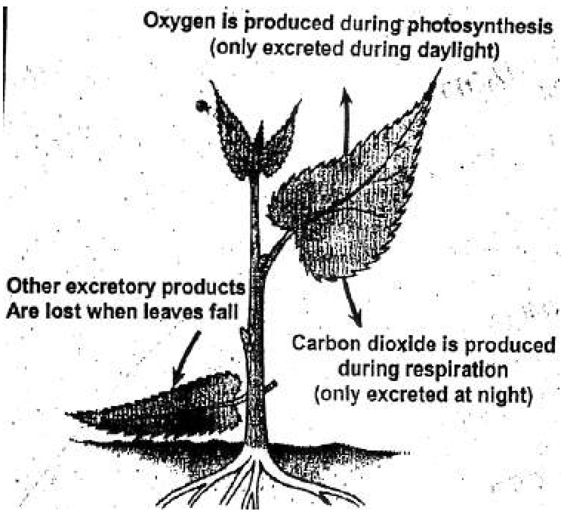


Fig. 50.5: Excretion in a plant

TEST QUESTIONS

1. a) i. Write a short note on excretion.
   ii. Distinguish between excretion and egestion.
   b) Tabulate the main excretory organs of mammals with the respective excretory product

2. a) Describe briefly how plants excrete each of the following substances.
   i) Excess water
   ii) Oxygen
   b) Describe the formation of urine in humans.

3. a) Mention four disorders of the urinary system of humans and their remedies.
   b) State two functions each of the following organs:
   i) lungs
   ii) kidney
   iii) skin
   iv) liver

4. a) Describe the functions of the dermis and the epidermis of the mammalian skin
   d) Briefly describe how a flowering plant eliminate it waste products.

3. a) Explain the following terms:
   i) Deamination;
   ii) Ultra-filtration;
   iii) Selective re-absorption
   (b) Explain why excretion in more important in animals than in plants.

4. (a) Briefly describe the functions of the mammalian skin as a:
   (i) an excretory organ;
   (ii) a thermoregulator;
   (iii) A protective organ;
   (iv) A sensory organ.
   (b) Explain four functions of the kidney.

5. (a) Describe the following processes of urine formation:
   (i) filtration
   (ii) reabsorption
   (iii) secretion
   (b) Mention three excretory products of mammal and explain how they are excreted.

30

REPRODUCTIVE SYSTEM AND GROWTH IN MAMMALS

Specific Objectives

After completing this chapter, you will be able to:

- Identify the reproductive organs of a mammal and their functions.
- State the advantages and disadvantages of male and female circumcision.
- Outline the processes leading to fertilization, zygote development and birth in humans.
- Describe the process of birth in mammals and ways to care for the young ones.
- Identify problems associated with reproduction in humans.
- Mention types of STIs including HIV/AIDS, how they are transmitted and their effects.
- Identify the phases of growth and development in humans and the associated changes.

INTRODUCTION

Reproduction is the process by which organisms make more organisms like themselves.

It is one of the things that sets living things apart from non-living matter.

The group of plant or animal organs that are necessary for the reproductive processes is term as the ***reproductive system.***

In this chapter we shall consider the reproductive system of mammals with emphasis on humans.

STRUCTURE AND FUNCTION OF THE MAMMALIAN REPRODUCTIVE SYSTEM

The mammalian reproductive system is made up of male and female reproductive systems or parts.

Male reproductive system

The reproductive system of male consists of a pair of testes (situated in the scrotal sac), epididymis, vas deferens, seminal vesicles, ejaculatory ducts, prostate, bulbo-urethral glands and penis.

Testes

The two testes (singular, testis) develop inside the abdominal cavity and descend into the scrotum.

Each testis is about one and a half inch long and one inch wide. It is made up of coiled tubes called ***seminiferous tubules***.

The sperms develop inside the tubules. In between the seminiferous tubules are the cells of ***Leydig*** which secrete the male sex hormone, ***testosterone***. *Luteinizing hormone* secreted by the pituitary gland stimulates these cells to secrete testosterone.

The seminiferous tubules join together to form the ***epididymis***. It is a tube about 20 feet long coiled on the back of the testis. The sperms mature in the epididymis. The testes produce millions of sperms. Each milliliter of semen contains more than twenty million sperms.

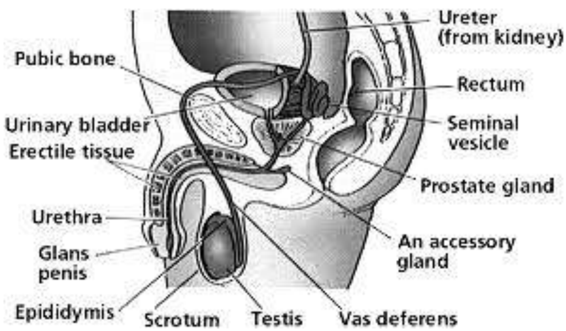


Fig. 50.6: *Male reproductive system*

The vas deferens

The vas deferens is the continuation of the epididymis. It starts at the lower pole of the testis and enters the abdominal cavity through the inguinal canal. It joins the tube from the seminal vesicle of that side to form the ejaculatory duct which joins that part of the urethra which is inside the prostate.

The seminal vesicles

The seminal vesicles are a pair of glandular structures situated behind the urinary bladder. They secrete a fluid which becomes part of the semen. It is rich in fructose which provides the energy for the sperms as they have to travel up the cervix and uterus to reach the ovum.

Ejaculatory ducts

There are two ejaculatory ducts; each is formed by the union of the vas deferens and the tube from the seminal vesicle. The two ejaculatory ducts open into the urethra.

The prostate gland

The prostate gland secretes and stores a clear fluid which is slightly alkaline in nature. This fluid along with sperms and the fluid from the seminal vesicles constitute the semen. The alkalinity of semen neutralizes the acidity of the vagina and prolongs the life of sperms. The prostate gland also contains smooth muscles which help expel semen during ejaculation.

Penis

The penis contains three cylinders of tissue that run parallel to the urethra. During sexual arousal, these tissues become engorged with blood and expand, causing the penis to enlarge and become erect. Men do not have a penis bone or a muscle that causes erection, as do some other animals.

Structure of the sperm

Each sperm cell consists of a head and tail. The tail with its whipping movements helps to propel the sperm forwards to meet the ovum in the fallopian tube.

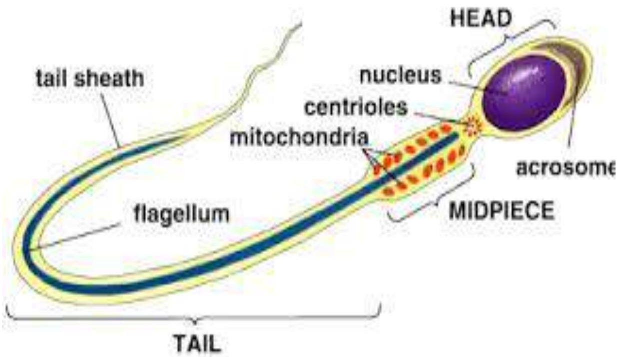


Fig. 50.7: Structure of a sperm

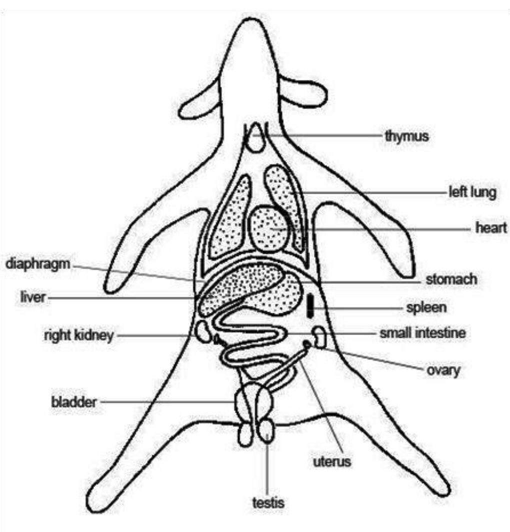


Fig. 50.8: Reproductive system of a male rabbit

Female reproductive system

The female reproductive organs are the vagina, uterus, fallopian tubes, cervix and ovary.

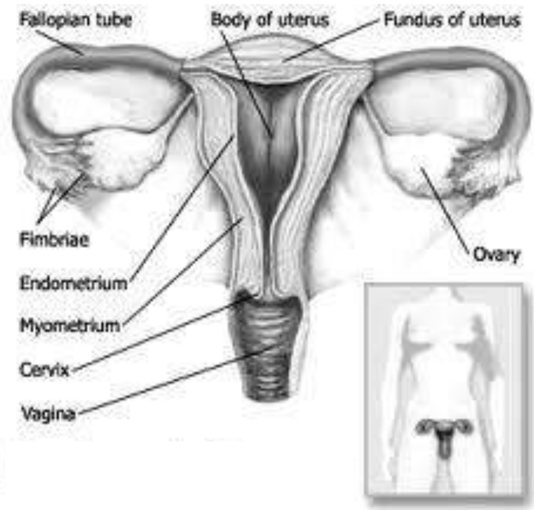


Fig. 50.9: Female reproductive system

Vagina

The vagina is a fibro muscular tubular tract leading from the uterus to the exterior of the body in female mammals. The vagina is the place where semen from the male is deposited into the female's body at the climax of sexual intercourse, commonly known as ejaculation. Around the vagina, pubic hair protects the vagina from infection and is a sign of puberty. The vagina has a thick layer outside and it is the opening where the baby comes out during delivery.

Cervix

The cervix is the lower, narrow portion of the uterus where it joins with the top end of the vagina. It is cylindrical or conical in shape and protrudes through the upper anterior vaginal wall. Approximately half its length is visible, the remainder lies above the vagina beyond view. The cervix is also called the neck of the uterus.

Uterus

The uterus or womb is a pear-shaped muscular organ. Its major function is to accept a fertilized ovum which becomes implanted into the endometrium, and derives nourishment from blood vessels which develop exclusively for this purpose. The fertilized ovum becomes an embryo, develops into a foetus and gestates until childbirth. If the egg does not embed in the wall of the uterus, a woman begins menstruation and the egg is flushed away.

Fallopian tubes

The Fallopian tubes or oviducts are two tubes leading from the ovaries of female mammals into the uterus. On maturity of an ovum, the follicle and the ovary's wall rupture, allowing the ovum to escape and enter the Fallopian tube. There it travels toward the uterus, pushed along by movements of cilia on the inner lining of the tubes. This trip takes hours or days. If the ovum is fertilized while in the Fallopian tube, then it normally implants in the endometrium when it reaches the uterus, which signals the beginning of pregnancy.

Ovaries

The ovaries are small, paired organs that are located near the lateral walls of the pelvic cavity. These organs are responsible for the production of the ova and the secretion of hormones. The process by which the ovum is released is called ***ovulation*****.** The speed of ovulation is periodic and impacts directly to the length of a menstrual cycle.

After ovulation the ovum is captured by the oviduct, after travelling down the oviduct to the uterus, occasionally being fertilized on its way by an incoming sperm, leading to pregnancy and the eventual birth of a new human being.

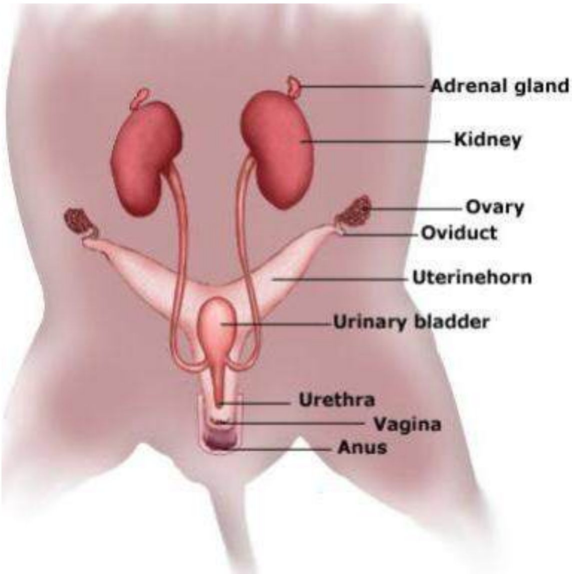


Fig. 60.0: Reproductive system of a female rabbit

MALE AND FEMALE CIRCUMCISION

Circumcision is the surgical removal of all or part of the foreskin of the human male penis.

In females, it is the removal of the skin prepuce covering the clitoris or the clitoris itself and sometimes the lip of the vagina.

In some cases the *labia manora* and *labia majora* are stitched together. Female circumcision is illegal in Ghana, though some communities secretly practice it.

Circumcision, although done for various reasons, (including medical, cultural and religious reasons), has its positive and negative impacts.

Advantages of male circumcision

1. Smegma, a sebaceous gland which collects under the foreskin of the penis is believed to cause diseases such as cancer. Therefore, removing the foreskin eliminates smegma.
2. Circumcision is believed to reduce penile cancer. This is because germs that cause penile cancer are eliminated during circumcision.
3. The foreskin is very soft and may easily get irritated or swollen with little pressure.
4. Research findings reported in 2006 indicate that circumcision could lower by more than half the risk of contracting HIV/AIDS among men who engaged in heterosexual intercourse. This does not suggest that circumcised men could safely engage in unprotected sex.

Disadvantages of male circumcision

1. The wound inflicted may be easily infected by germs and cause serious infections.
2. It may result in excessive haemorrhage (bleeding) from the wound, which can lead to death.
3. If unsterilized instruments are used for the operations, they may carry disease causing pathogens that can infect the person circumcised.
4. The operation is normally performed without any anaesthetic, thus, very painful.
5. Unsterilized instruments used may be infected with HIV/AIDS or other blood related diseases.

Advantages of female circumcision

The advantages presented in favour of female circumcision have no scientific basis, but proponents believe that the removal of the clitoris:

- ❖ allows free outflow of urine;
- ❖ allows direct dilation of the vagina during birth, making delivery easier;

- ❖ makes penetration of the vagina by the penis during sexual intercourse easier.

Disadvantages of female circumcision

1. The vaginal wall, the vulva and the bladder can be damaged.
2. Sexual intercourse becomes painful and not enjoyable.
3. Childbirth may become difficult because of the stitched labia minora and labia majora.
4. The wound inflicted may be easily infected by germs and lead to infections.
5. It may result in excessive haemorrhage from the wound.
6. If unsterilized instruments are used for the operations, they may carry disease causing pathogens that can infect the person circumcised.
7. The operation is normally performed without any anaesthetics, thus, very painful.
8. Unsterilized instruments used may be infected with HIV/AIDS.

FERTILIZATION, DEVELOPMENT OF THE ZYGOTE AND BIRTH IN HUMANS

Fertilization is the process in which the male and female gametes fuse together, producing a single cell that develops into an organism.

Fertilization begins when the sperm contacts the outer surface of the egg and it ends when the sperm's nucleus fuses with the egg's nucleus. Fertilization is not instantaneous, it takes about 2 to 3 days to complete. This is because of the stages involved.

Stages in fertilization

Approach of sperm

The first step is the sperm approaching the egg. In humans it is evident that sperm are attracted to the fluid surrounding the egg so the sperm just swim randomly toward the egg

Attachment of sperm

The second step of fertilization is the attachment of several sperm to the egg's surface coat. This attachment step may last for just a few seconds or for several minutes.

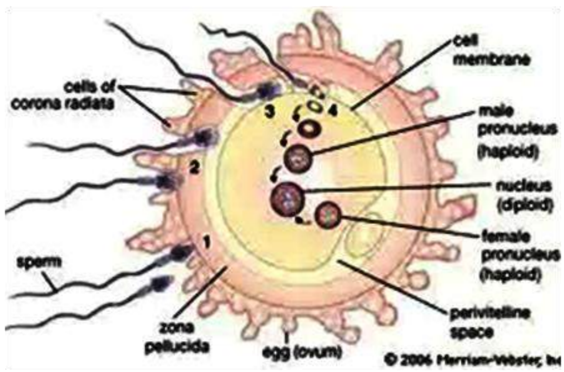


Fig. 60.1: Attachment of sperm to the egg

Penetration of sperm

The third step is a complex process in which the sperm penetrate the egg's surface coat. The head, or front end, of the sperm contains an ***acrosome***, a membrane-enclosed compartment. The acrosome releases proteins that dissolve the surface coat of the egg of the same species. In mammals, a molecule of the egg's surface coat triggers the sperm's acrosome to explosively release its contents onto the surface coat, where the proteins dissolve a tiny hole.

A single sperm is then able to make a slit-like channel in the surface coat, through which it swims to reach the egg's cell membrane. When more than one sperm enters the egg, the resulting zygote typically develops abnormally.

Fusion of sex cells

The next step in fertilization is the fusion of sex cells. When the membranes fuse, a single sperm and the egg become one cell. This process takes only seconds. A specific protein on the surface of the sperm induces this fusion process.

After fusion the male sex cell becomes motionless. The female gamete pulls the male gamete into itself. Filaments called ***microtubules*** begin to grow from the inner surface of the female gamete inward toward the centre. As the microtubules grow, the male gamete and the female gamete's nuclei are pushed toward the female gamete's centre.

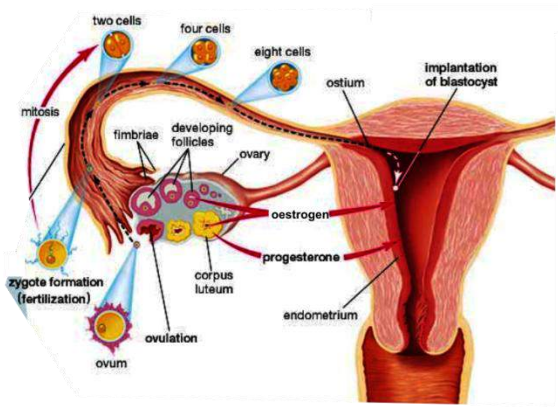
Finally, both gametes nuclear envelopes (outer membranes) fuse, permitting their chromosomes to mix within a common space.

After nuclear fusion, the fertilized egg is called a ***zygote***. When the zygote divides to a two-cell stage, it is called an ***embryo***.

Pregnancy and childbirth

Implantation

After fertilization, the egg undergoes cell division, or cleavage. Thus, one cell divides into two; the daughter cells, called ***blastomeres,*** and then cleave into four; these cleave into eight, and so on. When the embryo consists of a hundred or more cells it may form a solid mass, called a ***blastocyst***. The blastocyst enters the uterus through the Fallopian tube and ***implants*** itself in the lining of the uterus (uterine wall) and continues development into an embryo.



implantation

Pregnancy

Pregnancy starts right after implantation. The pregnancy period is divided into three stages known as ***trimesters***.

First trimester: During the first three months or trimester of pregnancy the embryo develops with recognizable human features, measuring about 9 cm from crown to rump. The ***placenta*** develops in the uterus to pass nutrients and oxygen from the mother to the foetus through the ***umbilical cord***; it also removes waste products from the foetus. In the fourth week the heart begins to beat, and by the eighth week the cardiovascular system (heart and blood vessels) becomes fully functional. By the end of the first trimester all internal organs are functional, the genital organs may be visible, and blood-cell formation begins in the bone marrow. The unborn baby is called a ***foetus***.

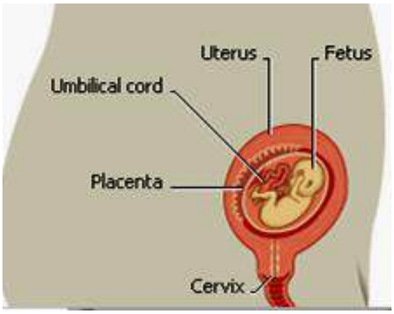


Fig. 60.2

Second trimester: In the second trimester the foetus's eyes start to blink and the lips perform sucking motions. By week 17 the foetus moves into a foetal position in which the body lies curled up on one side with the head bowed and the legs and arms drawn in toward the chest. By week 20 the mother can feel the foetus moving. At the end of the second trimester, the foetus has reached about 19 cm in length.

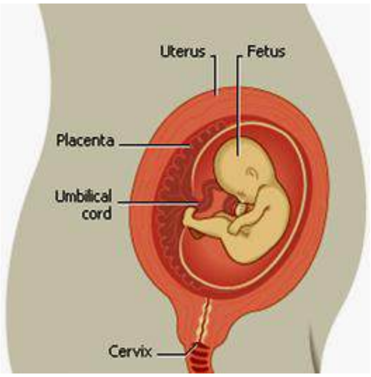


Fig. 60.3

Third trimester: In the third trimester the foetus prepares for survival outside of the mother's uterus. The internal organs mature, temperature regulation develops, and the lungs begin to produce ***surfactant***, a foamy fluid that prevents the lungs from collapsing when the infant exhales. Some of the mother's antibodies pass through the placenta to the foetus, establishing a primitive immune system that protects the foetus from disease. Nails form on the fingers and toes. Toward the last weeks of pregnancy, the baby may assume a head-down position as it prepares for birth. Newborns normally range from 2.5 to 4.5 kg in weight and from 46 to 56 cm in length from crown to toe.

After the third trimester or 9 months, the baby is ready to be born.

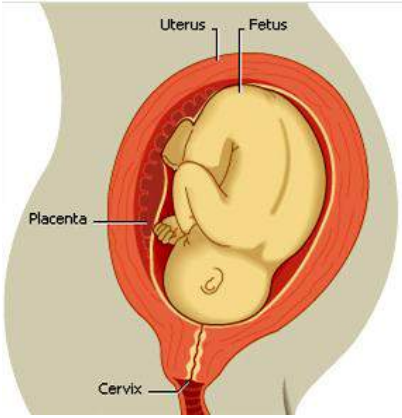


Fig. 60.4

The process of birth

The mother undergoes labour. Labour involves rhythmic uterine contractions which grow progressively stronger. Each contraction shortens the muscle fibres in the uterus, pulling the cervix (uterine opening) wider. In this early stage of labour the cervix dilates (opens) to about 4 cm.

As the first stage of labour ends, the labour pattern changes. Contractions become more painful and occur closer together. As labour progresses the cervix opens to its full width of 10 cm. The baby's head begins to rotate to fit through the birth canal.

After the cervix becomes fully dilated, contractions become very intense and usually last a minute or longer. With each contraction the baby continues its descent through the birth canal.

If a baby's position is head first during delivery, the mother's vagina fits like a crown around the baby's head, making the head visible as it emerges from the birth canal.

As the head emerges, the neck flexes and the baby rotates to the side. This enables the shoulders to manoeuvre around the pubic bone. One shoulder emerges, quickly followed by the other shoulder and the rest of the body.

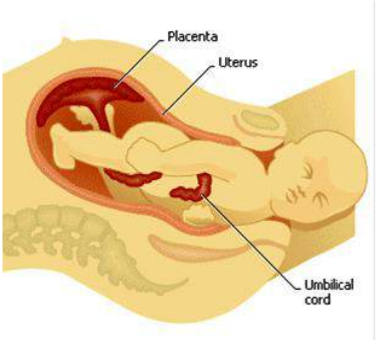


Fig. 60.5

In the final stage of labour, the uterus continues to contract and the placenta detaches from the uterus and is expelled.

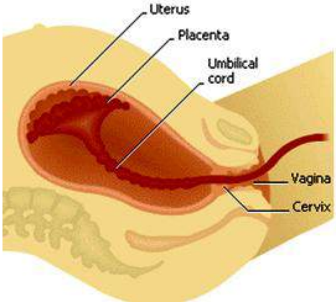


Fig. 60.6

Formation of twins

Twins or multiple birth is a birth of more than one offspring at a time. Multiple births in human arise either from the simultaneous impregnation of more than one ovum or from the impregnation of a single ovum that divides into two or more parts, each of which develops into a distinct embryo.

Identical twins

Identical twins are formed when plural offspring develop from a single egg. They are always of the same sex, resemble one another very closely, and have similar fingerprints and blood types.

Fraternal twins

Fraternal twins are offspring produced from separate ova. They are not necessarily of the same sex and may have the usual family resemblance of brothers and sisters.

Multiple births may be achieved artificially by implanting in the uterus several fertilized ova.

The prenatal and infant mortality rate in multiple pregnancies is higher than that in single gestations. The danger of premature birth increases progressively with the number of offspring involved.

Siamese twins

Siamese twins, also known as conjoined twins, are a set of identical twins that are joined together. Identical twins form when a fertilized egg starts to split into two parts during its early development. Normally the splitting process is completed and each part develops into a separate embryo, resulting in identical twins. On very rare occasions, however, the splitting is not completed properly and the embryos remain linked, resulting in conjoined twins.

Conjoined twins may be attached only by a cord of tissue, or they may be fused at some part of the body, such as the head, chest, or leg. Sometimes they share one or more internal organs, such as the heart or liver, or they may have a common circulatory system. Surgical separation of conjoined twins has been successful, although if the twins share important organs, surgery may be risky for one or both twins.

Fig. 60.7*:* Siamese twins

Parental care (Care for the young)

Caring for the young has three main stages:

- Ante-natal or pre-natal care
- Post-natal care
- Child care

Ante-natal care

During pregnancy the mother undergoes a lot of changes, such as a missed menstrual period, breast tenderness and swelling, fatigue, nausea or sensitivity to smells, increased frequency of urination, mood swings, and weight gains. Some women also experience cravings for unusual substances such as ice, clay, or cornstarch; this condition, called pica, can indicate a dietary deficiency in iron or other nutrients. The first few months of pregnancy are the most critical for the developing infant, because during this period the infant's brain, arms, legs, and internal organs are formed. For this reason a pregnant woman should be especially careful about taking any kind of medication except on the advice of a physician who knows that she is pregnant. This means that it is advisable for a pregnant woman to visit the doctor regularly. X-rays should also be avoided (as it can cause miscarriage), and pregnant women should avoid smoking and alcohol consumption.

Importance of pre-natal care

1. Pre-natal care reduces the number of maternal and foetal death.
2. It reduces complications during labour.
3. It improves the general health of both mother and child.
4. It helps the mother to have a safe delivery.
5. It diagnoses abnormalities for early treatment.

Post-natal care

It is recommended that the mother after birth undertakes regular exercises prescribed by the doctor or midwife. This is to ensure that her muscles which were over-stretched during childbirth are restored. The mother must also bath regularly with soap and use antiseptics and disinfectants to help the wound left by the placenta's attachment on the uterine wall to heal. The mother must also eat a well balanced diet to enable her get the necessary nutrients she needs.

Importance of post-natal care

1. Post-natal care helps restore the mother to her usual form.
2. It helps prevent post-delivery complications.
3. It helps the mother to get healed from the wound left by the placenta's attachment on the uterus.

Child care (parental care)

Humans are the only group of mammals who keep their young ones with them for a longer time. The baby needs much care during the early stages of its life. At these stages, breast feeding is very essential, especially during the first six months. Bottle feeding can be introduced as time goes on. The baby should also be bathed regularly to keep it clean, insulated from cold by properly wrapping it up, immunized, vaccinated etc.

Importance of parental care

1. It enables the baby to adapt itself to the outside conditions.
2. It prevents the baby from contracting diseases.
3. It keeps the bond between the mother and the baby.

PROBLEMS ASSOCIATED WITH REPRODUCTION IN HUMANS

Miscarriage

Miscarriage is the term for spontaneous abortion; the unintentional termination of a pregnancy before the foetus is capable of independent life.

Causes of miscarriage

1. Abnormal development of the embryo or of the placental tissue, which links the embryo to the mother.
2. Severe vitamin deficiencies.
3. Hormone deficiencies.
4. Acute infectious diseases
5. Systemic diseases such as nephritis (kidney disease) and diabetes, and severe trauma.
6. Uterine malformations, including tumours, are responsible in some instances.
7. Extreme anxiety and other psychological disturbances.

Prevention of miscarriage

1. Treatment for threatened miscarriage usually consists of bed rest. Almost continuous bed rest throughout pregnancy is required in some cases of repeated miscarriage.
2. Vitamin and hormone therapy also may be given.
3. Surgical correction of uterine abnormalities may be needed in certain cases.

Ectopic pregnancy

This is the development of a fertilized egg outside the womb.

Causes of ectopic pregnancy

1. ***Cervical abnormalities:*** If the cervical mucus thickens it may cause the fertilized egg to get stuck in the fallopian tube and not get to the womb.
2. ***Abdominal pregnancy***: This is the condition where the zygote is expelled into the abdominal cavity through the ovarian end of the fallopian tube.

3. ***Abnormal egg development:*** If the morula (a ball of divided zygote) develops rapidly than normal, it will implant itself in the fallopian tube.
4. ***Ovarian pregnancy:*** This problem arises when fertilization occurs in the Graafian follicle (a small fluid-filled sac in the ovary). This may cause the ovary to rupture.

Effects of ectopic pregnancy

1. Infection of dead foetus may be poisonous and have severe complications.
2. The growth of a foetus in the fallopian tube may weaken it and cause it to move out of place.
3. Ectopic pregnancy can also result in infertility on the part of the mother

Infertility

Infertility is the inability to conceive or carry a child to term. People who suffer from infertility can seek medical advice to identify the cause of infertility and undergo treatment.

Causes of infertility in men

Low sperm count is the most frequent cause of male infertility. Although ultimately only one sperm is required for fertilization, men whose semen contains less than 20 million sperm per millilitre frequently have infertility problems.

The quality of sperm may affect male fertility. Physicians determine sperm quality according to its motility (ability to move) and its physical structure. Poor motility will prevent sperm from swimming the long distance from the woman's vagina to the fallopian tubes to fertilize an egg. Sperm that have structural problems will also have problems penetrating an egg.

Other causes of infertility in men include:

- ❖ Premature ejaculation
- ❖ Impotence
- ❖ Absence of vas deferens (sperm duct) due to diseases or infection.

Causes of infertility in women

Hormonal deficiency: Normally one egg will be released each month about midway through the menstrual cycle, under the direction of several hormones. If any of these hormones are not functioning, ovulation will occur irregularly or perhaps not at all. Disorders of the endocrine system, including thyroid disease, diabetes mellitus, and polycystic ovarian syndrome may cause infertility in women.

Harmful chemicals can affect hormonal levels and adversely affect fertility. For instance, marijuana use can shorten the menstrual cycle. Cigarette smoking reduces some types of hormone production and may deplete egg supply.

Cervical problems: Once inside the female's cervix, sperm may encounter obstacles. The cervical mucus (thick fluid that protects the cervix and uterus from infection) may be too thick for the sperm to penetrate, or it may be chemically hostile to the sperm. A fertilized egg may become stuck in the fallopian tube and result in an ectopic pregnancy.

Infertility in women is also caused by being born with a malformed ***cervical canal***. An impaired cervical canal can prevent passage of sperm from the vagina to the uterus as the sperm travel toward the fallopian tubes. If a woman is able to conceive, problems with the cervical canal can lead to miscarriage.

Other causes of infertility in women include:

- ❖ Blockage in the fallopian tube.
- ❖ Constantly working near fire.
- ❖ Exposure to x-rays.
- ❖ Excessive hard work – prevents ovulation.
- ❖ Dietary deficiency – not getting the needed vitamins, protein or minerals.
- ❖ Sexually transmitted diseases – STDs such as syphilis and gonorrhoea lead to the infection of the reproductive organs.

Treatment of infertility

Once the cause or causes of infertility are determined, doctors devise a strategy for the couple to increase their fertility. Structural problems, such as varicoceles or blocked ejaculatory ducts in men and fallopian tube obstruction in women, can be treated by surgery. When no structural problems are identified, infertility treatments usually begin with non-invasive measures. Sometimes only small adjustments in the frequency and timing of sexual intercourse are required to bring about pregnancy. Couples are instructed on how to identify when a woman is ovulating so that they can plan sexual intercourse around her most fertile time. Practices that temporarily result in lowered sperm counts or damaged sperm can be curtailed, such as the use of certain medications, alcohol, marijuana, and hot tubs. If these non-invasive measures are unsuccessful, a doctor may recommend fertility drugs or assisted reproductive technologies.

Impotence

This is the inability of the penis to erect. Since the penis cannot erect it cannot penetrate the vagina, thus no sexual intercourse.

Causes of impotence

- ❖ Damage to the part of the spinal cord where impulses which help in sexual act play

- Damage to parasympathetic nerve supplies to the penis.
- Damage to the sensory nerve supply to the penis, epithelium or scrotum which leads to loss of sensory sensation.

Fibroid

A seemingly harmless growth or tumour composed of fibrous muscle tissue, especially one that develops in the walls of the uterus and is associated with painful and excessive menstrual flow.

Causes of fibroid

- ***Artificial hormones:*** Some women with an intact uterus who take oestrogen are more likely to develop fibroids. Other side effects from taking oestrogen, include headaches, and swelling and tenderness of the breasts
- ***Hereditary:*** Fibroid can be passed on from mother to daughter. If a woman's mother has or was diagnosed with fibroid, she also has a good chance of contracting it.

Effects of fibroid

- Fibroid located in the uterine cavity may prevent fertilization by blocking the entrance of the fallopian tube.
- It may prevent implantation, causing infertility in some women.
- It may grow and occupy much of the space in the uterus, leaving a little space for the foetus. This may result in miscarriage.

Fibroids can be removed with a special surgical operation known as ***myomectomy***. The woman may then regain her fertility.

Ovarian cyst

Ovarian cyst is a benign (non-cancerous) or harmless growth or tumour in a fluid-filled sac called a cyst. Many women, especially younger women, develop ovarian cysts. They do not usually display any symptoms unless they grow so large as to cause a visible swelling of the abdomen. Ovarian cysts sometimes disappear without treatment; in other cases, they must be removed surgically.

Causes of ovarian cyst

- ***Infected follicle***: Sometimes when the follicle becomes infected, it develops into an ovarian cyst.
- ***Hormonal imbalance***: An imbalance in the hormones produced by the pituitary gland can cause ovarian cyst.

SEXUALLY TRANSMITTED INFECTIONS STIs

Sexually Transmitted Infections (STIs) are infections passed from one person to another primarily during sexual contact. Despite the prevalence of STIs, studies show that many people are unaware of

Table 6.9: Sexually transmitted diseases

Disease and pathogen	Mode of transmission	Symptom	Prevention and control
Acquired Immune Deficiency Syndrome-AIDS (*virus – HIV*)	• Sexual intercourse with infected person • Mother to baby in the womb • Mothers to babies through breast milk • Prick or cut by sharp infected instruments; • Transfusion with infected blood. • Transplant with infected organ • Transfer of infected body fluids, for example, bone marrow transplant	• Weakened immune system resulting in high susceptibility to diseases such as diarrhoea, pneumonia, skin cancer etc. • Loss of weight and emaciated body	• Have protected sexual intercourse, • Avoid using sharp, unsterilized instruments such as syringes, hair clippers etc. • Infected mothers should not breastfeed • Blood should be tested and certified before transfusion. • Organ donors should be tested for HIV. • No immediate cure; anti-retroviral drugs can be used to minimize impact of virus.
Gonorrhoea (*Coccus bacterium – Neisseriae gorrnohoea*)	• Sexual intercourse with an infected person	• Burning sensation in the genitals during urination • Discharge of white fluid from the genitals • Untreated gonorrhea can cause pelvic inflammatory disease (PID) in women. • Babies born to mothers with gonorrhea are at risk of infection during childbirth; such infections can cause eye disease in the newborn.	• Avoid casual and unprotected sexual intercourse • Treatable with several antibiotics
Syphilis (*spiroghaete bacterium – Treponema pallidum*)	• Sexual intercourse with an infected person	• In the early stage of syphilis, a genital sore develops shortly after infection and eventually disappears on its own.	• Avoiding casual sex • Using protection during sexual intercourse • Syphilis is easily treated with penicillin.

		• If the disease is not treated, the infection can progress over years, affecting the vertebrae, brain, and heart, and resulting in such varied disorders as lack of coordination, meningitis, and stroke. Syphilis during pregnancy can be devastating to the foetus, causing deformity and death.	
Hepatitis B (*virus*)	• Unprotected sexual intercourse with an infected person, • Sharing of infected needles or other sharp instruments that break the skin. • Hepatitis B can also spread during childbirth.	• Hepatitis B attacks liver cells, sometimes leading to cirrhosis and cancer of the liver. • In most cases hepatitis B is incurable	• Practicing safe sex • Arduous chemotherapy can eliminate the virus in some patients. • There is a safe, effective vaccination for hepatitis B.
Genital warts (*virus - papillomavirus (HPV*)	• Having unprotected virginal and anal sex.	• Genital warts grow on the penis and in and around the entrance to the vagina and anu. • Certain types of HPV that cause genital infections can also cause cervical cancer. s	• Practice safe sex. • Genital warts are treatable with topical medications and can be removed with minor surgical procedures. • Regular Pap test screenings can detect abnormalities at an early stage, when they can more easily be treated to prevent cancer developing.
Trichomoniasis (protozoa - *Trichomonas vaginalis)*	• Unprotected sex with infected person	• Burning, itching, and discomfort in the vagina in women and the urethra in men.	• Avoid unprotected sex. • Trichomoniasis is easily treated with a single dose of antibiotics.

Genital herpes (*Herpes simplex virus (HSV*).	• Sexual intercourse with an infected person	• Genital herpes causes recurrent outbreaks of painful sores on the genitals, although the disease often remains dormant with no symptoms for long periods.	• Have protected sex. • HSV can be treated with antiviral drugs, such as acyclovir, but HSV cannot be eradicated from the body—it is incurable.

their risks for contracting an STI or the serious, and sometimes deadly, health consequences that may result from an untreated infection. Table 6.9 gives some sexually transmitted disease, the mode of transmission, symptoms and preventive or control measures,

Prevention of STIs

1. Since STIs are generally contracted through sex, the primary way to prevent it is abstinence from sex.
2. Having protected sex.
3. Maintain one sex partner, preferably wife or husband.
4. Thorough sex education, especially for boys and girls.
5. Avoid sharing hypodermic syringes or blade.
6. Seek medical advice in all cases of STIs.

CASE STUDY:

A childless couple thinks their problem is due to witchcraft. Use scientific method and reasoning to explain the possible causes of the infertility.

PHASES OF GROWTH AND DEVELOPMENT IN HUMANS

All living things increase in size and weight due to the addition of new protoplasm (content of cell). This increase in size is referred to as growth and is normally irreversible.

In humans, growth is accompanied by development, which is the continuous change from one stage of life to another.

The stages or phases of growth and development in humans are:

- Infancy
- Childhood
- Adolescence
- Adulthood
- Old-age

Infancy

Infancy is the period from birth to about 2 years of age. This first stage of life is an important time, characterized by physical and emotional growth and development.

Sensory acuity develops rapidly during the first three months of life. By two days after birth infants can discriminate odours.

Within three months they can distinguish colour and form.

Newborns perform motor movements, many of which are reflexive. Soon after birth they gain voluntary control of movements. The major stages of locomotion are crawling, creeping and walking. The average infant walks between 13 and 15 months of age.

Normal infants possess neurological systems that detect and store speech sounds, permit reproduction of these sounds, and eventually produce language. Infants utter all known speech sounds, but retain only those heard regularly. Word-like sounds occur at 12 months and have meaning at about 18 months. Early words generally include naming objects and describing actions, for example, “fall floor.” Acquisition of complex language after 18 months is very rapid.

During the first 24 months the average child makes considerable gains in height and weight, begins teething, develops sensory discrimination, and begins to walk and talk.

Childhood

The childhood stage is the period from age two to ten; but due to some distinct changes that occur, the childhood stage can be divided into early childhood and later childhood.

Early childhood

Also known as the pre-school age, is the period from age two to age six (2 -.6 years). At this stage the child’s physical and mental abilities develop. The child would have a better control over the language and could express itself more clearly with a stock of words. It would also engage in more physical activities such as running, jumping and climbing.

Later childhood

This stage is from age six to ten (6 – 10 years). During this stage, the child’s mental capacity increases considerably. They can solve simple problems and influence or be influenced by their peers.

Girls at this stage grow much faster than boys. Boys and girls recognize their gender difference and therefore do not play together as often as they used to.

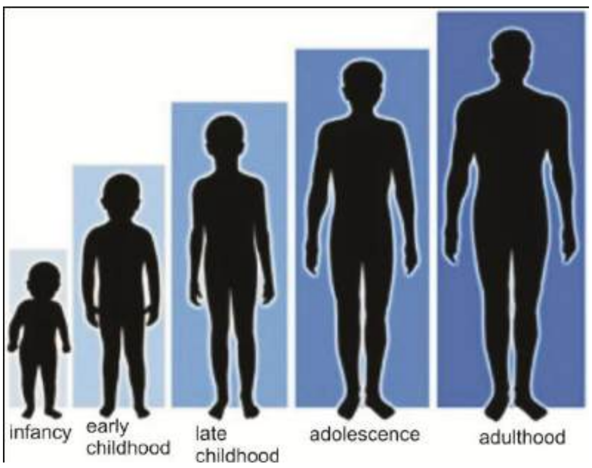


Fig. 60.8: Phases of growth

Adolescence

Adolescence is the stage of maturation between childhood and adulthood. The term denotes the period from the beginning of puberty to maturity. It usually starts at about age 12 in males and age 10 in females. The transition to adulthood varies among cultures, but it is generally defined as the time when individuals begin to function independently of their parents.

Dramatic changes in physical stature and features are associated with the onset of pubescence. The activity of the pituitary gland at this time results in the increased secretion of hormones, with widespread physiological effects. Growth hormone produces a rapid growth spurt, which brings the body close to its adult height and weight in about two years. The growth spurt occurs earlier among females than males, also indicating that females mature sexually earlier than males. During adolescence, the individuals develop some characteristics known as ***secondary sexual characteristics.***

Secondary sexual characteristics in males

1. Growth of facial, bodily, and pubic hair.
2. Deepening of voice
3. Enlargement of penis
4. Body becomes muscular
5. Chest broadens
6. Production of sperms starts. Boys at this stage normally have wet dreams.

Secondary sexual characteristic in females

1. Development and enlargement of breasts
2. Growth of pubic hair
3. Broadening of hips
4. Ovulation and menstruation start.

Adulthood

From the age of 20 upward marks the adulthood stage in a person's life. At this stage, physical growth begins to cease. Increase in size in most people is due to presence of extra fats deposited under the skin. There is a strong sense of maturity and independence as the person ventures out to try new things.

Old age

The final stage of growth and development in humans is old age, otherwise known as the ***senescence***. During this stage several general changes take place in the human body: hearing and vision decline, muscle strength lessens, soft tissues such as skin and blood vessels become less flexible, and there is an overall decline in body tone.

Most of the body's organs perform less efficiently with advancing age. The immune system also changes with age. A healthy immune system protects the body against bacteria, viruses, and other harmful agents by producing disease-fighting proteins known as antibodies. A healthy immune system also prevents the growth of abnormal cells, which can become cancerous. With advancing age, the ability of the immune system to carry out these protective functions is diminished.

Menopause

From about age 45 to age 55 women undergo a stage known as menopause. This is the stage where menstruation stops. At this stage, women develop many bodily as well as psychological changes. Some of these changes are rather problematic.

Problems associated with menopause

1. Women become depressed.
2. They become restless.
3. They experience hot flashes.
4. They normally have night sweats.
5. They experience disturbed sleep.
6. Their mood changes.

TEST QUESTIONS

1. a) i. What is *reproduction*?
   ii. List three reproductive structures each of male and female and state their functions.

   b) i. What are *male* and *female circumcision*?
   ii. Mention three advantages and three disadvantages each of male and female circumcision.

2. a) i. Explain the term *fertilization*.
   ii. Briefly describe the process of zygote development and birth in humans.

   a) Describe the formation of:
   i. Identical twins
   ii. Fraternal twins
   iii. Siamese twins

3. a) Explain the following stages of childcare and state two importance of each stage:
   i. ante-natal
   ii. post-natal
   iii. childcare

   b) Mention five problems associated with reproduction in humans, their causes and remedies.

4. a) Describe the phases of growth and development in humans.

   b) i. Write a short note on four sexually transmitted diseases.

ii. Mention four ways by which sexually transmitted diseases can be prevented.

5. (a) i. What is menopause?

ii. State three signs shown by women in their menopause.

(b) State three secondary sexual characteristics shown by both boys and girls in their adolescence.

31

THE CIRCULATORY SYSTEM

Specific Objectives

After completing this chapter, you will be able to:

- Describe the structure and functions of the circulatory system.
- Describe the composition and functions of blood and lymph.
- Identify disorders associated with blood and the blood circulatory system.

INTRODUCTION

The circulatory system, also called the ***cardiovascular system*** in humans, is the combined function of the heart, blood, and blood vessels to transport oxygen and nutrients to organs and tissues throughout the body and carry away waste products to the various excretory organs.

Functions of the circulatory system

Among its vital functions,

1. the circulatory system increases the flow of blood to meet increasing energy demands during exercise and hard work.
2. It regulates body temperature.
3. When foreign substances or organisms invade the body, the circulatory system swiftly conveys disease-fighting elements of the immune system, such as white blood cells and antibodies, to regions under attack.
4. In case of injury or bleeding, the circulatory system sends clotting cells and proteins to the affected site, which quickly stop the bleeding and promote healing.

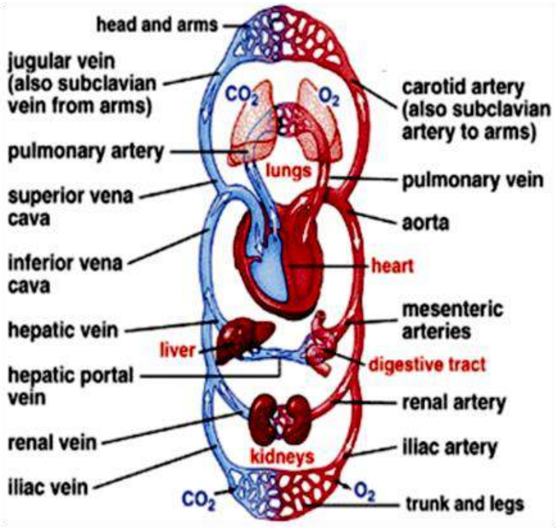


Fig. 60.9: The circulatory system

STRUCTURE AND FUNCTIONS OF THE CIRCULATORY SYSTEM

The circulatory system is divided into three major parts:
- ❖ The Heart
- ❖ The Blood
- ❖ The Blood Vessels

The Heart

The heart is the engine of the circulatory system. It is a muscle about the size of your fist. The heart is located in the centre of your chest slightly to the left. Its job is to pump blood and keep the blood moving throughout the body.

Structure, working and functions of the heart

The heart is divided into four chambers: the ***right atrium,*** the ***right ventricle,*** the ***left atrium,*** and the ***left ventricle.***

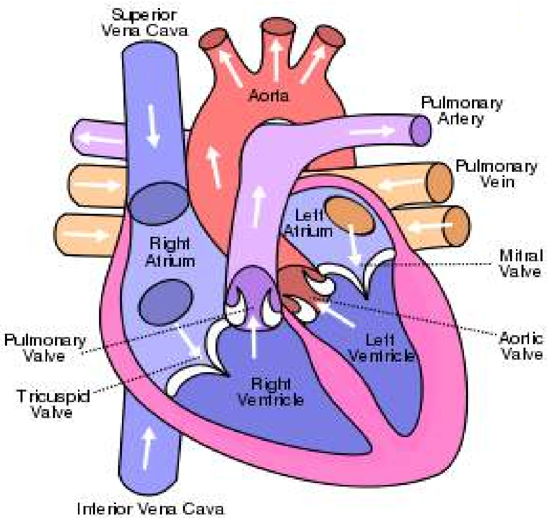


Fig. 70.0: Structure of mammalian heart

Circulation in the heart

- ❖ De-oxygenated blood from the body enters the heart from two large blood vessels, the ***inferior vena cava*** and the ***superior vena cava***, and collects in the ***right atrium***.
- ❖ When the atrium fills, it contracts, and blood passes through the ***tricuspid valve*** into the ***right ventricle***.
- ❖ When the ventricle becomes full, it starts to contract, and the tricuspid valve closes to prevent blood from moving back into the *atrium*.
- ❖ As the right ventricle contracts, it forces blood into the ***pulmonary artery,*** which carries blood to the lungs to pick up fresh oxygen.
- ❖ When blood exits the right ventricle, the ventricle relaxes and the ***pulmonary valve*** shuts, preventing blood from passing back into the ventricle.
- ❖ Blood returning from the lungs to the heart collects in the ***left atrium***. When this chamber contracts, blood flows through the mitral valve into the left ventricle. The ***left ventricle*** fills and begins to contract, and the ***mitral valve*** between the two chambers closes. In the final phase of blood flow through the heart, the left

ventricle contracts and forces blood into the aorta.

- After the blood in the left ventricle has been forced out, the ventricle begins to relax, and the ***aortic valve*** at the opening of the ***aorta*** closes.

Functions of the parts of the heart

Pulmonary vein – transports oxygenated blood from the lungs to the heart.

Pulmonary artery – transports de-oxygenated blood from the heart to the lungs.

Aorta – transports oxygenated blood from the heart to all parts of the body.

Vena cava (plural; ***venae cavae***) – transports de-oxygenated blood from the body to the heart.

Left auricle / atrium – relaxes and expands to receive oxygenated blood from the lungs through the pulmonary vein.

Right auricle/ atrium – relaxes and expands to receive blood from the vena cava; contracts to pump blood under pressure into the right ventricle.

Left ventricle – relaxes and expands to receive blood from the left auricle; contracts to pump oxygen-rich blood under pressure to all parts of the body through the aorta.

Right ventricle – relaxes and expands to receive blood from the left auricle; contract to pump blood into the lungs through the pulmonary artery.

Bicuspid valve – prevents blood from flowing back from the left ventricle into the left auricle.

Tricuspid valve – prevents blood from flowing back from the right ventricle into the right auricle.

Blood

About 55 percent of the blood is composed of a liquid known as ***plasma***. The rest of the blood is made of three major types of cells: red blood cells (also known as erythrocytes), white blood cells (leukocytes), and platelets (thrombocytes).

Plasma

Plasma consists predominantly of water and salts. It also contains small molecules of vitamins, minerals, nutrients, and waste products. The plasma is made in the liver. Due to proteins dissolved in it, such as albumin, gamma globulin, and clotting factors, the plasma is usually yellowish in colour.

Plasma that has had the clotting factors removed is called ***serum***.

Functions of the plasma

1. Albumin helps regulate the water content of tissues and blood.
2. Gamma globulin is composed of antibodies which neutralize or help destroy infectious organisms.
3. Clotting factors, such as fibrinogen, are involved in forming blood clots that seal leaks after an injury.

Red Blood Cells (erythrocytes)

Red blood cells are responsible for carrying oxygen and carbon dioxide. They are composed of protein and iron compounds called ***haemoglobin*** that captures oxygen molecules as the blood moves through the lungs, giving blood its red colour.

The haemoglobin picks up oxygen from the lungs and as blood passes through body tissues, then releases the oxygen to cells throughout the body.

After delivering the oxygen to the cells it gathers up the carbon dioxide and transports it back to the lungs where it is removed from the body when we exhale (breath out). There are about 5,000,000 red blood cells in one drop of blood.

White Blood Cells (*leukocytes*)

White blood cells help the body fight off germs. They are the primary defence mechanism against invading bacteria, viruses, fungi, and parasites. They often accomplish this goal through direct attack, which usually involves identifying the invading organism as foreign, attaching to it, and then destroying it. This process is referred to as ***phagocytosis***.

When you have an infection your body will produce more white blood cells to help fight an infection.

There are several varieties of white blood cells, including *neutrophils*, *monocytes*, and *lymphocytes*, all of which interact with one another and with plasma proteins and other cell types to form the complex and highly effective immune system.

Platelets

The smallest cells in the blood are the platelets, which are designed for a single purpose; to begin the process of coagulation, or forming a clot, whenever a blood vessel is broken.

As soon as an artery or vein is injured, the platelets in the area of the injury begin to clump together and stick to the edges of the cut. They also release messengers into the blood that perform a variety of functions such as constricting the blood vessels to reduce bleeding, attracting more platelets to the area to enlarge the platelet plug, and initiating the work of plasma-based clotting factors, such as fibrinogen. Through a complex mechanism involving many steps and many clotting factors, the plasma protein fibrinogen is transformed into long, sticky threads of fibrin. Together, the platelets and the fibrin create an intertwined meshwork that forms a stable clot. This self-sealing aspect of the blood is crucial to survival.

NB: The red blood cells, white blood cells and platelets are made by the bone marrow. Bone marrow is a soft tissue inside of our bones that produces blood cells.

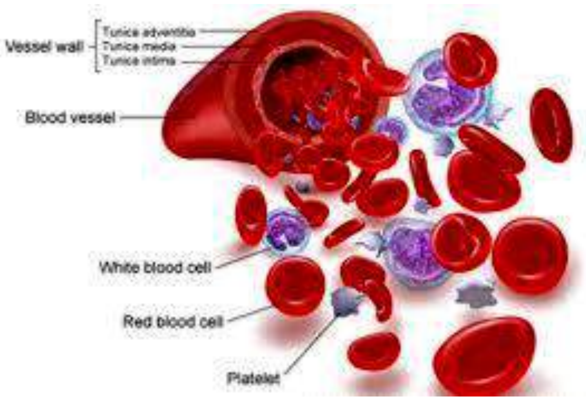


Fig. 70.1: Blood cells and platelets

Importance of the blood

The primary functions of the blood could be summarized into three categories – transport, protection or defence and reproduction.

Transport

1. The blood carries oxygen from the lung to the other parts of the body
2. It carries carbon dioxide to the lungs to be excreted.
3. It maintains the temperature of mammals by carrying heat or fluid through the body when the temperature is cold or warm.
4. It carries digested food substances to the tissues.
5. It carries waste substances to their respective excretory organs.
6. The blood transports hormones from the endocrine gland to other organs.

Protection or defence

1. White blood cells engulf and destroy foreign organisms which enter the body.
2. Blood clotting reduces blood loss and prevents foreign germs or poisonous substances from entering the body.
3. It also helps wound to heal fast.

Reproduction

The blood causes the penis to erect in mammal, so that sexual intercourse can take place.

Blood clotting

Blood clotting is the normal physiological response that prevents significant blood loss following an injury. This process is called ***haemostasis***.

How blood clotting occurs

1. The vessel constricts to reduce blood flow.
2. Circulating platelets adhere to the vessel wall at the site of the injury.
3. Platelets are activated and aggregate coupled with an intricate series of enzymatic reactions involving coagulation proteins.

4. This produces fibrin which forms a stable haemostatic plug that covers the surface of the cut, hence preventing loss of blood.

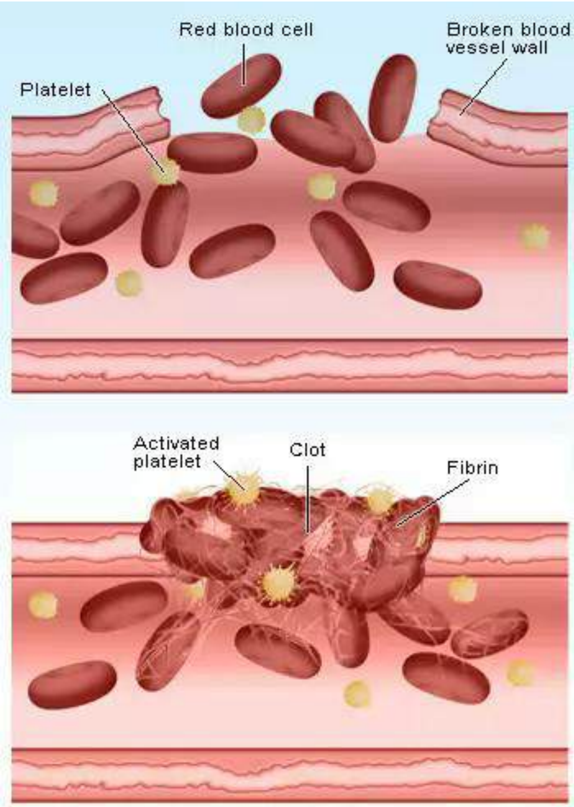


Fig. 70.2: Blood clotting

Important of blood clotting

1. Blood clotting prevent excessive loss of blood.
2. It enables the body tissues to heal quickly.
3. It prevents disease-causing pathogens from entering the body.
4. It also prevents toxic substances from entering the body.

The Blood Vessels

Three types of blood vessels form a complex network of tubes throughout the body - arteries, capillaries and veins.

Arteries

Arteries are blood vessels that carry oxygen rich or oxygenated blood away from the heart. Arteries have thick walls to withstand the pressure of blood being pumped from the heart.

Veins

Veins carry de-oxygenated blood back to the heart. Blood in the veins is at a lower pressure, so veins have one-way valves to prevent blood from flowing backwards away from the heart.

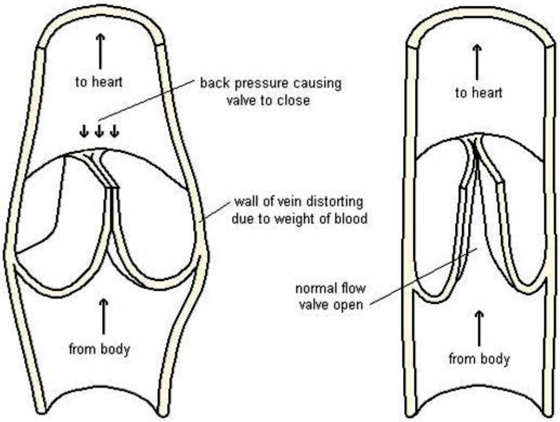


Fig. 70.3: Section through vein showing valves

Capillaries

Capillaries are tiny blood vessels; as thin as or thinner than the hairs on your head. Capillaries connect arteries to veins. Food substances or nutrients, oxygen and wastes pass in and out of the blood through the capillary walls.

The inner layer of blood vessels is lined with ***endothelial cells*** that create a smooth passage for the transit of blood. This inner layer is surrounded by connecting tissue and smooth muscle that enable the blood vessels to expand or contract.

Blood vessels expand during exercise to meet the increased demand for blood and to cool the body. Blood vessels contract after an injury to reduce bleeding and also to conserve body heat.

The arteries, veins, and capillaries are divided into two systems of circulation: ***systemic*** and ***pulmonary.***

The systemic circulation carries oxygenated blood from the heart to all the tissues in the body except the lungs and returns deoxygenated blood carrying waste products, such as carbon dioxide, back to the heart.

The pulmonary circulation carries spent blood from the heart to the lungs. In the lungs, the blood releases its carbon dioxide and absorbs oxygen. The oxygenated blood then returns to the heart before transferring to the systemic circulation.

The systemic and pulmonary circulation are known as ***double circulation***.

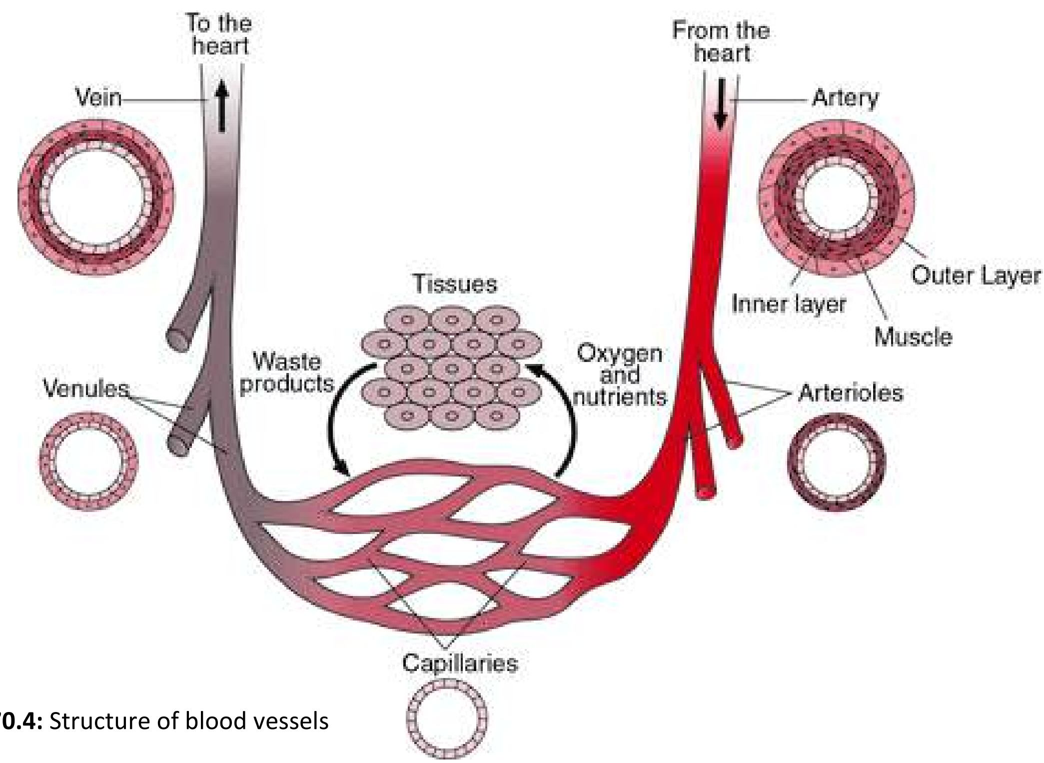


Fig. 70.4: Structure of blood vessels

Table 7.0: Comparison of the blood vessels

Arteries	Veins	Capillaries
Carry oxygenated blood	Carry de-oxygenated blood	Carry oxygenated blood at arteriole and de-oxygenated blood at venule end
Carry blood away from the heart	Carry blood to the heart	Carry blood between the arteries and veins
Valves present	Valves absent	Valves absent
Have smaller lumen	Have larger lumen	Have smaller lumen
Carry blood under high pressure	Carry blood under low pressure	Blood under relatively low pressure
Possess thick walls	Have thin walls	Have very thin walls
Have more elastic and muscle tissues	Have less elastic and muscle tissues	Have one cell-thin layer
Deeply seated in the body	Found on the surface of the body	Microscopic and in close contact with the cells and tissue fluid

LYMPH

Lymph is diluted blood plasma containing large numbers of white blood cells, especially *lymphocytes,* and occasionally a few red blood cells with less protein. Because of the number of living cells it contains, lymph is classified as a ***fluid tissue***.

The capillaries are not the only routes by which the lymph returns to the circulatory system. Some of it returns through the ***lymphatic system***.

The proteins in the fluid tissue are not able to re-enter the capillaries but can drain into blindly-ending, thin-walled vessels which are found between the cells. These lymphatics join up to form larger vessels which eventually unite into two main ducts and empty their contents into the large veins entering the atrium.

The lymph flow takes place in only one direction from the tissue to the heart with no special pump. The flow is brought about by the pressure of the lymph that accumulates in the tissues.

Lymph diffuses into and is absorbed by the lymphatic capillaries. The lymphatic capillaries run together to form larger ducts that intertwine about the arteries and veins.

Functions of the lymph

1. It transports waste from cells.
2. It transports blood cells through the body.
3. It makes white blood cells available in all the parts of the body

4. It helps the red blood cells carry oxygen to the body tissues.
5. Its fluid nature regulates the temperature of the body.

DISORDERS OF THE BLOOD AND THE CIRCULATORY SYSTEM

Disorders of the blood

Many diseases are caused by abnormalities in the blood. Some diseases which affect the blood are:

Anaemia

One of the most common blood diseases worldwide is anaemia, which is characterized by an abnormally low number of red blood cells or low levels of haemoglobin.

One of the major symptoms of anaemia is fatigue, due to the failure of the blood to carry enough oxygen to all of the tissues.

The most common type of anaemia is iron-deficiency anaemia which occurs because the marrow fails to produce sufficient red blood cells. When insufficient iron is available to the bone marrow, it slows down its production of haemoglobin and red blood cells.

The most common causes of iron-deficiency anaemia are certain infections that result in gastrointestinal blood loss and the consequent chronic loss of iron. Adding supplemental iron to the diet is often sufficient to cure iron-deficiency anaemia.

Sickle-cell anaemia

Sickle-cell anaemia is a genetic disease that occurs as a result of increased destruction of red blood cells. The red blood cells of sickle-cell patients assume an unusual crescent shape, causing them to become trapped in some blood vessels, blocking the flow of other blood cells to tissues and depriving them of oxygen.

Leukaemia (blood cancer)

Any disease in which excess white blood cells are produced, particularly immature white blood cells, is called leukaemia, or blood cancer.

Many cases of leukaemia are linked to gene abnormalities, resulting in unchecked growth of immature white blood cells. If this growth is not halted, it often results in the death of the patient. These genetic abnormalities are not inherited in the vast majority of cases, but rather occur after birth. Although some causes of these abnormalities are known, for example exposures to high doses of radiation or the chemical benzene, most remain poorly understood.

Treatment for leukaemia typically involves the use of chemotherapy, in which strong drugs are used to target and kill leukemic cells, permitting normal cells to regenerate.

In some cases, bone marrow transplants are effective.

Haemophilia

Haemophilia is a genetic bleeding disorder in which one of the plasma clotting factors is produced in abnormally low quantities, resulting in uncontrolled bleeding from minor injuries.

Although individuals with haemophilia are able to form a good initial platelet plug when blood vessels are damaged, they are not easily able to form the meshwork that holds the clot firmly intact. As a result, bleeding may occur for some time after the initial traumatic event.

Treatment for haemophilia relies on giving transfusions of the plasma clotting factor.

Other disorders of the blood include ***myeloproliferative disorders*** - a group of disorders caused by abnormal increase in the number of bone marrow cells; ***hypocalcaemia*** - an unusually low level of calcium in the blood and ***hypercalcemia*** – too much calcium in the blood.

Disorders of the circulatory system

Disorders of the circulatory system include any injury or disease that damages the heart, the blood, or the blood vessels. The commonest circulatory diseases include hypertension, hypotension, arteriosclerosis, septal effect and atherosclerosis.

Hypertension

Hypertension or high blood pressure, develops when the body's blood vessels narrow, causing the heart to pump harder than normal to push blood through the narrowed openings.

Hypertension that remains untreated may cause heart enlargement and thickening of the heart muscle. Eventually the heart needs more oxygen to function, which can lead to heart failure, brain stroke, or kidney impairment.

Some cases of hypertension can be treated by lifestyle changes such as a low-salt diet, maintenance of ideal weight, aerobic exercise, and a diet rich in fruits, vegetables, plant fibre, and the mineral potassium. If blood pressure remains high despite these lifestyle adjustments, medications may be effective in lowering the pressure by relaxing blood vessels and reducing the output of blood.

Hypotension or low blood pressure

Hypotension is a condition that reduces the amount of blood pumped by the heart. It is caused by slow heart rate, the weakening of the heart muscle with age, serious injuries or illness, diabetes mellitus or some medications

Low blood pressure is characterized by blurred vision, light-headedness, nausea, palpitations, temporary loss of consciousness, confusion and general weakness. Hypotension can result in heart failure or heart attack.

Arteriosclerosis

In arteriosclerosis, commonly known as hardening of the arteries, the walls of the arteries thicken, harden, and lose their elasticity. The heart must work harder than normal to deliver blood, and in advanced cases, it becomes impossible for the heart to supply sufficient blood to all parts of the body.

The causes of arteriosclerosis are not known, but heredity, obesity, smoking, and a high-fat diet all appear to play roles.

Atherosclerosis

Atherosclerosis, a form of arteriosclerosis, is the reduction in blood flow through the arteries caused by greasy deposits called plaque that forms on the insides of arteries and partially restricts the flow of blood.

Plaque deposits are associated with high concentrations of cholesterol in the blood. Reduction in blood flow can cause organ damage. When brain arteries become blocked and brain function is impaired, the result is a stroke. A heart attack occurs when a coronary artery becomes blocked and heart muscle is destroyed.

Risk factors that contribute to atherosclerosis include physical inactivity, smoking, diet high in fat, high blood pressure, and diabetes.

Some cases of atherosclerosis can be corrected with healthy lifestyle changes, aspirin to reduce blood clotting and inflammation, or drugs to lower the blood cholesterol concentration. For more serious cases, surgery to dilate narrowed blood vessels with a balloon, known as ***angioplasty***, or to remove plaque with a high-speed cutting drill, known as ***atherectomy***, may be effective. Surgical bypass, in which spare arteries are used to construct a new path for blood flow, is also an option.

Hole-in–heart (septal defect)

Hole-in-heart is a defect which affects the septum (partition) which separates the two sides of the heart.

Septal defect causes oxygenated blood to move from the left to the right of the heart through a hole in the septum or septal hole, so that too much blood passes into the lungs with little going into the tissues. Open-heart surgery can be used to correct septal defect.

TEST QUESTIONS

1. (a) What is the circulatory system?
   (b) Describe the functions of the following organs as components of the circulatory system:
   i. heart;
   ii. blood;
   iv. blood vessels

2. (a) Draw and label the mammalian heart.
   (b) Briefly describe blood circulation in the heart.

3. State the functions of the following parts of the heart:
   i. pulmonary vein;
   ii. pulmonary artery;
   iii. aorta
   iv. vena cava;
   v. left ventricle;
   vi. right ventricle;
   vii. bicuspid valve;
   viii. tricuspid valve

4. (a) State five importance of the blood.
   (b) How does the blood carry out its function in protecting the body.

5. (a) What is blood clotting?
   (b) Describe how blood clotting occurs.
   (c) Mention three importance of blood clotting.

6. (a) Describe the structure and functions of the following blood vessels:
   i. arteries;
   ii. veins
   iii. capillaries
   (b) List three differences between the veins and the arteries.

7. (a) What is the lymph?
   (b) State four functions of the lymph.

8. (a) Mention three diseases that affect the blood and the circulatory system.
   (b) Describe the effects and remedies of two of the diseases mentioned in (a).

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ELECTRICAL ENERGY

Specific Objectives

After completing this chapter, you will be able to:

- Describe static and current electricity.
- Name the various component of electric circuit and state the functions.
- Calculate resistance, current, potential difference and power.
- Identify the sources of electric power generation.

INTRODUCTION

Electrical energy is a form of energy associated with electric charges (protons and electrons).

Electrical activity takes place constantly everywhere in the universe. Electrical forces hold molecules together. The nervous systems of animals work by means of weak electric signals transmitted between neurons (nerve cells). Electricity is generated, transmitted, and converted into heat, light, motion, and other forms of energy through natural processes, as well as by devices built by people.

Electrical energy is an extremely versatile form of energy; it can be generated in many ways and from many different sources. It can be sent almost instantaneously over long distances. Electrical energy can also be converted efficiently into other forms of energy, and it can be stored.

Because of this versatility, electrical energy plays a part in nearly every aspect of modern technology. It provides light, heat, and mechanical power. It makes telephones, computers, televisions, and countless other necessities and luxuries possible.

Electric charges can be stationary, as in static electricity, or moving, as in electric current or current electricity.

NATURE OF STATIC AND CURRENT ELECTRICITY

Static electricity

Static electricity is produced as a result of attraction between oppositely charged particles.

It can be produced by rubbing together two objects made of different materials. Electrons move from the surface of one object to the surface of the other if the second material holds onto its electrons more strongly than the first does. The object that gains electrons becomes negatively charged, since it now has more electrons than protons. The object that gives up electrons becomes positively charged.

For example, if a plastic comb is run through clean, dry hair, some of the electrons on the hair are transferred to the comb. The comb becomes negatively charged and the hair becomes positively charged.

The following materials are named in decreasing order of their ability to hold electrons: rubber, silk, glass, flannel, and fur (or hair). If any two of these materials are rubbed together, the material earlier in the list becomes negative, and the material later in the list becomes positive. The materials should be clean and dry.

ACTIVITY

1. Run a plastic comb several time through your hair on a dry day. Bring it close to pieces of paper. What happens between the comb and the pieces of paper?
2. Rub a glass rod with silk or synthetic fibres of a paint brush. Electrons will be transferred from glass rod to the synthetic paint brush. This makes the glass rod positively charged. Bring it close to bit of papers

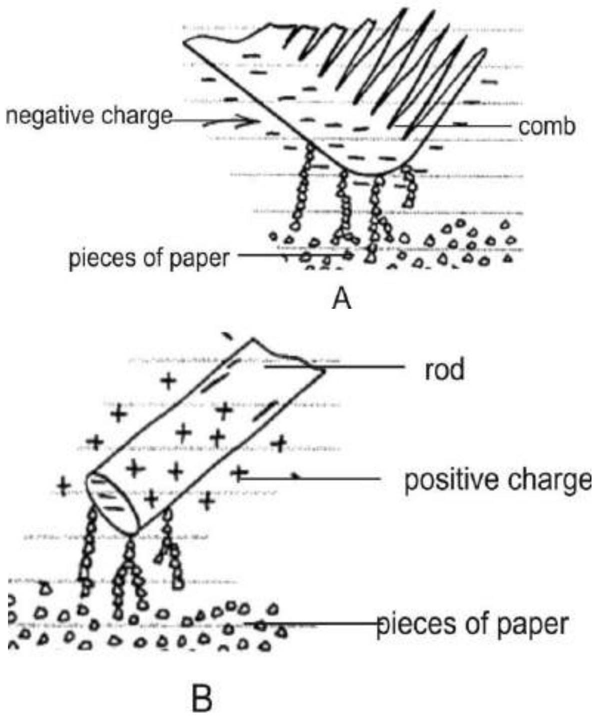


Fig. 70.4: (A) Charged comb and (B) rod attracting pieces of paper

Lightening

Lightening is caused by charges between clouds or between a cloud and the ground, resulting in atmospheric electrical discharges.

Clouds normally become charged as they rub against each other. The charges give rise to relative potential difference between the clouds and the earth. Since the clouds and the earth are of different charges, they attract each other, which is seen as lightening. This explains the fact that lightening is caused by electrostatics.

The flow of electricity from one discharge point to another also produces a sound wave heard as ***thunder***. Lightening is seen before thunder is heard. This is because light travels faster than sound.

Fig. 70.5: Lightening

Protection from lightening

Lightening could be dangerous; for this reason tall buildings are protected from lightning by providing them with metallic lightning rods, extending to the ground from a point above the highest part of the roof. These rods form a low-resistance path for the lightning discharge and prevent it from travelling through the structure itself. Power lines and radio sets with external aerials are protected against lightning by ***lightning arresters*** that consist of a small gas-filled gap between the line and ground wire.

This gap offers a high resistance to ordinary voltages.

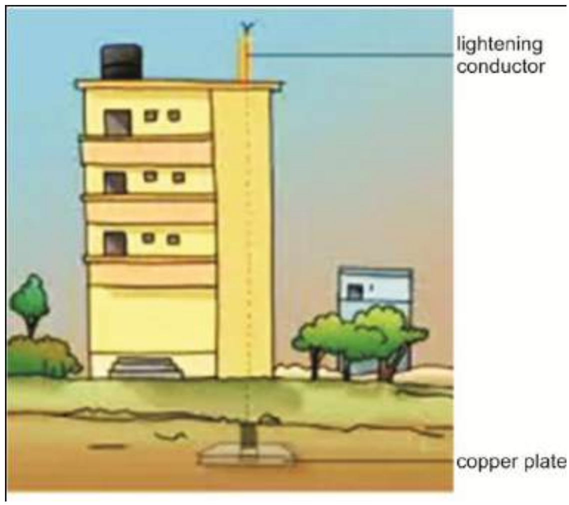


Fig. 70.6: Lightning conductor

Current electricity

An electric current is the movement of electrons through a conductor.

When two objects with different charges touch and redistribute their charges, an electric current flows from one object to the other until the charge is distributed according to the *capacitances* of the objects.

If two objects are connected by a conductor such as a copper wire, then an electric current flows from one object to the other through the wire.

Electric current can be demonstrated by connecting a small light bulb to an electric battery by two copper wires. When the connections are properly made, current flows through the wires and the bulb. causing the bulb to glow.

Electric current in measured in ***ampere*** (**Amp** or **A**). And the instrument used to measure it is an ***ammeter***.

Sources of electric current

- Cells – dry and wet cells
- Generators
- Solar cells
- Electrolytic cells
- Hydroelectric power

Generator

dry cell

Electrolytic cell

Solar cell

Fig. 70.7: Sources of electric current

Types of electric current (Direct and alternating currents)

Direct current

Direct current is the current that flows in one direction only.

Direct current is used in most battery-powered devices. An example of direct is the current in a battery-powered torchlight.

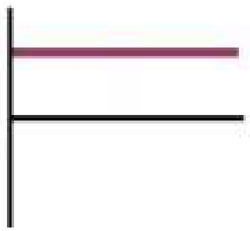
Alternating current

This is the current that flows back and forth, reversing direction again and again.

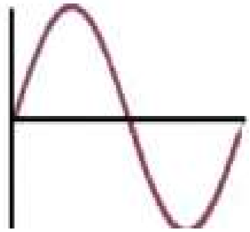
Alternating current is used in most devices that are plugged in to electrical outlets in buildings.

Properties of electric current

Other properties that are used to quantify and compare electric currents are the electromotive force and resistance.



direct current



alternating current

Table 7.1: Differences between direct and alternating currents

Direct current	Alternating current
Flows in one direction	Alternates – flows in forwards and backwards.
Does not induce emf in other materials in a magnetic field	Induces emf in other materials with a magnetic field
Produces a magnetic field with constant size and polarity	Produces a magnetic field with alternating size and polarity

Electromotive force (emf)

Electromotive force is the force that drives current through a conductor. It is commonly referred to as voltage. Emf is measured with a voltmeter, in voltage (V).

Resistance

Resistance is the opposition to the flow of current in a circuit. Electric resistance is measured in Ohm (Ω). In an electrical or electronic system, resistance is dependent on the number of resistors in the system.
The type of resistor that varies or continuously changes the resistance or current in a circuit is called a ***rheostat*** (or variable or adjustable resistor).

ELECTRIC CIRCUIT

Electric circuit is the path of an electric current.

The term is usually taken to mean a continuous path composed of conductors and conducting devices and including a source of electromotive force that drives the current around the circuit.

Closed and open circuits

A circuit in which the current flows freely is called a ***closed circuit***. This circuit has a load (resistor), a source of emf and a turned on switch or key.

Open circuit

A circuit in which the current path is not continuous is called an ***open circuit***. This circuit either has a break or the switch is turned off.

A ***short circuit*** is a closed circuit in which a direct connection is made, with no appreciable load (resistance, inductance, or capacitance) between the terminals of the source of electromotive force.

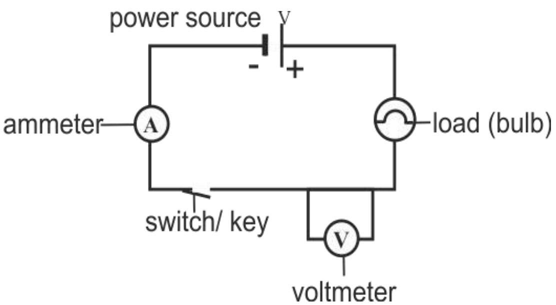


Fig. 70.8: A simple electric circuit

Circuit symbols

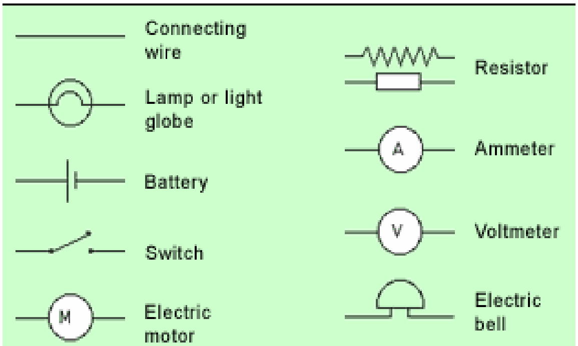


Fig. 70.9

Series and parallel arrangement of components in circuits

Series arrangement

A *series circuit* is one in which the devices or elements of the circuit are arranged in such a way that the entire current passes through each element without division.

In other words the components are arranged in a straight line with the same current flowing through each component in sequence.

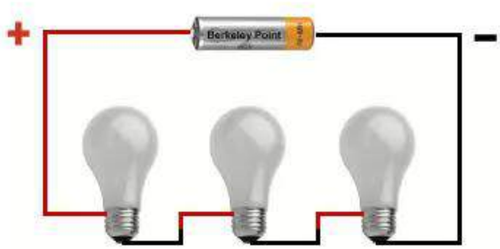


Fig. 80.0: Series arrangement

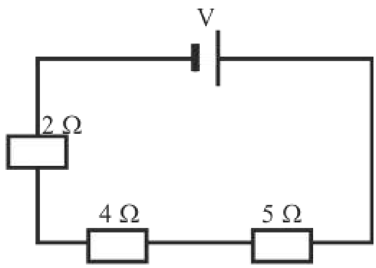
Resistors in series

When two or more resistors are in series in a circuit, the total resistance may be calculated by adding the values of such resistances as below.

$\boldsymbol{R_T = R_1 + R_2 + R_3}$ etc

where $\boldsymbol{R_1}$, $\boldsymbol{R_2}$ and $\boldsymbol{R_3}$ are the number of resistors in the circuit.

For example, calculate the total resistance of the circuit below.



Solution:

$R_T = R_1 + R_2 + R_3$

$2 + 4 + 5$

$R_T = 11\ \Omega$

Parallel arrangement

In a parallel circuit, electrical devices, such as incandescent lamps or the cells of a battery, are arranged to allow all positive (+) poles, electrodes, and terminals to be joined to one conductor, and all negative (-) ones to another conductor, so that each unit is on a parallel branch.

The value of two equal resistances in parallel is equal to half the value of the component resistances, and in every case the value of resistances in parallel is less than the value of the smallest of the individual resistances involved.

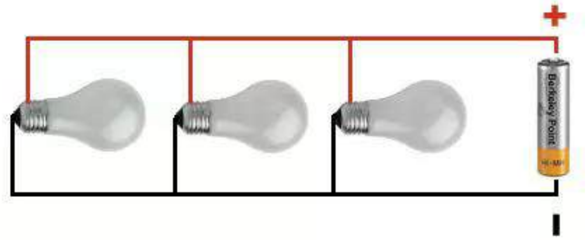


Fig. 80.1: Parallel arrangement

Resistors in parallel

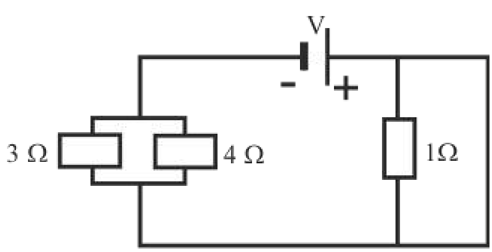
If the resistors are in parallel, the total value of the resistance in the circuit is given by the formula

$$\frac{\mathbf{1}}{\boldsymbol{R_T}} = \frac{\mathbf{1}}{\boldsymbol{R_1}} + \frac{\mathbf{1}}{\boldsymbol{R_2}} + \frac{\mathbf{1}}{\boldsymbol{R_3}} \text{ etc}$$

where $\boldsymbol{R_1}$, $\boldsymbol{R_2}$ and $\boldsymbol{R_3}$ equals the number of resistors in the circuit.

Example

What is the total resistance of the circuit below?



Solution

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_T} = \frac{1}{3} + \frac{1}{4} + \frac{1}{1}$$

$$\frac{1}{R_T} = \frac{4+3+12}{12} = \frac{19}{12}$$

$$\frac{RT}{1} = \frac{12}{19} = 0.632\ \Omega$$

Series-parallel arrangement

Many electrical circuits are both series and parallel providing a complete path from the power supply.

To determine the total resistance of resistors in a series-parallel circuit, first calculate those in series using the series formula, then calculate those in parallel using the parallel formula and then finally add the two answers together to get your total resistance.

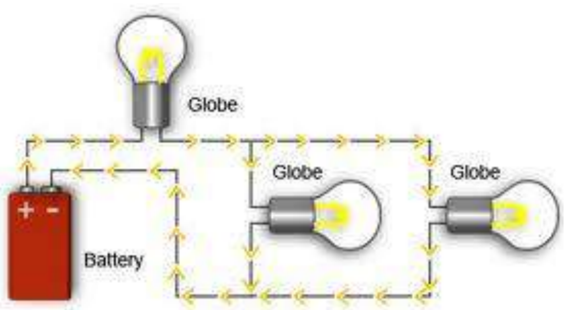


Fig. 80.2: Series-parallel arrangement

NB: In a circuit containing a voltmeter and an ammeter, the ammeter is connected is series while the voltmeter is connected in parallel.

CALCULATING CURRENT (OHM'S LAW)

Current flows in an electric circuit in accordance with several definite laws. The basic law of current flow is Ohm's law, named for its discoverer, the German physicist Georg Ohm.

Ohm's law states that ***the amount of current flowing in a circuit is directly proportional to the potential difference and inversely proportional to the total resistance of the circuit***.

Expressed by the formula:

$$\boldsymbol{I = \frac{V}{R}}$$

Where ***I*** is the current in amperes (A), ***V*** is the electromotive force in volts (V), and ***R*** is the resistance in ohms (Ω).

Ohm's law applies to all electric circuits; for both direct current (DC) and alternating current (AC).

Example

1. An electric circuit has a cell of 3V with three resistors of total resistance 10Ω. What is the current in the circuit?

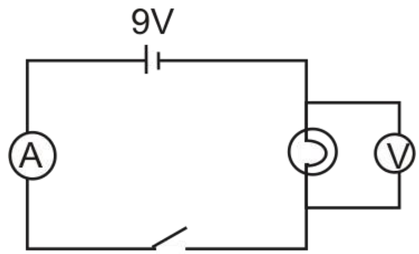
Solution

$$I = \frac{V}{R}$$

V = 3V, R = 10

$$\therefore \quad I = \frac{3}{10} = 0.3 \text{ A}$$

2. Josh wished to find the resistance of an electric bulb and set up a circuit as shown below.



The voltmeter reading was 9V. The resistance of the bulb was 6Ω.

(a) What would be the reading on the ammeter?

(b) He then added a 4Ω resistor in series. Find the reading on the ammeter now.

(c) If the 4Ω resistor was added in parallel to the lamp. What would be the reading on the ammeter?

Solution

(a) $I = \frac{V}{R}$ V = 9V, R = 6Ω

$$I = \frac{9}{6} = 1.5\text{A}$$

(b) When a 4Ω resistor is added in series, total resistance

$R_T = R_1 + R_2$

$= 6 + 4 = 10\Omega$

$$\therefore I = \frac{9}{10} = 0.9\text{A}$$

(c) If the 4Ω resistor is added in parallel, the total resistance

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_T} = \frac{1}{6} + \frac{1}{4}$$

$$\frac{1}{R_T} = \frac{2+3}{12} = \frac{5}{12}$$

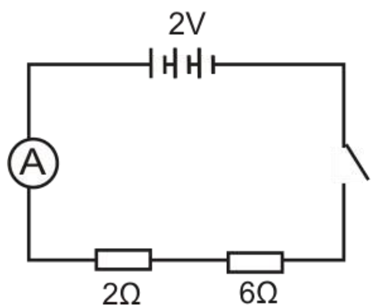
$$\frac{RT}{1} = \frac{12}{5} = 2.4\ \Omega$$

$$\therefore\ I = \frac{9}{2.4} = 3.75\text{A}$$

3. Three cells of emf 2V and negligible internal resistance are connected in series with an ammeter, two resistors of resistance 2Ω and Ω and a key.
   (a) Draw a circuit diagram for the connection above.
   (b) Calculate the current that flows through the circuit.

Solution

(a)



(b) $I = \frac{V}{R}$

V = 2V

R = 2 + 6 = 8Ω

$$\therefore I = \frac{2}{8} = 0.25\text{A}$$

Electric power

For an electrical device to work or produce, it must get power.

Power is the energy used or produced per unit time (second).

$$\mathbf{P} = \frac{E}{t}$$

The unit of power is watt (W).

In relation to current, voltage and resistance, power is expressed as:

❖ $P = VI$
❖ $P = I^2R$
❖ $P = \frac{V^2}{R}$

where P = power in watt (W)
V = voltage in volt (V)
I = current in ampere (A or Amp)
R = resistance in ohm (Ω)

Example

1. A mobile phone has a 4V battery. If the current passing through the phone is 1.2A, what is its power?

Solution

$P = VI$ $\quad V = 4\text{V}, I = 1.2\text{A}$

$P = 4 \text{ X } 1.2$

$P = 4.8\text{W}$

2. An electric iron is rated 240V, 1000W. Calculate:
(a) the current
(b) the resistance

Solution

(a) $P = VI \quad \Rightarrow I = \frac{P}{V}$

P = 1000W, V = 240V

$\therefore I = \frac{1000}{240} = 4.167A$

(b) $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$

$R = \frac{240^2}{1000} = 57.6\Omega$

Alternative method

$P = I^2R \Rightarrow R = \frac{P}{I^2}$

P = 1000W, I = 4.167A

$\therefore R = \frac{1000}{4.167^2} = 57.6\ \Omega$

Power rating of electrical appliances

Electrical appliances have ratings based on the amount of power they operate on. For example an electric light bulb and electric stove rated 20W and 750W respectively, use 20 and 750 watts of power.

It is essential to know how much power an appliance consumes when using it, since it will determine the amount of energy you are using and consequently the amount you have to pay the power company.

Cost of electricity supply

Energy is power used over a length of time. It is measured in ***joules.*** One joule is the equivalent of one watt of power, used for one second of time.
In electricity, energy is measured in *watt hour* (Wh) or the *kilowatt hour* (kWh). That is the power used over one hour.
For example, a 60-watt bulb will burn 60 Wh in an hour, or 1 Wh per minute. A 1000W electric cooker would burn 1 Wh in 1/1000 hour.
The power company has a fixed charge for every kWh of electricity used. To calculate the cost of electricity used by an appliance, multiply its wattage by the total time it was used and the fixed charged.

For example, if the cost of every kWh = Gh¢ 0.1, then the cost of electricity = the power (kW) X time (in hour) X Gh¢0.1.

Examples

1. An electric pressing iron of 750 W is used for 10 minutes, calculate the cost of usage if 1 kWh costs Gh¢ 0.2.

Solution

Power = 750 W = $\frac{750}{1000}$ = 0.75 kW

Time = 10 minutes = $\frac{10}{60}$ = 0.167 hour

Charge per kWh = Gh¢ 0.2.

$\therefore$ Cost of electricity = $0.75 \times 0.167 \times 0.2$

= Gh¢ 0.0251

2. Calculate the total amount a user of a television set rated at 60 W is supposed to pay if it is user for 8 hours a day for two weeks, assuming 1kWh is equal to Gh¢ 0.2.

Solution

Power = 60 W = 0.06 kW

Time = 8 x 14 (two weeks) = 112 hours

Charge per kWh = Gh¢ 0.2.

Cost of electricity = $0.06 \times 112 \times 0.2$

= Gh¢ 1.344

Power rationing

The introduction of energy saving appliances (such as energy saving bulbs) to replace the traditional power wasting appliances (e.g. Incandescent bulbs), is a step in the right direction. Power rationing enables minimum power to be used at a time. If electrical energy is misused, there will come a time when there will be no more. Imagine a world without electrical energy; everything will be at a standstill, and that will send our world back to the prehistoric era.

Power rationing methods

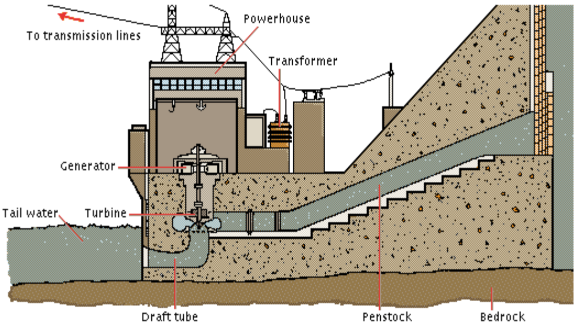
1. Use energy saving appliances.
2. Switch off appliances that are not being used.
3. Use appliances when necessary.
4. Try opening the windows to let in fresh air instead of air conditioning or fan.
5. Turn off electric lights during the day.
6. Do ironing and washing (with machine) in bulk.
7. Make sure the refrigerator doors are well closed.

SOURCES OF ELECTRIC POWER GENERATION

There are several different devices, systems and principles by which electrical energy can be generated. Some methods or sources of electric power generation are:

Hydro-electric power

In many countries, hydroelectric power is the dominant source of electric power which comes from dammed water. A dam generates electricity by releasing a controlled flow of high-pressure water from a reservoir through a channel called the ***penstock***, where it spins turbines that turn the generators, producing an electric current. The current then passes through a ***step up transformer***, which changes it from a large current at a low voltage to a small current at a high voltage. This current then goes over transmission lines to a substation, where the voltage is reduced for customers. The water exits the dam through a ***draft tube***.



A hydroelectric dam

Geothermal energy

Geothermal energy plants generate electricity and heat by harnessing the heat energy contained within the earth. The earth transfers its energy to deep-lying circulating water, which the plants access with wells and pumps. Geothermal energy is attractive because it creates almost no environmental pollution. However, the number of sites where geothermal energy can be economically extracted is limited.

Nuclear energy

Nuclear energy is the energy released during the splitting or fusing of atomic nuclei. Nuclear power is a controversial energy source. This is because radioactive substances released during accidents at nuclear power plants has caused deaths and environmental damage. However, in the absence of accidents nuclear energy is inexpensive and creates no air pollution.

Solar energy

Energy from the sun can be harnessed to produce electricity. Specially designed panels collect energy from sunlight and convert it directly into electricity. The solar panels contain semiconducting materials. When light strikes the material, electrons move from one layer of the material to another, forming an electric current.

Wind energy

Another environmentally friendly source of electrical energy is the wind energy. Wind turbine generators are used to convert wind power into electrical energy. They consist of a variety of components; the rotor converts the power of the wind to the rotating power of the shaft; a gearbox increases speed and a generator converts the shaft power into electrical power.

Fig. 80.3: A windmill farm

Tidal energy

Another form of waterpower in electric power generation is tidal energy. The incoming tide of the river flows through a dam, driving turbines, and then is trapped behind the dam. When the tide recedes, the trapped water is released and flows back through the dam, again driving the turbines. Such tidal power plants are most efficient if the difference between high and low tides is great

Biogas

Biogas is a mixture of methane and carbon dioxide produced by decomposing organic matter such as human or agricultural waste. The energy generated from biogas is mostly used for domestic purposes.

Thermal energy

Thermal energy is generated by burning petroleum products such as oil, natural gas, liquefied petroleum gas (LPG) and other highly flammables substances.

POWER TRANSMISSION

Transformers

A transformer is a device that consists of two coils wound around a common magnetic or laminated core. When the current through one coil (the primary winding) changes, a voltage is induced in the other winding (the secondary winding).

How the transformer works

Transformers have the ability to transform or change voltage and current to higher or lower levels. Transformers do not create power from nothing. If a transformer increases the voltage of a signal it reduces its current, and if it cuts the voltage of a signal it raises its current. In other words, the power flowing from a transformer cannot exceed the incoming power.

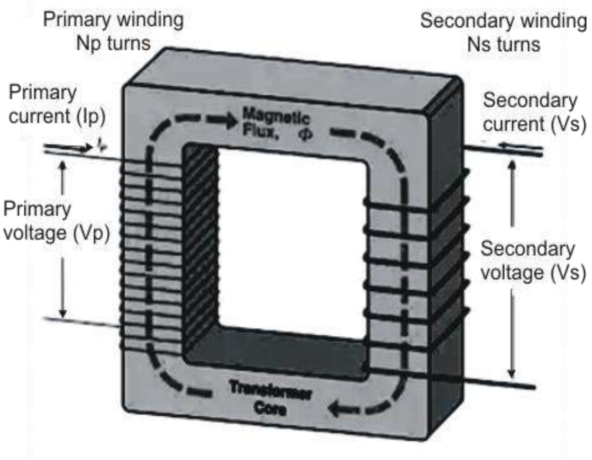


Fig. 80.4: A Transformer

Types of transformers

Transformers are categorized based on their turns ratio; that is the ratio of the number of windings of the primary and secondary coils. This is expressed as:

$$\frac{I_s}{I_p} = \frac{n_p}{n_s}$$

where I_p and I_s are the current in the primary and secondary coils and

n_p and n_s are the number of turns in the primary and secondary coils.

Current is proportional to voltage, therefore when the voltage is given it can be written as

$$\frac{V_s}{V_p} = \frac{n_s}{n_p}$$

where V is the voltage or potential difference.

Example

The primary coil of a step-up transformer has 80 turns. What is the total number of turns needed in the secondary coil if the transformer is needed to turn a 12V to 240V?

Solution

$$\frac{V_s}{V_p} = \frac{n_s}{n_p}$$

$$\frac{240}{9} = \frac{n_s}{80}$$

$$240 \times 80 = 9n_s$$

$$n_s = 2133.3 \text{ turns}$$

The ratio of primary to secondary determines a transformers voltage ratio.

The types of transformers are

Step-up transformers – The type of transformers used to convert low voltage into high voltage. They have more turns on the secondary coil than on the primary coil.

Step-down transformers – These type of transformers convert high voltage into low voltage. Step-down transformers have more turns on the primary coils than on the secondary coils.

Power distribution

In Ghana, Volta River Authority (VRA) generates and initiates the transmission of power to various homes; but they do not do that directly.

The power coming from VRA is as high as 400 kV or 400,000 V. This large amount of energy is transmitted to Ghana Grid Company Limited, who has a series of transformers which step down the voltage to about 30 kV, and then passes it on to Electricity Company of Ghana.

Heavy industries which have huge machineries may use the 30kV energy, while smaller industries may go in for about half or less. The maximum voltage used by household appliances is 240V, therefore ECG, with their numerous inter-city transformers bring the voltage way down for use at homes, offices, schools etc.

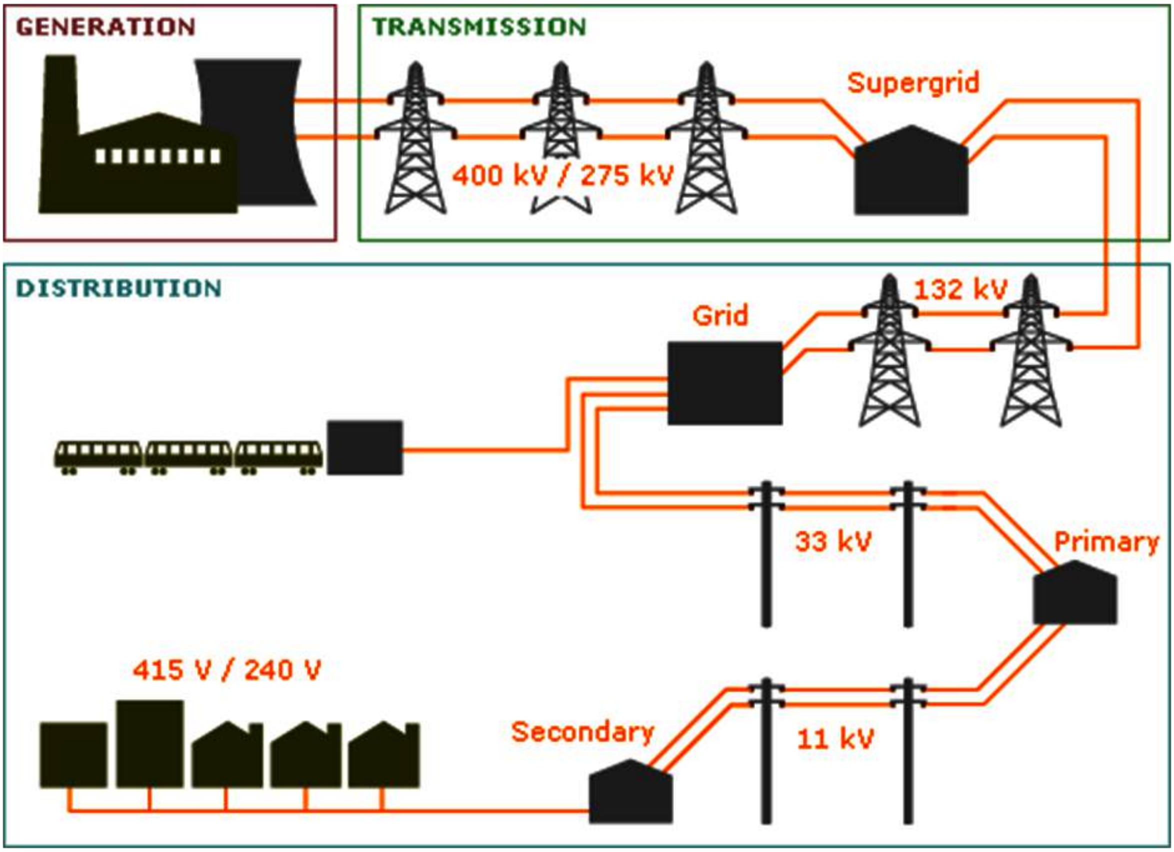


Fig. 80.5: An electrical network

Home wiring

The electricity which gets to our houses is carries by a thick overhead or underground cable. This cable called ***live***, though contains all the needed energy, however cannot work alone. A supporting cable called ***neutral***, which is ***earthed*** at the transformer, and thus has no potential, accompanies the live wire to the individual houses.

In the house, both cables pass through a meter (provided by ECG), get through the main switch and then the fuse box, which contain fuses. An additional cable, called the ***earth***, accompanies the two cables to the individual sockets in the house.

Wiring of a plug

The mains supply is at 240V A.C. Three connections are made at the usual domestic supply socket, labelled *live* (**L**), *neutral* **(N)** and *earth* (**E**)

Connection to domestic supplies is made by a three-pin plug; either the British Standard (BS) type or the International

Electrotechnical Commission (IEC) type of connector shown opened in fig. 80.6 below.

The BS plug is designed so that shutters within the socket are raised only when the plug is correctly inserted. In some cases the earth wire is ignored, especially when the appliance the plug connects to has a plastic casing. In this case there can be no connection with the outer body of the appliance and the main circuitry.

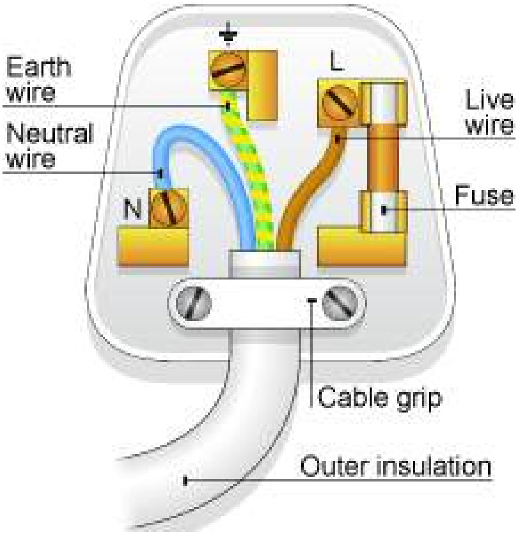


Fig. 80.6: The wiring of a plug

Wire colour code

Each of the wires used in the connection are colour coded. Table 7.2 below lists the colour coding of the wire connections to the plug. Although it is many years since the coding colours were changed, older equipment can still be found bearing the older colours (and any such equipment should be checked carefully to ensure that it conforms to modern requirements for insulation).

Table 7.2: Wire colour codes

	Live	Neutral	Earth
Modern	Brown	Blue	Green/ yellow
Old colours	Red	black	Green

How to correctly wire a plug

- The cable is firmly held and clamped without damaging the insulation or the conductors

- All connections are tight with no loose strands of wire; the ends of the cable can be coated with solder to prevent loose strands from separating, but the soldered end should not be used for clamping because the wire is more brittle and will loosen off after some time.

- The wires should be cut to length so that the live lead will break and pull free before the earth lead if there is excessive force on the cable. The design of some plugs can make this very difficult, and you should select plugs that permit the use of a live lead that will pull out before the earth lead becomes strained.

PROTECTIVE DEVICES IN HOUSEHOLD WIRING

Fuse

One important component in household and plug wirings is the fuse.

A fuse is a piece of soft wire that melts, breaking a circuit if the current exceeds a certain level.

Fuses are placed in series with the live wire or transformer primary (in the case of transformers). Any component failure, short circuit, or overload that might cause catastrophic damage (or fire) will burn the fuse out. Fuses are easy to replace.

If a fuse blows, it must be replaced with another of the same rating. If the replacement fuse is rated too low in current, it will probably blow out right away, or soon after it has been installed. If the replacement fuse is rated too high in current, it might not protect the equipment.

The standard fuse ratings for domestic equipment are 3 A, colour coded red and 13 A, colour coded brown.

A

B

C

D

E

Fig. 80.7: Different Types of fuses (A and D– ceramic fuses; B-glass fuse; C and E plug fuses)

Circuit breakers

The circuit breaker does the same job as fuses, except that a breaker can be reset by turning off the power supply, waiting a moment, and then pressing a button or flipping a switch. Some breakers reset automatically when the equipment has been shut off for a certain length of time.

Fig. 80.8: Air circuit breaker

Stabilizers

The voltage from the power company can sometimes surge or become more than normal. In situations like that appliances which require minimum or regular voltage can breakdown. Voltage (or power) surge can hardly be prevented, but it destructive impact on electrical appliances can be avoided. This can be achieved through the use of voltage stabilizers.

Voltage stabilizers are devices used to maintain constant voltage through an electric system.

Earthing

Another protective measure in household wiring and the wiring of appliances with metal casing is *earthing*. The earth wire is a special wire, so named because it connects to the earth or ground.

When using devices with metal casing, sometimes the live wire might accidentally touch the casing sending current to the case which can harm the user.
The earth wire is thus connected to the metallic casing so that in case there is a malfunction in the circuit and the casing of the appliance is earthed or becomes live, that current in the casing will be sent to the earth or ground-ward, thereby preventing the user from injury or harm.

Fig. 80.9: A voltage stabilizer

TEST QUESTIONS

1. (a) Explain in terms of electron movement what happens when a plastic comb is used to rub the hair.
   (b) Describe how lightening occurs.

2. (a) State the functions of the following components of an electric circuit;
   i. battery
   ii. switch
   iii. resistor
   (b) Distinguish between current electricity and static electricity.

3. (a) Resistors 2Ω and 6Ω are connected in (i) parallel and (ii) series to a 1.5 V cell. Calculate the current flowing in each arrangement
   (b) An electric appliance is rated 240V, 750W. Calculate
   (i) the current
   (ii) the resistance

4. (a) Discuss how electric power reaches your home from a hydro-electric power station.
   (b) An ammeter, a cell of 8 V and two resistors of resistances 4 Ω and 5Ω are connected in series. A voltmeter, V, is then connected across the 4Ω resistor.
      (i) draw a circuit diagram to illustrate the arrangement.
      (ii) calculate the current in the circuit.

5. (a) What is a direct current as used in electricity?
   (b) An electric furnace operating on 200 V, uses 3 kW of power . ` calculate the current that flows.

6. (a) What is an electric current?
   (b) State four source s of electric current.
   (c) Mention three differences between direct current and alternating current.

7. (a) What is an electric circuit?
   (b) Distinguish between an open circuit and a closed circuit.

8. (a) Explain the following arrangements of electric components:
      (i) series arrangement;
      (ii) parallel arrangement.
   (b) Explain why household wiring is done in parallel.

9. (a) State the Ohm's law.
   (b) Calculate the current that passes through an electric light bulb, a television set and a fan connected in parallel with resistances 3Ω, 10Ω and 6Ω, respectively, given that the voltage from the mains is 240V.

10. (a) Define the term electric power.
    (b) List five ways of conserving electric power.

11. Describe the following sources of electric power:
    (a) hydroelectric power;
    (b) solar energy;
    (c) thermal energy;
    (d) wind energy;

12. (a) Describe how a transformer works.
    (b) differentiate between a step-up and a step-down transformer.
    (c) The primary coil of a step-down transformer has 200 turns. calculate the total number of turns needed in the secondary coil if the transformer is needed to turn a 240 V to 24 V in a laptop adapter?

13. Describe the importance of the following devices:
    (a) fuse;
    (b) stabilizer

(c) earthing.

14. An electric kettle is rated 1000 W, 240V
 a) Explain the above statement.
 b) Calculate the maximum current the kettle can draw.

15. (a) Draw and label a diagram of a step-up transformer.
 (b) How do step-up and step-down transformers differ.

16. **Fig. 3** below is an illustrated of a part of the human body.
 Study the figure carefully and answer the questions that follow.

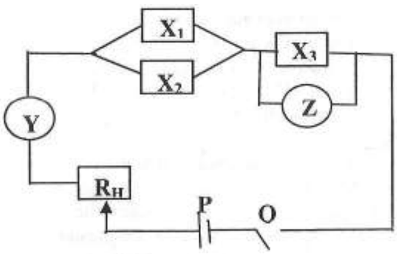


Fig. 2

 a) i) Identify the components labelled **Y**, **Z**, **X**$_1$, **P**, **Q** and **R$_H$**.
 ii) State the unit of measurement of readings on **Y** and **Z**.
 b) Describe briefly the arrangements of all the components in the circuit.
 c) Give the direction of the conventional current when the circuit is closed.
 d) State one function each of
 i) **R$_H$**;
 ii) **Q**;
 iii) **Y**;
 iv) **Z**.

17. Fig. **2a** is an illustration of an electrical circuit used to investigate the relationship between the potential difference, v, and length, L, of a wire.

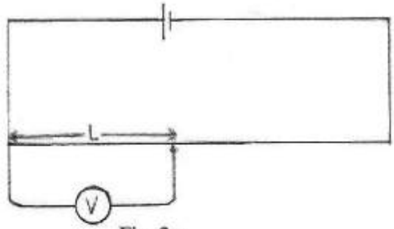


Fig. 2a

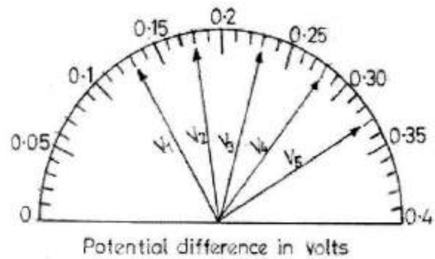


Fig. 2b

The potential difference $V = V_1$ across the corresponding length, $L = L_1$, of the wire was determined. The experiment was repeated for various lengths, L_2, L_3, L_4 and L_5 of wire and their corresponding potential difference V_2, V_3, V_4 and V_5 were read.
Fig. **2b** represents the potential difference $V = V_1$, V_2, V_3, V_4 and V_5 and Fig **2c** represents the length $L = L_1$, L_2. L_3, L_4 and L_5 of the wire.

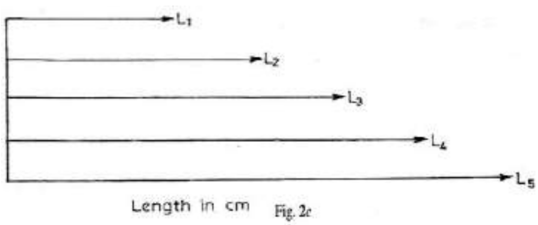


Fig. 2c

(a) (i) Measure and record the length, $\boldsymbol{L} = \boldsymbol{L}_1, L_2, \boldsymbol{L}_3, \boldsymbol{L}_4$ and $\boldsymbol{L}_5$

(ii) Read and record the corresponding potential difference , $\boldsymbol{V} = \boldsymbol{V}_1, \boldsymbol{V}_2, \boldsymbol{V}_3, \boldsymbol{V}_4$ and $\boldsymbol{V}_5$.

(iii) Tabulate results obtained in (i) and (ii) as shown below.

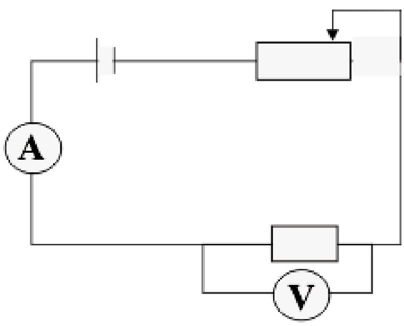
V/V					
L/cm					

(b) Plot a graph with potential difference, V, on the vertical axis and the length, L, of wire on the horizontal axis.

(c) (i) Determine the slope of the graph.

(ii) State the relationship between the potential difference, V, and length, L, of the wire.

18.



The diagram above is an identical circuit used to investigate a certain scientific law. Fig. 3 (a) shows five ammeter readings, while fig. 3(b) shows the corresponding voltmeter readings take during an investigation of a law.

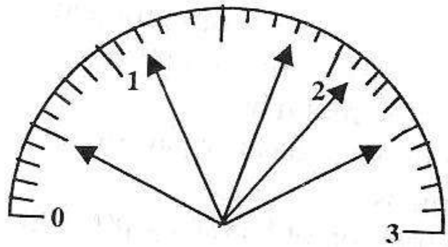


Fig. **3**(**a**): ammeter sowing actual reading in amperes

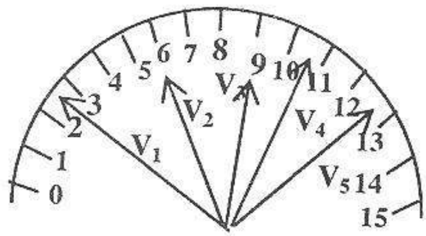


Fig. **3**(**b**); Voltmeter showing actual reading in volts

a) Read and record the ammeter readings $I = I_1, I_2, I_3$, I_4 and I_5 and their corresponding voltmeter readings $V = V_1, V_2, V_3, V_4$ and V_5 in the table below.

b) i) Plot a graph with V as ordinate and I as abscissa.

ii) Draw the best line through your points. From your graph, state your relationship between V and I.

33

ELECTRONICS II

Specific Objectives

After completing this chapter, you will be able to:

- Classify solid materials into conductors, insulator and semi-conductors.
- Explain the formation of P-type and N-type semi conductors.
- Describe the formation of a P-N junction diode.
- Explain the behaviour of a P-N junction diode in a d.c. and a.c. electronic circuits.

INTRODUCTION

A simple electronic circuit consists of three classes of solid materials – ***conductors***, ***insulators*** and ***semi-conductors***. In some materials, electrons move easily from atom to atom. In others, the electrons move with difficulty and in some materials, it is almost impossible to get them to move. All these three materials are essential in building circuits – whether simple or complex.

CONDUCTOR, INSULATOR AND SEMI-CONDUCTORS

Conductors

An electrical conductor is a substance that allows electricity to flow through it easily.

In such substances, the electrons are mobile and can move freely from one atom to another. In a circuit conductors are used to connect various components together.

The best conductor at room temperature is silver. Copper and aluminium are also excellent electrical conductors. Iron, steel, and various other metals are fair to good conductors of electricity.

In most electrical circuits and systems, copper or aluminium wire is used. Silver is impractical because of its high cost.

Some liquids are good electrical conductors. Mercury is one example. Salt water is a fair conductor.

Gases are, in general, poor conductors of electricity. This is because the atoms or molecules are usually too far apart to allow a free exchange of electrons. But if a gas becomes ionized, it is a fair conductor of electricity.

Electrons in a conductor do not move in a steady stream, like molecules of water through a tap. Instead, they are passed from one atom to another right next to it. This happens to countless atoms all the time. As a result, literally trillions of electrons pass a given point each second in a typical electrical circuit.

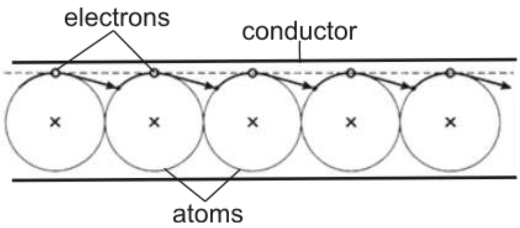


Fig. 90.0: Movement of electrons in a conductor

Insulators

Insulators are substances that prevent electrical currents from flowing.

Most gases are good electrical insulators. Glass, dry wood, paper, and plastics are other examples. Pure water is a good electrical insulator, although it conducts some current with even the slightest impurity. Metal oxides can be good insulators, even though the metal in pure form is a good conductor.

Insulators keep electrical charges apart, preventing the flow of electrons that would equalize a charge difference between two places. Excellent insulating materials can be used to advantage in certain electrical components such as capacitors, where it is important that electrons not flow.

Porcelain or glass can be used in electrical systems to keep short circuits from occurring.

These devices come in various shapes and sizes for different applications. You can see them on high-voltage utility poles and towers. They hold the wires up without running the risk of a short circuit with the tower or a slow discharge through a wet wooden pole.

Semi-conductors

These materials have the property of a conductor and an insulator. They allow some electrons to move through them while preventing the movement of others. Examples of such materials are silicon, selenium, germanium, gallium arsenide, silicon carbide and metal oxide.

A pure semiconductor is often called an ***intrinsic*** semiconductor. The electronic properties and the conductivity of a semiconductor can be changed in a controlled manner by adding very small quantities of other elements, called ***dopants***, to the intrinsic material. In crystalline silicon typically, this is achieved by adding impurities of boron or phosphorus to the melt and then allowing the melt to solidify into the crystal.

This process is called ***doping***. Semi-conductors are used in making diodes, transistors, integrated circuits (ICs), resistors etc.

P-TYPE AND N-TYPE SEMI-CONDUCTORS

P-type semi-conductors

These are semiconductors which has most of the charge carriers being protons.
This means that there are more positively charged particles in P-type semiconductors than negatively charged electrons.

N-type semi-conductors

These are semiconductors possessing more negatively charged electrons as charge carriers that the positively charged protons.

In a semiconductor, the more abundant type of charge carrier is called **the *majority carrier.*** The less abundant kind is known as the ***minority carrier.*** Hence in P-type semiconductors, the majority carrier is protons and the minority carrier is electrons whereas the inverse is true for N-type semiconductor.

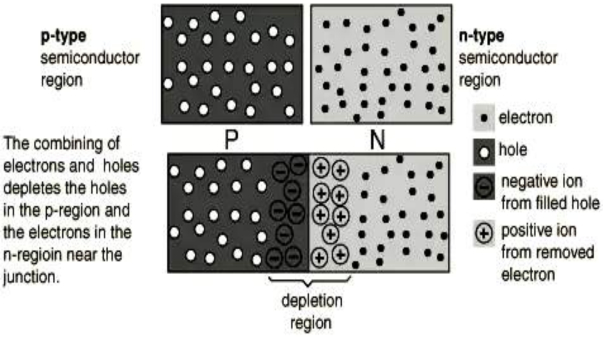


Fig. 90.1: P-N Semi-conductor

P-N JUNCTION DIODE FORMATION

When a P-type and N-type semiconductor regions are adjacent to each other, they form a semiconductor diode, and the region of contact is called a ***P-N junction***.

A diode is a two-terminal device that has a high resistance to electric current in one direction but a low resistance in the other direction.
A P-N junction diode will only allow current to flow in one direction. The electrons from the n-type material can pass to the right through the p-type material, but the lack of excess electrons in the p-type material will prevent any flow of electrons to the left.
The conductance properties of the P-N junction depend on the direction of the voltage, which can in turn be used to control the electrical nature of the device. Series of such junctions are used to make transistors and other semiconductor devices such as solar cells, P-N junction lasers, rectifiers, and many others.

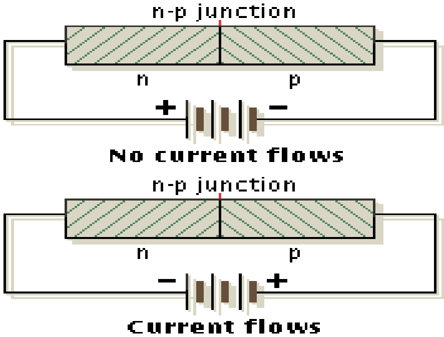


Fig. 90.2: A P-N junction diode

DOPING

Doping refers to the addition of impurities to a semiconductor.

The addition of impurities adds charge carrying elements to the semiconductor.

Types of doping

The two classes of doping are p-type and n-type doping.

N-type doping: This is the type of doping in which negative charged carriers (electrons) are introduced to the semiconductor.

For example, in silicon, each atom has four valence electrons; two are required to form a covalent bond. In n- type silicon, atoms such as phosphorus (P) with five valence electrons replace some silicon and provide extra negative electrons.

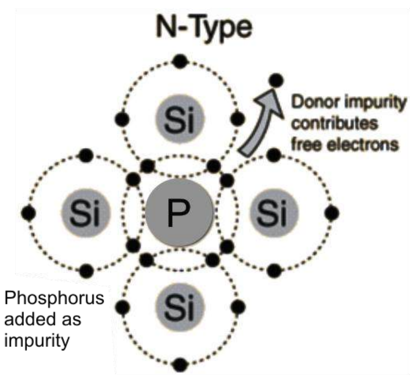


Fig. 90.3: N-type doping

P-type doping: This is the type of doping in which positive charge carriers called ***holes*** are introduced to the semiconductor.

By doping the same silicon semiconductor with boron, an electron is donated in order to fit neatly into the structure. The site which is now missing an electron represents a positive charge, and therefore the doping is p-type.

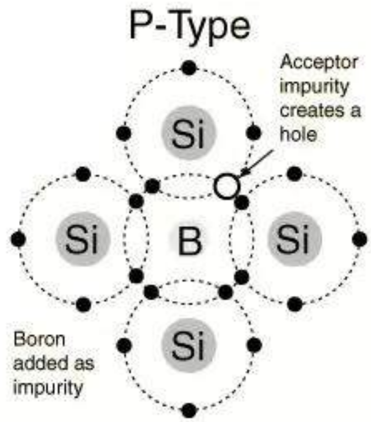


Fig. 90.4: P-type doping

Doping becomes important when p-doped and n-doped materials are connected to form a diode.

BEHAVIOUR OF A P-N JUNCTION DIODE

The most important property of a junction diode is its ability to pass an electric current in one direction only. If the diode is connected to a simple circuit consisting of a battery and a resistor, the battery can be connected in either of two ways.

Forward bias

When the p-type region of the p-n junction is connected to the positive terminal of the battery, current will flow. The diode is said to be on or under ***forward bias***. As the forward bias voltage is increased, the current through the junction becomes greater.

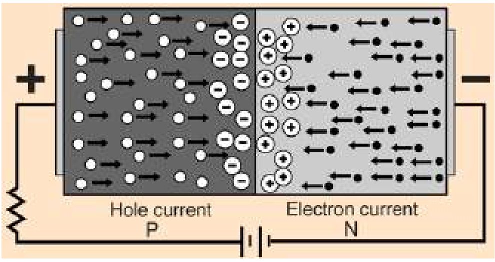


Fig. 90.5: Forward biased p-n junction diode

Reverse bias

When the battery terminals are reversed, the p-n junction almost completely blocks the current flow. This is called ***reverse bias***.

As the reverse voltage is increased, the current is reduced to almost zero. However, a very small reverse current does flow.

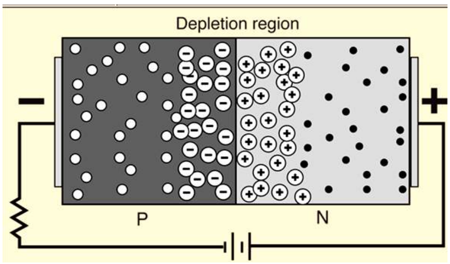


Fig. 90.6: Reverse biased p-n junction diode

If the diode is not connected at all, it is said to be ***open-circuited*** and no current can flow through the diode.

RECTIFICATION

Rectification is the process of converting an alternating current to direct current.

A device known as a ***rectifier***, which permits current to pass in only one direction effectively blocking its flow in the other direction, is inserted into the circuit for the purpose.

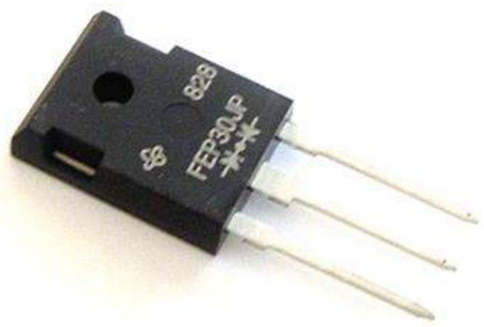


Fig. 90.7: A rectifier diode

Half-wave rectification

This is the type of rectification where only half of the alternating current signal is rectified or converted to direct current signal.

The other half is blocked off or lost. A half-wave rectifier is used to achieve half-wave rectification. In contrast to half-wave rectification is ***full wave rectification*** where all the signal or current in the circuit is converted or rectified.

TEST QUESTIONS

1. Describe the properties of the following materials:
   (a) conductor
   (b) insulator
   (c) semi-conductor

2. (a) Differentiate between p-type and n-type semi-conductors.
   (b) Describe the formation of p-n junction diode.

3. (a) Define the term doping.
   (b) Describe how p-type and n-type doping are achieved.

34

SOUND ENERGY

Specific Objectives

After completing this chapter, you will be able to:

- Explain the sources and nature of sound.
- Describe musical notes and distinguish musical notes from noise.
- Identify the parts of the human ear and describe their functions.

INTRODUCTION

Sound energy is a form of energy that is associated with vibration of matter.

Sound originates from the vibrations that result after an object applies a force to another object. It is a physical phenomenon that stimulates the sense of hearing.

Although sound is a form of energy, it is not measured in joule like the other forms of energy. This is because the amount of energy in a sound wave is far smaller and cannot be transformed into other energy forms.

Sound energy is instead measured in decibels (dB). The measurement of sound energy is related to its pressure; hence it can also be measured in pascal (Pa)

SOURCES OF SOUND

Humans and animals hear different sounds because as the sound (vibrations) enters the ear, the ear also vibrates.

All sounds come from something that vibrates. If you bang on a drum, the top of the drum vibrates. When you talk, the vocal chords in your throat vibrate.

Sources of sound

The sources of sound are categorised based on the mode of production of a particular sound. It is perceived that every object that produces sound is an instrument.

Hence the sources of sound are categorized as:

1. stringed instruments
2. percussion instruments
3. wind instruments

Stringed instruments

These are instruments which use the vibration of strings in air to produce sound. Examples of stringed instruments are harp, guitar, banjo and violin, etc.

Percussion instruments

These are instruments which produce sound as a result of the vibration of the skin or surface when struck.
Examples are drums, xylophone, gong, bell etc.

Wind instruments

These are instruments which vibrate the air around them to produce sound.
Examples are flute, trumpet, organ, accordion etc.

Guitar

Talking drum

Gong

Trumpet

Fig. 90.8: Musical instruments

90.9: *A drummer*

NATURE OF SOUND

Sound waves require a medium such as air or water in order to be heard. Therefore sound waves are classed as mechanical waves.

Mechanical waves

Mechanical waves are waves which need material medium for their transmission.

They are in two types – transverse waves and longitudinal waves. In both cases, only the energy of wave motion is propagated through the medium; no portion of the medium itself actually moves very far.

Transverse waves

In transverse waves, the material through which the wave is transmitted vibrates

perpendicular to the wave's forward movement.

As a simple example, if a rock is thrown into a pool of water, a series of transverse waves (ripples) moves out from the point of impact. A cork floating near the point of impact will bob up and down. That is, move transversely with respect to the direction of the wave motion.

Longitudinal waves

These are waves propagated in the same direction in which the particles of the medium vibrate.

A sound wave is a ***longitudinal wave***. As the energy of wave motion is propagated outward from the centre of disturbance, the individual air molecules that carry the sound move back and forth, parallel to the direction of wave motion. Thus, a sound wave is a series of alternate increases and decreases of air pressure. Each individual molecule passes the energy on to neighbouring molecules, but after the sound wave has passed, each molecule remains in about the same location.

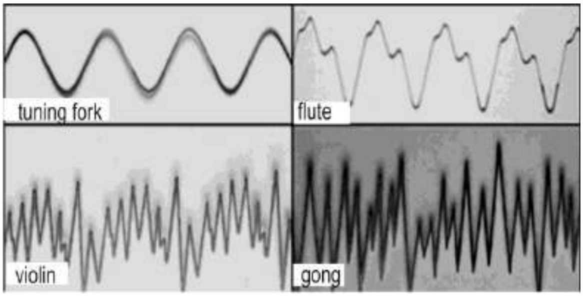


Fig. 90.9: Sound waves in different instruments

Since sound is a mechanical (longitudinal) wave, it needs a material medium for its transmission. That is sound cannot travel through a vacuum (a completely empty space).

Experiment to show that sound needs a material medium for its transmission

- Enclose an electric bell in a bell jar.
- Connect the bell jar to a vacuum pump.
- Switch on the electric bell.
- While the bell is ringing, use the vacuum pump to suck the air from the bell jar.
- This causes the sound of the bell to die down.
- An observer may see the bell ringing, yet will not hear any sound.

Conclusion

This shows that sound needs a material medium for its transmission.

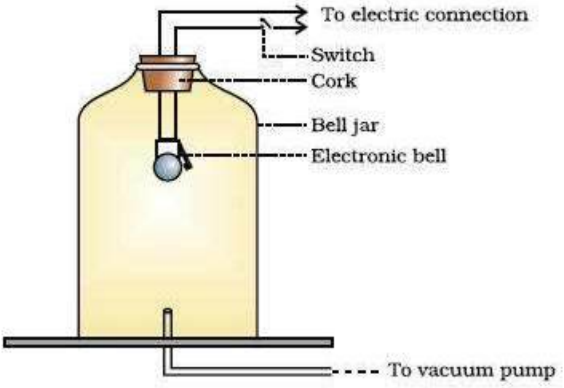


Fig. 100.0

Table: Differences between electromagnetic waves and mechanical waves

Electromagnetic waves	Mechanical waves
Travel in a vacuum	Travel through material media
Travel in straight lines	May travel in straight lines or branch off
Have high velocities	Have low velocities
Have high frequencies	Have low frequencies
Always transverse	May be transverse or longitudinal

Similarities between electromagnetic waves and mechanical waves

1. They can both be polarised.
2. They can both be interfered and diffracted.
3. They can both be reflected and refracted.

Characteristics of sound

The characteristics of sound are:

1. Frequency
2. Amplitude
3. Wavelength

Frequency

Frequency is the number of cycles or oscillations a sound wave makes per second.

We perceive frequency as low or high sounds. Frequency is measured in hertz (Hz). One hertz is equal to one cycle per second.

Amplitude

Amplitude is the maximum displacement of vibrating or distorted particle from it mean or equilibrium position.

It is the characteristic of sound waves that humans perceive as volume. The amplitude corresponds to the distance that air molecules move back and forth as a sound wave passes through them. As the amount of motion in the molecules is increased, they strike the ear drum with progressively greater force. This causes the ear to perceive a louder sound.

Wavelength

Wavelength is defined as the distance between two successive crests or troughs.

It can also be defined as the distance between any two refractions or compressions.

The distance at which a sound can be heard depends on its wavelength. The wavelength of sound is the average rate of flow of energy per unit area perpendicular to the direction of propagation. Wavelength is measured in metre (m).

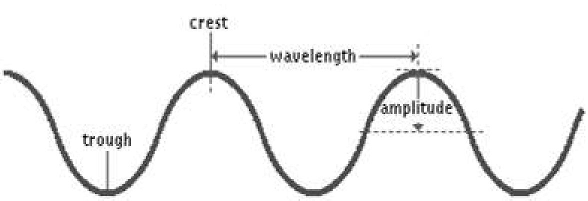


Fig. 100.1

VELOCITY OF SOUND

Velocity of a sound wave is the product of frequency and wavelength.

This is expressed mathematically as:

$$\textit{velocity} = \textbf{frequency} \times \textbf{wavelength}$$

Or

$$v = f\lambda$$

The unit of velocity is metre per second (ms^{-1}).

The velocity or speed of sound varies from one medium to another. The velocity of sound in a medium depends mostly on its density and elasticity.

Speed of sound in solids

Solids, which have greater density and elasticity, allow sound waves to travel through them much faster than liquids, which also provide relatively faster speed for sound waves than gases.

For example, the speed of sound in copper is about 3,350 m/s at normal temperatures. In steel, which is even more elastic, sound moves at a speed of about 4,880 m/s. Sound is propagated very efficiently in steel.

Speed of sound in liquids

The speed of sound in water is slightly less than 1,525 m/s at ordinary temperatures, almost five times as fast as in air.

Speed of sound in gases

The speed of sound in dry, sea level air at a temperature of 0 °C is 332 m/s. The speed of sound in air varies under different conditions. If the temperature is increased, for example, the speed of sound increases; thus, at 20 °C, the speed of sound is 344 m/s.

The speed of sound is different in other gases of greater or lesser density than air. The molecules of some gases, such as carbon dioxide, are heavier and move less readily than molecules of air. Sound progresses through such gases more slowly. Conversely, sound travels through helium and hydrogen faster than through air because the molecules of helium and hydrogen are lighter than the molecules of air.

Comparison of velocity of light and sound

In comparison to light, sound waves travel slower than light, this is the reason why if you watch an action at a distance, you see that action before hearing the accompanying sound.

Examples

1. In a medium, a wave of sound has a frequency of 32 Hz and a wavelength of 40 m. Find the velocity of the wave.

Solution

$$v = f\lambda$$

f = 32 Hz, λ = 40 m
v = 32 x 40 = 1,280 ms^{-1}

2. What is the wavelength of sound which has a frequency of 95 Hz and a speed of 1800 ms^{-1}?

Solution

v = f λ

f = 95Hz, v = 1800 ms^{-1}

$$\lambda = \frac{v}{f} = \frac{1800}{95} = 19.94 \text{ m}$$

3. A sound wave travelling at a velocity of 3,000 ms^{-1} has a wavelength of 20m. Calculate its frequency.

Solution

v = f λ

v = 3000ms^{-1}, λ = 20m

$$f = \frac{v}{\lambda} = \frac{3000}{20} = 60\text{Hz}$$

Refraction of sound waves

Refraction of sound is the bending of a sound wave from its original path when it travels from one medium into another.

When sound travels from hot air to cold air or from cold air to hot air it will refract. On a hot day the air near the ground is hot so the sound wave bends upwards from the hot air into the cold air

In polar (cold) regions, where the air close to the ground is colder than air that is higher, a rising sound wave entering the warmer less dense region, (in which sound moves with greater speed), is bent downward by refraction. The excellent reception of sound downwind and the poor reception upwind are also due to refraction.

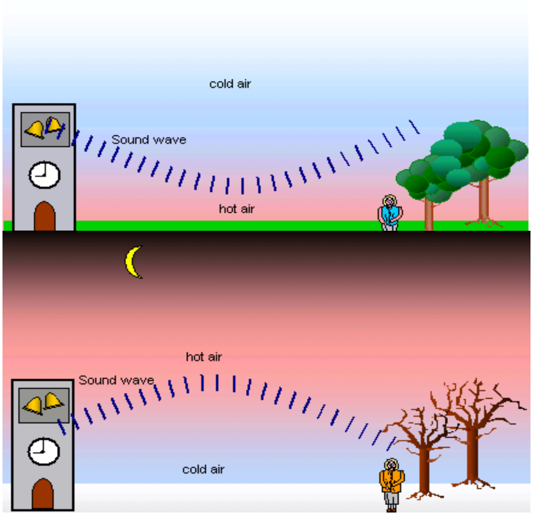


Fig. 100.2: Refraction of sound during the day and at night

Reflection of sound waves

This is the bouncing back of a sound wave when it hits a hard surface.

This means that sound waves obey the fundamental law that the angle of incidence equals the angle of reflection. An ***echo*** is the result of reflection of sound.

A megaphone is a funnel-like tube that forms a beam of sound waves by reflecting some of the diverging rays from the sides of the tube. A similar tube can gather sound waves if the large end is pointed at the source of the sound; an ear trumpet is such a device.

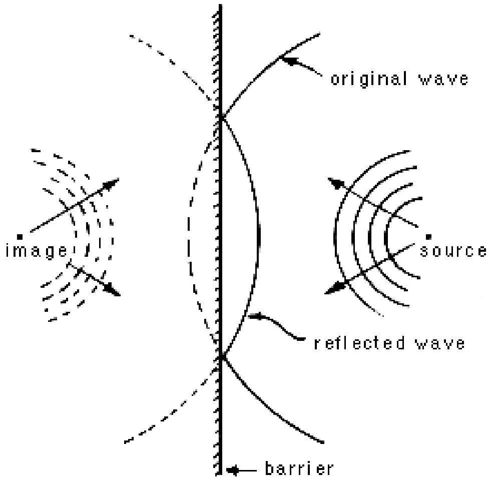


Fig. 100.3: Reflection of sound

Echo

An echo is the continuous sound heard after the original sound is reflected by a hard smooth surface.

The shorter the time-lapse between the original sound and the echo, the smaller the distance.

Uses of echo

1. An echo can be used to estimate the distance of an object, its size, shape and velocity, as well as the velocity of sound itself.
2. It is used to determine the depth of sea bed.
3. It helps in oil and gas exploration.
4. Echoes are used to estimate the thickness of polar icecap.
5. Bats use echoes to find their way around in the dark.
6. Dolphins communicate with each other through echoes.
7. It is used to find lost objects at sea.
8. It is also used in medical procedures to assess for blockages in the heart.
9. Ultra-sonic echoes are used to determine the size of a baby in the womb

Disadvantage of echo

In a big hall or auditorium, echoes interfere with the original sound making it difficult to hear.

MUSICAL NOTE AND NOISE

Musical note is produced by sound waves of regular frequencies.

At its simplest, music consists of a short, unaccompanied melody, known as ***monophony***.

Properties of music

1. Pitch

Pitch is the highness or lowness of a musical tone as determined by the speed of the vibrations producing it.

Changes in pitch are as a result of difference in frequency at which the sound wave vibrates.

2. Quality or timbre

The quality of a musical note distinguishes one musical note from others of the same pitch or loudness.

It is referred to as the colour or texture of the note.

3. Loudness or volume

Loudness is the strength with which an instrument or a musical note is played.

If one strikes a drum gently, it produces a sound with a specific loudness and pitch. If the same drum is struck harder, it will produce a louder sound but same pitch.

Noise

Noise is produced by sound waves of irregular frequencies or different pitches.

Unlike music, noise produces no harmony, and is often very unpleasant to listen to. The production of noise is mostly unintended and may come from different sources such as moving vehicles, aircrafts, machines, rainfall on roofing sheet etc.

Table Differences between musical note and noise

Musical note	Noise
Produces by sound of regular frequency	Produced by sound of irregular frequency
Produces harmony	Produces no harmony
Its production is intentional	It production is unintended
Pleasant to listen to	Unpleasant to listen to
Produced from specific sources	Produced from diverse sources

The Doppler Effect

Doppler Effect is the apparent variation in frequency of any emitted wave as the source of the wave approaches or moves away, relative to an observer.

Doppler Effect explains why, if a source of sound of a constant pitch is moving towards an observer, the sound seems higher in pitch, whereas if the source is moving away it seems lower. This change in pitch can be heard by an observer listening to the whistle of an express train from a station platform or another train.

Applications of Doppler Effect

1. It is used in radar speed detector used by traffic police for checking speed.
2. It is used to calculate the relative motion of the earth and stars.
3. It is used to determine approaching or receding sounds.
4. It is used in weather forecasting
5. It helps bats find their way around when flying in the dark.

THE HUMAN EAR

The human ear contains receptors which are sensitive to sound vibrations in the air.

It consists of three sections: the outer, middle, and inner ear.

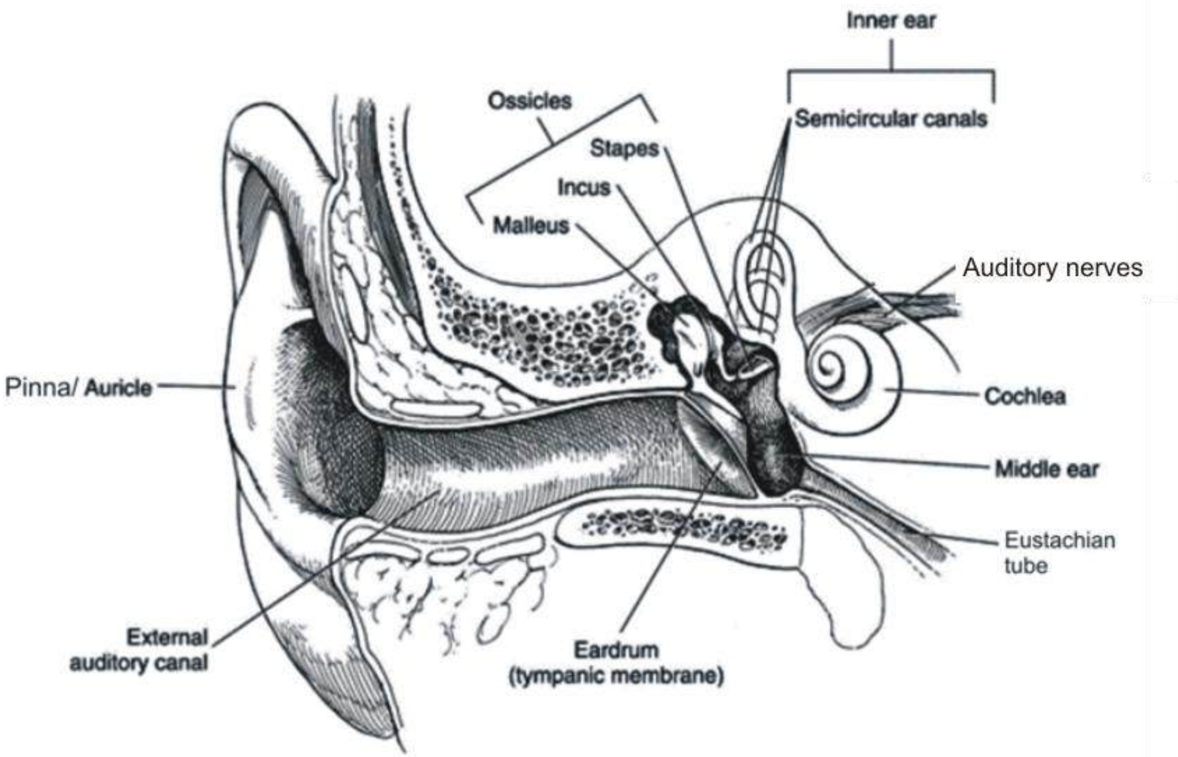


Fig. 100.4: **The human ear**

The outer ear

The outer ear is a tube opening on the side of the head and leading inwards to the ear-drum. At the outside end is an extension of skin and cartilages called ***pinna***, which helps to concentrate and direct the vibration into the ear and assists in judging the direction from which the sound came. In the innermost end of the outer layer is the ***ear-drum***, a membrane of skin and fine fibre which stretches across and closes the outer ear.

The middle ear

This part of the ear is an air-filled cavity in the skull. It contains the ***Eustachian tube***, which connects to the mouth cavity. The ear-drum is linked to the ***oval window***, (a small opening in the skull), of the middle ear by ***ossicles*** (three small bones – ***malleus, stapes*** and ***incus*** or ***anvil***) which lead to the inner ear.

The inner ear

The inner ear is filled with fluid and contains a coiled tube known as the ***cochlea***, with sensory endings in it. Sound vibrations are converted to nervous impulses in the inner ear.

How humans hear

- Vibrations in the air are made up of sound waves entering the outer ear and cause the ear-drum to vibrate.
- The vibration is transmitted through the three ossicles, which act as levers and cause the staples to vibrate against the oval window, increasing the vibration by about 22 times.
- The movement of the staples set the fluids in the inner ear to vibrate, particularly in the cochlea.
- Nerve impulses are transmitted from the cochlear to the brain through the auditory nerves.
- The brain interprets the nerve impulses as sound. These actions are very quick.

Effects of loudness and pitch on humans

a) Human ears are capable of perceiving an extraordinarily wide range of changes in loudness. The softest audible sound to humans is 0 decibels, while painful sounds are those that rise above 140 decibels.

b) Beside loudness, the human ear can detect a sound's pitch, which is related to a sound's vibration frequency; the greater the frequency, the higher the pitch.

c) Different parts of the cochlea respond to different pitches of sound. High-pitched sound stimulates the cochlea hairs close to the oval window, while low-pitched sounds stimulate hair further up the spiral.

d) The tiny hairs in the cochlea may be damaged by very loud sound, resulting in partial deafness.

e) Another characteristic of sound detected by the human ear is ***tone***. The ability to recognize tone enables humans to distinguish a violin from a guitar when both instruments are playing the same note.

f) The least noticeable change in tone that can be picked up by the ear varies with pitch and loudness.

Masking

In noisy places, high-pitched sounds make it difficult to hear lower-pitched sound. This is known as masking. To overcome the effects of masking in noisy places, people are forced to raise their voices.

Earring aids

To amplify sound waves, special devices are used. Megaphones are used to address small groups of people while microphones fitted to amplifiers are used to reach a large crowd of people.

Stethoscopes are used by doctors to listen to heartbeat, and a person suffering from partial deafness will require a hear trumpet to hear well. Radio waves are modulated with sound and transmitted across the world. These devices improve hearing.

There are some situations when sounds (especially noise) have to be reduced or snuffed out. When using electronic devices, it is easy to turn down the volume to an appreciable level.

In some other cases the sound or noise cannot be controlled, especially noise from machines or vehicles. These sounds can be reduced with ear-muffs and ear plugs. They absorb some of the sound energy so that less noise reaches the ear, keeping it from damage.

Fig. 100.5: Hearing aids

TEST QUESTIONS

1. Mention three examples of each of the following sources of sound:
   (i) stringed instruments;
   (ii) wind instruments;
   (iii) percussion instruments.

2. (a) What are mechanical waves?
   (b) Distinguish between longitudinal and transverse waves.

3. Describe an experiment to show that sound needs a material medium for its transmission.

4. (a) Define the term velocity of sound.
   (b) A sound wave moving at a velocity of 800 ms^{-1} possesses a wavelength of 110 m. Calculate its frequency.

5. Explain the following terms:
   (a) reflection of sound;
   (b) refraction of sound.

6. Define the following terms as applied to sound waves:
   (a) amplitude;
   (b) frequency;
   (c) wavelength.

7. Explain the following components of music:
   (a) pitch;
   (b) quality;
   (c) loudness.

8. Differentiate between:
   (a) Pitch and loudness
   (b) Loudness and quality of sound

9. (a) What is Doppler Effect?
   (b) State four applications of Doppler Effect.

10. (a) What is an echo?
    (b) Mention four uses of echo.

11. Draw and label the structure of the mammalian ear.

12. Describe the following parts of the human ear:
    (a) outer ear;
    (b) middle ear;
    (c) inner ear.

13. Describe the hearing mechanism in humans.

14. What is the effect of high pitch sound on humans?

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NUCLEAR ENERGY

Specific Objectives

After completing this chapter, you will be able to:

- Describe the nature of radioactivity.
- Describe the nature and uses of radioisotopes.
- State the uses of nuclear energy\outline ways for protecting people from the effects of radioactivity.
- Outline the problems associated with disposal of nuclear waste.

INTRODUCTION

Nuclear energy is the energy released during the splitting or fusing of atomic nuclei.

The energy of any system, whether physical, chemical, or nuclear, is manifested by the system's ability to do work or to release heat or radiation. The total energy in a system is always conserved, but it can be transferred to another system or changed in form.

RADIOACTIVITY

Radioactivity is the spontaneous disintegration of unstable atomic nuclei to form stable nuclei with the emission of radiations.

Types of radioactivity

Natural radioactivity – This is the type of radioactivity produced when naturally occurring unstable nuclei disintegrate with the emission of radiations.

Artificial radioactivity – This is the controlled bombardment and disintegration of unstable nuclei using nuclear particles to form stable nuclei with emission of radiations.

The radiations released during radioactivity are:

1. alpha particle,
2. beta particle and
3. gamma rays

Characteristics of the radiation emitted during radioactivity

Alpha particles (α)

1. They are made up of helium atoms.
2. Alpha particles are positively charged (+2).
3. They have mass number of 4 and atomic number of 2 ($^{4}_{2}He$)
4. They consist of two protons and two neutrons.
5. They have no electron.
6. They have small deflection in a magnetic field.
7. They are relatively slow-moving.
8. They have very high ionizing power.
9. They have low penetration power.
10. They undergo high fluorescence.

Beta particles (β)

1. They consist of a stream of electrons.
2. They are negatively charged ($^{0}_{-1}e$).
3. Beta particle move fast.
4. They have high ionizing power.
5. They have high penetrating power.
6. They are deflected by magnetic and electric fields.
7. They are lighter than alpha particles.
8. They undergo low fluorescence.

Gamma rays (γ)

1. Gamma rays have a very short wavelength (0.0005 to 0.1 nm)
2. They have a very high velocity (equal to that of light, $3.0 \times 10^8\ ms^{-1}$)
3. They are not deflected by either magnetic or electric fields.
4. They have very high penetrating power.
5. They have no charge (neutral).
6. They have lower ionizing power.
7. They have no mass.
8. They do not undergo fluorescence.

Summary of the characteristics of the radiations

Characteristic	Alpha Particles (α)	Beta Particles (β)	Gamma rays (γ)
Speed	Slow	Fast	Very fast
Charge	Positive	Negative	Neutral
Mass	Very heavy	Heavy	No mass
Penetration power	Low	High	Very high
Ionization power	Very high	High	low
Electric and magnetic field effect	Small	Big	No effect
Fluorescence	High	Low	Non

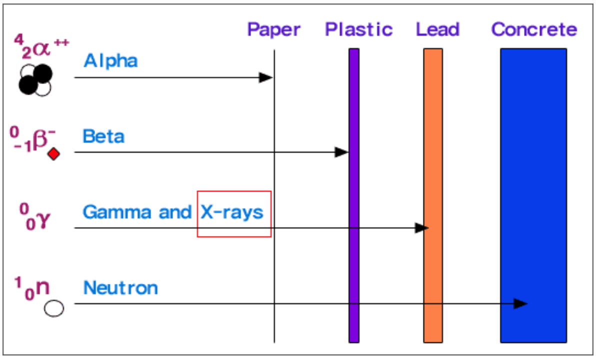

Fig. 100.6: Penetration power of the radiations

Similarities between alpha and beta particles

i. They are both charged.
ii. They can both ionize gas.
iii. They are both deflected by electric and magnetic field,
iv. They both have masses.
v. They both undergo fluorescence.

Nuclear reactions

Alpha particle

When an alpha particle is ejected from a nucleus, the mass number of the nucleus decreases by four units and the atomic number decreases by two units.

Examples

1. In alpha disintegration of Radium -226 to form Radon:
   $^{226}_{88}\mathrm{Ra} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{222}_{86}\mathrm{Rn}$

2. Copy and complete the following alpha disintegration of Uranium -238 to form Thorium:

$$^{238}\mathrm{U} \rightarrow {}_{2}\mathrm{He} + {}_{90}\mathrm{Th}$$

Solution

- ❖ The mass number of thorium is 90. Since uranium had to lose 2 atomic numbers, its total atomic number = 90 + 2 = 92
- ❖ The mass number of helium atom = 4
- ❖ Also, uranium had to lose 4 units of its mass number to form thorium;
  Hence 238 – 4 = 234

$$\therefore {}^{238}_{92}\mathrm{U} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{234}_{90}\mathrm{Th}$$

Beta particles

When a beta particle is ejected, a neutron in the nucleus is converted to a proton, so the mass number of the nucleus is unchanged, but the atomic number increases by one unit.

Example

1. $^{20}_{5}\mathrm{X} \rightarrow {}^{0}_{-1}\mathrm{e} + {}^{20}_{6}\mathrm{Y}$

2. Find the values of x and y in the nuclear reaction below:
   $^{214}_{x}\mathrm{Pb} \rightarrow {}^{0}_{-1}\mathrm{e} + {}^{y}_{83}\mathrm{Bi}$

Solution

$x = 83 - 1 = 82$
$y = 214 – 0 = 214$

Gamma rays

The emission of gamma radiation results from an energy change within the atomic nucleus. Gamma emission changes neither the atomic number nor the atomic mass. Alpha and beta emission are often accompanied by gamma emission, as an excited nucleus drops to a lower and more stable energy state.

Half-life ($t_{½}$)

Half-life is defined as the time taken for half the number of radioactive nuclei to undergo disintegration.

The time rate at which a radioactive change takes place is usually expressed by the time required for the decay or disintegration of one-half of the atom.
The half-life of ***P*** is 2 years means out of a given mass of ***P*** half of it will did integrate in 2 years.

Nuclear fission and fusion

Nuclear fission is the splitting up of a heavy nucleus into two smaller nuclei with the release of large amount of energy.

Nuclear fusion is the combination of two light nuclei under high temperature to produce a heavy nucleus and high amount of energy.

Stable and unstable atoms

A stable atom is the one whose probability of undergoing disintegration is so small that no decay of the atom is observed.

An unstable atom is the one which has a tendency to disintegrate to form a new atom of different element

Causes of instability

Radioactive atoms have large atomic numbers. There is therefore a large number of protons. Hence, the force of repulsion within the nucleus is so great that the atom disintegrates so as to become a stable atom.

NATURE AND USES OF RADIOISOTOPES

Radioisotope is an isotope that has an unstable nucleus and emits radiations during its decay to a stable form.

For example, the most stable isotope of uranium, U-238, has an atomic number of 92 and mass number of 238. U-235, has three fewer neutrons, also with an atomic number of 92 but mass number of 235. The chemical behaviour of U-235 is identical to all other forms of uranium, but its nucleus is less stable, giving it higher radioactivity and greater susceptibility to the chain reactions that power both atomic bombs and nuclear fission reactors.

Uses of radioisotopes

1. Food preservation or irradiation

Food irradiation is a method of treating food in order to make it safer to eat and have a longer shelf life. This is done by either slowing down or eliminating the actions of disease-causing microorganisms or those that cause spoilage

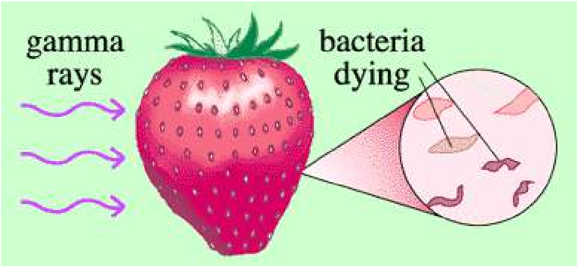

Fig. 100.7: Food treatment with radioisotopes

2. **Medicine**
   a) It is used to studying the cellular functions and bone formation in mammals.
   b) It is used to treat cancerous tumours.
   c) Used to measure correct patient dosages of radioactive pharmaceuticals
   d) Used in research in red blood cell survival studies
   e) Used as a tracer to diagnose pernicious anaemia
   f) Used in cancer treatment
   g) Used to sterilize surgical instruments
   h) Used to diagnose thyroid disorders and other metabolic disorders including brain function
   i) Different chemical forms are used for brain, bone, liver, spleen and kidney imaging and also for blood flow studies.

3. **Industry**
   a) Used to measure and control the liquid flow in oil pipelines
   b) Used to ensure the right fill level for packages of food, drugs, and other products. (The products in these packages do not become radioactive.)
   c) Used to ensure uniform thickness in rolling processes like steel and paper production
   d) Used to gauge the moisture content of soil in the road construction and building industries
   e) Used in mining to analyze materials excavated from pits
   f) Used to test the integrity of pipeline welds, boilers and aircraft parts
   g) Used to detect explosives
   h) Used to locate leaks in industrial pipe lines
   i) Used as fuel for nuclear power plants

4. **Education and Research**
   a) Important aid to biomedical researchers studying the cellular functions and bone formation in mammals.
   b) Helps in research to ensure that potential new drugs are metabolized without forming harmful by-products.
   c) Used in research in red blood cell survival studies
   d) Used in protein studies in life science research.
   e) Used in survey meters by the military and emergency management authorities.
   f) Major tool for biomedical research

5. **Agriculture**
   a) They are used to treat disease resistant varieties of crops.
   b) Used to improve crop yield

c) Used to obtain information about the nutrition values of plants
d) Used to sterilize male tsetse fly to discourage the rise in tsetse fly population

3. **Other uses**
   a) Used in many smoke detectors for homes and businesses
   b) Used to inspect airline luggage for hidden explosives
   c) Used in indicator lights in appliances such as clothes washers and dryers, stereos, and coffeemakers
   d) Helps fluorescent lights last longer
   e) Used to produce fluorescent glassware, a variety of coloured glazes and wall tiles

USES OF NUCLEAR ENERGY

Nuclear energy is one of the safest energies that can be used to achieve many desired results, provided it is used safely with due precautions. The following are some of its uses.

Food and Agriculture

The use of isotopes and radiation techniques in agriculture comes under this category. Leading organizations have been working on the technology to increase agricultural production, improve food availability and quality, reduce production costs and minimize pollution of food crops.

Human Health

One very common application is in the treatment of cancer (radiotherapy). Also, small amounts of radioisotope tracers are used for diagnostic and research purposes. The radioisotopes aid in measuring the concentration of various enzymes, some drugs, hormones and many other substances that are present in the human blood. These techniques have also helped in monitoring the levels of toxic substances in food, air and water.

Fig. 100.8: Radiotherapy treatment

Sterilization

Gamma emissions can be used for the sterilization of medical supplies like cotton, bandages, gloves used for surgery, syringes, burn dressings, etc.

Tracing Pollutants

Radioisotopes can be actively used for tracing the pollutants present in air. The dangerous residues of the radioisotope present even in small amounts in air can be very harmful to humans (can cause health effects such as kidney disease, etc.). Hence, the tracing quality helps to detect the residue easily, thereby ensuring a healthy and safe environment.

Detecting Leaks in Pipelines

The gamma rays emitted by radioactivity can be used to check welds of gas and oil pipelines. In this the radioactive source is placed inside the pipe and the film outside the welds. This being convenient, can successfully be used in place of X-ray equipment, which was earlier used to detect leakage in pipelines.

Power Sources

While decaying, radioactive elements emit lots of energy which is used to control the heart pacemaker. This energy also provides power to the beacons and satellites used for navigation.

Determination of Age (carbon dating)

Another interesting use of the nuclear energy is that it can be used by archaeologists, geologists and anthropologists in determining the age of structures, rocks, insects, bones, etc.

Use in Space

Both fission and fusion of nuclear power is actively used in providing power for the missions in space. It generates high velocities that increase the speed of rockets. This high generation of velocity is due to the higher density reactions that take place and is around 7 magnitudes more than the chemical reactions, which is used to power the current generation of rockets.

Generating Electricity

With so many different uses, the use of nuclear energy for the production of electricity is the most important. The energy released by the fission that takes place in a nuclear reactor of the nuclear power plant is converted and generated into electricity.

Advantages of Nuclear Energy

1. It is a very reliable source of energy. The average life span of a nuclear reactor is approximately 40 years which can be extended up to 60 years.
2. Nuclear power stations are usually very compact compared to thermal stations.
3. Although the initial capital cost of building a nuclear plant is high, the maintenance and running costs are relatively low.
4. It produces very less amount of pollution (when no accident occurs).

5. Very small amount of raw material is required to generate huge amount of nuclear energy.
6. Since they are required in small quantities, atomic materials can be easily transported to far-off places.
7. If nuclear power stations are operated up to their full capacity they can produce cheap electricity and gain from other benefits of nuclear energy.
8. The quantity of nuclear waste produced is relatively small.

Disadvantages of Nuclear Energy

1. Radioactive minerals are unevenly distributed around the world and are found in limited quantities.
2. Supply of high quality uranium, one of the raw materials, will last only for few decades.
3. Nuclear waste from nuclear power plant creates thermal (heat) pollution which may damage the environment.
4. Disposal of nuclear waste is a major problem.
5. There is the danger of accidental discharge of radioactivity.
6. Starting a nuclear plant requires huge capital investment and advanced technology.
7. Nuclear plants are opposed on moral grounds, by many groups, because of their close linkage with development of nuclear weapons.
8. There are a number of restrictions on the export or import of nuclear technology, fuels etc.
9. Nuclear power stations are always at risk from terrorist attack.
10. Proliferation of nuclear technology increases the risk of nuclear war.
11. The waste produced remains 'active' over many years and disposing it safely is an issue which needs to be addressed properly.
12. Nuclear power is not a renewable source of energy.

EFFECTS OF RADIOACTIVITY

The unit used to measure radiation dosage is the ***rem***, which stands for ***roentgen equivalent in man.*** It represents the amount of radiation needed to produce a particular amount of damage to living tissue. The total dose of rems determines how much harm a person suffers.

Doses above 100 rems cause the first signs of radiation sickness including:

a) nausea
b) vomiting
c) headache
d) diarrhoea
e) fever
f) some loss of white blood cells
g) miscarriage in pregnant women

Doses of 300 rems or more cause

h) loss of hair

i) burns
j) damage to nerve cells and the cells that line the digestive tract
k) leukaemia (cancer of the blood)
l) lung cancer
m) thyroid cancer, breast cancer, and cancers of other organs can appear due to the radiation received.
n) severe loss of white blood cells, which are the body's main defence against infection; this makes radiation victims highly vulnerable to disease.
o) Radiation also reduces production of blood platelets, which aid blood clotting, so victims of radiation sickness are also vulnerable to haemorrhaging.

Half of all people exposed to 450 rems die, and doses of 800 rems or more are always fatal.

Protection from the effects of radiation

As of yet, there is no effective treatment for radiation exposure. It is hence ideal to protect oneself against it.

The following are some protective measures to undertake:

1. Wear protective clothing always.
2. Spend less time at the radiation area or device.
3. Have medical checks regularly.
4. Wear protective shields.
5. Assess radioactive materials through the use of robots and other remote-controlled devices.
6. Keep the body away from radioactive devices.

NUCLEAR WASTE DISPOSAL

After their usage mostly for energy production, nuclear wastes must be properly disposed of because of the following dangers and health risks posed to humans and the environment:

- ***Toxic*** - nuclear materials are poisonous or unhealthy if ingested, inhaled, or touched.
- ***Corrosive*** - nuclear materials cause significant damage to other materials that come in contact with them.
- ***Radioactive*** - they emits radiation which may cause harm to living tissue
- ***Explosive/Heat Risk*** - the materials may explode under certain circumstances, or is significantly exothermic (i.e. produce heat on their own)
- ***Chemically Reactive*** - the materials are prone to producing volatile, toxic, or other dangerous compounds when exposed to the normal environment.
- ***Militarily Valuable*** - nuclear materials have military value; that is, they are important ingredients in creating some weapons.

Problems associated with nuclear waste disposal

The major problem of nuclear waste is what to do with it. In fact, one of the biggest (and perhaps the single biggest) expenses of the nuclear power industry could eventually be the storage of nuclear waste. Currently there are several ways in which nuclear waste is stored. Most of these methods are temporary. In most cases a viable long-term solution for waste storage has yet to be found. This is because the time period for storage is so incredibly long, (on the order of thousands of years).

Proper nuclear waste disposal

1. Toxic materials must be either rendered chemically non-toxic, or isolated from exposure to the environment.
2. Corrosive materials must be rendered chemically inert, or stored in special containers resistant to the corrosive effects.
3. Radioactive materials must be stored in shielded containers until the amount of radiation emitted by the material is at a safe level.
4. Explosive and exothermic materials must either be chemically neutralized, or sealed in a non-reactive container and isolated from any outside contact.
5. Chemically reactive materials must either be chemically destroyed (i.e. turned into a compound which is not reactive), or stored in special chemically inert, sealed containers.
6. Militarily valuable materials must either be guarded from theft, or reformulated into materials that are impractical for making into a weapon.

Other effective methods of disposing of nuclear wastes include:

7. burying the wastes under the ocean floor,
8. storing them underground
9. shooting them into space

TEST QUESTIONS

1. (a) Explain the term radioactivity.
   (b) Name the three radiations produced by radioactivity.
   (c) Arrange the radiations named in (ii) in order of increasing penetration power.

2. (a) Differentiate between natural and artificial radioactivity.
   (b) Distinguish the characteristics of the radiations in terms of speed, penetration power and ionizing power.

3. (a) Explain the term radioisotope.
   (b) List six uses of radioisotopes.

4. (a) Describe four uses of nuclear energy.

(b) Mention four advantages and four disadvantages of nuclear energy.

5. (a) State five harmful effects of radioactivity.
   (b) List four ways of protecting oneself from radiation.

6. (a) State four dangers of nuclear waste.
   (b) Mention five ways of proper disposal of nuclear waste.

7. (a) Explain the term nuclear decay.
   (b) Copy and complete each of the following radioactive reactions by providing the missing mass and atomic numbers.
   (a) $^{234}_{90}Th \rightarrow {}_{91}Pa + {}_{-1}e$
   (b) ${}_{88}Th \rightarrow {}^{222}Pa + {}^{4}_{2}He$

8. Describe the uses of radioisotopes in the following fields:
   (a) food preservation;
   (b) medicine;
   (c) education;
   (d) industries;
   (e) agriculture.

9. Describe the effects of radioactivity in term of doses on humans.

10. How can nuclear energy be used for the following:
   (a) treatment of diseases;
   (b) detecting leaks in pipes;
   (c) sterilization;
   (d) electricity generation.

11. Explain the problems associated with nuclear waste disposal

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MAGNETISM

Specific Objectives

After completing this chapter, you will be able to:

- Distinguish between magnetic and non-magnetic materials.
- Explain magnetic field
- Outline the processes of magnetization and demagnetization

INTRODUCTION

The phenomenon of magnetism has been known since ancient times. The mineral lodestone, an oxide of iron that has the property of attracting iron objects, was known to the Greeks, Romans, and Chinese. When a piece of iron is stroked with lodestone, the iron itself acquires the same ability to attract other pieces of iron. The magnets thus produced are ***polarized***—that is, each has two sides or ends called north-seeking and south-seeking poles.

Magnetism or magnetic force is a force which tends to attract certain materials.

This force is produced by the motion of electrons. Materials which are responsive to magnetic forces are known as ***magnetic materials*** and those which are not affected by magnetic forces are referred to as ***non-magnetic materials.***

MAGNETIC AND NON-MAGNETIC MATERIALS

Magnetic materials

Magnetic materials are elements which can be attracted by a magnet.

Examples of magnetic materials include iron, nickel, cobalt steel and *alnico* (an alloy of aluminium, nickel and cobalt.

Ferromagnetic materials *are magnetic materials which can easily be magnetised* (i.e turned to magnets).

Examples include nickel, steel, cobalt, iron etc.

Testing for a magnet

To test whether an object is a magnet, bring it closer to a known of magnet. If it is repelled by one of the poles then that object is a magnet.

Non-magnetic materials

These are materials which are neither attracted nor repelled by a magnet.

Examples of non-magnetic materials are plastic, copper, gold, wood, fabric etc.

Law of magnetism

The law of magnetism says that the like poles of two magnets repel one another, and the unlike poles attract.

This law means that when the same pole (e.g. north-north) of two magnets are brought together, they push each other apart, whereas if different pole (north-south) are brought together, they attract each other.

TYPES OF MAGNETS AMD THEIR USES

Magnets are classified as temporary magnets and permanent magnets, based on the ferromagnetic material they are made from.

Temporary magnets

Temporary magnets are magnet from materials that are easy to magnetize but lose their magnetism easily..

These are magnets made are from soft irons, such as iron rod or iron nails. Temporary magnets have low ***retentivity*** and thus lose their magnetism easily.

Retentivity is the ability of a magnet to keep it magnetic property over time.

Permanent magnets

Permanent magnets are magnets from ferromagnetic materials are difficult to be magnetized, but once they are magnetized, retain their magnetism for a long time.

Permanent magnets have high retentivity. Steel and alnico can be magnetized this way.

Characteristics of magnets

1. Magnets have two poles – north and south poles
2. Like poles, such as North to North or South to South poles repel each other.
3. Unlike poles, such as North to South attract each other.
4. Magnets have magnetic lines of force around them.

Uses of magnets

1. Magnets are used in many electrical and electronic appliances, such as microphones, television sets etc.
2. They are used in industries to lift pieces of iron.
3. They are used in refrigerator and freezer doors.
4. Magnets are used in loud speaker.
5. They are used in magnetic compass to show directions.
6. The index of minimum and maximum thermometers is reset with a magnet.
7. They are used to remove pieces of magnetic materials swallowed accidentally or fallen into the eye.
8. Cobblers, shoemakers, seamstresses and tailors use magnets to pick and keep small nails, pins and needles.
9. Magnets are used to post information on metallic (iron or steel) notice board and refrigerators.
10. Cassettes, magnetic tapes (CDs and DVDs) and floppy disks are coated with a thin film of magnet.

Function of magnet in a telephone earpiece

A telephone set contains a receiver that amplifies sound from an incoming call. The receiver of a telephone set is made from a flat ring of magnetic material with a short cuff of the same material attached to the ring's outer rim. Underneath the magnetic ring and inside the magnetic cuff is a coil of wire through which electric current, representing the sounds from the distant telephone, flows. A thin diaphragm of magnetic material is suspended from the inside edges of the magnetic ring so it is positioned between the magnet and the coil.

The magnetic field created by the magnet changes with the current in the coil and makes the diaphragm vibrate. The vibrating diaphragm creates sound waves that replicate the sounds that were transformed into electricity by the other telephone.

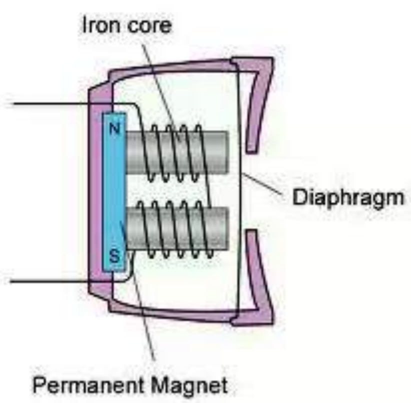


Fig. 100.9: *Telephone earpiece*

Function of magnet in a loud speaker

A loudspeaker is a device that converts an electrical signal into sound. A solenoid is placed inside a hollow, cylindrical permanent magnet and attached to a paper or plastic cone that is held steady at the open end by a metal, plastic, or wooden ring. When current flows through the solenoid, the interaction between the permanent magnet's magnetic field and the solenoid's changing magnetic field causes the solenoid to move, making the cone vibrate.

When the cone vibrates, it produces the original sound. The vibration of the cone is proportional to the current in the solenoid, which, in turn, represents the original sound converted into an electric signal by the microphone.

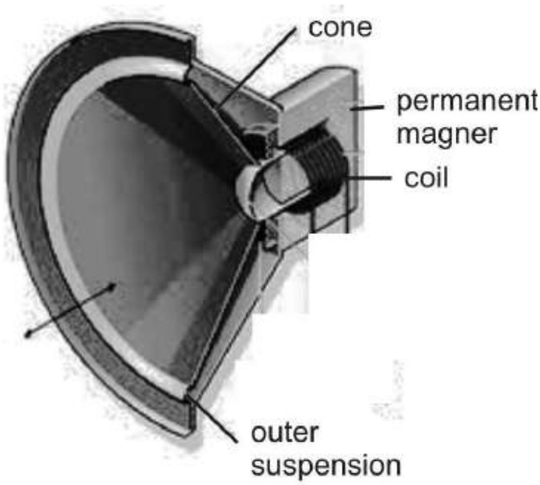


Fig. 101.1: Loud speaker

Function of magnet in a microphone

A microphone is a device used to transform sound energy into electrical energy. In microphones, a thin metallic ribbon is attached to a diaphragm and placed in a magnetic field. When sound waves strike the diaphragm and vibrate the ribbon, a small voltage is generated in the ribbon by electromagnetic induction.

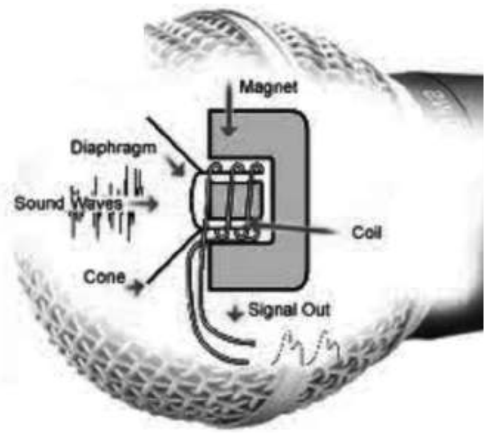


Fig. 101.2: Microphone

Function of magnet in magnetic compass

A compass is an instrument that indicates direction, used by mariners, pilots, campers, hunters, and other travellers to enable them to get from one place to another. A magnetic compass indicates direction by means of a needle pointing in the direction of the magnetic North Pole. It consists of a magnetized needle mounted on a pivot at the centre of a fixed graduated card so as to permit the needle to swing freely in the horizontal plane.

Fig. 101.3: Magnetic compass

Function of magnet in electric metre

Electric metres measure and indicate the magnitude of electrical values such as current, voltage, power etc. Metres use the force existing between a magnetic field and a pivoted, current-carrying coil within the field which causes an observable deflection of the coil. Because the deflection is proportional to the current, a calibrated scale is employed to measure the electric current.

Function of magnets in refrigerators

When opening the door of refrigerators and freezers, you will need to apply a little more effort to it than you will in opening an ordinary door of the same size. This is because refrigerator doors are lined with magnetic strips. These give refrigerators air-tight closure.

Function of magnet in electricity generation

When a conductor moves through the gap between the poles of a magnet, the negatively charged electrons in the conductor will experience a force along the length of the conductor and will accumulate at one end of it, leaving positively charged atomic nuclei, partially stripped of electrons, at the other end. This creates a potential difference, or voltage, between the ends of the conductor. If the ends of the wire are connected by a conductor, a current will flow around the circuit.

MAGNETIC FIELD

Magnetic field is the region or area around a magnet where magnetic effects can be experienced or felt.

Magnetic field is often represented by magnetic lines of force. The lines of force are imaginary lines used to show the size, strength and shape of the magnetic field.

In a magnetic field, electrons move from the North Pole to the South Pole. The lines of force do not cross or run over each other, and are closer and stronger at the pole of a magnet.

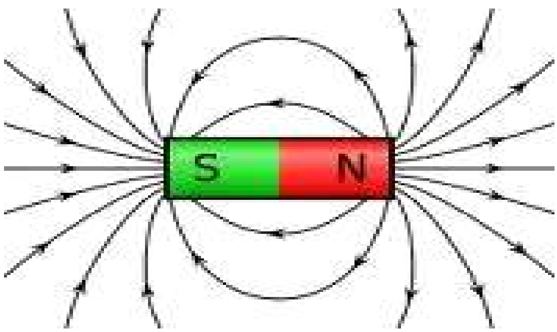


Fig. 101.4: *Magnetic lines of force*

Two adjacent magnets

If two magnets are placed beside each other, electrons will move from the North Pole of one magnet to the South Pole of the other.

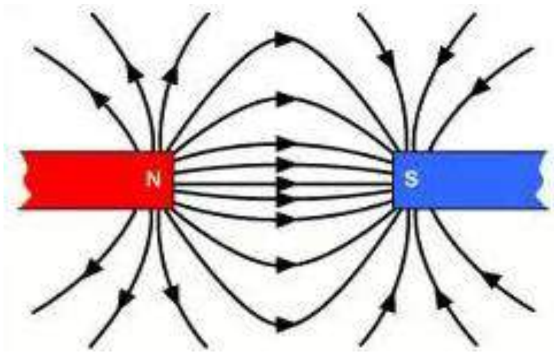
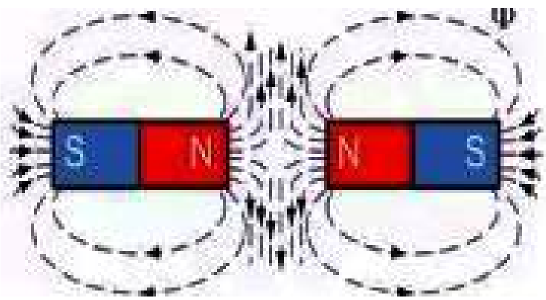


Fig. 101.5: Two magnets with unlike poles facing

If two bars of magnets are placed horizontally with the like poles facing each other, no lines of force pass through it resulting in a zero point. This point is known as the ***neutral point***.



Two like poles together repel

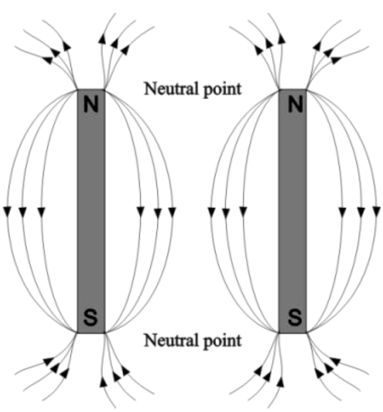


Fig. 101.6: Neutral point

Demonstrating the effect of magnetic field

- ❖ Place a bar of magnet on a horizontal surface.
- ❖ Sprinkle some iron filings on it.
- ❖ Note the pattern formed by the iron filings around the magnet.

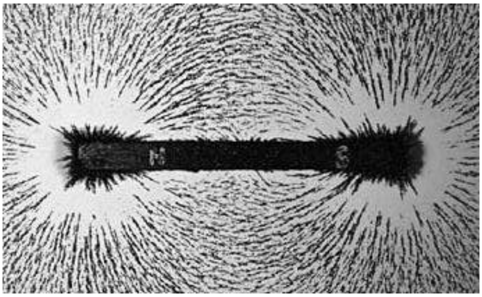


Fig. 101.7: Iron filings around a bar of magnet

The earth's magnetic field

The Earth's magnetic field is believed to be generated by charged particles that circulate in the Earth's liquid outer core, along with forces caused by its rotation. An electric current is generated, which in turn creates a magnetic field.

The magnetic field protects the Earth by deflecting high-energy particles from the Sun. Without it, life on Earth would be impossible.
The earth's magnetic field is tilted at angle 11^{o} to the spin axis.

Unlike a bar of magnet, the Earth's North Pole is actually the South Pole of a magnet. This is why the compass needle always points towards the north but not the south.

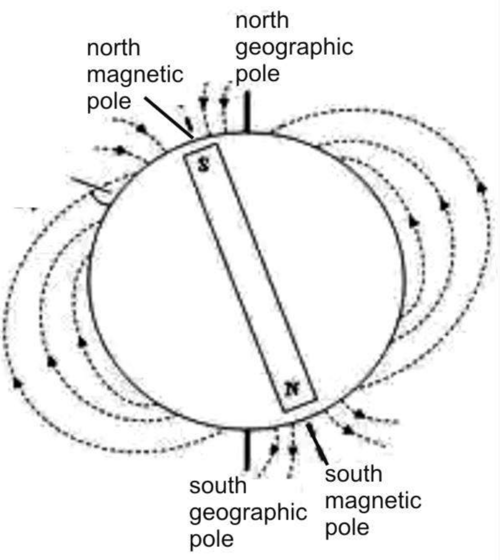


Fig. 101.8: The earth as a magnet

MAGNETIZATION AND DEMAGNETIZATION

Magnetization

Magnetization is the process of turning a magnetic material into a magnet.

There are various ways of magnetizing a material based on the characteristics of the ferromagnetic material and the type of magnet expected. Some methods of magnetization are:

The electrical method

- ❖ Wound a piece of insulated wire around the magnetic material (such as an iron nail).
- ❖ Connect the two opposite ends of the wire to a power source (such as a battery).
- ❖ Close the circuit for a short time and then open it.
- ❖ Bring the magnetized material close to other magnetic materials.

Observation

It would be noted that the magnetized material attracts other magnetic materials. Bring it close to a known magnet whiles turning it around. It will repel or push away the magnet in one of the poles.

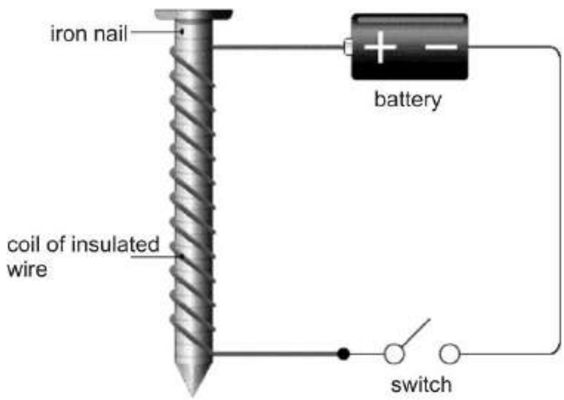


Fig. 101.9: Electrical method

The stroke method

There are two methods of magnetizing a material using the stroke method – the single stroke or touch and the double stroke or touch.

The single stroke method

- ❖ Place the material to be magnetized (e.g. nail), on a horizontal surface.
- ❖ Stroke the North Pole of a known bar of magnet continuously in the same direction on the magnetic material.
- ❖ Turn the North Pole of the magnet closer to the end of the nail where the stroking started. The nail would be repelled.
- ❖ Bring the magnet to the opposite end of the nail; the nail gets attracted strongly.

Conclusion

This shows that the end where the stroking started is the North Pole while the other end is the South Pole.

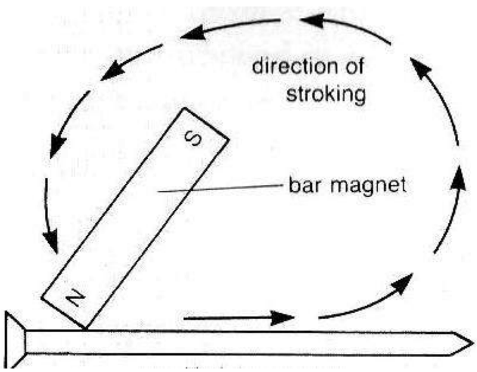


Fig. 102.0: Magnetization by the stroke method

The double stroke method

- ❖ Put the magnetic material on a horizontal surface.
- ❖ Stroke continuously from the middle to one end with the North Pole of a magnet.
- ❖ Repeat the procedure with the South Pole of the magnet to the other end of the magnetic material.

Conclusion

This would result in the formation of consequent poles in the middle of the magnetic material as well as the ends.

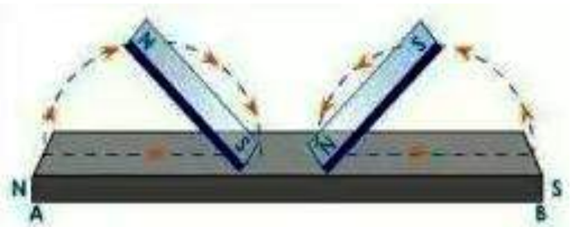


Fig. 102.1: Double stroke magnetization

Demagnetization

Demagnetization is the process by which a magnet loses it magnetic properties.

In other words, the magnet will not be able to attract magnetic materials nor repel other known magnets.

Methods to demagnetize a magnet

The electrical method

- ❖ Wound a magnet in an insulated wire.
- ❖ Connect the ends of the wire to a power source.
- ❖ Close and open the circuit.

Heating

- Heat the magnet to red hot
- Cool it quickly

Hammering

- Put the magnet on a hard surface.
- Hammer it repeatedly for sometime

Dropping

- Drop or throw the magnet several times on the floor.

ELECTROMAGNETISM

When electric and magnetic forces are combined, a new phenomenon is created known as electromagnetism.

Electromagnetism is produced when electric current moves through a magnetic field.

Magnets produced through electromagnetism, are very essential in the functioning of many electrical and electronic devices.

Uses of electromagnets

1. Electromagnets are used in relays and circuit breakers.
2. Loudspeakers used the principles of electromagnets.
3. They are used in telephone receivers
4. They are used in electric bells.
5. Electromagnetic clutches and breaks in automobiles functions with it.
6. Electric motors work on the principle of electromagnetism.
7. They are used in generators.
8. They are used in metal detectors.
9. Electromagnets are used in television sets.

TEST QUESTIONS

1. (a) Define the term magnetism.
   (b) Differentiate between the following magnetic and non magnetic materials and give three examples each.

2. Explain the following types of magnets:
   (a) temporary magnets;
   (b) permanent magnets.

3. (a) Mention three characteristics of magnets.
   (b) List five uses of magnets.

4. Describe the function of magnets in the following devices:
   (a) telephone earpiece;
   (b) loudspeaker;
   (c) microphone;
   (d) refrigerator door.

5. (a) What is magnetic field?
   (b) Draw and label a diagram of a bar of magnet showing the lines of force around it.

(a) Define the term neutral point.

(b) Two bars of magnets are placed with the same poles facing each other. Draw the lines of forces around them and indicate the neutral point.

6. Describe the earth as a magnet.

7. (a) Define the term magnetization.

(b) Describe two methods of magnetizing an iron nail.

8. (a) What is demagnetization?

(b) Mention four ways of demagnetizing a magnet.

9. (a) Define electromagnetism.

(b) State five uses of electromagnets.

37

FORCE, MOTION AND PRESSURE

Specific Objectives

After completing this chapter, you will be able to:

- Define force and outline various types.
- State the Archimedes Principle and the law of flotation.
- Explain the terms, distance, displacement, speed, velocity, acceleration and momentum.
- Explain types of motion and the Newton's laws of motion.
- Define centre of gravity and distinguish between stable, unstable and neutral equilibrium.
- Explain moment of a force.
- Define pressure and describe the effects and application of pressure in solids, liquids and gases.

INTRODUCTION

Have you tried to change the position or shape of an object? If you have then you probably pushed or pulled it. When you apply a push or pull on an object, it causes the object to move (sets it in motion), change shape (deforms) or change speed (accelerate or decelerate). This action of pulling or pushing the object is referred to as a ***force***.

FORCE

A force is a push or pull on an object which causes it to change position or shape or speed.

Or

A force is an influence that changes the velocity of or deforms an object.

Force is the product of ***mass*** and ***acceleration due to gravity*** (the earth's pull on an object).

Force = mass × acceleration due to gravity

OR

$$F = mg$$

The unit of force is ***newton (N).***

Force is a ***vector quantity***, which means that it has both ***direction*** and ***magnitude***.

When several forces act on an object, the forces can be combined to give a *net force* which is equal to zero. In this case the object remains either motionless or moves at a constant velocity.

Categories of forces

There are different types of forces, and they can be collectively classified as contact and non-contact forces.

Contact force

These are the forces between two or more objects which are in contact with each other.

Examples are frictional force, viscosity, tensional force etc.

Non-contact or field force

These are the forces which are between objects which are not in contact or touching.

Examples are gravitational force, magnetic force, electrostatic force etc.

Types of forces

Frictional force

This is the force which opposes relative motion between objects or surfaces which are in contact.

To investigate the effect of friction, take a brick and slide it on a surface. The brick moves a short distance and then stops. This is because the surface applies a force on it which slows it down until it finally comes to a halt.

The friction between stationer bodies is called ***static friction***, whiles the one between moving bodies is referred to as ***dynamic friction***.

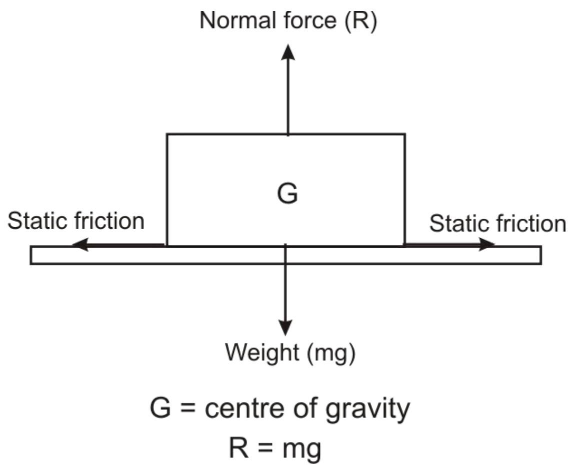


Fig. 102.3: A stationary block with a number of forces acting on it

Coefficient of friction

To determine the amount of friction between two bodies, we find the coefficient of friction (μ), given mathematically as:

$$\textbf{\textit{Coefficient of friction}}\ (\boldsymbol{\mu}) = \frac{\boldsymbol{frictional\ force\ (F)}}{\boldsymbol{normal\ force\ (R)}}$$

Examples

1. A mug of weight 20 N rests on a table. If a static frictional force of 30 N acts on it, find it coefficient of friction.

Solution

Coefficient of friction (μ) = $\frac{F}{R}$

Frictional force (F) = 30 N

Normal force (R) = W = 20 N

$\therefore \mu = \frac{30}{20} = 1.5$

2. An object of mass 50 kg is placed on a rough horizontal floor. If the object is moved by the application of a horizontal force of 320 N, calculate the
   (a) Force exerted by the floor on the object
   (b) Coefficient of friction
   [take $g = 10\ ms^{-2}$]

Solution

(a) Normal force (R) = weight (W)
$W = mg = 50 \times 10 = 500$ N
$\therefore R = 500$ N

(b) Coefficient of friction (μ) = $\frac{F}{R}$

$$\mu = \frac{320}{500}$$

Normal force

This is an upward force which is exerted on an object by the surface it rests on.

The normal force is equal the weight of the object.

Magnetic force

This is the force possessed by some objects (magnets) which tends to attract other objects (magnetic materials).

As we discussed earlier, magnets have two poles – north and south poles. Like poles repel each other and unlike poles attract each other.

When electricity is introduced in the magnetic field, an ***electromagnetic force*** is produced.

Gravitational force (force of gravity)

This is the force that pulls all objects towards the surface of the earth.

Take a piece of stone and throw it vertically upwards. What happens? The stone moves at a height up and then falls back down. The stone behaves that way because the earth exerts a force on it which pulls it towards the surface of the earth. The force of gravity (g) on earth has a value of 9.8 ms^{-1}; approximately 10 m s^{-1}.

Weight

Weight is the effect of gravity on a body.

When the body rests on a surface, the force of gravity and the mass of the body gives it weight.

Weight = mass x acceleration due to gravity

Or

$$W = mg$$

Viscosity

Viscosity is the force which prevents motion in fluids.

It is also referred to as fluid friction. If you move an object through a thick liquid, the object moves at a slower rate as opposed to the object moving in the air. This is because thick fluids are more viscous, while thin fluids are less viscous.

Electrostatic force

This is the force which exists between two bodies with opposing charges.

Electrostatic force can be demonstrated by rubbing a plastic comb in the hair (which gives it a negative charge) and using it to attract pieces of positively charged paper. Another example of electrostatic force is lightening, which occurs as a result of the attraction between the negatively charged crowds and the positively charged earth.

Tensional force

This is the force that exists in a stretched material.

Materials such as springs and ropes become stretched when their ends are pulled in the opposite directions; this is because of the tensional force existing in them.

Centripetal force

This is the force that pulls a rotating object towards the centre or axis.

For example, supposing a ball is tied to a string and swung around in a circle at a constant velocity. The ball moves in a circular path because the string applies a centripetal force to the ball.

Centrifugal force

This is the force which keeps and object in a circular path pulls it away from the centre of rotation.

Take for instance the earth, which revolves round the sun. As it moves round, Gravitational force from the sun attracts it. However, the reason why the earth does not crush into the sun is because of centrifugal force which pulls the earth from the sun (centre of rotation).

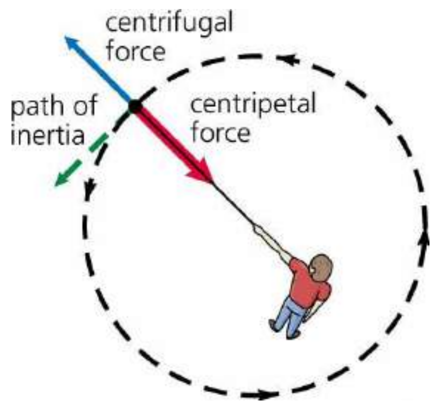


Fig. 102.4: Centripetal and centrifugal force

Upthrust

Upthrust is the upward force that prevents a floating body from sinking to the bottom of a fluid.

Take an inflated ball and try forcing it under water. The ball forces it way upwards no matter how much force you apply to it. This is because the water applies a force which keeps the ball from sinking into it.

ARCHIMEDES' PRINCIPLE AND THE LAW OF FLOATATION

The Archimedes' principle states that a body immersed in a fluid displaces an amount of the fluid which is equal to its weight.

The Archimedes' principle explains what happens to an object which is totally or partially immersed in a fluid. The body experiences an upthrust which pushes it out of the fluid.

A body in a fluid has two main forces acting on it – its weight and upthrust. If the upthrust is greater than or equal to the weight of the body, it will float. On the other hand, if the weight of the body is greater than the upthrust, it will sink either fully or partially based on the amount of fluid it displaces.

An experiment to verify Archimedes' Principle

- Tie a thread to a solid object such as a piece of stone.
- Wrap the thread around the hook of a spring balance, and hence determine the weight of the stone in air as W_1.
- Fill a Eureka can with water and allow it to stop dripping from the spout.
- Place a beaker of known weight W_2 under the spout of the Eureka can.
- Gently lower the solid object hanging on the spring balance into the water in the Eureka can.
- When the object is fully submerged, record the reading on the spring balance as W_3.
- Allow the water dripping into the beaker from the spout of the Eureka can to stop.
- Record the weight of the beaker and the water as W_4.

- ***Calculations***
- Weight of object in air = W_1
- Weight of object in water = W_3
- Loss of weight of object = $W_1 - W_3$
- Weight of empty beaker = W_2
- Weight of beaker and water = W_4
- Weight of displaced water = $W_4 - W_2$

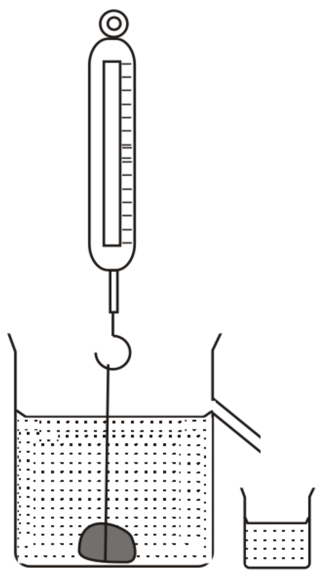
Observation

It would be observed that the loss in weight of object $W_1 - W_3$ equals the weight of displaced water = $W_4 - W_2$

$W_1 - W_3 = W_4 - W_2$ = upthrust of the (object)

Conclusion

This concludes that the amount of fluid displaced by an object equals its weight. This is known as the law of flotation.



Stone suspended on spring balance

Examples

A solid weighs 250 N in air and 170 N when fully immersed in water. Calculate the

(a) weight loss of the solid;
(b) upthrust of the solid;
(c) volume of the solid;
(d) mass of the solid;
(e) density of the solid.
(f) Why is the weight of solid in air greater than that in water?
[take g = 10 m s^{-2}]

Solution

(a) Weight of solid in air W_1 = 250 N
Weight of solid water W_2 = 170 N
Weight loss of solid = $W_1 - W_2$
= 250 – 170 = 80 N

(b) Upthrust = weight loss = 80 N

(c) Volume of the solid = weight of displaced water = volume of displaced water = 80cm^3

(d) Mass = $\frac{weig\,ht\ of\ solid\ in\ air}{g} = \frac{250}{10}$ = 25g

(e) Density = $\frac{mass}{vlume} = \frac{25}{80}$ = 0.3125gcm^{-3}

(f) This is because a part of the weight of the solid is used to overcome upthrust exerted by the water.

The law of flotation

The law of flotation states that a floating body displaces its own weight of the fluid it floats in.

The higher the density of the body, the further it sinks. But, the higher the volume, the higher it floats. Remember, density is equal to the ratio of mass and volume. Therefore, by increasing the volume of the body, its density lessens, though its mass remains the same.

Applications of Archimedes' principle and the law of flotation

The flight of birds

Birds have streamline bodies, which allow them to move through the air with little air resistance. In flight, birds extend their wings; this increases their volumes increasing their upthrusts or the upward force of the air under their wings, making them less dense than the air around them; thus being able to float.

The flight of aeroplanes

The principle underlying the design of aeroplanes was discovered from birds. Aeroplanes have wings so in flight air builds up under the wings giving them the lift upwards. They are also hollow; this increases their volumes, thereby making them less dense than the surrounding air. This causes them to afloat in the air. The heavier the aeroplane, the higher it should be so as to allow more air pressure under the wings to increase it volume and reduce its density.

Fig. 102.5: Wing of an aeroplane

Boats in water

Boats float in water because their hollow nature increases their volumes, giving them less density than that of the water. This reduces the weight of the boat causing it to displace some water which is equal to its weight, which is submerged in the water, making it able to float.

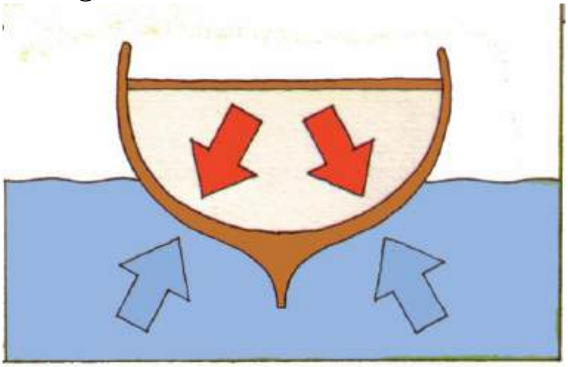


Fig. 102.6: A floating boat

Flotation of ships

The principle governing the flotation of ships is similar to that of boats. They are hollow, thus, high volume, less density or weight. Ships normally have Plimsoll line which indicates the safe-depth to which they should sinks. If they are supposed to sink further, they are given more loads to increase their weight or density and vice versa.

Operation of submarine

Submarines have the ability to submerge in water and also rise up to the surface. For a submarine to sink, water is pumped into its ballast tanks, increasing its average density. For it to afloat, air is pumped into the ballast tanks to displace the water and decreasing its average density, hence floating.

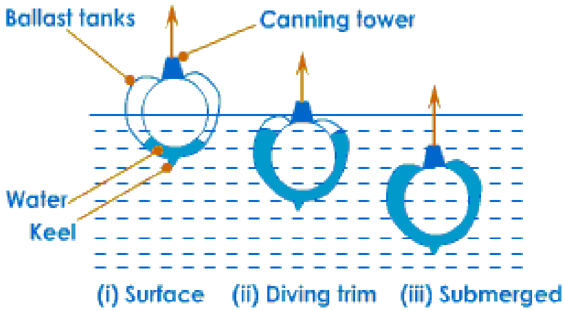


Fig. 102.7: Flotation of submarine

GROUP ACTIVITY:

1. Observe and record what happens when different metallic and wooden objects are placed in water.
2. Place a fresh egg first in water and then in concentrated salt solution and explain the observation in each case.

TERMS TO NOTE

Distance – ***This is the interval between any two points.***
It has only magnitude, making it a scalar quantity. It is measured in metres (m).

Distance travelled = speed × time taken

Displacement – This is the distance covered in a specific direction. Displacement is a vector quantity because it has both magnitude and direction. It is also measured in metres (m).

Displacement = velocity × time

Speed – ***Speed is the rate of change of distance with time.***
In other words, speed is the distance covered in a specific time. It is a scalar quantity and is measured in meter per second (ms^{-1}).

$$\textit{Speed} = \frac{\boldsymbol{distance\ (m)}}{\boldsymbol{time\ (s)}}$$

Velocity – ***It is the rate of change of displacement with time.***
Velocity is a vector quantity and is measure in ms^{-1}.

$$\textit{Velocity} = \frac{\boldsymbol{displacement\ (m)}}{\boldsymbol{time\ (s)}}$$

Average velocity: This is the total displacement covered with time.

$$\textit{Average velocity} = \frac{\boldsymbol{total\ displacement}}{\boldsymbol{time}}$$

For a body with constant acceleration,

$$\textit{Average velocity} = \frac{\boldsymbol{sum\ of\ velocities}}{\boldsymbol{2}} = \frac{\boldsymbol{v + u}}{\boldsymbol{2}}$$

where v = final velocity
u = initial velocity

Constant or uniform velocity: This is the rate of change of displacement with constant time.

Instantaneous velocity: It is the velocity at any given time.

Acceleration – ***This is the change in velocity with time.***
Acceleration is a vector quantity with the SI unit ms^{-2}. In a situation where a uniformly body changes its velocity with respect to time, the velocity with which it starts moving is the initial velocity (u), whiles the velocity with which it ends is its final velocity (v).
Mathematically,

$$\textit{Acceleration} = \frac{\boldsymbol{change\ in\ velocity}}{\boldsymbol{time\ taken}}$$

or

$$\textit{Acceleration} = \frac{\boldsymbol{final\ velocity - initial\ velocity}}{\boldsymbol{time\ taken}}$$

or $$a = \frac{\boldsymbol{v - u}}{\boldsymbol{t}}$$

Constant or uniform acceleration: This is the rate of change of velocity with constant time. An object moving with constant velocity has no acceleration.

Deceleration or retardation: It is the decrease in velocity with time. Deceleration is always negative.

$$\textit{Deceleration} = \frac{\textit{change in velocity}}{\textit{time}}$$

Constant or uniform deceleration: This is the decrease in velocity with constant time

Examples

1. A car moves with a velocity of 19 ms^{-1} and increases to 25m^{-1} in 5 seconds. Determine it acceleration.

Solution

$$\text{Acceleration} = \frac{\textit{change in velocity}}{\textit{time taken}}$$

$$= \frac{25 - 19}{5} \qquad = \frac{6}{5} \qquad = 1.2 \text{ ms}^{-2}$$

2. An object of mass 4 kg moving with initial velocity of 20 ms^{-1} accelerates for 10 s and attains a final velocity of 60 ms^{-1}. Calculate the acceleration.

Solution

$$\text{Acceleration } (a) = \frac{\boldsymbol{v - u}}{\boldsymbol{t}}$$

$$a = \frac{60 - 20}{10} \qquad = \frac{40}{10} \qquad = 4 \text{ ms}^{-1}$$

GRAPHS OF MOTION

The motion a body undergoes can be represented on a graph and calculated. this is referred to as graph of motion. The two principal graphs of motion are:

- displacement–time graph
- velocity-time graph

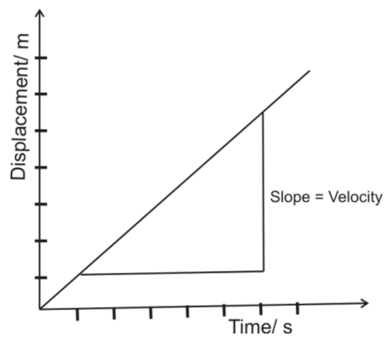
Displacement-time graph

In this graph, the displacement (or distance) travelled by a body is plotted against time. From this graph, the velocity at anytime can be deduced from the slope of the graph at that time.

Features of displacement-time graph

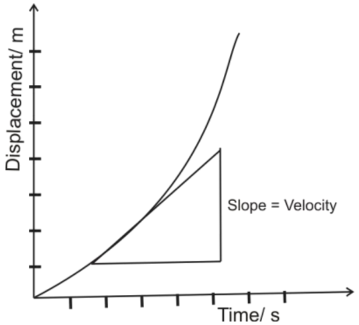
Uniform motion

If the motion is uniform, the slope is constant throughout the motion and the graph is a straight line. Velocity is the slope or gradient of the line.



Increasing velocity

When the slope of the line increases with time to give a curved line indicate that the body is increasing its velocity, thus accelerating.

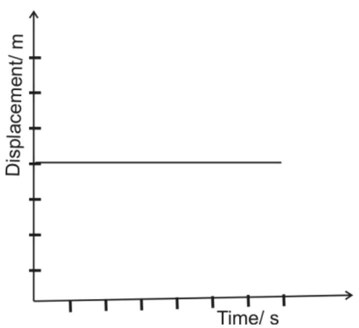


Uniform deceleration

When the graph is curved with decreasing slope, the body is slowing down or breaking. This is called uniform retardation or deceleration.

Uniform velocity

When equal distances are covered in equal time, the velocity is uniform and acceleration is zero.



Examples

1. The data below represent the distance moved by particle with time recorded at 10 seconds interval.

Distance/m	0	20	40	60	80	100	120	140
Time/s	0	10	20	30	40	50	60	70

(a) Use the above data to plot a graph of distance against time.
(b) Use your graph to calculate the slope of the particle.
(c) What does the slope represent?

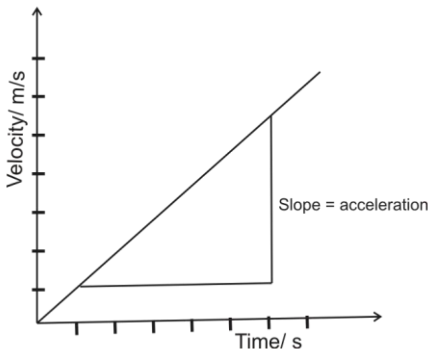
Velocity-time graph

In this graph, velocity is plotted against time. The slope of the graph at anytime represents the acceleration of that time.

Features of velocity-time graph

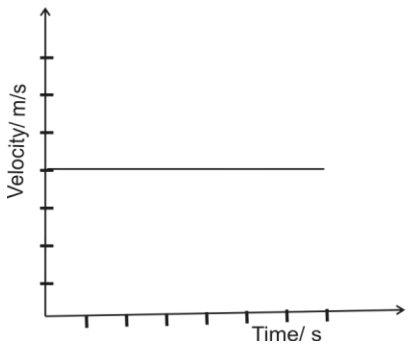
Constant or uniform acceleration

Constant acceleration is indicated by a straight diagonal line. This shows that the body is moving faster from its current velocity.



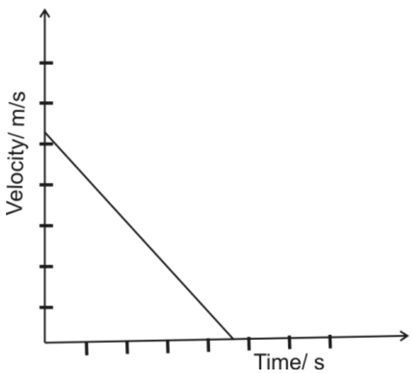
Uniform or constant velocity

This is shown with a straight line parallel to the time line (horizontal axis. Uniform velocity shows that the body maintains its velocity in the given time. This means that there is no acceleration.



Constant deceleration or retardation

This is shown with downward diagonal line. It indicates that the body is slowing down or stopping. If the body decelerates to a stop, the diagonal line touches the time line, indicating zero velocity.

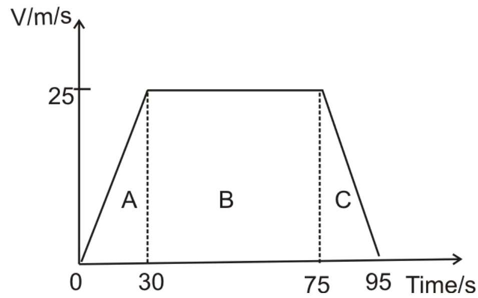


Examples

1. A car starts from rest and move with a velocity of 25 ms^{-1} in 30 seconds. It maintains it speed for 45 seconds after which it slowed to a stop in 20 seconds.
   (a) Sketch the velocity-time graph to represent the motion of the car.
   (b) Calculate the total distance is covers;
   (c) What is the acceleration of the car?

Solution

(a)



(b) Total distance (s) covered = area of trapezium

$s = \frac{1}{2}(l_1 + l_2)h$

$l_1 = 45$; $l_2 = 95$; $h = 25$

$\therefore s = \frac{1}{2}(45 + 95) \times 25$

s = 1210 m

2. A bus starts from rest and accelerates uniformly at 2 ms^{-2} for 10 s. It maintains the maximum speed attained for further 10 s and decelerates at 1 ms^{-2} gradually to rest.
   (a) Draw the velocity- time graph of the motion of the bus
   (b) Use your graph to determine
      (i) the maximum velocity attained
      (ii) the time taken
      (iii) the total distance covered
      (iv) the average velocity.

Momentum – The momentum of a body in motion is the product of its mass and the velocity with which it moves. This is represented as:

***Momentum* (*p*) = mass (*m*) × velocity (*v*)**

Momentum is a vector quantity and is measured in newton-second (Ns) or kilogram-metre per second ($kgms^{-1}$).

Example

Calculate the momentum of a ball of 2kg rolling at 3m\s.

Solution

Momentum = mass × velocity

$p = 2 \times 3 \quad = 6$ kg-m/s

MOTION

Whenever a body gains velocity, it changes position or direction. In this case the body is said to be set in motion.

Motion is the change in position or direction of a body.

Types of motion

Linear/ rectilinear/ translational motion

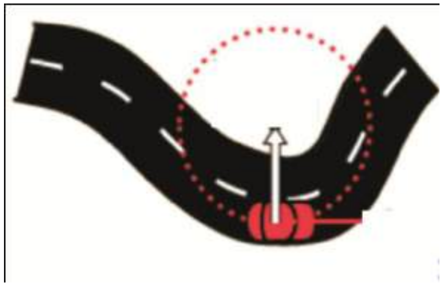
When a body moves along a straight line or path, it is said to have undergone linear motion. Examples include:

a) Motion of a toy car pushed along a floor.
b) Motion of a train on straight tracks.
c) Motion under gravity.

Circular motion

The movement of a body in a circle is termed as circular motion. Examples of circular motion include:

a) The motion of the moon around the earth
b) A motorist negotiating a curve.
c) Revolution of the Earth around the moon



Rotational/ spin motion

When a body moves around on its axis, it is termed as rotational motion. Examples are

a) The spin of a car tyre.
b) The rotation of a fan.
c) The rotation of the Earth.

Oscillatory motion

When a body moves but returns to a specific periodic point continuously, it is said to have undergone oscillatory motion. Examples of oscillatory motion are:

a) The motion of weight hanged on an elastic string.
b) Motion of water in a displaced container.
c) Motion of loaded test tube on water.

Random motion

Random motion is the motion of a body or bodies in all directions. This type of

motion does not follow any pattern, plan, system or connection. Examples include:
a) Motion of dust particles.
b) Motion of the molecules of gas.
c) Movement of scattered balls.

NEWTON'S LAWS OF MOTION

Sir Isaac Newton, an 18th century English scientist proposed three laws to describe the motion of bodies.

Newton's first law of motion

It states that an object at rest will stay at rest, and an object in motion will remain in motion at constant velocity, unless acted upon by an unbalanced force.

In other words, a stationary object will remain stationary until a force is applied on it. An object in motion will continue to travel at constant velocity until a force is applied to it.
This means that if an object is standing still and is not contacted by any forces, it will continue to remain motionless. It also means that if an object is moving along, untouched by a force of any kind, it will continue to move along in a perfectly straight line at a constant speed. This is referred to as ***the law of inertia.***

Inertia

Inertia is the reluctance of a body to change its velocity.

Effects of inertia

Inertia is felt when one is in a car or on a bus or train. The next time you stand on a bus, observe your actions. When the bus stops suddenly, you lurch forward.
This is because when the bus was travelling at the previous speed, say 70 ms^{-1}, your body was also travelling at the same speed. However, as soon as the bus stops at 0 ms^{-1}, since your body is not attached to the bus, it keeps moving at 70 ms^{-1}. Passengers on buses and trains have to grab hold of something firm in order to prevent themselves from falling.

The opposite effect is felt when the bus moves suddenly. The passengers will be thrown backwards, since the bus will gain speed at a short notice while the passengers' speed remain at 0 ms^{-1}.

Similarly, when the bus negotiates a curve the bodies of the passengers will move to the direction where the bus was going previously.

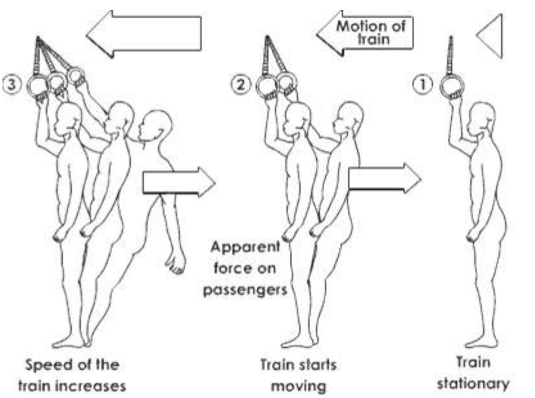


Fig. 102.9: Effects of inertia

The need for seatbelts

When a car is brought to a sudden stop, for example, during a collision, the effect of inertia is severe. This could cause passengers to hit their heads against something right in front of them, or in worst cases, be thrown out through the windscreen of a car. This could be very fatal.

To prevent this, vehicles such as cars, trains, buses and aeroplanes are fitted with seatbelts.

Seatbelts keep passengers in their seats so that in the event of a sudden stoppage, the body would be restrained from moving further forward, hence saving the passenger from injuries.

Newton's second law of motion

This law states that the rate of change of momentum of a body is directly proportional to the force acting on it and occurs in the direction of the force.

The second law of motion is also known to as the ***law of momentum***.

This can be represented as

$$Force = \frac{change\ \ in\ momentum}{time}$$

Remember that

Momentum **(p) =** ***mass*** **(*m*) ×** ***velocity*** **(*v*)**

$\therefore$change in momentum

$\boldsymbol{\Delta p = m(v - u)}$

Hence, $\boldsymbol{Force = \frac{m(v-u)}{t}}$

But, $\boldsymbol{\frac{v-u}{t}}$ = ***acceleration (a)***

$\therefore Force = mass \times acceleration$

Or

$\boldsymbol{F = ma}$

For instance, Kofi's car, which has a mass of 1,000 kg, is out of fuel. Kofi is trying to push the car to a petrol station, and he makes the car go 0.05 m/s^2. Using Newton's Second Law, you can compute how much force Kofi is applying to the car.

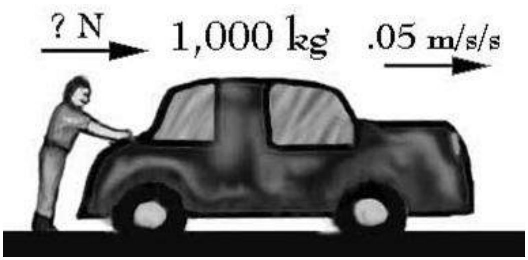


Fig. 103.0:

Force = mass* x *acceleration

F = 1000 X 0.05 = 50 N

Example

1. A force of 80 N is applied on a body to cause it to rise to a certain height. Calculate the mass of the body. [Take g = 10 m s^{-2}]

Solution

F = 80 N, a = g = 10 m s^{-2}, m = ?

$F = ma$

$m = \frac{F}{a} = \frac{80}{10} = 8N$

2. A certain particle of mass 250 g moved with a uniform acceleration of 4 m s^{-2}. Find the force with which it moves.

Solution

$F = ma$

$m = 250g = \frac{250}{1000} = 0.25kg$

$a = 4ms^{-2}$

$\therefore F = 0.25 \times 4 = 1.0N$

Newton's third law of motion

Newton's third law of motion states that to every action there is an equal and opposite reaction.

This means that whenever a body exerts a force F on another body, the second body exerts a force −F on the first body. F and −F are equal in magnitude and opposite in direction. For example, if you push on a wall, it will push back on you as hard as you are pushing on it.
This law is sometimes referred to as the ***action-reaction law***, with F called the "action" and −F the "reaction". The action and the reaction are simultaneous.

Application of the third law

- ***Rocket design:*** The third law of motion was applied in designing rockets. The rocket's action is to push down on the ground with the force of its powerful engines, and the reaction is that the ground pushes the rocket upwards with an equal force.

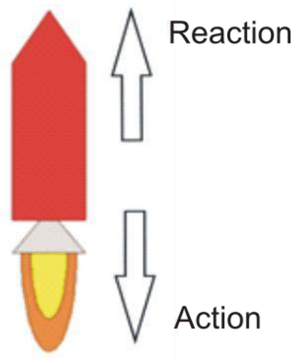


Fig. 103.1: A rocket

- ***Flight of birds:*** A bird flies by using its wings. The wings of a bird push air downwards. In turn, the air reacts by pushing the bird upwards.
The size of the force on the air equals the size of the force on the bird; the direction of the force on the air (downwards) is opposite the direction of the force on the bird (upwards).
Action-reaction force pairs make it possible for birds to fly.

CENTRE OF GRAVITY AND STABILITY OF OBJECTS

The centre of gravity of an object is the point in the object where its entire weight appears to act.

The centre of gravity is the point where if a pivot is located can create a turning effect on a body.

Determining the position of the centre of gravity of regular lamina bodies

A lamina body is a shape cut out from a flat sheet. To determine the centre of gravity of a regular shaped lamina (e.g. square or triangle), draw plumb lines across its centre from different points on its edges. The point where the plumb lines cross each other is the centre of gravity of the lamina. Examples of regular laminas with their centres of gravity are shown below.

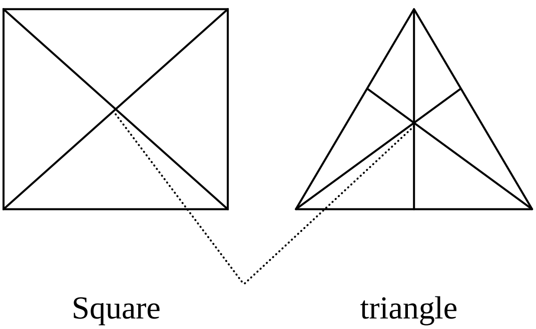


Fig. 103.2: Centre of gravity

The centre of gravity of irregular-shaped lamina

- Cut out an irregular shape from a cardboard.
- Punch three holes near its edges.
- Hung the shape on a support through one of the holes.
- Suspend a plumb line from the hole holding the body and mark the line of the string across the cardboard.
- Hung the shape by the other holes at the edges and repeat the last step.
- The lines will cross each other at a particular point on the card. That point is the centre of gravity of the body or shape.

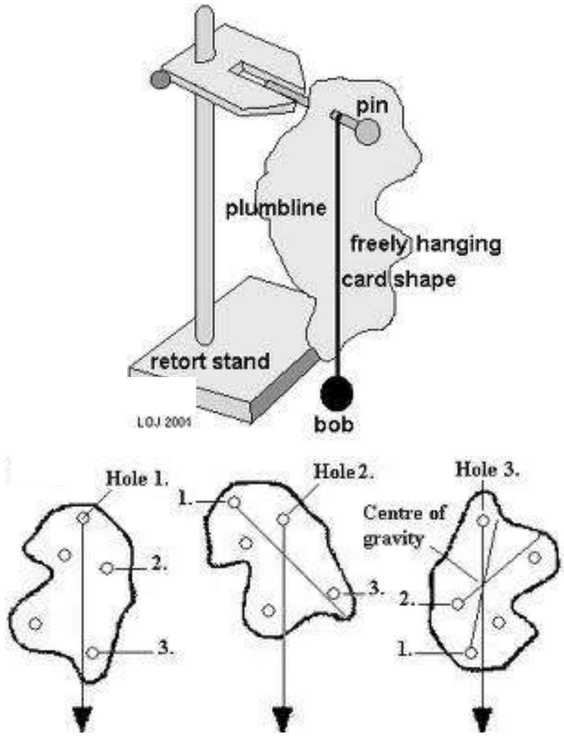


Fig. 103.3: Plumbline method

STABILITY OF OBJECTS

States of equilibrium

A body is said to be in equilibrium when it returns to its usual position after a nudge or slight displacement.

States of equilibrium

There are three states of equilibrium – stable, unstable and neutral equilibrium.

Stable equilibrium

A body in a stable equilibrium returns to its usual or former position after a slight tilt or displacement.

Such body has its centre of gravity close to its base.

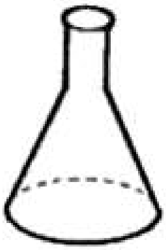


Fig. 103.4: Stable equilibrium

Unstable equilibrium

A body is said to be in an unstable equilibrium if it falls over and does not return to its former or usual position after a slight displacement.

This body has its centre of gravity above the base.

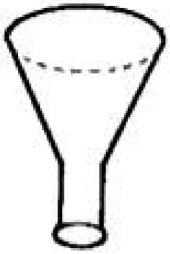


Fig. 103.5: Unstable equilibrium

Neutral equilibrium

A body in neutral equilibrium assumes a new position after slightly displaced. The centre of gravity is neither raised nor lowered and remains above the base or point of support.

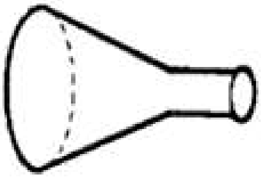


Fig. 103.6: Neutral equilibrium

Stability of objects

The position of the centre of gravity of a body determines its stability. Stable bodies standing on a support have their centre of gravity at their bases.

To determine the strength of the centre of gravity of a body, draw a vertical line from the centre of gravity to the base. If the two halves are the same, then the object is stable. That is the body will return to its former position when released. On the other hand, if the vertical hand does not bisect the body evenly, then its stability is weak and will fall over when nudged.

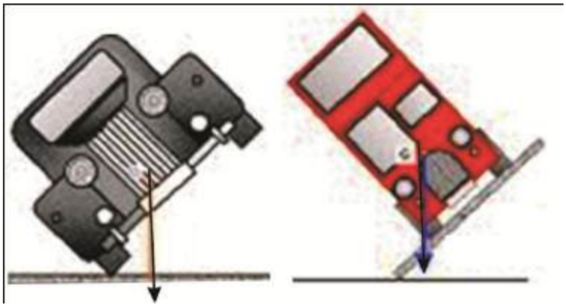


Fig. 103.7: Comparing the stability of a double-decker bus and a car

Stability of an acrobat on rope

The stability of an acrobat on a stretched rope is less. This means that the acrobat could easily fall over. To improve his stability, an acrobat on a stretched rope carries a long pole or stick which he balances in his hands. That is the central point of the pole must be on the rope. This lowers his centre of gravity. If he feels heavy at one point, and is about to fall, he shifts the pole to the opposite point so that the weight would be distributed evenly, maintaining his centre of gravity.

Fig. 103.8: An acrobat on a rope

Spinning a pan with a finger

A pan spinning on a finger is less stable. This is because the centre of gravity of the pan is above the base. To bring the centre of gravity a little closer to the base, the spinner brings the weight of his body to where the pan might appear to fall over; this increases the stability of the pan.

Similarly, a person on stilt has his centre of gravity raised and might easily fall.

Reason why the loading compartments of buses are found at the base

When the load is located at the base of a bus, it brings its centre of gravity closer to the base making it more stable. On the other hand, if the load is located at top of the bus, its centre of gravity is raised, making it less stable. This may cause the bus to easily fall over or overturn easily.

MOMENT OF A FORCE

The moment of a force about a point is the product of the force and its perpendicular distance to the line of force.

Moment = force × perpendicular distance

Or

$$p = Fs$$

Moment of a force is referred to as the turning effect of the force. That is the moment of a force causes a pivoted body to turn. The S.I unit of moment is Nm or kg.ms^{-2}.

Applications of moment of force

1. ***Screwing and unscrewing with a screw driver*** – the force applied to the screw driver causes the screw to turn.
2. ***Riding a bicycle*** – force applied by the rider on the pedals produces a resultant turning effect on the hind wheel.

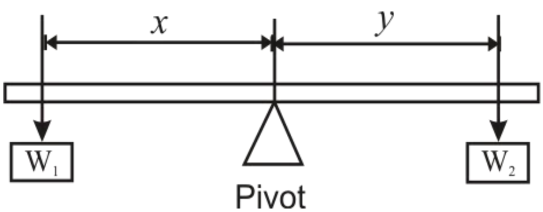
3. ***Wheel and axel*** – wheels of automobiles are attached to axels. When the axel turns, the wheel turns.
4. ***Opening and closing doors and gates*** – the force applied on the door causes it to turn due to the turning effect exerted on the hinges.

Principle of moments

The principle of moment states that the sum of clockwise moment about a point (pivot) on a body in equilibrium is equal to the sum of the anticlockwise moment about the same point.

Clockwise moment = anticlockwise moment

Consider the metre rule balanced on a knife edge, as illustrated below.

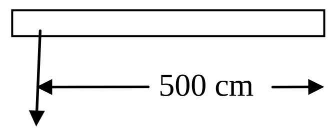


Supposing a weight W_1 is hanged x cm from the pivot and another weight W_2 is hanged y cm from the pivot at the other side of the beam. Then by applying the principle of moment:
clockwise moment = anticlockwise moment

$$W_1 \times x = W_2 \times y$$

Conditions of equilibrium
When a number of parallel forces are in equilibrium:
i. the sum of clockwise moment about a point is equal to the sum of anticlockwise moment about the same point.
ii. the sum of the forces in one direction is equal to the sum of forces in the opposite direction.

Example
1. Calculate the moment of the figure below



15 N

Solution

Moment (p) = force (F) x distance (s)

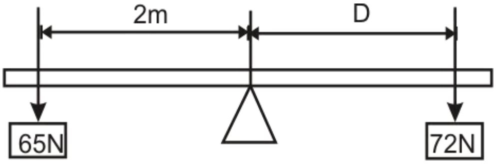
F = 15N

$s = \frac{500}{100} \quad = 5 \text{ m}$

p = 15 x 6 = 75 Nm

2. A little girl of weight 65 N sits 2 m from the pivot of a see-saw. Find the distance from the other end that a boy who weighs 72 N should sit.

Solution
Let's represent the distance from the other end by *D*.

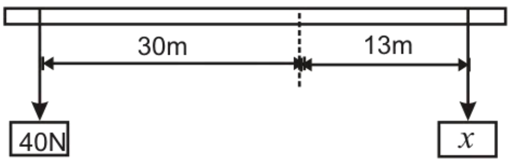


According to the principle of moment:
clockwise moment = anticlockwise moment

$\therefore$ 65 x 2 = 72 x D

$D = \frac{130}{72} = 1.8$ m

3. Consider the bar below.



Find the value of x

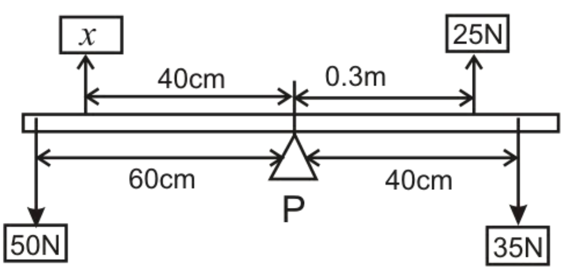
Solution

According to the principle of moment,

40 x 30 = x x 13

$x = \frac{1200}{13} = 92.3$ N

4. Calculate the value of x below if $F_1 = x$, F_2 = 25, F_3 = 35N, F_4 = 50N, S_1 = 40cm, S_2 = 3m, S_3 = 40cm and S_4 = 60cm.



Solution

From to the principle of moment,

$F_1S_1 + F_2S_2 = F_3S_3 + F_4S_4$

S_1 = 40cm = 0.4m
S_2 = 0.3m
S_3 = 40cm = 0.4m
S_4 = 60cm = 0.6m

$\therefore 0.4x + 25.03 = 35 \times 0.4 + 50 \times 0.6$

$0.4x + 7.5 = 14 + 30$

$0.4x = 44 - 7.5$

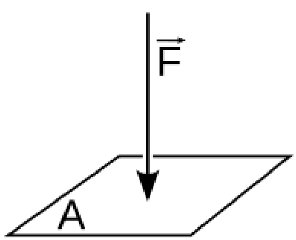
$x = \frac{36.5}{0.4} = 91.25$N

PRESSURE

Pressure is defined as the force per unit area at a particular region of a substance.

It is expressed mathematically as:

Pressure, (P) = $\frac{\boldsymbol{Force\ (F)}}{\boldsymbol{Area\ (A)}}$



The standard unit for pressure is the **Pascal** (**Pa**), which is Newton per square meter (**Nm^{-2}**).

Sharp pointed devices such as nails can easily be driven into other materials with little force as opposed to broad pointed devices. This is because sharp pointed devices have little area in contact. That is the smaller the area, the smaller the force needed and the greater the pressure.

Example

1. A cement block weighs 50N and a side area of $0.2m^2$, rests on a table. Calculate the pressr exerted on the table by the block.

Solution

$$\text{Pressure, (P)} = \frac{\text{Force (F)}}{\text{Area (A)}}$$

$$\text{(P)} = \frac{50}{0.2}$$

$$P = 250Nm^{-2}$$

2. A boy presses an inflated ball into a rectangular basin filled to the brim with water, with a force of 40 N. If the lenth and breadth of the basin are 7m and 5m respectively, find the pressue of in the water.

Solution

$$\text{Pressure, (P)} = \frac{\text{Force (F)}}{\text{Area (A)}}$$

Force (F) = 40 N

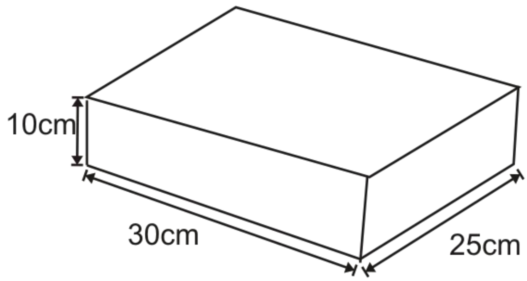
Area (A) of the rectangular basin = l x b

$$A = 7 \times 5 = 35m^2$$

$$\text{Pressure, (P)} = \frac{40}{35} = 1.14 \text{ Pa}$$

3. A 600 g book of dimensions, 30 cm × 25 cm × 10 cm, rests on a table.
   a. What is the maximum and minimum pressure exerted on the table by the book?
   b. How does increase in area affect pressure? [$g = 10 ms^{-2}$]

Solution



a. Mass of book = 600 g = 0.6 kg

Force = ma = 0.6 × 10 = 6 N

Area $A_1 = 0.3 \times 0.25 = 0.075 \text{ m}^2$

Area $A_2 = 0.3 \times 0.1 = 0.03 \text{ m}^2$

Area $A_3 = 0.25 \times 0.1 = 0.025 \text{ m}^2$

$$\text{Pressure, } P_1 = \frac{F}{A_1} = \frac{6}{0.075} = 80 \text{ Pa}$$

$$\text{Pressure, } P_2 = \frac{F}{A_2} = \frac{6}{0.03} = 200 \text{ Pa}$$

$$\text{Pressure, } P_1 = \frac{F}{A_3} = \frac{6}{0.025} = 240 \text{ Pa}$$

Maximum pressure is 240 Pa and minimum pressure is 80 Pa.

b. Increase in area decreases the pressure produced.

Pressure in liquids

Pressure transmitted through liquid is known as ***hydrostatic*** pressure. In liquid, pressure depends on:

- ❖ Depth of the liquid;
- ❖ Density of the liquid an
- ❖ Acceleration due to gravity.

Pressure increases as the depth and density of the liquid increase. The pressure at one point of a liquid is the same at every point of the liquid.

An experiment showing that pressure increases with depth.

- ❖ Get a water tank with three spouts, labelled 1, 2, 3 along its vertical height.
- ❖ Fill it with water and maintain the water level with a continuous flow from a tap running into the tank.

Observation

It would be observed that water comes out of spout 3 with the highest speed showing that pressure is greatest there. Followed be spout 2 and 1, in order of decreasing pressure.

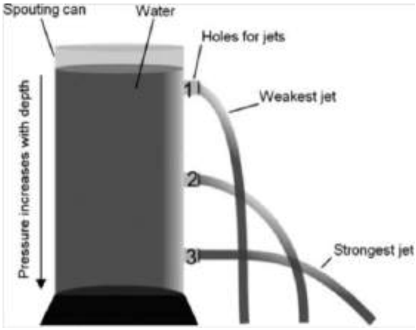


Fig. 103.9

The Pascal's Principle

Pascal's principle states that pressure at any part of a fluid equals pressure at all the parts in the fluid.

Applications of Pascal's Principle

The hydraulic press

In a hydraulic press, a small force applied to the master or effort piston, generates pressure which is equally transmitted throughout the liquid to the slave or load piston**.** This produces a large force at the slave or load piston and a heavy mass is raised by the force produced by the pressure acting on the piston.

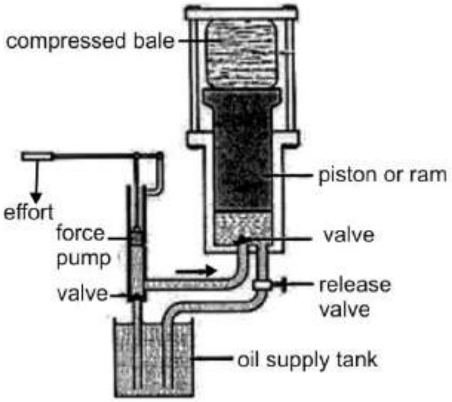


Fig. 104.0: The hydraulic press

Bicycle pump

A bicycle pump has a leather washer which is cap-shaped, fixed to the end of a piston rod which moves up and down the barrel of the pump. Air flows into the barrel through a small hole between the barrel and the leather washer, when the handle is pulled up. When the piston is pressed down the valve closes to prevent the air that has entered the barrel from escaping. The air is then compressed through the hole at the nozzle into the football or bicycle tyre that needs to be inflated. This is done continuously until the football or tyre is fully inflated.

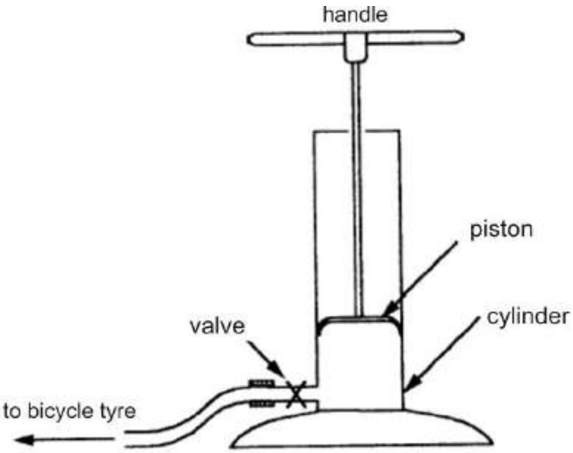


Fig. 104.1: A bicycle pump

Air compressor

Air compressors are of two general types: reciprocating and rotating.

In a ***reciprocating***, or ***displacement compressor***, which is used to produce high pressures, the air is compressed by the action of a piston in a cylinder. When the piston moves to the right, air flows into the cylinder through the intake valve; when the piston moves to the left, the air is compressed and forced through an output-control valve into a reservoir or storage tank.

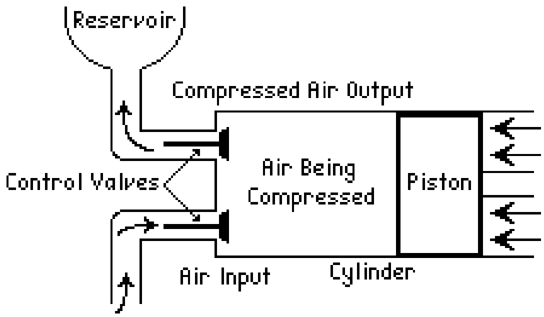


Fig. 104.2: Air compressor

A ***rotating air compressor*** consists of a bladed wheel that spins inside a closed circular housing. Air is drawn in at the centre of the wheel and accelerated by the centrifugal force of the spinning blades. The energy of the moving air is then converted into pressure in the diffuser, and the compressed air is forced out through a narrow passage to the storage tank.

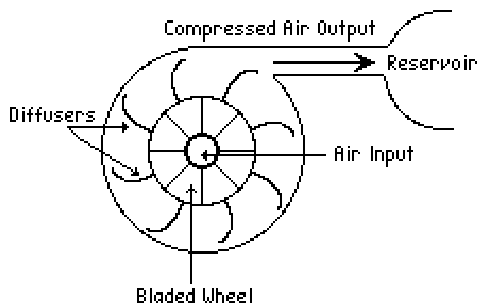


Fig. 104.3:

Water/ force pump

Water pumps are devices for moving water from one location to another, using tubes or other machinery.

Two types of modern pumps used to move water are the positive-displacement pump and the centrifugal pump.

Positive-displacement pumps use suction created by a vacuum to draw water into a closed space. An example of this type of pump is the lift or force pump. The lift pump is operated by raising a handle that is attached to a piston encased in a pipe. Lifting the piston creates a partial vacuum beneath it in the pipe, causing water to be drawn from a well below, through the pipe, and into a chamber in the pump. Subsequent pumps of the piston pull more water into the chamber, which eventually overflows, spilling water out of a spout.

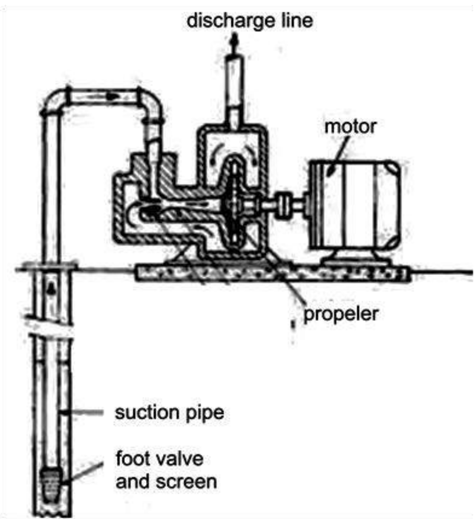


Fig. 104.4: A centrifugal water pump

Centrifugal pumps use motor-driven propellers that create a flow of water when they rotate. The blades of the propeller are immersed in the water to be pumped. As the propeller turns, water enters the pump near the axis of the blades and is swept out toward their ends at high pressure.

Operation of siphons

The siphon is a tube used to remove liquid mostly from a container. The siphon tube is bent over the edge of the container, one end in the liquid and the other outside end at a lower level than the surface of the liquid in the container. If the tube is once filled, a flow of liquid from the container through the tube will be set up.

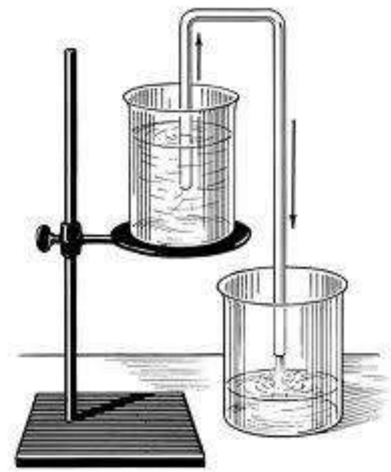


Fig. 104.5: Siphoning a liquid from one beaker into another

Siphons operate by atmospheric pressure. The container from which the liquid is siphoned must therefore be open to the air. When the tube is filled, the liquid will run out of the lower end. As the liquid starts to flow, the fluid pressure at the top of the tube is lowered. A liquid always flows from an area under higher pressure to an area of lower pressure. The liquid in the container (under atmospheric pressure) flows up into the tube, an area of lowered pressure). This liquid in turn will flow out the outside end of the tube, again lowering the pressure at the top of the tube.

TEST QUESTIONS

1. (a) What is a force?
   (b) Distinguish between contact a force and field force.

2. Write a short note on the following types of forces and categorize them as contact or field forces
   (a) frictional force;
   (b) magnetic force;
   (c) electrostatic force
   (d) tensional force
   (e) gravitational force

3. (a) State three effects of force on a body.
   (b) Explain the term upthrust.
4. (a) State the law of floatation.
   (b) Describe the application of flotation in the following:
      (i) flight of airplanes;
      (ii) flotation of a ship on the sea;
      (iii) the operation of submarines

5. Explain why a piece which weighs 0.005 kg sinks to the bottom of water but a ship of 5000 kg floats.

6. (a) Define the following terms:
      (i) distance;
      (ii) displacement;
      (iii) velocity;
      (iv) acceleration.
   (b) A man of mass 80 kg, runs across a field with an initial velocity of 10 ms-1 and accelerates for 120 s and attains a final velocity of 35 ms^{-1}. What is his acceleration?

7. (a) Define the term momentum.
   (b) Determine the momentum of a metal ball of mass 17 kg, moving at 8 ms^{-1}.

8. (a) What is motion?
   (b) State and explain three types of motion. Give one example in each case.

9. State and explain Newton's three laws of motion.

10. (a) Define the term inertia.
    (b) Why is it important to wear a seatbelt in a moving car.

11. (a) What is the centre of gravity of a body?
    (b) Describe how you will determine the centre of gravity of the following bodies:
       (i) a rectangular lamina
       (ii) a triangular lamina
       (iii) an irregular lamina

12. (a) When is a body said to be in equilibrium?
    (b) Explain the following types of equilibrium:
        (i) stable equilibrium;
        (ii) unstable equilibrium;
        (iii) neutral equilibrium.

13. (a) Explain the term moment of a force.
    (b) State the conditions of equilibrium.

14. (a) What is pressure?
    (b) Describe an experiment to show that pressure in liquid increases with depth.

15. Explain the danger associated with loading the top carriers of vehicle heavy.

16. (a) State the Pascal's principle.
    (b) Describe the operation of the following:
        (i) hydraulic press;
        (ii) bicycle pump;
        (iii) siphon.

17. Explain why a carpenter will require more force to hammer a blunt nail into a wood compared to a pointed nail.

18. A cement block of weight 50.0 N and one side of area 0.2 m^2 rests on a table. Calculate the pressure exerted on the table by the cement block

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SAFETY IN THE COMMUNITY

Specific Objectives

After completing this chapter, you will be able to:

- Identify appliances used at home and how to use them properly.
- Use first aid methods to treat accidents at home.
- Identify hazardous substances at the work place and their effects on humans.
- Identify some common hazards in the community.
- Outline the role of organizations working to improve the health of mankind and the factors which promote Public Health.

INTRODUCTION

Safety is the act of being protected from danger or accidents.

Accidents are unintended and unforeseen event, usually resulting in personal injury or property damage.

The basic causes of accidents are, in general, unsafe conditions of machinery, equipment, or surroundings, and the unsafe actions of persons that are caused by ignorance or neglect of safety principles.

APPLIANCES USED AT HOME

There are wide ranges of appliances which are used at home for various purposes such as entertainment, food preparation and preservation, communication, utility, personal upkeep, etc.

The appliances used for these purposes include television sets, radio, computers, phones, DVD players, electric cookers, microwave ovens, rice cookers, refrigerators, electric irons, electric fans, electric lamps etc.

Proper ways of using appliances at home

All appliances and gadgets need to be handled and used with care and precaution. The following are some proper ways of using home appliances:

1. Appliances usually come with instructional manuals, therefore make sure to read and understand them before using the appliance.

2. You should make sure that the appliances are usable. If they are faulty, get a qualified person to repair them.
3. The environment where they are used must be considered. Such appliances normally work safely in a dry environment.
4. The electric outlet where you plug the appliances should be checked regularly.
5. Check all electrical connections to be sure that none is bare or unsafe.
6. Iron clothes in bulk.
7. Do not leave refrigerator and freezer doors open.
8. Keep a distance between you and the television.
9. Do not overload electric sockets.
10. Turn off all unused appliances.
11. Turn off all electric sockets when not in use or when leaving the house.
12. Avoid scattering and trailing electric cables on the floor.
13. Do not open appliances when still plugged in.
14. Regularly brush or dust the appliances.
15. Do not immerse any appliance in water or pour water on them.

Some common accidents at home

a) Electric shock
b) Burns
c) Suffocation
d) Fall
e) Cuts
f) Scalds
g) Food poisoning
h) Gas poisoning
i) Drowning
j) Fire outbreak

How to prevent accidents at home

1. Keep working electrical appliances from water.
2. Avoid touching hot plates, pans etc. with the bare hands.
3. Keep plastics and cables from hot electric plates.
4. Turn off gas right after use.
5. Keep sharp tools and instruments on racks and other appropriate places.
6. Keep drugs and non-edible substances out of the reach of children.
7. Wipe off all spilled substances from the floor.
8. Keep combustible substances from fire or sources of heat.
9. Throw away spoiled food and drinks.
10. Do not dip the finger in a bucket of water with an electric heater.
11. Maintain ventilation in the room.
12. Repair all faulty pipes and electrical appliances.
13. Keep children from swimming pools or supervise them when near swimming pools.

FIRST AID

First aid is an emergency medical treatment given to a sick or injured person before more thorough medical attention can be obtained.

First aid may prevent a victim's condition from worsening and provide relief from pain. First aid must be administered as quickly as possible. In the case of the critically injured, a few minutes can make the difference between complete recovery and loss of life.

First-aid measures depend upon a victim's needs and the provider's level of knowledge and skill. Knowing what not to do in an emergency is as important as knowing what to do. Improperly moving a person with a neck injury, for example, can lead to permanent spinal injury and paralysis.

Methods of administering first aid

Injury to the head

Injuries to the head may involve the scalp, skull, or brain. If the victim has a head wound, the first-aid provider should not apply pressure to it, as this may damage the brain. If the victim has a seizure (a sudden spasm of the body) the head must be protected with cushions to prevent further injury. All individuals with head injuries should be evaluated by a physician.

Injury to the eyes

Medical attention should be sought for all eye injuries as well. In the case of foreign material in the eye, especially *caustic* substances, or those that can burn, corrode, or dissolve tissues, the eye should be flushed immediately with a cool, sterile, saline solution - if available, or plain tap water for 5 to 10 minutes. The first-aid provider should not attempt to remove embedded objects from the eye.

Injury to the nose

The most common injuries to the nose involve nosebleeds, objects lodged in the nasal passages, and broken nasal bones. The victim of a simple nosebleed should sit down, tilt the head backwards, and gently pinch together the soft part of the nose for 15 minutes. A cold compress can also be placed on the bridge of the nose. If material lodged in the nose cannot be forced out by gently blowing the nose, the victim should request medical help. In the case of a broken nose, the first-aid provider should apply a cold compress to the bridge of the nose and seek medical attention.

Sprains and fractures

A sprain is the painful stretching or tearing of ligaments (tissues that connect bones at joints), occurs when a bone is suddenly wrenched at the joint. A fracture, break or crack in a bone, is caused by sudden, violent pressure against the bone. Great pain and swelling characterize both sprain and fracture, but the inability to move the affected part, a deformed appearance, and pain or tenderness at a specific point usually indicate a fracture. Sprains and fractures should be treated in the same way by a first-aid provider, since it can be difficult to diagnose a fracture without an x-ray of the affected bone.

Because the slightest movement of the affected part may cause the injured person great pain and increase the damage, no attempt should be made to straighten or move sprained or broken limbs until medical help arrives. If the injured person must be transported to a hospital, rigid splints should be used to immobilize the broken part and adjacent joints or bones. Splints can be improvised from light, smooth boards or folded cardboard and tied to the broken part with wide strips of cloth or improvised material.

Fig. 104.6: Securing broken arm

Electric shock

In giving first aid to an electric-shock victim, a caregiver must not touch the victim with bare hands until the source of electricity has been removed safely or the power source shut off. If the victim is not breathing, mouth-to-mouth or mask-to-mouth resuscitation is necessary.

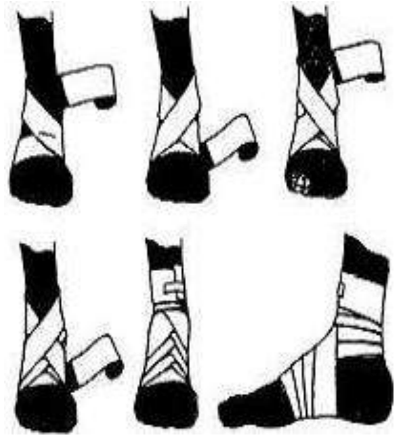

Fig. 104.7: Twisted ankles should be gentle bandaged

If the person also has no pulse, ***cardiopulmonary resuscitation*** (***CPR***) should be administered until professional emergency help arrives.

Fig. 104.7: *Safely remove the source of the electricity*

Burns

A burn is an injury to the skin caused by exposure to fire, hot liquids or metals, radiation, chemicals, electricity, or the sun's ultraviolet rays.

First aid for burns involves removing the source of the burn as soon as possible. The burn should be cooled immediately with cold water. A clean, cold wet towel or dressing can be placed on less serious burns to ease pain and protect the burn from contamination.

Continuously bathe chemical burns with running water for at least 20 minutes to dilute the substance. Any powder should be carefully brushed off with gloved or protected hands before washing. Wet dressings or ointments should never be used for burns. Instead, the first-aid provider should gently apply dry, sterile dressings held in place by bandages and seek immediate medical attention.

Poisoning

A poisonous substance introduced into the body through the mouth or nose causes symptoms such as nausea, cramps, and vomiting. Poisons include toxic medications, herbicides, insecticides, household disinfectants, and noxious gases. Try to identify the poison, either by questioning the victim or searching for suspicious containers. Containers of many poisonous substances list the antidote or remedy on the label.

Burns or stains on the skin or a characteristic odour on the breath may also help the first-aid provider to recognize the poison.

Before clearing the victim's mouth of any obstructions, the provider should first put on clean first-aid gloves or wrap a cloth around his or her fingers. If the person who ingested the poison is unconscious, the airway, breathing, and circulation should be checked and CPR started if necessary.

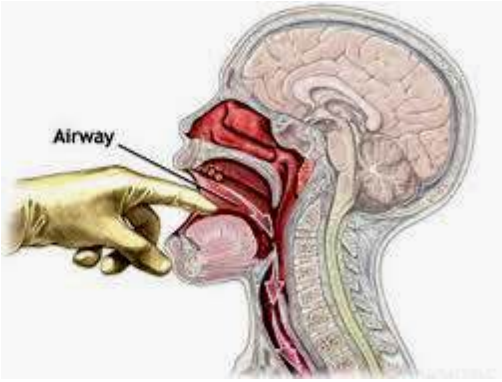

Fig. 104.8: *Clearing victim's airway*

Drug overdose

A drug overdose occurs when an individual takes too large a dose of a drug or takes a dose that is stronger than the person can tolerate.

A drug overdose can be difficult to diagnose because the signs and symptoms vary widely and often mimic other illnesses or injuries. Symptoms of a drug overdose include unusually dilated or contracted pupils, vomiting, difficulty in breathing, hallucinations, and in severe cases unconsciousness and slow, deep breathing. If an overdose is not treated, the individual may die. Victims of overdose should be taken immediately to a hospital emergency room.

How to perform cardiopulmonary resuscitation (CPR)

Mouth-to-mouth ventilation

- Open the airway of the casualty by raising the chin and pressing the forehead backwards.
- Take a deep breath and exhale, cover the casualty's mouth with yours and exhale, while pinching his nose shut. Do this until the patient starts breathing.
- Place the casualty in a recovery position to ensure that his airway remains open ad that the tongue cannot fall backwards into the throat to choke him. It also ensures that any fluid from the patients mouth drains freely

External chest compression

- Place the casualty on his back on a flat surface, and with one hand covering the other, place both hands on the lower part of his chest.
- Press downwards sharply and release 14 to 15 times at a rate of 80 compressions per minute.
- Pulse compression and give two mouth-to-mouth ventilation. Check for pulse and heart beat. Continue the process if nothing happens.

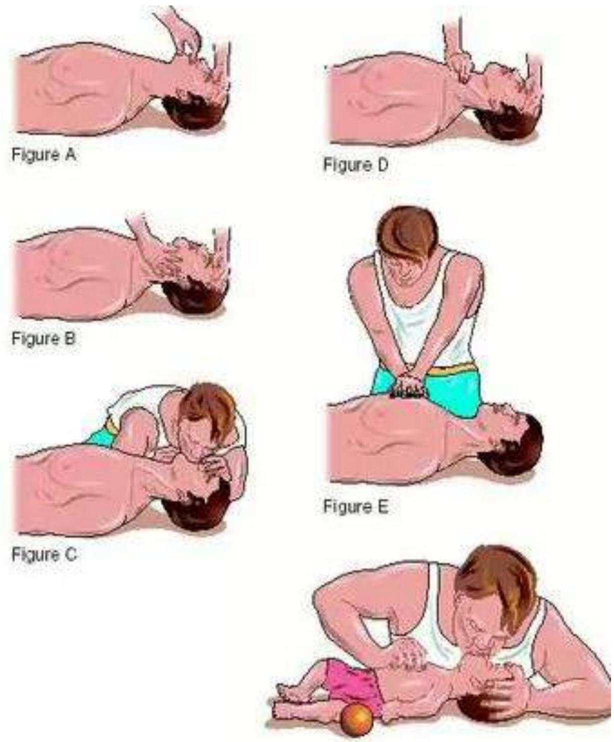

Fig. 104.9: Administering CPR

ACTIVITY:

1. Invite a Fire Service Officer or Red Cross Personnel to talk to class on accidents prevention and first aid.
2. Students to make a first aid box for the class.

HAZARDOUS SUBSTANCES AND THEIR EFFECTS

Hazardous substances are solids, liquids or gaseous substances that can cause death, illness, or injury to people or destruction of the environment if improperly treated, stored, transported, or discarded.

Substances are considered hazardous if they are ***ignitable*** (capable of burning or causing fire), ***corrosive*** (able to corrode steel or harm organisms because of extreme acidic or basic properties), ***reactive*** (able to explode or produce toxic cyanide or sulphide gas), or ***toxic*** (containing substances that are poisonous).

Sources of hazardous substances

Industrial waste

Hazardous substances are generated by nearly every industry. Various industries such as the chemical manufacturing plants, petroleum industry, metal production and fabrication plants etc. produce various wastes which normally go untreated. These wastes are often discarded in the environment and in water bodies, resulting in the destruction of terrestrial and aquatic habitats as well as posing danger to humans.

Agricultural waste

Industries are not alone in generating hazardous wastes. Agriculture produces such wastes as pesticides and herbicides and the materials used in their application. Fluoride wastes are by-products of phosphate fertilizer production. Even soluble nitrates from manure may dissolve into groundwater and contaminate drinking-water wells; high levels of nitrates may cause health problems.

Domestic waste

Household sources of hazardous wastes include toxic paints, flammable solvents, caustic cleaners, toxic batteries, pesticides, drugs, etc.

Renovations of older homes may cause toxic lead paint to flake off from walls. Insulation material on furnace pipes may contain asbestos particles, which can break off and hang suspended in air; when inhaled, they can cause lung disease and cancer.

Fig. 105.0: Domestic waste

Medical waste

Hospitals use special care in disposing of wastes contaminated with blood and tissue, separating these hazardous wastes from ordinary waste. Hospitals must be especially careful with needles, scalpels, and glassware, called "sharps." Pharmacies discard outdated and unused drugs; testing laboratories dispose of chemical wastes. Medicine also makes use of significant amounts of radioactive isotopes for diagnosis and treatment, and these substances must be tracked and disposed of carefully.

Effects of hazardous substances

Water pollution

Some wastes are directly discharged into water bodies. This normally results in the death of organism in those water bodies; interfering with life in the aquatic habitat. Some waste and poisonous substances get into drinking water, causing harm to people who may drink it. Oil spills are known to cause a great deal of damage to water bodies.

Fig. 105.1: A pelican covered in spilled oil

Air pollution

Air may become contaminated by direct emission of hazardous wastes. Evaporation of toxic solvents from paints and cleaning agents is a common problem. The air above hazardous waste may become dangerously contaminated by escaping gas.

Land contamination

Agricultural wastes often lead to the contamination of productive land. These often reduce crop yield produced by those lands. Other pollutants which may contaminate the land include plastics and scraps of metal.

Plastics remain in the soil for a very long time while scraps of metal may corrode and leak toxic chemicals into the soil, killing soil living organisms in the process.

Radioactive radiation

Some industrial waste products are radioactive. Radioactive waste can be active for a very long time and cause damage to animal and plant lives. While some nuclear or radioactive waste may cause immediate damage, other may slowly deteriorate a community or township for decades.

Spread of diseases

When hazardous wastes get into humans and animals, they interfere with the normal functioning of the body rendering the person or animal sick. Some of the diseases may be easy to treat, other may be fatal.

Control of hazardous substances

Recycling

Recycling is the collection, processing, and re-use of waste materials. Materials ranging from precious metals to broken glass, from old newspapers to plastic spoons, can be recycled. The recycling process reclaims the original material and uses it in new products.

Using recycled materials to make new products costs less and requires less energy than using new materials. Recycling can also reduce pollution, either by reducing the demand for high-pollution alternatives or by minimizing the amount of pollution produced during the manufacturing process. Recycling decreases the amount of land needed for trash dumps by reducing the volume of discarded waste.

Treatment

Wastes may be made less hazardous by physical, chemical, or biological treatment. Sulphuric acid wastes, if not recycled, can be treated with ammonia wastes from the same plant, forming ammonium sulphate, a fertilizer.

Solidification of wastes involves melting them and mixing them with a binder, a substance that eventually hardens the mix into an impenetrable mass. One suggested treatment of radioactive waste involves turning it into a glass through a process known as ***vitrification***.

Sometimes waste can be stabilized on-site; simple remedies such as covering the waste may be sufficient. Other stabilization methods involve building a barrier around the waste. This barrier can be of plastic, steel, concrete, clay, or even glass.

Source reduction

The best way to eliminate hazardous waste is not to generate it in the first place. For example, improvements have been made in the production of integrated circuits: The toxic chlorinated hydrocarbons commonly used in the 1970s were replaced in the 1980s by less toxic glycol ethers and in the 1990s by low-toxicity esters and alcohols.

Disposal

Surface impoundment (placing liquid or semi-liquid wastes in unlined pits) keeps waste in long-term storage, but it is not considered a method of final disposal. About 8 percent of hazardous waste is injected into deep wells; 21 percent enters *landfills* (large, unlined pits into which solid wastes are placed) as its ultimate resting place.

OCCUPATIONAL HAZARD

Occupational hazard is any danger or risk associated with a particular job or profession.

Any disease that arises as a result of occupational hazard is referred to as ***occupational disease.***

Different companies, establishments and even institutions have specific products they bring out. However, before those products are brought out, other products have to be used. Either the products brought or used or both could be hazardous. In this case, worker of such companies are at risk of occupational hazards of various forms.

Examples of hazards that workers are exposed to

(a) corrosive substances
(b) explosion and fire
(c) toxic fumes
(d) radiation
(e) gas poisoning
(f) electric shock and electrocution
(g) dust
(h) general poisoning from ingestible.

Physical hazards include traumatic injuries and noise.

Trauma arising from unsafe environments accounts for a large proportion of preventable human illness, and noise in the workplace is responsible for the most prevalent occupational impairment - hearing loss or permanent deafness.

One hazard suffered by teachers who use chalk is dust. Such teachers, and sometimes students experience respiratory problems. A worker in a nuclear power plant may be exposed to radiation, and so on.

Protection against occupational hazards

1. Workers should be educated about the possible hazard they may be exposed to in a particular field.
2. They should be given appropriate protective clothing such as overalls, safety booths, goggles, ear muffs etc.
3. Regular medical screening and check-up should be conducted for them.
4. Hazard signs should be mounted to show the various dangers in that work environment.
5. Hazardous substances should be well labelled.
6. Safety signs and symbols should be mounted in various places.

Fig. 105.2: Protective clothing must be worn

Fig. 105.3: Hazard signs

Fire and its management

Fire is said to be “a good servant but a bad master”. This is because while fire helps humankind in so many ways, its destructive impact is very huge. Fire has caused many losses of lives and destruction of properties.

Knowing how to stop a small fire will help in preventing a huge, destructive and uncontrollable fire.

Some methods of controlling fire include the use of fire extinguishers, sprinklers sand and fire blankets

Fig. 105.4: Fire extinguishers

Table 7.4: Fire prevention methods

Fire prevention method	Application	Mode of operation	Where normally used
Sprinklers	A sprinkler system is an integrated system of underground and overhead piping, connected to one or more automatic water supplies. The system is usually activated by heat from a fire, and the sprinkler heads then discharge water over the fire area.	Lowers the temperature and prevents oxygen from getting into contact with the fire	In huge buildings and offices and large fires. Sprinkler systems are nearly 100 % effective.
Fire blanket	Cover the fire with the blanket. If it is a person on fire, wrap him/ her with the blanket.	Prevents oxygen from reacting with the fire	Small fires and burning clothes or persons
Sand	Pour sand over the fire	Covers the fire and prevents contact with air	Small fires
Fire extinguishers			
Carbon dioxide extinguisher	Splash a continuous stream of carbon dioxide on the base of the fire	CO_2, which is heavier than oxygen and does not support burning, displaces oxygen from the fire.	Electrical and medium-sized fires
Foam extinguisher	Spray foam directly onto the fire	Prevents oxygen from contacting the fire	Small and medium-sized fires
Water extinguisher	Spray continuous flow of water to the base of the fire	Lowers the surrounding temperature and prevents air from the fire	Not suitable for electrical fires since water has some appreciable conductivity
Dry powder extinguisher	Spray on the base of the fire and then spread to the rest of the fire	Engulfs the fire and prevents contact with oxygen	Suitable for all fires
Soda and gas cartridge extinguisher	Direct the mixture of CO_2 and water at the base of the fire	Water lowers the temperature whiles CO_2 displaces air	Suitable for all fire except electrical fires

HAZARDS IN THE COMMUNITY

Some common hazards in the community include diseases, pest and parasite outbreak, insanitary conditions, traffic problems, pollution and waste generation.

Table : Hazards in the community and their effects

Hazard	Effect
Disease, pest and parasite outbreak	• Reduction in labour force and productivity • High expenses on treatment and medication
Insanitary conditions	• Disease outbreak, e.g. cholera • Increase in the rate of mosquito breeding • Flooding due to choked gutters
Traffic problems	• Increase in accidents and fatalities • Air pollution • Congestion • Stress • Decreased productivity
Pollution and waste generation	• Diseases • Discomfort • Suffocation • Hearing impairment • Mutation • Low crop yield • Destruction of aquatic life

PUBLIC HEALTH

Public health is the protection and improvement of the health of entire populations through community-wide action, primarily by governmental agencies.

Aims of public health authorities

The goals of public health authorities are to:

1. prevent human disease, injury, and disability;
2. protect people from environmental health hazards;
3. promote behaviours that lead to good physical and mental health;
4. educate the public about health;
5. ensure availability of high-quality health services.

Activities of public health authorities

Public health workers may engage in activities outside the scope of ordinary medical practice. These include:

1. inspecting and licensing restaurants.
2. conducting rodent and insect control programs.
3. checking the safety of housing, water, and food supplies.
4. in assuring overall community health, public health officials also act as advocates for laws and regulations—such as drug licensing or product labelling requirements.

Ways of improving public health

1. Providing waste and refuse containers
2. Providing good and safe drinking water
3. Proper house planning
4. Cleaning public places
5. Public education
6. Proper siting of refuse dumps

7. Provision of waste disposal facilities in cities, towns and villages
8. Cleaning and desilting of gutters
9. Proper house planning to improve ventilation
10. Effective cleaning of homes and public toilets

The role of health service organizations

To ensure effective delivery of health services, various health service organizations have been set up.

In Ghana, local health service organizations such as Ghana Health Service, Environmental Protection Agency, Ghana Standards Board and the Food and Drug board as well as international health service organizations like the WHO, FAO, UNICEF, Red Cross, Red Crescent and Blue cross all contribute in ensuring efficient health service delivery.

The role of Ghana Health Service

1. Training of health personnel.
2. Advising the government on health issues.
3. Carrying out vaccinations.
4. Educating the public on health matters.
5. Monitors and prevents outbreaks of diseases.
6. Provides national health statistics.

The role of the Environmental Protection Agency (EPA)

1. Controls air, water and land pollution.
2. Regulates noise level in communities.
3. Checks radiation levels in industries.
4. Educate on the proper use of pesticides, and other toxic substances.
5. Check waste disposal by industries.
6. Research into the effects of water and other contaminants.

Ghana Standards Boards

1. Promotes development and efficiency in industries
2. Ensures high quality of goods produces in Ghana
3. Regulates importation of goods
4. Promotes industrial and public standards
5. Checks the safety of manufactured goods

Food and Drugs Board

1. Ensures that foods are pure and wholesome and produced under sanitary conditions
2. Checks the safety and effectiveness of drugs and therapeutic devices
3. Makes sure that cosmetics are safe and made from appropriate ingredients
4. Ensures that labels and packaging of products are truthful, informative, and not deceptive.
5. Ensures proper labelling and safety of chemical products, toys, and other articles used in the home.

The role of the World health Organization (WHO)

1. Works to reduce human disease
2. Funds medical research
3. Provides emergency aid during disasters,
4. Aims to improve nutrition, housing, sanitation, and working conditions in developing countries.
5. Manages international disease prevention and control efforts,
6. Involved in training medical personnel,
7. Educates the world populations about public health issues including widespread diseases, nutrition, population control, and the benefits of environmental sanitation.

The role the Food and Agriculture Organization

1. Collects, analyzes, and distributes information about nutrition, food, and agriculture
2. Fosters conservation of natural resources
3. Develops basic soil and water resources
4. Controls animal diseases and plant diseases;
5. Provides needy member nations of technical assistance in such fields as nutrition, food preservation, irrigation, soil conservation, and reforestation.
6. Establish monitoring networks to warn of possible food shortages
7. Facilitates improved worldwide food security.
8. In recent years, the FAO has worked to develop new plant mutations by using radioactive materials, to aid developing nations in cultivating fast-growing varieties of crops such as rice and wheat

The role of the United Nations International Children's Emergency Fund (UNICEF)

1. UNICEF has helped provide millions of children with clean drinking water and sanitary living conditions.
2. By training educators to develop effective school programs, the agency has enabled children around the world to benefit from a primary school education.
3. UNICEF also provides a relief network for children and their parents or other caregivers in the aftermath of disasters, such as floods, earthquakes, and droughts.
4. It has worked extensively with children from war-torn countries to help alleviate their suffering.
5. The organization works to prevent child abuse, child labour, sexual exploitation of children, and the use of children as soldiers.
6. UNICEF has provided immunizations to millions of children against

potentially fatal diseases, such as diphtheria, tetanus, whooping cough, measles, polio, and tuberculosis.

7. UNICEF's promotion of basic health-care delivery systems and treatments, such as rehydration therapy for children suffering from diarrhoea, has also contributed to dramatic reductions in child mortality.
8. UNICEF works to prevent the transmission of AIDS and human immunodeficiency virus (HIV) in young people and to obtain medicine for infected individuals.
9. It also helps communities care for the millions of children orphaned by the death of their parents from AIDS.

Fig. 105.5: A UNICEF official giving a vaccine

The role of the International Red Cross (IRC) and Red Crescent Movement

1. Provides first aid in time of war
2. Trains personnel
3. Provides support during disasters such floods, earthquakes, fire outbreak etc.
4. Maintains maternal and child welfare clinics in deprived areas.
5. Provides emergency aid to those in distress
6. Provides supports for refugees

SANITATION AND PERSONAL HYGIENE

Sanitation is measures to protect public health through proper solid waste disposal, sewage disposal, public and personal hygiene and cleanliness during food processing and preparation.

Waste disposal

Solid waste disposal is the disposal of normally solid or semi-solid materials, resulting from human and animal activities that are useless, unwanted, or hazardous.

Types of solid waste

1. ***Garbage***: decomposable wastes from food
2. ***Rubbish:*** non-decomposable wastes, either combustible (such as paper, wood, and cloth) or non-combustible (such as metal, glass, and ceramics)
3. ***Ashes***: residues of the combustion of solid fuels
4. ***Large wastes***: demolition and construction debris and trees
5. ***Dead animals***

6. ***Sewage-treatment solids:*** material retained on sewage-treatment screens, settled solids, and biomass sludge
7. ***Industrial wastes****:* such materials as chemicals, paints, and sand
8. ***Mining wastes****:* slag heaps and coal refuse piles
9. ***Agricultural wastes****:* farm animal manure and crop residues

Waste disposal methods

Selecting a disposal method depends mostly on the type of waste and costs. Waste disposable methods include:

Landfill

Sanitary landfill is the cheapest and the most satisfactory means of disposal, but only if suitable land is within economic range of the source of the wastes. Typically, collection and transportation account for 75 percent of the total cost of solid waste management.

In a modern landfill, refuse is spread in thin layers, each of which is compacted by a bulldozer before the next is spread. When about 3 m of refuse has been laid down, it is covered by a thin layer of clean earth, which also is compacted. Pollution of surface and groundwater is minimized by lining and contouring the fill.

Fig. 105.6: A landfill site

Incineration

In incinerators, refuse is burned on moving grates in refractory-lined chambers; combustible gases and the solids they carry are burned in secondary chambers. Combustion is 85 to 90 percent complete for the combustible materials.

In addition to heat, the products of incineration include the normal primary products of combustion—carbon dioxide and water—as well as oxides of sulphur and nitrogen and other gaseous pollutants; non-gaseous products are fly ash and unburned solid residue

Compostin*g*

Composting operations of solid wastes include preparing refuse and degrading organic matter by aerobic micro-organisms. Materials that might have salvage value or cannot be composted are removed, and are ground up to improve the efficiency of the decomposition process. The refuse is placed in long piles on the ground or deposited in mechanical systems, where it is degraded biologically to humus with a total nitrogen, phosphorus, and potassium content of 1 to 3 percent, depending on the material being composted. After about three weeks, the product is ready for curing, blending with additives, bagging, and marketing.

Sewage Disposal

Sewage or wastewater disposal is the various processes involved in the collection, treatment, and sanitary disposal of liquid and water-carried wastes from households and industrial plants.

Methods of disposing of sewage

Direct discharge into water

Direct discharge into a receiving stream or lake is the most commonly practiced means of sewage disposal; the water body is then treated. The treatment process involves conventional primary and secondary treatment followed by lime clarification to remove suspended organic compounds. During this process, an alkaline condition is created to improve the process.

In the next step, ***re-carbonation*** is used to bring the pH level to neutral. Then the water is filtered through multiple layers of sand and charcoal, and ammonia is removed by ionization. Pesticides and any other dissolved organic materials still present are absorbed by a granular, activated-carbon filter. Viruses and bacteria are then killed by ***ozonization.***

At this stage the water should be cleansed of all contaminants, but, for added reliability, second-stage carbon absorption and reverse osmosis are used, and chlorine dioxide is added to attain the highest possible water standard.

Septic tanks

A sewage treatment process commonly used to treat domestic wastes is the septic tank: a concrete, cinder block or metal tank where the solids settle and the floatable materials rise. The partly clarified liquid stream flows from a submerged outlet into subsurface rock-filled trenches through which the wastewater can flow and percolate into the soil where it is oxidized aerobically. The floating matter and settled solids can be held from six months to several years, during which they are decomposed anaerobically.

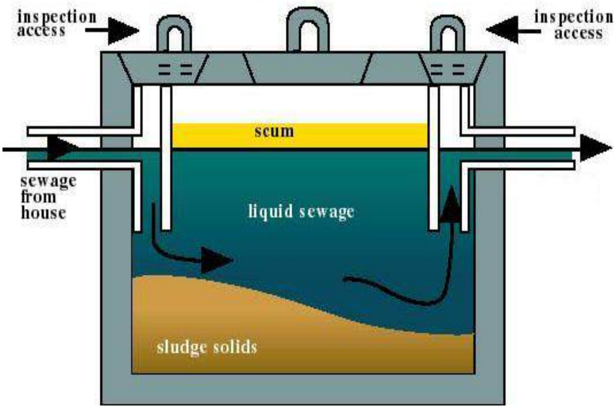

Fig. 105.7: A septic tank

Public and personal hygiene

Hygiene is the practice or principles of cleanliness thereby preserving life.

Diseases like cholera, dysentery etc. are acquired as a result of insanitary and unhygienic lifestyles. To improve quality of life, both environmental and personal hygiene must be observed.

How to observe public and environmental hygiene

1. Cleaning public places regularly.
2. Desilting gutters.
3. Maintenance of clean roads and streets.
4. Weeding weed-infested public places.
5. Providing refuse dump and containers.
6. Proper disposal of solid and liquid wastes.
7. Proper town and village planning.
8. Provision of clean water supply.
9. Availability of public toilet.
10. Proper ventilation in house planning.

How to observe personal hygiene

1. Regular bathing or washing of the body with soap.
2. Washing the hands with soap after visiting the toilet.
3. Changing and washing dirty and odorous clothes.
4. Cleaning the teeth regularly.
5. Keeping the hair tidy by combing.
6. Washing the hand before touching food or cooking
7. Wearing shoes or foot ware to protect the feet

Ventilation

Ventilation is the removal of hot, stale and odorous air from a room or an enclosed area and replacing it with clean, fresh air.

Buildings in which people live and work must be ventilated to replenish oxygen, dilute the concentration of carbon dioxide and water vapour, and minimize unpleasant odours.

How to ensure proper ventilation in a room

Natural method

1. Adequate number of windows and air vents.
2. Large windows raised above the ground level.
3. Windows should face each other so that fresh air will enter through one and force warm air through the other.
4. Windows, doors and air vents should face the direction of the wind.
5. Planting of trees around or near the house.
6. Siting houses on hilltops and flatlands instead of valleys.
7. Avoid overcrowding in houses and buildings

8. Siting air vents in the ceiling and high on the walls

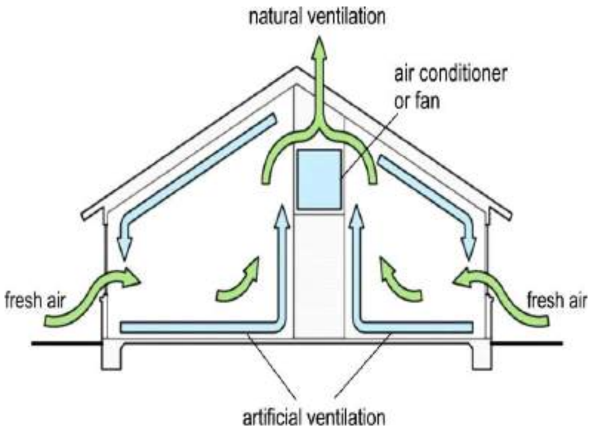

Fig. 105.8: Proper ventilation

Artificial methods

1. Provision of electric fans (standing, ceiling or wall fans)
2. The use of hand fans
3. Availability of air-conditioners or air-conditioning units.

Importance of ventilation

1. It replaces warm, stale, humid air with fresh one.
2. It increases evaporation of sweat from the body.
3. It reduces droplet infections by bacteria and viruses.
4. It prevents headaches and drowsiness as a result of overheating and high concentration of carbon dioxide.
5. It prevents pest such as cockroaches and bedbugs from encroaching rooms.
6. It removes body odour.
7. It prevents suffocation.

TEST QUESTIONS

1. Explain six proper ways of using appliances at home.

2. (a) Mention five common accidents at home.
   (b) State five ways of preventing accidents at home.

3. (a) What is first aid?
   (b) List four importance of first aid.

4. Describe the methods of administering first aid to the following accident:
   (a) head injury;
   (b) sprain
   (c) electric shock
   (d) poisoning

5. Describe how you will perform cardiopulmonary resuscitation to an unconscious person.

6. (a) What are hazardous substances?
   (b) Explain four sources of hazardous substances.

7. (a) List four effects of hazardous substances.
   (b) Describe four ways of controlling hazardous substances.

8. Some hazard symbols usually displayed on chemical containers are listed below.


A B C D E

(a) What does each symbol signify?
(b) Give one name of a chemical which can be associated with each symbol.

9. (a) Define the term occupational hazard.
   (b) Mention six common occupational hazards.

10. (a) List five ways of controlling occupational hazards.
    (b) Why is it important to know the basic hazard symbols?

11. Describe the appropriate methods of controlling the following fires:
    (a) bush fire;
    (b) small fire;
    (c) large factory fire.

12. State two effects of each of the following hazards in the community:
    (a) disease;
    (b) insanitary conditions;
    (c) traffic problems.

13. (a) What is public health?
    (b) State three aims of public health authorities.
    (c) Mention three methods used by public health authorities in preventing disease outbreak.

14. List four roles played by each of the following bodies in ensuring proper health care.
    (a) Ghana Health Service;
    (b) Environmental Protection Agency;
    (c) Foods and Drugs Board;
    (d) World Health Organization;
    (e) United Nations International Children's Emergency Fund.

15. (a) Briefly explain the term sanitation
    (b) Explain four waste disposal methods.

16. (a) What is personal hygiene?
    (b) List five ways of ensuring personal hygiene.

17. (a) Explain the term ventilation.
    (b) State six methods of ensuring proper ventilation.
    (c) Mention five importance of ventilation.

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ENDOGENOUS TECHNOLOGY

Specific Objectives

After completing this chapter, you will be able to:

- Distinguish between science and technology.
- Outline the significance of science and technology to society.
- Explain the term endogenous technology.
- Outline problems associated with application of some technology.
- Identify some small scale industries and their importance.
- Identify the scientific principles underlying the operation of some small scale industries.

INTRODUCTION

Technology has been a cumulative process at the centre of human experience. It is perhaps best understood in a historical context that traces the evolution of early humans from a period of very simple tools to the complex, large-scale networks that influence most of contemporary human life.

The prehistoric man made tools out of stone, mostly for hunting. As time went on, they discovered fire by striking flint against pyrites to produce sparks.

The rise in civilization introduced man to new forms of tools, materials, designs structures, etc. In the Renaissance era, more complex tools and machineries were invented as humans continued to understand science better.

Our modern period has brought about even more sophisticated tools, gadgets, vehicles, structures etc. Humans are still developing and the development is measured primarily on the technologies invented and introduced.

SCIENCE AND TECHNOLOGY

The meanings of the terms science and technology have changed significantly from one generation to another. More similarities than differences, however, can be found between the terms.

Science

Science is the systematic study of anything that can be examined, tested, and verified.

From its early beginnings, science has developed into one of the greatest and most influential fields of human endeavour. Today different branches of science investigate almost everything that can be observed or detected, and science as a whole, shapes the way we understand the universe - our planet, ourselves, and other living things.

Knowledge gained in science accumulates as time goes by, building on work performed earlier. Some of this knowledge, such as our understanding of numbers, stretches back to the time of ancient civilizations, when scientific thought first began. Other scientific knowledge, such as our understanding of genes that cause cancer, dates back less than 50 years. However, in all fields of science, old or new, researchers use the same systematic approach, known as the ***scientific method***.

Technology

Technology is the study, development and application of devices, machines and techniques for manufacturing and productive purposes.

The term technology also refers to the processes by which humans develop tools and machines to increase their control and understanding of the material environment.

Technology is an essential condition of advanced, industrial civilization. The rate of technological change has developed its own momentum in recent centuries.

Table 7.5: Distinction between science and technology

Science	Technology
Study of how things work	Uses science to make things work
Systematic study of things	Process of converting things or materials into useful products
Aims to learn, to know and to understand.	Aims to create, to build, to accomplish
Pursues knowledge for knowledge sake	Applies science to achieve specific ends
Focuses on understanding natural phenomena	Focuses on understanding the made environment
Experimental and logical skills needed	Design, construction, testing, planning, quality assurance, problem solving, decision making, interpersonal and communication skills needed
Drawing correct conclusions based on good theories and accurate data	Taking good decisions based on incomplete data and approximate models
Corresponding Scientific Processes	Key Technological Processes
Discovery (controlled by experimentation)	Design, invention, production

Similarities between science and technology

1. Both science and technology imply a thinking process;
2. Both are concerned with causal relationships in the material world;
3. Both employ an experimental methodology that results in thorough demonstrations that can be verified by repetition

Significance of science and technology

Communication

Over the years communication has been much easier through science and technology. The invention of mobile phones, which allow people to speak to others kilometres away, are a key example. In Ghana, about 60% to 70% of the populace own or use mobile phones.

The internet with the *World Wide Web* offers people the chance to chart through various social networking sites such as “Facebook”, “Twitter”, “Myspace”, “LinkedIn”, etc. Skype, for instance enables people to talk over the internet while seeing each other on the computer screen.

Transportation

Before the invention of automobiles, trains and aircrafts, people had to walk for miles, wasting much needed strength and productive hours. Nowadays, a person can travel over thousands of miles in a matter of hours, thanks to science and technology.

Health

Today with continuous scientific studies and new technologies, medical research is being done every day to help find cures or vaccines for devastating diseases such as malaria, cancer, HIV and AIDS, leukaemia, etc. People are now assured of long lives because of better available healthcare and facilities.

Education

Through the internet, teachers and students have a huge stock of information to help in their research and learning processes. Students can sit in their houses and study everything they ought to know through computer-based distance learning. Other multimedia tools such as videos, slides with projector and even audio help educators to reach wide audiences.

Agriculture

More scientific and technological processes have helped farmers to improve on the quality, quantity and yield of their produce. Fertilizers improve crop yield while “weedicides” kill weeds thereby reducing competition for nutrients. Machineries such as ploughs, planters and harvesters help farmers to grow more crops with little manual labour. Animal breeding techniques enable farmers to raise all kinds of farm animals.

Economy

All the benefits listed lead to one major field – economic growth. Computers and mobile phones are used extensively in industries for communication and other purposes such as report writing, accounting and database. Transportation helps save valuable, productive time; education help expose people to many fields of the economy; when people are healthy they can work more hours and faster; and finally, with agriculture being the number one employer in Ghana and Africa at large, advances in it invariably improves the economy of the country.

ENDOGENOUS TECHNOLOGY

Endogenous technology covers technologies that make the best usage of the local resources of a particular country or region. It may include foreign or exogenous technology, but such technologies must be adapted to suits the particular needs, conditions and resources of the developing country.

The development of endogenous technology almost always requires innovation and is therefore as much a challenge for well-qualified personnel as in the development of some of the most sophisticated new technologies.

Industries which make use of endogenous technology in Ghana

1. The akpeteshie or local gin distillation industry
2. The kente weaving industry
3. Local soap manufacturing industry
4. Black smith
5. Wood carving industry
6. Small-scale mining or ‘galamsey’ industry
7. Herbal medicine production
8. Food preparation and processing
9. Batik tie and dye industry

Problems associated with endogenous technology development in Ghana

The development of endogenous technology in Ghana suffers huge setbacks. The following are some problems with endogenous technology in Ghana.

Imports of foreign goods

Imported goods, especially those from Eastern and Southern Asia (chiefly China and India) usually have low prices compared to those locally made. This gives those goods better patronage than those made in the country.

Lack of technological development

Most local manufacturers rely on the age-old technologies handed down from their ancestors. Those technologies are often less productive as they require more manual labour. The wood carving industry, for instance requires the carver to use axe to shape the wood instead of faster means such as the use of chain saws or other mechanized means.

Lack of highly qualifies of skilled personnel

Endogenous technology requires highly qualified and skilled engineers or technicians, since a high degree of competence is needed to determine how certain technologies can be adapted to suit different environments in countries. However, that is not the situation on the ground. Most of the people who work at the various fields lack the expertise needed to make the industry going.

Lack of capital

Though endogenous technology demands relatively lesser capital, some basic tools and materials have to be purchased. Most of the endogenous technology industries are located in rural areas where there is little or no financial institutions to assist. People therefore have to depend on the little resources they have. This results in less production.

Transportation problems

Most of the roads in the rural areas where endogenous technology industries are based are in a deplorable state. People find it extremely difficult conveying their raw materials and products to their production sites and marketing centres. Most of the roads are virtually non-motorable during the rainy season.

SMALL-SCALE INDUSTRIES

A great majority of industries which rely on endogenous technology are small-scale.

Features of small-scale industries

1. Small-scale industries normally operate on little resources.
2. They rely on low starting capital because of lack of access to credit.
3. They normally employ fewer people, in some cases a nuclear family may be the only workers on the field.
4. The workers usually lack technical know-how in their chosen fields.
5. The workers have little or no formal education.
6. The products are only known and patronized by a few people.
7. They are labour intensive, employing more labour per unit of capital than large enterprises.

Examples of small-scale industries

a) Soap making
b) Vegetable oil extraction
c) Palm oil extraction
d) Bread making
e) Salt extraction
f) Pomade making
g) Herbal medicine extraction

Importance of small-scale industries in Ghana

1. Small-scale industries are the major employers in Ghana.
2. They promote indigenous technological know-how.
3. They are a huge contributor to the economic growth of the country.
4. They alleviate or reduce poverty in Ghanaian societies.
5. They reduce rural-urban migration.
6. They are able to compete (but behind protective barriers).
7. They cater for the needs of the poor.
8. They adapt easily to customer requirements.
9. They serve as a source of foreign exchange.
10. They help in developing deprived areas of the country.

Preparation of soap

Soap is a combination of animal fat or plant oil and caustic soda. When dissolved in water, it breaks dirt away from surfaces. Through the ages soap has been used to cleanse, to cure skin sores, to dye hair, used as perfume and skin ointment.

Raw materials needed

Soap requires two major raw materials: ***fats*** and ***alkali***.

Alkalis used in soap making

1. Sodium hydroxide
2. Potassium hydroxide or potash (ash of dried plantain peel).
3. Ash of coconut husk
4. Ash of dried cocoa pod

Fats used in soap making

1. Palm oil
2. Olive oil
3. Palm kernel oil
4. Coconut oil
5. Animal fat

The Manufacturing Process

- Fats and alkali are melted in a container.
- After boiling, the mass thickens as the fat reacts with the alkali (saponification), producing soap and glycerine.
- The mixture is treated with salt, causing the soap to rise to the top and the glycerine to settle to the bottom.
- The glycerine is extracted from the bottom of the kettle.
- To remove the small amounts of fat that have not saponified, a strong caustic solution is added to the kettle.

- The next step is called ***pitching***. The soap in the container is boiled again with added water.
- The mass eventually separates into two layers. The top layer is called ***neat soap***, which is about 70% soap and 30% water. The lower layer, called ***nigre***, contains most of the impurities in the soap such as dirt and salt, as well as most of the water.
- The neat soap is taken off the top. The soap is then cooled.

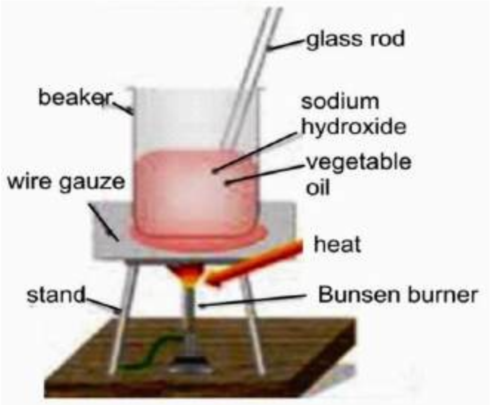

Fig. 105.9: Soap preparation

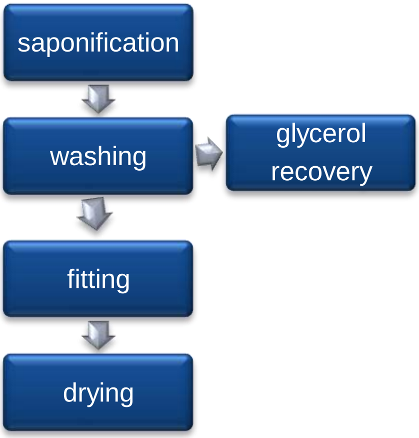

Fig. 106.0: Flow chart of soap preparation

Types of soap

Soft soap – These are soaps which are very soluble in water. They are produced from potassium-based alkalis.

Hard soap – These are soaps which are not readily soluble in water. These soaps are produced from sodium-based alkalis.

Palm oil production

Only fully ripe palm fruits are harvested and extracted into oil.

Procedure

- Boil the palm nut fruits until they are soft and cracking open. The flesh of the fruits should be soft and tender.
- Grind the fruits with mortar and pestle. Do not break the nut, but grind the palm nuts with the pestle so the juices flow out of it. You will have a big mush when you are finished.
- Mix the mush with water and filter the solution through a sieve.
- Boil the palm nut fruit juice. As you boil the juice, the oil will separate and float on the surface of the water.
- Use a spoon to scrape the oil off the surface of the water. Ladle it into an airtight container.

Fig. 106.1: Summary of palm oil extraction

Fig. 106.2: People preparing palm oil

Palm kennel oil production

The palm kennel extracted from the palm fruit is used for the production of palm kennel oil.

Procedure

- ❖ Clean dried kernels with water. Make sure they are free of any dirt or gravel.
- ❖ Shell the kernels with a nutcracker to separate the shells from the kernels.
- ❖ Heat the nuts in an oven until roasted.
- ❖ Grind in a blender or pound with a hand mortar until it is a pasty consistency.
- ❖ Mix the paste with water heated to boil in a bowl. Boiling water separates the oil from the palm paste. The oil will float to the top.
- ❖ Skim the palm kennel oil off the top of the cooled water with a spoon.
- ❖ Pour the oil in the jar and close tightly.

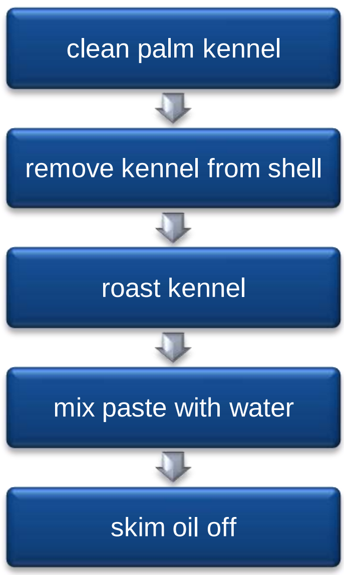

Fig. 106.3: Palm kennel oil extraction

Scientific processes involved in some industries

Salt production

There are two main scientific processes involved in salt production – ***evaporation*** and ***crystallization***.

Procedure

- Brine or seawater, which has high salt concentration, is pumped into a shallow pond.
- With the help of the sun and dry wind, the water is evaporated from the salt.
- Insoluble impurities such as sand and clay and slightly soluble impurities such as calcium carbonate settle to the bottom as evaporation begins.
- The remaining brine is moved to yet another pond where the salt settles out as evaporation continues.
- Iodine, which helps prevent the thyroid disease (goitre), is added to the solution. The water is finally allowed to evaporate and the salt crystallized.
- The final product is then bagged.

Fig. 106.4: Shovelling crystallized salt

Yoghurt production

Yoghurt is a dairy product, which is made by blending fermented milk with various ingredients that provide flavour and colour. Main scientific processes involved in the production of yoghurt include ***homogenization*** and ***pasteurization***.

Raw Materials

- Milk
- Sugars
- Stabilizers
- Fruits and flavours
- Bacterial culture

Procedure

- ***Blend Ingredients***: The ingredients are mixed or adjusted.
- ***Pasteurization:*** The milk mixture is pasteurized (heated) at 85°C for 30 minutes.
- ***Homogenization:*** The blend is homogenized to mix all ingredients thoroughly to improve yogurt consistency.
- ***Cooling:*** The milk is cooled to 42°C to bring the yogurt to the ideal growth temperature for the starter culture.
- ***Inoculation with bacteria cultures***
- The starter cultures are mixed into the cooled milk.
- ***Incubation:*** The milk is held at 42°C until a pH 4.5 is reached. This allows the fermentation to progress to form

a soft gel and the characteristic flavour of yogurt. This process can take several hours.

- ***Stirring and cooling:*** The yogurt is stirred and cooled to 7°C to stop the fermentation process.
- ***Addition of Fruit & Flavours:*** Fruit and flavours are added at different steps depending on the type of yogurt. The yogurt is pumped from the fermentation vat and packaged as desired.

TEST QUESTIONS

1. Explain, with examples, how science and technology have helped to improve the living standards of human kind

2. (a) State five differences between science and technology.
   (b) Explain four significance of science and technology.

3. (a) What is endogenous technology?
   (b) Mention five industries which use endogenous technology in Ghana.

4. Explain four problems associated with endogenous technology in Ghana.

5. Briefly describe the production of soap.

6. How is palm oil extracted?

7. Describe the production of palm kennel oil.
8. The main processes involved in salt production are evaporation and crystallization. Describe how they occur.

9. Describe the following steps involved in yoghurt production:
   (a) pasteurization;
   (b) homogenization
   (c) incubation

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BIOTECHNOLOGY

Specific Objectives

After completing this chapter, you will be able to:

- Explain the term biotechnology.
- Give examples of industrial bases biotechnology.
- Explain the meaning of genetic engineering and list examples of its application.
- Explain the term tissue culture and give examples of it application.

INTRODUCTION

For thousands of years, humankind has used biotechnology in agriculture, food production and medicine. Whenever a farmer puts a seed into the soil, he is practicing the concept of basic cloning; whenever corn dough is left overnight before using it, we are practicing fermentation and whenever a herbalist collects different plants and put them together as one medicine, he or she is using a form of biotechnology.

These processes are traditional biotechnological methods.

On the large scale, biotechnology may be used for various industrial processes. The question is what is biotechnology?

MEANING OF BIOTECHNOLOGY

Biotechnology is the use of living systems and organisms to develop or make useful products.

It involves the practice of using plants, animals and micro-organisms such as bacteria, as well as biological processes - such as the ripening of fruit or the bacteria that break down compost, to some benefit.

Biotechnology is not only limited to biological fields, but links biology to the other fields of science. It can thus be defined as utilizing the sciences of biology, chemistry, physics, engineering, computers, and information technology to develop tools and products that hold great promise.

Both traditional and modern biotechnology share the same foundation - the use of living organisms to enhance crops, fuels, medical treatments and a host of other tools that can help humans.

EXAMPLES OF INDUSTRIES BASED ON BIOTECHNOLOGY

Biotechnology has applications in most major industrial areas, including:

1. healthcare (medical),
2. crop production and agriculture,
3. non food (industrial) uses of crops and other products (e.g. biodegradable plastics, vegetable oil, biofuels),
4. environmental uses.

Importance of yeast in bread making

Yeast is a tiny plant-like microorganism which serves as a catalyst in the process of fermentation. In the production of bread, yeast is a key ingredient and serves three primary functions:

Production of carbon dioxide

Carbon dioxide is generated by the yeast as a result of the breakdown of fermentable sugars in the dough. The evolution of carbon dioxide causes expansion of the dough as it is trapped within the protein matrix of the dough.

Causes dough maturation

This is accomplished by the chemical reaction of yeast to produce alcohols and acids on protein of the flour and by the physical stretching of the protein by carbon dioxide gas. These result in the light, airy physical structure associated with yeast leavened products.

Development of fermentation flavour

Yeast imparts the characteristic flavour of bread and other yeast leavened products. During dough fermentation, yeast produces many secondary metabolites such as ketones, higher alcohols, organic acids, aldehydes and esters.

Some of these alcohols for example, escape during baking. Others react with each other and with other compounds found in the dough to form new and more complex flavour compounds. These reactions occur primarily in the crust and the resultant flavour diffuses into the crumb of the baked bread.

Fig. 106.5: Gaps left by escaped CO_2 in bread

Uses of yeast in brewing industry

- ❖ It metabolises the sugars extracted from grains, which produces alcohol and carbon dioxide, and thereby turns wort into beer.

- In addition to fermenting the beer, yeast influences the character and flavour. The dominant types of yeast used to make beer are *Saccharomyces cerevisiae*, known as ale yeast, and *Saccharomyces uvarum*, known as lager yeast.
- If the brewer wishes to increase the alcohol content of his beer, he only has to add more yeast of a specific variety.

Penicillium Sp. in drug and food industries

Penicillium is a genus of *Ascomycetous* fungi of major importance in the natural environment as well as food and drug production.

Several species of the genus Penicillium play a central role in the production of cheese and of various meat products. *Penicillium nalgiovense* is used to improve the taste of sausages and hams, and to prevent colonization by other moulds and bacteria.

In addition to their importance in the food industry, species of Penicillium and Aspergillus serve in the production of a number of biotechnologically produced enzymes and other macromolecules, such as gluconic, citric, and tartaric acids, as well as several pectinases, lipase, amylases, cellulases, and proteases.

Fig. 106.6: Penicillium

Lactobacillus Sp. in food industry

Lactobacillus is a major part of the lactic acid bacteria group, named as such because most of its members convert lactose and other sugars to lactic acid.

Some Lactobacillus species are used for the production of yogurt, cheese, sauerkraut, pickles, beer, wine, cider, cocoa, and other fermented foods, as well as animal feeds, such as silage.

Sourdough bread is made using a starter culture, which is a symbiotic culture of yeast and lactic acid bacteria growing in a water and flour medium. Lactobacilli are some of the most common beer spoilage organisms.

The species operates by lowering the pH of the fermenting substance by creating the lactic acid, neutralising it to the desired extent.

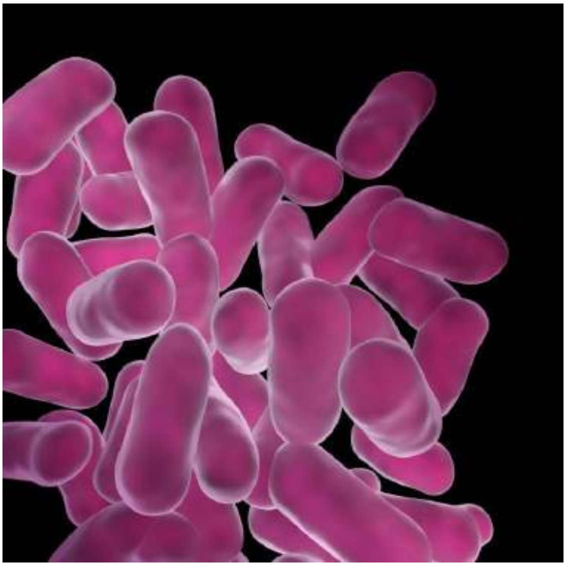

Fig. 106.7: Lactobacillus

GENETIC ENGINEERING

Genetic engineering is the manipulation of the genetic makeup of an organism in order to produce an organism of a desired trait.

This is done using techniques that remove heritable material or that introduce DNA prepared outside the organism either directly into the host or into a cell that is then fused or hybridized with the host.

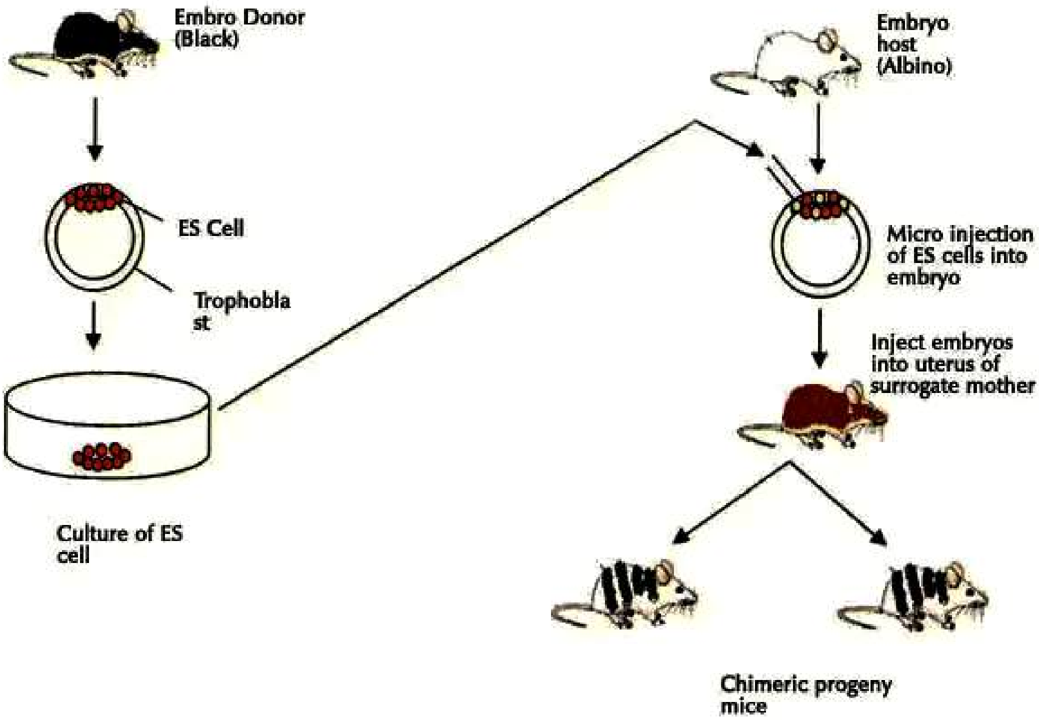

Fig. 106.8: Genetically engineering a mouse

Applications of genetic engineering

Genetic engineering techniques have been applied in numerous fields including research, agriculture, industrial biotechnology and medicine.

Applications in Agriculture

1. Genetically modified (GM) foods are foods which have been engineered to produce higher yield both in sizes and quantities.
2. Fertilizers and bio synthesizers which help in proper growth of crops at the same time killing the harmful bacteria have helped the agriculture sector.
3. When nutrients are added to the soil, the produce is of better quality and of higher quantity, both of which are very beneficial.
4. Soils which have been treated with organic matter, such as humus, do not lose their potential to grow more crops, which allows the agriculture process to be carried out all through the year.

Applications in Medicines

1. Majority of the findings of genetic research have proved to be of great significance to the medical world.
2. With different kinds of vaccines, antibodies and vitamins developed and easily available on the market, many diseases are now under control and can be treated. These elements can be injected into the bodies of the patient.
3. Chemotherapy and Radiology, which are very prominently used in cases of terminal diseases, are a gift of nothing but genetic research.
4. Treatment of heredity diseases too is possible by manipulating the genes in the human body before birth.
5. Helps in giving birth to babies of desired traits.

Applications for Humans

1. Scientists are pondering over the possibility of making children with only the desirable traits are possible after the successes in cloning.
2. It helps prevent some diseases and infections.
3. Babies who have deficiency could be treated with additions being done to their genetic structure.
4. people who cannot reproduce due to medical complications have also seen positive results with surrogate parent's concept becoming available.

Applications to the Environment

Organisms have been known to help in the bio-degradation of waste materials. However, there are some materials like plastics which cannot be degraded by them. To help such causes, genetic research has produced modified micro-organisms which not only have the capability of doing this but are also more efficient due to the speedy process.

Oceanophillis bacteria were popularly used in the 2010 BP oil spill in the Gulf of Mexico to eat up the spilled oil.

Applications in Industries

1. Synthetically produced items are now available on the markets which are used as raw materials by the industrialists.
2. Commercially viable items can also be produced by using the biological procedures like fermentation (in bakeries).
3. Genetic research has been able to tell exactly what percentage and what quantities of items should be used for optimum results making everything calculated and risk free.

TISSUE CULTURE

Tissue culture is a process that involves exposing plant tissue to specific nutrients, hormones, and lights under sterile conditions to produce many new plants, each a clone of the original mother plant, over a very short period of time.

This method is known as ***micro-propagation.***

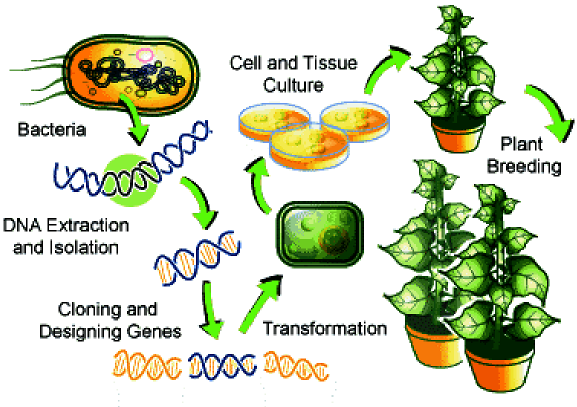

Fig. 106.9: Plant tissue culture

Advantages of tissue culture

1. It can create a large number of clones from a single seed or plant.
2. It is easy to select desirable traits directly from the culture setup (in vitro), thereby decreasing the amount of space required for field trials.
3. The time required is much shortened, no need to wait for the whole life cycle of seed development.
4. For species that have long generation time, low levels of seed production, or seeds that do not readily germinate, rapid propagation is possible.
5. It overcomes seasonal restrictions for seed germination.
6. It enables the preservation of pollen and cell collections form which plants may be propagated (like a seed bank).
7. It allows for the international exchange of sterilized plant materials (eliminating the need for quarantine.)

8. It helps to eliminate plant diseases through careful stock selection and sterile techniques.
9. It enables cold storage of large number of viable plants in a small space.
10. If there is plant with partially infected tissue, it is possible to produce a new plant without infection.

Disadvantages of tissue culture

1. Costs of the equipments are very expensive in large-scale production.
2. The procedure is very variable and it depends on the type of the species so sometimes it needs trial-and-error type of experiments if there is not any review about that species.
3. The procedure needs special attention and diligently done observation.
4. There may be error in the identity of the organisms after culture.
5. Infection may continue through generations easily if possible precautions are not taken
6. Decrease genetic variability

TEST QUESTIONS

1. Explain the term biotechnology.

2. Describe the importance of biotechnology in the following industries:
   (a) bread making industry;
   (b) brewing industry;
   (c) food industry.

3. (a) What is genetic engineering?
   (b) Explain the application of genetic engineering in the following fields:
      (i) agriculture;
      (ii) medicine;
      (iii) environment.

4. (a) Define the term tissue culture.
   (b) State six advantages and four disadvantages of tissue culture.

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WORK AND MACHINES

Specific Objectives

After completing this chapter, you will be able to:

- Explain work, energy and power.
- Identify simple machines.
- Explain the terms mechanical advantage, velocity ratio and efficiency of machines.
- Explain the effects of friction and methods of reducing friction in machines.

INTRODUCTION

Since prehistory, humans have been working day-in and day-out for various reasons. To make the work they do easier and effective, they have invented numerous machines.

The dawn of civilization brought with it even more complex machines. Every work, no matter how simple or complex, has got a machine with which the work is done efficiently and with ease.

WORK, ENERGY AND POWER

Work

***Work is done on an object when an applied force moves it through a distance in the direction of the force*.**

$$\textbf{\textit{Work}} = \textbf{\textit{Force}} \times \textbf{\textit{distance}}$$

Or

$$\boldsymbol{W = F \times s}$$

Work is a scalar quantity. The SI unit for work is the ***joule* (J)**, which is newton-meter (Nm).

When work is done on a body, there is a transfer of energy to the body, and so work can be said to be ***energy in transit***.

If, for example, an object is lifted from the floor to the top of a table, work is done in overcoming the downward force of gravity, and the energy imparted to the body as work will increase its potential energy.

Examples

1. A boy applies a force of 30N to move a wooden box across the floor over 2m. Calculate the work done.

Solution

Work (W) = force (F) x distance (s)

W = 30 N x 2 m = 60 J

$\therefore$ Work done = 60 J

2. A bag of cement of mass 50 kg is raised to a height of 4 m. Find the work done in raising the cement.
   [g = 10 m s^{-2}]

Solution

W = F x s

F = m x g

m = 50 kg, g = 10 m s^{-2}

F = 50 x 10 = 500 N

s = 4 m

W = 500 x 4 = 2000 J = 2 kJ

3. 1.2kJ of work is done in pushing a load over a distance of 80m. Determine the mass of the load.
   [Take g = 10ms^{-2}]

Solution

$$Mass = \frac{Force}{g}$$

$$Force = \frac{Work}{distance}$$

Work = 1.2kJ = 1.2 x 1000 = 1200J

Distance = 80 m

$$\therefore Force = \frac{1200}{80} = 15 \text{ N}$$

$$Mass = \frac{15}{10} = 1.5 \text{ kg}$$

Energy

Energy is the ability to do work.

Energy can also be defined as the capacity of matter to perform work as a result of its motion or its position in relation to forces acting on it.

Like work, energy is also measured in ***joule*, J.**

NB: The various forms of energy have been treated in chapter 15.

Mechanical energy

Mechanical energy is defined as the energy possessed by a body by virtue of its position or motion

Or

Mechanical energy is the sum of potential and kinetic energies of a body or system.

Potential energy is the energy an object or a system has because of it position or condition.

Kinetic energy is the energy possessed by an object due to its motion.

Potential and kinetic energies are treated in detail in chapter 15.

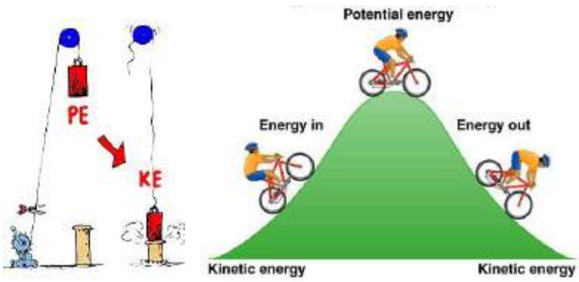


Fig. 107.0: Examples of PE and KE

Power

Power is the rate of doing work.

Or

Power is the rate of transferring energy.

$$Power = \frac{work\ done}{time\ taken}$$

Or

$$Power = \frac{energy\ transfered}{time\ taken}$$

Its S.I unit is the ***watt*** (**W**).
1 watt is equivalent to 1 joule per second (**Js^{-1}**).

Example

1. A man pulls a load with a force of 80N across a distance of 32m in 15s. Determine the
   i. work done;
   ii. power of the man.

Solution

(i) $Work\ done = \text{force} \times \text{distance}$

$= 80 \times 32$

$= 2560$ J

(ii) $$Power = \frac{work\ done}{time\ taken}$$

$$P = \frac{2560}{15} = 170.67\text{ W}$$

2. An electric motor is fed with 1200W of power. If it is allowed to run for two minutes, calculate the total work it can produce.

Solution

$$Power = \frac{work\ done}{time\ taken}$$

Work done = power x time

Power = 1200W

Time = 2 minutes = 2 ×60 = 120s

Work done = 1200 × 120 = 144000J

Total work produced = 144kJ

3. An engine does 40kJ of work when it was supplied with 2.0kW of power. What is the total time with which the engine does the work?

Solution

$$Power = \frac{\text{work done}}{\text{time taken}}$$

$$Time\ taken = \frac{\text{work done}}{\text{power}}$$

Work done = 40kJ = 40000J

Power = 2.0kW = 2000W

$$\text{Time take} = \frac{40000}{2000} = 20\text{ s}$$

SIMPLE MACHINES

A simple machine is any device that enables work to be done more easily and effectively.

A force called ***effort*** is applied to the machine which responses by providing a greater force to move, lift or break a ***load***.

The work input to a simple machine is expressed as:

Work input = effort × distance moved by effort

The work done or work output of a simple machine is given as:

Work output = load × distance moved by load

Types of simple machines

The types of simple machines include:

- the lever,
- the pulley,
- the inclined plane,
- the wheel and axle,
- the screw,
- the wedge,
- gear.

Each machine affects the direction or the amount of effort needed to do work. Most mechanical machines, such as automobiles or power tools, are complex machines composed of many parts. However, no matter how complicated a machine is, it is composed of the combination of some of the simple machines.

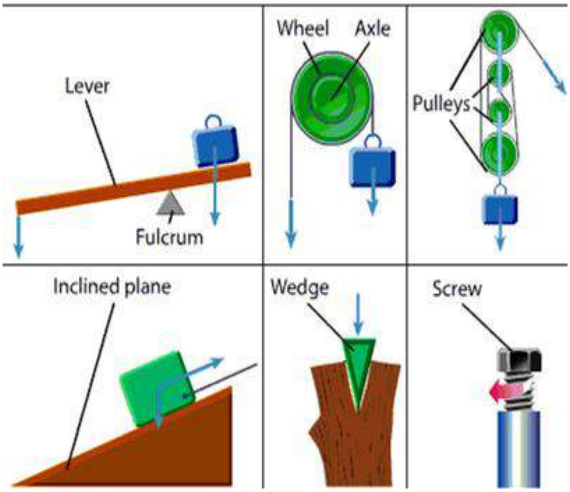


Fig. 107.1: Types of simple machines

Lever

A lever is a simple machine consisting of a rigid bar that rotates about a fixed point, called a ***fulcrum*** or ***pivot.*** Levers affect the effort needed to do a certain amount of work, and are used to lift heavy objects.

To move an object with a lever, a force (effort) is applied to one end of the lever, and the object to be moved (load) is usually located at the other end of the lever, with the fulcrum somewhere between the two.

The distance from the effort to the fulcrum is called the ***effort distance or arm*** and the distance from the load to the fulcrum is called the ***load distance or arm***.

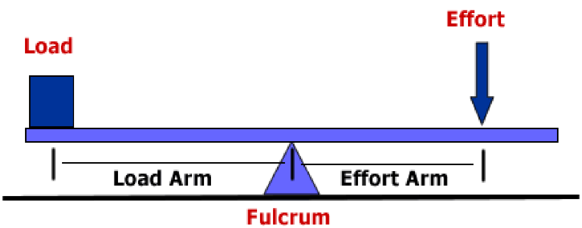


Fig. 107.2: Parts of a lever

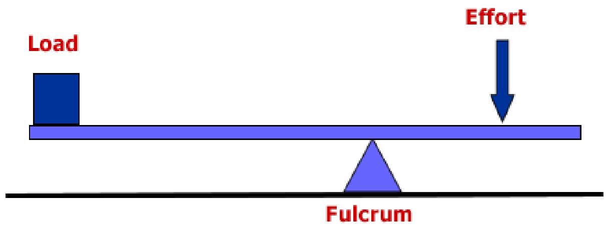


Fig. 107.4: First-class lever

The velocity ratio (VR) of a lever is the ratio effort distance to the load distance.

$$\text{VR} = \frac{effort\ \ distance}{load\ \ distance}$$

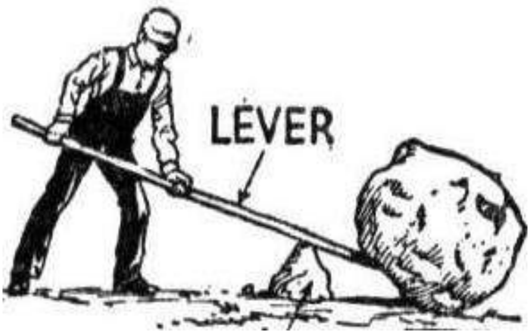


Fig. 107.3: Man lifting a rock with a lever

Types or classes of levers

There are three types or classes of levers, according to where the load and effort are located with respect to the pivot.

First-class lever

A first-class lever has the ***pivot*** placed between the ***effort*** and ***load***. If the effort goes upwards, the load goes downwards and vice versa.

Examples of first-class levers

- Scissors
- Pliers
- Seesaw
- Pincers
- Claw hammer
- Shears
- Beam balance
- Catapult

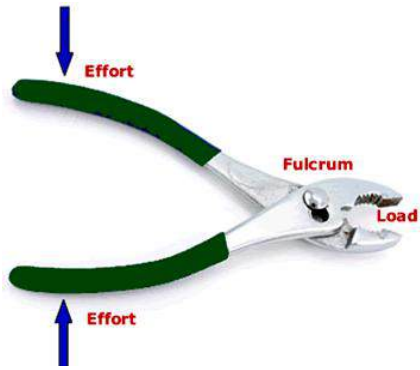


Fig. 107.5: Pliers is an example of first-class lever

Second-class lever

A second-class lever has the ***load*** in between the ***effort*** and the ***pivot.*** In this type of lever, the movement of the load is in the same direction as that of the effort.

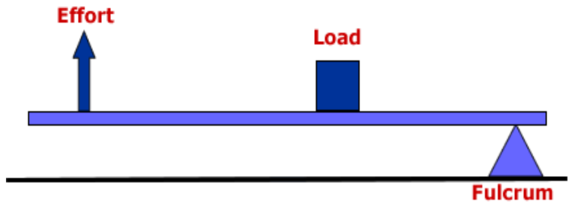


Fig. 107.6: *Second-class lever*

Examples of second-class levers

- Wheelbarrow
- Crowbar
- Nut cracker
- Bottle opener

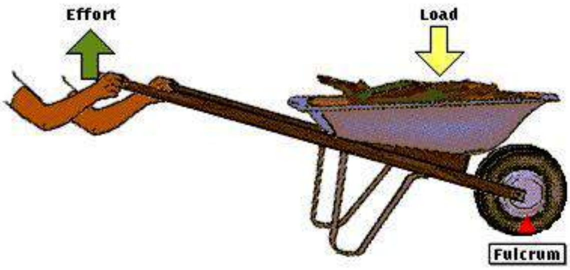


Fig. 107.7: *Wheelbarrow, an example of second-class lever*

Third-class lever

A third-class lever has the ***effort*** in between the ***load*** and the ***pivot.*** Both the effort and load are in the same direction.

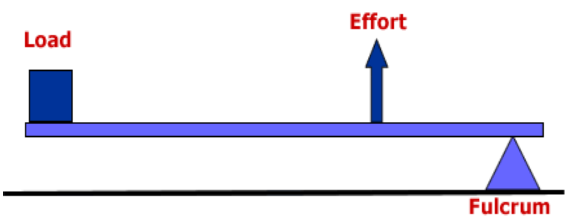


Fig. 107.8: Third-class lever

Examples of third-class levers

- Tweezers
- Stapler
- Mousetrap
- Broom
- Hockey stick

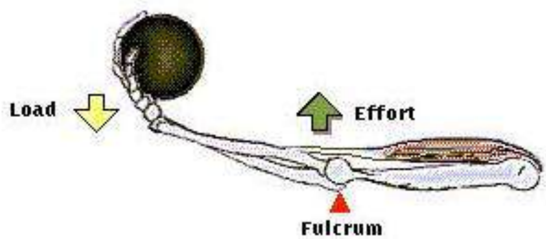
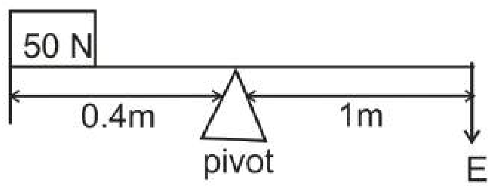


Fig. 107.9:

Example

1. A uniform bar 200 cm long is pivoted at the centre of gravity. A load of 50 N is hung 40 cm from the end of the bar. What is the magnitude of the effort placed at the 200 cm mark?

Solution



Moment about the pivot:

Effort x effort arm = load x load arm

Effort = ?

Effort arm = 40 cm = 0.4 m

Load = 50 N

Load arm = 100 cm = 1 m

$\therefore E \times 1 = 50 \times 0.4$

$E = \frac{20}{1} = 20$ N

2. A crowbar is used to lift a load of 200 N acting at a distance of 90 cm from the pivot.
   a) What is the value of the effort applied in at a distance of 2.3 m raising the load?
   b) Draw a diagram to show the force on the crowbar.
   c) Calculate the mechanical advantage of the crowbar
   d) What is the velocity ratio of the crowbar?

Solution

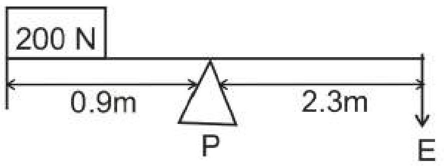
a) Moment about P:

$200 \times 0.9 = 2.3 \times E$

$180 = 2.3E$

$E = 78.26$ N

b)



c) Mechanical advantage of the crowbar

$MA = \frac{Load}{Effort} = \frac{200}{78.26}$

$MA = 2.56$

d) $VR = \frac{effort\ distance}{load\ distance}$

$VR = \frac{2.3}{0.9} = 2.56$

Pulley

A pulley is a simple machine used to lift objects. A pulley consists of a grooved wheel or disk within a housing, and a rope or cable threaded around the disk. The disk of the pulley rotates as the rope or cable moves over it. Pulleys are used for lifting by attaching one end of the rope to the object, threading the rope through the pulley (or system of pulleys), and pulling on the other end of the rope.

Pulley systems

Two common types of pulley systems are the chain hoist and the block and tackle systems.

Chain hoist system, also called ***fixed pulley*** is usually operated by hand. Take a look at a flagpole. That is an example of a fixed pulley. Because of the pulley at the top, the person raising the flag can stand on the ground and hoist the flag by pulling down on the rope.

Fig. 108.1: Fixed pulley

The velocity ratio of a fixed pulley is given by the ratio of the radius of the axel (r) to the radius of the wheel (R).

$$VR = \frac{R}{r}$$

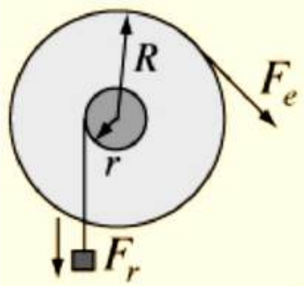


Fig. 108.2:

Block and tackle system

Block and tackle system is often used with an engine or motor. This type of pulley system is called a 'block and tackle', where 'block' refers to the pulleys and 'tackle' is the chain that the person is pulling to lift the load.

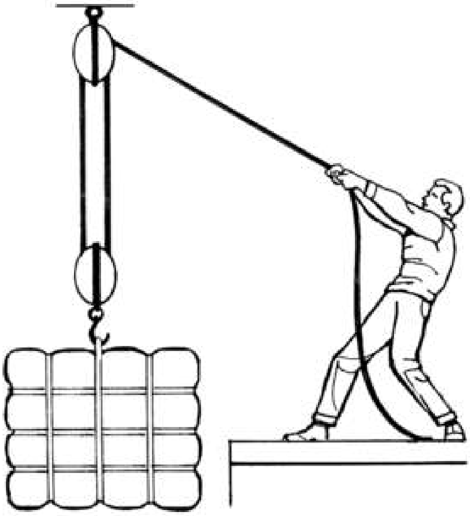


Fig. 108.3: A man pulling a load with a block and tackle system

The V.R. of the block and tackle system is equal to the number of pulleys present in the system.

For example if there are 2 pulleys present, the V.R. is 2; if there are 4 pulleys, the V.R. is 4 and so on.

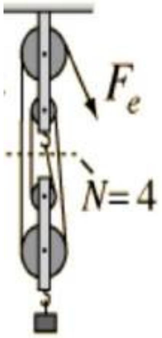


Fig. 108.4: Pulley system with VR = 4

Inclined plane

Inclined plane is a simple machine consisting of a flat surface that is at an angle to the load.

An inclined plane makes it easier to lift heavy objects by enabling a person to apply the necessary force over a greater distance. The same amount of work is accomplished in lifting the object with or without the inclined plane, but because the inclined plane increases the distance over which the force is applied, the work requires less force.

Examples of inclined planes are ramp for wheelchairs and stare case.

$$\textit{Velocity ratio} = \frac{\textit{length of plane}}{\textit{height of plane}}$$

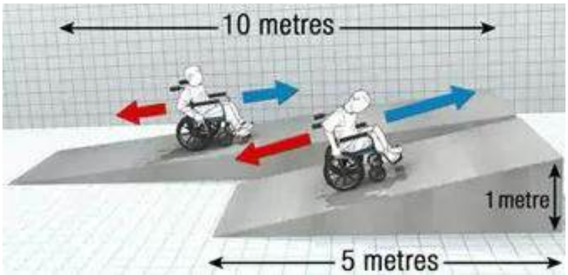


Fig. 108.5: *Inclined plane*

Wheel and axel

Wheel and axle is a simple machine, consisting of a circular object (the wheel) with a shaft (the axle) running through and attached to the centre of the wheel.

A round doorknob and a round water tap are both examples of wheels and axles. The much larger handle turns a much smaller axle to move a door latch, in the case of a doorknob, or open a water valve, in the case of a tap. A wheel and axle makes work easier by changing the amount and direction of the force applied to move (or in this case turn) an object.

$$\textit{Velocity ratio} = \frac{\textit{Radius of wheel}}{\textit{Radius of axel}} = \frac{R}{r}$$

Wedge

A wedge is a form of inclined plane. A wedge is essentially a double inclined plane, where two planes are joined at their bases. The joined inclined planes form a blunt end that narrows down to a tip. Wedges are often used to split materials such as wood or stone. Since

The main benefit of the wedge is changing the direction of effort to help split or cut through an object. A knife and an axe are forms of wedge. The wedge shape of their edges helps the user cut through materials.

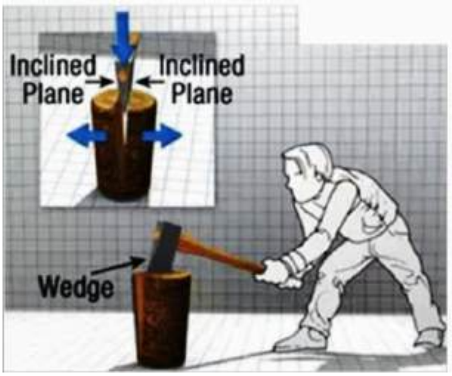


Fig. 108.6: Axe, an example of a wedge

Screw

A screw is an inclined plane wrapped around an axis, or pole. The edge of the inclined plane forms a helix, or spiral, around the axis. Screws are often used to raise objects.. The screw requires a lot of turning, which equates with effort applied over a long distance; this allows heavy loads to be lifted with a small amount of effort. Screws are also useful as fastening devices.

Screws driven straight into wood or other materials, as well as threaded nuts and bolts take advantage of the friction that results from the contact between the inclined plane and other objects. These devices use friction to hold things together.

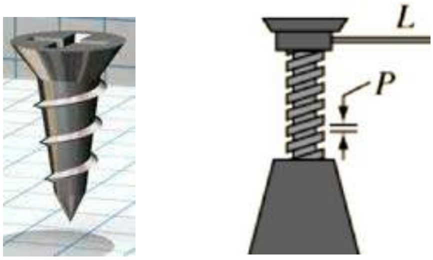


Fig. 108.7: A screw

The mechanical advantage of a screw is related to the circumference of the screw divided by the *pitch* of the threads.

$$VR = \frac{2\pi L}{P}$$

The ***pitch*** of a thread is the distance along the axis of the screw from one thread to the next.

Gears

Gears are a type of simple machine consisting of wheels with ***teeth***. The teeth fit together so that when one gear turns it also turns the other gear. Sometimes the gears fit right together, and sometimes they work together through a chain or a belt. Gears are modified form of the pulley and the wheel and axel.

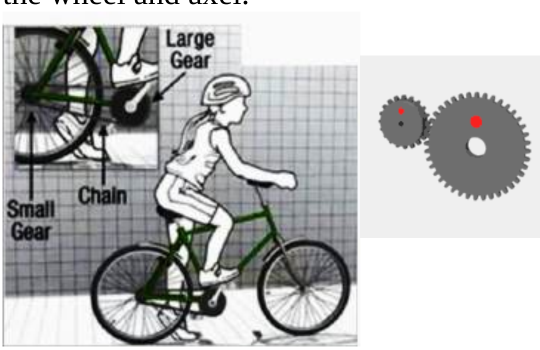


Fig. 108.8: Gear system of a bicycle

$$VR = \frac{number\ of\ teeth\ in\ driving\ gear}{num,ber\ of\ teeth\ in\ driven\ gear}$$

MECHANICAL ADVANTAGE, VELOCITY RATIO AND EFFICIENCY OF A MACHINE

Mechanical advantage

Mechanical advantage (MA) is the ratio of load to effort.

$$MA = \frac{load}{efort}$$

The mechanical advantage (MA) of simple machine measures how much the machine magnifies the effort applied to it.

Example

A force of 200N is applied to a machine to raise a load of mass 30kg. Calculate the mechanical advantage of the machine.
[Take g = 10ms^{-2}]

Solution

MA = $\frac{load}{efort}$

Effort = 200N

Load = mg = 30 x 10 = 300N

MA = $\frac{300}{200} = 1.5$

Velocity ratio

Velocity ratio (VR) is the ratio of the distance moved by effort to the distance moved by load.

$$VR = \frac{distance\ moved\ by\ effort}{distance\ moved\ by\ load}$$

Or

$$VR = \frac{effort\ distance}{load\ distance}$$

Example

An effort of 180N is applied 2m away from a load of 260N to move it onto a truck 1.3m high. Determine the

(a) velocity ratio

(b) mechanical avantage.

(c) efficiency

Solution

(a) $VR = \frac{effort\ \ distance}{load\ \ distance}$

Effort distance = 2m

Load distance = 1.3m

$VR = \frac{2}{1.3} = 1.54$

(b) $MA = \frac{load}{efort}$

Load = 260N, effort = 180N

$MA = \frac{260}{180} = 1.44$

(c) $Efficiency = \frac{mechanical\ \ advantage}{velocity\ \ ratio}$ x 100%

$MA = \frac{1.44}{1.54}$ x 100%

MA = 93.5%

Efficience

Efficiency is the ratio of mechanical advantage to velocity ratio expressed in percentage.

$$Efficiency = \frac{mechanical\ advantage}{velocity\ ratio} \times 100$$

Or

Efficiency is the percentage ratio of work output to work input of a machine.

$$Efficiency = \frac{work\ output}{work\ input} \times 100$$

Since work and energy are relative,

$$Efficiency = \frac{energy\ output}{energy\ input} \times 100$$

NB: The efficiency of a machine is always less than 100%. This is because part of the work input or effort is used to overcome some other forces such friction, gravity, drag and viscosity.

Examples

1. A simple machine is provided with 2000J of work. If it produces a total work of 1500J of work, calculate its efficiency.

Solution

$Efficiency = \frac{work\ \ output}{work\ \ input}$ x 100

Work output = 1500J

Work input = 2000J

$Efficiency = \frac{1500}{2000}$ x 100%

Efficiency = 75%

2. The efficiency of an earth-moving device is 82%. If its velocity ratio is 2.8, determine its mechanical advantage.

Solution

$Efficiency = \frac{mechanical\ advantage}{velocity\ \ ratio}$ x 100

Efficiency = 82%

VR = 2.8

MA = ?

$\therefore 82 = \frac{MA}{2.8}$ x 100

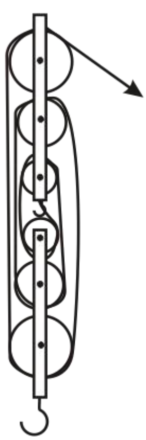
82 x 2.8 = 100 MA

MA = $\frac{229.6}{100}$ = 2.3

3. (a) Draw a pulley system with a velocity ratio of 6.
   (b) If the pulley system is used to raise a load of 5000 N with the application of a force of 3000 N. What is the efficiency of the pulley system.

Solution

(a)



(b) $Efficiency = \frac{mechanical\ advantage}{velocity\ \ ratio}$ x 100

VR = 6

MA = $\frac{load}{efort}$

Load = 5000N

Effort = 3000N

MA = $\frac{5000}{3000}$ = 1.67

$Efficiency = \frac{1.67}{6}$ x 100%

= 27.8%

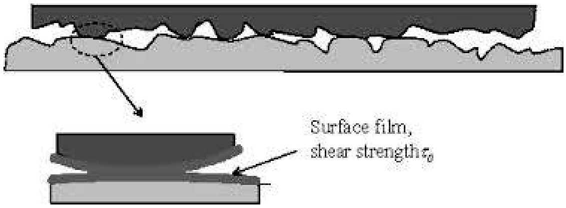
FRICTION

Friction is the force that resists the relative motion of two bodies in contact.

In other words, the force which prevents two objects in contact from sliding against each other is called friction.

Causes of friction

Friction occurs in part because rough surfaces tend to catch on one another as they slide past each other. Even surfaces that are apparently smooth can be rough at the microscopic level. They have many ridges and grooves. The ridges of each surface can get stuck in the grooves of the other, effectively creating a type of mechanical bond, or glue, between the surfaces.



Advantages or uses of friction

1. It prevents humans from slipping when walking or running.
2. It enables nails and screws to fasten pieces of objects together.
3. It prevents car and bicycle tyres from sliding on the road.
4. It enables automobiles to stop when the breaks are applied.
5. It causes the hand to grip firmly to objects.
6. It helps to light match.
7. It helps grind food into required pieces.
8. It enables sharpening of tools.
9. It helps humans to write or produce desired marks with pens and pencils

Effects or disadvantages of friction

1. It slows down motion.
2. It causes unnecessary heat, especially in machines.
3. It causes vehicle tires to wear.
4. It shrinks and tears the sole of shoes.
5. It reduces efficiency of a machine.
6. It leads to melting, warbling and breaking of machine parts as a result of overheating.

Methods of reducing friction

Reducing the amount of friction in a machine increases the machine's efficiency. Less friction means less energy lost to heat, less noise, and less wearing down of material.

There are two main methods of reducing friction. The first method involves reducing the roughness of the surfaces in contact. For example, sanding two pieces of wood lessens the amount of friction that occurs between them when they slide against one another.

Applying a lubricant to a surface can also reduce friction. Common examples of lubricants are oil and grease. They reduce friction by minimizing the contact between rough surfaces. The lubricant's particles slide easily against each other and cause far less friction than would occur between the surfaces.
Lubricants such as machine oil reduce the amount of energy lost to frictional heating and reduce the wear damage to the machine surfaces caused by friction.

The use of ball bearings is also an ideal method of reducing friction.

In summary, friction can be reduced by:

1. polishing
2. oiling
3. greasing

4. ball-bearing
5. powdering
6. streamlining (reduces fluid friction or viscosity)

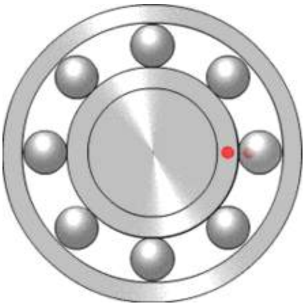


Fig. 108.9: Ball bearing

TEST QUESTIONS

1. Explain the following terms:
   (a) work;
   (b) energy;
   (c) power.

4. (a) Differentiate between potential energy and kinetic energy.
   (b) An orange falls to the ground from a branch which is 4 m above the ground. If the mass of the orange is 20 g and rolls on the ground at 12 ms^{-1}, calculate the:
   (i) potential energy of the Orange;
   (ii) kinetic energy of the orange.
   [take g = 10 ms^{-2}]

5. (a) What are simple machines?
   (b) Discuss the following types of simple machines:
   (i) lever;
   (ii) pulley;
   (iii) inclined plane;
   (iv) wheel and axle.

6. Explain the following classes of lever and give two examples each:
   (a) first-class lever;
   (b) second-class lever;
   (c) third-class lever.

7. Explain why the efficiency of a machine is always less than 100%.

8. (a) Define the term friction.
   (b) Explain the causes of friction.

9. (a) Mention five advantages and four disadvantages of friction.
   (b) Describe the three methods used to reduce friction in machines.

10. A boy who weighs 120 N carries a load of 50 N up a fright of stair which is 5 m high in 2 minutes. Calculate:
    (a) the work he does;
    (b) his power

11. Define the following terms:
    (a) mechanical advantage
    (b) velocity ration
    (c) efficiency

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METALS AND NON-METALS

Specific Objectives

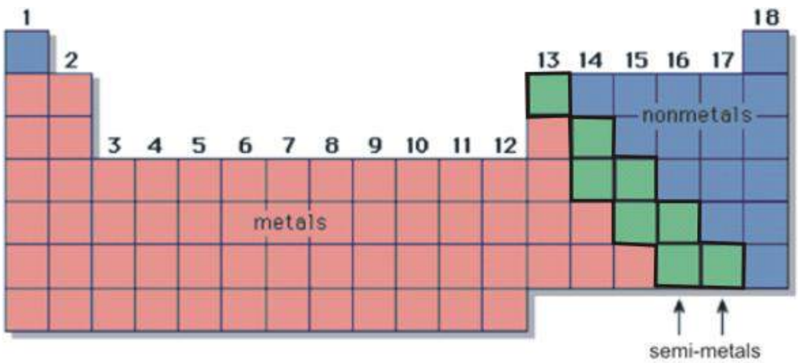
After completing this chapter, you will be able to:

- Classify chemical elements into metals, semi-metals and non-metals and relate them to their physical and chemical properties.
- Relate the properties of metals, non-metals and semi-metals to their uses in the home and other places.
- Explain what alloys are, their properties and uses

INTRODUCTION

All elements found on the periodic table fall under two main categories – ***metals*** and ***non-metals***. In between these two categories is another known as ***semi-metals***. Different elements fall under each of the categories based on their characteristics.

Metals and non-metals are separated on the periodic table by a diagonal line of elements. Elements to the left of this diagonal are metals, and elements to the right are non-metals. Elements that make up this diagonal have both metallic and non-metallic properties and are known as semi-metals. Metallic elements can combine with one another and with certain other elements, either as compounds, as solutions, or as intimate mixtures.



[See the back cover for the full list of the elements on the periodic table]

CLASSIFICATION OF ELEMENTS INTO METALS, SEMI-METALS AND NON-METALS

Metals

A metal is a chemical element that is a good conductor of both electricity and heat and forms cations and ionic bonds with non-metals.

In a metal, atoms readily lose electrons to form positive ions (cations). Those ions are surrounded by delocalized electrons, which are responsible for the (electricity and heat) conductivity. The solid thus produced is held by electrostatic interactions between the ions and the electron cloud, which are called ***metallic bonds***.

Metals are elements that undergo chemical reaction by donating electrons in other to be stable.

Examples of metals are sodium, potassium, calcium, lithium and magnesium.
These are known as ***reactive metals***. Other metals such as gold, copper, silver and lead are less reactive metals.

Properties of metals

Physical properties

1. Metals are good conductors of heat and electricity.
2. Metals have high melting and boiling points.
3. They are solid at room temperature except mercury.
4. They have high tensile strength. Thus can be stretched without breaking.
5. Metals are sonorous. That is they produce sharp ringing sound when struck.
6. Metals are malleable. (Can be beaten to any shape).
7. They are ductile. (Can be drawn into thin wires).
8. They are usually hard except alkali metals such as sodium, potassium, calcium etc.
9. They have high density except alkali metals such as lithium.
10. Metals are lustrous (or shiny) in their pure state except gold and copper.

Chemical properties of metals

1. Metals react with chlorine to form solid metal chlorides, which can conduct electricity in their molten state.
   $Ca + Cl_2 \rightarrow CaCl_2$
   $2Na + Cl_2 \rightarrow 2NaCl$

2. Metals react with oxygen to form oxides over changing timescales
   Examples:
   $4\ Na + O_2 \rightarrow 2\ Na_2O$ (sodium oxide)
   $2\ Ca + O_2 \rightarrow 2\ CaO$ (calcium oxide)
   $4\ Al + 3O_2 \rightarrow 2\ Al_2O_3$ (aluminium oxide)

3. Metals react with water to form

hydroxides. Examples:
$Na + H_2O \rightarrow NaOH_2$
$Ca + H_2O \rightarrow CaOH_2$

4. Metals react with dilute acids to form salt and hydrogen. Examples:
$K + 2HCl \rightarrow KCl + H_2$
$Ca + H_2SO_4 – CaSO_4 + H_2$

5. Metals typically have low ionization potentials. This means that metals react easily by loss of electrons to form positive ions, or cations. Thus, metals can form salts (chlorides, sulphides, and carbonates, for example) by serving as reducing agents or electron donors.

6. Metals react chemically with non-metals to form compounds such as NaCl, KOH, CaCl etc.

Non-metals

Metals are said to be electron donor. When metals donate electrons, the class of elements which accepts or gain the electrons are non-metals.

Non-metals are defined as elements which undergo chemical reaction by accepting electrons in order to be stable.

Examples of non-metals are carbon, oxygen, chlorine, hydrogen, nitrogen, sulphur etc.

These elements are mostly located on the right hand side of the periodic table.

Properties of non-metals

Physical properties

1. Non-metals are poor conductors of electricity except carbon (graphite).
2. They are poor conductors of heat
3. They usually have low melting and boiling points.
4. They usually have low densities.
5. They are mostly brittle.
6. Non-metals are not sonorous.
7. They are not lustrous (do not have shiny surfaces).

Chemical properties of non-metals

1. Non-metals react with metals to give salt. Example,
   $Cl + K \rightarrow KCl$
2. Non-metals react with oxygen to give various oxides. Phosphorus, for example reacts with oxygen to give phosphorus oxide, a white crystalline solid which is used as a reducing agent.
   $P_2 + O_5 \rightarrow P_2O_5$

   Carbon reacts with oxygen in the presence of heat to give carbon dioxide (CO_2) or carbon monoxide (CO) in limited supply of oxygen.
   $C + O_2 \rightarrow CO_2$

   $C + \frac{1}{2} O_2 \rightarrow CO$

Heated sulphur reacts with oxygen to give sulphur dioxide (SO_2), a pungent smelling gas.

$S + O_2 \rightarrow SO_2$

3. Some non-metals such as carbon react with water (steam) to give a mixture of carbon monoxide and hydrogen gas known as ***water gas***. This gas is used for heating in industries.

$C + H_2O \rightarrow CO + H_2$

Chlorine gas reacts with water to produce oxochlorate acid, a bleaching and antiseptic solution

$Cl_2 + H_2O \rightarrow HOCl + HCl$

4. Most non-metals do not react with dilute acids, owing to the fact that they cannot donate protons or displace hydrogen from dilute acids.

5. Some non-metals react with hydrogen to form hydrides. For example phosphorus and nitrogen react with hydrogen to give hydride of phosphorus and nitrogen respectively.

$P + H_3 \rightarrow PH_3$

$N + H_3 \rightarrow NH_3$ (ammonia)

Semi—metals

Elements with characteristic in between metals and non-metals are known as semi-metals or ***metalloids***. Because of their unique properties, semi-metals are used primarily in the manufacturing of electronic components. Examples of semi-metals are boron, silicon, germanium, arsenic, antimony, tellurium, polonium, and astatine. These elements form a diagonal line from the upper right of the periodic table.

CHARACTERISTICS AND USES OF SOME COMMON METALS

Aluminium (Al)

Aluminium is the commonest metal in the earth's crust. It is a good conductor of heat, has fairly low density, does not corrode, can easily be mould into shape. This makes it ideal for making cooking utensils and other household objects, roofing sheets and overhead electric cables. It also has a high tensile strength therefore it is used in making ladders. Air craft bodies are made with aluminium. This is because of its lightness and the ability to form aluminium oxide, a protective film which prevents it from rusting upon reaction with oxygen.

Fig. 109.0: Aircraft body made of aluminium

Gold (Au)

Gold is a soft, dense, bright yellow metallic element. It is the most malleable and ductile of all the metals. It can easily be beaten or hammered and could be drawn into a thin wire. It is one of the softest metals and is a good conductor of heat and electricity. It is unaffected by air, heat, moisture, and most solvents and therefore does not corrode or rust.

Due to its properties, gold is used mostly for making jewellery and ornaments such as necklaces, wrist watches, earrings, bracelets etc. It is also used in electrical and electronic equipments because of its high conductivity, and electroplating because of its ability to withstand corrosion.

Fig. 109.1: The Ashanti King arrayed in gold regalia

Iron (Fe)

Iron is a metal with high density and tensile strength. It is good conductor of electricity and can easily be magnetized at normal temperature. Iron is used for construction work, tools, engines and automobile bodies. This is because of its high density and tensile strength; therefore, things made of iron are quite durable. Iron has a disadvantage of being very corrosive.

Copper (Cu)

Because of its many desirable properties, such as its conductivity of electricity and heat, its resistance to corrosion, its malleability and ductility, and its beauty, copper has long been used in a wide variety of applications.

Copper is very ductile; hence, it can be drawn into wires of any diameter. Because of its high tensile strength and conductivity; it can be used in outdoor power lines and cables, as well as in house wiring, lamp cords, and electrical machinery such as generators, motors, controllers, signalling devices, electromagnets, and other communication equipments.

Copper has been used for coins throughout recorded history and has also been fashioned into cooking utensils, and ornamental objects.

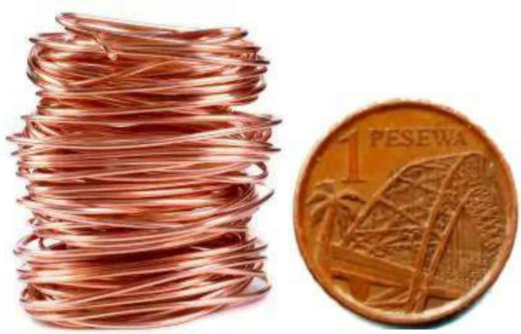


Fig. 109.2: Common uses of copper

Zinc (Zn)

Pure zinc is a crystalline metal, insoluble in hot and cold water and soluble in alcohol, acids and alkalis. It is extremely brittle at ordinary temperatures, but becomes malleable between 120°C and 150°C. Zinc is unaffected by dry air; in moist air it is oxidized and becomes coated with a carbonate film that protects it from further corrosion.

The metal is used principally as protective coating, or galvanizer, for iron and steel; as an ingredient of various alloys, especially brass; as plates for dry electric cells; and for die castings.

Zinc oxide, known as zinc white or Chinese white, is used as a paint pigment. It is also used as filler in rubber tires and is employed in medicine as an antiseptic ointment. Zinc chloride is used as a wood preservative and as a soldering fluid. Zinc sulphide is useful in applications involving electroluminescence, photoconductivity and semi-conductivity and has other electronic uses. It is employed as a phosphor for the screens of television tubes and in fluorescent coatings.

Silver (Ag)

Silver is a white, lustrous metallic element that conducts heat and electricity better than any other metal. It is highly malleable and ductile; softer than copper but harder than gold. It does not react with oxygen or water at ordinary temperatures.

Silver is used in jewellery and coinage. It is used to coat smooth glass surfaces for mirrors. Silver is also widely used in the circuitry of electrical and electronic components.

Fig. 109.3: Silverware

Nickel (Ni)

Nickel is a white magnetic metal which is hard, malleable, ductile and capable of taking a high polish. It resists corrosion.

Nickel is used as a protective and ornamental coating for metals, particularly iron and steel, that are susceptible to corrosion.

It is used as catalyst in many processes, including the hydrogenation of oils. Nickel is used chiefly in the form of alloys. It imparts great strength and corrosion resistance to steel. Nickel steel is used in automobile parts such as axles, crankshafts, gears, valves, and rods; in machine parts; and in armour plate.

Tin (Sn)

Tin is a highly ductile and malleable metal which resists corrosion. Tin is a widely sought metal and is used in hundreds of industrial processes.
In the form of tinplate, it is used as a protective coating for copper vessels, various metals used in the manufacture of tin cans, and similar articles.

Tin is important in the production of alloys such as bronze (tin and copper), solder (tin and lead), and type metal (tin, lead, and antimony). It is also used as an alloy with titanium in the aerospace industry and as an ingredient in some insecticides.

CHARACTERISTICS AND USES OF SOME COMMON NON-METALS

Oxygen (O)

Oxygen is a colourless, odourless, tasteless, slightly magnetic gaseous element. It is very combustive and is thus used in oxy-hydrogen torches for welding, and as a fuel for rockets. It is also used in metal fabrication industries. Animals require oxygen for respiration.

Carbon (C)

Carbon occurs in nature in nearly pure form in diamond and graphite. It is also the major component of coal, petroleum, asphalt, limestone, and most materials made by plants and animals. Allotropes (different forms) of carbon are used in industrial processes.

Diamond is used for making jewellery and because of its hard nature, used for cutting steel, glass and ceramics. Graphite is used as lubricant in machines and for electrodes. It is also used in controlling reactions in nuclear reactants. Coal is used as fuel for industrial and domestic purposes.

Hydrogen (H)

Hydrogen is a combustive gas which is lighter than air. It is used as fuel in rocket and space shuttles.
It is also used in oxy-hydrogen torches for welding, and for the manufacture of ammonia. Because of its light nature, hydrogen is used in filling balloons.

Nitrogen (N)

Nitrogen is a colourless, odourless, tasteless, nontoxic gas. It can be condensed into a colourless liquid, which can in turn be compressed into a colourless, crystalline solid.

Most of the nitrogen used in the chemical industry is obtained by the fractional distillation of liquid air. It is then used to synthesize ammonia.

From ammonia produced in this manner, a wide variety of important chemical products are prepared, including fertilizers, nitric acid, urea, hydrazine, and amines. In addition, an ammonia compound is used in the preparation of nitrous oxide (N_2O) a colourless gas popularly known as ***laughing gas***.

Mixed with oxygen, nitrous oxide is used as an anaesthetic for some types of surgery. Used as a coolant, liquid nitrogen has found widespread application in the field of cryogenics, and as a refrigerant.

Sulphur (S)

Sulphur is a tasteless, odourless, light yellow flammable solid. The most important use of sulphur is in the manufacture of sulphur compounds, such as sulphuric acid, sulphites, sulphates, and sulphur dioxide.

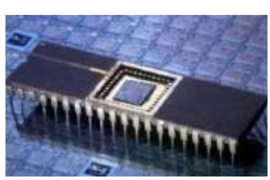
Medicinally, it has assumed importance because of its widespread use in sulpha drugs and in many skin ointments. Sulphur is also employed in the production of matches, vulcanized rubber, dyes, and gunpowder. In a finely divided state and frequently mixed with lime, sulphur is used as a fungicide on plants.

When combined with various inert mineral fillers, sulphur forms special cement used to anchor metal objects, such as railings and chains in stone.

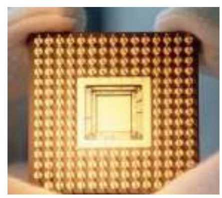
CHARACTERISTICS AND USES OF SEMI-METALS

At low temperatures, pure semiconductors behave like insulators. Under higher temperatures or with the addition of impurities, however, the conductivity of semiconductors can be increased dramatically, reaching levels that may approach those of metals.

Due to these properties of semi-metals, they are used mostly in electronic components such as transistors, diodes, microchips and integrated circuits (ICs). They are also used for making solar panels for trapping and converting solar energy into electrical energy.



An integrated circuit



A computer microchip

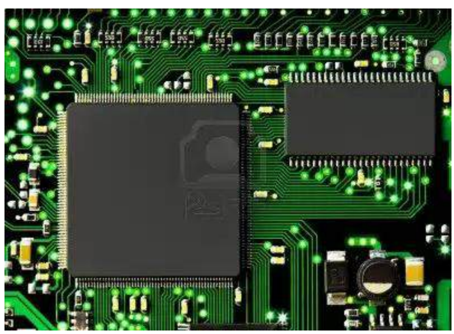


Fig. 109.4: Uses of semi-conductors

ALLOYS, PROPERTIES AND USES

An alloy is a substance composed of a uniform mixture of two or more metals or a metal and a non-metal.

The properties of alloys are frequently far different from those of their constituent elements. Alloys are built to possess some characteristics which are not already possessed by the constituent elements. In most cases, a property lacking in one element may be provided by another.

General properties or importance of alloys

1. Alloys have more tensile strength than their constituent elements.
2. They are more corrosive resistant.
3. They are often more durable.
4. They are usually cheaper than the constituent metals.
5. They suit the purpose for which they are used.
6. Alloys are able to sustain very high temperatures.

Table 7.6: Properties and uses of some alloys

Alloy	Constituent elements	Properties	Uses
Brass	Copper and zinc	Lustrous, non-corrosive, sonorous	Musical instruments, ornaments, screws,
Steel	Iron and carbon	High tensile strength, high density	Vehicle parts, constructing of buildings and bridges, machine parts
Stainless steel	Iron, carbon and chromium	High tensile strength, high density, resists corrosion	Cutlery set, food processing materials, surgical instruments
Bronze	Copper and tin	Non-corrosive, sonorous, lustrous	Ornaments, medals, machine parts, statues
Cupronickel	Copper and nickel	Non-corrosive, high tensile strength, lustrous	Machine part, coinage, ornaments
Solder	Lead and tin	Low melting point, soft	Joining electric circuit parts, joining metals
Duralumin	Aluminium and copper	Non-corrosive, strong, light	Aircraft parts, parts of automobile and railway carriages
Nichrome	Nickel and chromium	Resists corrosion, resistant to electric shock	Electroplating, resistant wires, tools

Fig. 109.5: An alloyed wheel rim

TEST QUESTIONS

1. Mention four physical properties and three chemical properties of metals.

2. List three physical properties and three chemical properties of non-metals.

3. (a) What are semi-metals?
   (b) Mention four semi metals.
   (c) Discuss the uses of semi metals

4. Describe the characteristics and uses of the following metals:
   (a) gold;
   (b) aluminium;
   (c) silver;
   (d) copper.

5. Describe the characteristics and uses of the following non-metals:
   (a) oxygen;
   (b) carbon;
   (c) hydrogen;
   (d) nitrogen.

6. (a) What are alloys?
   (b) Explain the importance of alloys.

7. Copy and complete the following table:

Alloy	*Constituent elements*	*Uses*
Solder		
Bronze		
Steel		
Brass		
Nichrome		

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EXPLOITATION OF MINERALS

Specific Objectives

After completing this chapter, you will be able to:

- Outline the effects of mineral exploitation on the environment.

INTRODUCTION

A mineral can be defined as a naturally occurring inorganic solid that possesses an orderly internal structure and a definite chemical composition.

Minerals can be described by variable physical properties, which relate to its chemical structure and composition. Common distinguishing characteristics include crystal structure and habit, hardness, lustre, diaphaneity, colour, streak, tenacity, cleavage, fracture, parting, and specific gravity.

More specific tests for minerals include reaction to acid, magnetism, taste or smell, and radioactivity.

Characteristics of a mineral

1. A mineral must occur naturally.
2. It must be inorganic.
3. It must be a solid.
4. It must possess an orderly internal structure; that is, its atoms must be arranged in a definite pattern.
5. It must have a definite chemical composition that may vary within specified limits.

Examples of minerals include gold, diamond, bauxite, manganese, etc. All these minerals are mined in various parts of Ghana.

Fig. 109.6: Gold bars and a diamond

MINERAL EXPLOITATION

Mineral exploration is the process of finding ore (commercially viable concentrations of minerals) to mine.

Mineral exploration activities are undertaken to determine the presence of geological formations which may contain deposits such as precious metals, base metals, gemstones, coal or other minerals, as well as to determine the extent, geometry and grade of such deposits.

Drilling, pitting, trenching and surface stripping are common activities undertaken during mineral exploration. Temporary work camps and docks are often established to support mineral exploration activities.

Mineral exploration methods vary at different stages of the process depending on size of the area being explored, as well as the density and type of information or mineral sought.

Fig. 109.7: Panning for diamond

In Ghana, mineral exploration is done by large registered companies such as Anglogold Ashanti as well as small-scale exploration called Galamsey.

Fig. 109.8: Large scale mining

Fig. 109.9: Galamsey operators at work

EFFECTS OF MINERAL EXPLOITATION

1. ***Destruction of landscape*** – Pits dug during mining are left uncovered and that destroys the structure of the land.

2. ***Air pollution*** – Dust which rises during excavation and milling pollutes

the air. Sulphur oxides cause acid rain which destroys vegetation whiles arsenic is poisonous to humans.

3. ***Pollution of water bodies*** – Chemicals used for the extraction process such as cyanides and mercury leak into nearby water bodies and pollute them. Cyanides prevent the human cells from using oxygen.

4. ***Land pollution*** – Solid wastes generated are normally left or dumped on the surface of the land; these could harm vegetation.

5. ***Mercury poisoning*** – Mercury vapour which is given off is very poisonous to humans and animals.

6. ***Noise pollution*** – Heavy machineries used for the mining and extraction produce lots of noise which can affect hearing.

7. ***Destruction of ecosystem*** – Habitats of organisms are normally disturbed. Aquatic life is destroyed when chemicals leak into water bodies, whiles plants and other small organisms on and inside the soil are destroyed.

8. ***Formation of slag*** – Slag such as red mud, which is high in alkaline, is produced from the filtration stage. Slag destroys natural vegetation.

Fig. 110.0: Lake polluted by mining activities

How to minimize the negative effect of mineral exploitation

1. Solid waste generated should be properly disposed of.
2. Chemicals should be used in moderation.
3. Chemicals should be prevented from leaking or running off into water bodies.
4. Strict enforcement of mining laws.
5. Communities where mining takes place should be educated about the negative effects of mining.
6. Mining companies should be made to pay compensation and royalties to mining communities.
7. The Environmental Protection Agency should monitor the activities of mining companies.
8. Mining communities should be provided with treated water instead of

drinking from water bodies which may be contaminated.
9. Reforestation of degraded forests and vegetation.
10. Mining companies should be made to periodically sprinkle mining areas with water to control the rise of dust particles.
11. Small-scale miners (Galamsey) should be educated on the right methods to extract minerals.

Fig. 110.1: Land left desolate after mining activities

Social and economic impact of mineral exploitation

1. Provision of income and foreign exchange
2. Provision of employment.
3. Attraction of investors to the country.
4. Provision of social amenities such as schools, hospitals, good roads, electricity etc. to mining communities.
5. Rise in living standard of mining communities and townships.

TEST QUESTIONS

1. What are minerals?
2. Mention four characteristics of minerals.
3. List four minerals mined in Ghana.
4. What is mineral exploitation?
5. Explain five effects of mineral exploitation.
6. Mention five methods of reducing the negative effects of mineral exploitation.

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EXTRACTION OF MINERALS

Specific Objectives

After completing this chapter, you will be able to:

- Describe the chemical extraction of aluminium and gold from their ores

INTRODUCTION

Most metals appear naturally as compounds in their ores. These ores are not useful for anything, therefore, in order to work with them, the pure metal have to be extracted from their ores.

Ores are naturally occurring rock containing high concentrations of one or more metals that can be profitably mined.

Ore minerals are the minerals within ores that contain the metal. Ores occur as large bodies of rock called ***ore deposits***, which are metal-bearing mineral deposits.

EXTRACTION OF ALUMINIUM

Aluminium is extracted from bauxite. Bauxite contains the hydrated oxide of aluminium ($Al_2O_3.3H_2O$), silica (SiO_2) and iron (II) oxide impurities.

Stages in the extraction of aluminium

There are two stages involved in the extraction of aluminium

1. Purification of bauxite to get alumina
2. Electrolysis of alumina

Purification of bauxite

- Dry and crush bauxite to powder.
- Dissolve or digest the powdered bauxite in hot concentrated sodium hydroxide (NaOH).
- The hydrated oxide of aluminium, reacts with sodium hydroxide to give sodium aluminate, while silica reacts with NaOH to form sodium trioxosilicate (IV). The reaction equations are:

- $Al_2O_3 + 3H_2O + 2NaOH \rightarrow 2NaAl(OH)_4$
- $SiO_2 + 2NaOH \rightarrow Na_2SiO + N_2O$

- Iron (II) oxide, on the other hand does not react with sodium hydroxide, therefore it is filtered off.
- The solution is seeded with crystals of pure aluminium oxide, $Al(OH)_3$ to precipitate out solid aluminium hydroxide.
- The solid aluminium hydroxide is washed, filtered and heated.
- This causes the hydroxide to lose water leaving pure aluminium oxide, Al_2O_3, called ***alumina***.

➢ $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$

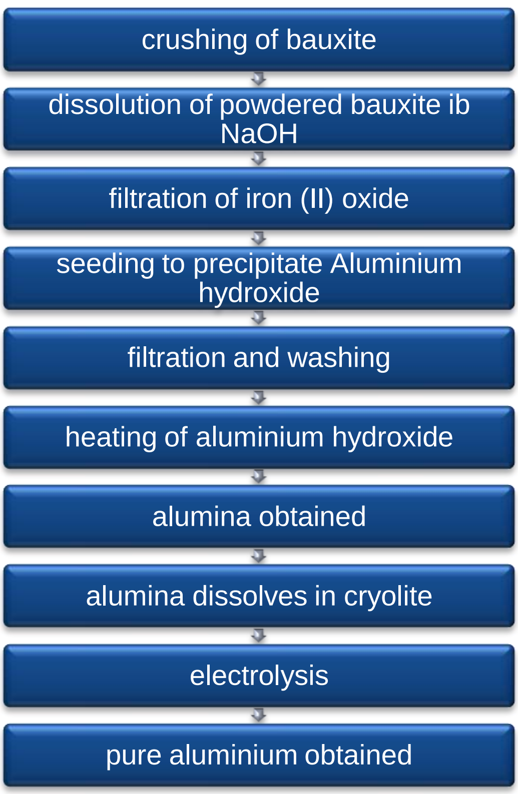

Fig. 110.2: Flow chart of diamond extraction

Electrolysis of alumina

Because aluminium is very reactive, it cannot be extracted from alumina by heating with carbon. It is, therefore, extracted by electrolysis. That is passing an electric current through alumina using graphite electrode in a process known as ***Hall process***, using Hall cell.

The electrolysis process

- The alumina is dissolved in liquid cryolite (sodium aluminum fluoride), Na_3AlF_6.
- Aluminium oxide is decomposed into aluminium ions, Al^{3+} and oxide ions, O^{2}.
- Aluminium ions are extracted to the negative electrode (cathode) where they gain electrons to become aluminium atoms. $Al^{3+} + 3e^{-} \rightarrow Al$
- Oxide ions are extracted to the positive electrode (anode) where they lose electrons to become oxygen molecules; $2O^{2-} - 4e^{-} \rightarrow O_2$

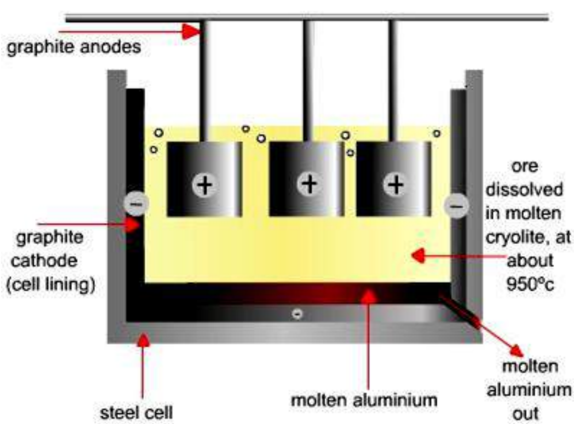

Fig. 110.3: Hall cell

EXTRACTION OF GOLD

Unlike aluminium, gold is un-reactive: thus, does not form compounds with other elements in its ore. This means that the gold ore is in its native or un-combined form.

The extraction process

- The gold ore is ground into fine powder.
- Water is added to form slurry (a watery mixture of insoluble ore particles).
- The slurry is dissolved or leached in a solution of sodium cyanide and oxygen to form a soluble gold cyanide called ***dicyanoaurate(I)***, $Au(CN)_2$.

$$4Au + 8NaCN + O_2 + 2H_2O \rightarrow 4NaAu(CN)_2 + 4NaOH$$

- The gold ion is absorbed or attached to the surface of carbon and then desorbed or detached by washing with sodium cyanide solution. Absorption and desorption remove impurities and increase the concentration of the gold
- The gold is electrolysed in a process known as ***electrowinning***.

Electrowinning is the process of using electricity to recover pure gold from its ore.

Fig. 110.4: Smelting of gold

- The reaction that takes place during the electrolysis is:

$$[Au(CN)_2]^- + e^- \rightarrow Au + 2CN^-$$

- The gold is deposited at the steal cathode.
- The crude gold is heated without melting in order to eliminate moisture and other impurities in a process known as ***calcinations***.
- The calcined gold is finally smelted in which the gold is mixed with chemicals called ***flux*** and then melted into a yellow liquid and then allowed to solidify into gold bars.

crushing of gold
↓
leaching
↓
absorption and desorption
↓
electrolysis (electrowinning)
↓
calcination and smelting

Fig. 110.5: Gold extraction process

TEST QUESTIONS

1. Bauxite is the ore of aluminium. Briefly describe the extraction of aluminium from bauxite.

2. Draw and label the structure of a hall cell.

3. Briefly describe the extraction of gold.

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RUSTING

Specific Objectives

After completing this chapter, you will be able to:

- Explain the process of rusting.

INTRODUCTION

When reactive metals come into contact with oxygen and moisture, oxides of those metals are produced. This process is known as ***oxidation***.

More reactive metals show stronger oxidation effects. The oxidation reaction causes them to wear off. One metal which shows stronger oxidation reaction is iron. When iron reacts with oxygen or moisture, in a process known as ***corrosion***, iron (III) oxide (hydrated iron oxide) is formed. This iron (III) oxide is a thin reddish-brown layer that forms on the surface of the iron called ***rust***.

The oxide is a solid that retains the same general form as the metal from which it is formed but, porous and somewhat bulkier, is relatively weak and brittle.

PROCESS OF RUSTING

Corrosion and rust

Corrosion is the chemical reaction which causes metals to wear away when it reacts with air and moisture.

When corrosion persists for a long time, it results in rust.

Rust is the reddish-brown coating on the surface of iron or steel that forms when the metal is exposed to air and moisture.

Fig. 110.6: *A* rusting car

Conditions necessary for corrosion or rusting of iron

- Air or oxygen
- Water or moisture or vapour

NB: Both conditions must be available for rusting to occur.

Experiment to show that air and water are necessary for rusting of iron

- Collect three test tubes, labelled **A**, **B** and **C**, with stoppers, clean iron nails, freshly boiled water (with the dissolved air in it removed), silica gel and liquid paraffin.
- Place a nail in each test tube.
- Fill test tube **A** with ordinary tap water and stopper it. (Tap water contains dissolved oxygen)
- Fill test tube **B** with the freshly boiled water and cover the water with a layer of the oil and stopper it. (The layer of oil prevent air from entering the water)
- Put a small quantity of the calcium chloride in test tube **C** and stopper it. (The calcium chloride absorbs moisture and keeps the atmosphere dry).
- Leave the test tubes for 4 to 6 days.

Observation

After four days, it would be observed that the nail in test tube **A** has begun rusting whiles the nails in test tubes **B** and C show no sign of rusting.

Conclusion

Test tube **A**, which contain both water and air, supports rusting while test tubes **B** and **C**, which have water only and air only respectively, do not support rusting. Therefore, both water and air are necessary for rusting.

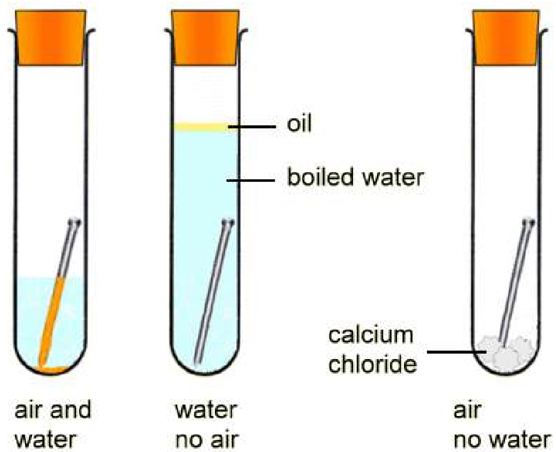

Fig. 110.7

Reactive metals which do not rust

Some metals, such as aluminium, although very active chemically, appear not to corrode under normal atmospheric conditions.

Actually, aluminium corrodes rapidly, and a thin, continuous, transparent layer of oxide forms on the surface of the metal, protecting it from further rapid corrosion.

Lead and zinc, although less active than aluminium, are protected by similar oxide films. Copper, a comparatively inactive metal, is slowly corroded by air and water in the presence of such weak acids as carbonic acid, producing a green, porous, basic carbonate of copper. Green corrosion products, called ***verdigris*** or ***patina***, appear on such copper alloys as brass and bronze, as well as on pure copper.

Non-reactive metals which do not rust

Some metals, called ***noble metals***, are so inactive chemically that they do not suffer corrosion from the atmosphere; among them are silver, gold and platinum.

A combination of air, water, and hydrogen sulphide will act on silver, but the amount of hydrogen sulphide normally present in the atmosphere is so small that the degree of corrosion is negligible except for the black discoloration, called ***tarnishing***, produced by the formation of silver sulphide.

PREVENTION OF RUSTING

Rusting and corrosion destroy iron and steel structures. The most effective method of preventing iron or steel from rusting is to keep it away from water and air.

The following are some methods of keeping iron from water and air.

Galvanizing

Galvanizing is the process of coating the iron or steel with a thin layer of zinc. Since zinc does not rust, the zinc layer protects the base iron from rusting.

Some examples of commonly galvanized products are garbage cans, corrugated sheets for roofing, iron pipe, and fence wires.

Enamelling

Enamelling involves coating the surface of the metal with enamel. Enamel is a special paint which consists of zinc oxide and lithopone in brown linseed oil and high-grade varnish.

Products which are normally enamelled are refrigerators, gas cookers, auto mobiles etc.

Fig. 110.8: Enamelled ware

Painting

Oil-based paints when applied to the surfaces of irons harden and become affixed, thereby preventing air and water from having direct contact with the iron.

Tinplating

Cans and tins, especially those for commercial products are often tinplated. Tin plating involves coating the surface of the iron or steel with a thin film of tin (which is resistant to rusting).

Electroplating

Cadmium, chromium, copper, gold, nickel, silver and tin are electroplated on irons or steel. This is because they resist rusting.

In electroplating, the iron is placed in a solution and is connected to the negative terminal with the coating metal connected to the positive terminal of an electricity source. Current is then allowed to pass through which causes atoms from the coating metal to move to cover the surface of the base iron.

Alloying

One effective method of preventing corrosive metals from rusting is chemically mixing them with metals which do not rust. Iron, for example, is normally alloyed with carbon and chromium or nickel to give a rust-proof metal known as ***stainless steel***. Objects which are made from stainless steel include cutlery set, surgical instruments etc.

Cladding

Cladding is the process of bonding a non-corroding metal to a corrosive metal.

One example of a clad metal is rolled gold, which consists of an iron or steel core with a layer of gold on the outside. Cladded aircraft components may have a thick layer of strong aluminium alloy in the centre and thin outer layers of corrosion-resistant pure aluminium sheets. The various layers or plies of metal are usually heated and rolled together.

Other cladding methods include casting, welding, and pouring molten metal around a hardened core. In addition to sheets and strips, clad metals are produced in wires, bars, and tubing.

Oiling and greasing

Corrosive metals can be oiled or greased to protect their surfaces from rusting. One disadvantage of oiling or greasing is, it does not last long and the metal will have to be oiled or greased regularly.

Plastic coating

Some metals are also coated with a thin film of plastic. Such metals are prevented from contact with fire or heat as that will cause the plastic coating to melt off exposing the underlying corrosive metal.

TEST QUESTIONS

1. Explain the following terms:
   (a) corrosion;
   (b) rust.

2. State the conditions necessary for rusting of iron.

3. Describe an experiment to show the conditions necessary for rusting of iron.

4. Differentiate between reactive and non-reactive metals.

5. Describe four ways of preventing rusting.

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ORGANIC AND INORGANIC COMPOUNDS

Specific Objectives

After completing this chapter, you will be able to:

- Relate the nucleus, chromosomes
- Classify chemicals as organic and inorganic compounds.
- Identify chemicals as organic compounds.
- Distinguish between neutralization and estrification.

INTRODUCTION

Organic chemistry

Organic chemistry is the scientific study of the structure, properties, composition, reactions, and preparation of carbon-based compounds, hydrocarbons, and their derivatives.

These compounds may contain any number of other elements, including hydrogen, nitrogen, oxygen, the halogens as well as phosphorus, silicon, and sulphur. Organic compounds are structurally diverse.

The range of application of organic compounds is enormous. They either form the basis of, or are important constituents of many products including plastics, drugs, petrochemicals, food, explosives and paints. They form the basis of almost all earthly life processes (with very few exceptions).

Inorganic Chemistry

Inorganic Chemistry is the study of the structure, properties, and reactions of the chemical elements and their compounds.

Inorganic chemistry does not include the investigation of ***hydrocarbons*** (compounds composed of carbon and hydrogen).

ORGANIC AND INORGANIC COMPOUNDS

Organic compounds

Organic compounds can be defined as compounds which contain hydrocarbon.

Organic compounds are classified as natural or artificial.

The physical structures of all living things are composed mainly of organic compound.

Examples of natural organic compounds

1. Carbohydrates
2. Protein
3. Vitamins
4. Enzymes
5. Hormones
6. Herbs
7. Fats and oils

Examples of artificial organic compounds

1. Plastic
2. Insecticides
3. Pesticides
4. Soap
5. Dyes
6. Drugs etc.

Characteristics of organic compounds

1. Carbon can forms four bonds and is described as ***tetravalent***.
2. Carbon can form a ring or cyclic compounds.
3. Carbon can bond with other atoms to form stable compounds.
4. Carbon atoms can combine with each other and hydrogen atom to form open, straight and branched chains.
5. Carbon can bond with another carbon atom through stable, double and triple bonds. No other atom can form stable multiple bonds.
6. Organic compounds have low melting and boiling points
7. They occur in all the states of matter (liquid, gas and solid).
8. They are highly volatile.

Inorganic compounds

Inorganic compounds are compounds which do not contain hydrocarbons.

Examples of inorganic compounds

1. Chalk
2. Salt
3. Washing soda
4. Carbon monoxide, CO
5. Carbon dioxide, CO_2
6. Trioxocarbonate (IV), CO_3^{2-}

Characteristics of inorganic compounds

1. They are mostly ionic.

2. They are insoluble in organic solvents.
3. They have quick rate of reaction.
4. They are not volatile.
5. They have high melting and boiling points.

Table 7.8: Differences between organic and inorganic compounds

Organic compounds	Inorganic compounds
Mostly covalent	Mostly ionic
Have low boiling and melting points	Have high boiling and melting points
Generally soluble in organic solvents	Generally insoluble in organic solvents
Slow reaction rate	Fast reaction rate
Highly volatile	Normally not volatile

Types or families of organic compounds

Organic compounds are divided into families known collectively as ***homologous series.***

Homologous series is a group of organic compounds made up of a series of compounds in which all members can be represented by the same general formula but different structural formulas.

Properties of the homologous series

1. They can be represented by a general formula.
2. They possess different structural formulas.
3. They can be prepared by a general method.
4. They have similar chemical properties because they have the same general formula.
5. They have different physical properties.

Types of organic compounds

- Hydrocarbons
- Alkanol
- Alkanoic acids
- Alkanoates
- Fats and oils.

Functional groups

Functional groups are atoms or group of atoms or bonds which give a family of organic compounds it characteristic chemical properties.

Table 7.9: Organic compounds with their functional groups

Organic compound	Functional group	General formula
Alkanes	$-\overset{\vert}{\underset{\vert}{C}}-$	C_nH_{2n+2}
Alkenes	$>C=C<$	C_nH_{2n}
Alkynes	$-C\equiv C-$	C_nH_{2n-2}
Alkanols	$-OH$	$R-OH$
Alkanoic acids	$-COOH$	$R-COOH$
Alkyl alkanoates	$-CO\text{-}O-$	$-CO\text{-}OR$

A functional group does not exist independently but forms a part of a molecule. The properties of any family of organic compound in the homologous series depend on the functional group as well as the nature of the carbon chain.

Hydrocarbons

Hydrocarbons are organic compounds, composed entirely of carbon and hydrogen.

They are the organic compounds of simplest composition and may be considered the parent substances from which all other organic compounds are derived.

Types of hydrocarbons

Hydrocarbons are conveniently classified into two major groups, ***open-chain*** and ***cyclic***.

Open-chain hydrocarbons

In open-chain compounds containing more than one carbon atom, the carbon atoms are attached to each other to form an open chain; the chain may carry one or more side branches.

They are made up of ***aliphatic compounds*** such as alkanes, alkenes and alkynes which are hydrocarbons with single, double and triple bonds respectively.

Cyclic compounds

In cyclic compounds the carbon atoms form one or more closed rings as in ***aromatic compounds***.

The two major groups are subdivided according to chemical behaviour into ***saturated*** and ***unsaturated*** compounds.

Saturated hydrocarbons: These are those compounds which contain hydrogen and carbon atoms only, with single covalent bonds. Alkanes are saturated hydrocarbons.

Petroleum contains a great variety of saturated hydrocarbons, and petroleum products such as gasoline, kerosene, heavy fuel oil, lubricating oils, petroleum jelly, and paraffin consist principally of mixtures of paraffin hydrocarbons, which range from the lighter liquid members to the solid members.

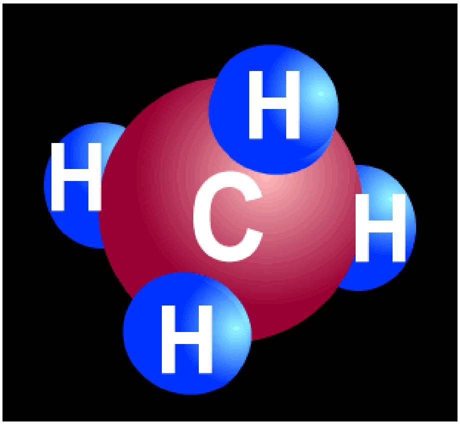


Fig. 110.9: A single bonded hydrocarbon

Unsaturated hydrocarbons: These are those with double and triple covalent bonds. Alkenes and alkynes are unsaturated hydrocarbons.

Alkanes

These are single-bonded hydrocarbons, containing carbon atoms surrounded by a maximum of four hydrogen ions.

Alkanes have the general formula $\mathbf{C_nH_{2n+2}}$, where n is the number of carbon atoms in the molecule.

The structure and naming of alkanes

Hydrocarbons are named based on the number of carbon atoms contained in the molecule.

- If the molecule contains only one carbon atom, it is prefixed **meth-**;
- if there are two carbon atoms, it is prefixed **eth-** and
- if the molecule contains ten carbon atoms, it is prefixed **dec-**.
- Each of the members of alkane is suffixed with **–ane**. Among the members of the series are methane $\mathbf{CH_4}$; ethane, C_2H_6; propane, $\mathbf{C_3H_8}$; and butane, $\mathbf{C_4H_{10}}$.

There are two ways of writing a condensed structural formula. For example, butane may be written as:

$\mathbf{CH_3CH_2CH_2CH_3}$

or

$\mathbf{CH_3(CH_2)_2CH_3}$.

Table 8.0: The members of alkane, their molecular and structural formulas

Alkane	Name	Molecular Formula	Structural Formula
1	Methane	CH_4	CH_4
2	Ethane	C_2H_6	CH_3CH_3
3	Propane	C_3H_8	$CH_3CH_2CH_3$
4	Butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$
5	Pentane	C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$
6	Hexane	C_6H_{14}	$CH_3(CH_2)_4CH_3$
7	Heptane	C_7H_{16}	$CH_3(CH_2)_5CH_3$
8	Octane	C_8H_{18}	$CH_3(CH_2)_6CH_3$
9	Nonane	C_9H_{20}	$CH_3(CH_2)_7CH_3$
10	Decane	$C_{10}H_{22}$	$CH_3(CH_2)_8CH_3$

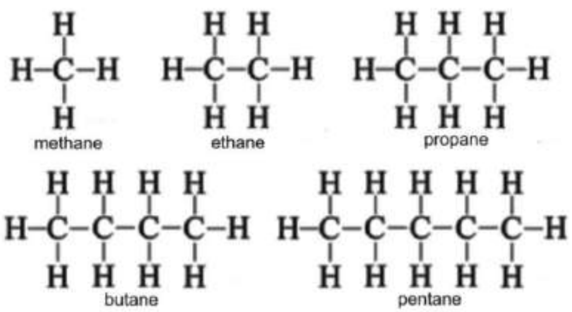


Fig. 111.0: Structure of the first five members of alkane

Properties of alkanes

Physical properties

1. The first four members of the series are gases at ordinary temperature and pressure
2. Intermediate members are liquids; and the heavier members are semi-solids or solids.
3. Alkanes are insoluble in water.
4. They are soluble in non-polar solvents such as ether and benzene.

5. They are colourless.

Chemical properties

1. Alkanes are unreactive; that is, they do not react readily at ordinary temperatures with such reagents as acids, alkalies, or oxidizers.

2. They undergo reaction in the presence of excess oxygen to give carbon dioxide and water.
   $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$

3. They react in limited supply of oxygen to give carbon monoxide.
   $2CH_{4(g)} + 3O_{2(g)} \rightarrow 2CO_{(g)} + 4H_2O_{(g)}$

4. Alkanes react with halogens in the presence of sunlight to give haloalkanes and the corresponding halide. Methane, for example, reacts with chlorine to give chloromethane and hydrogen chloride.
   $CH_{4(g)} + Cl_{2(g)} \rightarrow CH_3Cl_{(g)} + HCl_{(g)}$

Sources of alkanes

- petroleum (crude oil)
- natural gas
- plants and animal remains and wastes

Uses of alkanes

1. Alkanes are used mostly as sources of fuels either domestically or industrially.
2. Methane, for instance, is used in the house for cooking
3. Butane is used in lighters.
4. Higher alkanes such as octane are used in petrol and other fuels.

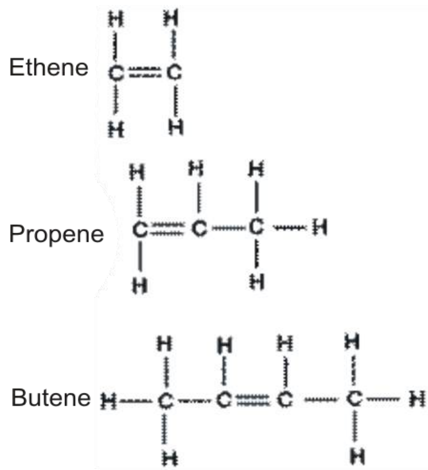
Alkenes

Alkenes, also known as ***olefins***, are formed when an alkane misses two hydrogen atoms to form a carbon-carbon double bond (**C=C**)

Alkenes have the general formula **C*n*H₂*n***.

The structure and naming of alkenes

- Alkenes have the same prefixes as alkanes, but are suffixed –**ene** instead of **-ane**.
- If the molecule contains two carbon atoms (**C_2**), for example, it is called **ethene** (**C_2H_4**);
- If there are three carbon atoms – **propene** (**C_3H_6**)
- If there are ten carbon atoms – **decene** (**$C_{10}H_{20}$**).



Properties of alkenes

Physical properties

1. Alkenes have relatively low boiling and melting points than alkanes.
2. They are insoluble in water.
3. They are soluble in certain non-polar organic solvents such as petroleum.
4. At normal temperature and pressure, the first four members are gases, the middle members, liquids and the last members are semi-solids and solids.

Chemical properties

1. Alkenes react with water to give alkanol (alcohol).

   $C_2H_4 + H_2O \rightarrow C_2H_5OH$

2. They undergo additional reaction with hydrogen, halogens and bromine water. For example, they undergo hydrogenation with hydrogen in the presence of Pb, Ni or Pt catalyst to give the corresponding alkane.

   $C_2H_4 + H_2 \rightarrow C_2H_6$

3. Alkenes react with excess oxygen to produce carbon dioxide and water.

   $C_2H_4 + 3O_2 \rightarrow 3O_2 + 2H_2O$

4. Alkenes can undergo self addition reaction to form large molecules under high temperature and pressure in a process known as ***polymerisation***. For example, ethene can polymerise to produce low-density polyethene.

Uses of alkenes

1. Ethene is used to manufacture detergents.
2. It is used to manufacture plastics.
3. It is used in the process of fruit ripening.
4. It is used in the industrial preparation of ethanol.
5. It is used in the manufacture of glycerol.
6. Propene is used to make polypropene.
7. It is used to make synthetics fibres and Buna-n-rubber.

Fig. 111.2: Buna-n-rubber

Alkynes

Alkynes are triple-bonded **(-C≡C-)** hydrocarbons with the general formula $\mathbf{C_2H_{2n-2}}$**.** Where n is the number carbon atoms present in the molecule.

Structure and naming of alkynes

Just like alkanes and alkene, alkynes are named based on the corresponding number of carbon atoms in the molecule, with the suffix **-yne**, as in ethyne (**C_2H_2**), which is the most important member of the alkyne group.

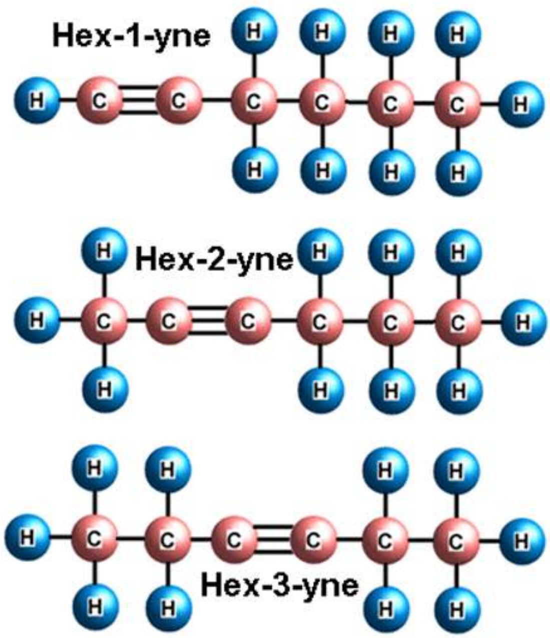


Fig. 111.3: Examples of alkynes

Properties of alkynes

Physical properties

1. They are insoluble in water.
2. They are quite soluble in the usual organic solvents of low polarity (e.g. ligroin, ether, benzene, carbon tetrachloride, etc.).
3. They are less dense than water.
4. Their boiling points show the usual increase with increasing carbon number.
5. They are very nearly the same as the boiling points of alkanes or alkenes with the same carbon skeletons.

Chemical properties

❖ Alkynes burn in excess oxygen to give carbon dioxide and water.

$$2C_2H_2 + 5O_2 \longrightarrow 4CO_2 + 2H_2O$$

❖ Akynes do not undergo polymerization reaction as readily as alkenes. However, ethyne may be polymerised under different conditions.

$$3C_2H_2 \xrightarrow{\text{copper tube}} C_6H_6$$

Uses of alkyens

Ethyne which is the most useful member of the series may be used to manufacture:

1. polyvinylchloride (PVC) plastics;
2. synthetic fibres;
3. neoprene rubber;
4. triochloroethane;
5. Buna-N-rubber etc.

Alkanols

Akanols have the functional group **-OH**. They are not hydrocarbons because they contain oxygen (O) atoms in their compounds.

They have the general formula **$C_nH_{2n+1}OH$**, where n is the number of carbon atoms.

Structure and naming of alkanols

Alkanols are named by replacing the last letter of the name of the parent alkane with the ending **-ol**. For example, **methanol** (**CH_3OH**) is an alkanol derived from methane. Others members in the series are ethanol (C_2H_5OH), propanol (C_3H_7OH) etc.

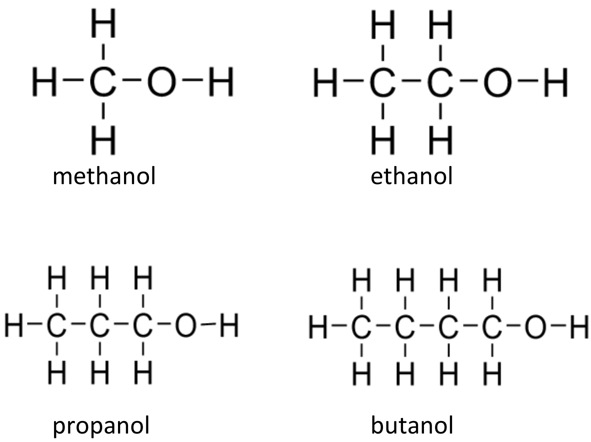


Fig. 111.4: Examples of alkanols

Properties of alkanols

Physical properties

1. Alkanols are generally soluble in water.
2. They are colourless.
3. They are mostly liquids or gases at room temperature.
4. Ethanol has a boiling point of 78^oC and freezes at -114^oC.

Chemical properties

1. Alkanols burn in air to produce carbon dioxide and water.
   $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

2. They undergo dehydration in the presence of concentrated H_2SO_4 to produce the corresponding alkene and water.
   $C_2H_5OH \rightarrow C_2H_4 + H_2O$

3. They react with carboxylic acids to produce esters and water in a reversible reaction called ***esterification reaction***.

4. They are oxidized to acids with acidified potassium dichromate.
   C_3H_7 (with $K_2C_2O_7$ H) C_3H_5OOH

Uses of alkanols

1. Alkanols are used for sterilizing instruments.
2. They are used as fuels.
3. They are used as solvents for paints, vanish, stains etc.
4. They are used to prepare alcoholic beverages such as wines, beers, spirits etc.
5. alkanols (95% ethanol and 5% methanol) are used in the preparation of industrial methylated spirit.

Isomers

Isomerism is the phenomena whereby certain compounds, with the same molecular formula, exist in different forms owing to their different organisation of atoms.

Isomers are atoms with the same molecular formula but different structural formulas.

The concept of isomerism illustrates the fundamental importance of molecular structure and shape in organic chemistry.

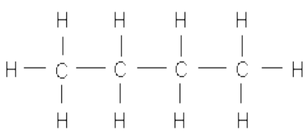
Structural Isomers

Structural isomers have different structural formulas because their atoms are linked together in different ways.

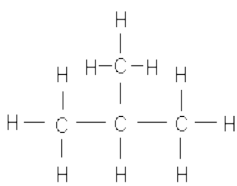
This arises owing to:

1. *Arrangement of Carbon skeleton*

e.g. The formula C_4H_{10} represents two possible structural formulae, butane and mythylpropane



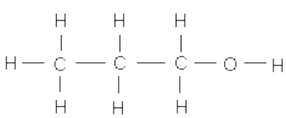
butane



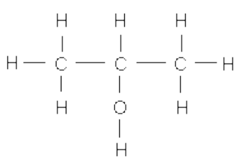
methylpropane

2. *Position of Functional group*

   e.g. **propan-1-ol** and **propan-2-ol**



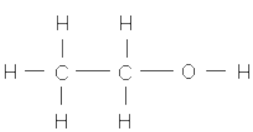
propan-1-ol



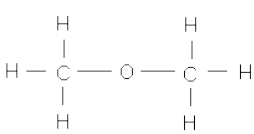
propan-2-ol

3. *Different Functional groups*

   e.g. the molecular formula C_2H_60 represents both ethanol and methoxymethane

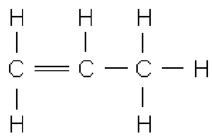


ethanol

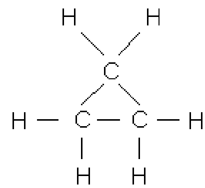


methoymethane

Cyclic alkanes are isomeric with alkenes, e.g. **cyclopropane** and **propene**



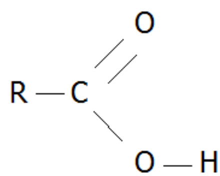
propene



cyclopropane

Alkanoic acids

These are organic compound with the functional group -COOH and the general formula $\mathbf{C_nH_{2n+1}COOH}$, where n is the number of carbon atoms.



Naming of alkanoic acids

Alkanoic acids are named by replacing the last letter '**-e**' of the parent alkane with the ending '**-oic**'. If the number of carbon atoms is 3 the formula is $\mathbf{C_3H_7COOH}$.

Since this compound contains four carbon atoms, its name is butanoic acid. Remember that the general formula already contains a carbon (C) atom.

Alkanoic acids are commonly known as ***fatty acids***. This is because many of them were obtained from fat sources. The table below gives some alkanoic acids and their natural sources.

Table 8.1: Sources of alkanoic acids

Alkanoic acid	Formula	Source
Methanoic acid (Formic acid)	$HCOOH$	Bee and ant stings
Ethanoic acid (acetic acid	CH_3COOH	Palm wine, vinegar, sour wine
Citric acid	$C_3H_4(OH)(COOH)$	Orange, lemon, grapefruit
Lactic acid	$CH_3CH(OH)COOH$	Sour milk, animal muscle
Stearic acid (octadecanoic acid)	$CH_3(CH_2)_{16}COOH$	Animal and plant fats and oils
Oleic acid	$CH_3(CH_2)_7CH=CH(CH_2)_7COOH$	Olive oil, pork fat, groundnut
Palmitic acid	$CH_3(CH_2)_{14}COOH$	Palm oil, animal and plant fats and oils

Properties of alkanoic acids

1. Alkanoic acids undergo neutralization reaction with alkalis to produce salt and water.

$$CH_3COOH + NaOH \rightarrow CH_3COON + H_2O$$

2. They react with reactive metals to give a salt and a hydrogen gas.

$$2CH_3COOH + Zn \rightarrow (CH_3COO)_2Zn + H_2$$

3. They react with trioxocarbonates (IV) to give carbon dioxide and water.

$$Na_2CO_3 + 2CH_3COOH \rightarrow 2CH_3COON + CO_2 + H_2O$$

Uses of methanoic acid

1. It is used to remove calcium trioxocarbonate (IV) (chalk) deposits from containers.
2. It is used as drying agent in making textiles.
3. It is used as starting chemical in the production of other chemicals.

Uses of ethanoic acid

1. It is used as solvent to dissolve other carbon compounds.
2. It is used as vinegar in flavouring food.
3. It is used for cleaning dirty metal surfaces at home.
4. It is used in laundries for brightening colours.
5. It is used in drug production.
6. It is used to make cellulose ethanoate for use in vanishes, lacquers, cinema films and some synthetic fibres such as rayon.

Alkyl alkanoates (esters)

Esters are produced when an alkanol reacts with an alkanoic acid.

$$\text{alkanol} + \text{alkanoic acid} \rightleftharpoons \text{ester} + \text{water}$$

This process is known as ***esterification***. Esters have a characteristic pleasant smell. Esters occur naturally in fruits (such as banana, pineapple etc.) and flowers (such as queen of the night, frangipani, thumbergia etc), giving them their pleasant smell.

They have the general formula **RCOOR'**, where **R** and **R'** belong to the same or different alkyl groups.

Structure and naming of esters

- To write the correct structure of the ester, the alkanoic acid loses the **–OH** group and the alkanol loses the **–H** atom.

- The naming is done by combining the alkanol and the alkanoic acid.

- The alkyl of the alkanol is stated first, followed by the acid name with the –**oic** replaced by **–oate**. For example if methanol, CH_3OH, reacts with ethanoinc acid, CH_3COOH, the ester, methyl ethanoate CH_3COOCH_3, is formed.

$CH_3OH + CH_3COOH \rightarrow CH_3COOCH_3 + H_2O$

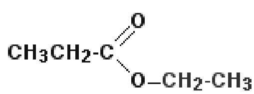
- The alkyl group of methanol is methyl and the acid part of ethanoic acid becomes ethanoate, hence the name, methyl ethanoate to the resulting ester.

- The reaction takes place in the presence of heat, with concentrated H_2SO_4, acting as a catalyst to speed up the reaction.

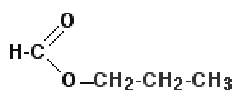
Fruits

Flowers

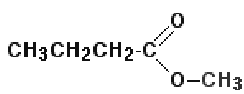
Fig. 111.5: Fruits and flowers are scented by esters



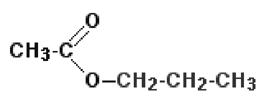
ethyl propanoate



propyl methanoate



methyl butanoate



propyl ethanoate

Fig. 111.6: Examples of esters

Properties of esters

Physical properties

1. Esters have pleasant smell.
2. They are usually colourless.
3. Esters with short chains are soluble in water.

Chemical properties

❖ Since the esterification reaction is reversible, an ester can be hydrolysed to form an alkanol and alkanoic acid.

$$CH_3COOH_2CH_3 \rightleftharpoons CH_3COOH + CH_3CH_2OH$$

❖ The addition of lithium aluminium hydride can reduce an ester to ethanol.

$$CH_3COOCH_3 \xrightarrow{LiAIH} CH_3CH_2OH + CH_3OH$$

Uses of esters

1. Esters are used for making artificial flavours and essences.
2. These are used in cold drinks, ice-creams, sweets and perfumes.
3. They are used as solvents for oils, fats, gums, resins, cellulose, paints, varnishes, etc.
4. Esters are used as plasticizers (added to plastics or other materials to make or keep them soft or pliable).

Fats and oils

Fats and oils are formed when one or more of the hydrogen atoms of the glycerol group are replaced by a long chain ester (acid alkanoate). This is also known as ***glyceride***. Esters with all the three hydrogen atoms replaced are called ***triglyceride*s** or ***triester*** of glycerol.

Both fats and oils contain acid and alkanol parts, which make them esters of the glycerol, long-chain fatty acids and propanetriol.

CH_2 – OH
|
CH - OH
|
CH_2 - OH

Fig. 111.7: Structure of glycerol

Butyric acid (butanoic acid) is one of the saturated short-chain fatty acids responsible for the characteristic flavour of butter

$CH_3CH_2CH_2COOH$ or $CH_3(CH_2)_2COOH$

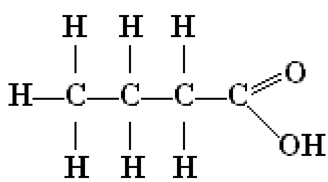


Fig. 111.9: Butanoic acid

Table 8.2: Differences between fats and oils

Fats	Oils
Usually have Saturated hydrocarbon chain	Have unsaturated hydrocarbon chain
Solid at normal temperature	Liquid at normal temperature
Have high melting point	Low melting point

Fig. 111.8: Bar of butter (fat) and oil

Uses of fats and oils

1. Dietary fats supply energy, carry fat-soluble vitamins (A, D, E, K), and are a source of antioxidants and bioactive compounds.
2. Fats are incorporated as structural components of the brain and cell membranes.
3. Fats and oils are used in the manufacture of soap, drugs, margarine etc.
4. They are used to make candles, paints, glycerols etc.

Esterification and neutralization

Esterification is the reaction between alkanols and alkanoic acids to produce esters and water.

This reaction is reversible, and the catalyst, concentrated H_2SO_4 is used to dehydrate the ester. The following example is an esterification reaction.

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOH_5C_2 + H_2O$$

NB: The acid is named by counting up the total number of carbon atoms in the chain - including the one in the -COOH group.

For example, CH_3CH_2COOH is propanoic acid, and CH_3CH_2COO is the propanoate group.

Since esterification reaction is reversible, the ester produced can be broken up into the constituent alkanol and alkanoic acid. This process is known as ***saponification***.

Saponification is the reaction between an ester and an alkali to produce soap and glycerol.

Water may be produced but in small amount.

fat or oil + sodium hydroxide→soap + glycerol

Esterification and neutralization (reaction of acid and base) have some differences as well as a similarity.

Table 8.2: Differences between esterification and neutralization

Esterification	Neutralization
Reaction is reversible	Reaction is irreversible
Reaction is slow	Reaction is fast
Heat involved	No heat involved
Catalyst needed	No catalyst needed
Products are ester and water	Products are salt and water
Reactants are alkanol and alkanoic acid	Reactants are acid and base (alkali)

Similarity between esterification and neatralization

One similarity between esterification and neutralization is water is produced as a by-product in both reactions.

PETROCHEMICALS

Petrochemicals are chemicals made from petroleum (crude oil) and natural gas. Petroleum and natural gas are made up of hydrocarbon molecules, which are comprised of one or more carbon atoms, to which hydrogen atoms are attached.

Oil and gas are the main sources of the raw materials because they are the least expensive, most readily available, and can be processed most easily into primary petrochemicals, such as olefins (ethylene, propylene and butadiene) aromatics (benzene, toluene, and xylenes); and methanol.

Uses of petrochemicals

Primary petrochemicals can be converted chemically to form more complicated derivative products such as

1. vinyl acetate for paint, paper and textile coatings,
2. vinyl chloride for polyvinyl chloride (PVC),
3. ethylene glycol for polyester and textile fibres,
4. styrene which is important in rubber and plastic manufacturing.
5. Petrochemicals have had a dramatic impact on our food, clothing, shelter and leisure.
6. Some synthetics, tailored for particular uses, actually perform better than products made by nature because of their unique properties.

PVC pipes

Oil

Fig. 112.0: Some end products of petrochemicals

Cracking

Cracking is the process of breaking long-chain hydrocarbons into short ones.

The rate of cracking and the end products are strongly dependent on the temperature and presence of catalysts.

Oil refinery cracking processes allow the production of light products such as liquefied petroleum gas (LPG) and petrol from heavier crude oil distillation fractions such as gas oils and residues. Fluid catalytic cracking produces a high yield of petrol and LPG, while hydro-cracking is a major source of jet fuel, diesel, naphtha and LPG.

Fractional distillation of petroleum

Petroleum refining is the process of separating the many compounds present in crude petroleum. The principle which is used is that the longer the carbon chain, the higher the temperature at which the compounds will boil.

Fractional distillation process

- The crude petroleum is heated and changed into a gas.
- The gas is passed through a distillation column which becomes cooler as the height increases.
- When a compound in the gaseous state cools below its boiling point, it condenses into a liquid.
- The liquids may be drawn off the distilling column at various heights.

Many kinds of compounds including alkenes are made during the cracking process.

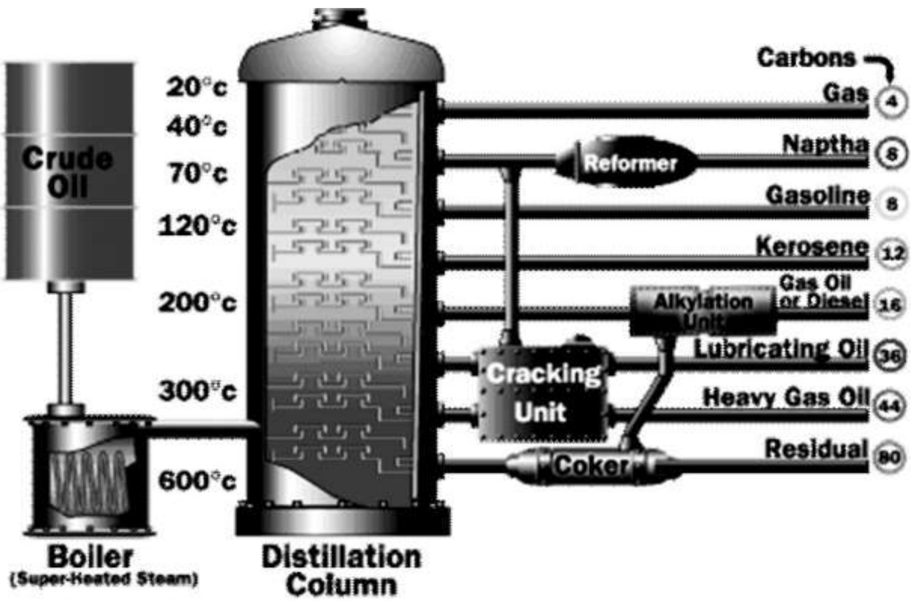


Fig. 112.1: Fractional distillation process

Table 8.3: Uses of the various fractions of distillation

Fraction	Uses
Liquefied Petroleum Gas (LPG)	• As fuel for cooking • Used to provide heat during fractional distillation
Naphtha	• Used to produce high-octane elements of petrol. • Used within the rubber industry and in dry cleaning operations • Used as fuel in camping stoves, lanterns, etc.
Petrol	• As fuel for automobiles • As fuel for generators • Used as solvent
Kerosene	• As fuel for jet planes • As fuel for lanterns and camping stoves
Gas oil or Diesel	• As fuel for automobiles • As fuel in diesel generators
Lubricating oil	• Used to lubricate machinery • Used for power generation
Heavy gas-oil	• As fuel in ships • Undergoes cracking to produce more petrol
Residue/ bitumen	• Used to surface roads • Used to make roof felting

TEST QUESTIONS

6. Write a short note on the following:
   (a) organic compounds;
   (b) inorganic compounds.
   (c) Mention four differences between organic and inorganic compounds.

7. (a) What is the homologous series?
   (b) State four properties of the homologous series.

3. (a) What are functional groups?
   (b) Draw t he functional groups of the following organic compounds:
   (i) alkanes
   (ii) alkenes
   (iii) alkynes
   (iv) alkanols
   (v) alkanoic acids

4. (a) Briefly explain the term hydrocarbons.
   (b) Differentiate between saturated hydrocarbons and unsaturated hydrocarbons.

5. Write briefly on the fooling types of hydrocarbons.
   (a) alkanes;
   (b) alkenes;
   (c) alkynes.

6. Draw the molecular and structural formulae of the first five members of alkane.

7. (a) Mention four physical and four chemical properties of alkanes.
   (b) Explain the uses of alkanes.

8. (a) Write a brief note on alkenes
   (b) State five uses of alkenes.

9. (a) Mention three physical and three chemical properties of alkenes.
   (b) State two differences between alkanes and alkenes.

10. (a) List four physical properties of alkynes.
    (b) State three uses of alkynes.

11. (a) Mention three uses of alkanols.
    (b) Describe the structure of alkanols.

12. (a) Define the term isomerism.
    (b) Draw the structural formulae of the following isomers:
        (i) propan-2-ol
        (ii) methylpropane.

13. Name the natural sources of the following alkanoic acids:
    (a) methanoic acid;
    (b) ethanoic acid;
    (c) citric acid;
    (d) lactic acid;
    (e) palmitic acid.

14. (a) What are esters?
    (b) State three uses of esters.

15. (a) Define the term esterification.
    (b) List four differences between esterification and neutralization.

16. (a) What are petrochemicals?
    (b) Name five products derived from petrochemicals

17. (a) Explain the term cracking.
    (b) Describe the fractional distillation of petroleum.

18. Name five products produced from petroleum distillation and state one use each.

19. Consider the following organic compounds:
    C_3H_6; C_3H_8; C_2H_5OH; CH_3COOH and CH_3COOCH_3
    Select one compound which will
    (i) undergo polymerization,
    (ii) give a pleasant smell,
    (iii) liberate carbon dioxide with sodium trioxocarbonate (IV),

(iv) decolorize bromine water.

20. State the main functional group in each of the following organic compounds:
(α) CH_3COOH;
(β) $CH_3C{=}CCH_3$;
(γ) $HCOOCH_3$.

21. (a) Write the formula of each of the following compounds:
i) hex-2-yne;
ii) butan-2-ol;
iii) 2, 2-dimetylpentane.
b) Give an example each of
i) an organic acid;
ii) a naturally occurring alkanoate.

22. Write the chemical formulae of each of the following compounds:
(a) ethanoic acid;
(b) 2, 2 – dimethylbutane;
(c) 3, methyl pent – 2 – ene;
(d) 2, methylbutan – 2 – ol.

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LIFE CYCLE OF PESTS AND PARASITES

Specific Objectives

After completing this chapter, you will be able to:

- Identify some common pests and parasites of farm animals and humans.
- Explain the term life cycle and its relevance in the life of some plants and animals.

INTRODUCTION

Pests

Pests are living organisms that cause damage to livestock, crops, humans or land fertility.

Pests cause damage to crops and livestock by feeding on them and thereby cause diseases to them. This is because most pests serve as transmitters of disease-causing pathogens from a sick person or animal to a healthy person or animal. Such pests are known as ***vectors***. Examples of pests are mosquitoes, house flies, tsetse flies, etc.

Parasites

Parasites are organisms which live in or on another organism, called a host, usually causing it some harm.

A parasite is generally smaller than the host and of a different species. Parasites are dependent on the host for some or all of their nourishment.

For example, tapeworm, a flattened worm that lives in the gastrointestinal tract of mammals, lacks an intestine of its own and must absorb pre-digested food from the intestine of its host. This food is the tapeworm's only energy source for growth and reproduction.

Types of parasites

There are two types of parasite. They are endoparasites and ectoparasites.

Endoparasites: These are parasites which live in the bodies of their hosts.
Examples of endoparasites are tape worm, liver fluke, guinea worm, round worm etc.

Ectoparasites: These are parasites which live on the bodies of their host.
Examples of ectoparasites are flea, louse, tick, mite etc.

TYPES OF PEST AND PARASITES OF HUMAN, FARM ANIMALS AND CROPS

Humans are subjected to numerous protozoan, worm, and insect-related parasites. Two of the most damaging human parasites are the protozoan ***plasmodium*** that causes malaria and the flatworm ***schistosoma*** that causes schistosomiasis.
There are an estimated 400 million to 600 million cases of malaria each year and 200 million cases of schistosomiasis worldwide.

Anopheles mosquito (Malaria)

Malaria is an infectious disease caused by a one-celled parasite known as *Plasmodium*. The parasite is transmitted to humans by the bite of the female ***Anopheles mosquito***. The Plasmodium parasite spends its life cycle partly in humans and partly in mosquitoes.

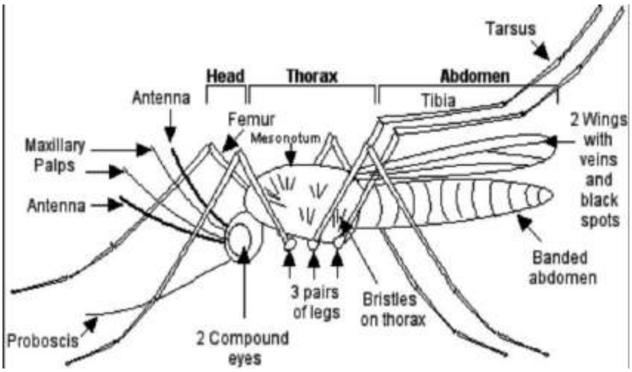


Fig. 112.2: Female anopheles mosquito

Transmission of malaria

- Mosquito infected with the malaria parasite bites human, passing cells called ***sporozoites*** into the human's bloodstream.
- Sporozoites travel to the liver. Each sporozoite undergoes asexual reproduction, in which its nucleus splits to form two new cells, called ***merozoites***.
- Merozoites enter the bloodstream and infect red blood cells. In red blood cells, merozoites grow and divide to produce more merozoites, eventually causing the red blood cells to rupture. Some of the newly released merozoites go on to infect other red blood cells.
- Some merozoites develop into sex cells known as male and female ***gametocytes***.

- ❖ Another mosquito bites the infected human and ingests the gametocytes. In the mosquito's stomach, the gametocytes mature. Male and female gametocytes undergo sexual reproduction, uniting to form a zygote. The zygote multiplies to form sporozoites, which travel to the mosquito's salivary glands.
- ❖ If this mosquito bites another human, the cycle begins again.

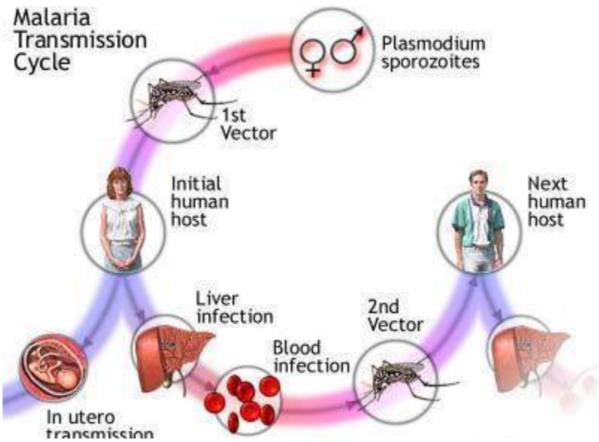


Fig. 112.3: Life cycle of the malaria parasite

Symptoms and treatment of malaria

The characteristic symptoms of malaria include.

- ❖ Cold, headache and weakness
- ❖ Shivering
- ❖ High fever
- ❖ Excessive sweat as a result of drop in temperature
- ❖ Prolonged infection results in anaemia and enlarged spleen

Drugs such as chloroquine, camoquine, atesunate and amodiaquine can be used to prevent infection in the blood. An infected person should seek a health professional's advice immediately.

Control and prevention of malaria

1. The use of insecticides and mosquito repellents to kill and drive mosquitoes from an area.
2. Destroying the mosquito larvae and pupae by spraying oil on their breeding areas such as pools, marshes etc. (The oil causes the larvae and pupae to sink as it lowers the surface tension of the water)
3. Draining mosquito breeding ground.
4. Introducing fish such as top minnow into the water to feed on the larvae and pupae.
5. Preventing mosquito bites by sleeping under treated nets, applying repellent cream on the body, covering windows with fine-mesh to keep out mosquitoes.
6. Taking prescribed anti-malarial drugs regularly.

Schistosomiasis or Bilharziasis

Schistosomiasis is a widespread disease caused by the infestation of the human body by flukes commonly called blood flukes, of the genus ***Schistosoma***. These flukes cause serious diseases.

Fig. 112.4: Intestinal Schistosoma mansoni, seen under light micrograph

Fig. 112.5: A leg infected with Schistosomiasis

Life cycle of Schistosoma

- Eggs discharged from the host hatch into larval forms in fresh water; from the water, the larvae, ***miracidia,*** invade the snail that acts as an intermediate host.
- The larval form of the parasite undergoes partial maturation in the snail, and then escapes back into the water, as mature larvae called ***cercariae***.
- Humans are infected with *Schistosoma* when they enter water containing infected snails. The larval stages of this flatworm develop in the tissues of infected snails and eventually release fork-tailed cercariae into the water.
- The cercariae penetrate human skin, lose their forked tails, enter the blood, and migrate to major veins in the liver, intestine, or urinary bladder.
- Within about six weeks of infection, the juvenile worms develop into sexually mature adults measuring 1 to 2cm in length.
- The males and females mate and produce microscopic eggs, some of which migrate to the liver and cause a condition known as ***cirrhosis***.
- Other eggs move into the intestines and are passed out in the faeces. When untreated human sewage enters waters containing the snail hosts, the eggs hatch and start a new cycle.

Control and prevention

1. Avoid swimming in or drinking from infected water.
2. Wear protective clothing such as water-proof boots and gloves when entering infected water.

3. Apply special ointments to block penetration of the larvae into the skin.
4. Use drugs such a molluscicides to kill infected water snails.
5. The drug praziquantel has proved effective in killing *Schistosoma* in humans.

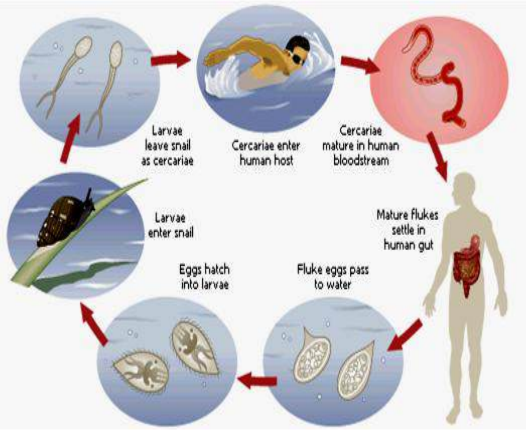


Fig. 112.6: Life cycle of blood fluke, parasite of schistosomiasis

Tapeworm

Tapeworms are flattened worms ranging in length from about 13mm to about 9m. The adult tapeworm is characterized by the presence of a head, or scolex, equipped with a crown of hooklets for attachment to the intestinal lining of its host. At the rear end of the scolex is a narrow neck, from which body segments, or ***proglottids***, are budded off asexually.

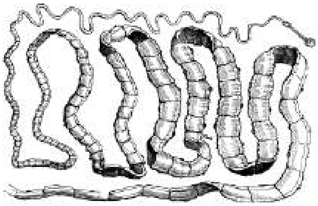


Fig. 112.7: *Tape worm*

Life cycle of tapeworm

- Tapeworms may have a few or thousands of proglottids. The proglottids contain organs of sexual reproduction, each with both testes and ovaries; the segments farthest from the head mature most rapidly and, when ripe, separate from the main body of the worm and pass out with the faeces of the host animal.
- These newly detached proglottids contain numerous eggs, and each egg contains an embryonic tapeworm.
- When the living segment is ingested by another primary host, the proglottids regenerate a new scolex, which attaches itself to the intestinal wall, and the tapeworm resumes its growth by budding.
- When eggs are ingested, they hatch in the intestinal tract and release larval forms, which burrow into the tissues of the host and form ***cysts***
- The larvae often exhibit specific selection of tissues in encysting; for example, one species attacks the liver in humans and dogs, whereas another attacks the brain in sheep, causing the disease known as *gid* or *staggers*.
- When larvae are ingested by a primary host, usually in the form of encysted meat of the intermediate host, they

are stimulated by the gastric juice to develop into adult tapeworms.

- The adults attach themselves to the intestinal wall and absorb partially digested food through their body surface; tapeworms have no mouths or digestive canals.

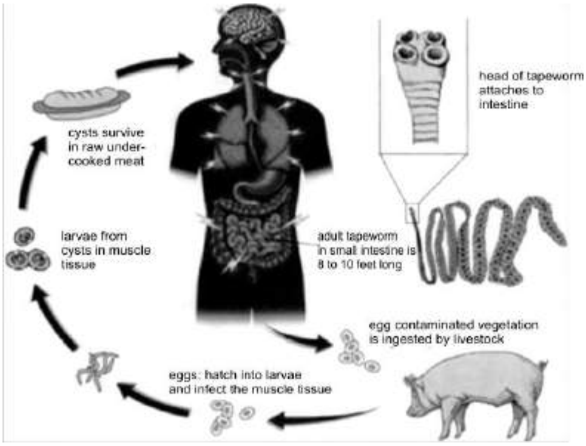


Fig. 112.8: Life cycle of tapeworm

Control of tapeworms

1. Cooking meat properly before eating
2. Maintaining good sanitary conditions
3. Regular de-worming or taking anti-parasitic and anti-inflammatory drugs to destroy parasites

Roundworm

Roundworms, also called nematodes are responsible for various diseases in humans as well as farm animals. An ascarid nematode, a species of roundworm is responsible for the disease ascariasis in humans, and it is the largest and most common parasitic worm in humans

Fig. 112.9: Roundworm

Life cycle of roundworm

- Roundworm normally gets into humans when an ingested fertilised egg becomes a larval worm that penetrates the wall of the duodenum and enters the blood stream.
- From there it gets to the liver and heart and enters pulmonary circulation to break free in the alveoli, where it grows In 3 weeks, the larvae pass from the respiratory system to be coughed up, swallowed, and thus returned to the small intestine, where it matures to adult male and female worms.
- Fertilization can now occur and the female produces as many as 200,000 eggs per day for a year. These fertilized eggs become infectious after 2 weeks in soil; they can persist in soil for 10 years or more.

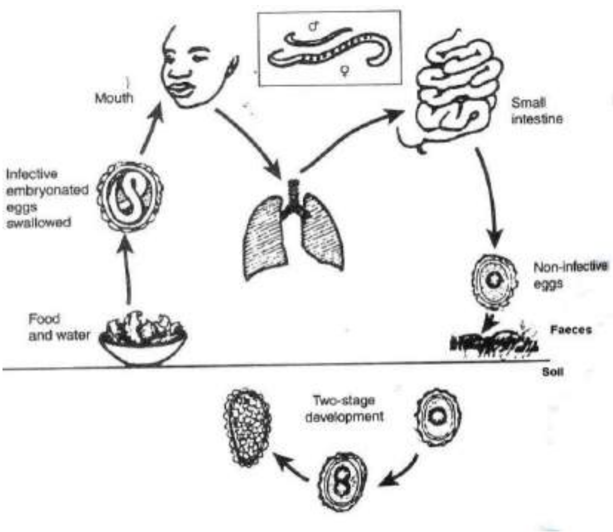


Fig. 113.0: Life cycle of roundworm

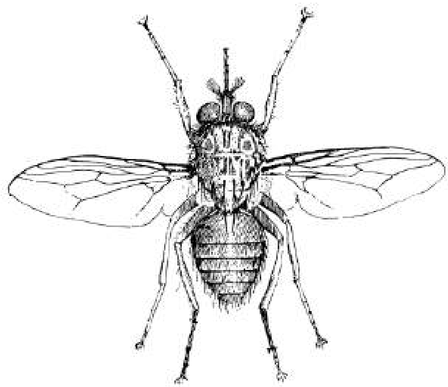


Fig. 113.1: Tsetse fly

Trypanosomiasis

Trypanosomiasis, also sleeping sickness, is an endemic and sometimes epidemic chronic disease caused by a protozoan blood parasite, genus ***Trypanosoma***. Trypanosomiasis is very prevalent in Sub-Saharan Africa. It affects both humans and farm animals.

Causes of trypanosomiasis

The parasite trypanosoma is carried by a blood-sucking insect known as ***tsetse fly***. They also transmit a similar disease called ***nagana*** in domestic animals. Tsetse flies have mandibles modified into bladelike structures used to pierce skin. They readily feed on the blood of humans, domestic animals, and wild game.

Life cycle of tsetse fly

- Female tsetse fly mates just once. After 7 - 9 days she produces a single egg which develops into a larva within her uterus.
- About nine days later, the mother produces a larva which burrows into the ground where it pupates.
- The mother continues to produce a single larva at roughly nine day intervals for her entire life.
- The adult fly emerges from the pupa in the ground after about 30 days.
- Over a period of 12-14 days it matures, mates and, if it is a female, deposits its first larva.
- Thus 50 days elapse between the emergence of one female fly and the subsequent emergence of the first of its progeny.

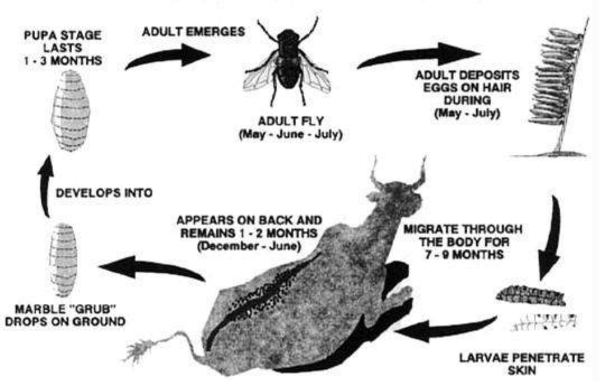


Fig. 113.2: Life cycle of tsetse fly

Control of trypanosomiasis

1. Destruction of the wild game upon which the flies feed.
2. Clearing of woodlands, and periodic burning to prevent the growth of brush.
3. Trapping of tsetse flies, control by natural parasites.
4. Spraying of insecticides in affected areas.
5. Exposing male tsetse flies to gamma radiation to make them sterile or sexually unproductive.
6. Introducing a large population of sterilized male tsetse flies into a wild population.

Mites

Mites are a group of parasitic organisms which normally bite or cause irritation to humans, animals and plants. They normally cause asthma as well as allergies in humans.

Mites often have three pairs of legs in the larval stage and four pairs of legs in the nymph and adult stages. The mouthparts are adapted for piercing.

Life cycle of mites

The dust mite is a common mite found in human populations. They normally prefer humid areas. The average life cycle for a male house dust mite is 10 to 19 days. A mated female house dust mite can last up to 70 days, laying 60 to 100 eggs in the last 5 weeks of her life. In a 10-week life span, a house dust mite will produce approximately 2,000 faecal particles and an even larger number of partially digested enzyme-covered dust particles

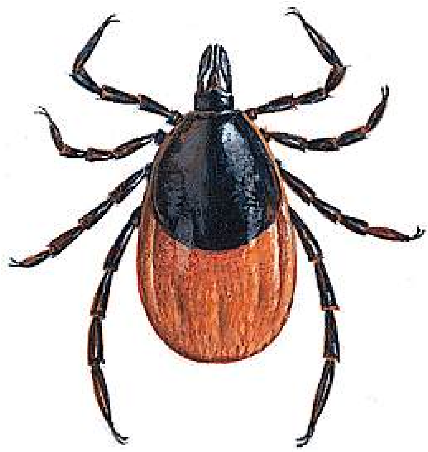


Fig. 113.3: A mite

Control of mites

1. Using a recommended insecticide to control mites.
2. Destroying their hiding places. Mostly under carpets, in pillows and mattresses, etc.
3. Regularly cleaning or sweeping rooms.

4. Regularly washing beddings such as bed sheets and pillow covers.
5. Using air-conditioners to reduce the temperature and humidity of a room.
6. Dusting farm animals with recommended insecticides.

Tick

Ticks are bloodsucking parasites found mostly in woods, tall grass, and shrubby vegetation where they climb onto plants and wait to jump on a passing host.
They are particularly sensitive to carbon dioxide and movement—signals that a host is nearby. Their grasping forelegs allow them to climb on a host. They quickly find a protected spot on the host's body, sink their mouthparts into the flesh, and begin to feed. When full, they drop off the host.
Several diseases are transmitted to humans and domestic animals through tick bites or tick excrement. Some of these are spotted fever, relapsing fever, Lyme disease, tularemia, some forms of encephalitis, and Texas cattle fever.

Fig. 113.4: Sheep tick

Control of tick

1. Keeping the surroundings clean.
2. Practicing rotational grazing
3. Spraying of animals with recommended chemical or dipping to destroy tick
4. Regularly changing animal bedding
5. Culling infected animals
6. Picking ticks from the bodies of host animals

Flea

Fleas are bloodsucking, wingless insects.
Fleas lay their eggs under carpets, in the folds of tapestry, in refuse piles, and in other places that provide safety and adequate nutrition.

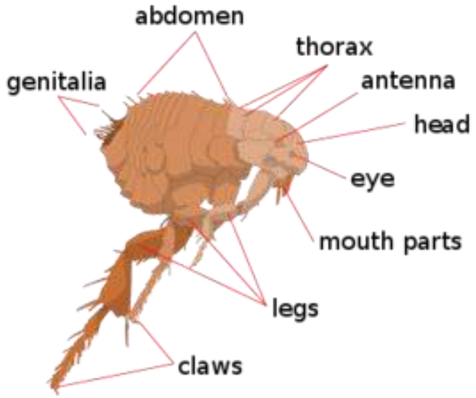


Fig. 113.5: Flea

Life cycle of fleas

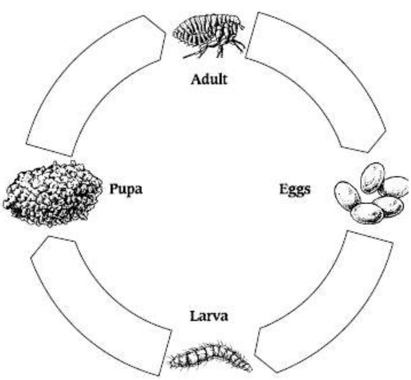
In 6 to 12 days the eggs hatch, becoming larvae with biting mouth parts. After a few days of voracious feeding upon organic refuse, the larvae spin cocoons and enter a pupal stage. The adult flea emerges from the cocoon in a few weeks.

Adult fleas, which are slightly more than 0.3 cm long, have broad, rather flat bodies, short antennae, and piercing and sucking mouth parts; their eyes are either minute or absent. Their long, powerful legs enable them to leap relatively high into the air.

Several flea species infest household pets and domestic animals. The dog flea and the cat flea are two of the most common species, both of which are parasites also on human beings, poultry, and livestock.

Fleas are intermediate hosts of a common cat and dog parasite, the ***cucumber tapeworm***. Tapeworm eggs are deposited in feacal matter, and some of these eggs may cling to the hair of the primary host. Fleas swallow the eggs, which then undergo some development in the flea. If an animal or person accidentally swallows an infected flea, an adult tapeworm develops in the new host.

Dog eczema is usually associated with the presence of fleas.



Life cycle of flea

Control of fleas

1. Destroying the adults with insecticides
2. Making breeding places unsuitable for larval life.
3. Adult fleas are destroyed by bathing the host with strong soap and by applying insecticides or petroleum.

Louse (plural **Lice)**

Lice are wingless parasitic insects which normally thrive in conditions of filth and overcrowding.

Effects of lice infestation

1. Lice usually bite their hosts and feed on loose skins.
2. Lice are carriers of typhus and relapsing fever.
3. Heavy infestations of lice may cause intense skin irritation, and scratching for relief may lead to secondary infections.
4. In domestic animals, rubbing and damage to hides and wool may also occur, and meat and egg production may be reduced.
5. In badly infested birds, the feathers may be severely damaged.
6. One of the dog lice is the intermediate host of the dog tapeworm, and a rat louse is a transmitter of murine typhus among rats.

Control of louse

1. Avoid overcrowding of humans and animals.
2. Avoid sharing combs and clothing with an infected person.
3. Infected animals should be culled.
4. Farm animals should be dusted with insecticides.
5. Maintain good sanitary conditions.

COMMON PEST AND PARASITES OF PLANTS

Some plants are parasitic on other plants while some other living organism only rely on plants for their survival. They include mistletoe, dodder, weevils, leaf hopper, etc

Mistletoe

Mistletoe is a widespread flowering plant parasite that grows on oak, apple, juniper, pine and other trees. Most mistletoe species are actually *hemiparasites*, or partial parasites.

Their leaves, found on the surface of branches and trunks, produce sugar through photosynthesis, but their roots penetrate into the host tree's tissue and absorb its nutrients. Some species of mistletoe lack leaves altogether and rely solely on their host tree to provide all their nourishment.

Dodder

Dodder, also called love vine, strangle-weed, goldthread, and hell-bind is one of

Fig. 113.5: *Mistletoe*

the few higher plants that are parasitic on other plants. It has yellowish or reddish threadlike twining stems and no leaves or chlorophyll.

During germination, the plant attaches itself to surrounding vegetation by means of sucking organs called ***haustoria***, through which it takes nourishment from the host.

Dodder is especially harmful to clover, alfalfa, and flax and is also found on ornamental plants. Commercial clover and alfalfa seed is always specially treated to avoid contamination by dodder seed.

Fig. 113.6: *Dodder*

Weevils

Weevils are among the most destructive of pests that attack crops. They feed entirely on plant life, causing much damage to crops.

Adult weevils lay their eggs in stalks or seeds of crops such as cotton, wheat, rice, and alfalfa. As larvae, weevils then feed on these plant tissues, extensively damaging the plant in the process. There are about 30,000 to 40,000 species of weevils some families or subfamilies of weevils include fungus weevils, primitive weevils, snout weevils, leaf-rolling weevils, grain weevils, flea weevils, acorn and nut weevils.

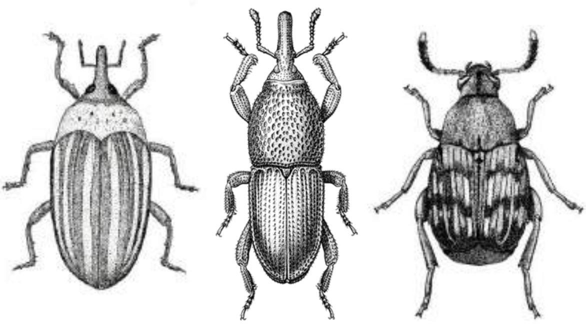


Fig. 113.7: Some species of weevils

Control of weevils

Weevils can be controlled by

1. Burning infested fruits, nuts, or stems;
2. Ploughing the ground in which the insects pupate;
3. Spraying crops with insecticides.

Leafhoppers

Leafhoppers have piercing sucking mouthparts for feeding on the juices of vegetation. In doing so, they may also transmit viral and fungal diseases from plant to plant and can cause extensive damage to shrubs, field crops, and fruit trees.

Sweet liquid called *honeydew* composed of unused sap and other excretions that they exude from their anus is eaten by ants and other insects. Leafhoppers lay eggs in leaves and stems.

Fig. 113.8: Leafhopper

TEST QUESTIONS

7. Briefly explain the following terms:
   (a) pest;
   (b) parasite.

8. Describe how female anopheles mosquitoes transmit malaria.

9. (a) Mention four symptoms of malaria.
   (b) State four methods of controlling malaria.
10. Describe the life cycle of schistosoma.

5. (a) state three effects of schistosomisis
   (b) List five ways of controlling schistosomiasis.

6. (a) Describe the life cycle of tapeworm
   (b) State the control measures for tapeworm.

7. Describe the life cycle of tsetse flies and their effects on livestock.

8. Describe the following pests and how they are controlled:
   (a) tick;
   (b) mite;
   (c) flea;
   (d) louse.

9. Describe the following pests of plants:
   (a) dodder;
   (b) mistletoe;
   (c) weevils.

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CROP PRODUCTION

Specific Objectives

After completing this chapter, you will be able to:

- Describe the general principles of crop production
- Cultivate a crop up to harvesting stage

INTRODUCTION

Crop production is the extensive cultivation of plants to yield food, feed, or fibre; to provide medicinal or industrial ingredients or to grow ornamental products.

Crop production developed in ancient times as hunters and gatherers of the Stone Age turned to the cultivation of favoured species.

Modern crops were gradually derived from their wild ancestors through continual selection for larger seed size, improved fruit and other desired traits.

Classification of crops

There are various classes of crops based on their uses and method of cultivation. Some classes of crops include:

1. **Grains and cereal -** e.g. maize, rice, sorghum, wheat, millet
2. **Vegetables** – e.g. tomatoes, onion, pepper, cabbage
3. **Stem and Root tubers** – eg. cassava, yam, potatoes, cocoyam
4. **Fruits** – eg. citrus fruits, pineapple, mango, banana
5. **Grain legumes** – eg. groundnut, bambara beans, cowpea, soya beans
6. **Beverage crops** – eg. cocoa, coffee, tea
7. **Industrial crops** – eg. cotton, tobacco, rubber, sugar cane
8. **Oil crops** – eg. oil palm, coconut, shea butter tree, groundnut
9. **Ornamental crops** – eg. roses, lawns, sunflower
10. **Fibre crops** – eg. cotton,

Other classes of crops

Most crops are seasonal; that is they have a specific period they are planted.

Some seasonal crops are:

- ***annual*** or ***arable*** crops: Grown, reproduce and die in a year;
- ***biennial crops***: have two-year life span;
- ***perennial crops***: have life span of 3 to 5 years;
- ***permanent crops***: last for 10 to 30 years.

Uses of crops

Crops are used for many purposes including:

Food

The major importance of crops is food; and the crops produced as food for human consumption are known as ***food crops***. Some food crops are hugely used by some people in a community either because of its availability or quality. Such crops are their ***staple food*** crops.

Examples of staple food crops in Ghana are rice, maize, millet, cassava, yam, etc.

Animal feed

Farm animals feed mostly on vegetation such as grasses and forage legumes as well as food crops such as root tubers and grains.

Raw materials

Some crops are grown to feed the agro-industries. Crops like cocoa, sugar cane and tobacco are used to make chocolate, sugar and cigarette respectively. Fibre crops such as cotton are used to make clothing of all kinds for humans.

Drugs

Some crops are cultivated and used as alternative medicines or herbal drugs. In West Africa, most people rely on herbal drugs to either stay fit or recover from diseases.

Ornaments

Lawns, roses and other ornamental crops are planted to beautify a particular area. Hedges both appeal and also enclose an area.

Shelter

Some tree crops are cut and used to build houses. Live tree crops serve as shelter from the sun and other elements. Grasses and leaves are sometimes used as roofing materials.

GENERAL PRINCIPLES OF CROP PRODUCTION

For healthy growth, development and yield of crops, a crop farmer should consider the following guidelines:

a) Selection of appropriate varieties

b) Site selection and land preparation
c) Method of propagation and planting methods
d) Cultural practices
e) Pests and disease control
f) Harvesting, processing, storage and marketing

a) Selection of appropriate varieties

Once the crop farmer is certain on the type of crop he wants to cultivate, the next step is selecting the variety of that species of crop.
For instance, if the farmer wants to grow cassava, he may choose to go in for any of the following varieties – Obaatanpa, Dobidi or Okomasa.

Factors to consider when selecting a crop variety

1. Resistance to pests and disease
2. Drought tolerance
3. Climatic conditions (e.g. Rainfall, sunlight, temperature, wind)
4. Soil requirement (e.g. pH, moisture, nutrient, texture etc.)
5. Maturity period
6. Growth habit
7. Yield
8. Market

b) Site selection and land preparation

The type of land used to cultivate a particular variety of crop is very important. Before the farmer selects a piece of land for growing his crop, he should consider:

The type of soil

The soil should be fertile, have good pH value, contain enough moisture, and have good texture and structure. All these should be known with respect to the selected crop variety.

Climatic condition

How often does it rain on the land, how hot or cold is the temperature, how hard does the sunshine or the wind blow? These are the questions that the farmer should ask himself.
These climatic factors affect the way the crop grows and yields. While some crops like more water, others do better with little water and so on.

Source of water

The farmer will be required to irrigate his crops, especially, during the dry season to stabilize the soil texture. Nearness to a river, stream or lake will help the farmer irrigate his crops without so much labour.

Topography

Land with low level or gentle slope topography is mostly preferred to steep-sloped land. This is because such lands can easily be tilled mechanically and have low incidence of erosion.

If the land available is hilly, the farmer can adopt soil conservation methods like terracing, contour ploughing and strip cropping.

Availability of labour

Large-scale farming is usually labour-intensive. It involves weeding, planting and harvesting. Locating the farm to where labour is available is a good practice.

Accessibility to farm

Being accessible means the farmer can get to the farm with ease. That is, either the farmer lives on or closer to the farm or there is an accessible road to the farm. Having accessible road will ensure efficient transportation of heavy implements to the farm as well as transporting the farm produce to the market.

Land preparation

There are three major stages in land preparation – land clearing, soil tillage and seedbed preparation.

Land clearing

Land clearing is the removal of vegetation cover from the land, either manually or mechanically.

Land clearing gives way for soil tillage, planting of crops and other cultural practices. The cleared vegetation could be used as mulch or gathered and burnt.

Soil tillage

Soil tillage is the physical manipulation of the soil to gain condition favourable for crop production.

Soil tillage creates optimum environmental conditions for plant growth.

It involves breaking up, stirring and turning the surface of the soil. Primary tillage is the initial breaking up of the surface of the land and the subsequent tillage is the secondary tillage.

Tillage can be done manually with hoe or mechanically using disc plough or mould-board plough driven by a tractor.

Fig. 113.9: A farmer tilling a land

Importance of soil tillage

1. Soil tillage improves soil structure.
2. It loosens the soil for water to pass through easily.
3. It improves root penetration.
4. It cleans the soil of stubbles.
5. It controls soil erosion.

6. Soil tillage is used to mix fertilizers into the soil.
7. It levels the surface of the land and aids in mechanical sowing.
8. It increases the aeration of the soil.
9. It exposes the larvae and pupae of soil-borne crop pest to the sun, thereby destroying them

Seedbed preparation

A seedbed is a place where seeds germinate and from which established plants get nourishment and support through their roots.

Seedbeds increase the depth of soil surface, limit erosion and provide good drainage. They provide the basis for the expansion of root and stem tubers of crops as well as adequate supply and retention of soil water. Seedbeds for vegetable production are normally raised and rectangular in shape.

Characteristics of a good seedbed

1. Fine and granular surface soil
2. Firm underneath to ensure satisfactory movement of moisture to sown seeds
3. Free from weeds
4. Retains adequate moisture but porous enough to ensure good drainage and aeration.

(c) Method of propagation and planting methods

There are two main methods of plant propagation –

- sexual propagation and
- asexual propagation.

Sexual propagation

Sexual propagation is the use of seeds to propagate crops.

Characteristics of seeds used for sexual reproduction

1. Ability to germinate into normal seedlings under normal soil conditions.
2. Free from seed-bed diseases and pest damage.
3. Free from seed weeds and seeds of other crop varieties.
4. Should have high germination percentage.
5. Should have uniform shapes and sizes
6. Should not be mouldy, cracked or shrunken.

A farmer may obtain good seeds from certified agricultural agencies or grow his own from laid down procedures. The seeds should be chemically treated to avoid attacks from pests and parasites.

Planting of seeds

Depending mostly on the sizes and shapes of the seeds, they could either be planted or sown. Large seeds are planted while small seeds are sown.

Differences between planting and sowing of seeds

Planting: Seeds are planted (at stake or in situ) by making holes in the soil and putting the seeds in them after which it is covered and firmed.

Examples of seeds which are normally planted are maize, groundnut and beans. Seeds may be planted by drilling, precision planting.

Sowing: Seeds are sown by scattering them on the soil and covering them with a layer of soil to protect them from direct sunlight, pests, or being washed away by run-off water.

It is advisable to sow seeds in a nursery to produce seedlings and then transplanting them onto a prepared field.

Examples of seeds sown are rice, tomatoes, and millet. Seeds are sown by broadcasting and random planting.

Fig. 114.0: Planting of seeds

Time of planting

Planting time of crops is normally influenced by factors such as rainfall, temperature, maturity period, day length, availability of labour, diseases and pest. Crops are generally planted at the beginning of the rainfall season when there is adequate moisture and temperature for seed germination and for the growth and development of crops.

Some crops such as maize and millet are planted at a chosen time within the ideal cropping season in order for them to mature immediately after the rains. This provides suitable conditions for harvesting and processing the produce.

Plant population and spacing

The number of plants per hectare of farmland is referred to as plant population.

Very high plant population reduces crops yield because of competition for nutrients, while excessively low plant population cuts down yield because of unused land and its resources.

Plant spacing varies from plant to plant since each plant has its own growth habit.

Growth habit is the relative spread of plant canopy and root system.

Plant population is the product of the number of crop stands per hectare of land

and number of plants per stand. The number of plants is determined by dividing one hectare of land (10 000 m^2) by stands or feeding area of a plant in square metres.

Mathematically,

$$\text{Number of stands} = \frac{Plant\ \ population}{No.of\ \ plants\ \ per\ \ stand}$$

If the area of farmland is more or less than a hectare, the area of the farmland is calculated in metres and used in the formula in place of hectare.

Example

A maize farmer plants three seeds per hole at a spacing of 40 cm by 80 cm on 2 hectares of land. What is the plant population if there is 98 % germination?

Solution

$$\text{No. of crop stands} = \frac{area\ \ of\ \ farm}{feeding\ \ area\ \ per\ \ stand}$$

$$= \frac{2\ hectare\ s}{0.4 \times 0.8} = \frac{2 \times 10\ 000}{0.32}$$

$$= 62\ 500 \text{ stands}$$

Plant population =
No. of stands x No. of plants per stand

$$= 62\ 500 \times 3 = 187\ 500$$

At 98% germination plant population

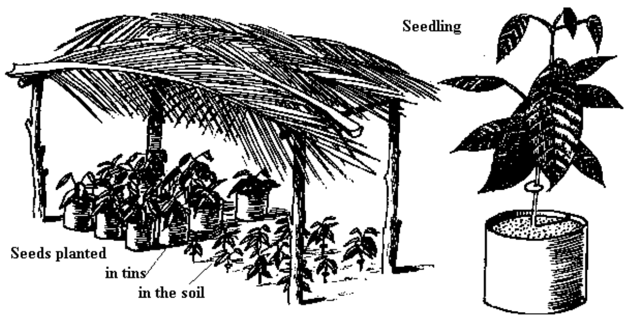
$$= \frac{98}{100} \times 187500 \quad = 183\ 450 \text{ plants}$$

Plant nursery and nursery beds

A plant nursery is a specially selected site where seedlings are raised.

In nurseries, plants are grown on nursery beds, or containers. Seeds are sown in a nursery by drilling or broadcasting method and then covered with a layer of soil. It is then mulched with grass or other mulching materials.

A light shade is raised above the bed to protect the weak plant from excessive heat and the impact of heavy rainfall. Nursery beds are regularly watered until germination. The next stage after germination is prickling-out.



Plant nursery and seedling

Pricking-out is the transfer of seedlings onto a larger nursery bed or seed boxes at a spacing of about 4 cm.

Importance of pricking out

1. It eliminates competition for nutrients by seedlings.
2. It helps farmers to get seedlings of uniform sizes for transplanting.

3. It helps to eliminate late germinators since only active seedlings are pricked out.
4. It gives the farmer a good account of the number of viable seedlings.
5. It reduces the incidence of diseases because of the elimination of overcrowding.

Characteristics of a good nursery bed

1. A nursery bed should be free from disease pathogens, weeds and pests.
2. It should be rich in vitamins and free draining.
3. It should be sited in open places with enough sunlight and free air flow.
4. It should be close to a reliable source of clean water to make regular watering of seedlings easy.

Hardening-off

Before seedlings are transplanted, they are hardened-off.

Hardening-off is the act of preparing seedlings for transplanting by gradually exposing them to harsher conditions.

Hardening off involves inducing seedlings to higher temperature, higher light intensity, etc.

Importance of hardening off

- It helps the seedlings to accumulate carbohydrate in their tissues which enables them to produce roots quickly after transplanting.
- It enables transplants to withstand harsher environmental conditions such as dry winds, intense sunlight and temperature.

Fig. 114.1: A nursery bed

Transplanting

***Transplanting is the transfer of seedlings from the nursery for planting*.**

Transplanting is normally done early in the morning or late in the afternoon when light intensity is low, and for that matter the rate of transpiration is low.

Before transplanting, seedlings should be watered around the roots in order to soften the soil and reduce damage to it during removal. It is also advisable to heavily water seedlings after transplanting in the morning so that the high afternoon temperatures do not affect their establishment.

Vegetative propagation (asexual reproduction)

Vegetative propagation is the use of non-reproductive parts of plants to induce the formation of roots and the regeneration of plants.

Depending on the extent of human interference, vegetative propagation may be described as natural or artificial.

Natural vegetative propagation

This method involves the use of naturally modified vegetative structures such a stem tuber, bulb, rhizome, corm and off-shoots (slips, suckers and crowns).

These vegetative structures produce new plant and in the process renew themselves because they are naturally endowed with food reserves to help them survive harsh environmental conditions.

Artificial vegetative propagation

This method of vegetative propagation is influenced by humans in regulating and inducing the growth of new plants from prepared vegetative structures.

Examples of artificial vegetative propagation methods include layering, budding, cottage (stem, root or leaf cutting) and grafting. Another artificial propagation method is ***tissue culture***.

Plant tissue culture

Tissue culture is the cultivation of plant cells, tissues, or organs on specially formulated nutrient media.

Plant tissue culture relies on the fact that many plant cells have the ability to regenerate a whole plant (*totipotency*). Single cells, plant cells without cell walls (protoplasts), pieces of leaves, stems or roots can often be used to generate a new plant on culture media given the required nutrients and plant hormones.

Advantages of tissue culture

1. High quality of produce
2. Rapid propagation or maturity of plant
3. Better production programming
4. Production of disease-free plants
5. Economic use of planting materials
6. The production of exact copies of plants that produce particularly good flowers, fruits, or have other desirable traits.
7. The production of multiples of plants in the absence of seeds.
   necessary pollinators to produce seeds.
8. The production of plants in sterile containers that allows them to be moved with greatly reduced chances of transmitting diseases, pests, and pathogens.
9. The production of plants from seeds that otherwise may have very low chances of germinating and growing.

10. it cleans particular plants of viral and other infections and quickly multiplies these plants as ***cleaned stock*** for horticulture and agriculture.

(d) Cultural practices (Post-planting activities)

After planting the crops, there are some basic activities that are performed until the time of harvesting. These activities are known as cultural practices. Cultural practices in crop production include:

Irrigation/ watering

Irrigation is the artificial means of providing or replenishing the moisture of the soil to support plant growth.

It is normally done right after planting or sowing the seeds. Some methods of irrigation used on large-scale farms include sprinkler irrigation, trickle or drip irrigation and furrow irrigation.

Importance of irrigation

1. It regulates soil temperature.
2. It provides adequate moisture to the plants.
3. It keeps the temperature of the plants cool through transpiration.
4. It provides a suitable medium for soil nutrients to dissolve in for absorption by plant roots.
5. It maintains the turgidity of the plants to avoid wilting and lodging.

Mulching

This is the use of suitable materials to cover the surface of the soil. A material used for mulching is known as mulch. Examples of suitable mulches are grass, rice straw, sawdust, paper and corn cob.

Importance of mulching

1. It maintains or conserves soil moisture.
2. It controls soil erosion.
3. It regulates soil temperature.
4. It suppresses weeds.
5. It increases water infiltration into the soil.
6. The decay of the mulches improves the organic matter content in the soil.

Thinning-out

Thinning-out is the removal of extra seedlings from a stand.

It is ideal for each stand to have at most two plants, but during seed planting more than two seeds may be planted per hole to compensate for the incidence of some seeds not germinating.

Importance of thinning-out

1. It reduces competitions among seedlings for nutrients.
2. It control incidence of diseases among seedlings.
3. It prevents misshaping in carrots and onions.

Filling-in or supplying

This is the replanting of seeds at specific spots where planted seeds failed to germinate or died immediately after germination.

It is advisable to do filling-in before two weeks of first planting in order to avoid variation in seedling size and the shading of younger seedlings by older ones which results in poor growth.

Importance of filling-in

It helps the farmer to get the total number of plants he wants.

Fertilizer application

In fertilizer application, the farmer supplies fertilizers or organic manure to the soil to supplement its nutrient content. A nutrient-rich soil ensures proper growth and development of crops.

Weed control

Weed control is the use of *mechanical* (cutlasses, hoes), *biological* (living organisms), *chemical* (weed-killers) or *cultural* (crop rotation, mulching) methods in preventing the growth of weeds on a farmland.

Importance of weed control

1. It controls competition for nutrients between growing crops and weeds.
2. It checks incidence of disease in plants.
3. It improves aeration for growing crops.
4. It helps the farmer to easily estimate the number of crops on his farm.
5. It creates easy access through the farm.

Pruning

Pruning is the removal of unwanted parts of a plant.

Pruning may be done on any part of the plant to achieve horticultural or agronomic objectives. Pruning may be done with a cutlass, shears, saw, knife or secateurs.

Importance of pruning

1. It gives plants desired shapes and forms.
2. It removes diseased or damaged parts of plants.
3. It promotes fruit formation.
4. It improves aeration within the plant.
5. It controls the size of plants.
6. It enhances uniform ripening of fruits.
7. It promotes the penetration of sunlight onto the plant.
8. It helps in carrying out other cultural practices due to space creation.

Staking

Staking is the use of wooden or metallic pole to physically support plants which have week stems to grow on.

Qualities of a good staking pole or material

1. It should not interfere with the growth of the plant.
2. It should be strong enough to support the plant.
3. It should last for a long time.
4. It should not be too fresh to sprout quickly.
5. It should be free from thorns and pricks.

Importance of staking

1. It keeps plants erect.
2. It reduces rotting of fruit vegetables.
3. It provides good aeration to the staked plant.
4. It enables plants to display their leaves to receive sunlight for photosynthesis.
5. It makes harvesting of produce easier.
6. It facilitates weeding on the farm.

(e) Pest and disease control

Pests and disease-causing parasites cause low yield in crops and could even destroy the crops. They normally reduce the market value and at the same time increase the cost of production.

Methods to control pests and diseases

1. Physical methods

These include using devices and means that destroy pest and pathogens by affecting their physical environment.

Examples of physical methods of pest and disease control

i. Fencing farm to keep out rodents
ii. Using barriers to protect crops against large pests as well as microscopic pathogens
iii. Screening crops to keep off birds and insects
iv. Using traps and scarecrows
v. Heat sterilization of the soil.

2. Chemical methods

These methods are the use of chemicals known as ***pesticides*** to control pest and disease pathogens. Pesticides are normally in liquid forms and are sprayed on farms.

The table below gives types of pests and pathogens and the pesticides used to control them.

Table 8.5: Pesticides and pest or parasites they are used to control

Pesticide	Pest/ pathogen used on
Insecticides	Insects
Fungicides	Fungi
Rodenticides	Rodents
Bactericides	Bacteria
Avicides	Bird
Nematicides	Nematodes

3. Biological methods

This is the use of some living organisms to control pests and pathogens by mostly feeding on them.

Examples of biological methods

i. Use of dogs and cats to catch and kill rodents
ii. Use of chicken to control cotton stainers in cotton fields
iii. Planting of African marigold to control nematodes

4. Cultural methods

These include the use of agronomic activities to reduce the incidence of pest and pathogen attacks on crops.

Examples of cultural practices

i. Use of resistant varieties of crops.
ii. Early planting and harvesting to control pests and pathogens attack.
iii. Heavy organic-manuring to control nematodes.
iv. Eliminating diseased seeds and plant.
v. Use of clean and healthy panting materials.
vi. Crop rotation.
vii. Removing infected part of a plant through pruning.
viii. Soil tillage to expose soil borne pests and parasites.

5. Integrated Pest Management

This is the use of environmentally friendly measures in controlling pests and diseases. The main objective of this method is to keep the population of pest very low so that they will not be able to do much damage to crops.

Some Integrated Pest Management techniques include preventing pests from entering the farm, physically removing pests from farm etc.

Importance of Integrated Pest Management

i. It leads to the production of good quality produce.
ii. It increases crop yield.
iii. It is cost effective.
iv. It has little or no adverse effect on the environment.

6. Prohibitions

Prohibitions involves restricting crops or their planting materials suspected to be carrying pests and pathogens from entering a country or being moved from one place to another within a country.

(f) Harvesting

The final stage in the field operations with crops is harvesting. Harvesting is done when crops mature. Different crops have different maturity timelines. It is advisable for the farmer to harvest his produce on time so as to prevent pests and diseases from attacking them.

The method of harvesting should also be taken into consideration. The choice of harvesting method depends on the type of crop and the scale of cultivation. While manual harvesting methods such as the use of cutlasses, hoes, hands etc. may be used in small-scale farming, heavy-duty machinery such as combine harvesters, cassava lifter etc. are often employed in large-scale farming.

Processing

In their raw form, most farm produce cannot be used; they, therefore, have to be processed. Thus, have their forms changed, either manually or with machines, in order to make them usable and also easy to store. The method of processing used is different from plant to plant. Cereals are threshed, shelled, dehusked or winnowed with equipments such as threshing machines, shellers and winnows.
Cassava may be grated with cassava graters while palm fruits are digested with palm fruit digesters.

Storage

After harvesting, the crops could be kept from damage by storing them properly. Rice, for example is sun-dried after harvesting to remove moisture, after which it is stored in sacks. Yam is stored in a cool and well ventilated shed, while cassava can be store in water or underground for a short period.

In large quantities, farm produce may be stored in silos or barns. Some produce which are highly susceptible to spoilage may be stored in refrigerators and freezers. Stored crops are prevented from contact with pests, rain, mould, fungus or excessive heat.

Marketing

Cash crops, such as cocoa and coffee are sold to the manufacturing industries through agents, while food crops are sold in various open markets. Most communities have market days where farmers bring their produce to sell.

In order to successfully produce crops, a farmer needs to follow the general principles explained above. The principles could differ based on the crops cultivated.

FARM TOOLS

Farm tools are various equipments and machineries used in the farm to help cultivate crops successfully. Different farm tools are used for different fields of work on the farm. Table 8.6 below gives some farm tools and their uses.

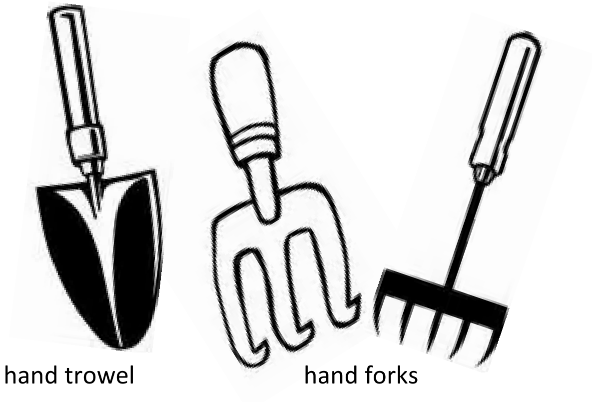


Table 8.6: Farm tools and their uses

Farm tool	Uses
Cutlass	Weeding, pruning, sharpening pegs and stakes
Hoe	Weeding, bed preparation, marking out
Hand fork	Turning/ stirring-up soil, making moulds
Secateurs	Cutting hardwood, pruning bushes
Watering can	Irrigating crops
Spade	Turning, digging and lifting soil
Axe	Pruning and hardwood cutting
Rake	Gathering rubbish, levelling the soil
Budding knife	Making budding and grafting
Pruning saw	Pruning branches
Measuring tape	Measuring out
Sickle	Harvesting rice, wheat, millet
Shovel	Turning and lifting soil
Hand trowel	Digging, cutting roots and removing stumps
Dibber	Making, stirring and loosening beds
Shears	Cutting and trimming hedges
Knapsack sprayer	Spraying chemicals such as pesticides, insecticides, weed-killers
Wheel barrow	Transporting produce, tools and fertilizers over short distances
Digging mattock	Digging out and removing stumps
Pick axe	Uprooting stumps, making trenches
Long-handle sickle	Harvesting cocoa and oil palm
Plough	Used to till the soil

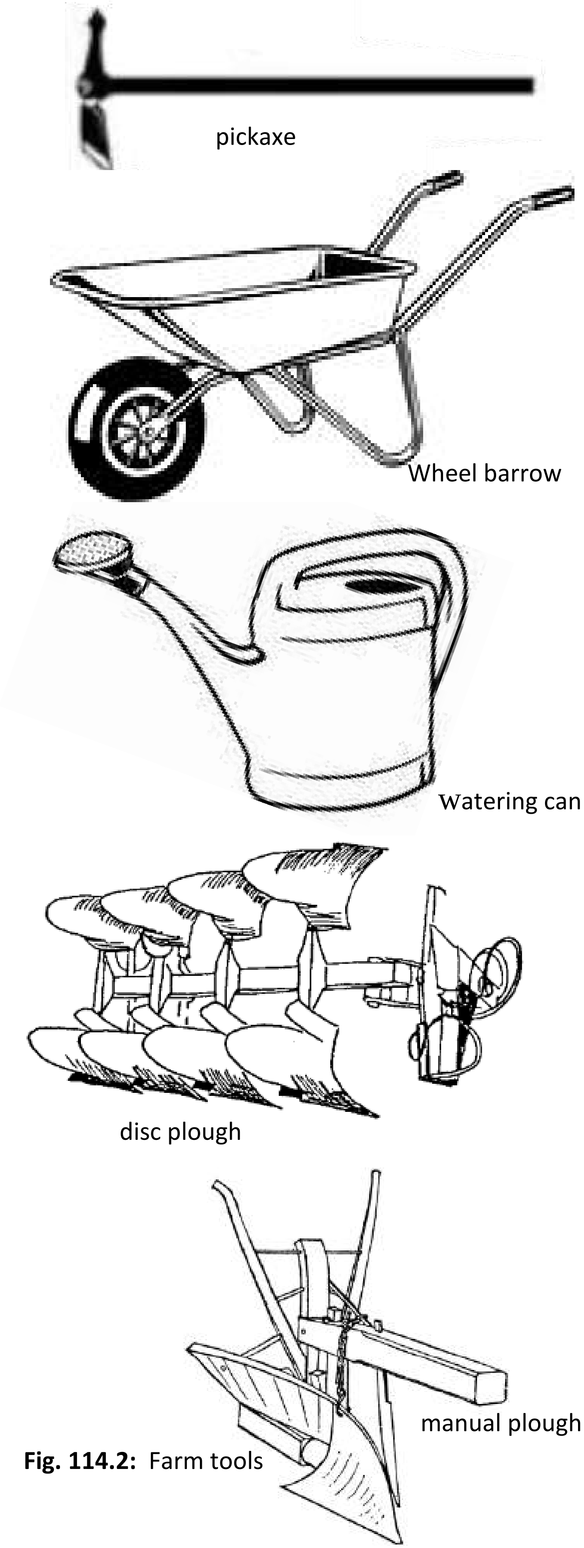


Fig. 114.2: Farm tools

PRODUCTION OF CROPS

Production of vegetables

Vegetables are the edible products of herbaceous plants (i.e. plants with soft stems).

Vegetables can be grouped according to the edible part of each plant: leaves (lettuce), stalks (celery), roots (carrot), tubers (potato), bulb (onion) and flowers (broccoli). In addition, fruits such as tomato and seeds such as pea are considered vegetables. Vegetables may be eaten raw or cooked

Importance of vegetables

1. Vegetables are valuable sources of vitamins.
2. They provide the body with energy.
3. They provide vital minerals needed by the body such as calcium, iodine, iron etc.
4. They are important sources of roughage which helps in digestion and prevents constipation.
5. They are low in fat and calories therefore, keeping the body healthy.

Okro/ Okra (*Hibiscus esculentus or Abelmoschus esculentus*)

Okro (okra), also called gumbo is an annual herb which is cultivated for its long, many-seeded pod, and when young and green, is used to thicken soups and stews and as a cooked vegetable. The okro bears large, yellow flowers, similar to those of the closely related hibiscus, and is occasionally planted in flower gardens.

Fig. 114.3: okro

Soil and climate

Okra prefers a well-drained soil and requires annual rainfall of 800 mm to 1000 mm and an average temperature of 25^{o}C. It tolerates acidic conditions but does not do well in heavy clays.

Planting

Okro is planted at the beginning of the rainy season, as a sole plant or an interplant crop. Two or three viable seeds are planted at stake on a bed or field to a depth of 3 – 4 cm with a spacing of 30 – 40cm and 50 – 70cm within and between rows. The seedlings are thinned-out a week after germination to leave one plant per stand.

Cultural practices

The crop is constantly watered up to fruiting stage, especially if rainfall is inadequate. To stimulate reasonable vegetative growth for efficient photosynthesis, the okro crop is top-dressed with fertilizers such as NPK.

Pest and disease control

Viral diseases which affect okro are leaf mosaic and leaf curl. These diseases are best controlled by crop rotation and the use of resistant varieties. The vector of leaf mosaic, white fly, can be destroyed with insecticides.

Fungal diseases such as powdery mildew and black leaf mould attack both leaves and fruits of okro. They are controlled through crop rotation, clean husbandry and the use of fungicides such as Dinocap and Benomyl.

Leaf and fruit-eating caterpillars, which are the major pests of okro, eat and kill seedlings, eat leaves and fruits of older plants. These pests are controlled by spraying with Cypermethrin and Carbaryl.

Harvesting

After maturity (8 to 12 weeks after planting) okro fruits are harvested either by hand picking or by cutting with a knife. Fruits should be harvested on time to prevent them from toughening and hardening.

Production of cereals and grains

Cereals and grains are various species of the grass family, cultivated for their seed, which is used as food. The most extensively cultivated grains are maize, rice, wheat, oats, rice, millet, and the grain sorghums known as durra or guinea corn.

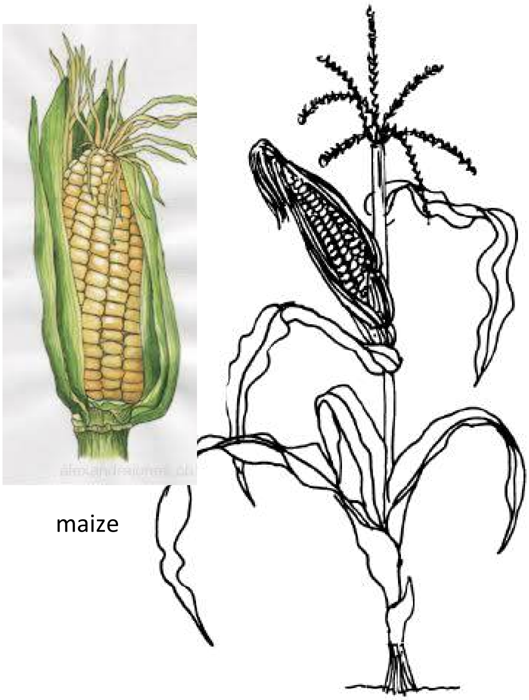
Maize (*Zea mays*)

Common varieties

The varieties of maize in Ghana include Dobidi, Obaatanpa, Safita, Temale Yellow etc.

Soil and climate

Maize grows well in a well-drained loam soil, which has temperature of about 20^{o}C and pH of 5 to 7. It requires consistent



maize

Maize plant

Fig. 114.4

rainfall or regular irrigation. Maize needs a dry weather for maturity and harvesting.

Planting

Maize seedbed should be weed-free and be finely tilled. Seed is planted three seeds per hole at a depth of 3cm – 5cm. Seeds are normally planted at the onset of rain, in March and August.

Cultural practices

Organic manure may be ploughed with the soil before planting. A base dress of compound fertilizer may be applied two weeks before planting. Nitrate top-dressing can be applied when the crop reaches a height of 40 – 50cm. Weeds can be controlled using either physical or cultural methods. Week plants can be removed to give room for healthy ones.

Pest and disease control

Common maize pests include weevils and stem borers. They can be controlled by spraying with insecticides. Birds and rats can be kept off the farm or storage by Integrated Pest Management methods. Maize diseases such as smut and mildews can be controlled with fungicides.

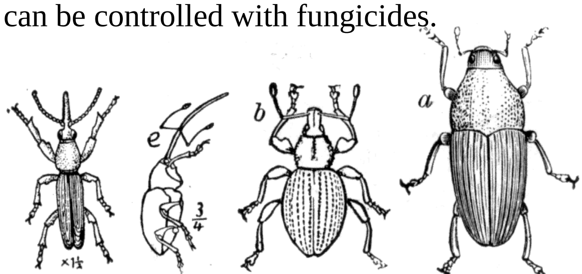


Fig. 114.5: Maize weevils

Harvesting

Maize matures between 12 to 15 weeks after planting, depending on the variety. Its maturity can be recognized by yellowing of the leaves, hardness of grain and papery husk. The cobs are broken off by hand and left in the sun for further drying.

In large-scale farming, maize may be harvested with a combine harvester. A combine harvester is a machine that picks the ears from the stalks, removes the husks, and shells the kernels from the cobs.

Fig. 114.6: Harvesting maize by hand

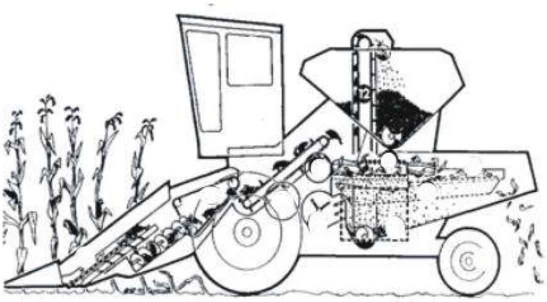


Fig. 114.7: A combine harvester

Storage

Maize may be stored with grains on cob or shelled (grains removed). In either case, it should be dried well and treated with recommended insecticides. The storage room should be free of rodents and birds.

Processing

The grain is usually grounded and made into several dishes such as kenkey of all kinds, porridge, banku etc. It can also be roasted and ground to make “winiemix”, a form of porridge.

Marketing

The grains can be exported, sold in the open market, schools and institutions as well as other governmental agencies such as the Ghana National Procurement Agency.

Fig. 114.8

Production of legumes

Legumes, also known as grain legumes are crops characterized by a single-chambered, flattened seedpod with two sutures. It usually splits open along the two sutures, as in the common bean. The seeds are attached along one of the sutures.

The legume may be indehiscent (not splitting), as in the groundnut, which matures underground; or explosively dehiscent, as in cowpea.

It also may range from only a few millimetres long to more than 30cm and may be single or many seeded and brightly or dully coloured.

A common feature in the family is the presence of root nodules containing bacteria of the genus ***Rhizobium***. These bacteria are capable of converting atmospheric nitrogen, which cannot be used by the plants, into nitrate (NO_3^-), a form that can be used. Legumes are often planted specifically to renew nitrogen supplies in soils.

Cowpea (*Vigna spp.)*

Fig. 114.9: Cowpea plant

Common varieties

Common locally cultivated varieties of cowpea include amatin, soronko, valienga and asontem. These varieties are characterized by their distinctive colour as well as their maturity periods.

Soil and climate

Cowpea tolerates a wide range of soils. It is drought resistant but cannot withstand prolonged water-logging. It is mostly grown in dry savannah areas to temperatures up to 35^{o}C.

Planting

With a spacing of 52 – 30cm in the row and 75 – 90cm between rows, two or three seeds are planted at a depth of 3 – 4cm. Planting is done at the beginning of the rainy season.

Cultural practices

Depending on the soil fertility, single superphosphate and sulphate of potash can be applied in amounts. Weeds can be controlled by hoeing or hand pulling. The first weeding is normally done 2 – 3 weeks after planting with the next done when flowering starts.

Pest and disease control

Pests such as shield bugs, aphids and other leaf-eating beetles can be controlled with insecticides. Fungal attack is controlled with regular application of fungicides.

Harvesting

Since all pods do not mature at the same time, several hand pickings may be required.

Storage

Pods are sundried and threshed. The stored crop must be protected from pests. Various recommended insecticides can be used to keep insect pest away.

Processing and marketing

Cowpea is either cooked and eaten or is grounded and added to other foods. It has high protein content and provides enough energy. It can also be used as feed for farm animals. Marketing is mostly done on the open market.

Production of root and stem tubers

Root tubers are modified underground stems that store food and support tiny new plants that bud off their surfaces.

Young plants developing from tuber buds are nourished from starch stored in the tubers until mature enough to develop root systems.

Examples of stem and root tubers include cassava, yam, sweet potato and cocoyam. These foods have high reserves of carbohydrates and normally serve as staples in some communities.

Cassava (*Manihot spp.*)

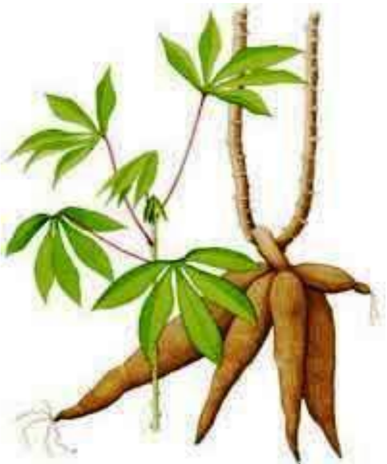


Fig. 115.0: Cassava tubers

Common varieties

Varieties of cassava include Enyan Abasa, Katawere, Ankars, Okwenyonmko and Katawia. These could be classed based on their sweetness or bitterness.

Soil and climatic conditions

Cassava does well in soils which are well drained, deep and rich in organic matter. They require annual rainfall of 500 – 750mm with a temperature 25^{o}C – 30^{o}C.

Method of propagation

Cassava is cultivated asexually by stem cuttings, with each stem being 20 to 30 cm long and containing 3 – 5 buds. The stems are planted either fully or partially in the soil with a spacing of 100 cm x 100 cm.

Cultural practices

The soil may be ploughed with organic fertilizer to improve its fertility. The crop can be side-dressed or broadcast with NPK fertilizer two weeks after germination occurs. Weeds can be controlled by hoeing or with recommended weed killers.

Pest and disease control

Mealy bugs, grasshoppers and green spiders can be controlled with recommended insecticides.

White flies, which are the vectors of mosaic virus, are controlled with insecticides or using resistant varieties.

Harvesting

Cassava matures 8 – 18 months after planting. When mature, large tubers are carefully dug up leaving smaller ones to grow. Smaller tubers can be harvested by pulling or lifting from the soil. Cassava lifters can be effectively used to harvest the produce.

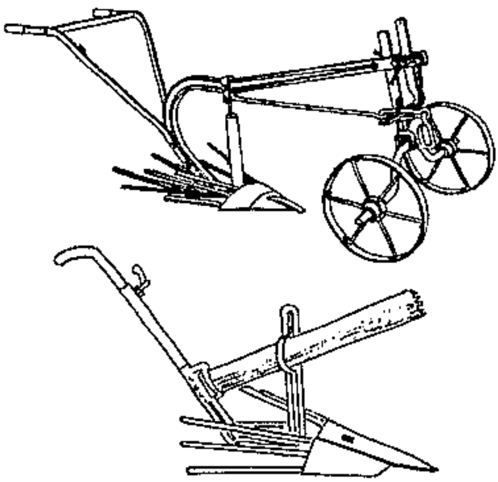


Fig. 115.1: Cassava lifters

Storage

Cassava cannot be stored for a long time. It can be stored underground or in water from 1 to 2 weeks.

Processing and uses

Cassava can be cooked and eaten (ampesi) or pounded into fufu. It can also be grated into gari. For the textile industry, cassava can be used to prepare starch. The leave of cassava plants and peal of tubers can serve as feed for livestock.

Marketing

Like most farm produce, cassava or its products can be sold on the open market. It can be sold in large quantities to industries for making starch.

Production of Fruits

Most fruits grow on trees. Fruits have an important reserve of vitamins and minerals. Fruits such as orange and pineapple are fibrous and serve as roughage which increases digestion in humans and prevents constipation.

Fruits provide the body with energy. Some fruits serve as ingredient for the production of beverages.

Examples of fruits are pawpaw, citrus fruits, mango, pineapple, etc.

Citrus fruits (Citrus spp.)

Citrus is a common name for several related evergreen trees and shrubs of the rue family. The plants are characterized by wing-like appendages on the leaf stalks, white or purplish flowers, and fruit (classified as a kind of berry) with a spongy or leathery rind and a juicy pulp divided into sections. The leaves, flowers, and rind of the fruit abound in volatile oil and emit a sharp fragrance. Many citrus plants have thorny branches.

Common varieties

- ❖ Sweet orange (*Citrus sinensis*);
- ❖ Sour orange (*Citrus aurantium*);
- ❖ Tangerine (*Citrus reticulate*);
- ❖ Grapefruit (*Citrus maxima*);
- ❖ Lemon (*Citris limon*);
- ❖ Lime (*Citrus aurantifolia*).

Soil and climatic conditions

Citrus fruits require well-drained soils with modest level of fertility. They do well with annual rainfall of 1000 mm, and an average temperature of 20^{o}C.

Method of propagation

Citrus fruits are mostly propagated by budding on to sour orange stock which has been raised from seed, with spacing of 1m x 1m. When the seedlings are about 40 cm tall, they are budded 25 – 30cm. The trees are replanted 5 months after budding to coincide with the start of the rainy season.

Pest and disease control

Insect pests such as aphids, mites, mealy bugs, caterpillars and scales are controlled with insecticides. The virus Tristeza and the fungal disease gummosis can be controlled by using resistant root stocks.

Harvesting and storage

Citrus fruits are picked by hand when ripe. Fruits can be stored for 5 to 8 days. It is advisable to harvest fruits only when they are ready to be used or transported.

Processing and uses

Citrus fruits could be eaten raw or used for other domestic purposes. They can be canned or have the juice extracted. Pulped whole fruit can be used to make marmalades. The outer skin has high oil reserve and it can be extracted for other uses.

TEST QUESTIONS

10. Mention two examples of each of the following classes of food"
    (a) cereals;
    (b) vegetables;
    (c) fruits;
    (d) grain legumes;
    (e) stem and root tubers.

11. Describe four uses of crops.

12. (a) Mention four factors to be

considered when selecting a site for crop production.

(b) What is soil tillage?

(c) State five importance of soil tillage.

13. (a) What is plant population?

(b) Describe how to determine the population of crops on a farm

14. (a) What is a nursery bed?

(b) Mention five characteristics of a good nursery bed.

6. (a) What is plant tissue culture?

(b) state four advantages of plant tissue culture.

7. Explain the importance of the following post-planting cultural practices:

(a) irrigation;

(b) mulching;

(c) staking;

(d) pruning.

8. Describe the following methods of pest and disease control:

(a) physical methods;

(b) chemical methods;

(c) biological methods;

(d) cultural methods.

9. Name the pests and pathogens controlled by the following chemicals:

(a) bactericides:

(b) rodentmicides;

(c) avicides;

(d) nematicides:

(e) fungicides.

10. State the uses of the following farm tools and machinery

Farm tool	*Uses*
Hand fork	
Secateurs	
Shovel	
Shears	
Plough	

11. State four cultural methods and four biological methods used to control pests and parasites.

12. (a) Name four examples of vegetables.

(b) Briefly describe how you will grow a known vegetable to the harvesting stage.

13. (a) What is the scientific name of the following crops:

(i) maize;

(ii) cassava;

(iii) cowpea;

(b) Mention four varieties of maize.

14. Outline the main cultural practices involved in a cereal crop production.

15. (a) Name four varieties of cassava.

(b) Describe how cassava is harvested and stored.

16. In an experiment to determine the population of two plant species **A** and **B** in a field, a device was thrown at random and the number of each of the species of the plant enclosed by the device was determined at each throw of the device. The device was thrown ten times. The data obtained are given in the table below.

Study the table carefully and answer the questions that follow.

Throw of device	**Species A**	**Species B**
1^{st}	4	7
2^{nd}	9	11
3^{rd}	12	7
4^{th}	4	16
5^{th}	5	7
6^{th}	9	13
7^{th}	7	8
8^{th}	5	9
9^{th}	8	7
10^{th}	9	11

a) Name the device used in the experiment.

b) Determine the total number of each of the species A and B for the ten throws.

c) Calculate the population density of species B given that the area of the device used is 1 m^2

d) State one precaution to be taken when using the device.

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NERVOUS SYSTEM

Specific Objectives

After completing this chapter, you will be able to:

- Describe the structure and functions of the nervous system of humans.
- Identify the causes and effects of damages to the brain and spinal cord.
- Classify stimulus responses as voluntary or involuntary actions.
- Describe the endocrine system and its functions.

INTRODUCTION

Have you wondered why your heart beats faster when you run, or why you pull your hand away when you touch a hot objects? When a person runs, more oxygen is used up; therefore, a message is sent to the heart to replenish the used oxygen in the cells. Hence, it beats faster. The mechanism in the body which sends this message to the heart is known as the ***nervous system***.

The nervous system is a group of elements within an organism that is concerned with the reception of stimuli, the transmission of nerve impulses, or the activation of muscle mechanisms.

STRUCTURE AND FUNCTIONS OF THE NERVOUS SYSTEM OF HUMANS

Neurones (Nerve Cells)

The nervous system is made up of units called nerve cells or ***neurons***. Neurons are responsible for the transmission and analysis of all electrochemical communication within the nervous system. Each neuron is composed of a cell body called a ***soma***, a major fibre called an ***axon***, and a system of branches called ***dendrites***.

Functions of the parts of neurones

Axons, also called nerve fibres, convey electrical signals away from the soma.

Dendrites convey electrical signals toward the soma. They are shorter than axons, and are usually multiple and branching.

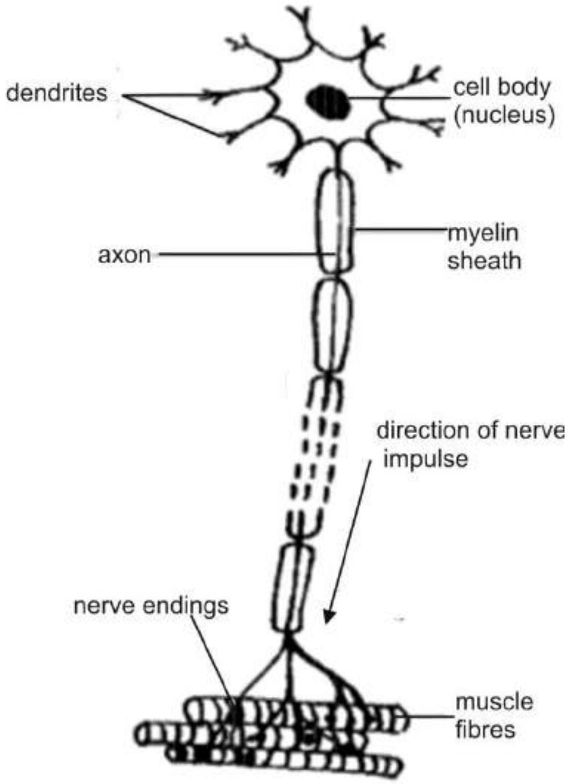

Fig. 115.2: A motor neuron

Types of neurons

1. The ***motor neurones*** – transmit information to the muscles.
2. The ***sensory neurones*** – transmit information to and from the sense organs.
3. The ***relay neurones*** – connect the motor sensory neurones.

Other parts of the nervous system

The nervous system is made up of two other systems –

❖ the central nervous system and
❖ the peripheral nervous system.

The central nervous system

The central nervous system is made up of the brain and the spinal cord. These organs communicate with the body through nerves.

The brain communicates with the body through (ten pairs of) cranial nerves while the spinal cord communicates with the body through (31 pairs of) spinal nerves.

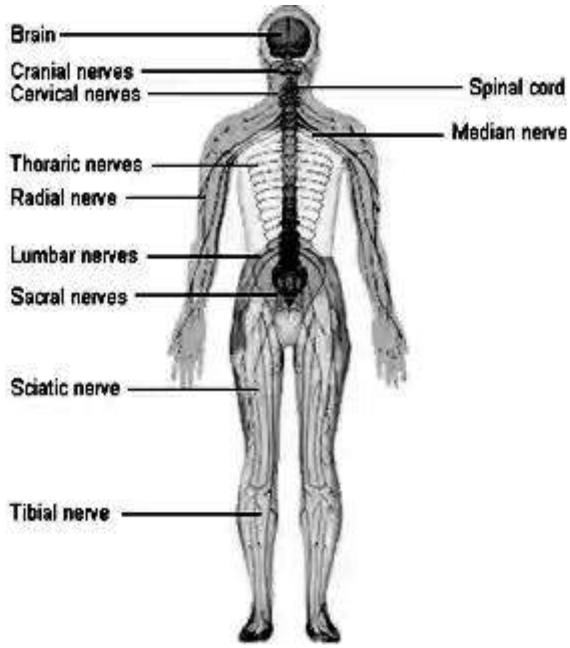

Fig. 115.3: The central nervous system

The brain

The brain consists of millions of neurons. There are three main distinguishable parts of the brain –

❖ the cerebrum,
❖ the cerebellum and
❖ the medulla oblongata.

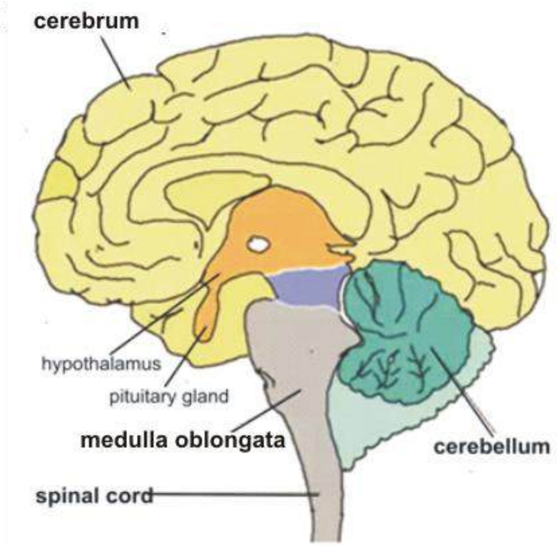

Fig. 115.5: Parts of the human brain

The cerebrum (fore-brain)

The cerebrum is the largest part of the brain. It is the region where most high-level brain functions take place.

It consists of two ***cerebral hemispheres***, responsible for intelligence, memory and consciousness; and ***cerebral cortex***, responsible for vision, hearing, speech, emotions, language, and other aspects of perceiving, thinking, and remembering.

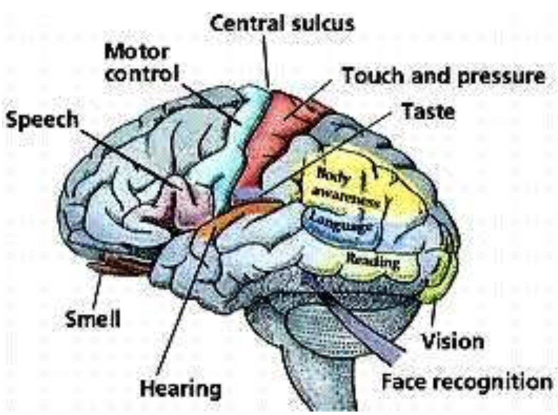

Fig. 115.4: Parts of the cerebrum

The cerebellum (Middle Brain)

The cerebellum is located at the lower part of the hind brain. It controls and coordinates body movements.

The cerebellum also maintains posture and balance by controlling muscle tone and sensing the position of the limbs. All motor activities such as kicking a football, running, jumping etc. depend on the cerebellum.

Medulla oblongata (Hind-Brain)

This part of the brain, which is located at the foot of the hind-brain, is responsible for all involuntary actions such as sneezing, heart-beat, yawning, blinking, etc.

The spinal cord

The spinal cord is composed of large number of neurons grouped into a cylindrical mass. It runs from the brain to the tail. It is protected by the bone of the spinal column. From between the vertebrae, spinal nerves emerge and run to all parts of the body.

The spinal nerves carry sensory impulses to the brain and motor impulses from the brain to the muscles and other organs.

The spinal cord is also concerned with reflex actions and the conduction of nervous impulses to and from the brain.

DAMAGE TO THE CENTRAL NERVOUS SYSTEM

The central nervous system is a delicate system. Damage done to any part of the brain or spinal cord could result in irreversible physical or psychological conditions. This is because damaged nerve cells cannot heal.

A person with defect in the central nervous system may experience problems in:

1. looking after his/ her welfare;
2. forming and sustaining relationship with others;
3. making rational decisions;
4. adjusting to stress;
5. planning for the future;

Causes of damage to the central nervous system

There are many activities, lifestyles and conditions that cause damage to the central nervous system. These include:

(a) Accidents

Accidents of all sorts, such as traffic accidents, accidents at home, school or in the workplace could cause a person to hit his/ her head or fall heavily. Since the brain and the spinal cord are not designed to hit anything hard, such accidents may damage the nerve cells and cause the brain or the spinal cord to malfunction.

(b) Diseases

Diseases of the nervous system include genetic malformations, poisonings, metabolic defects, vascular disorders, inflammations, degeneration, and tumours. These involve either nerve cells or their supporting elements.

Vascular disorders, such as cerebral haemorrhage or other forms of stroke, are among the most common causes of paralysis.

Some diseases exhibit peculiar geographic and age distribution. In Temperate zones, multiple sclerosis is a common degenerative disease of the nervous system, but it is rare in the Tropics.

The nervous system is subject to infection by a great variety of bacteria, parasites, and viruses. For example, meningitis, or infection of the meninges infesting the brain and spinal cord, can be caused by many different agents.

Some viruses causing neurological illnesses affect only certain parts of the nervous system. For example, the virus causing poliomyelitis commonly affects the spinal cord; viruses causing encephalitis attack the brain.

(c) Drug abuse

A drug is a chemical substance used for medical or veterinary purposes.

If a person misuses drugs, he/ she is said to be abusing drug.

Drug abuse is characterized by taking more than the recommended dose of prescription drugs without medical supervision, or using government-controlled substances such as marijuana, cocaine, heroin, or other illegal substances. Legal substances, such as alcohol and nicotine are also abused by many people. Abuse of drugs and other substances can lead to physical and psychological dependence.

Psychotropic drugs, such as marijuana and ecstasy, often taken by people to escape from unpleasant realities or make them feel good, actually affect the way the nerve cells transmit information, thereby disrupting the person's thinking process and changing their states of mind and their perception about things around them. They often interfere with the person's judgement and may induce them to do things normal human beings would not do.

Causes of drug abuse

1. Peer pressure
2. Curiosity
3. Ignorance
4. Performance enhancement
5. Frustration
6. Easy access
7. Loneliness
8. Boredom

Effects of drug abuse

Drug abuse can cause a wide variety of adverse physical reactions.

1. Long-term drug use may damage the heart, liver, and brain.
2. Drug abusers may suffer from malnutrition if they habitually forget to eat, cannot afford to buy food, or eat foods lacking the proper vitamins and minerals.
3. Individuals who abuse injectable drugs risk contracting infections such as hepatitis and HIV from dirty needles or needles shared with other infected abusers.
4. One of the most dangerous effects of illegal drug use is the potential for overdosing—that is taking too large or too strong a dose for the body's systems to handle. A drug overdose may cause an individual to lose consciousness and to breathe inadequately. Without treatment, an individual may die from drug overdose.
5. Drugs are expensive, which means the individual wastes a lot of money in buying them.
6. It often results in metal disorders as a result of the malfunction of the brain cells.
7. Drug abuse may result in serious accidents because of lack of proper judgement by abusers.

(d) Depression

Depression is a mental illness in which a person experiences deep sadness and diminished interest in nearly all activities.

Depression could be as a result of changes in the way the brain cells work, such as change in the balance of chemicals produced by the synapses of nerves cells.

This is called ***endogenous depression***; or a person's response to the environment, such as death of loved one, loss of job, stress, frustration etc. This is referred to as ***reactive depression***.

The peripheral nervous system

The peripheral nervous system consists of 12 pairs of cranial nerves extending from the cerebrum and brain stem; a system of other nerves branching throughout the body from the spinal cord; and the ***autonomic nervous system***, which regulates vital functions not under conscious control, such as the activity of the heart muscle, smooth muscle (involuntary muscle found in the skin, blood vessels, and internal organs), and glands.

VOLUNTARY AND INVOLUNTARY ACTIONS

Voluntary or Conscious Action

Voluntary actions are actions which are under the direct control of the brain.

How do you go about performing your favourite activities? You think about them, either over a short time or long time before carrying them out. Before a footballer kicks a ball, he thinks about who or where to kick it to, how to kick it and when to kick it. Other activities such as singing, running, walking or waving of the hand are all done consciously and can be controlled.

Involuntary or Reflex Action

A reflex action is a quick, automatic response to a stimulus by an organism or a system of organs, which does not involve the brain.

For example, the iris of the eye can contract or dilate the pupil with change in light intensity without our being aware that it is happening.
More commonly, we are aware of a reflex occurring but are unable to control it. For example, blinking of the eye when a foreign object touches the cornea; the jerking of the knee when it is struck just above or below the knee-cap or sneezing when dust particles enter the nose are all reflex action which we are aware of, but cannot control or prevent.

One similarity between voluntary and involuntary actions is that they are both controlled by the ***autonomic nervous system***.

The table below gives the differences between voluntary and involuntary or reflex actions.

Table 8.7: Difference between voluntary and involuntary actions

Voluntary action	Involuntary action
Involves the brain	Involves the spinal cord
Delayed response	Immediate response
Controlled consciously	Not controlled consciously
Response is in the skeletal muscle only	The response is in skeletal or an internal involuntary muscle
Longer path of the nerve impulse	Shorter path of the nerve impulse
Impulse travels from the brain to the spinal cord	Impulse travels only up or down the spinal cord

The Autonomic Nervous System

Autonomic Nervous System is one of the two main divisions of the nervous system, supplying impulses to the body's heart muscles, smooth muscles, and glands.

The autonomic system controls the action of the glands; the functions of the respiratory, circulatory, digestive, and urogenital systems; and the involuntary muscles in these systems and in the skin.

Controlled by nerve centres in the lower part of the brain, the system also has a reciprocal effect on the internal secretions, being controlled to some degree by the hormones and exercising some control, in turn, on hormone production.

Types of the autonomic nervous system

- sympathetic nerves and
- parasympathetic nerves.

Table 8.8: Differences between the sympathetic and parasympathetic nerves

Sympathetic nerves	Parasympathetic nerves
Speed up heart beat	Slow down heart beat
Constrict arteries	Dilate arteries
Dilates pupils	Constrict pupils
Slow down peristalsis	Increase peristalsis
Decrease flow of saliva	Increase flow of saliva

The reflex arc

The reflex arc is the pathway along which nerve impulses travel when they are stimulated.

Many of the actions of the nervous system can be explained on the basis of reflex arcs, which are chains of interconnected nerve cells, stimulated at one end and capable of bringing about movement or glandular secretion at the other. Actions such as the pulling away of the hand when one touches a hot object and the jerk of the knee are all initiated by the reflex arc.

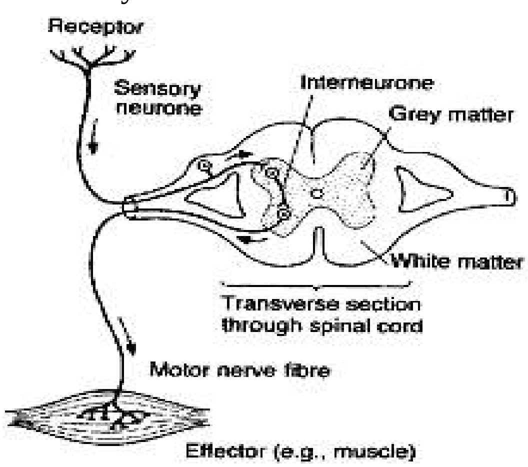

Fig. 115.6: Reflex arc

Pulling away of the hand when one touches a hot or sharp object

Heat or pain receptors in the skin are stimulated and release impulses which travel along a sensory neurone and enter the spinal cord via the dorsal root.

The impulse then moves over a synapse and stimulates a relay neurone and at the other end of the relay neurone to stimulate a motor neurone.

It travels out of the spinal cord via the ventral root to the muscles and the muscles contract to pull the hand away.

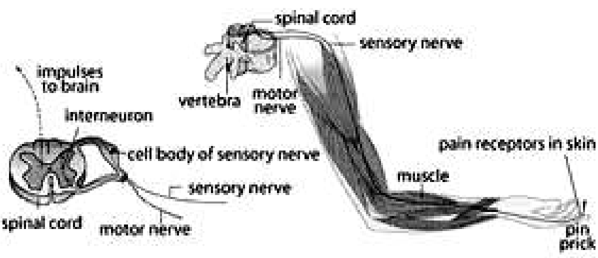

Fig. 115.7

How knee-jerk occurs

The knee jerks when the sensory fibres make a synapse directly with the motor fibre (no relay fibre involved). Striking the tendon below the knee cap stimulates stretched receptors in the leg extensor muscle. The impulse travels along the reflex arc and causes the same muscle to contract.

Foreign matter in the eye

When a person cuts an onion, for example, the vapour arising from the cut onion passes into the nose and eyes and stimulate the olfactory nerve cells. When the vapour dissolves in the eye, fluid is discharged from the tear glands to dilute the dissolved vapour.

Types of reflex

1. ***Cranial reflex*** is the reflex action which takes place in the brain and involves the organs of the head.

2. ***Spinal reflex*** are reflex actions which take place below the head. In animals such as frogs, spinal reflex action occurs even if the brain is destroyed.

THE ENDOCRINE SYSTEM

The endocrine system consists of a group of specialized organs and body tissues that produce, store, and secrete chemical substances known as ***hormones***.

Endocrine organs are called ***ductless glands*** *because they have no ducts connecting them to specific body parts.*

The hormones they secrete are released directly into the bloodstream.

In contrast, the ***exocrine glands***, such as the sweat glands or the salivary glands, release their secretions directly to target areas—for example, the skin or the inside of the mouth. Some of the body's glands are described as ***endo-exocrine glands*** because they secrete hormones as well as other types of substances. Even some non-glandular tissues produce hormone-like substances. For example, nerve cells produce chemical messengers called ***neurotransmitters***.

Hormones

Hormones are chemical substances which are produced by an endocrine gland or some nerve cells that regulates the function of a specific tissue or organ.

Hormones are responsible for regulating the body's growth and development, controlling the function of various tissues, supporting pregnancy and other reproductive functions, and regulating metabolism.

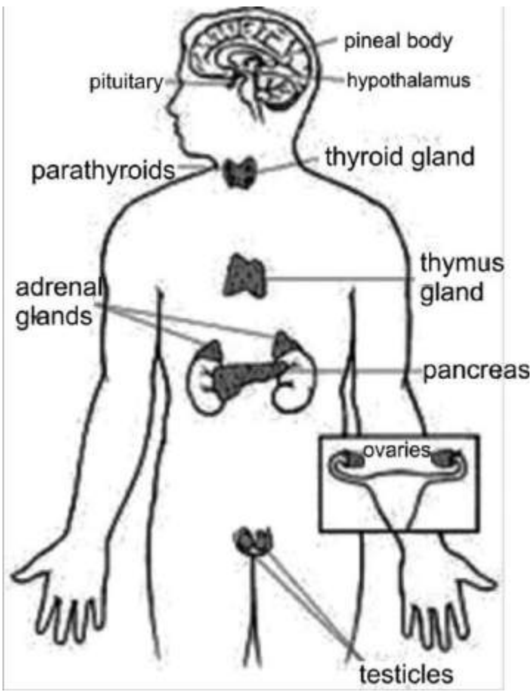

Fig. 115.8: The endocrine system

Components of the endocrine system

The primary glands that make up the human endocrine system are the ***hypothalamus***, ***pituitary***, ***thyroid***, ***parathyroid***, ***adrenal***, ***pineal body***, the ***pancreas*** and ***reproductive glands*** (the ovary and testis).

Some non-endocrine organs are known to actively secrete hormones. These include the brain, heart, lungs, kidneys, liver, thymus, skin, and placenta.

Almost all body cells can either produce or convert hormones, and some secrete hormones. For example, glucagon (a hormone that raises glucose levels in the blood when the body needs extra energy) is made in the pancreas but also in the wall of the gastrointestinal tract. However, it is the endocrine glands that are specialized for hormone production.

They efficiently manufacture chemically complex hormones from simple chemical substances—for example, amino acids and carbohydrates—and they regulate their secretion more efficiently than any other tissues.

Table 9.0: Endocrine glands and their secreted hormones

Gland	Location in the body	Secreted hormone	Function of hormone	Effects of under or over secretion
Thyroid gland	Neck	Thyroxin	• Controls growth rate and development; • Controls rate of chemical activities in adult humans	• Under secretion causes cretinism in children and myxoedema in adults; • over secretion causes thinness and over-activity
Adrenal gland	Just above the kidneys	Adrenaline	• Increases heartbeat and blood glucose level, • dilates coronary arteries. (controls response for fight or flight)	• Under secretion causes delayed response to stimulus; • Over secretion causes high blood pressure
Pituitary gland	Base of the fore-brain	Anti-diuretic hormone (ADH)	• Stimulates re-absorption of water by the kidney tubules	
		Thyroid-stimulating hormone (TSH)	• Stimulates production of thyroxin	
		Luteinising hormone (LH)	• Stimulates ovulation and corpus luteum development in females	
		Testis-stimulating hormone (TSH)	• Stimulates production of sperm in males	
		Follicle-stimulating hormone (FSH)	• Maturation of Graafian follicles in females	

		Prolactin	• Stimulates production of milk	
		Oxytocin	• Stimulates contraction of uterus during childbirth	
Islets of Langerhans or Pancreas	Pancreas	Insulin	• Controls blood sugar level	• Under-secretion results in diabetes mellitus
		Glucagon	• Stimulates the liver to convert stored glycogen into glucose when the blood glucose level is low	
Testes	Inside the scrotum	Testosterone	• Controls development of male secondary sexual characteristics	• Under-secretion results in under-development of male secondary sexual characteristics
Ovaries	Behind the kidneys	Oestrogen	• Controls development of female secondary sexual characteristics	• Under-secretion results in under development of secondary sexual characteristics in females
		Progesterone	• Development of uterine wall for implantation	• Under-production results in poor development of uterine wall
Parathyroid gland	Atop the thyroid gland in the neck	Parathormone	• Stimulates release of calcium from bones into blood	• Under-secretion results in low level of calcium in the blood

Table 9.1: Differences between the nervous system and the endocrine system

Nervous system	Endocrine system
Response is quick	Response is slow
Response e is localised (only specific parts of the body)	Response is widespread
Impulse transmitted through the neurons	Impulse transmitted through hormones
Messages are electrical	Messages are hormonal
Messages are controlled by brain and spinal cord	Messages controlled by endocrine glands
Effect is short	Effect last longer
Linking mechanisms are neurons and the central nervous system	Linking mechanism is the circulatory system

The importance of iodated salt

Iodine is an important component of human diet. It is required by the thyroid to produce thyroxin, which controls chemical activities in the body as well as growth and development in children.
Low level of iodine results in Iodine Deficiency Disorders which include enlarged thyroid (goitre), cretinism, mental retardation etc.

Iodated salt is table salt treated with iodine and a small amount of potassium or sodium to replenish the iodine content in the body.

Fig. 115.9: *A woman suffering from goitre*

TEST QUESTIONS

1. (a) State the functions of the following neurons:
   (i) motor neuron
   (ii) sensory neuron
   (iii) reflex neuron

   (b) Draw and label a motor neuron.

2. Describe the structure of the human brain.

3. Explain four causes of damage to the central nervous system.

4. (a) Mention five causes of damage to the central nervous system.
   (b) Explain four effects of damage to the central nervous system.

5. (a) What are voluntary actions?
   (b) Mention four differences between voluntary action and involuntary action.

6. (a) What are reflex actions?
   (b) List four differences between sympathetic and parasympathetic nerves.

7. Explain the following:
   (a) A boy quickly withdraws his hand when he accidentally touches a hot metal.
   (b) How the knee jerks.

8. (a) What are hormones?
   (b) State the effects of under-secretion and over-secretion of the following hormones:
      (i) adrenalin;
      (ii) insulin;
      (iii) testosterone;
      (iv) oestrogen.

9. Mention four differences between the nervous system and the endocrine system.

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LIGHT ENERGY

Specific Objectives

After completing this chapter, you will be able to:

- Review reflection and refraction of light and the characteristics of images formed by plane mirrors.
- Draw the mammalian eye and describe the functions.
- Describe and explain how to separate white light into its component colours.
- Differentiate between primary colours and secondary colours.
- Outline the components of the electromagnetic spectrum.

INTRODUCTION

Light is a form of energy that makes vision possible.

We say that light makes vision possible because without it humans or animals cannot see their way round. During the day, the sun provides us with enough light energy. At night, we get light from the moon, which actually reflect the sun's light energy onto earth. Nevertheless, the energy we get from the moon is not strong enough therefore man has devised alternative sources of light.

Sources of light

The sources of light are grouped into natural and artificial sources.

Natural sources: These are lights which are provided by nature.

Examples of natural sources of light

1. Sun
2. Moon
3. Stars
4. Fireflies

Artificial sources: These are light which are provided by man-made devices and mechanisms.

Examples of artificial sources of light

1. Fluorescent lamps
2. Incandescent lamps
3. Kerosene lamps
4. Candles
5. Lanterns
6. Light emitting diodes (LEDs)

Luminous and non-luminous bodies

Luminous body: This is a body which produces it own light.
Examples are the sun, stars, electric lamps, candles, etc.

Non-luminous body: This is a body which is not capable of producing it own light. It produces light by reflecting light energy from other sources.
Examples are the moon, mirror, water, glass etc.

Types of light energy

The sources of light listed above can be put into three other groups. They are luminescent, incandescent and phosphorescent types or sources of light.

Luminescent light

This is the type of light which is produced without the production of heat. This type of is normally produced from chemical or electrical sources.
Examples of luminescent light are florescent lamps, LCD screens, fireflies, moon, etc.

Incandescent light

This is the type of light whose production depends on the production of heat. It is produced as a result of the movement of electrons from one level to another.
Examples of incandescent lamps are incandescent lamps, the sun, stars, CRT screens, lanterns, etc.

Phosphorescent light

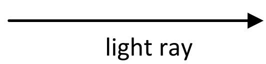
This is the type of light produced by a material after the main source of light has been removed. The material absorbs the light energy from a source and reproduces it in a different colour or frequency.
Examples of phosphorescent light are light produced by advertising boards, road signs, road markings, etc.

Characteristics of light

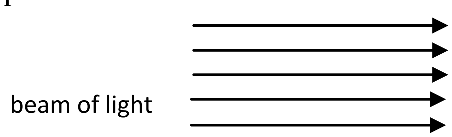
1. Light travels in a straight line or path. This is known as ***rectilinear propagation of light.***
2. Light travels at a speed of $\mathbf{3.0 \times 10^{8}}$ $\mathbf{ms^{-1}}$.
3. Light can be reflected, refracted, diffracted, interfered or scattered.
4. Light is an electromagnetic wave.
5. Light is composed of photons.

How light travels

Ray: This is the path along which light travels. A ray of light is represented by an arrowed line.



Beam: A group of rays moving in a particular direction form a beam of light.



Movement of light through substances

Transparent substances: These are substances or materials that allow light to pass through them.
Examples of transparent objects are glass, water, quartz etc.

Opaque substances: These are substances that do not allow light to pass through them.
Examples of opaque substances are wood, human body, stone, metal etc.

Translucent substances: These are substances that allow some amount of light to pass through them.
Examples of translucent substances are frosted glass, smoke, murky or coloured water, fog, etc.

An experiment to show that light travels in a straight line

(Rectilinear propagation of light)

- Obtain three square cardboard or screens of the same size.
- Punch a hole in the middle of each cardboard.
- Align them up so that the holes are in a straight line by threading a string through the holes and pulling it taut.
- Place a lamp or lighted candle at one end of the arranged screens.
- The light can be seen through the three holes by an observer viewing at the other end of the light source.
- Shift one of the cardboards so that the holes are no longer in a straight line.
- The observer will no longer be able to see the light through the last cardboard.
- This shows that light travels in straight lines.

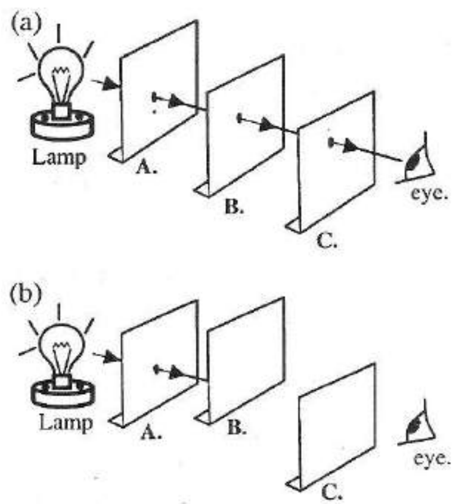


Fig. 116.1

NB: The principle of rectilinear propagation of light states that light travels in a straight path.

Shadow formation

When a ray of light is blocked by an opaque object, a darkened image of the object is formed behind it. This darkened image is known as a ***shadow***.

A shadow is a dark image formed when a ray of light is blocked by an opaque object.

In the experiment on rectilinear propagation of light, you could see that when the card is shifted, shadow of it is thrown on the card behind it.

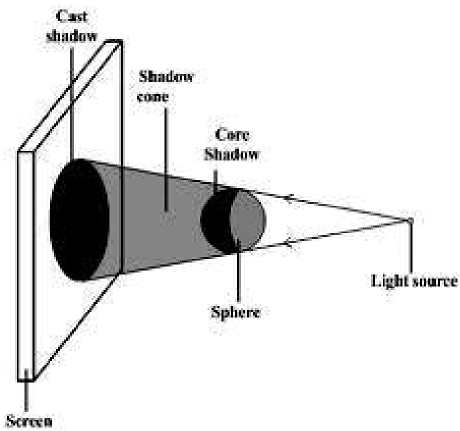


Fig. 116.2: Shadow formation

Types of shadow

There are two types of shadow, based on the type and source of light producing it. They are umbra and penumbra.

Umbra

This is the type of shadow with total darkness cast by an opaque object between a point (small) source of light and a screen. Umbra shadows are sharp, very dark and well-defined.

Penumbra

Penumbra is a shadow formed as a result of an opaque object placed in between a large source of light and a screen.

Penumbra is less dark, large and indistinct.

Penumbra normally surrounds a core of umbra.

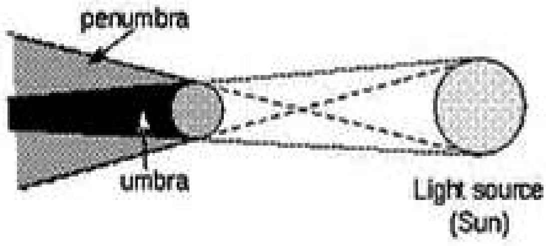


Fig. 116.3: Umbra and penumbra

Eclipses

An eclipse is the complete or partial hiding of either the sun or the moon from the earth when one moves in between the other and the earth.

Eclipse occurs when the sun or the moon disappears from the earth because of the movements and relative positions of the earth and the moon.

An eclipse could be ***total*** (umbra) or ***partial*** (penumbra).

Eclipse occurs because of rectilinear propagation of light.

Types of eclipse

Two kinds of eclipses involve the earth:

- ❖ eclipse of the sun, or solar eclipses;
- ❖ eclipse of the moon, or lunar eclipses.

Solar eclipse or eclipse of the sun

Solar eclipse occurs when the moon is between the sun and the earth and its shadow moves across the face of the earth.

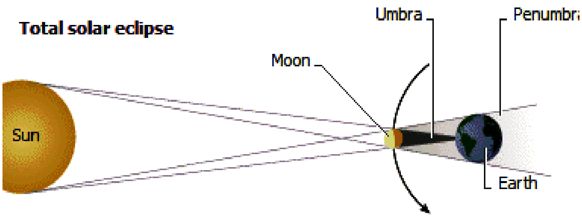


Fig. 116.4: Solar eclipse

Total solar eclipse

During a total solar eclipse, the moon completely blocks out the sun for as long as seven to eight minutes. This type of eclipse is visible from the area on the earth within the umbra, the inner part of the moon's shadow. The umbra is formed by tangents to the sun and moon, and the outer portion.

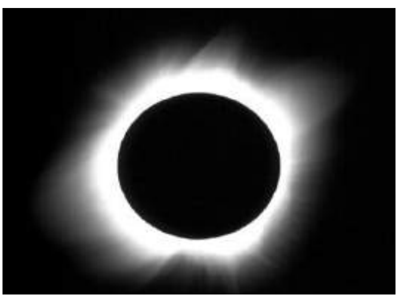


Fig. 116.5: Solar eclipse seen from the earth

Partial solar eclipse

Partial solar eclipse can be seen when part of the Earth is within the penumbra, or the outer portion of the Moon's shadow.

The penumbra is formed by tangents that intersect between the sun and the moon.

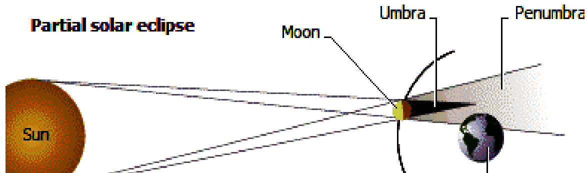


Fig. 116.6

Annular solar eclipse

An annular eclipse can be observed when the moon is in between the earth and the sun but is far from the earth that the umber does not fall on the earth.

The sun's outer edges are still visible and form a ring around the moon.

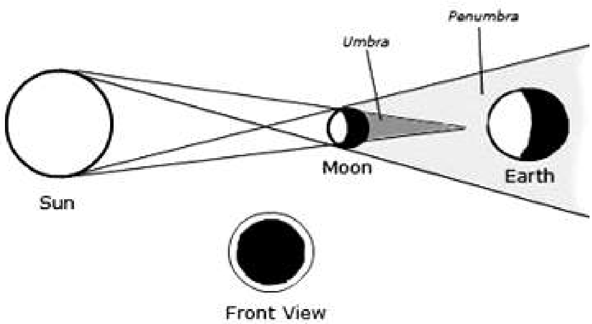


Fig. 116.4

Lunar eclipse or eclipse of the moon

A lunar eclipse occurs when the earth is between the sun and the moon and its shadow darkens the moon.

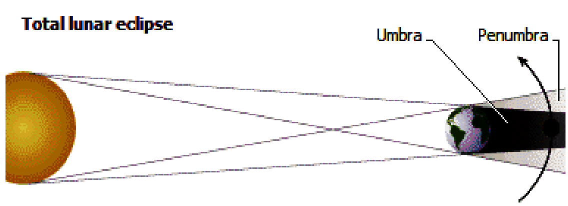


Fig. 116.6: Lunar eclipse

T***otal lunar eclipse*** occurs when the moon passes completely into the umbra of the earth's shadow.

Partial lunar eclipse occurs when only a part of the Moon enters the umbra of the Earth's shadow, leaving a portion of the Moon in total darkness.

The pinhole camera

A pinhole camera is a basic form of camera with a tiny hole for the aperture, with no lens. Light passes through the hole to form an inverted image on a screen. The construction of the pinhole camera is based on the principle of rectilinear propagation of light.

A pinhole camera can be built by making a pinhole (small hole) in a box. Light passes through the hole and forms an inverted, backwards image of the subject on the back of the box.

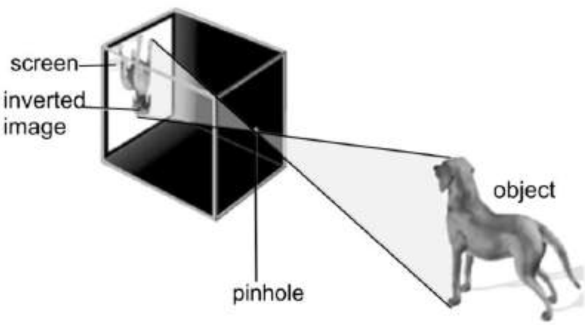


Fig. 116.7: Pinhole camera

Characteristics of image formed by pinhole camera

1. The image formed is inverted.
2. It is smaller than the object.
3. It is dim or blur.
4. It is real (i.e. can be formed on a screen).
5. The image formed is out of focus (i.e. the smaller the hole, the sharper the image.

REFLECTION OF LIGHT

Reflection is the bouncing back of light rays after hitting a hard, smooth and shiny surface.

Types of reflection

There are two types of reflection – regular an irregular reflection.

Regular or specular reflection

Reflection is termed regular when it occurs on a smooth, shiny surface.

The images produced in regular reflection are clear with all the reflected rays parallel to each other.

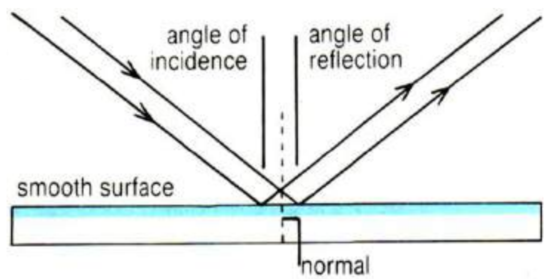
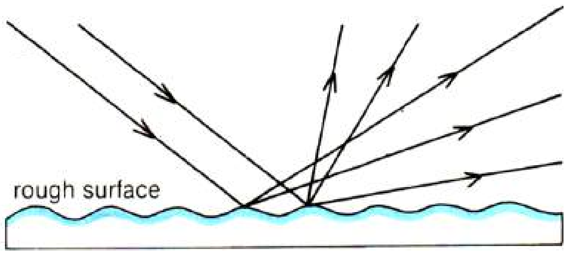


Fig. 116.8: *Regular reflection*

Irregular or diffused reflection

Irregular or diffused reflection is any reflection on a rough surface. When the surface is rough, it reflects the rays not as a parallel set but in various directions.

Most of the light rays are absorbed resulting in the production of diffused or unclear image.



Irregular reflection

Fig. 116.8:

Laws of reflection

Regular reflection is governed by the law that both the ***incident***, or striking rays and the ***reflected*** rays travel in directions making equal angles with the ***normal***, a line perpendicular to the reflecting surface at the point of incidence; and that the rays lie in the same plane as the normal.

These angles are called the ***angle of incidence*** and ***the angle of reflection***.

The laws of reflection state that:

1. At the point of incidence, the incident ray, the reflected ray and the normal all lie in the same plane.
2. The angle of incidence (i) is equal to the angle of reflection (r). [i.e. $i = r$]

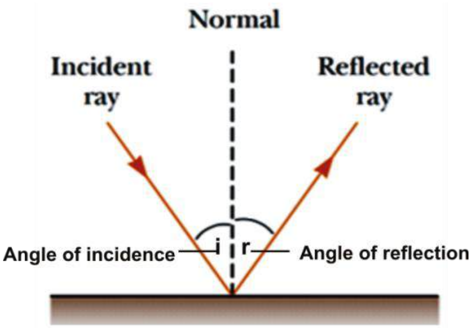


Fig. 116.9: Reflection on a plane mirror

Terms to note

❖ ***Incidence ray:*** This is the original ray from the source of light.

❖ ***Reflected ray:*** This is the ray produced when the incident ray strike the surface.

❖ ***Normal:*** This is an imaginary line in between the incident ray and the reflected ray, and perpendicular to the surface.

❖ ***Angle of incidence:*** This is the angle the incidence ray makes with the normal. It is represented by the letter i.

❖ ***Angle or reflection:*** This is the angle the reflected ray makes with the normal. It is represented by the letter r.

Formation of images

Light rays converging from an object onto a shiny surface produce an image of the object. There are two types of images – real and virtual images.

Real image

An image is said to be real if it can be formed on a screen. Real images are formed as a result of actual intersection of light rays. Thus, they have the same direction as the object. Real images can be magnified or diminished. Images produced by the pinhole camera and concave (converging) mirror are real.

Virtual image

Images which cannot be formed on a screen are referred to as virtual. Such images are formed by the apparent intersection of rays which have backward direction. They appear to be behind the mirror or screen. Plane mirrors and diverging (convex) mirrors produce virtual images.

Types of mirrors

There are three main types of mirrors:

- ❖ Plane mirrors – e.g. dressing mirrors
- ❖ Concave or converging mirrors – e.g. shaving mirrors
- ❖ Convex or diverging mirrors – e.g. driving mirrors

Plane mirrors

A plane mirror is simply a flat mirror on which objects can be reflected with little change, (except lateral inversion). In a plane mirror, a straight line drawn from a part of the object to the corresponding part of its image makes a right-angle with the surface of the mirror. An example of plane mirror is dressing mirror.

Characteristics of image formed by a plane mirror

1. It is erect or upright.
2. It is of the same size as the object.
3. It has the same shape as the object.
4. It is virtual (meaning that the light rays do not come from the image)
5. It is laterally inverted. (The right side appears as the left side and vices versa).
6. It appears behind the mirror.
7. It has the same distance behind the mirror as the object in front of it.
8. It is on the same normal to the mirror as the object. I.e. a line joining the mirror will cross at right angle.

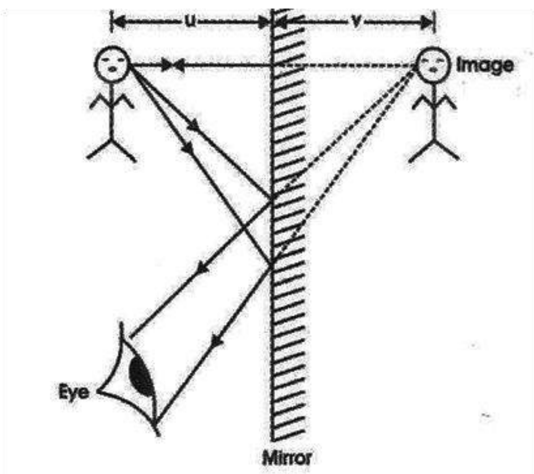


Fig. 117.0: Formation of an image on a plane mirror

Uses of a plane mirror

1. Plane mirrors are used in shops and supermarkets to detect shoplifters.
2. They are used in periscopes.
3. Microscopes and cameras use plane mirrors to reflect light onto an objective lens.
4. They are used for dressing.
5. Barbers and hairdressers use mirrors to enable their customers see how they look.
6. They are used in kaleidoscopes.
7. They are used in meters.
8. They are used as mirror tiles.

Differences between image formed by a plane mirror and a pinhole camera

Plane mirror	Pinhole camera
Virtual	Real
Same size as the object	Diminished
Upright	Upside down
Sharp	Dim or blur
Same distance as the object	Distance between image and object may be different
Laterally inverted	Not laterally inverted

Concave or converging mirrors

A concave or converging mirror curves inward like the inside surface of a hollow sphere. Light striking the surface of a concave mirror reflects inward, or converges.

Image formed by concave mirror

The image formed on a concave mirror can be either real or virtual, depending on the size and position of the object and the focal point of the mirror, or the place where light rays converge.

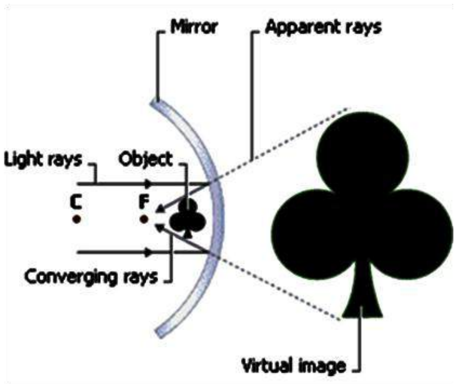


Fig. 117.1: Formation of image on concave mirror

Characteristics of image formed on concave mirror

1. It is enlarged (larger than the object).
2. It is upright or erect.
3. Image formed can be real or virtual.
4. Image distance is closer than object distance.

Uses of converging mirrors

1. They are used to bring light rays to a focus in a solar furnace.
2. They are used as reflectors behind torch bulbs and electric fire elements.
3. They are used as shaving or make-up mirrors.
4. They are used for making reflecting telescopes.

Convex or diverging mirrors

A convex mirror curves outward like the outer surface of a ball. Light striking the surface of a diverging mirror reflects outward, or diverges.

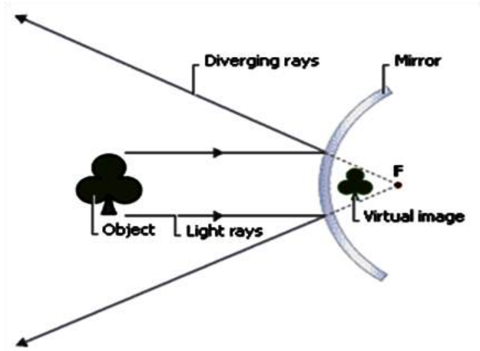


Fig. 117.3: Formation of image on convex mirror

Characteristics of image formed by convex mirror

1. It is erect.
2. It is diminished (smaller than the object).
3. Image distance is longer than object distance.
4. Image formed is virtual.

Uses of diverging mirrors

1. Used as a driving mirror
2. Placed at awkward bends in a road
3. Used in shops and supermarkets to keep an eye on customers.
4. Used in some automated teller machines (ATM) to enable the user to see what is going on behind them.
5. Attached to some mobile phone cameras to allow the user to correctly aim the camera when taking a self-portrait.

Diverging mirrors distort the distance involved in images; hence, images seen in a convex rear mirrors of vehicle appear farther away than they are.

Some terms used in curved mirrors

Principal axis – This is a long straight line which passes through the poles of a mirror perpendicular to the mirror.

Principal focus (F) – This is a point on the principal axis where rays close and parallel to the principal axis converge.

Focal length (f) – This is the distance between the mirror and the principal focus.

Centre of curvature – This is the centre of the sphere of which the mirror is part.

Pole – This is the centre of a mirror through which the principal axis passes.

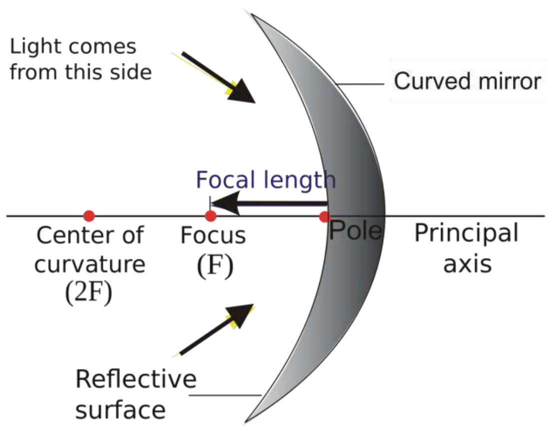


Fig. 117.4:

The mirror formula

The formula which gives the relationship between the focal length, the image distance and the position of an object for all spherical mirrors is given as:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Where,

f is the focal length of the mirror,

u is the distance of the object from the pole of the mirror.

v is the distance of the image from the pole of the mirror.

Examples

1. A body is placed 14 cm from the pole of a mirror. Determine the distance of the image from a converging mirror of focal length 18 cm.

Solution

Using the formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Object distance u = 14

Focal length f = 18

$$\frac{1}{v} = \frac{1}{18} - \frac{1}{14}$$

$$\frac{1}{v} = \frac{7-9}{126} = \frac{-2}{126}$$

$$\frac{v}{1} = \frac{126}{-2}$$

v = - 63cm

Alternative method

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$uv = fv + fu$

$uv - fv = fu$

$v(u - f) = fu$

$$v = \frac{fu}{u-f}$$

$$v = \frac{18 \text{ x } 14}{14-18} = \frac{252}{-4} \text{ -63cm}$$

2. A diverging mirror of focal length 15 cm produces an image on its principal axis, 10 cm from the mirror. Find the position of the object.

Solution

Using the formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$uv = fv + fu$

$uv - fu = fv$

$u(v - f) = fv$

$$u = \frac{fv}{v-f}$$

v = 10 cm

f = 15 cm

$$\text{u} = \frac{15 \text{ x } 10}{15-10} = \frac{150}{5} = 30\text{cm}$$

3. A pedestrian stands 2 m from a convex mirror of a bus and appears 4 m away. Calculate the focal length of the mirror.

Solution

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$f = \frac{uv}{v+u}$$

u = 2 m

v = 4 m

$$f = \frac{2 \text{ x } 4}{2+4} = \frac{8}{6} = 1.33\text{m}$$

Magnification by spherical mirrors

Magnification is the enlargement or reduction of the actual size of the image formed by a mirror.

In spherical mirrors, magnification is referred to as linear, lateral or transverse magnification. It is represented mathematically as:

Magnification (m) = $\frac{\boldsymbol{height\ of\ image\ (h_1)}}{\boldsymbol{height\ of\ object\ (h_0)}}$

OR

Magnification (m) = $\frac{\boldsymbol{image\ distance\ (v)}}{\boldsymbol{object\ distance\ (u)}}$

OR

Magnification (m) = $\frac{\boldsymbol{image\ size\ (v)}}{\boldsymbol{object\ size\ (u)}}$

Magnification has no unit.

Examples

1. An object 4 cm in height is placed 10 cm in front of the mirror. The image formed is 20 cm behind the mirror. Find the
   (a) magnification of the image
   (b) size of the image.

Solution

(a) Magnification (m) = $\frac{image\ \ distance}{object\ \ distance} = \frac{v}{u}$

u = 10 cm, v = 2 0cm

$$m = \frac{20}{10} = 2$$

(b) The size of the image

$$m = \frac{h_1}{h_0}$$

$h_0 = 4$, $h_1 = ?$

$h_1 = m \times h_0$

$h_1 = 2 \times 4 = 8$ cm

2. A ball of diameter 20 cm appear 17 cm in a concave mirror when placed before it.
   (a) What is magnification of the image?
   (b) Calculate the focal length of the Mirror.

Solution

(a) Magnification (m) = $\frac{image\ \ distance}{object\ \ distance} = \frac{v}{u}$

u = 20 cm, v = 17 cm

$$m = \frac{17}{20} = 0.85$$

(b) $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$$f = \frac{uv}{v+u} = = \frac{20 \text{ x } 17}{17+20} = \frac{340}{37} = 9.2\text{cm}$$

REFRACTION OF LIGHT

When a light ray travels from one substance or medium to another of different optical density, it changes direction. This change in direction is termed refraction.

The light gets refracted on entering or leaving the second medium.

For example,

- ❖ when light travels from air into water, its path is refracted as it enters the water.
- ❖ If it leaves the water and emerges back into the air it is refracted again.

Fig. 117.4: Refraction of a stick from air to water

Refraction of light is defined as the change in direction of a light ray as it moves from one medium to another of different optical density.

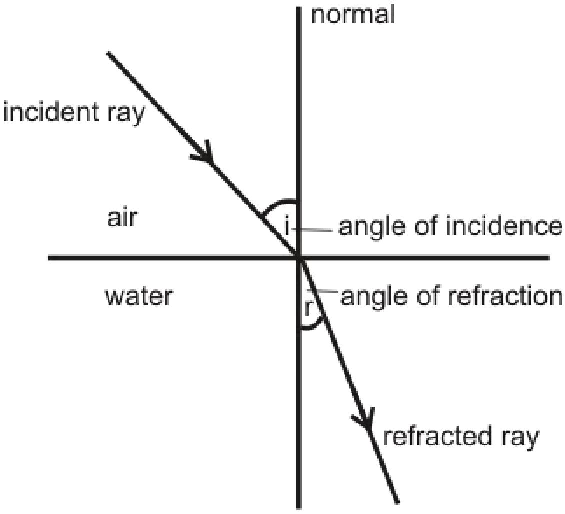


Fig. 117.6: Reflection and refraction of light by glass

Types of refraction of light

There are two types of refraction depending on the order of the densities of the media.

1. If the second medium is denser than the first, the light is refracted towards the normal.
2. If the first medium is denser than the first medium, the light is refracted away from the normal.

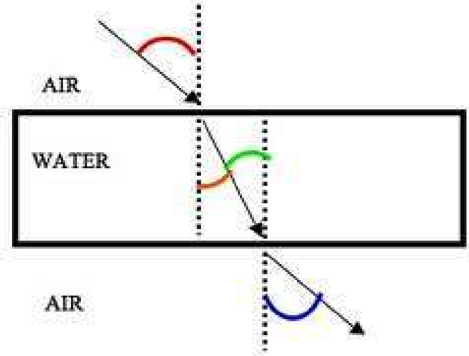


Fig. 117.5

Laws of refraction

Refraction of light is governed by two laws. The laws of refraction state that:

1. The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.

2. The ratio of the sine of the angle of incidence in one medium to the sine of the angle of refraction in another medium is constant for the two media. This is known as the ***Snell's law***.

i.e. $\frac{\sin\ i}{\sin\ r} = \mu$

This constant is referred to as the ***refractive index*** of the two media. Therefore, according to Snell's law, for two media **A** and **B**, the refractive index is given as

$$_A\mu_B = \frac{\sin\ i}{\sin\ r}$$

Where ***i*** is the angle in medium A and ***r*** is the angle in medium B.

The refractive index is also given as the ratio of the speed of light in one medium to the speed of light in another medium. The refractive index from medium A to medium B is expressed as $_A\mu_B$ where

$$_A\mu_B = \frac{speed\ \ of\ \ light\ in\ medium\ \ A}{speed\ \ of\ \ light\ in\ medium\ \ B}$$

In terms of depth, the refractive index can be calculated by measuring the real depth and the apparent depth in the liquid or glass.

$$\mu = \frac{real\ \ depth}{apparent\ \ \ depth}$$

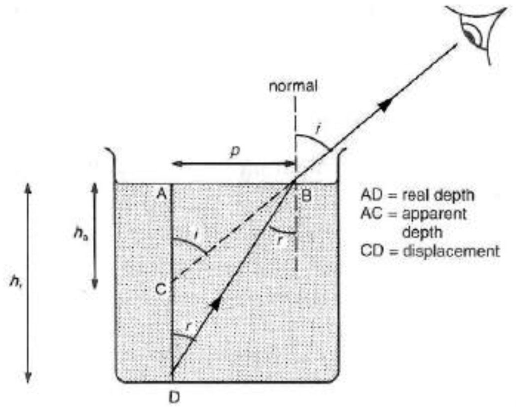


Fig. 117.7: Refractive index

Examples

1. A coin in a bucket of water appears to be 20cm when viewed vertically from above. What is the real depth of the coin if its refractive index of the water is 1.5cm?

Solution

$$\eta = \frac{real\ \ depth}{apparent\ \ \ depth}$$

Real depth = η x apparent depth

= 1.5 × 20 = 30 cm

2. A ray of light travelling through the air strikes a pool of water at an angle of 45° to the normal. If it moves through the water at an angle of 25° to the

normal, calculate the refractive index of the water.

Solution

$$\mu = \frac{\sin i}{\sin r}$$

angle of incidence (i) = 45^o

angle of refraction (r) = 25^o

$$\therefore \mu = \frac{\sin 45^o}{\sin 25^o} = \frac{0.707}{0.42}$$

$$\mu = 1.68$$

3. A ray from a flash light travelling at a velocity of $3.0 \times 10^8 ms^{-1}$ in the air slows down to $1.9 \times 10^8 ms^{-1}$ when it passed through glass. Determine the refractive index of the glass.

Solution

$$\mu = \frac{velocity\ \ of\ light\ in\ air}{velocity\ \ of\ light\ in\ glass}$$

velocity of light in air = $3.0 \times 10^8 ms^{-1}$

velocity of light in glass = $1.9 \times 10^8 ms^{-1}$

$$\mu = \frac{3.0\ x\ 10^8}{1.9\ x\ 10^8} = 1.58$$

THE MAMMALIAN EYE

The eye is a light-sensitive organ of vision in animals. The eyes of various species vary from simple structures that are capable only of differentiating between light and dark to complex organs, such as those of humans and other mammals that can distinguish minute variations of shape, colour, brightness, and distance.

The actual process of seeing is performed by the brain rather than by the eye. The function of the eye is to translate the electromagnetic vibrations of light into patterns of nerve impulses that are transmitted to the brain.

Structure and functions of the parts of the eye

The amount of light entering the eye is controlled by the pupil, which dilates and contracts accordingly. The cornea and lens, whose shape is adjusted by the ciliary body, focus the light on the retina, where receptors convert it into nerve signals that pass to the brain. A mesh of blood vessels, the choroid, supplies the retina with oxygen and sugar.
Lacrimal glands secrete tears that wash foreign bodies out of the eye and keep the cornea from drying out. Blinking compresses and releases the lacrimal sac, creating a suction that pulls excess moisture from the eye's surface.

The ***Ciliary body*** is the thickened edge of the choroid in the region around the lens. It contains blood vessels and muscle fibres. The ciliary body contracts or relaxes to alter the shape of the lens.

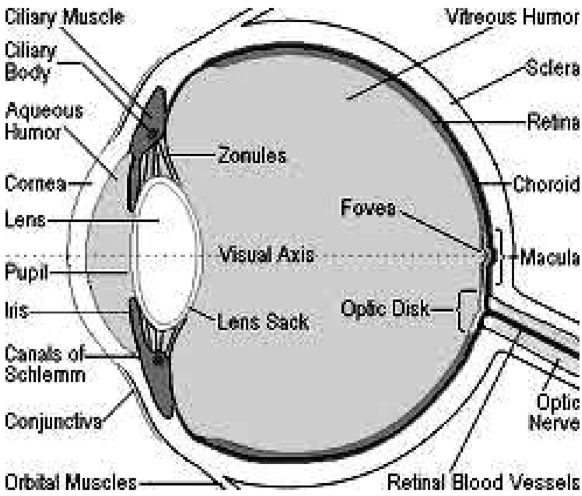


Fig. 117.8: Structure of the mammalian eye

The ***Aqueous humour*** is a fluid in front of the lens which carries nutrients to the lens and the cornea and helps to maintain the shape of the eye.

The ***Cornea*** is the transparent disc in the front of the sclera. The cornea allows light to pass through it to the eye. Its curved surface enables the refraction and convergence of light

The ***Lens*** is a flexible, transparent body, situated behind the iris and focuses light on the retina.

The ***Iris*** consists of an opaque disc of tissue. In its centre is the pupil. The tissues contract to regulate the size of the pupil. This controls the amount of light entering the eye.

The ***pupil*** is the dark opening at the centre of the iris which controls the amount of light entering the eye.

The ***Choroid*** is a layer of tissue lining the inside of the sclera. It contains a network of blood vessels which supplies food and oxygen to the eye. It contains a black pigment which reduces the reflection of light within the eye.

The ***Eye muscles*** contract to move the eye from up and down or side to side.

The ***Conjunctiva*** is a thin membrane which lines the inside of the eyelids, the front part of the sclera and is continuous with the membrane of the cornea.

The ***Sclera*** is a tough, non-elastic, fibrous coat around the eye-ball.

The ***Retina*** is located at the back of the eye and contains rods and cones, which are light sensitive cells connected to the brain by the optic nerve.

The ***Vitreous humour*** is a jelly-like material behind the lens which maintains the shape of the eyeball.

The ***Fovea*** is a small depression in the centre of the retina. It is also known as ***yellow spot*** and is the most sensitive part of the eye because of the concentration of sensitive cells there. This enables it to give detailed description of colour and form.

The ***blind spot*** is where the nerve fibres leave the eye to enter the optic nerve. It contains no light sensitive cells therefore images which fall on it are not visible.

Image formation and vision

- Light from an external object enters the eye.
- The curved surface of the cornea, the lens and the humours refract the light and focus on the retina.
- The image thrown onto the retina is real, diminished and upside down.
- Impulses are sent to the brain through the optic nerve.
- The brain interprets the nature, colour, size and distance of the object.
- The inversion of the image on the retina is corrected in the optical centre of the brain to form the impression of an upright object.

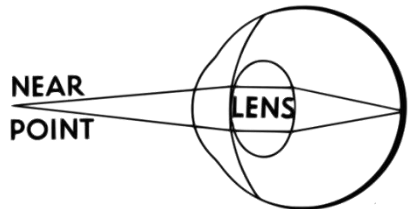
Accommodation

Accommodation is the ability of the eye to focus both near and distant objects.

The eye achieves accommodation as a result of the change in shape of the lens. Accommodation is automatic and instantaneous, and occurs as soon as the person looks near or distance object.

Focus on near objects

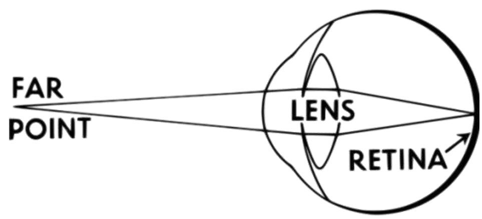
When the eye focuses on near objects the ciliary muscle contracts around the lens and decreases it size. The suspensory ligaments relax, causing the lens to form a more spherical shape.



Focus on near and distant objects

Focus on distant objects

When the eye focuses on distant objects the suspensory ligaments pull the edges of the lens towards the ciliary body, causing the lens to flatten.



Colour vision

The cones and rods are responsible for colour vision in the eye. There are three types of cones and rods in the retina. They are sensitive to red, green and blue light respectively.

The cones and rods are stimulated by their own particular wavelength of light so that, according to the kind of cell and the number which are stimulated, the brain receives an impression of colour.

How the eye controls light intensity

- ❖ Intensity in the eye is controlled by the iris and pupil.
- ❖ In dim light, the iris dilates (expands), causing the pupil to expand to allow more light into the eye.
- ❖ In bright light, the iris contracts (becomes smaller), resulting in the contraction of the pupil to allow less light into the eye.
- ❖ The contraction of the pupil protect the retina from damage by light of high intensity, and in poor light the widened pupil helps to increase the brightness of the image.

EYE DEFECTS

The proper functioning of the eye may be affected by a number of factors. Depending on the part affected, defects of the eye may be classified as:

- ❖ Long-sightedness (hypermetropia)
- ❖ Short-sightedness (myopia)
- ❖ Astigmatism
- ❖ Lack of accommodation (presbyopia)

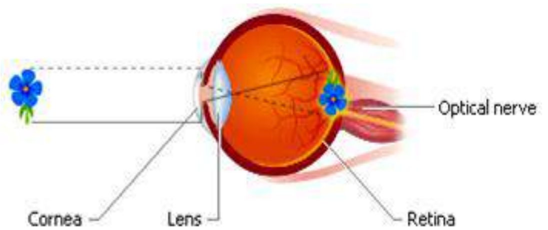


Fig. 117.9: Correct vision

Long sightedness or hypermetropia

Long sightedness is caused by small or short eyeballs or the inability of the muscles of the eye to alter the shape of the lens to focus light rays accurately. Light from a close object would be brought to focus behind the retina blurring the image of the object. People with long sight can see distant objects clearly but not near objects.

Long sightedness can be corrected by wearing converging lenses.

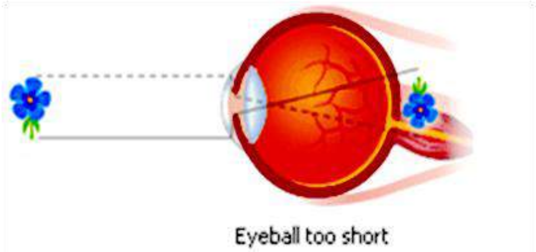


Fig. 118.0

Short sightedness or myopia

Short sightedness is caused by the eyeball being too long or large. Light from distant objects is focused in front of the retina blurring the image of the object. People with short sight can see near objects clearly but not distant objects.

Short sightedness can be corrected by wearing diverging lenses.

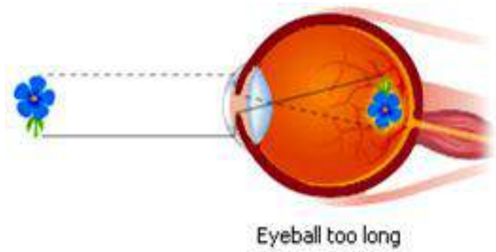


Fig. 118.1: Short sightedness

Astigmatism

Astigmatism is a defect in the outer curvature on the surface of the eye that causes distorted vision.

In the normal eye, light rays coming from a single point are bent, or refracted, toward each other by the cornea. As the rays pass through the inner parts of the eye, the lens bends the rays still further, focusing them to a point on the retina.

In a patient with astigmatism, the cornea or sometimes the lens of the eye is curved abnormally. This causes light rays to refract unevenly inside the eye.

While some light rays focus on the retina, other light rays focus in front of or behind the retina, resulting in blurred vision.

Astigmatism may be corrected by wearing cylindrical lenses to change the focal length of vertical or horizontal planes.

Presbyopia (lack of accommodation)

Presbyopia occurs with age as the lens of the eye gradually loses its elasticity. This reduces the ability of the lens to focus for near vision.

The first indication of presbyopia usually is difficulty with reading. Large print appears clearly, but small print is difficult to read except at arm's length. Eventually the lenses of the eyes have little or no focusing ability.

Simple reading eyeglasses with convex lenses correct most cases of presbyopia. These convex lenses reduce the distances of both near and distant points.

Formation of mirage

Mirage is an optical illusion where objects appear to be at a place when in actual sense they are not.

Mirages appear because differences in air temperature cause light rays from an object to take different paths to a viewer's eye. Warm, less dense air near the ground bends light, so when the light reaches the viewer's eye, the ray seems to point into the ground.

This produces a second image that looks like a reflection of the object.

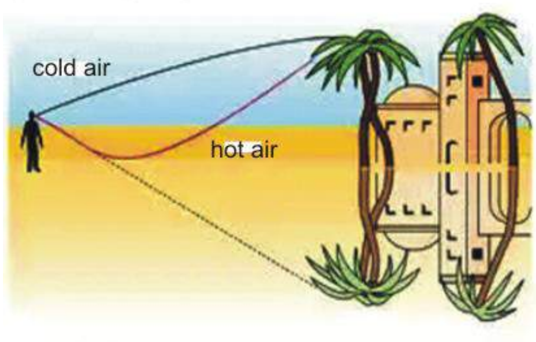


Fig. 118.2: Mirage formation

LENSES

A lens is a glass or other transparent substances so shaped that it will refract the light from any object and form a real or virtual image of the object.

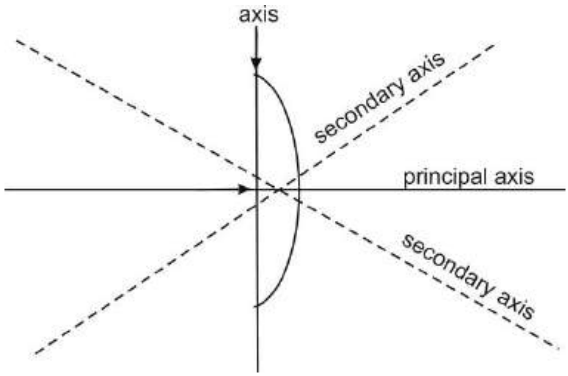


Fig. 118.3: Parts of a lens

Types of lenses

Lenses come in two forms:

- ❖ Converging or convex lenses
- ❖ Diverging or concave lenses

Converging or convex lenses

A convex lens curves outward; it has a thick centre and thinner edges. Light passing through a convex lens is bent inward, or made to converge. This usually causes a real image of the object to form on the opposite side of the lens.

The size of the image and the place where it is in focus depends upon the size and position of the object and the focal point (F) of the lens (the place where light rays converge).

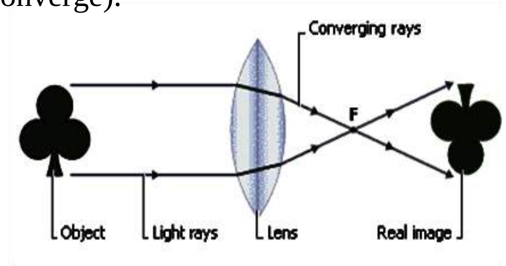


Fig. 118.4: Converging lens

Diverging or concave lenses

A diverging lens is curved inward, with a thin centre and thicker edges. Light passing through a diverging lens bends outward, or diverges.

Diverging lenses produce only virtual images; that is, light rays do not actually come from the virtual image, but the brain perceives the diverging rays as though they do.

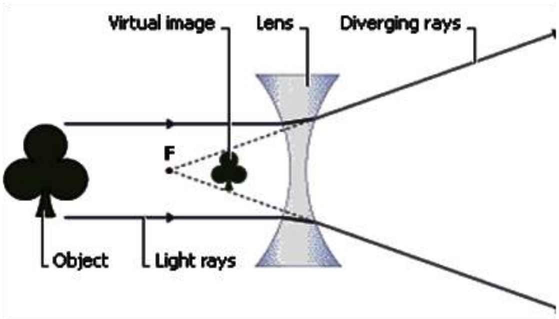


Fig. 118.5: Diverging lens

Uses of lenses

1. Lenses are used in contact lenses and eyeglasses to correct visual defects.
2. Lenses are used in cameras
3. They are used in microscopes to magnify specimen.
4. They are used in telescopes to make distance object clearer.
5. They are used in magnifying glasses to make objects look bigger.

DISPERSION OF LIGHT

Dispersion of light is the separation of white light or other electromagnetic waves into different wavelength.

OR

Dispersion is the separation of visible light into its component colours.

Dispersion occurs as a result of the refraction of light when it travels from one optical medium to another.

White light is dispersed when it passes through media such as a triangular prism, water or cloud.

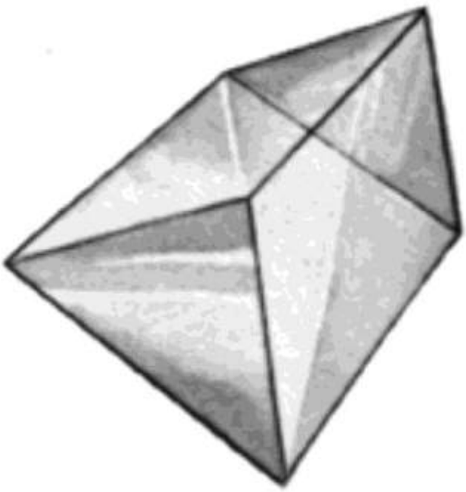


Fig. 118.6: A glass prism

Dispersion of white light

When white light is dispersed, its shows a spectrum of seven visible colours, namely, red, orange, yellow, green, blue, indigo and violet (ROYGBIV).

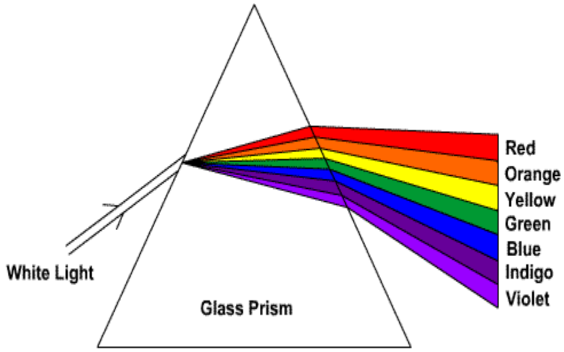


Fig. 118.7: Dispersion of white light

Other electromagnetic radiations displayed after dispersion of light are ultraviolet radiation, X-rays and gamma rays, which are above the violet colour, and infrared radiation, microwave and radio wave, which are below the red colour.
These colours are displayed because they have different wavelength and speed in other materials other than a vacuum.

Formation of rainbow

Sunlight is a white light made up of many colours. When a ray of sunlight enters a raindrop, the light refracts, or bends. Different colours bend slightly different amounts, so the colours spread out.
Much of the light reflects from the back of the raindrop, and then refracts again as it exits the raindrop.
Red light refracts the least; violet light refracts the most. The other colours would fan out between red and violet to form the familiar rainbow.

PRIMARY COLOURS OF LIGHT

Cones, which are the light-sensitive cells in the eye are in three types – seeing three types of colours – red, green and yellow (RGB). These colours are known as primary colours. They cannot be formed by mixing any other colour. They occur naturally.

The three primary colours combined will produce a white light.

Two primary colours combined will produce a secondary colour.

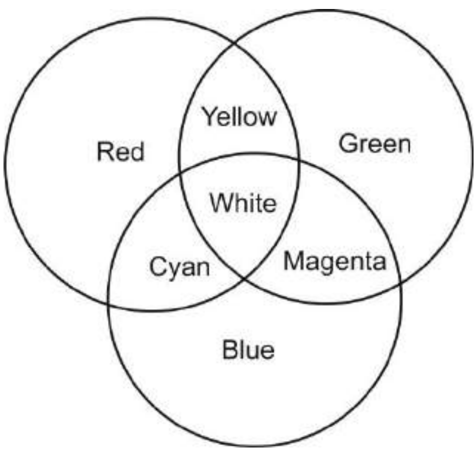


Fig. 118.8: Primary and secondary colours

Secondary colours

These are colours produced when two primary colours are combined. The three secondary colours obtained when each of the primary colours is mixed with the other are yellow, cyan and magenta.

Table 9.2 below shows the mixture of primary colours to obtain secondary colours.

Table 9.2

Primary colours	Secondary colours
Red + Blue	Magenta (purple)
Red + Green	Yellow
Green + Blue	Cyan (blue-green)

Complementary colours

When certain primary colours are mixed with secondary colours, they produce white light. These colours are known as ***complementary colours***.

Examples are.

Red	+	Cyan	=	White light
Green	+	Magenta	=	White light
Blue	+	yellow	=	White light

Behaviour of objects under different coloured light

When white light falls on a white object all the wavelengths are reflected back, hence, we see white.

On the other hand, when white light falls on a coloured object, for example, blue the object absorbs all the colours except blue, which it reflects, hence, we see blue colour.

If the blue object is viewed in red light it will appear black. This is because the blue object will absorb the red light and not reflect any light.

ELECTROMAGNETIC SPECTRUM

An electromagnetic wave is an energy waves produced by the oscillation or acceleration of an electric charge.

Electromagnetic waves have both electric and magnetic components. They travel in a vacuum at a speed of 3×10^8 ms^{-1}.

The electromagnetic spectrum consists of a group of electromagnetic waves arranged according to their frequencies and wavelengths.

On the spectrum, waves extend extremely high frequency and short wavelength to extremely low frequency and long wavelength. White light, which is the only component visible to the human eye, is only a small part of the electromagnetic spectrum.

In order of decreasing frequency, the electromagnetic spectrum consists of gamma rays, X-rays, ultraviolet radiation, visible light, infrared radiation, microwaves, and radio waves.

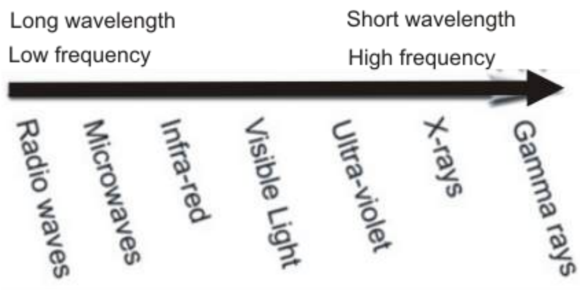


Fig. 118.9: The electromagnetic spectrum

Application of electromagnetic radiation

X-rays

An X-ray is a penetrating electromagnetic radiation, having a shorter wavelength than light, and produced by bombarding a target, usually made of tungsten, with high-speed electrons.

There are two types of X-rays –

- ❖ soft X-rays have long wavelengths with less penetrating power.
- ❖ hard X-rays have short wavelengths with high penetrating power.

Uses of X-rays

1. X-rays are used to detect fractured boned in the human body.
2. They are used to destroy cancer cells in the body.
3. They are used to destroy tumours in the body.
4. X-ray photographs can detect swallowed metal
5. X-rays are used to detect cracks in metals and welded joints
6. They are used to study crystal structures of atoms in a substance.
7. X-rays are used to diagnose diseases

Dangers or harmful effects of x-rays

In the above uses, X-rays are used in controlled amounts. Over-exposure to x-rays is very dangerous as it poses many health hazards, which include:

a) X-rays can cause skin burns.

b) They can cause cancer.
c) They can damage cells and tissues in the body.
d) They affect genetic functions in humans.
e) They cause leukaemia (blood cancer).
f) They cause mutations of living cells.
g) X-rays cause miscarriage in pregnant women.
h) They can ionise the fluid in humans which can interfere with the chemical activities in the body.

Radio waves

Radio waves have two related properties known as frequency and wavelength. Frequency refers to the number of times per second that a wave oscillates, or varies in strength. Low-frequency radio waves have long wavelengths, whereas high-frequency radio waves have short wavelengths.

Uses of radio waves

1. Radio waves are used in television, FM and AM broadcast.
2. They are used in wireless communication e.g. mobile phones
3. They are used by air-traffic controllers to communicate with pilot.
4. They are used in radars to detect other signals.
5. They are used in robotics.

Gamma rays

Gamma rays have the shortest wavelength, greatest energy, and most penetrating power of any form of electromagnetic radiation.

Like most other forms of electromagnetic rays, gamma rays have no mass or charge, carry discrete amounts (quanta) of energy, and can behave like either waves or particles.

Gamma rays are emitted in normal radioactive decay, in nuclear explosions, in matter-antimatter annihilation, in nuclear fusion in the cores of stars, and in high-energy astronomical events such as exploding stars called supernovas, or mergers of neutron stars and black holes.

Uses of gamma rays

1. Gamma rays are used in nuclear medicine for diagnoses purposes.
2. They are used to detect cracks in metals.
3. They are used to sterilize medical equipments.
4. They are used in food storage to kill bacteria.
5. They are used to treat some forms of cancer.
6. They are used in radiation therapy and CT scans.

Dangers of gamma rays

a) They can cause cancer.

b) They can interfere with the genetic make-up of an organism.
c) They can cause skin burns.
d) They can destroy cells and tissues in the body.

Ultraviolet (UV) radiation

Ultraviolet radiation is often divided into three categories based on wavelength, UV-A, UV-B, and UV-C. Natural ultraviolet radiation is produced principally by the sun. Ultraviolet radiation is produced artificially by electric-arc lamps. In general shorter wavelengths of ultraviolet radiation are more dangerous to living organisms.

Uses of ultraviolet radiation

1. Ultraviolet lamps are used to sterilize medical and laboratory equipments.
2. They are use to detect fake money note, and documents.
3. They are used to control pests.
4. Ultraviolet radiation is used in skin tanning
5. It is used in food processing.
6. UV is used to brighten dyes and paints.

Dangers of ultraviolet radiation

a) Over-exposure can cause sunburns
b) High intensity UV light can cause defect in the eye.
c) Ultraviolet radiation can damage polymers, pigments and dyes.

Infrared (IR) radiation

Infrared radiation is an electromagnetic wave in the portion of the spectrum just beyond visible light. The wavelengths of infrared radiation are shorter than those of radio waves and longer than those of light waves. Infrared radiation may be detected as heat, and instruments such as a bolometer are used to detect it.

Uses of infrared radiation

1. It provides vitamin D to humans.
2. It provides medium for the radiation of heat energy.
3. It provides warmth to living organisms
4. Infrared devices enable sharpshooters to see their targets in total visual darkness.
5. It is used in thermal imaging to remotely determine position of objects. This is mostly used in industrial and military operations.
6. It is used to track launched missiles.

Dangers of Infrared radiation

a) Over-exposure can damage the eye.
b) It may lead to skin burns.
c) It can cause skin irritations.
d) It can cause low blood pressure in elderly patients.
e) It can cause dehydration.

Microwave

Microwaves are short, high-frequency radio waves lying between infrared waves and conventional radio waves Microwaves thus range in length from about 1 mm to 30 cm. They are generated in special electron tubes or by special oscillators or solid-state devices.

Uses of microwaves

1. Microwaves are used in radio and television broadcasting.
2. They are used in radars for communication
3. They are used to detection and measure distance of objects.
4. They are used in meteorology to forecast the weather.
5. They are used in satellite communications.
6. They are used in research into the properties of matter.
7. Microwave ovens are used to cook and heat food.

Dangers of microwaves

Exposure to microwaves is dangerous mainly when high densities of microwave radiation are involved.

a) They can cause burns to the skin
b) They can cause cataracts (an eye defect)
c) Microwaves can damage the nervous system.
d) They can cause sterility in organisms.

Visible light

The visible light refers to the range of frequencies that can be seen by humans. The frequencies of these waves are very high, about 5×10^{14} to 7.5×10^{14} Hz and wavelengths ranging from 400 to 700 nm.

Importance of visible light

1. It enables humans and animals to see.
2. It helps humans to differentiate colours.
3. It helps human to judge distances.
4. Plants use visible light for photosynthesis.
5. Lack of visible light (sunlight) can cause depression and brain damage.

***Dangers of* visible light**

a) Over exposure can cause eye defects.
b) It can damage pigments and dyes (photodegradation).
c) It can cause breakdown of some plastics and polymers.

TEST QUESTIONS

3. (a) Differentiate between natural and artificial sources of light.
   (b) Name two examples each of natural and artificial sources of light.

2. Explain the following terms:
   (a) transparent objects:
   (b) opaque objects;
   (c) translucent objects.

3. (a) Mention three characteristics of light.
   (b) Describe an experiment to show

that light travels in a straight path.

4. (a) What is a shadow?
   (b) Distinguish between umbra and penumbra.

5. (a) Define the term eclipse.
   (b) Briefly explain:
      (i) solar eclipse;
      (ii) lunar eclipse.

6. Describe how a pinhole camera works.
7. (a) Define the term reflection of light.
   (b) Explain the following types of reflection:
      (i) regular reflection;
      (ii) diffused reflection.

3. A girl applies make-up100cm in front of a spherical mirror and notices her reflection 80cm in the mirror.
   (a) What is magnification of the image?
   (b) Calculate the focal length of the mirror
8. (a) State the laws of reflection.
   (b) Draw a ray diagram of light incident at an angle of 45^o with the normal on the surface of a plane mirror.

9. Explain:
   (a) real image;
   (b) virtual image.

10. (a) Mention one example of the following types of mirror:
      (i) plane mirror;
      (ii) concave mirror;
      (iii) convex mirror.

11. (a) List four uses of plane mirror.
    (b) Mention four characteristics of plain mirror.
12. (a) Describe the nature of images formed by concave and convex mirrors.
    (b) Mention three uses of each of concave and convex mirrors.
13. Explain the following terms:
    (a) principal axis;
    (b) principal focus;
    (c) focal length;
    (d) centre of curvature.

14. (a) A man stands 50cm from a plane mirror. What is the distance of his image if the focal point is 85cm?
    (b) An object of height 2.5cm is placed 20cm from of a converging mirror of focal length 10cm. find the height of the image formed.

15. (a) What is refraction of light?
    (b) State the laws of refraction.

16. Describe how the eye:
    (a) forms an image;
    (b) controls light intensity.

17. Explain the following defects of the eye.
    (a) hypermetropia;
    (b) myopia;
    (d) astigmatism.

18. Describe the formation of mirage.

19. (a) What is a lens?
    (b) Describe the following lenses:
        (i) converging lens;
        (ii) diverging lens.

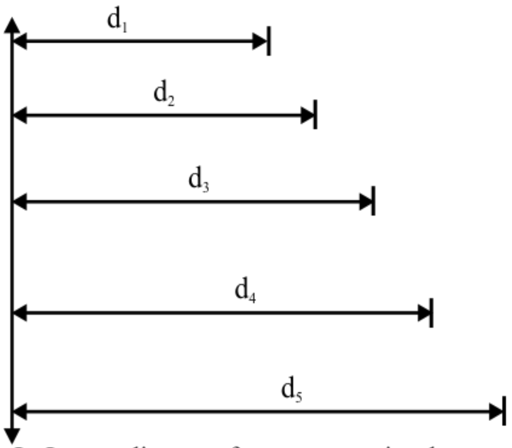
20. (a) Explain the term dispersal of light.
    (b) Briefly describe the formation of rainbow.

21. (a) Distinguish between primary colours and secondary colours.
    (b) indicate the primary colours which form the following secondary colours:
        (i) magenta;
        (ii) cyan;
        (iii) yellow.

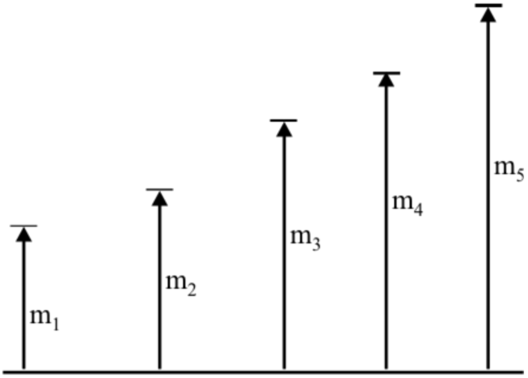
22. (a) Define the term electromagnetic spectrum.
    (b) State three uses of the following electromagnetic radiations:
        (i) X ray;
        (ii) gamma ray;
        (iii) radio waves;
        (iv) microwave;
        (v) visible light.

23. In an experiment to determine focal length , f of a converging lens, the image distance $d = d_1, d_2, d_3, d_4$ and d_5 and the corresponding magnifications $m = m_1, m_2, m_3, m_4$ and m_5 were determined.

In the figure below, **I** shows the iamge distances $d = d_1, d_2, d_3, d_4$ and d_5 while **II** shows the corresponding magnifications $m = m_1, m_2, m_3, m_4$ and m_5 produced.



I. Image distance form converging lens
Scale: 1 cm represents 5 cm



II. Magnification produced
Scale: 1 cm represents 0.5

a) Measure and record the image distances $d = d_1, d_2, d_3, d_4$ and d_5.
b) Measure and record the corresponding magnifications $m = m_1, m_2, m_3, m_4$ and m_5.
c) Convert the raw value obtained in (a) and (b) above into actual values. Present the result in a tabular form.
d) Plot a graph with d as the vertical axis and m as the horizontal axis.
e) Determine the slope of the graph and the intercept on the d-axis.
f) What do the slope and the d-intercept represent?

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HEAT ENERGY

Specific Objectives

After completing this chapter, you will be able to:

- Describe the modes of heat transfer.
- Explain the concept of temperature and how it is measured.
- Discuss the effects of heat.
- Explain how heat causes change of state of matter.

INTRODUCTION

Heat is one of the most important and commonest forms of energy on earth. It is generated by most activities usually as a result of friction.

Heat energy is usually the final energy in most energy conversions, and can itself be converted to other forms of energy such as electricity and light.

Almost all chemical changes take effect because of heat energy. Living organisms rely on heat to keep them warm. Humans depend on it for most of their day-to-day activities.

NATURE AND SOURCES OF HEAT

Nature of heat

Heat is the transfer of energy from one part of a substance to another or from one body to another.

Heat is energy in transit; it always flows from a substance at a higher temperature to a substance at a lower temperature, raising the temperature of the latter and lowering that of the former substance until they reach ***thermal equilibrium*** (the same temperature) provided the volume of the bodies remains constant.

Since heat is energy, it is measured in **joule (J).**

Sources of heat

1. The sun
2. Nuclear reaction
3. Electricity
4. Friction
5. Chemical reaction
6. Fossil fuel
7. Fire

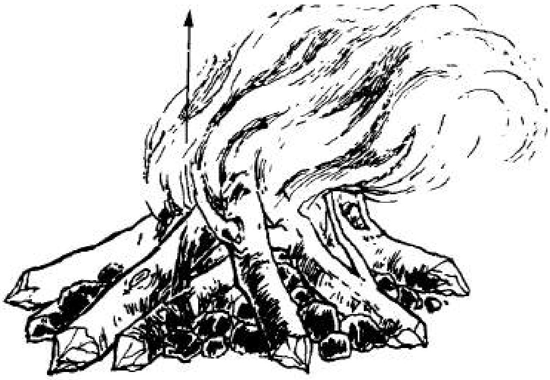


Fig. 119.0: Heat from fire

Uses of heat energy

1. Provision of warmth to humans and animals.
2. Cooking and heating of food.
3. Burning of fuels in automobile and aeroplane engines.
4. Extracting metals from their ores.
5. Cutting, shaping and softening of metals and other materials.
6. Drying substances such as food, clothes, etc.

MODES OF HEAT TRANSFER

Heat can be transferred by three processes:

- ❖ Conduction,
- ❖ Convection,
- ❖ Radiation.

These different methods are used to transmit heat in different states of matter.

Conduction

Heat conduction is the transfer of heat energy from one region of a solid to another or from one solid to another.

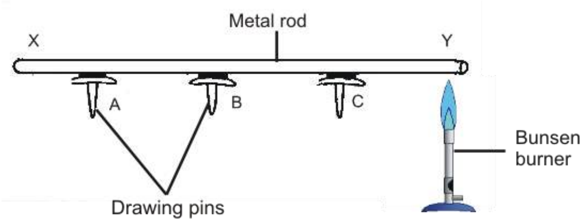
Heat transfer in metals is stronger and faster than through non—metals. This is because metals have free moving electrons which can readily transfer heat from one area of the metal to another. Thus metals are good ***conductors*** of heat.

On the other hand, non-metals are bad conductors of heat, and are referred to as ***insulator*** (e.g. plastic, wood, glass etc.).

It is conduction that makes the handle of a metal spoon hot when we use it to stir hot soup. It is actually conduction that makes the pan containing the soup hot as heat from the fire is easily transferred to it.

Experiment to show the Transfer of Heat Energy by Conduction

- ❖ Light the Bunsen burner.
- ❖ Place a lump of Vaseline and a drawing pin on an aluminium stick.
- ❖ Place the stick about 3 inches above the flame and start the stopwatch.
- ❖ After a few seconds, the drawing pin will fall off.
- ❖ This is because the iron rod has heated up by conduction and melted off the Vaseline.



Rate of conduction

The rate of conduction depends on:

1. The nature of solid
2. The temperature gradient or difference in temperature between two points in the solid
3. The cross-sectional area of the solid

Reason why cooking utensils have wooden or plastic handles

Wood and plastic are insulators, which are bad conductors of heat. This prevents the conduction of heat from the cooking utensils into them. Therefore, the utensils can be lifted up by their handles without getting the hand burnt.

Fig. 119.1

Convection

Convection is the transfer of heat through the exchange of hot and cold molecules.

Conduction happens only in fluids (i.e. liquids and gases); hence can be defined as the transfer of heat in fluids.

How conduction occurs

It is through convection that water in a kettle becomes uniformly hot even though only the bottom of the kettle contacts the flame.

This occurs because the molecules in the water gain kinetic energy and move about more quickly, causing the water to expand and decrease in density. The less dense water rises up from the bottom of the kettle and is replaced by cool, denser water from the top. This cycle continues until there is thermal equilibrium.

The movement of hot fluid to replace cold one is called ***convection current***.

An experiment to demonstrate convection current in fluid

- Fill a beaker or flask with water.
- Place a crystal of potassium permanganate, $KMnO_4$ (purple colour) at a corner of the bottom of the beaker.
- Gently apply heat to the beaker at where the crystal of potassium permanganate is located.

Observation

- It would be observed that an upward movement of purple-coloured water from the region of the crystal to the top.
- When the coloured stream reaches the top it spreads out and after a while circulates down the sides of the beaker.

Conclusion

- Heat is carried from one place to another in the water by the movement of the water itself shown by the movement of the colour. This is due to convection.

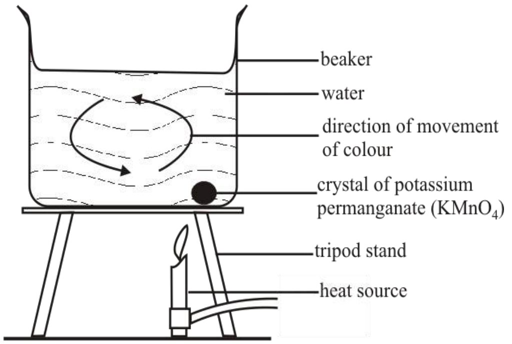


Fig. 119.2

Radiation

Radiation is the transfer of heat through electromagnetic radiation.

Radiation does not need a medium for its transmission; hence, occurs in a vacuum. The sun's energy gets to the earth through radiation. Radiant energy is emitted in the form of waves or rays, since it is electromagnetic.

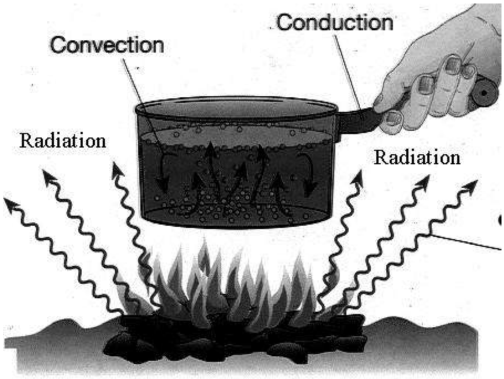


Fig. 119.2: heat transfer around a source pan

The vacuum flask

The vacuum flask, also known as the ***thermos flask***, is an insulating storage vessel which is used to maintain the temperature of a substance.

Structure of the vacuum flask

- It is made up of two glass-walled bottles, one placed in the other and joined together at the neck.
- The gap between the bottles is partially evacuated to create a near vacuum which prevents heat transfer by conduction or convection.
- The silvery surfaces of the glass walled bottles prevent heat transfer by thermal radiation.
- The insulating cork, which covers the opening, prevents heat transfer by conduction.

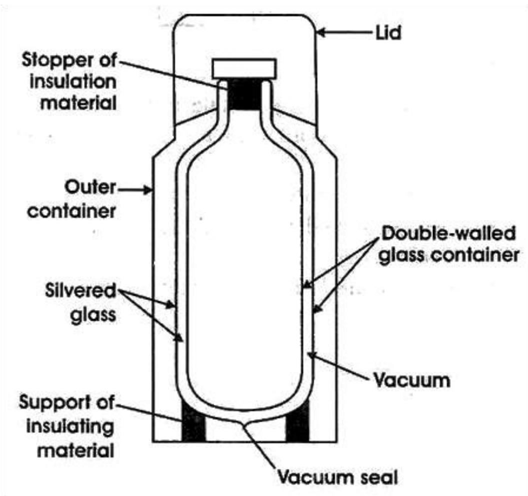


Fig. 119.3: A vacuum flask

Uses of vacuum flasks

1. Vacuum flasks are used domestically to keep food warm.
2. In laboratories, they are used to store liquids which become gaseous below ambient temperature, such as oxygen and nitrogen.
3. They are used in industries, especially food industries to maintain the temperature of products.
4. They are used to store standard cells and ovenized Zener diodes as well as their printed circuit boards.
5. They are used to store certain types of rocket fuels.

TEMPERATURE

Temperature is a measure of how hot or cold a substance is.

Temperature tells the amount of heat or thermal energy in a body. It is a vector quantity and is measured in Kelvin, K or degree Celsius.

Table 9.3: Differences between heat and temperature

Heat	Temperature
Measured in joules, J	Measured in Kelvin, K
Can be transmitted from one body to another	Cannot be transmitted from one body to another
Is a form of energy	Is not a form of energy
Measured with calorimeter	Measured with thermometer
Does not depend of external body	Depends on external body
Amount of kinetic energy in a body	Measure of the amount of kinetic energy in a body

Temperature scales

There are a couple of temperature scales available, which makes determining temperature of bodies easier.

Some temperature scales available include the Celsius scale, the Kelvin scale, the Fahrenheit scale and the Rankine scale. The Celsius and the Kelvin scales are the commonest scales used in Ghana.

Celsius Scale or Centigrade Scale

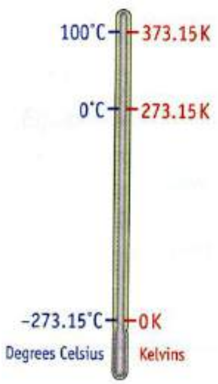
On this scale, the unit of temperature is degree Celsius (^oC). The lower fixed point is represented by 0 ^oC and the upper fixed point by 100 ^oC.

The lower fixed point refers to the temperature at which freezing of water takes place at standard atmospheric pressure. It is also known as the ice point

The upper fixed point refers to the temperature of water at which its vaporization takes place at standard atmospheric pressure. It is also known as the steam point.

Kelvin scale

The Kelvin scale is also known as the thermodynamic scale. The temperature on this scale is measured in Kelvin (K). The lowest temperature of a body is represented as 0 K, (equivalent to -273.15 ^{o}C) which is called the absolute zero temperature. Also, the lower fixed point is represented as 273.15 K and upper fixed point 373.15 K.



Conversions between the Celsius and the Kelvin scales

To convert temperature in Kelvin to Celsius, subtract the absolute zero temperature, -273, from the figure given.

This can be expressed as:
C = K – 273

Examples

Convert the following to temperatures to ^{o}C:

i. 373K
ii. 500K
iii. 25K

Solution

C = K – 273

i. C = 373 – 273
= 100^{o}C

ii. C = 500 – 273
= 227^{o}C

iii. C = 25 – 273
= -248^{o}C

Conversion from degree Celsius to Kelvin

Inversely, to convert from degree Celsius to Kelvin, add 273 to the given value. This can be expressed as:
K = C + 273

Example

Convert the following temperatures to Kelvin:

i. 100 ^{o}C
ii. 0^{o}C
iii. 12^{o}C

Solution

K = C + 273

i. K = 100 + 273
= 373 K

ii. K = 0 + 273
= 273 K

iii. K = 12 + 273
= 285 K

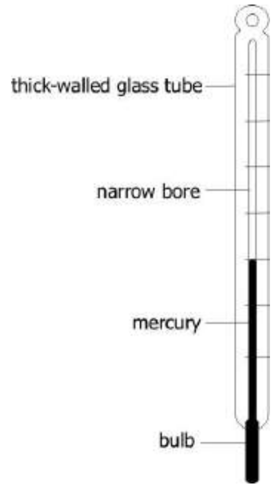
TEMPERATURE MEASUREMENT - THERMOMETERS

A thermometer is an instrument use to measure the temperature of a body. The various types of thermometer include:

- Liquid-in-glass thermometer
- Bimetallic thermometer
- Gas thermometer
- Pyrometer
- Resistance thermometer
- Thermocouple

Liquid-in-glass Thermometer

This type of thermometer consists of a narrow, sealed glass tube filled with liquid such as mercury or alcohol at the lower end. As the temperature rises, the liquid expands and rises in the tube. A drop in temperature causes the liquid to contract so that its level in the tube falls.



Thermometric liquids

These are liquids that increase or decrease in depth with temperature change. Mercury and alcohol are usually preferred as thermometric liquid because of the following reasons.

Reasons why mercury is preferred to alcohol in liquid-in-glass thermometer

1. Mercury has low specific heat and hence absorbs little heat from body.
2. Mercury is comparatively a good conductor of heat.
3. Mercury can be seen in a fine capillary tube conveniently, alcohol is colourless.
4. Mercury does not wet the wall of the tube.
5. Mercury has a uniform coefficient of expansion over a wide range of temperature and remains liquid over a large range as its freezing and boiling

points are –39 °C and 357 °C respectively.

Disadvantages of mercury as a thermometric liquid

a) Mercury cannot be used to measure low temperature. This is because it freezes at -39 ^{o}C.
b) It cannot be used in cold or temperate regions.
c) It is poisonous, hence harmful if the bulb breaks
d) It is relatively expensive

Advantages of alcohol as a thermometric liquid

1. Alcohol can be used to measure extremely low temperatures. (It freezes at -122 ^{o}C).
2. It can be used in cold regions.
3. It has a very high expansion rate.
4. It is very cheap.
5. It is not poisonous.

Disadvantages of alcohol as a thermometric liquid

a) It is not easily visible
b) It is a poor conductor of heat and responds slowly.
c) It boils at 78 ^{o}C and cannot be used to measure high temperatures.

Reason why water is not a good thermometric liquid

1. The expansion of water is not uniform.
2. Its boiling and freezing temperature range is very small.
3. It is not easily visible.
4. It vaporizes easily.
5. It wets glass (Clings to the glass tube when falling).

Reason why it is not advisable to sterilize alcohol thermometer in boiling water

Alcohol expands at 78 ^{o}C, while boiling water has a temperature of 100 ^{o}C. Therefore, if an alcohol thermometer is sterilized in boiling water, the alcohol will over expand and cause the capillary tube to break.

Clinical thermometer

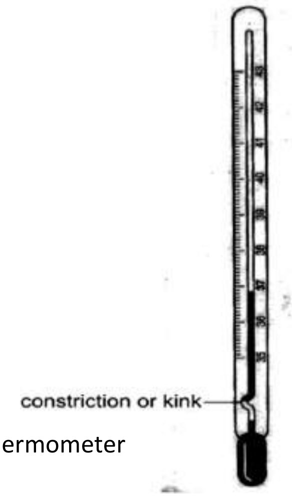
The clinical thermometer is a type of liquid-in-glass thermometer (with mercury as the thermometric liquid). It is specially designed to measure human temperature. It is so named because it is mostly used in clinics, hospital and other health centres or homes to determine the temperature of patients.

Structure of the clinical thermometer

The clinical thermometer has a constriction or kink which stops the mercury from falling back into the bulb when removed from the body. This enables the reading of high body temperatures with ease.

Range

It has a graduation range from 35 ^{o}C to 43 ^{o}C. This range is centred on the normal body temperature of 36.9 ^{o}C. The short range enables the scale to be divided into smaller intervals (0.1 ^{o}C) for greater accuracy.



Clinical thermometer

Using the clinical thermometer

The thermometer is placed under the armpit or tongue of the patient. The mercury level rises in correspondence to the patient's body temperature, (remember, the normal human body temperature is 36.9 ^{o}C). After reading the temperature, the mercury can be returned to the bulb by vigorously shaking the thermometer towards the bulb.

Reason why the clinical thermometer should not be sterilized in boiling water

The clinical thermometer has a maximum temperature range of 43 ^{o}C, whiles boiling water has a temperature of 100 ^{o}C. Hence, if the clinical thermometer is sterilized in boiling water, the mercury in the capillary tube will over-expand and cause the tube to break.

THERMAL EXPANSION

Objects generally expand when heated; whether, solid, liquid or gas. This is due to the fact that when heated, the molecules in the objects gain kinetic energy which causes them to move about more strongly, bumping into each other and the walls of the object. This constant bumping causes the objects to become larger.

When cooled, they contract (become smaller). This is because the molecules lose kinetic energy and attract each other.

Experiment to demonstrate that heat causes expansion
(The ball and ring method)

- ❖ Pass a metal ball through a metal ring. The diameter of the ball should be slightly smaller than that of the inside of the ring.
- ❖ Heat the ball over a heat source for a few minutes.
- ❖ Try passing it through the ring again.

Observation

- ❖ It would be observed that the metal ball will not be able to enter the ring after heating.

Conclusion

- This shows that upon heating, the metal ball has expanded or increased in size.

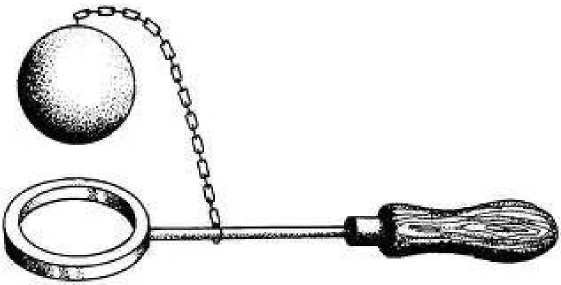


Fig. 119.9: *A ball and ring*

Experiment to investigate expansion in liquid due to heat

- Fill a test tube half way with water and mark the level of the water on the test tube.
- Heat the test tube gently over a heat source.

Observation

It would be observed that the level of the water in the test tube rise pass the mark.

Conclusion

This show that the heat has caused the molecules to be vibrant, and that has in turn caused the level of the water to rise.

Practical application of expansion

Uneven expansion and contraction of matter could be problematic to structures. Therefore, to prevent or reduce the effects of heat some measures have been observed to reduce the damages they cause.

Overhead cables

Overhead cables are in pairs or more; and are not suppose to touch each other. They are, hence, laid with enough spacing in between them.

Another issue that is taken into consideration is the effect of thermal expansion which causes the cables to sag and drop down during the hot day. The cables are normally lay very high so that when they drop they will not pose danger to passers-by. They are also allowed to drop slightly to compensate for contraction.

Building construction

Some buildings have holes in the walls which serve as expansion gaps to control expansion. Terrazzo works have soft materials such as plastic in between the slabs; and tiles are bevelled to prevent them from cracking when they expand or contract.

Glass tumblers and bottles

Glass tumblers with thick walls are not used to store hot liquid. This is because when a hot liquid is poured into them, the inner walls heat faster than the outer ones. This causes uneven expansion in the walls resulting in the cracking of the tumblers.

It is, therefore, ideal to store hot liquids in thin bottles which have even expansion and will not break.

Grooves in roofing sheets

On hot afternoons cracking sounds are normally head from metal roofing sheets. These are a result of the expansion of the roofing sheets.

To control the effects of expansion, roofing sheets are built with grooves and also allowed to overlap so that when they expand or contract, they would not deform.

Fig. 120.0: A corrugated roofing sheet

Bursting of inflated hot lorry tyres

When an automobile moves, only a part of the tyre touches the road at a time. This mean when the road is heated up by the sun, only a part of the tyre gets heated up by the road through conduction. This causes that part of the tyre to expand more than the other parts, resulting in uneven expansion and the bursting of the tyre.

Gaps in railway lines

Railway lines are lay with gaps in between the ends of each line. This is to prevent the lines from bulging up or undulating when they expands, which could derail a train.

Fig. 120.1: Undulated railway lines

Bimetallic strip

A bimetallic strip is made up of two different metals of the same length, riveted together. These two metals have different expansion rates. For example, if iron and brass are used; for a given temperature, the brass will react more than the iron.

So if the strip is heated the brass will expand more which will cause the strip to bend toward the iron because it would be shorter.

At normal atmospheric temperature, the strip retains its straight shape. When the temperature drops the brass contracts more, hence, becomes shorter; causing the strip to bend towards it.

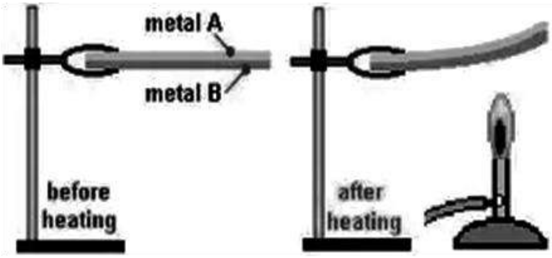


Fig. 120.2: Bimetallic strip

Bimetallic thermostats

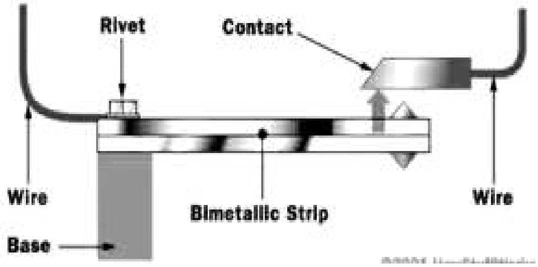
A thermostat is a device used to control or regulate temperature in electrical and mechanical devices.

Examples of devices which use thermostats are refrigerators freezers, electric ovens gas cookers, pressing irons, electric cookers, blenders and stoves, electric and automatic flashers etc.

How the Bimetallic Thermostat works

A bimetal thermostat uses a special strip of metal to open and close a circuit as temperature fluctuates. Two metals with different expansion rates are bonded to make the strip.

The thermostat is arranged so that when the metals are hot, the strip bends upward (toward the metal with the lower expansion rate) and disconnects the circuit.



Gas oven and cooker thermostats

Thermostats are sometimes used to regulate gas ovens. It consists of a gas-filled bulb connected to the control unit by a slender copper tube. The bulb is normally located at the top of the oven. The tube ends in a chamber sealed by a diaphragm. As the thermostat heats up, the gas expands applying pressure to the diaphragm which reduces the flow of gas to the burner.

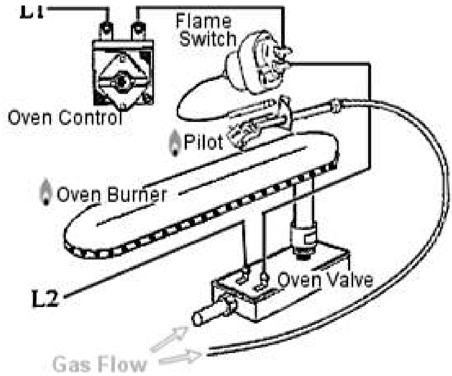


Fig. 120.3: A thermostat

CHANGE OF STATE OF MATTER

As we have discussed earlier under matter in chapter 4, matter exists in three states – solid, liquid and gas.

All matter are made up of molecules and a change in the molecular structure of the matter, caused by change in temperature, could cause that matter to change from its state to another state.

One particular matter which can exist in all three states is water, which can exist as ice (solid), water (liquid) and vapour (gas).

When ice is heated, the molecules gain more kinetic energy and begin to break off each other. This causes the ice to melt into water. When the water is heated further, the molecules become more energetic because of more kinetic energy induced.

This causes them to break even further and move faster, resulting in vapour.

When the vapour is cooled, the molecules loose kinetic energy. This causes the gas to condense into water again. Further cooling decreases the kinetic energy further and solidifies the water into ice. This shows that the change of state of water is reversible.

Physical and chemical changes

Matter undergoes changes in characteristics. That is one object in one state of mater can change and have the properties of another state.

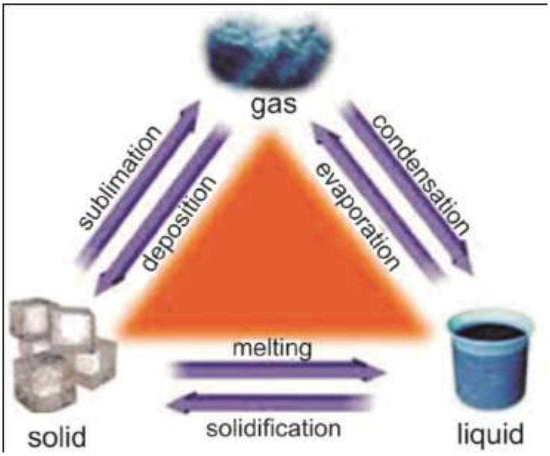


Fig. 120.4: *Change of state of matter*

There are two types of change of state of matter. They are:

i. Physical change
ii. Chemical change

Characteristics of Physical Change

1. It is a change which no new substances are formed.
2. Chemical properties of the substances do not change
3. It is easily reversible.
4. It is not accompanied by great amount energy change.

Examples of physical changes

a) Freezing water to ice
b) Boiling water to vapour
c) Melting ice to water
d) Melting of candle
e) Melting or metal
f) Esterification

Characteristics of Chemical Change

1. New substances are formed.
2. Chemical properties of the objects are changed
3. It is not easily reversible.
4. It results in change of temperature.

Examples of chemical change

a) Rusting of iron
b) Burning of wood
c) Cooking of food
d) Decomposition (when a molecule breaks down into two or more substances)
e) Neutralisation (reaction of acid and base).

TERMS TO NOTE

Sublimation

It is the process of changing directly from the solid state into the gaseous state, without the substance passing through the liquid state.

Examples of substances that sublime are:

- Camphor (Naphthalene balls)
- Solid iodine
- Ammonium chloride
- Ice
- Solid carbon dioxide

When a gas changes into solid without being liquid first, it is called ***deposition*** or ***reverse sublimation***.

Melting

Melting is the process of changing from the solid state to the liquid state (by heating).

The molecules gain energy when heated, and they vibrate more and more quickly. They break from each other and move around. When this happens the solid melts to form a liquid.

The temperature at which a solid melts is its ***melting point***.

Examples of substances that melt are ice, plastic, wax/candle, metal, etc.

Evaporation / vaporization

Evaporation is the process of changing from the liquid state into the gaseous state (by heating). The liquid molecules move around each other more quickly when heated. Some of the molecules near the surface of the liquid gain enough energy to escape from the liquid into the air or space. The temperature at which a liquid evaporates is its ***boiling point***.

Boiling is the process whereby a liquid turns into vapour at a given or fixed temperature. The fixed temperature is called the boiling point.

Table: Differences between evaporation and boiling

Evaporation	Boiling
Occurs only at the surface of the liquid	Occurs throughout the liquid
Occurs at all temperatures	Occurs only at the boiling point
Does not involve huge amount of heat energy	Involves huge amount of heat energy
Temperature is inconsistent	Temperature is consistent
An extensive property	An intensive property

Condensation

Condensation is the process of changing from the gaseous state into the liquid state (by cooling). When the gas is cooled the temperature drops slowly, the molecules lose energy to bounce off each other as they collide.

Freezing / solidification

Freezing is the process of changing from the liquid state into the solid state (by cooling). Reducing the temperature cools the liquid. This causes the molecules to slow down. The particles vibrate and stick together as they come together. The temperature at which liquids freeze is their ***freezing points***.

HEAT CAPACITY

The heat capacity of a substance is the amount of heat energy required to change its temperature by one degree Celsius (1 oC) or 1 Kelvin.

Heat capacity C $= \frac{\textbf{heat energy } Q}{\textbf{temperature change } \theta}$

Or

$\boldsymbol{C = \frac{Q}{\theta}}$

The unit of heat capacity is J $^oC^{-1}$ or J K^{-1}.

Examples

1. 3500J of heat energy applied on a piece of metal changes it temperature from 26^{o}C to 180^{o}C. Calculate the heat capacity of the metal.

Solution

$Heat\ capacity\ C = \frac{heat\ energy\ Q}{temperature\ \ change\ \theta}$

Q = 3500J

$\theta = 180 - 26 = 154^oC$

$\therefore C = \frac{3500}{154} = 22.7\ J\ ^oC^{-1}$

2. What is the amount of energy required to change the temperature of a water from the melting point to the boiling point if the heat capacity of water is 4.18 J $^oC^{-1}$

Solution

$C = \frac{Q}{\theta}$

C = 4.18 J $^oC^{-1}$

Boiling point of water = 100^{o}C

Melting point of water = 0^{o}C

$\theta = 100 - 0 = 100^oC$

$\therefore Q = C \times \theta$

$Q = 4.18 \times 100 = 418J$

Specific heat capacity (c)

This is the amount of heat required to change the temperature of one unit mass of a substance by one degree.

$\mathbf{c} = \frac{\textbf{heat capacity } (C)}{\textbf{mass } (m)}$ OR $\mathbf{c} = \frac{\boldsymbol{C}}{\boldsymbol{m}}$

but $C = \frac{Q}{\theta}$

hence $\mathbf{c} = \frac{\boldsymbol{Q}}{\boldsymbol{m\theta}}$

The unit of specific heat capacity is joule per kilogram per degree Celsius or Kelvin. (J kg^{-1} oC^{-1} or Jkg^{-1} K^{-1})

For instance, specific heat of water is 4200 J $kg^{-1}K^{-1}$. This means that it takes 4200 J to raise the temperature of 1 kg of water by 1 K.

Differences between heat capacity and specific heat capacity

Heat capacity	Specific heat capacity
Amount of heat energy needed to change temperature of a body by 1K	Amount of heat needed to change the temperature of one unit of mass of a substance by 1K
Measured in JK^{-1}	Measured in $Jkg^{-1}K^{-1}$
An extensive variable	An intensive variable

Calorimetery

Calorimetery is the measure of the amount of heat energy in a body or system.

A ***calorimeter*** is an instrument for measuring the amount of heat released or absorbed in physical and chemical processes.

Consider two bodies at different temperatures, insulated from all other bodies, yet able to transfer heat to each other. The heat gained by the initially cooler body is equal to the heat lost by the initially hotter one.

$m_1c_1\,(\Delta\theta_1) = m_2\,c_2\,(\Delta\theta_2)$

where m_1, c_1 and $\Delta\theta_1$ are the mass, specific heat and temperature of the cooler body and m_2, c_2 and $\Delta\theta_2$ are the mass, specific heat and temperature of the hotter body.

Examples

1. Explain the statement; the specific heat capacity of water is 4200 J $kg^{-1}K\text{-}^{1}$.

2. Water of mass 400.0 g at 30.0°C is mixed with 200.0g of water at 80.0°C. Neglecting heat losses, calculate the temperature of the mixture. [specific heat capacity of water = $4200Jkg^{-1}K^{-1}$]

Solution

According to the law of heat transfer,

Heat loss = heat gained

$m_1c_1\,(\theta - \Delta\theta_1) = m_2\,c_2\,(\theta - \Delta\theta_2)$

m_1 = 400g = 0.4 kg

$\Delta\theta_1$ = 30°C

m_2 = 200g = 02kg

$\Delta\theta_2$ = 80°C

c = $4200Jkg^{-1}K^{-1}$

$\therefore 0.4 \times 4200\,(30 - \theta) = 0.2 \times 4200\,(80 - \theta)$

$1680(\theta - 30) = 840(\theta - 80)$

$1680\theta - 50400 = 67200 - 840\theta$

$1680\theta + 840\theta = 67200 + 50400$

$2520\theta = 117600$

$\theta = \frac{117600}{2520} = 46.7^{o}C$

Temperature of the mixture = 46.7°C

LATENT HEAT

Latent heat is the hidden heat used to change the state of a substance without a change in its temperature.

Before water turns to vapour, its temperature must reach the boiling point of 100^oC.

At this temperature, only the temperature of the water changes but not the state. To change the state of the water, an additional heat energy which will not increase the temperature of the water, but increase the kinetic energy of the molecules.

This extra energy needed to convert the water from the liquid state into vapour is called ***latent heat of vaporization.***

In similar manner, water freezes into ice at 0^oC. To melt ice into water, an extra hidden energy is required to change the ice into water while the temperature is still 0^oC. This energy is known as ***latent heat of fusion***. The latent heat of fusion breaks the bond between the molecules of the solid ice, giving them kinetic energy to move about.

Specific latent heat

Specific latent heat is the quantity of heat required to change 1kg of a substance from one state to another.

Specific latent heat (L) $= \frac{Q}{\Delta m}$

where Q = quantity of heat energy

Δm = mass of the substance

Specific latent heat is measured in is joules per kilogram (Jkg^{-1}).

Change of heat energy = temperature change x mass x specific heat capacity

OR

$$Q = \Delta Tmc$$

Specific latent heat of fusion

This is the amount of heat needed to change 1kg of a substance from the solid state into the liquid state without a change in its temperature.

Specific latent heat of vaporization

This is the amount of heat needed to change 1kg of a substance from the liquid state into gaseous state without a change in it temperature.

Examples

1. If the specific latent heat of vaporization is 2.26 J kg^{-1}, calculate the amount of energy required to vaporize 200g of water at its normal boiling point.

   Solution

   $Q = L \text{ x } \Delta m$

   $Q = 2.26 \text{ x } 200$

   $Q = 452$ J

2. How much heat is required to melt 500 g of ice at ^{o}C. The specific latent heat of fusion of water is 336 kJ kg^{-1}.

   Solution
   $Q = L \times \Delta m$
   L = 336 kJ kg^{-1}; m = 500 g = 0.5 kg
   Q = 336 x 0.5 = 168 J

3. Calculate the amount of heat given out when an 8 kg bar of metal cools from 75^{o}C to 28^{o}C. (take specific heat capacity of metal = 380 Jkg^{-1}K^{-1})

   Solution
   Q = ΔTmc
   Temperature change ΔT = 75^{o}C –28^{o}C
   = 47 ^{o}C
   Q = 47 x 8 x 380
   Q = 142880 J = 142.88 kJ

Reason why steam from boiling water inflicts more burns than the boiling water

Boiling water has a temperature of 100^{o}C and is still liquid. When the water changes into steam, it means its latent heat of vaporization has given the molecules extra heat energy. This causes the steam to have more heat energy that the boiling water, hence its ability to inflict more burns.

How water in earthenware or clay pot cools

An earthenware pot has pores through which warm water evaporates. When warm water evaporates it carries with it latent heat thereby keeping the remaining water cool.

Reason why a fan rotating in a room may make you feel colder even though the temperature remains the same

The moving air dries the moisture or sweat on the body. As the moisture or sweat evaporates from the body, it takes latent heat of evaporation from the body, reducing the amount of heat energy in the body even though the temperature does not change.

TEST QUESTIONS

1. (a) What is heat?
   (b) Name five sources of heat.
   (c) Mention five uses of heat.

2. Describe the following modes of heat transfer:
   (a) conduction;
   (b) convection;
   (c) radiation.

3. Explain why cooking utensils have plastic or wooden handles.

4. (a) Draw and label a vacuum flask.

(b) Give four uses of vacuum flasks.

5. (a) Define the term temperature.
   (b) Write a brief note on the following temperature scales:
      (i) Celsius scale;
      (ii) Kelvin scale;
   (c) Distinguish between ice point and steam point.

6. (a) Draw and label a liquid-in-glass thermometer.
   (b) Describe how a clinical thermometer is used.

Explain the reason why mercury is often preferred to alcohol in liquid-in-glass thermometers.

7. Explain the reason why it is not advisable to sterilize alcohol thermometers in boiling water.

8. Explain the reason why clinical thermometers should be sterilized in boiling water.

9. Describe an experiment to demonstrate that heat causes expansion in metals.

10. Explain:
   (a) Why roofing sheets are corrugated;
   (b) Thick glass tumblers break when a hot liquid is poured in them but thin ones do not break.

11. (a) Describe a bimetallic strip.
   (b) Describe how the bimetallic thermostat works

12. Mention three characteristics and three examples each of physical change and chemical change.

13. Explain the following term:
   (a) sublimation;
   (b) condensation;
   (c) melting;
   (d) solidification;
   (e) evaporation.

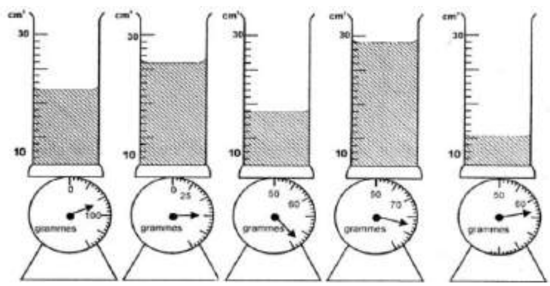
14. (a) What is latent heat?
   (b) Water of mass 1.5 kg is heated from $20^{o}C$ to $70^{o}C$. Calculate the amount of heat absorbed.[Specific heat capacity of water = 4200 J kg^{-1}]

15. Explain the following observations:
   (a) steam from boiling water inflict more burns that the boiling water itself;
   (b) rotating fan make you feel colder even though the temperature remains the same.
   (c) water in an earthenware cools.

16. In an experiment five different volumes of water at $62^{o}C$ were allowed to cool to $27^{o}C$. They were then weighed separately. Figure 4 shows the five volumes V = V_1, V_2, V_3, V_4 and V_5 of water at $27^{o}C$ and corresponding

masses M = M_1, M_2, M_3, M_4 and M_5 of water and the measuring cylinder.

Study the diagram carefully and answer the questions that follow.



A **B** **C** **D** **E**

a) Read and record the volumes V = V_1, V_2, V_3, V_4 and V_5 of water shown in the figure.

b) Read and determine the masses of the volumes of water in **A**, **B**, **C**, **D** and **E** given that the masses of the empty measuring cylinder is 50 g.

c) If the specific heat capacity of water is 4200 J kg^{-1} K^{-1}, calculate the heat losses in kilojoules in each case.

d) Tabulate your result in (a), (b) and (c).

e) Plot a graph of heat loss on the vertical axis against volume of water on the horizontal axis.

f) Draw the line of best fit for the point you have plotted.

g) Determine the heat loss for water whose volume is 30 cm^3.

17. (a) Define *specific heat capacity*.

(b) Some quantity of water of mass 3.0 kg is heated from 26°C to 76°C. If the specific heat capacity of water is 4200 J/kg °C, calculate the amount of heat used.

18. The rise in temperatures θ_1, θ_2, θ_3, θ_4 and θ_5 of a heated metal and the corresponding time t_1, t_2, t_3, t_4 and t_5 are shown below in fig. 5 (a) and Fig. 5 (b) respectively.

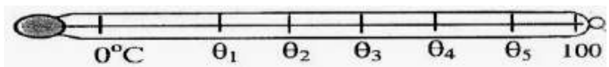


Fig. 5(a): An ungraduated thermometer drawn to scale.

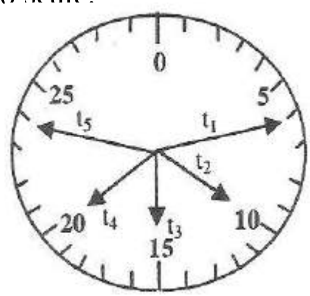


Fig. 5 (a): time in seconds

a) Determine and record the temperature θ = θ_1, θ_2, θ_3, θ_4 and θ_5 and the corresponding time t = t_1, t_2, t_3, t_4 and t_5 in the following table.

Temperature θ (°C)					
Time, t (s)					

b) i) Plot a graph with temperature, θ on the vertical axis against time, t on the horizontal axis.

ii) Calculate the slope of your graph.

iii) What is the relationship between the rise in temperature and the time?

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ELECTRONICS III

Specific Objectives

After completing this chapter, you will be able to:

- Describe the structure of a transistor and investigate its uses.

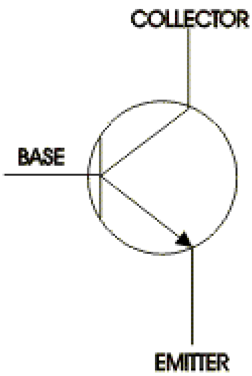


Fig. 120.5: Transistor symbol

INTRODUCTION

Transistors, as we have considered already, play a vital role in all electronic devices and gadgets. In this chapter we are going to consider transistors as electronic switches, both for high-power applications such as switched-mode power supplies and for low-power applications such as logic gates.

TRANSISTOR AS A SWITCH

The following diagram shows an NPN transistor which is often used as a type of switch. A small current or voltage at the ***base*** allows a larger voltage to flow through the other two leads (from the ***collector*** to the ***emitter***).

The circuit shown in fig. 120.6 below is based on an NPN transistor. When the switch is pressed a current passes through the resistor into the *base* of the transistor. The transistor then allows current to flow from the 9V to the 0V, and the lamp comes on.

The transistor has to receive a voltage at its '*base*' and until this happens the lamp does not light. The resistor is present to protect the transistor as it can be damaged easily by too high a voltage or current.

The transistor is acting as a switch, and this type of operation is common in digital circuits, such as computers, calculators, mobile phones, digital clocks and watches etc., where only "***on***" and "***off***" values are relevant.

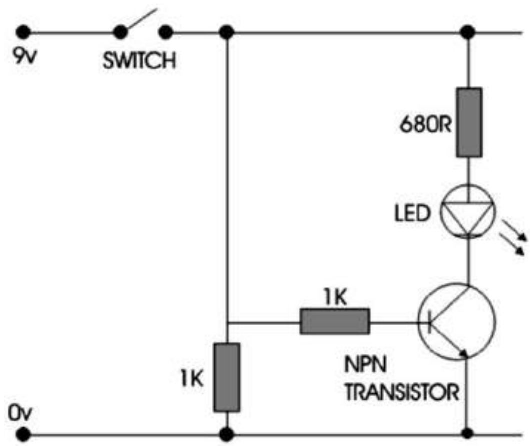


Fig. 120.6

Darlington pair

A Darlington pair is a combination of two transistors in a circuit.

In Darlington pair, two bipolar transistors (either integrated or separated devices) connected in such a way that the current amplified by the first transistor is amplified further by the second one.

This configuration gives a much higher common-emitter current gain than each transistor taken separately and, in the case of integrated devices, can take less space than two individual transistors because they can use a shared collector.

Integrated Darlington pairs come packaged singly in transistor-like packages or as an array of devices (usually eight) in an integrated circuit.

The circuit below is a 'Darlington Pair' driver. The first transistor's emitter feeds into the second transistor's base and as a result the input signal is amplified by the time it reaches the output.

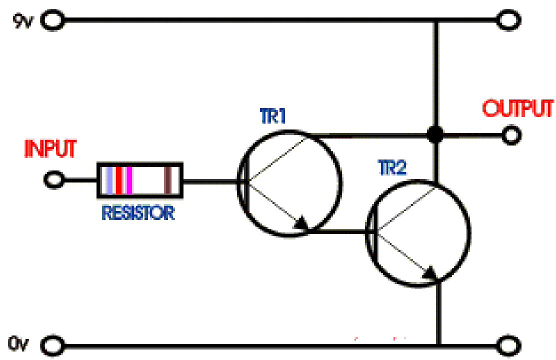


Fig. 120.7:

The important point to remember is that the Darlington Pair is made up of two transistors and when they are arranged as shown in the circuit they are used to amplify weak signals.

LOGIC GATE

A logic gate is a circuit in which transistors act as high-speed switches.

Digital and telecommunication devices use logic gates for their operations. There are two inputs with two levels of voltage known as **1** and **0**. 1 means ***on*** or ***high*** voltage level, while 0 means ***off*** or ***low*** voltage level.

Combinations of logic gates in open or closed positions can be used to represent and execute operations on data.

A series of logic gates together form a ***logic circuit.***

The output of a logic circuit can provide input to another circuit or produce the result of an operation.

The output derived is dependent on the input. In circuits, logic gates are referred to as electronic gates or simply gate. Electronic gates require a power supply. Gate **INPUTS** are driven by voltages having two nominal values, e.g. 0V and 5V representing logic 0 and logic 1 respectively. The **OUTPUT** of a gate provides two nominal values of voltage only, e.g. 0V and 5V representing logic 0 and logic 1 respectively. In general, there is only one output to a logic gate except in some special cases.

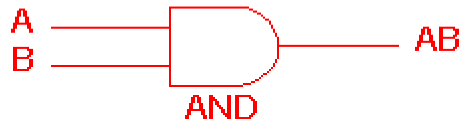
Types of logic gates

There are seven basic gates used in digital circuits. They are

- AND gate
- OR gate
- NOT gate
- NAND gate
- NOR gate
- XOR gate
- XNOR gate

AND gate

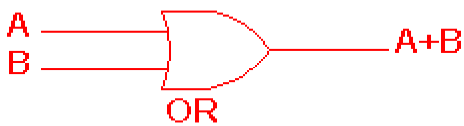
The AND gate is an electronic circuit that gives a **high** output (1) only if **all** its inputs are high. A dot (.) is used to show the AND operation i.e. A.B. The dot is sometimes omitted i.e. AB



2 Input AND gate		
A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

OR gate

The OR gate is an electronic circuit that gives a high output (1) if **one or more** of its inputs are high. A plus (+) is used to show the OR operation.

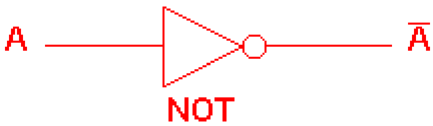


2 Input OR gate		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate

The NOT gate is an electronic circuit that produces an inverted version of the input at its output. It is also known as an ***inverter***. If the input variable is A the inverted output is known as NOT A. This is also shown as A', or Ā, as shown at the outputs.

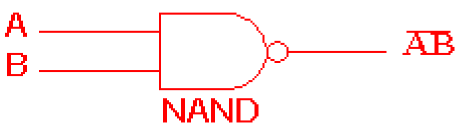
The diagrams below show two ways that the NAND logic gate can be configured to produce a NOT gate. It can also be done using NOR logic gates in the same way.



NOT gate	
A	$\overline{A}$
0	1
1	0

NAND gate

This is a NOT-AND gate which is equal to an AND gate followed by a NOT gate. The outputs of all NAND gates are high if **any** of the inputs are low. The symbol is an AND gate with a small circle on the output. The small circle represents inversion.

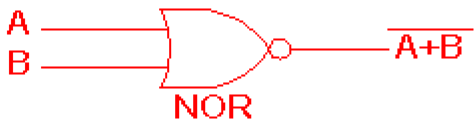


2 Input NAND gate		
A	B	$\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate

This is a NOT-OR gate which is equal to an OR gate followed by a NOT gate. The outputs of all NOR gates are low if **any** of the inputs are high.

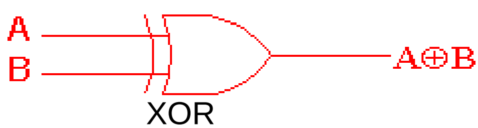
The symbol is an OR gate with a small circle on the output. The small circle represents inversion.



2 Input NOR gate		
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Exclusive OR (XOR) gate

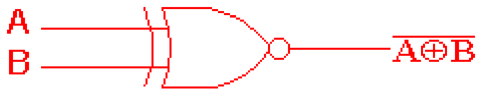
The 'Exclusive-OR' gate is a circuit which will give a high output if either, but not both, of its two inputs are high. An encircled plus sign ($\oplus$) is used to show the XOR operation.



2 Input EXOR gate		
A	B	$A\oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive NOR (XNOR) gate

The 'Exclusive-NOR' gate circuit does the opposite to the EOR gate. It will give a low output if either, but not both, of its two inputs are high. The symbol is an EXOR gate with a small circle on the output. The small circle represents inversion.



ENOR

2 Input EXNOR gate		
A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

Fig. 120.8 is the 7400 chip, containing four NAND gates. The two additional pins supply power (+5 V) and connect the ground.

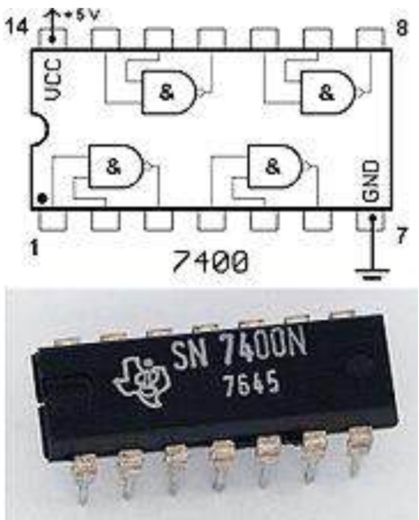


Fig. 120.8

Uses of transistors

- Transistors are used as switches
- The amplify signals

TEST QUESTIONS

1. (a) Describe the structure of a transistor.
   (b) List two uses of transistors.

2. Describe how transistors act as switches in circuits.

3. Explain the term Darlington pair.

4. (a) What is a logic gate?
   (b) Write a short not on four logic gates.

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VARIATION, INHERITANCE AND EVOLUTION

Specific Objectives

After completing this chapter, you will be able to:

- Relate the nucleus, chromosomes and genes as a sequence in inheritance.
- Explain the causes of and consequences of variation.
- Explain the concept of inheritance in organisms.
- Explain how sex is determined in humans.
- Discuss the inheritance of the various blood groups and Rh-factors and outline their importance.
- Explain how sickle cell gene is inherited and sickle cell anaemia is acquired.
- Explain evolution and discuss some theories of evolution.
- Cite some evidence of evolution.

INTRODUCTION

Living organisms are classified based mainly on their similarities. For example, organisms with a single cell are known as ***unicellular organisms***, whereas those with multiple cells ***are multi-cellular***. Also, all animals are put into kingdom animalia while plants fall into kingdom plantae. The smallest category of organisms is called ***species***, where members are so similar that they can interbreed; human, for example fall into the species ***sapiens***.

Living organisms transmit traits from their parents making them similar to the parents, in a process known as ***inheritance*** or ***hereditary***.

Nevertheless, no two creatures are exactly the same, even twins. Every organism has got its own distinctive characteristics which makes it different from other organisms of the same kind. This is called ***variation***.

The study of the hereditary and variations is called ***genetics.***

CHROMOSOMES AND GENES

A chromosome is structure found in a cell nucleus that carries the hereditary materials that determines the sex and characteristics of an organism inherited from its parents.

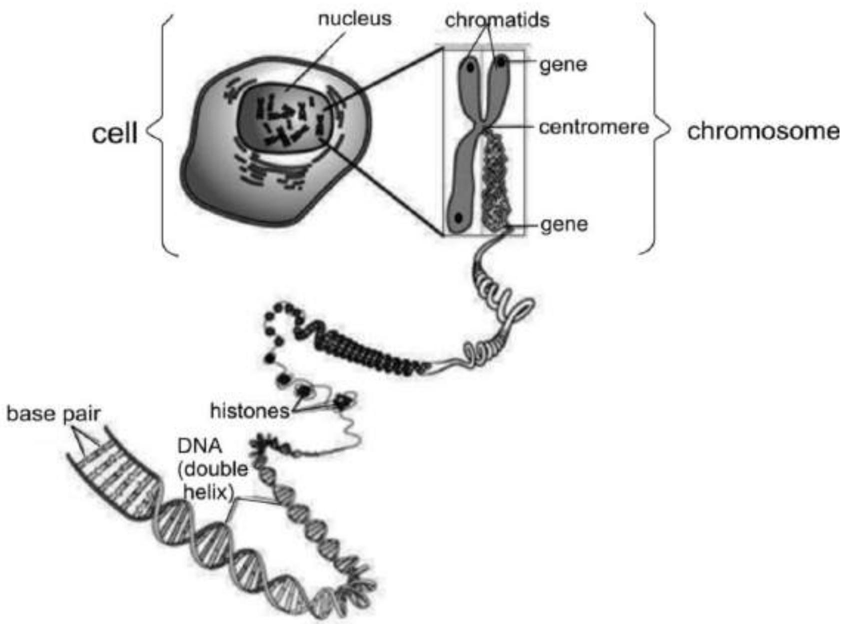


Fig. 120.9: A cell showing the location of a chromosome and gene

Chromosomes consist chiefly of proteins and deoxyribonucleic acid (DNA). Tiny chemical subunits called *nucleotide* bases form the structure of DNA. A sequence of bases along a DNA strand that codes for the production of a protein is known as a ***gene.***

Gene

A gene is a molecular unit of heredity of a living organism.

Genes occupy precise locations called ***locus*** on the chromosome. A group of different genes which occupy the same locus and perform the same functions is known as ***allele***.

Genes hold the information to build and maintain an organism's cells and pass genetic traits to offspring.

All organisms have many genes corresponding to various biological traits, some of which are immediately visible, such as eye colour or number of limbs, and some of which are not, such as blood type or increased risk for specific diseases, or the thousands of basic biochemical processes that comprise life.

Terms to know

Genotype – This is the genetic makeup of an organism.

Phenotype – This is the physical characteristics of an organism.

Heterozygote – An organism or cell with two different genes or alleles at the same location.

Homozygote – An organism or cell with two identical genes or alleles at the same place.

Dominant character – One of the different pair of genes in a heterozygote that is easily distinguished or expressed. A dominant character is represented by a capital letter. A gene which controls a dominant character is called a ***dominant gene***.

Recessive character – On of the pair of genes in a heterozygote that is not easily distinguished in the organism. It is represented by a small letter. A gene which controls a recessive character is called a ***recessive gene***.

Hybrid – A hybrid is an organism produced from parents that are not genetically identical.

F_1 generation (First filial generation) – The first filial generation offspring is produced by crossing two parental line.

F_2 generation (second filial generation) – Offspring produced by crossing members of F_1 generation.

MENDEL'S LAWS OF INHERITANCE

Mendel's laws of inheritance are statements about the ways certain characteristics are transmitted from one generation to another in an organism. The laws were derived by Gregory Mendel, an Austrian Mont.

Mendel's approach was to transfer pollen from the anther of one pea plant to the stigma of a second pea plant, resulting in a ***hybrid***.

As a simple example of this kind of experiment, suppose that one takes pollen from a pea plant with red flowers and uses it to fertilize a pea plant with white flowers.

Among the traits Mendel wanted to known were the colour of the plant's flowers, their location on the plant, the shape and colour of pea pods and seeds, and the length of the plant's stem,

In a second series of experiments, Mendel studied the changes that occurred in the second generation. That is, suppose two offspring of the red/white mating (*cross*) are themselves mated. What colour will the flowers be in this second generation of plants?

As a result of these experiments, Mendel was able to state three generalizations (or laws) about the way characteristics are transmitted from one generation to the next in pea plants.

The laws are:

1. The law of segregation of characteristics;
2. The law of independent assortment;
3. Law of dominance.

In our discussion, we are only interested in the first law.

The law of segregation of characteristics

This law states that every individual possesses a pair of alleles for any particular trait and that each parent passes a randomly selected allele of only one of these to its offspring.

This law describes what happens to the alleles that make up a gene during formation of gametes. For example, suppose that a pea plant contains a gene for flower colour in which both alleles code for red. One way to represent that condition is to write RR, which indicates that both alleles (R and R) code for the colour red. Another gene might have a different combination of alleles, as in Rr. In this case, the symbol R stands for red colour and the r for "not red" or, in this case, white.

The first law says that the alleles that make up a gene separate from each other, or segregate, during the formation of gametes. That fact can be represented by simple equations, such as:

$$\mathbf{RR \rightarrow R + R}$$

or

$$\mathbf{Rr \rightarrow R + r}$$

Application of the sequence of inheritance in cloning and stem cells

Cloning is the process of creating an exact copy of a single gene, cell, or organism.

The copies produced through cloning have identical genetic makeup and are known as ***clones***. Scientists use cloning techniques in the laboratory to create copies of cells or organisms with valuable traits. Their work aims to find practical applications for cloning that will produce advances in medicine, biological research, and industry. Gene cloning, for example, is often used to study human disease.

As part of the cloning process, scientists coax embryos to divide and grow. The embryos contain cells that can transform into any cell type that an organism needs during its development, such as blood cells, skin cells, and all the specialized cells that make up body tissues.

These cells are encourage to develop to form embryo stem cells, which have the ability to form any cell type.

Although all cells can divide to make copies of themselves, only stem cells can create new cell types.

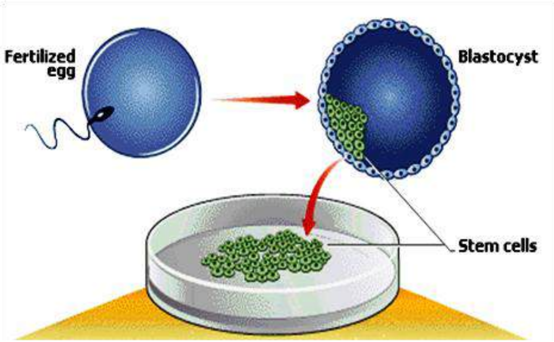


Fig. 121.0: Stem cell cloning

VARIATION

Variation is the difference in characteristics between organisms of the same species.

The differences among people most easily and commonly noticed include those in skin colour, body shape, the shape of the face, facial features and hair colour and texture.

Types of variation

1. Continuous variation
2. Discontinuous variation

Continuous variation

This is the type of variation where there are intermediate forms between the organisms under consideration.

When one tries to classify individuals according to height or weight, the decision becomes more difficult; this is because there are not merely two classes of people, tall or short, but a whole range of intermediate sizes differing from each other by measurable distances. It is known as continuous variation because the particular characteristic under consideration keeps changing from time to time.

Examples of continuous variation are weight, height, intelligence, feet length, finger length, scar, language, etc.

Continuous variation may be influenced by the environment or genetically controlled.

Discontinuous variation

Discontinuous variation is the variation where there is a clear-cut difference between organisms.

This type of variation has no intermediate forms. For example, if white and black mice are bred together, they will produce black or white offspring. There are no intermediate colours and no difficulties arise in deciding the colour categories to place the individual. It is not possible to arrange the mice in continuous range of colour from white to black. Another example is the inheritance of sex; one is either a male or a female. The feature of discontinuous variation is determined genetically; they cannot be changed during the lifetime of the individual.

Examples of discontinuous variation are eye colour, blood group, sex, Rhesus factor, sickle cell anaemia, albinism, tongue rolling, baldness, etc.

Causes of variation

The two prime factors responsible for variation are environmental factor and genetic factors.

Environmental factors

Environmental factors are external conditions that influence the life processes and cause differences between organisms of the same species. These factors cannot be passed down from parent to offspring (not inherited).

Common environmental factors of variation

1. ***Food/ diet*** – Living organisms depend on food for growth and development. Eating a well-balanced diet is a sure way to keep the growth pattern normal. However, if some nutrients are deficient in the body, it may give rise to some diseases like rickets, kwashiorkor or marasmus.

2. ***Wind velocity*** – Plants which grow in windy areas normally have stronger root systems than those of the same species growing elsewhere with low wind velocity. This is because the plants with time adopt a growth habit which may slope it away from the force of the wind.

3. ***Light intensity*** – Plants which have enough light tend to do better than those in the shade. This is because light provides plant with conditions that help to photosynthesize. The food prepared through photosynthesis is mostly used by the plants themselves.

4. ***Altitude*** – The higher one goes, the lesser the oxygen he is expected to get. This is because the amount of oxygen in the air decreases with increase in altitude. To compensate for the minimal level of oxygen, inhabitants of highlands have more red blood cells per cubic centimetre than those in lowlands.

5. ***Heat*** – Plants which grow in deserts, humid areas and areas much susceptible to drought have the ability to live longer with little water as compared to those in mashes.

6. ***Other organisms*** – The competition for nutrients in plant-world may cause some plants to grow and develop better than the others. Disease causing organisms infect other organisms with diseases which may leave permanent marks on them.

7. ***Chemicals*** – Some chemicals are toxic and when one is exposed or over-exposed to them may interfere with their structure or growth pattern.

8. ***Exposure to radiations*** – exposure to radioactivity may also result in changing the structure of an organism. X-ray, for example, may interfere with the development of a foetus if the mother is exposed to it.

Genetic factors

Genetic factors are concerned with heredity. These are the characteristics the offspring inherits from their parents. They are passed down to the offspring through the genes.

Genetic factors which cause variation

1. Mutation
2. Segregation
3. Polygenic characteristics
4. Co-dominance (incomplete or partial dominance)
5. Independent assortment
6. Albinism

MUTATION

Mutation is the spontaneous change in a gene or chromosome which may produce a change in the characteristic under its control.

Mutations occur during DNA replication when the chemical structure of genes undergoes random modifications. Once a change has occurred, the altered genes continue to replicate in their changed form unless another mutation occurs.

Types of mutation

1. Chromosome mutation
2. Gene mutation

Chromosome mutation

Chromosome mutations result in the change in the number or functions of chromosomes incorporated into cells.

A child produced as a result may have, for instance, an extra chromosome, or an extra part of a chromosome attached to the normal set in a condition known as ***Down's syndrome,*** which is caused by having 47 chromosomes instead of the normal 46 per cell.

Radiation damage can also affect an entire chromosome, disrupting the function of many genes.

Gene mutation

Genes themselves are constantly being modified through mutation, changing the structure of the DNA in an individual's cells. Mutations can occur during replication, the process in which a cell splits itself into two identical copies known as ***daughter cells***. Normally each daughter cell receives an exact copy of the DNA from the parent cell. Occasionally, however, errors occur, resulting in a change in the gene. Such a change may affect the protein that the gene produces and, ultimately, change an individual's traits.

Causes of mutation

While some mutations occur spontaneously, others are caused by factors in the environment, known as mutagens.

Examples of mutagens that affect human DNA include:

1. Ultraviolet light
2. Chemicals, such as asbestos, cigarette smoke and nitrous acid
3. High-energy radiation, e.g. X rays
4. Error during DNA replication

Consequences of variation

Variation promotes natural selection.

Natural selection is the process by which organisms better adapted to their environment tend to survive and reproduce more offspring.

The characters that inhibit reproductive success decrease in frequency from generation to generation. The resulting increase in the proportion of reproductively successful individuals usually enhances the adaptation of the population to its environment.

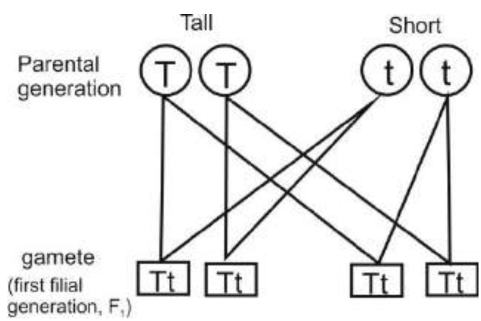
Natural selection thus tends to promote adaptation by maintaining favourable adaptations in a constant environment (***stabilizing selection***) or improving adaptation in a direction appropriate to environmental changes (***directional selection***).

INHERITANCE IN PLANTS AND ANIMALS

Genetic inheritance refers to the passing or handing-on of characteristics of a parent to its offspring.

Two parents, (male and female parent), are involved in sexual reproduction. The offspring produced possesses characteristics of both parents. This is because each parent passes half the number of its chromosomes to the offspring. In humans, for example, the offspring inherits 23 chromosomes from each parent, adding up to 46 chromosome (haploid).

If for example, a tall parent (TT) and a short parent (tt) are crossed, the offspring will inherit characteristics from both parents as shown below.



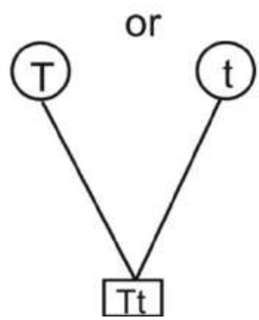


Fig. 121.1: Inheritance in living organisms

The first filial generation (F_1) is the offspring produced after crossing the parental generation.

When the offspring of F_1 are inter-crossed they form the second filial generation (F_2).

F_2 is made up of homozygous (two similar allele) tall (TT), homozygous short (tt) and heterozygous black (two dissimilar allele) (Tt).

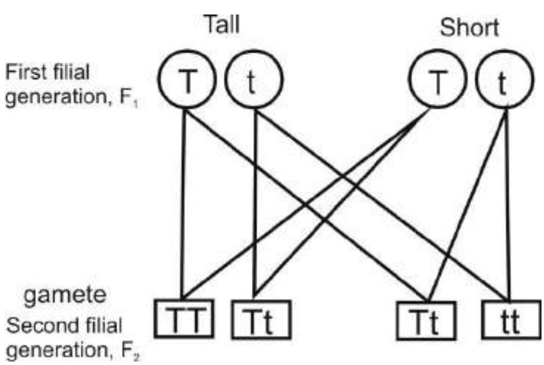


Fig. 121.2

The crossing is also similar in plants.

Heritable and non-heritable characters in humans

Heritable characters

Heritable or sex-linked characters are traits or characters which are easily passed-on from parents to offspring.

Examples of heritable characters

1. Blood group
2. Rhesus factor
3. Haemophilia
4. Baldness
5. Tongue rolling
6. Eye colour
7. Colour blindness

Non-heritable characters

Non-heritable characters are characters which may be possessed by both or one of the parents but may not be passed-on to the offspring.

Such characters are normally influenced by the environment.

Examples of non-heritable characters

1. Weight
2. Height
3. Intelligence
4. Scar
5. Skill
6. Skin colour
7. Temper

SEX DETERMINATION

Most chromosome pairs consist of identical or homologous partners. In many species, including humans, there is one pair of chromosomes in which the partners noticeably differ from each other.

These are called the sex chromosomes because they determine the differences between males and females. Genes located on the sex chromosomes display different patterns of inheritance than genes located on other chromosomes.

Sex determination in humans

In human females, the sex chromosomes consist of two **X** chromosomes, while males have an **X** chromosome and a shorter **Y** chromosome with many fewer genes.

In males the **X** chromosome contains many genes that have no corresponding gene on the **Y** chromosome. A male's **X** chromosome may contain a recessive allele associated with a genetic disorder, such as haemophilia or Duchene muscular dystrophy. In this case, males do not have a normal second copy of the gene on the **Y** chromosome to mask the effects of the recessive gene, and disease typically results.

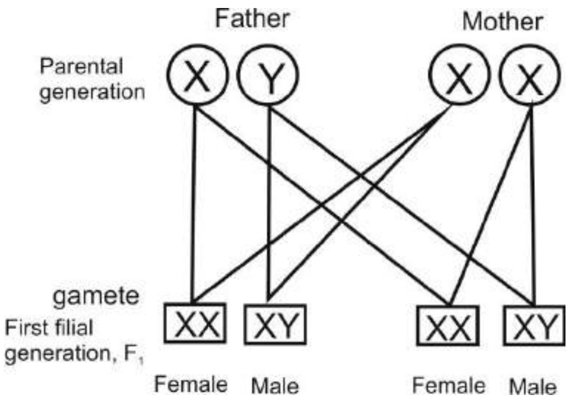


Fig. 121.3: Sex determination

BLOOD GROUPS

Blood group or blood type is a classification of blood based on the presence or absence of certain protein molecules called ***antigens*** and ***antibodies***.

The antigens are located on the surface of the red blood cells as well as other cells and tissues, while the antibodies are in the blood plasma. These antigens may be proteins, carbohydrates, glycoproteins, or glycolipids, depending on the blood group system.

Blood types are inherited and represent contributions from both parents. Individuals have different types and combinations of these molecules. The blood group you belong to depends on what you have inherited from your parents.

There are more than 20 genetically determined blood group systems known today, but the ***ABO*** and ***Rh systems*** are the most important ones used for blood transfusions.

Not all blood groups are compatible with each other. Mixing incompatible blood groups leads to blood ***clumping*** or ***agglutination***, which is dangerous for individuals.

The ABO blood grouping system

According to the ABO system, there are four groups of blood. They are **A**, **B**, **AB** and **O**.

Blood group **A** contains red blood cells that contain antigen **A** on their surface. This group of blood also contains an antibody directed against substance **B**, found on the red blood cells of persons with blood group **B**.

Group **B** blood contains the reverse combination. Serum of blood group **AB** contains neither antibody, but red cells in this type of blood contain both **A** and **B** substances.

In type **O** blood, neither substance is present on the red cells, but the individual is capable of forming antibodies directed against red cells containing substance **A** or **B**.

If blood type **A** is transfused into a person with **B** type blood, anti-**A** antibodies in the recipient will destroy the transfused **A** red blood cells.

Because **O** type blood has neither substance on its red cells, it can be given successfully to almost any person. Persons with blood type **AB** have no antibodies and can receive any of the four types of blood; thus blood group **O** is called ***universal donor*** whereas group **AB** is called ***universal recipient***.

Characteristics of the blood groups

	Group A	Group B	Group AB	Group O
Red blood cell type	A	B	AB	O
Antibodies in Plasma	Anti-B	Anti-A	None	Anti-A and Anti-B
Antigens in Red Blood Cell	A antigen	B antigen	A and B antigens	None

Blood group and pregnancy

Many pregnant women carry a foetus with a blood type different from their own, and the mother can form antibodies against foetal ***RBCs***.

Sometimes these maternal antibodies are **IgG**, a small immunoglobulin, which can cross the placenta and cause haemolysis of foetal RBCs, which in turn can lead to haemolytic disease of the newborn, an illness of low foetal blood counts that ranges from mild to severe.

Fig. 121.5 shows the blood groups of offspring of parents with blood groups AB and OO.
Because blood groups A and B are allele, if a person is AO or BO, the O has no effect hence the person is A or B.

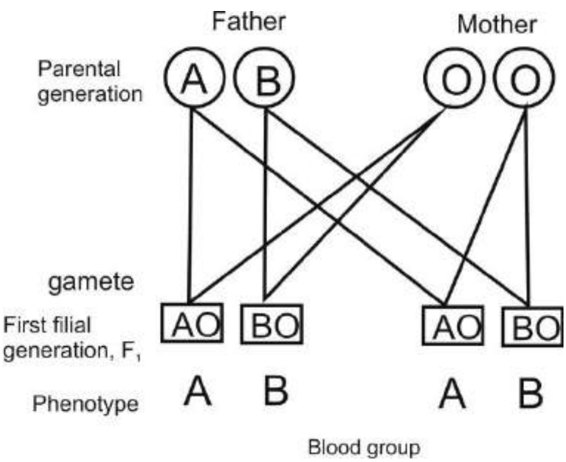


Fig. 121.5

RHESUS FACTOR

The Rhesus factor, also known as the Rh factor, is an antigen that exists on the surface of red blood cells in most people.

People who have the Rhesus factor are considered to have a "positive" (+) blood type, such as A+ or B+. Those who don't are considered to have a "negative" (-) blood type, such as "O-" or "AB-".

The Rh blood grouping system involves more than 50 antigens that are found on the surface of red blood cells. These antigens are proteins that, when introduced into a body that does not have the same type, can cause the person's immune system to respond by producing antibodies that attack the proteins. The Rh factor, Rh+ and Rh-, usually refers specifically to the presence or absence of one of these proteins — the **D** antigen. The **D** antigen tends to cause an especially strong immune response in people who do not have it.
There are two alleles, or genetic variants, of this antigen: **D** and **d**. A person who is Rh- has two recessive variants, **dd**. Anyone who has at least one **D** — **DD** or **Dd** — is Rh+. As with most genetic traits, one allele is inherited from each parent.

Rh Type and Pregnancy

A person's Rh type is generally most relevant with respect to pregnancies. During pregnancy, an Rh+ foetus developing in the womb of an Rh- woman runs the risk of developing Rhesus disease, also called Rh disease or haemolytic disease of the newborn. Only Rh- women risk having children with this disease; an Rh+ woman can carry an Rh- child without developing this condition.

For an Rh- woman to have an Rh+ child, the father must have been Rh+. An Rh+ man has at least a 50% chance of passing on the Rhesus factor to the child; a **Dd** father could pass either the **D** or **d** to his child. If the father is **DD**, there is a 100% chance that the child will be Rh+.

If the mother is Rh- and the child is Rh+, and if the child's blood enters the woman's bloodstream during pregnancy, labour, or delivery, the woman's immune system might respond by producing antibodies to fight off the child's antigens, which are foreign to the woman's system.

This may cause some health problems for the body including jaundice, anaemia, and brain or heart damage. In severe cases, Rh disease can be fatal to the infant.

SICKLE-CELL ANAEMIA

Sickle cell anaemia is a disorder of the blood caused by an inherited abnormal haemoglobin (an oxygen-carrying protein within the red blood cells).

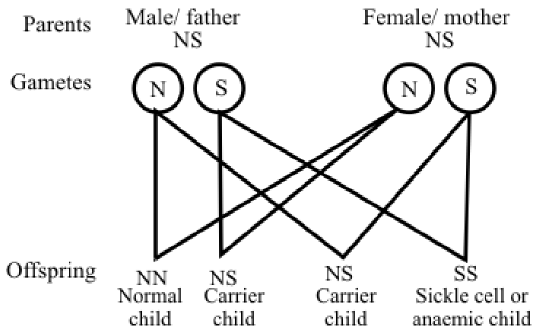
The abnormal haemoglobin causes distorted (sickled) red blood cells. The sickled red blood cells are fragile and prone to rupture. When the number of red blood cells decreases from rupture (haemolysis), anaemia is the result. This condition is referred to as ***sickle-cell anaemia***. The irregular sickled cells can also block blood vessels causing tissue and organ damage and pain.

How sickle cell anaemia is inherited

Sickle cell anaemia is inherited as an ***autosomal recessive condition*** (meaning that the gene is not linked to a sex chromosome) whereas sickle cell trait is inherited as an ***autosomal dominant trait***. This means that the gene can be passed on from a parent carrying it to male and female children. In order for sickle cell anaemia to occur, a sickle cell gene must be inherited from both the mother and the father, so that the child has two sickle cell genes.

The inheritance of just one sickle gene is called sickle cell trait or the ***carrier state***. Sickle cell trait does not cause sickle cell anaemia. Persons with sickle cell trait usually do not have many symptoms of sickle cell anaemia.

When two carriers of sickle cell trait mate, their offspring have a one in four chance of having sickle cell anaemia.



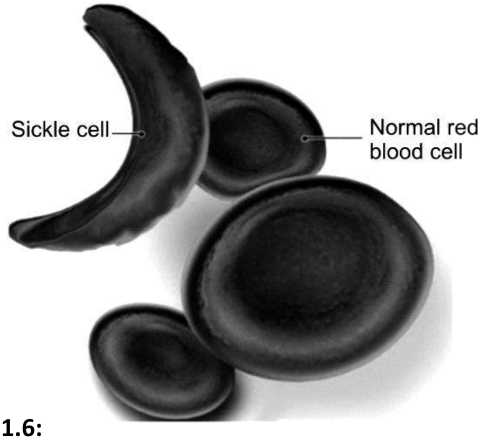


Fig. 121.6:

EVOLUTION

Evolution is a complex process by which the characteristics of living organisms change over many generations as traits are passed from one generation to the next.

The theory of evolution seeks to understand the biological forces that caused ancient organisms to develop into the tremendous and ever-changing variety of life seen on Earth today. It addresses how, over the course of time, various plant and animal species branch off to become entirely new species, and how different species are related through complicated family trees that span millions of years.

Development of theory of evolution

In 500s BC, the Greek philosopher, ***Anaximander,*** believed that the Earth first existed in a liquid state. He also believed that humans evolved from fishlike aquatic beings that left the water once they had developed sufficiently to survive on land.

Another Greek scientist called ***Empedocles*** speculated in the 400s BC that plant life arose first on Earth, followed by animals. Empedocles proposed that humans and animals arose not as complete individuals but as various body parts that joined together randomly to form strange, fantastic creatures. Some of these creatures, being unable to reproduce, became extinct, while others thrived.

Aristotle's theory

Aristotle, also a Greek philosopher and scientist who lived in 300 BC, used a chart he called "***ladder of nature***", a progression of life forms from lower to higher rank with humans occupying the top rank. Aristotle acknowledged that some organisms are incapable of meeting the challenges of nature and so cease to exist. As he saw it, successful creatures possessed a gift, or ***perfecting principle***, that enabled them to rise to meet the demands of their world.
Creatures without the perfecting principle died out. In Aristotle's view it was this principle (not evolution) that accounted for the progression of forms in nature.

Lamarck's theory

In 1809, a French biologist, Jean Baptiste Lamarck, proposed an evolutionary theory which says that plants and animals respond to their environments by becoming better adapted to them. This was based on the observation that parts of the body frequently used become more developed and parts less used degenerate. This trait is passed on to offspring. The classic example used to explain this concept is the elongated neck of the giraffe.

According to Lamarck's theory, a given giraffe could, over a lifetime of straining to reach high branches, develop an elongated neck.

A major downfall of his theory was that he could not explain how this might happen, though he discussed a "natural tendency toward perfection."

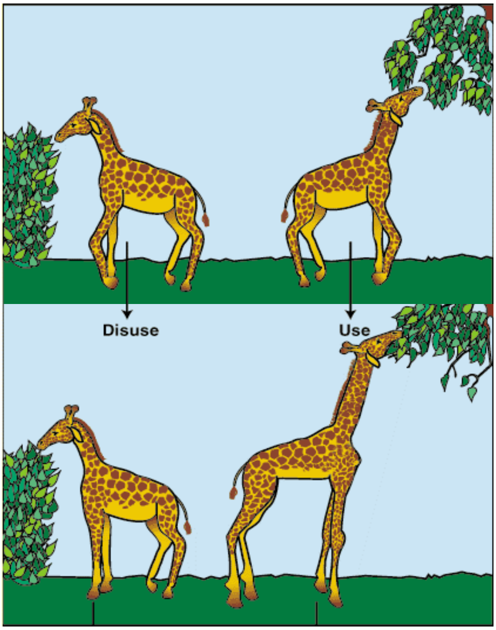


Fig. 121.7: Lamarck's theory of use and disuse

Another example Lamarck used was the toes of water birds. He proposed that from years of straining their toes to swim through water, these birds gained elongated, webbed toes to better their swimming.

These two examples demonstrate how use could change a trait. By the same token, Lamarck believed that disuse would cause a trait to become reduced. The wings of penguins, for example, would be smaller than those of other birds because penguins do not use them to fly.

Darwinism

In 1859, the man whose name is almost synonymous to evolution, a British naturalist called Charles Darwin, proposed his theory of evolution.

Darwin's theory asserts that all life descended from a common ancestor. His general theory presumes the development of life from non-life and stresses a purely natural "descent with modification." Basically, Darwin theorizes that complex creatures evolved from more simplistic ancestors, and that beneficial genetic mutations are preserved because they aid survival in a process he called ***natural selection***. That is, complex creatures naturally evolved over time from more simplistic ancestors. Beneficial mutations accumulated and eventually resulted in an entirely different creature. He theorized that evolution is a slow, gradual process.

Darwin's main ideas on evolution

Darwin's based his evolutionary theory on the following observations:

Variation

In every species there is variation. This variability is apparent even within related organisms. Even siblings will vary in colour, height, weight, number of offspring and other characteristics.

Organisms in some geographic areas may be related to those in other parts of the world. However, due to very specific conditions in their surroundings, these species evolve very distinct characteristics.

Competition

Most species produce more offspring each year than the environment can support. This high rate of growth results in competition among the local species for the limited natural resources available. The consequence of the struggle for resources is an increasing mortality rate within a species.

Inheritance

Each species has traits that are strongly influenced by inheritance. Other characteristics are affected more strongly by environmental factors and are not considered inherited.

Inherited traits are passed directly from parent to offspring in a consistent manner. Heritable traits that are related to a particular disease may be manipulated through gene therapy.

Those traits related to environmental factors often respond best to behavioural changes, such as dietary restrictions, increasing exercise and quitting smoking. Isolating traits based on their heritability has enormous implications for health care in humans.

Differential survival

Some individuals will survive the struggle for resources. These individuals will reproduce adding their genes to the succeeding generations. The traits that helped these organisms to survive will be passed on to their offspring. This process is known as natural selection.

The conditions in the environment result in the survival of individuals with specific traits which are passed through heredity to the next generation. Today we refer to this process as "survival of the fittest".

EVIDENCE OF EVOLUTION

Though evolution occurs over a long time frame – over hundreds, thousands or millions of time – evidence can be observed in numerous ways, including

1. fossil record
2. distribution of species (both geographically and through time)
3. comparative anatomy
4. embryology.

Fossil records

Remains of animals and plants found in sedimentary rock deposits give us an indisputable record of past changes through vast periods of time. This evidence attests to the fact that there has been a tremendous variety of living things.

Some extinct species had traits that were transitional between major groups of organisms. Their existence confirms that species are not fixed but can evolve into other species over time.

archaeopteryx fossil *archaeopteryx recreation*

Fig. 121.8:

Geographical distribution

It is evident that some organisms mutate in order to fit into their environment. They develop different structures which enable them to be stronger, faster, etc.

An example is the speeches of Galápagos' finch found on the Galápagos Archipelago. These birds, though from a common ancestor, have evolved by an adaptive radiation that diversified their beak shapes to adapt them to different food source.

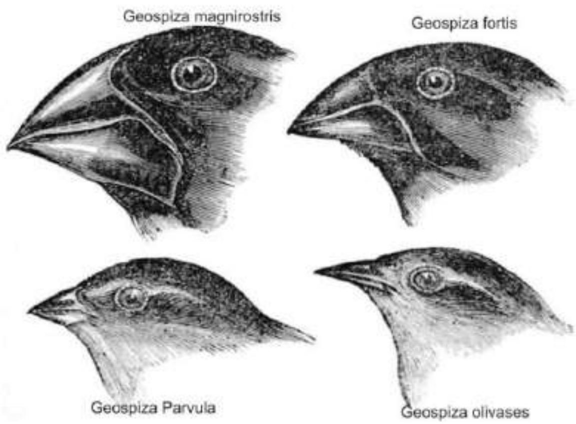


Fig. 121.9: Species of Galapagos finch

Comparative Embryology

It has been noted that in the earliest stages of development, the embryos of organisms that share a recent common ancestor are very similar in appearance. As the embryos develop, they grow less similar.

For example, the embryos of dogs and cats, both members of the mammal order Carnivora, are more similar in the early stages of development than just before birth. The same is true of human and ape embryos.

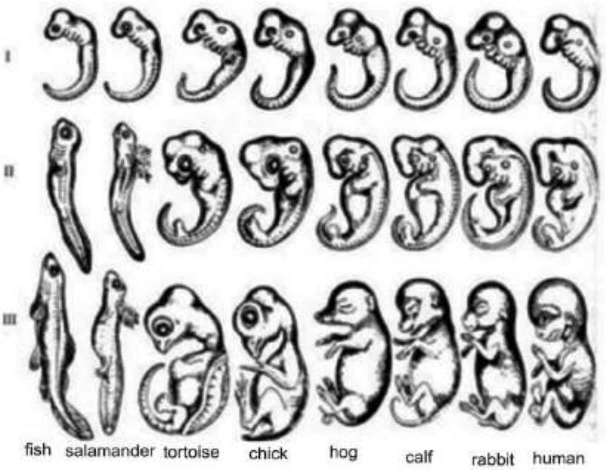


Fig. 122.0: Comparative embryology

Molecular similarity

Despite the enormous variety of form and function seen in living things, the underlying genetic code, the molecular building material of life, displays a striking uniformity.

Almost all living organisms have DNA, and in each case it consists of different pairings of the same building blocks: four nucleotide bases called ***adenine***, ***thymine***, ***guanine***, and ***cytosine***.

Using different combinations of these bases, DNA directs the assembly of amino acids into functional proteins. The same uniform code operates within all living things.

Anatomical similarity (Comparative anatomy)

Study of the internal and external features of different living organisms also provides a wealth of information about evolution. The arm of a human, the flipper of a whale, the foreleg of a horse, and the wing of a bird have different forms and are adapted to different functions. Yet they correspond in some way, and this correspondence extends to many details. In the case of the above structures, for example, each appendage shows a similar bone structure.

The study of comparative anatomy has revealed many instances of correspondence within various groups of organisms and these bodily structures are said to be ***homologous*** (having the same origin)**.** These homologous structures are said to originate from a common ancestor. The differences arose as each group diverged from the common ancestor and adapted to different ways of life.

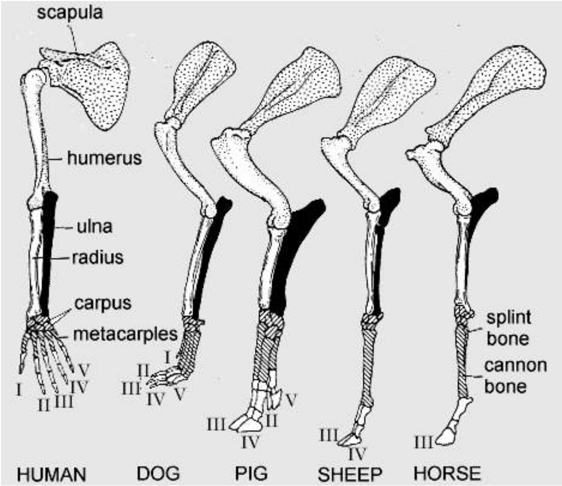


Fig. 122.1: Comparative anatomy of the limbs selected animals

TEST QUESTIONS

3. Briefly explain the following terms
   (a) chromosome;
   (b) gene;
   (c) phenotype;
   (d) heterozygote;
   (e) homozygote;
   (f) dominant character;
   (g) recessive character.

4. Write briefly on Mendel's first law of inheritance.
3. The offspring of a tall man and a short woman were all found to be tall. With the aid of an appropriate crosses, illustrate the above observation.

4. (a) What is variation?
   (b) Distinguish between continuous and discontinuous variations.

4. Describe four environmental factors that cause variation.

5. (a) What are genetic factors of evolution?
   (b) Mention four genetic factors of evolution.

6. (a) Define the term mutation.
   (b) Briefly explain chromosome mutation and gene mutation.

7. Explain the consequences of evolution.
8. Distinguish between heritable and non-heritable characteristics in humans.

9. Give four examples each of heritable and non-heritable characteristics in humans.

10. With the aid of a diagram, describe how sex is determined in humans.

11. Describe the ABO blood group system.

12. (a) What is *rhesus factor*?
    (b) Explain why it is not advisable for a rhesus negative woman to marry a rhesus man.

13. Explain how sickle-cell anaemia is inherited.

14. Why is the occurrence of sex-linked characters a more common feature in male than in female?

15. A woman with one sickle cell gene was married to a man of one sickle-cell gene. Using the appropriate crosses, explain the genetic constitution of their children.

16. (a) Define the term evolution,
    (b) Compare the evolutionary theories of Lamarch and Darwin.

17. State the explain four evidence of evolution.

18. The offspring of a black rabbit and white rabbit were all found to be black. With the aid of an appropriate crosses, illustrate the observation.

19. Copy and complete the table below on A, B, O blood system.

Blood group	*Antigen present on cells*	*Antibodies present in plasma*
A		
B		
OB		
O		

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EXAMINATION-TYPE QUESTIONS

(OBJECTIVE TYPE)

1. Which of the following instruments is used to measure the radius of an egg?
A. Spherometer
B. Micrometer screw gauge
C. Vernier calliper
D. Opisometer

2. Which of the following gases is not a constituent of air?
A. Oxygen
B. Water vapour
C. Carbon (IV) oxide
D. Sulphur (IV) oxide

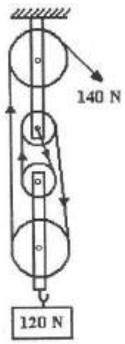
3. To which of the following kingdoms do protozoa belong?
A. Prokaryotae
B. Fungi
C. Plantae
D. Protoctista

4. Bodies are weightless in space because
A. the acceleration due to gravity of the earth balances that of the sun.
B. of the absence of air.
C. of the absence of the force of gravity.
D. the weight of the body exactly balances the force of gravity.

5. It is advisable to add sodium trioxocarbonate (IV) to water when washing with soap. This is because of the
A. presence of bacteria in water.
B. presence of dissolved magnesium salt.
C. acidity of water.
D. dirt in the water.

6. The organisms belonging to the same species
A. always live in the same habitat.
B. can produce fertile offspring.
C. cannot communicate with each other.
D. have the same lifespan.

7. What is the velocity ration in the simple machine shown below?
A. 4 B. 3 C. 2 D. 1



8. Which of the following statements is not correct about acids?

A. Acids contain replaceable hydrogen.
B. Acids turn blue litmus red.
C. Substances which contain hydrogen are acids.
D. Acids are neutralised by bases.

9. Which of the following compounds is used in the manufacture of soap?

A. $Ca(OH)_2$ B. Na_2SO_4
C. NaOH D. $Mg(OH)_4$

10. The following are all characteristics of living things **except**

A. excretion. B. nutrition.
C. respiration. D. secretion.

11. Which of the following is used to measure temperature of 100 ^oC?

A. Alcohol thermometer

B. Pyrometer
C. Thermosistor
D. Mercury thermometer

12. Chromosomes are found in the

A. cytoplasm. B. nucleus.
C. ribosomes. D. genes.

13. Atoms which defer in mass number but have the same proton number are referred to as

A. Allotropes. B. isomers.
C. Isotopes. D. nuclides.

14. A cell wall consists **mainly** of

A. lignin. B. cutin.
C. wax. D. cellulose.

15. A steel ball of mass 20.0 kg cools from 120 ^oC to 80 ^oC. If the specific heat capacity of steel is 435 J/kg, calculate the quantity of the heat given out.

A. 3.48×10^5 J B. 4.32×10^5J
C. 8.43×10^5 J D. 4.83×10^5J

16. The muscle is an example of

A. a system. B. an organ.
C. a tissue. D. an organelle.

17. Which of the following statements is/are correct? In the thermos flask

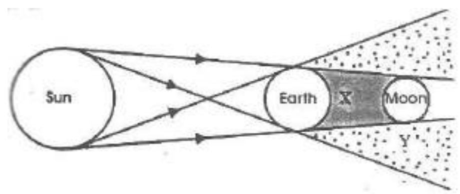
I. A vacuum is created between the two walls.
II. The outside faces of the glasses are silvered.
III. The heat loss by conduction is reduced by the use of a cork stopper.

A. I only B. I, II and III
C. III only D. I and II only

18. In the formation of an ionic compound, atoms of one of the combining elements must
A. convert neutrons to electrons.
B. lose protons.
C. gain protons.
D. gain electrons.

19. The gas given of during photosynthesis can be identified by using
A. potassium hydroxide.
B. a glowing splint.
C. lime water.
D. litmus paper.

20. Which of the following statements about the diagram below is correct?



A. Portion X is called penumbra.
B. Illustrates an eclipse of the sun.
C. Portion Y is called umbra.
D. Illustrates an eclipse of the moon.

21. Which of the following statements is/are correct? The particles associated with conduction of electricity are
I. Protons II. electrons III. ions

A. I only B. II only
C. I and II only D. II and III only

22. The new cell formed after fertilization is known as
A. an embryo. B. a foetus.
C. a zygote. D. a seed.

23. Glycerol is used for the manufacture of
A. dyes. B. stain removers.
C. detergents. D. dynamites.

24. Which of the following statements are corrects? In a concave mirror, when the object is placed at the centre of curvature,
I. The image formed is real.
II. The magnification is one.
III. The image formed is inverted.

A. I and II only B. I and III only
C. II and III only D. I, II and III

25. The rate of chemical reaction is increased when the
A. then concentrations of reactants are decreased.
B. concentrations of reactants are increased.
C. reactants are in large lumps.
D. concentrations of products are increased.

26. Which of the following substances is not carried in the plasma?

A. Urea B. Carbon (IV) oxide

C. Oxygen D. Glucose

27. An object is placed 25.0 cm from a converging lens of focal length 20.0 cm. calculate the magnification of the image formed.

A. 4 B. 3 C. 2 D. 1

28. Alkanols contain

A. carbon, oxygen and hydrogen only.
B. carbon, nitrogen and hydrogen only.
C. carbon, oxygen and nitrogen only.
D. carbon and oxygen only.

29. Gestation period is the time between

A. fertilization and birth.
B. fertilization and implantation.
C. ovulation and fertilization.
D. implantation and birth.

30. For any system to produce sound, it must

A. possess potential energy.
B. be vibrating.
C. both kinetic and potential energy.
D. Possess kinetic energy.

31. The tricuspid valve is formed between the

A. right ventricle and pulmonary artery.
B. left auricle and let ventricle.
C. right auricle and right ventricle
D. left ventricle and aorta.

32. Estrification is the reaction between

A. acids and bases.
B. alkanoic acids and ammonia.
C. organic bases and alkanools
D. alkanoic acids and alkanols.

33. Which of the following is an example of ferromagnetic material?

A. Silver B cobalt

C. Plastic D. Mercury

34. Which of the following is not found in the lymph?

A. Red blood cells
B. Antibodies
C. Glucose
D. White blood cells

35. When a plastic comb is used to comb a dry hair and it is brought quickly near small pieces of paper,

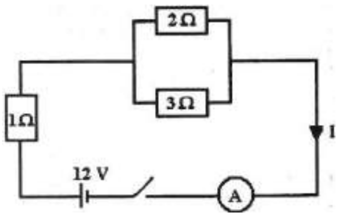
A. the papers get heated.
B. there is repulsion between paper and comb.
C. there is attraction between the comb and the pieces of paper.
D. the charges on the comb are neutralised.

36. Which of the following gases is used in welding?

A. Ethane B. Ethane

C. Methane D. Ethyne

37. Pollutants are substances
A. from industries only.
B. in the atmosphere.
C. which breed harmful organisms.
D. which make the environment harmful to life.



38. Calculate the value of the current I in the circuit above.
A. 3.5A B. 4.5A C. 5.5A D 6.5A

39. Muscles are attached to bones by
A. Tendons. B. ligaments.
C. Cartilage. D. tissues.

40. A baby's sex is determined by
A. mother at conception.
B. father at conception.
C. food eaten by mother.
D. drugs taken by mother.

41. Two resistors of values 2 Ω and 3 Ω are connected in parallel. Calculate the effective resistance.
A. 6 Ω B. 5 Ω C. $\frac{5}{6}$ Ω D. $\frac{6}{5}$ Ω

42. Which of the following not a vertebrae?
A. Axis B Atlas
C. Scapula D. Coccyx

43. In an electrical circuit, the function of the fuse is to
A. regulate the voltage in the circuit.
B. prevent lightening from destroying the appliances.
C. regulate the flow of current in the circuit.
D. protect the circuit against excess current.

44. Viruses can only reproduce when they are present in a
A. culture solution.
B. living organism.
C. dead matter
D. sugary medium.

45. All the following minimize soil erosion **except**
A. planting of trees.
B. creating lawns.
C. weeding premises with hoe.
D. creating gutters for running rain water.

46. In the wiring of houses, appliances are connected in parallel to enable
A. each appliance t have the same voltage.
B. power to be saved.
C. each appliance to have the same current.
D. the power to be stepped up.

47. Night blindness is caused by lack of
A. vitamin C. B. vitamin A.
C. vitamin D. D. vitamin E.

48. Which of the following methods would improve the quality of life in human beings?
A. Deforestation
B. Taking of unprescribed drugs
C. Uncontrolled birth
D. Family planning

49. A pressing ion is rated 100 W, 249 V. this means that the
I. Iron cannot work on 220 V.
II. Iron consumes 100 A and hour.
III. Maximum current the iron can take is 0.42 A.
IV. Iron can use a 5 A fuse.

A. I and II only B. II and III only
C. III and IV only C. I and III only

50. Which of the following is extensively used in medicine?
A. α-particle. B. γ-rays.
C. X-ray D. β-rays

51. All of the following processes occur during respiration in a human body except
A. the contraction of diaphragm muscles.
B. the contraction of intercostals muscles.
C. the increase in the volume of the chest cavity.
D. the increase in the volume of lungs.

52. The function of a step-down transformer is to
A. protect the power supply to an area.
B. reduce the voltage to the desired value.
C. change the a.c to d.c.
D. decrease the resistance in the cable.

53. Soap may itch the skin when it contain excess
A. glycerol B. oil
C. alkali D. acid

54. Movement of water into plant cell causes the cell to
A. become turgid.
B. become flaccid.
C. burst its cell wall.
D. become plasmolyzed.

55. Liquid petroleum gas, which is used as a domestic fuel, belong to the group of
A. alkenes. B. alkanes.
C. alkanols. D. alkanoates.

56. Which of the following statements is not true of X-rays?
A. They cause fluorescence.

B. They are deflected by electric field.
C. They can produce ionization.
D. They can be diffracted.

57. A person who needs glucose can take
A. ethanol. B. glycerol
C. starch. D. citric acid.

58. Which of the following is not involved in a reflex action?
A. Sensory neurone
B. Spinal cord
C. Cerebellum
D. Motor neurone

59. Which of the following is used by satellite in space?
A. Dry cells
B. Alkaline batteries
C. Solar cells
D. Lead acid batteries

60. The mode of nutrition involving the use of dead organic matter is referred to as
A. saprophytic. B. symbiotic.
C. parasitic. D. halophytic.

61. A virus is basically composed of
A. protein, DNA and RNA.
B. protein and either RNA or DNA.
C. protein, DNA and nucleic acid.
D. protein, RNA and nucleic acid.

62. The following are all abiotic factors **except**
A. wind. B. light.
C. bacteria. D. rainfall.

63. A plastic windscreen for a car is made of
A. nylon. B. Teflon.
C. Perspex. D. polythene.

64. All of the following substances can pollute the air **except**
A. sulphur dioxide. B. fertilizer.
C. exhaust fumes. D. dust.

65. Which of the following pairs of diseases is caused by insanitary surrounding?
A. Dysentery and measles
B. Bilharzia and dysentery
C. Cholera and chicken pox
D. Malaria and cholera

66. How many moles are there in 4.2 g of $NAHCO_3$?
(Na = 23, O = 16, C = 12, H = 1)
A. 0.05 mol B. 0.5 mol
C. 0.005 mol D. 5.0 mol

67. Which of the following statements about viruses is not true? They are
A. akaryotic.
B. non cellular
C. prokaryotic.
D. able to form crystals

68. Absolute ethanol can be prepared from 95% of ethanol by

A. fractional distillation of 95 % ethanol

B. crystallisation of 95 % ethanol

C. extraction of 95 % ethanol with benzene

D. distilling a calculated amount of mixture of benzene and 95 % ethanol

69. Which of the following sets of elements can be extracted from their ores by reduction with carbon?

A. Ca, Al and Pb B. Fe, Pb and Zn

C. La, Mg and Na D. Mg, Cu and Fe

70. During bread making, yeast is added as an ingredient because it

A. gives the bread a peculiar flavour.

B. helps to preserve the bread.

C. reduces the sugar content to an acceptable level.

D. produces CO_2, which causes the dough to rise.

71. In the preparation of local soap, the ash of plantain peels or cocoa pods is added to palm oil to

A. thicken the oil.

B. give the colour of the soap.

C. provide alkali to saponify the oil.

D. give the soap a present smell.

72. Which of the following statements is not a reason why family planning is encouraged?

A. It controls the rate of childbirth.

B. It discourages couples from having children.

C. It helps the mother to restore her health before another child is born.

D. It helps couples to have the number of children they can look after.

73. The left ventricle of the heart has the thickest wall because

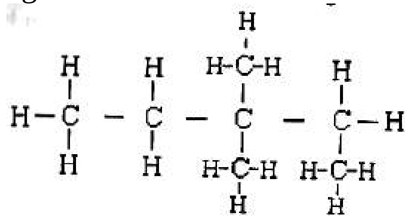
A. it is the largest camber of the heart.

B. it requires much pressure to pump blood to the heart.

C. the blood, which gets into the left ventricles, is under low pressure.

D. the left ventricle has to pump blood to all parts of the body.

74. In the hydrocarbon structure below, how many carbon atoms are in the longest chain?



A. 3 B. 4 C. 5 D. 7

75. In certain breed of dogs, brown fur is dominant over white fur. What is the possible genotype of a dog with brown fur, which produces a white puppy?

A. BB B. BW C. Bw D. Bb

76. The part of the eye responsible for changing the size of the pupil is called
A. Iris. B. lens.
B. Retina. D. cornea.

77. Which of the following parts of the brain is responsible for memory?
A. Hindbrain
B. Cerebrum
C. Cerebellum
D. Medulla oblongata

78. All the following are continuous variations in human being which can be measured **except**
A. weight of pupils in a class.
B. height of pupils in a class.
C. number of red blood cells per cubic millimetres of blood.
D. ability of pupils to roll their tongues.

79. The following are all methods of maintaining sanitation in a community **except**
A. cleaning of gutters.
B. weeding around houses.
C. draining of stagnant water.
D. fitting of mosquito nets on houses.

80. Cretinism in children result from the deficiency of the hormone
A. insulin. B. thyroxin.
C. adrenaline. D. testosterone.

81. Antagonistic muscles are known to
A. react fast to stimuli.
B. relax at the same time.
C. alternately contract and relax.
D. avoid resistance to stimuli.

82. In a simple electrical circuit, the voltage supply is 3.0 V. if the total resistance in the circuit is 6 Ω, calculate the current in the circuit.
A. 0.5 A B. 2.0 A
B. 12.0 A D. 18.0 A

83. The following statements are all reasons why air is a mixture **except**
A. there is no chemical formula for air.
B. the constituents of air are not in a fixed proportion.
C. air has weight and occupies volume.
D. the constituents of air can be separated by physical means.

84. The velocity of sound in steel is 500 ms^{-1}. If the wavelength of the sound is 0.2 m, calculate its frequency.
A. 100 Hz B. 250 Hz
C. 1000 Hz D. 2500 Hz

85. Which of the following is not a magnetic material?
A. Iron B. Nickel

C. Cobalt D. Magnesium

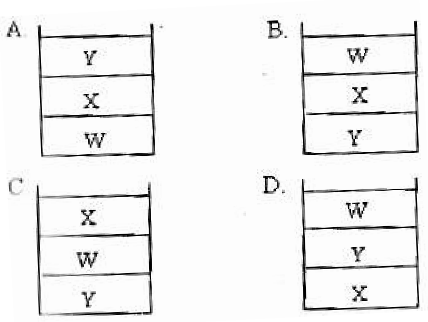
86. Copper sulphate solution and few drops of sodium hydroxide solution were added to food substance and shaken. A violet colour was observed. Which food substance was present?

A. Fat B. Sugar
C. Starch D. Protein

87. A tissue is defined as

A. a group of cells performing a function.
B. different types of cells performing the same function.
C. a group of cells specialized to perform the same function.
D. a group of similar cells, which perform the same function.

88. Two liquids X and Y of relative densities 0.8 and 0.6 respectively are mixed with water W. which of the following mixture arrangements will occur?



89. Which of the following statements are true of the equation below?

$A_{2(g)} + 3B \rightarrow 2AB_{3(g)}\ \Delta H = -X\ kJ\ mol^{-1}$

I. The equation is exothermic
II. Increase in pressure increases the rate of forward reaction.
III. Introduction of more $A_{2(g)}$ decreases the concentration of $AB_{3(g)}$

A. I and III only
B. II and III only
C. I and II only
D. I, II and III

90. When white light is dispersed, the following colours are observed **except**

A. Orange. B. yellow.
C. Green. D. magenta.

91. Which of the following instruments is not used for the purpose stated for it?

A. Spring balance measures the mass of a body.
B. Thermometer is used to measure the degree of hotness or coldness of a body.
C. Vernier callipers are used to measure the internal and external diameters of cylindrical objects.
D. Micrometer screw gauge is used to measure small diameters.

92. Which of the following is classified as an energy-giving food?

A. Egg B. Rice

C. Beef D. Fish

93. Which of the following are stages in the manufacture of soap?

I. Saponification
II. Salting out
III. Washing with water
IV. Distillation

A. I and II only
B. I and IV only
C. II and IV only
D. I, II and III only

94. A body of mass 2.5 kg is raised 4.0 m above the ground. Calculate the work done. [$g = 10\ ms^{-2}$]

A. 1.0 J B. 10.0 J
C. 25.0 J D. 100.0 J

95. Which of the following instruments does not depend on pressure for its operation?

A. Siphon B. Force pump
C. Thermometer D. Barometer

96. Which of the following is not true of the image formed by an object placed in front of a plane mirror? The image is formed by

A. refraction, its virtual, inverted and of the same size as the object.
B. reflection, its real, erect and of the same size as the object.
C. refraction, its virtual, erect and of the same size as the object.
D. reflection, its virtual, erect and of the same size as the object.

97. Which of the following is/ are source(s) of electricity?

I. Dynamo
II. Battery
III. Solar panel

A. I only B. II only
C. III only D. I, II and III

98. Which of the following devices operate by going through the following energy transformations?
Electrical energy → magnetic energy →kinetic energy → sound energy

A. Electric fan B. Electric kettle
C. Electric bell D. Transformer

99. All the following may produce acid **except**

A. muscles.
B. grapefruit.
C. fats and oils.
D. ash from cocoa husk.

100. In which of the following instruments are lenses not employed?

A. Microscope
B. Spectacles
C. Terrestrial telescope
D. Pinhole camera

101. Which of the following statements about the components of a thermos flask is true?

A. The silvery surface of the glass reduces heat loss by radiation.
B. The glass material reduces heat loss by convection.
C. The stopper reduces heat loss by radiation.
D. The vacuum reduces heat loss by radiation.

102. Isotopes of a particular elements have the same

A. mass number
B. atomic mass.
C. atomic number.
D. number of neutrons.

103. A substance which has a pH of 1.5 is considered as a

A. strong acid. B. weak acid.
C. strong base. D. weak base.

104. It took 5 minutes for a lump of calcium carbonate to react with HCl but 3 minute for the same mass of powdered calcium carbonate to undergo same reaction. Which of the following factors accounts for the differences in the reaction rates?

A. Temperature of the system
B. Volume of the system
C. Concentration of calcium carbonate
D. Surface area of calcium carbonate

105. The main disadvantage in the use of iron in the construction industry is that it

A. is too soft. B. is too hard
C. is too bulky D. corrodes easily

106. Consider the equilibrium reaction below:

$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$ + heat energy.

Which of the following will happen when the temperature of the reaction is lowered?

A. The equilibrium will shift to the right.
B. The equilibrium will stop completely.
C. The equilibrium will shift to the left.
D. The equilibrium will remain the same.

107. The alloy duralumin contains

A. copper and aluminium.
B. B. zinc and aluminium.
C. chromium and aluminium.
D. D. tin and aluminium.

108. Semiconductors are used in the manufacture of

A. magnets. B. transformers.
C. transistors D. electric cables.

109. Fats and oils are the esters formed from the reaction between

A. glycerine and long chain fatty acids.
B. glycerol and long chain fatty acids.
C. glycerine and sodium hydroxide.
D. glycerol and sodium hydroxide

110. Which of the following is not a fractional distillation of petroleum?

A. Ethanol B. Kerosene
C. Diesel D. Butane

111. The following are all synthetic polymers except

A. nylon. B. polythene.
C. cellulose. D. polystyrene.

112. A body of mass 4 kg absorbs 2.0 x 10^5 J of heat. If its temperature is raised by 25 ^oC, calculate the specific heat capacity of the body.

A. 2.0 x 10^3 $Jkg^{-1}K^{-1}$
B. 5.0 x 10^4 $Jkg^{-1}K^{-1}$
C. 8.0 x 10^6 $Jkg^{-1}K^{-1}$
D. 1.3 x 10^6 $Jkg\text{-}^1K^{-1}$

113. Which of the following confirms that light travels in a straight line?

I. Absorption of light
II. Eclipse of the sun
III. Shadow formation

A. I only B. III only
C. I and II only D. II and III only

114. Light ray passes from glass to air at a certain angle of incidence. If the refracted ray lies along the glass-air interface, the angle of incidence is called

A. contact angle.
B. angle of deviation.
C. critical angle
D. angle of emergence.

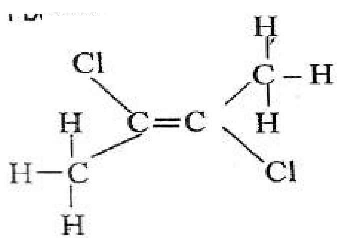
115. The distance between two successive compressions of sound wave is its

A. pitch. B. frequency.
E. wavelength. D. amplitude.

116. Which of the methods listed below is not used in the purification of water?

A. Boiling B. Distillation
C. Sterilization D. Fermentation

117. What is the systematic name of the compound whose structure is given below?



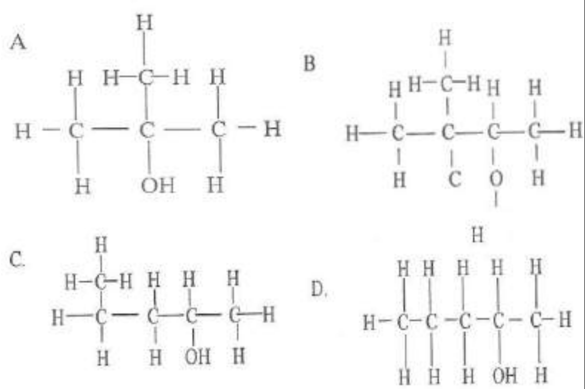
A. 1,2 - dichlorobutene
B. B. 1.2 – dichlorobut-2-ene

C. Trans–1,2dichlorobut-2-ene
D. D. cis1,2 dichlorobut-2-ene

118. Which of the following factors will not change the position of an equilibrium reaction?
I. Temperature
II. Concentration
III. Nature of reactants
A. III only B. I and II only
B. II and III only D. I, II and III

119. Water was added to a quantity of soil and the mixture shaken and filtered. When red litmus solution was added to the filtrate, the filtrate turned blue. The result shows that the soil is
A. acidic B. alkaline
C. clayey D. neutral

120. The structural representation of 3-methylbutan-2-ol is



121. Which of the following conditions is a symptom of food poisoning?
A. Worm infestation
B. Stomach ache
C. Presence of blood in urine
D. Reduction in body temperature

122. The number of positive charges on an alpha particle is
A. 2. B. 3. C. 4. D. 6.

123. A farming practices that is most effective for the maintenance of soil fertility is
A. monocropping. B. ploughing.
C. terracing. D. weeding.

124. Which of the following compounds is a hydrocarbon?
A. C_2H_6 B. $C_6H_{12}O_6$
C. C_2H_5OH D. CH_3COOH

125. A chemical substance that can be used in the laboratory to distinguish an alkaline from alkene is
A. hydrochloric acid.
B. bromine water.
C. cobalt chloride paper.
D. anhydrous copper sulphate.

126. A particle changed its velocity from 10 ms^{-1} to 15 m^{-1} in 8 seconds. Determine its acceleration.
A. 0.63 ms^{-2} B. 1.25 ms^{-2}
C. 1.66 $ms-^{2}$ D. 4.00 ms^{-2}

127. Which of the following bones form the axial skeleton in humans?

I. limb and limb girdle.
II. The vertebral column
III. The skull
IV. Ribs and sternum

A. I and II only B. I and III only
C. II and III only D. II and IV only

128. A farming method that is regarded as environmentally friendly is

a. bush burning
b. land rotation.
c. organic farming.
d. shifting cultivation.

129. Which of the following statements about the absorption of heat by a substance are correct?

I. The quantity of heat in the substance increases.
II. The molecules of the substance move faster.
III. The volume of the substance changes.

A. I and II only B. I and III only
C. II and III only D. I, II and III

130. A homozygous black rabbit is mated to a white mouse. If the black colour is dominant to white colour, determine the colour of the young rabbit.

A. All will be black.
B. All will be white.
C. Some will be black and some will be black.
D. They will be partly white and partly black.

131. A simple electrical circuit has a voltage supply of 12 V. If the total resistance in the circuit is 6 Ω, determine the current in it.

A. 0.5 A B. 2.0 A
C. 6.0 A D. 18.0 A

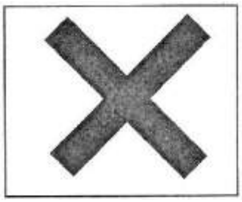
132. Steel is preferred to pure iron for constructional work because

A. steel is more shiny that iron.
B. iron possess more health hazard than steel.
C. steel is malleable while iron is brittle.
D. steel is cheaper than iron.

133. Which of the following relationships represents a food chain in a fish pond?

A. Phytoplankton→zooplankton→tilapia→catfish
B. Zooplankton→phytoplankton→tilapia→catfish
C. Phytoplankton→zooplankkton→catfish→tilapia
D. Zooplankton→phytoplankton→catfish→tilapia

Use the illustration of the hazard symbol below to answer questions **134** *and* **135.**



134. If the hazard symbol is shown on a bottle of a regeant bottle, it implies that the content is

A. corrosive. B. irritant.
C. oxidizing. D. radioactive.

135. An example of a chemical substance that is associated with this symbol is

A. ethanol.
B. calcium oxide.
C. mercury.
D. potassium hydroxide.

136. A sharp knife cuts deeper into a piece meat than a
blunt one because the

A. sharp knife exerts less pressure.
B. sharp knife has a greater area of contact.
C. blunt knife has a smaller area oof contact.
D. blunt knife exerts less pressure.

137. If the cock to hen ratio on a farm is 1:15, determine the number of cocks required to serve 60,000 hens.

A. 840 B. 3,700 4,000 D. 9,000

138. Heat is generated in the human body during the process of

A. defraction. B. expiration.
C. shivering. D. sweating.

139. Water is heated in a sauce pan from 20 ^{o}C to 80 ^{o}C. The heating results in increase in the

A. rate of evaporation.
B. mass of water.
C. density of water.
D. number of water molecules.

140. Fingerlings are stored for 24 hours before transportation in order to

A. reduce faecal contamination.
B. reduce overcrowding.
C. ensure healthy growth.
D. enhance pond fertilization.

141. An observer sees a distant flash of lightening before hearing the accompanying thunder because

A. light travels at a faster speed than sound.
B. light waves have lower frequency than sound Waves
C. sound waves experience resistance in the cloud.
D. the source of the thunder is farther away from the observer.

142. Which of the following substances is present in both glomerular filtrate and urine of a healthy person?
A. amino acids B. blood protein
C. glucose D. urea

143. A solution of 250 cm^3 HCl has a concentration of 2 mol dm^{-3}. Determine the number oof moles of the acid.
A. 0.25 moles B. 0.50 moles
C. 1.00 moles D. 1.50 moles

144. A soft magnet is the type which
A. maintains its magnetism permanently.
B. can be demagnetize with ease.
C. can easily be cut into smaller pieces.
D. attracts other magnetic materials.

145. Which of the following conditions in humans is a symptom of schistosomiasis?
A. Blood in urine B. Cataract
C. chest pains D. Constipation

146. Seed is the planting material for the propagation of all the following crops **except**
A. citrus. B. cocoa.
C. oil palm. D. plantain.

147. Which of the following substances is a anaturally occurring weak acid?
A. Ammonia B. Carbon dioxide
C. citric acid D. Hydrochloric acid

148. The level of classification that embraces the largest number of organisms is the
A. class. B. genus.
C. order. D. phylum.

149. All the following characteristics of a poultry egg are determined by the feed except the
A. shape. B. shell colour.
C. size. D. yolk colour.

150. Carbon monoxide is poisonous to humans when inhaled because it
A. enters the brain and kills the nerves.
B. enters the stomach and induces vomiting.
C. blocks the path of oxygen with the blood.
D. causes severe headache.

151. When ammonium salt is heated gently, it changes from
A. gas to liquid. B. liquid to gas.
C. solid to gas. D. solid to liquid.

152. An important factor that influences the choice of site for a fish pond is
A. Nature of the soil.
B. Gradient of the landscape.
C. availability of shade.

D. nearness to the settlement.

153. A quantity of water of mass 4.0 g at a temperature 50 °C was mixed thoroughly with 6.0 g water at 15 °C. Determine the final temperature of the mixture.
[Specific heat capacity of water is 4200 J kg^{-1} $°C^{-1}$]
A. 29°C B. 35°C C. 44°C D. 55 °C

154. Which of the following statements about anaerobic respiration is correct?
A. Carbon dioxide and water are produced.
B. Oxygen gas is released.
C. Oxygen gas is used up.
D. Glucose is broken down.

155. Soap may produce an etching effect on the skin when it contains excess
A. alkali. B. alkanol.
C. glycerol. D. oil.

156. The level of classification that embraces the largest number of organisms is the
A. class. B. genus.
C. order. D. phylum.

157. Aluminium is used in
A. the extraction of gold.
B. making coins.
C. the manufacture of ammonia.
D. making household utensils.

158. An example of biotechnological process is
A. esterification. B. fermentation.
C. hydrogenation. D. neutralization.

159. Which of the following layers of soil contains the largest population of living organisms?
A. Top soil B. Subsoil
C. Parent material D. Bedrock.

160. If the temperature of a substance is 30°C, determine the equivalent temperature on the Kelvin scale.
[Take absolute temperature to be 273K]
A. 240K B. 270K C. 300 K
D.330K

161. The part of the brain responsible for involuntary activities is
A. cerebral cortex. B. hypothalamus.
C. optic lobe. D. pituitary gland.

162. Which of the following conditions is likely to cause an epidemic in a community?
I,. Poor ventilation
II. living in a room with pests
III. poor sanitation

A. I and II only B. I and III only
C. II and III only D. I, II and III

163. Water is heated in a sauce pan from 20 °C to 80 °C. The heating results in increase in the

A. rate of evaporation.
B. mass of water.
C. density of water.
D. number of water molecules.

164. When a potted seedling is stored in a cupboard for 12 hours, it loses all the starch in its leaves because
A. the starch is converted to glucose in the dark.
B. the starch is oxidized to liberate energy.
C. heat in the cupboard causes the starch to melt.
D. darkness destroys the starch completely.

165. Immunization of farm animals results in the production of
A. antibodies.
B. antigens.
C. red blood cells.
D. white blood cells

166. Which of the following factors will affect the boiling and freezing points of water?
A. Adhesive forces molecules and container
B. density of water
C. atmospheric pressure
D. surface tension

167. The **main** reason for conducting blood tests on patients before carrying out blood transfusion is to
A. Select blood which has enough antibodies.
B. determine the percentage of active red blood cells.
C. avoid giving them incompatible blood.
D. diagnose the disease in them.

168. When a cube of sugar is treated with concentrated sulphuric acid, a black product is formed due to
A. dehydration. B. esterification.
C. oxidization. D. neutralization.

169. Poor sanitation enhances the spread of the disease known as the
A. asthma. B. bilharzias.
C. cholera. D. onchocerciasis.

170. Which of the following services are provided by farm animals?
I. Traction for farmwork
II. raw material for shoe factory
III. raw material for herbicides

A. I and II only B. I and III only
C. II and III only D. I, II and III

171. A wave of wavelength 40.0 m moves at a speed of 440 ms^{-1}. Determine the frequency of the wave.

A. 0.01 Hz B. 11.00 Hz
C. 99.00 Hz D. 440.00 Hz

172. Which of the following diseases is associated with the liver in humans?
A. Appendicitis B. Arteriosclerosis
C. Bronchitis D. Cirrhosis

173. A metal block is raised to a height above ground level. The type of energy possessed by the block at that height is
A. chemical. B. kinetic.
C. heat. D. Potential.

174. Given that Avogadro constant is 6.02×10^{23}, determine the number of atoms in 0.01 mole of a substance.
A. 6.02×10^{20} atoms
B. 6.02×10^{21} atoms
C. 6.02×10^{23} atoms
D. 6.02×10^{26} atoms

175. The gestation period in the reproduction cycle refers to the period between
A. ejaculation and fertilization.
B. fertilization and birth.
C. birth and lactation.
D. fertilization and implantation.

176. Which of the following compounds is a hydrocarbon?
A. C_2H_6 B. $C_6H_{12}O_6$
C. C_2H_5OH D. CHCOOH

177. Which of the following conditions is a symptom of food poisoning?
A. Worm infestation
B. Stomach ache
C. Presence of blood in urine
D. Reduction in body temperature

178. The hedge plant can be made to grow more lateral branches by
A. watering the plant regularly.
B. applying fertilizer to the plant.
C. erecting a fence to support the plant.
D. pruning the plant regularly.

179. Which of the following characteristics is an example of continuous variation?
A. Blood group B. Sickle cell
C. Skin colour D. Tongue rolling

180. The electron configuration of sodium is
A. 2, 1, 8 B. 2, 2, 7
C. 2, 7, 2 D. 2, 8, 1

181. The group to which reptiles is an example of is
A. an order. B. a family.
C. a class. D. a genus.

182. The chemical compound that can remove both types of hardness in water is
A. $NaCO_3$ B. $NaHCO_3$
C. $Ca(OH)_2$ D. $Na2SO_4$

183. A liquid has relative density of 12.2. Calculate its density. (Density of water = 1.0×10^3 kgm^{-3})

A. 1.220 kgm^{-3} B. 12.20 kgm^{-3}
C. 122.00 kgm^{-3} D. 12200 kgm^{-3}

184. The correct order used in classifying organisms is

A. phylum, species, class, genus.
B. phylum, class, genus, species.
C. phylum genus, species, class.
D. phylum, class, species, genus.

185. The component of air used by welders is

A. N_2. B. C_2H_2. C. CO_2. D. O_2.

186. When a hydrometer floats in a liquid, it means that the

A. weight of the liquid displaced is equal to the weight of the hydrometer.
B. density of hydrometer is equal to the density of the liquid.
C. upthrust on the hydrometer is less than the volume of the hydrometer.
D. volume of the portion of the hydrometer in the liquid is equal to the volume of the liquid displaced.

187. The catalyst used in the manufacture of H_2SO_4 in the contact process is

A. platinum.
B. iron (III) oxide.
C. vanadium (V) oxide.
D. nickel.

188. An object has a mass of 220.0 kg. Calculate its weight. (g = 10.0 ms^{-2})

A. 2200.0 N B. 220.0 N
C. 2.20 N D. 0.22 N

189. The causative organism of poliomyelitis is

A. worm. B. bacterium.
C. protozoan D. virus.

190. Ash from cocoa pod or baobab tree fruit, tastes bitter because it contains

A. acid. B. alkali.
C. alcohol. D. salt.

191. Objects are weightless in space, because

A. the force of gravity exactly balances weight of the objects.
B. there is no air.
C. there is negligible force of gravity.
D. the upthrust on the objects is equal to the weight of the objects.

192. Germs enter the human body through all the following **except**

A. cuts in the skin. B. breathing.
C. bites of animals. D. sneezing.

193. The centre of gravity of a body is the point

A. where its original weight acts.
B. where the body can be stable.
C. where the total mass of body acts.

D. which divides the body into two equal parts.

194. A reversible reaction attains equilibrium when
A. the forward reaction is faster.
B. both the forward and backward reactions are at the same rate.
C. the backward reaction is faster.
D. a catalyst is used.

195. Lack of thyroxin in children results in
A. albinism. B. gigantism.
C. cretinism. D. obesity.

196. Which of the following statements is correct?
A. Wheel barrow is an example of first class lever.
B. Sugar tong is an example of third class lever.
C. In the second class lever the fulcrum is in the middle.
D. The higher the centre of gravity of a the more stable it is.

197. Large scale preparation of NH_3 is based on the reaction:

$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

Which of the following will increase the yield of NH_3?
A. Decreasing the pressure
B. decreasing the volume of H_2
C. Using catalyst
D. Decreasing the volume of the reaction vessel.

198. Gastric juice contains the enzyme known as
A. pepsin. B. lipase.
C. ptyalin. D. erepsin.

199. The pressure at any point in a liquid
A. is not affected by acceleration due to gravity.
B. depends on its depth below the surface of the liquid.
C. is proportional to the cross sectional area of the liquid at that point.
D. depends on the shape of the container.

200. The ore of aluminium is called
A. magnetic. B. aluminate.
C. bauxite. D. duralumin.

The table below shows the functions of four cell organelles E, F, G and H.

Structure	*Function*
E	Connected with oxidation of food in the cell
F	Site for protein synthesis
G	Provides support
H	Site for the synthesis of carbohydrate

Use it to answer questions 201 *to* 203.

201. The letter in the table which represents mitochondria is
A. E B. F C. G D. H

202. Which letter in the table represents chloroplast?

A. H B. G C. F D. E

203. Which of the structures from the table are found only in a plant cell?

A. E and H B. E and F
C. G and F D. G and H

204. A body moves with a velocity of 30.0 ms^{-1}. This speed is increased to 40.0 ms^{-1} in 4.0 s. Calculates the average acceleration of the body.

A. 17.5 ms^{-2} B. 8.8 ms^{-2}
C. 4.4 ms^{-2} D. 2.5 ms^{-2}

205. The following equation represents the laboratory preparation of oxygen from $KClO_3$ using MnO_2 as catalyst.

$2KClO_3 \rightarrow 2KCl + 3O_2$

The oxygen produced comes from the

A. KCl produced.
B. The reaction between $KClO_3$ and MnO_2.
C. $KClO_3$ used.
D. MnO_2 used.

206. Which of the following components is not found in the bicycle pump?

A. Washer B. plunger
C. Barrel D. Pivot.

207. One useful bi-product from large scale production of chlorine is

A. sodium hydroxide. B. ammonia.
C. potassium hydroxide. D. methane.

208. A leaf was heated in alcohol before testing it for the presence of starch. This was to

A. give the leaf a large surface area.
B. remove chlorophyll.
C. soften the leaf.
D. 'kill' the leaf.

209. A boy pushes a stationary car. The car moves reluctantly. This is because the car has

A. a momentum. B. a moment.
C. an inertia. D. an impulse.

210. The general formula for alkanes is

A. CnH_2n B. CnH_{2n-2}
C. CnH_{2n+1} D. CnH_{2n+2}

211. The following features adapts a leaf for photosynthesis **except**

A. possessing of sunken stomata.
B. having many stomata.
C. being thin.
D. Being flat and broad.

212. A hydrocarbon contains

A. hydrogen and any other element.
B. only carbon and hydrogen.
C. carbon, hydrogen and oxygen.
D. carbon and calcium.
E.

213. The main process by which living things release the energy from food is known as

A. assimilation. B. digestion.
C. oxidation. D. fermentation.

214. A machine does 212.0 J of work in 40.0 s. calculate the power developed.

A. 8480.0 W B. 5300.0 W
C. 8.40 W D. 5.30 W

215. Which of the following is produced from vegetable oil?

A. Cheese B. Ethanol
C. Margarine D. Vinegar

216. A major constituent of haemoglobin is

A. iron. B. magnesium.
C. phosphorus. D. sodium.

217. A lever system has a velocity ratio 4. If an effort of 100.0 N is applied to the system to raise a load of 200.0 N, calculate the efficiency of the system.

A. 50.0% B. 25.0%
C. 8.0 % D. 2.0 %

218. Which of the following is a plant pigment?

A. Bile B. Chlorophyll
C. Haemoglobin D. Melanin

219. Bush fire can increase the amount of atmospheric

A. carbon (IV) oxide.
B. chlorofluorocarbons.
C. hydrogen.
D. oxygen.

220. The purpose of the kink in the clinical thermometer is to

A. allow the expansion of the mercury.
B. compensate for the expansion of the gas.
C. prevent the mercury from falling back into the bulb.
D. prevent the mercury from forcing the glass to crack.

221. Which of the following apparatus is used to investigate a sample of various plant species growing in a field?

A. Pooter B. Quadrant
C. Secchi disk D. Wicker-work trap

222. Which of the following is not found in the soil?

A. Air B. Ammonia
C. Mineral D. Water

223. Sand of mass 0.4 kg is heated from 0^oC to 75^oC. Calculate the amount of heat stored in the sand.
(specific heat capacity of sand 800 $Jkg^{-1}\,^oC^{-1}$)

A. 2.4×10^7 J B. 2.4×10^6 J
C. 2.4×10^5 J D. 2.4×10^4 J

224. The structure of capillary blood vessel is related to its function by being

A. Thick walled. B. thick and elastic.
C. Thin walled. D. Very large.

225. Which of the following statements is/ are correct?

I. Mercury thermometers cannot be used below -40 ^{o}C.

II. Alcohol thermometers cannot be used in conditions below -40 ^{o}C.

III. Mercury expands six times as alcohol.

A. I only B. II only
C. III only D. I and II only

226. Burning wood in limited supply of air is likely to produce

A. carbon (II) oxide.
B. carbon (IV) oxide.
C. nitrogen dioxide.
D. sulphur (IV) oxide.

227. Which of the following methods is not internationally recommended for the control of population growth?

A. Use of condom
B. Abortion
C. Intra-uterine devise
D. Coitus interrupt

228. At times, during the eclipse, the tip of the umbra fails to reach the earth's surface. In such a case we have

A. eclipse of the sun.
B. eclipse of the moon.
B. partial eclipse.
C. annular eclipse.

229. A ball and socket joint in the body allows for

A. circular movement.
B. gliding movement.
C. side of side movement.
D. up and down movement only.

230. Nitrogen from the atmosphere gets into the soil by the action of

A. burning. B. fungi.
C. microbes D. rain.

231. A object is placed vertically on the axis of a converging lens of focal length 20.0 cm. If the object distance is 30.0 cm, calculate the magnification of the image.

A. 1 B. 2 C. 3 D. 4

232. Carbon dioxide is transported in the blood in the form of

A. bicarbonate ions.
B. carbon (IV) oxide.
C. carbonate ions.
D. carboxy-haemoglobin.

233. Liquefied petroleum gas (LPG) is obtained from crude petroleum by

A. chromatography.
B. crystallization.
C. distillation.
D. filtration.

234. The result of mixing red light and green light is

A. yellow light. B. cyan light.

C. blue light. D. magenta light.

235. The main advantage of sexual reproduction over asexual reproduction is that sexual reproduction
A. is quicker than asexual reproduction.
B. provides more offspring than asexual reproduction.
C. brings about more variation than asexual reproduction.
D. takes place at all times.

236. Which of the following statements is not correct?
A. Te resistance of a semiconductor varies with temperature.
B. Semiconductors can be used as transistors.
C. Semiconductors do not obey Ohm's law.
D. Copper is an example of semiconductor.

237. A by-product obtained from the manufacture of soap is
A. glycerol. B. nylon.
C. polythene. D. vegetable oil.

238. Which of the following statements about x-ray is not correct? They
A. do not affect photographic.
B. travel in straight line.
C. can detect cracks in metal.
D. cause substances to fluoresce.

239. Which of the following is a property of X-ray that is used in medical diagnosis?
A. Ionizing power
B. Magnetizing power
C. Penetrating power
D. Diffracting power.

240. Which of the following statements about fertilization in humans is true?
A. Only one egg is fertilized at a time.
B. It can occur at anytime
C. Many sperms fertilize the ovum.
D. It always occurs inside the female.

241. Capping is practised in yam production in order to
A. Prevent damage to the yam.
B. Reduce soil temperature in the yam mould.
C. Prevent the vines from tailing on the ground.
D. Increase evaporation of excess water from the yam mould.

242. Which of the following devices is indigenous?
A. Gear wheel B. Loom
C. Pulley D. Wheel barrow

243. An adaptation of leaves which ensures efficient photosynthesis in plants is the possession of
A. A small number of chlorophyll in the mesophyll cells.

B. Narrow leaves with a small surface area for the absorption of light.
C. Thick leaves to store sufficient an-mount of water.
D. Thin leaves to allow easy diffusion of carbon dioxide into the mesophyll cells.

244. A block and tackle pulley system has three fixed pulleys and two moving ones. Determine the velocity ration of the system.
A. 2 B. 3 C. 5 D. 6

245. Information is fed into the computer with the help of
A. hard disk drive. B. keyboard.
C. memory. D. program

246. Which of the following structures can be classified as organs?
A. Leaf B. Muscle
C. Palisade layer D. Spermatozoa

247. Which of the following factors may lead to early parenthood?
A. High literacy
B. Improved medical care
C. Poverty
D. Sex education

248. It is not advisable to cover a glass bottle containing sodium hydroxide with a glass stopper because the
A. bottle will crack.
B. bottle will warm up.
C. stopper would get stuck.
D. stopper would prevent the sodium hydroxide from absorbing moisture.

249. The crowding of chick around the sources of heat in a brooder house is an indication that
A. There is an outbreak of disease.
B. The temperature of the brooding house is low.
C. A foreign object has come into the brooding house.
D. There is insufficient illumination in the brooding house.

250. An example of a biotic factor is
A. altitude. B. humidity.
C. predation. D. rainfall.

251.A human diet rich in roughage helps prevent
A. constipation. B. rickets.
C. scurvy. D. skin rashes.

252. The contraction of the biceps
A. strengthens the arm.
B. bends the arm.
C. rotates the arm.
D. raises the arm.

253. The S.I unit of work is
A. A B. J C. N D. W

254. A disadvantage of artificial insemination in livestock production is that
A. the growth and development of offspring is retarded.
B. there could be high rate of vulnerable diseases.
C. the cost of storage and transportation of semen may be high.
D. the cost of transporting animals to the farm may be high.

255. Which of the following organisms can act on carbohydrate to produce alcohol?
A. Algae B. Bacteria
C. Maggot D. Yeast

256. Which of the following substances is a common household chemical?
A. Chloroform
B. Hydrochloric acid
C. Iodine tincture
D. Potassium chloride

257. The main function of gizzard in a domestic fowl is to
A. store the feed.
B. synthesize vitamin B
C. grind the feed.
D. secret digestive juice.

258. The products of photosynthesis are
A. Carbon dioxide and water.
B. glucose and oxygen.
C. Oxygen and carbon dioxide.
D. water and glucose.

259. A collection of data that can immediately be assessed and operated by a data processing system for a specific purpose is called
A. database. B. data series.
C. data store D. data time.

260. Large scale tree planting can be used to reduce air pollution brought about by
A. CO_2. B. CO.
C. dust particles. D. sulphur dioxide.

261. A body of mass 2 kg floats on water. If the density of water is 1000 kgm^{-3}, calculate the volume of the body immersed.
A. $2.0 \times 10^{-3}\ m^3$ B. $2.0 \times 10^{-2}\ m^3$
B. $5.0 \times 10^{3}\ m^3$ D. $2.0 \times 10^{2}\ m^3$

262. Which of the devices listed below are used for harvesting fish?
I. Seine net II. Basket
III. Hook and line IV. Scoop net

A. I, II and III only
B. I, III and IV only
C. II, III and IV only
D. I, II, III and IV

263. An atom is said to be electrically neutral if
A. Neutrons are in the nucleus.
B. Protons are in the nucleus.
C. The number of protons and neutrons in the nucleus are equal.
D. The number of protons and electrons are equal.

264. Under which of the following conditions will maize germinate? If they are placed on
A. moist cotton wool in a test tube.
B. dry cotton wool in a test tube.
C. cotton wool covered with water having oil at the surface.
D. moist cotton wool in a test tube immerses in ice cubes.

265. Exhaled air may readily put off a burning candle because it contains high levels of
A. oxygen and nitrogen.
B. nitrogen and heat.
C. heat and oxygen.
D. carbon dioxide and moisture.

266. A gas which gives off a ‘pop sound with a glowing splint is
A. ammonia. B. carbon dioxide.
C. oxygen. D. hydrogen.

267. The substance present in food which makes the tooth strong is
A. calcium. B. carbohydrate.
C. protein. D. vitamin B.

268. Blinking of the eye is a reflex action because it is
A. A delayed response.
B. initiated from the cerebrum.
C. An automatic response.
D. under the control of the cerebellum.

269. An example of a plant used for hedge is
A. eucalyptus.
B. pride of Barbados.
C. purple bauhinia.
D. royal palm.

270. One disadvantage in the use of steal for construction is that it
A. is ductile. B. is malleable.
C. rusts. D. shines.

271. Which of the following pieces of evidence supports evolution?
A. Body structure of organisms.
B. existence of amphibians.
C. Fossil records of organisms
D. legends of the different races.

272. An athlete of mass 60 kg runs at a velocity of 8 ms^{-1}. Calculate the kinetic energy of the athlete.
A. 4.90×10^2 J B. 1.92×10^3 J
C. 7.84×10^3 J D. 1.44×10^{24} J

273. An important characteristic usually considered when selecting grass for lawn is its ability to
A. Withstand regular cutting.

B. withstand flooding.
C. Produce many seeds.
D. resist pest attack.

274. The rate at which green plants release carbon dioxide during the night is higher than that during the day because
A. Light inhibits respiration.
B. stomata are closed during the day.
C. Darkness is required for respiration.
D. The carbon dioxide produced during the day is used up.

275. Which of the following food substances will react with iodine solution to produce blue-black colour?
A. Cassava B. Meat
C. Orange D. Vegetable oil

276. The binomial system of nomenclature was developed by
A. Aristotle. B. Linnaeus
C. Mendel D. Watson.

277. The directory of a computer contains
A. names, sizes and dates of files.
B. names, sizes and uses of files.
C. names make-up and dates of files.
D. names, make-up and uses of files.

278. Starch is an example of a natural polymer because it
A. consists of naturally occurring monomers.
B. is found in most local food substances.
C. is not easily digestible.
D. can be converted into glucose.

279. Which of the following solvents may be used as an antiseptic?
A. Ammonia solution
B. Kerosene
C. Methylated spirit
D. Turpentine

280. The head of the humerus is connected to the scapula by
A. hinge joint.
B. pivot joint.
C. gliding joint.
D. ball and socket joint.

281. Which of the following materials is magnetic?
A. Brass B. Copper
C. Glass D. Nickel

282. Which of the following factors is usually considered when selecting a site for a fish pond?
A. Presence of tall trees to provide shade for the fish.
B. Presence of loam in the soil at the site.
C. Proximity to a source of market.
D. Proximity to a feed mill.

283. The frequency of a sound wave is 20 Hz. If it has a wavelength of 0.4 m, calculate the speed of the wave.

A. 50.0 ms^{-1} B. 40.0 ms^{-1}
C. 20.0 ms^{-1} D. 8.0 ms^{-1}

Use the information below to answer questions **284** *and* **285**.

The following are metabolic waste products:

I. water II. Mineral salt
III. Urea IV. Carbon dioxide

284. Which of the products above are excreted through the skin?

A. I and II only
B. I and III only
C. I, II and III only
D. II, III and Iv only

285. The product resulting from the breakdown of protein is

A. I B. II C. III D. IV

286. When detecting the odour of a substance it is not advisable to bring the substance very close to the nose because some chemicals

A. are highly flammable.
B. are poisonous.
C. are volatile.
D. have some pungent smell.

287. Which of the following diseases can be controlled by injecting patient with insulin?

A. Asthma B. Diabetes
C. Goitre D. Pneumonia

288. The main reason for providing vents, high in the walls of buildings is to allow

A. a warm humid air in the room to escape.
B. fresh cool air to flow into the room.
C. more light to enter the room.
D. bad odour in the room to escape.

289. In the computer science, DOS stands for

A. digital operating system.
B. direct operating system.
C. disk operating system.
D. distinct operating system.

290. Which of the following methods is a chemical means of birth control?

A. Condom B. Diaphragm
C. The pill D. Vasectomy

291. The function of the cervical vertebrae is to

A. allow the sliding movement.
B. protect the heart and lung.
C. provide support for the skull.
D. provide surface for attachment of thoracic muscles.

292. Which of the following industries can be termed as a large scale industry?
A. Leather tanning
B. Local soap production
C. Kente weaving
D. production of tetraoxosulphate (VI) acid

293. Which of the following statements about mercury as a thermometric liquid is true?
A. It has a low heat capacity.
B. It is colourless.
C. It does not vaporize.
D. It has high density.

294. Which of the following tools is a second class lever?
A. A pair of tongs
B. Bread knife
C. Claw hammer
D. Nut cracker

295. When diluting concentrated tetraoxoxulphate (VI) acid, it is advisable to add the acid slowly to the water while stirring continuously because
A. the acid does not dissolve easily in water.
B. the acid and water mixture can explode.
C. heat is absorbed during the reaction.
D. the water has a low density.

296. To which of the following kingdoms do microscopic unicellular organism with nucleus enclosed in a membrane belong?
A. Animalia B. Plantae
C. Prokaryotae D. Protoctista

297. Which of the following are the reasons for growing ornamental plants?
A. Provision of shade
B. Serving as wind break
C. Improvement of the nutrient status of the soil
D. Beautification of the environment

298. The kidneys are connected to the urinary bladder by the
A. renal artery. B. sphincter muscle
C. ureter D. urethra

299. Which of the following statements about artificial satellites is true? They are
A. Used to transmit visible light to other countries.
B. Used occasionally as weapons of mass destruction.
C. Used to transmit information from one place to another.
D. Filled occasionally with fossil fuel.

300. Which of the following cultural practices conserves the soil?
A. Cover cropping

B. Crop rotation
C. Mechanised farming
D. Shifting cultivation

301. Which of the following activities is a reflex action?
A. Reading a book
B. Shedding of tears
C. Singing
D. Talking

302. The leghorn is a breed of
A. cattle. B. fowl.
C. pig. D. rabbit.

303. Which of the following factors affect the loudness of sound?
A. Amplitude B. Frequency
C. Velocity D. Wavelength

304. A suitable method of reducing erosion on hilly lands is by
A. applying organic manure.
B. erecting barriers
C. creating wind breaks.
D. terracing.

305. One characteristic of fungi is the possession of
A. chlorophyll. B. mycelia.
C. roots. D. stem.

306. Anaemia in piglets is caused by deficiency in
A. calcium. B. iodine.
C. iron. D. sodium.

307. Silicon is an example of elements that can be used for
A. electrolytes. B. insulators.
C. thermostats. D. transistors

308. Hairs found on the epidermis of leaves are meant for
A. absorbing water vapour.
B. breathing.
C. preventing water loss.
D. resisting bush fires.

309. Which of the following electromagnetic radiations has the lowest frequency?
A. γ – ray
B. X – ray
C. Ultra-violet ray
D. Radio wave

310. Which of the following diseases is caused by bacteria?
A. Cerebrospinal meningitis
B. Measles
C. Poliomyelitis
D. Yellow fever

311. All the following statements in Dalton's atomic theory have been modifies in recent years **except**
A. All elements are made up of tiny indivisible particles called atoms.
B. Atoms cannot be created, destroyed or sub-divided.

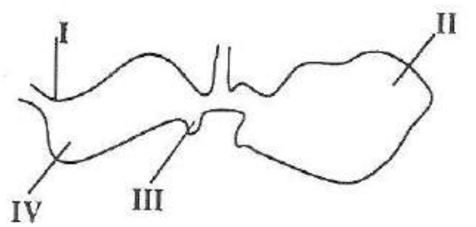
C. Atoms of the same element have the same mass.
D. Atoms of different elements have different properties and different masses.

312. When ammonium chloride salt is heated gently, it changes from
A. gas to liquid. B. liquid to gas.
C. solid to gas. D. solid to liquid.

313. Land breeze is a result of
A. Conduction. B. convection.
C. Evaporation. D. radiation.

314. The type of erosion in which large channels form on the soil surface is referred to as
A. Gully erosion.
B. rill erosion.
C. Sheet erosion.
D. splash erosion.

*The diagram below illustrates a part of the digestive system of a goat. Study it carefully and answer the questions 3***15** *to 3***18**.



315. The part labelled **III** is the
A. abomasums B. omasum.
C. reticulum D. rumen.

316. The true stomach is labelled
A. I B. II C. III D. IV

317. The function of the part labelled **II** is to
A. mix the ingested food with saliva.
B. digest cellulose in blood.
C. absorb the digested food.
D. store excess carbohydrates as glycogen.

318. An open place around a house when grass is established is referred to as
A. bed. B. border.
C. garden. D. lawn

319. Isotopes of the same element have all the following characteristic in common except having the same
A. number of protons.
B. number of electrons.
C. number of neutrons.
D. atomic number.

320. During the formation of rainbow, the rain droplets that hung in the sky serve as
A. a screen against poisonous gas.
B. a medium of dispersing light.
C. a screen against ultra-violet rays.
D. the source of colour for the rainbow.

321. Mastitis disease of cattle is caused by

A. bacteria. B. fungi
C. protozoa. D. viruses.

322. A fruit in which the pappus has been modified into a parachute is likely to be dispersed by
A. Explosion. B. mammal.
C. Water. D. wind.

323. Which of the following services are supplied by farm animals?
I. Tractor for farm work.
II. Raw material for shoe industry.
III. Raw material for herbicides.
A. I and II only B. I and III only
C. II and III only D. I, II and III

324. Which of the following substances is formed when an alkanoic acid reacts with an alkanol?
A. Alkane B. Alkanoate
B. Alkene D. Alkyne

325. Anthrax disease is caused by
A. bacteria. B. fungi
C. protozoa D. Viruses

326. Programmes that sneakily copy themselves onto disks used in the floppy disk drive and cause damage to other programmes are referred to as
A. database. B. graphics.
C. spreadsheet D. viruses.

327. It is advisable to use the slit method of fertilizer application in very humid regions in order to
A. facilitate the absorption of nutrients by plants.
B. meet with the seasonal rainfall.
C. reduce the rate of fertilizer leaching.
D. ensure early maturation of crops.

328. Which of the following devices employ lenses for its function?
A. Driving mirrors
B. Microscope
C. Periscope
D. Pinhole camera

329. Which of the following substances is an example of naturally occurring weak acid?
A. Ammonia
B. Citric acid
C. Hydrochloric acid
D. Sulphuric acid

330. Fast moving vehicles overturn when negotiating curves because of
A. adhesive force.
B. centripetal force.
C. frictional force.
D. tensional force.

331. Which of the following substances is present in both glomerular and urine of a healthy person?

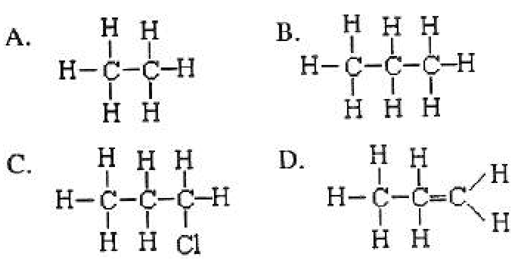
A. Amino acids B. Blood proteins
C. Glucose D. Urea

332. Which of the following peripherals of a computer is an input device?
A. Scanner B Speaker
C. Plotter D. Printer

333. Which of the following reasons does not account for the ability of carbon to form numerous compounds?
A. The exhibition of allotropes
B. The exhibition of isomerism.
C. Formation of covalent linkage with other elements.
D. Formation of carbon-carbon linkage.

334. Which of the following relationships represent food chain in a fish pond?
A. Phytoplankton → zooplankton →tilapia → catfish
B. Zooplankton → phytoplankton → tilapia →catfish
C. Phytoplankton →zooplankton →catfish → tilapia
D. Zooplankton → phytoplankton → catfish → tilapia

335. The structural formula of propane is



336.A crane scoops up a tonne of gravels and moves it to a height of 40.0 m in 10 s. determine the power of the crane. [Take g = 10 ms^{-2}]
A. 0.40 kW B. 4.0 kW
C. 40.0 kW D. 400.0 kW

337. Which of the following compounds will accept hydrogen ion in solution?
A. H_2SO_4 B. CH_3COOH
C. NH_3 D. NaCl

338. Which of the following crops require a sufficient supply of shade for it young plantation?
A. Cocoa B. Oil palm
C. Pawpaw D. Pineapple

339. The mass of an object is 24.0 g on earth. What will be its mass on the moon, it the acceleration due to gravity on the moon is $\frac{1}{6}$ that on the earth?
A. 114.0 kg B. 96.0 kg
C. 24.0 kg D. 4.0 kg

340. The concentration of industries in urban areas may
A. lead to decreased supply of goods and services.
B. lead to high population growth in those areas.
C. affect the energy requirement of the country
D. disturb the ecosystem of the country.

341. By what process is aluminium extracted from its ore?
A. Distillation
B. Electrolysis
C. Reduction
D. Washing under gravity

342. A step-down transformer has
A. equal number of coils in the secondary coil and primary coil.
B. a greater number of coils in the primary coil than the secondary coil.
C. a greater number of coils in the secondary coil than the primary coil
D. the number of primary coli reduced by half.

343. In an experiment to determine the amount of water retained by a soil sample, 50.0 cm3 of water was poured on the soil. It was observed that 38.0 cm3 of water drained through the soil. Determine the percentage water holding capacity of the soil.
A. 24 % B. 38 %
B. 76 % D. 88 %

344. The pressure in any part of a fluid depends on the
A. weight and volume of the liquid.
B. height and density of the liquid.
C. mass and volume of the liquid.
D. surface area and depth of the liquid.

345. Which of the following solvents is best for removing dry ink stain from a fabric?
A. Lemon juice
B. Kerosene
C. Oxalic acid
D. Warm water

346. A coin placed in a beaker filled with water appears to be raised 3.0 cm from the bottom. If the refractive index of water is 4, determine the depth of the water in the beaker.
A. 4.0 cm B. 6.0 cm
C. 8.0 cm D. 12.0 cm

347. A floating balloon may eventually burst because
A. its density has increased.
B. the buoyant force on it increases.
C. the magnetic force on it due to surroundings increases
D. its internal pressure become greater than that of the surroundings.

348. Which of the following factors influence soil formation?
I. Soil organisms
II. Topography
III. Climate
A. I and II only

B. I and III only
C. II and III only
D. I, II and III only

349. Calculate the mass of sodium chloride (NaCl) required to prepare 0.25 M NaCl solution in 100.0 cm3 solution. [Na = 23, Cl = 35.5]
A. 1.46 g B. 5.85 g
C. 58.5 g D. 58.75 g

350. The silvery surfaces in thermos flasks prevent heat loss by
A. Conduction. B. convection.
C. Evaporation. D. radiation.

351. Using a mating ration of 1: 50, determine the number of male goats required to serve 800 female goats.
A. 6 B. 15 C. 16 D. 60

352. The need to provide fuse and earth wire when wiring houses is to
A. protect ourselves and equipment.
B. allow current to flow in the wires.
C. control current flow in the wires.
D. enable electric bulb to light brightly.

353. Combustion, respiration and rusting are similar processes because they all involve
A. burning. B. corrosion
C. oxidation D. reduction.

354. The bending of light at the boundary between two optically different media is caused by the
A. difference in the speed of light in the two media concerned.
B. inability of the light to travel from the first media to the second media.
C. rejection of the light when entering the second media.
D. non-existence of a permanent boundary between the two media.

355. Stagnant water bodies with weeds are potential source of the vector of the organism that cause
A. cholera.
B. river blindness.
C. schistosomiasis.
D. typhoid fever.

356. An object is placed 40.0 cm from a plane mirror. If the mirror is moved 10.0 cm toward the object, through what distance does the object move?
A. 40.0 cm B. 30.0 cm
C. 20.0 cm D. 10.0 cm

357. Which of the following conditions are likely to cause an epidemic in a community?
I. Poor ventilation
II. Living in rooms with pets
III. Poor sanitation
A. I and II only
B. I and III only

C. II and III only
D. I, II and III

358. Night blindness can be prevented by
A. providing enough carbohydrates.
B. taking in vitamin A.
C. injecting iron preparation.
D. vaccinating periodically.

359. The non-metal among the following element is
A. argon. B. manganese.
C. mercury. D. sodium.

360. Which of the following processes brings about the disintegration of a rock?
A. Dissociation
B. Metamorphosis
C. Moulting
D. Weathering

361. The organelle responsible for respiration is
A. endoplasmic reticulum.
B. mitochondrion.
C. plasma membrane.
D. ribosome.

362. An allow containing copper and zinc as it constituent elements is
A. brass. B. bronze.
C. duralumin. D. steel

363. Indigestion is caused by
A. drinking a lot of water.
B. taking unbalanced diet.
C. improper chewing of food.
D. inadequate intake of food.

364. The best way to get the cursor of the mouse to the next line when using the computer is to press the
A. enter key. B. escape key.
C. shift key. D. tab key.

365. An example of a biological polymer is
A. cellulose. B. Dacron.
C. nylon. D. polythene

366. Lime is added to soil mainly to
A. increase water permeability.
B. destroy micro-organisms.
C. remove carbonates.
D. reduce acidity.

367. The heart is divided into left and right valves by the
A. cardiac muscles.
B. papillary muscle.
C. septum.
D. tendon.

368. Which of the following characteristics determine the colour of light?
A. Amplitude B. Frequency
C. Intensity D. Velocity

369. An aqueous solution of an acid
A. liberates hydrogen with active acid.

B. turns red litmus blue.
C. has a soapy feel
D. has a bitter taste.

370. Sexual reproduction has more advantages compared with asexual reproduction mainly because it
A. is quicker than asexual reproduction.
B. provides more offspring than asexual reproduction.
C. brings about variation.
D. takes place all the time.

371. Calculate the kinetic energy of an athlete of mass 60 kg running at 8 ms^{-1}.
A. 2.4×10^3 J B. 2.6×10^3 J
B. 1.92×10^3 J D. 3.84×10^3 J

372. What is the value of *x* in the equation CaCO?
A. 1 B. 2 C. 3 D. 4

373. Exhaled air may readily put off a burning candle because it contains high levels of
A. carbon dioxide and moisture.
B. heat and oxygen.
C. nitrogen
D. oxygen and nitrogen.

374. Which of the following crosses is a test-cross?
A. WW x WW B. WW x Ww
C. Ww x ww D. Ww x Ww

375. Safety devices used in chemistry laboratory include
I. A pair of goggles
II. Nose guard
III. Crash helmet
IV. Hand gloves

A. I, II and III only
B. I, II and IV only
C. II, III and IV only
D. I, II, III and IV

376. All the following preparations are required by poultry farmer before his chickens arrive **except**
A. putting litter in brooder house.
B. putting feed in the feeder three days before the chickens arrive.
C. washing the brooder house.
D. cleaning and drying the equipment in the brooder house.

377. Organisms with their material diffused in the cytoplasm belong to the kingdom
A. animalia. B. fungi.
C. prokaryotae. D. protoctista.

378. The struggle for survival among individuals of the same species arises because
A. they exhibit cannibalism.
B. they share the same limited resources together.

C. each individual produces a large number of offspring.
D. each individual has an equal chance of survival.

379. Which of the following compounds does not show covalent characteristics?
A. Ammonia gas
B. Ethanol
C. Hydrogen chloride
D. Sodium chloride

380. Which of the following statements is not true about boiling and evaporation?
A. Boling takes place throughout the liquid.
B. Evaporation takes place at the surface of the liquid.
C. Both boiling and evaporation take place at specific temperatures.
D. Evaporation results in cooling.

381. The tissue which is likely to stain red when a herbaceous plant is placed in a red solution for about two hours is the
A. cambium. B. parenehyma.
C. phloem. D. xylem.

382. Lateral inversion is a phenomenon in the reflection of light where
A. an image is turned upside down a plane mirror.
B. an image is formed outside a plane mirror.
C. the left hand of an object becomes the right hand of its image in a plane mirror.
D. an image can be projected onto a screen using a converging lens.

383. The organisation that provides humanitarian assistance to victims of war is the
A. Food and agricultural organisation.
B. Red cross
C. United Nations Children's fund.
D. World health Organisation.

384. A fruit that has a fibrous mesocarp with air spaces is likely to be dispersed by
A. animals.
B. explosive mechanism.
C. water.
D. wind.

385. Which of the following statements is not correct about X-rays?
A. They cause fluorescence
B. They are deflected by electric field.
C. They can produce ionization.
D. They can be diffracted.

386. Fish can be stored longest when
A. salted wet.

B. salted and dried.
C. smoked.
D. frozen.

387. Ornamental plants are grown for all the following reasons **except**
A. beautification. B. pasture.
C. pleasure. D. shade.

388. Which of the following factors influences the availability of nutrients in the soil?
A. Soil depth B. Soil colour
C. Soil consistency D. Soil pH

389. Asbestos suits are used by fire fighter because they are
I. Heat-resistant
II. Non-combustible
III. Poor conductors

A. I and II only B. I and III only
C. II and III only D. I, II and III

390.A poultry stockman tests his eggs at the beginning of the incubation period in order to
A. eliminate infertile eggs.
B. know the stage of development of the embryo.
C. eliminate eggs that are underweight.
D. eliminate eggs that will not hatch in time.

391. In Ghana, laws and regulations which prohibit abuse of the environment include
A. legislation on land tenure.
B. legislation on soil conservation.
C. convention on the ozone layer.
D. laws on bush fires.

392. Ventilation is essential in the construction of houses because it
A. creates conditions that favour the growth of fungi.
B. allows cool air to circulate in the room.
C. replaces humid stagnant air with dry air.
D. allows bad odour in the room to escape.

393. The addition of lime $(CaOH)_2$ in the water treatment process is to
A. kill bacteria and other micro-organisms.
B. aid sedimentation of suspended particles.
C. adjust the pH of the water and partially remove hardness.
D. insure effective filtration of raw water.

394. Which of the following ingredients are sources of protein for poultry?
A. Soya bean cake and cassava meal.
B. Blood meal and soya bean cake.
C. Cassava meal and blood meal.

D. Wheat bran and fish meal.

395. The importance of a fuse in an electric circuit is to

A. provide protection against electric shock.
B. protect electrical appliances from being damaged.
C. control electrical leakage.
D. protect the house from thunder and lightening.

396. A brooder covering 15 m^2 can accommodate 100 chicks. How many chicks can be kept comfortably in a brooder of area 3000 m^2?

A. 300 B. 450
C. 20,000 D. 31,000

397. Which of the following processes make use of the beneficial effects of microbes?

I. Baking of bread
II. Manufacturing of antibiotics
III. Brewing of beer

A. I and II only B. I and III only
C. II and III only D. I, II and III

398. Determine the relative molecular mass of $Ca(OH)_2$.
[H = 1.0; Ca = 40.0; O = 16]

A. 42 B. 57 C. 74 D. 87

399. A reflex action

A. can be controlled.
B. depends on previous experience.
C. depends on memory
D. is spontaneous.

400. Which of the following statements is correct?

A. Light travels faster in water than in air.
B. Sound travels faster in air than in wood.
C. Sound waves have elastic rate

401. During a hot day, the receptors in human skin cause

A. dilation of superficial blood vessels.
B. constriction of superficial blood vessels.
C. lowering of hairs on the skin.
D. a decrease in metabolic rate.

402. A system of levers has a velocity ratio of 5. If an effort of 10 N overcomes a load of 40 N, what will be the efficiency of the system?

A. 25 % B. 60 %
C. 80 % D. 100 %

403. The phenotypic ratio selecting the offspring of a cross between homozygous tall and homozygous short orange plant is

A. 1:1 B. 3:1

C. 1:2:1 D. 1:1:2

404. A clinical thermometer is not sterilized in hot water because

A. a vector may contaminate it.
B. it has a narrow bore.
C. water has lower boiling point.
D. it has short range of temperature.

405. The eradication of malaria is proving almost impossible in the tropics because of the

A. wide area of breeding ground for vector.
B. ineffectiveness of insecticides in use against the vector.
C. absence of natural enemy against the vector.
D. presence of thick cuticle that protects the parasite.

406. The point in an object where all forces acting on its particle pass is the object's

A. centre of curvature.
B. centre of gravity.
C. centre of mass.
D. midpoint.

407. Which of the following chemical formulae represents lead (II) oxide?

A. Pb_2O B. PbO
B. PbO_2. D. Pb_2O_4

408. A block ad tackle system has fixed pulleys and four moving ones. What is the velocity ration of the system?

A. 1 B. 4 C. 5 D. 9

409. The pituitary gland is referred to as master gland because

A. it secretes a lot of hormones.
B. it secretes growth hormones.
C. it controls the other members of the endocrine system.
D. its hormones may cause giantism.

410. A body moves with a velocity of 30.0 ms^{-1}. If the speed is increased to 40.0 ms^{-1} in 4 seconds, calculate the acceleration of the body.

A. 2.5 ms^{-2} B. 4.4 ms^{-2}
B. 8.8 ms^{-2} D. 17.5 ms^{-2}

411. Which of the following substances can cause temporary hardness in water?

A. Calcium hydrogen trioxocarbonate (IV)
B. Calcium trioxocarbonate (IV)
C. Potassium trioxocarbonate (IV)
D. Sodium trioxocarbonate (IV)

412. Which of the following bores through grains?

A. Aphids B. Army worms
C. Mealy bugs D. Weevils

413. If 4.52 x 105 J of heat energy was used to vaporize 200.0 g of boiling water, calculate the specific heat of vaporization of the water.

A. 2.26 x 105 Jkg^{-1}
B. 2.26 x 106 Jkg^{-1}
C. 22.6 x 106 Jkg^{-1}
D. 226 x 106 Jkg^{-1}

414. Which of the following is not associated with a classification system?

A. Aristotle B. Linnaeus
C. Mendel D. Mendeleev

415. Which of the following methods cannot be used to protect iron against rust?

A. Enamelling B. Galvanisation
C. Painting D. Riveting

416. One advantage steel has over iron in construction is that it is

A. ductile. B. malleable.
C. elastic. D. shiny

417. The S.I unit of the moment of a force is

A. Kg/m B. kg m/s
C. N/m D. Nn

418. When the number of pulleys in the block and tackle system is increased the

I. Velocity ratio is increased.
II. Mechanical advantage is increased.
III. Efficiency is decreased.

Which of the statements above are correct?

A. I and II only B. I and III only
C. II and III only D. I, II and III

419. The photovoltaic module of a solar panel

A. traps heat from the solar radiation.
B. converts heat energy from solar radiation into electrical energy.
C. use radiant energy to heat water.
D. stores solar energy.

420. The main sources of greenhouse gases are

A. ammonia gas, water vapour and carbon monoxide.
B. carbon monoxide, carbon dioxide and nitrous oxide.
C. aerosol propellants, burning of fossil fuels and methane.
D. water vapour, methane and hydrogen sulphide.

421. chromosomes of living cells are located in the

A. cytoplasm. B. genes.
C. nucleus. D. ribosomes

422. friction in engines can be reduced through

A. lubrication B. overhauling
C. spraying D. vulcanizing

423. which of the following parts of the computer is an output device? The

A. plotter B. printer
C. monitor D. scanner

424. A farming practice that promotes soil erosion is

A. bush burning.
B. contour ploughing
C. mulching
D. strip cropping

425. structures in mammals used for gaseous exchange have

A. large volumes and well supplied with blood vessels.

B. moist surface and well supplied with blood vessels.

C. thin walls with small surface areas.

D. moist surface and small surface areas.

426. The electron configuration of chlorine is

A. 2, 7 ,8 B. 2, 7, 7

C. 2, 8, 7 D. 2, 8, 2

427. Which of the following groups of crops has been correctly matched to it examples?

A. fruits → pawpaw → cashew and carrot

B. Grain legumes → soyabeans → groundnut and millet

C. cereal → maize → rice and sorghum

D. tree crops → cocoa → cassava and coconut

428. The bending of light at the boundaries between two optically different media is caused by

A. inability of the light to travel.

B. difference in the speed of light in the two media concerned.

C. reflection of the light when entering into the second media.

D. non-existence of a permanent boundary between the two media.

429. Which of the following practices are required in the regular maintenance of lawns?

A. cutting, rolling and pruning

B. cutting, weeding and rolling

C. weeding, pruning and staking

D. staking, rolling and pruning

Use the activities listed below to answer questions **430** *and* **431**.

I. Incineration

II. biogas production

III. recycling of waste products

IV. Compost preparation

430. Which of the activities results in the production of manure?

A. I B. II

C. III D. IV

431. Electric power can be generated by

A. I only B. II only

C. I and III only D. II and III only

432. The malpighian body of the mammalian kidney consists of the

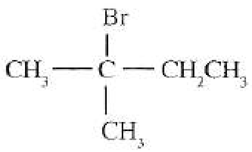
A. cortex and medulla.

B. glomerulus and Bowman’s capsule.

C. collecting tubules and pyramid.

D. glomerulus and loop of Henlé

433. The IUPAC name for the compound



A. 2 - ne.
B. 2 – bromo – 2 – ethylpropane.
C. 2 – methyl – bromobutane
D. 2 – methylbutane.

434. Which of the following factors would favour the siting of an industry in a locality?
I. ease of transportation
II. nearness to the market
III. availability of raw materials

A. I and II only B. I and III only
C. II and III only D. I. II and III only

435. During bread making, yeast is added to the dough as an ingredient because it
A. produces carbon dioxide, which causes the dough to rise.
B. gives the bread a peculiar flavour.
C. helps to preserve the bread.
D. reduces the sugar content to an acceptable level.

436. It is advisable to add washing soda to hard water when washing with soap because of the
A. dirt in the water.
B. acidity of the water.
C. presence of dissolved magnesium salts.
D. presence of bacteria in the water.

437. An ecosystem can be constituted by
A. animals, non-living elements and decomposers.
B. animals, plants and non-living elements.
C. plants, decomposers and non-living elements.
D. decomposers, living things and parasites.

438. A solution with pH 9 is a
A. strong acid. B. strong base.
C. weak acid. D. weak base.

439. Which of the following features must be present in a good fishpond?
I. an outlet
II. grassed walls
III. a screen

A. I and II only B. I and III only
C. II and III only D. I, II and III

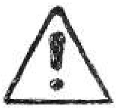
440. Offspring tend to have the same characteristics of their parents because
A. they feed on the same food as their parents.
B. a different appearance will result in extinction.
C. they inherit their genes for the traits.
D. they must replace their like.

441. Which of the following safety symbols warns against risk of electric shock?

A. 

B. 

C. 

D. 

442. Blood is returned from the lungs to the heart through the
A. hepatic artery.
B. hepatic portal vein.
C. pulmonary arteries.
D. pulmonary veins.

443. An alcohol thermometer is not suitable for the determination of boiling point of water because alcohol
A. vaporizes into the bore. B. molecules have high cohesive force.
C. is a transparent liquid. D has high specific heat capacity.

444. Consider the chemical equation:
$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ + heat energy.

I. increase in temperature
II. decrease in temperature
III. increase in pressure
IV. decrease in pressure
A. I and II only B. I and III only
C. II and III only D. II and IV only

445. During the formation of rainbow, the rain droplets that hung in the sky serve as
A. the source of colour for the rainbow
B. a screen against ultra-violet rays.
C. a medium of dispersing light.
D. a screen against poisonous gases.

446. Pollination is the
A. fusion of pollen grains and the stigma
B. transfer of pollen grains from the anther to the stigma.
C. formation of pollen grains in the anther.
D. attracting of insects and birds to the flower.

447. Which of the following nutrients improves the quality of the egg shell when added to the poultry feed?
A. calcium B. iron
C. Magnesium D. nitrogen

448. An effort of 100.0 N is applied to a machine to raise a load of 400.0 N. Calculate the mechanical advantage of the machine.
A. 0.25 B. 0.40

C. 2.50 D. 4.00

449. During land and sea breezes, warm air rises and cool air replaces it because warm air
A. is denser than cool air.
B. has the same density as cool air.
C. is less dense than cold air.
D. contains more water vapour than cold air.

450. Saliva performs the following major functions during digestion of food except
A. Making food soft and easy to swallow.
B. converting starch to maltose.
C. providing appropriate pH medium for salivary amylase.
D. enabling easy passage of food down the oesophagus.

451. Which of the following materials is magnetic?
A. Brass B. Copper
C. Glass D. Nickel

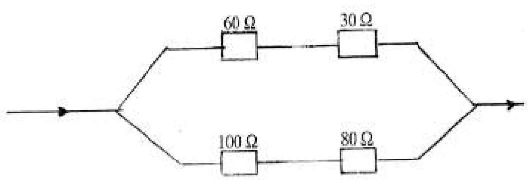
452. Screening of blood is important before transfusion in order to
A. determine the nutrient level of the blood.
B. avoid giving contaminated blood.
C. purify the blood.
D. determine the concentration of oxygen in the blood.

453. Which of the following pairs of diseases is caused by insanitary surroundings?
A. bilharziasis and dysentery.
B. dysentery and measles.
C. cholera and chicken pox.
D. malaria and cholera.

454. The use of soda lime in a photosynthesis experiment is to
A. de-starch the plant.
B. absorb carbon dioxide.
C. soften the plant.
D. decolourise the leaf.

455. A ball and socket joint in the skeletal system of humans allows for
A. circular movement.
B. gliding movement.
C. side and side movement
D. up and down movement.

456. In the electrical circuit illustrated, calculate the effective resistance of the resistors.



A. 60 Ω B. 90 Ω
C. 180 Ω D. 270 Ω

457. Which of the following organic compounds undergo polymerization?
A. C_2H_6 B. $C_2H_4Br_2$
C. C_2H_4 D. C_3H_8

458. Bronze is preferred to copper in making ornamental objects because it
A. is harder than copper.
B. is more resistant to corrosion.
C. has a higher density than copper.
D. is a better conductor of electricity.

459. Which of the following characteristics can be used to distinguish between the same note played on a piano and an organ?
A. Beat B. Loudness
C. Pitch D. Quality

460. The vertical section through the soil showing the distinct layers is known as
A. erosion. B. fertility.
C. profile. D. structure.

461. Which of the following statements about antagonistic muscles are correct? They
A. contract and relax alternatively.
B. are resistant to stimuli.
C. relax at the same time.
D. react fast to stimuli.

462. A force of 30.0 N moves a body through a distance of 1.5 m in the direction of the force. Calculate the work done.
A. 20.0 J B. 28.5 J
C. 31.5 J D. 45.0 J

463. Which of the following substances is necessary for the manufacture of starch by aquatic plants?
A. Carbon dioxide B. Mineral salts
C. Nitrogen D. Oxygen

464. The concentration of industries in urban areas of a country may
A. disturb the ecosystem.
B. lead to high population growth.
C. affect the country's energy requirement.
D. lead to a decrease in the supply of goods and services

465. Using the ting ration of 1:60, determine the number if male goats required to serve 720 female goats.
A. 6 B. 12 C. 15 D. 70

466. The number of hydrogen atoms used in the hydrogenation of ethyne to produce ethane is
A. 1. B. 2. C. 3. D. 4.

467. X-rays are produced when a metal target is hit by
A. fast-moving protons.
B. slow-moving electrons.
C. fast-moving electrons.
D. electromagnetic radiation.

468. A stagnant water body with weeds is a potential habitat of the vector that causes

A. cholera. B. river blindness.
C. schistosomiasis. D. typhoid fever

469. A quantity of sand of mass 0.6 kg is heated from 20°C to 75°C. Calculate the amount of heat stored in the sand. [Specific heat capacity of sand is 800 J kg^{-1} C^{-1}]
A. 30.2kJ B. 26.4kJ
C. 42.0kJ D. 50.2 kJ

470. Which of the following compounds is not ionic?
A. magnesium oxide
B. potassium trioxonitrate (V)
C. sodium tetraoxosulphate (IV)
D. sulphur (IV) oxide

471. A piece of evidence that support evolution is
A. body structure of organisms.
B. existence of amphibians.
C. fossil records of organisms.
D. legends of the different races.

472. A small fruit whose pericarp is expanded to form wings is likely to be dispersed by
A. birds. B. insects.
C. water. D. wind.

473. Which of the following properties is a characteristic of a pure metal?
A. Low melting point
B. Malleable
C. Non-lustrous
D. Poor thermal conductivity

474. The movement of water into a cell causes the cell to
A. become flaccid.
B. become turgid.
C. burst its cell wall.
D. become plasmolysed.

475. A boy stands 7 m away from a vertical plane mirror and sees his image. What is the distance between the boy and his image?
A. 3.5 m B. 7.0 m
C. 10.5 m D. 14.0 m

476. The part of the tooth that may be damaged when vry hot food is put into the mouth is known as
A. dentine. B. enamel.
C. Gum. D. pulp cavity.

477. Which of the following organs in human body are involved in excretion?
I. lungs
II. heart
III kidney
IV. skin
A. I and II only
B. II and III only
C. I, III and IV only
D. II, III and IV only

478. A floating balloon may eventually burst because
A. its density is increased.

B. the upthrust on it due to the surrounding increases.
C. its internal pressure becomes greater than that of the surroundings.
D. the buoyant force on it increases.

479. The unit of measurement of the amount of substance is the
A. gram. B. kilogramme.
C. molar mass. D. mole.

480. Diamond and graphite are
A. allotropes. B. hydrocarbons
C. isotopes. D. polymers

481. Which of the following structures do all cells possess?
A. Nucleus B. Centrioles
C. Cell wall D. Large vacuole

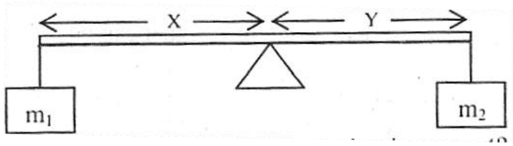
482. Which of the following is not a component of air?
A. Oxygen
B. Water vapour
C. Carbon (IV) oxide
D. Carbon (II) oxide

483. The statement, to every action there an equal and opposite reaction is
A. the law of inertia.
B. Newton's first law of motion.
C. Newton's second law of motion.
D. Newton's third law of motion.

484. Which of the following structures controls the passage of substances in and out of cell?
A. Nuclear membrane
B. Plasma membrane
C. Cell wall
D. Endoplasmic reticulum

485. Pipe borne water contains chlorine which
A. destroys bacteria.
B. makes the water clean.
C. gives taste to water.
D. neutralizes the base in water.

486.A uniform beam of rod is pivoted in horizontal equilibrium as shown below.



Which of the following equation is correct?
A. $m_1y = m_2x$ B. $m_1m_2 = xy$
C. $m_1x = m_2y$ D. $(x+y)m = (x-y)m$

487. Which of the following compounds is an acid salt?
A. $NaHSO_4$ B. Al2(SO4)$_3$
C. Mg(OH)Cl D. $KAl(SO_4).12H_2O$

488. Bacteria belongs to the kingsdom
A. animalia B. prokaryotae
C. protoctista D. Fungi

489. Which of the following devices does not employ the application of pressure?

A. Hydraulic press B. Siphon
C. Bicycle pump D. Hydrometer

490. A salt used as a fertilizer is

A. ammonium trioxocarbonate (IV).
B. potassium trioxosulphate (IV).
C. ammonium tetraoxosulphate (IV).
D. sodium chloride.

491. Which of the following statements is correct?

I. Inertia and momentum are identical.
II. The unit of inertia is kilogram metre per second.
III. Momentum is the product of mass and velocity.

A. I only B. II only C. III only
D. I and II only

492. In photosynthesis, light energy is converted to

A. chemical energy.
B. heat energy.
C. potential energy.
D. kinetic energy.

493. Which of the following statements is corrects?

A. Sound is a form of energy measured in Watts.
B. Kinetic energy is energy due to position.
C. Light energy is measured in hertz.
D. Electrical power is measured in Watts.

494. Combustion of organic fuel increases the

A. inert gases in the air.
B. carbon dioxide content in the air.
C. oxygen content in the air.
D. nitrogen content in the air.

495. Which of the following statements is/ are correct about equilibrium reaction?

I. Equilibrium is attained when the rate of both the forward and backward reactions is the same.
II. Equilibrium position shifts when one of the equilibrium factors change
III. A catalyst increases the rate of forward and backward reactions equally.

A. I, II and III B. III only
C. II only D. I only

496. A thermometer is used to measure

A. heat energy of a body.
B. specific heat capacity.
C. temperature.
D. latent heat of a body.

497. A student adds a substance to a reaction mixture and the rate of reaction increases without any chemical change in the substance. The substance added can be called

A. an oxidant. B. an reductant.
C. an indicator. D. a catalyst.

498. In ecological study, small animals like ants found on leaves and crevices can be collected by

A. sweep net. B. pooter.
C. wicker-work trap. D. transect.

499. Which of the following substances is a conductor of electricity?

A. Sulphur B. Graphite
C. Phosphorus D. Diamond

500. How much heat is needed to raise the temperature of 5.0 kg of water by 5.0 ^{o}C?
[specific heat capacity of water is 4200 $Jkg^{-1}\,^{o}C^{-1}$]

A. 50.0 kJ B. 85.0 kJ
C. 105.0 kJ D. 115.0 kJ

ANSWERS TO THE OBJECTIVE TEST

1.	**A**	47.	**B**	93.	**D**	139.	**A**	185.	**D**	231.	**B**	277.	**A**	323.	**A**	369.	**A**	415.	**D**	461.	**D**
2.	**D**	48.	**D**	94.	**D**	140.	**A**	186.	**A**	232.	**B**	278.	**A**	324.	**A**	370.	**C**	416.	**D**	462.	**D**
3.	**D**	49.	**D**	95.	**C**	141.	**A**	187.	**C**	233.	**C**	279.	**C**	325.	**A**	371.	**B**	417.	**D**	463.	**A**
4.	**C**	50.	**C**	96.	**B**	142.	**D**	188.	**A**	234.	**A**	280.	**A**	326.	**D**	372.	**B**	418.	**B**	464.	**B**
5.	**B**	51.	**B**	97.	**D**	143.	**B**	189.	**D**	235.	**C**	281.	**D**	327.	**C**	373.	**C**	419.	**D**	465.	**B**
6.	**B**	52.	**B**	98.	**C**	144.	**B**	190.	**B**	236.	**D**	282.	**C**	328.	**B**	374.	**D**	420.	**A**	466.	**B**
7.	**A**	53.	**A**	99.	**D**	145.	**A**	191.	**C**	237.	**A**	283.	**D**	329.	**B**	375.	**B**	421.	**C**	467.	**C**
8.	**C**	54.	**A**	100.	**D**	146.	**D**	192.	**D**	238.	**A**	284.	**C**	330.	**B**	376	**C**	422.	**A**	468.	**C**
9.	**C**	55.	**B**	101.	**A**	147.	**C**	193.	**A**	239	**C**	285.	**C**	331.	**D**	377.	**C**	423.	**B**	469.	**B**
10.	**D**	56.	**D**	102.	**C**	148.	**D**	194.	**B**	240.	**D**	286.	**D**	332.	**A**	378.	**B**	424.	**A**	470.	**D**
11.	**D**	57.	**C**	103.	**A**	149.	**A**	195.	**C**	241.	**B**	287.	**B**	333.	**C**	379.	**D**	425.	**B**	471.	**C**
12.	**B**	58.	**C**	104.	**D**	150.	**C**	196.	**B**	242.	**B**	288.	**B**	334.	**A**	380.	**C**	426.	**C**	472.	**D**
13.	**C**	59.	**C**	105.	**D**	151.	**C**	197.	**B**	243.	**D**	289.	**C**	335.	**B**	381.	**C**	427.	**D**	473.	**B**
14.	**D**	60.	**A**	106.	**C**	152.	**D**	198.	**A**	244.	**C**	290.	**C**	336.	**B**	382.	**C**	428.	**B**	474.	**B**
15.	**A**	61.	**B**	107.	**C**	153.	**D**	199	**B**	245.	**B**	291.	**C**	337.	**C**	383.	**B**	429.	**B**	475.	**B**
16.	**C**	62.	**C**	108.	**C**	154.	**A**	200	**C**	246.	**A**	292.	**D**	338.	**A**	384.	**C**	430.	**D**	476	**B**
17.	**B**	63.	**C**	109.	**B**	155.	**A**	201.	**A**	247.	**C**	293.	**B**	339.	**D**	385.	**D**	431.	**B**	477.	**C**
18.	**D**	64.	**B**	110.	**A**	156.	**D**	202.	**A**	248.	**C**	294.	**D**	340.	**B**	386.	**B**	432.	**B**	478.	**C**
19.	**B**	65.	**B**	111.	**C**	157.	**D**	203.	**D**	249.	**B**	295.	**B**	341.	**B**	387.	**B**	433.	**B**	479.	**D**
20.	**D**	66.	**A**	112.	**A**	158.	**B**	204.	**D**	250.	**C**	296.	**D**	342.	**B**	388.	**D**	434.	**D**	480.	**A**
21.	**D**	67.	**A**	113.	**D**	159	**A**	205.	**C**	251.	**A**	297.	**D**	343.	**A**	389.	**D**	435.	**A**	481.	**A**
22.	**C**	68.	**A**	114.	**C**	160.	**D**	206.	**D**	252.	**B**	298.	**C**	344.	**D**	390.	**A**	436.	**C**	482.	**D**
23.	**C**	69.	**B**	115.	**C**	161.	**B**	207.	**B**	253.	**B**	299.	**C**	345.	**B**	391.	**B**	437.	**B**	483.	**D**
24.	**D**	70.	**D**	116.	**D**	162.	**B**	208.	**D**	254.	**B**	300.	**B**	346.	**D**	392.	**B**	438.	**D**	484.	**B**
25.	**B**	71.	**C**	117.	**B**	163.	**A**	209.	**C**	255.	**D**	301.	**B**	347.	**D**	393.	**C**	439.	**B**	485.	**A**
26.	**C**	72.	**B**	118.	**A**	164.	**A**	210.	**D**	256.	**C**	302.	**B**	348.	**D**	394.	**B**	440.	**C**	486.	**C**
27.	**C**	73.	**D**	119	**B**	165.	**A**	211.	**A**	257.	**C**	303.	**B**	349.	**A**	395.	**B**	441.	**B**	487.	**A**
28.	**A**	74.	**B**	120.	**D**	166.	**C**	212.	**B**	258.	**D**	304.	**D**	350.	**D**	396.	**C**	442.	**D**	488.	**B**
29.	**A**	75.	**C**	121.	**B**	167.	**C**	213.	**A**	259.	**A**	305.	**B**	351.	**C**	397.	**D**	443.	**A**	489.	**D**
30.	**B**	76.	**A**	122.	**A**	168.	**C**	214.	**D**	260.	**A**	306.	**C**	352.	**A**	398.	**C**	444.	**B**	490.	**C**
31.	**C**	77.	**B**	123.	**B**	169.	**C**	215.	**C**	261.	**A**	307.	**D**	353.	**C**	399	**D**	445.	**C**	491.	**C**
32.	**D**	78.	**D**	124.	**C**	170.	**A**	216.	**A**	262.	**A**	308.	**C**	354.	**A**	400.	**C**	446.	**B**	492.	**A**
33.	**B**	79	**D**	125.	**B**	171.	**B**	217.	**A**	263.	**D**	309.	**D**	355.	**C**	401.	**C**	447.	**A**	493.	**D**
34.	**A**	80.	**A**	126.	**A**	172.	**D**	218.	**B**	264.	**A**	310.	**A**	356.	**B**	402.	**C**	448.	**D**	494.	**B**
35.	**C**	81.	**A**	127.	**C**	173.	**D**	219.	**A**	265.	**D**	311.	**C**	357.	**B**	403.	**A**	449.	**C**	495.	**A**
36.	**A**	82.	**A**	128.	**C**	174.	**B**	220.	**C**	266.	**D**	312.	**C**	358.	**B**	404.	**D**	450.	**C**	496.	**C**
37.	**D**	83.	**B**	129.	**D**	175.	**D**	221.	**B**	267.	**A**	313.	**B**	359.	**A**	405.	**A**	451.	**D**	497.	**D**
38.	**C**	84.	**D**	130.	**C**	176.	**A**	222.	**B**	268.	**C**	314.	**A**	360.	**D**	406.	**B**	452.	**B**	498.	**B**
39	**A**	85.	**D**	131.	**B**	177.	**B**	223.	**D**	269.	**B**	315.	**B**	361.	**B**	407.	**B**	453.	**A**	499	**B**
40.	**B**	86.	**D**	132.	**A**	178.	**D**	224.	**C**	270.	**C**	316.	**B**	362.	**A**	408.	**D**	454.	**B**	500.	**C**
41.	**D**	87.	**D**	133.	**A**	179.	**D**	225.	**A**	271.	**C**	317.	**D**	363.	**C**	409.	**A**	455.	**A**		
42.	**C**	88.	**A**	134.	**B**	180.	**D**	226.	**A**	272.	**B**	318.	**D**	364.	**A**	410.	**A**	456	**A**		
43.	**D**	89.	**C**	135.	**B**	181.	**C**	227.	**B**	273.	**A**	319.	**C**	365.	**A**	411.	**A**	457	**B**		
44.	**B**	90.	**D**	136.	**D**	182.	**A**	228.	**C**	274.	**D**	320.	**B**	366.	**D**	412.	**B**	458	**B**		
45.	**C**	91.	**A**	137.	**C**	183.	**D**	229.	**A**	275.	**A**	321.	**B**	367.	**C**	413.	**D**	459	**D**		
46.	**A**	92.	**B**	138.	**C**	184.	**B**	230.	**C**	276	**B**	322.	**D**	368.	**B**	414	**D**	460	**C**		